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Nicola Fusco · Paolo Marcellini
Carlo Sbordone

Mathematical Analysis

Functions of Several Real Variables and Applications

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Preface

There is a long and established tradition of printed manuals on *Mathematical Analysis I*. For lecturers in the faculties of sciences, engineering and architecture, it is therefore easy to select the most suitable ones, according to their own mathematical taste and depending on the objectives of the classes to be offered. There is, on the contrary, general agreement as regards the lack of a similar availability of textbooks on *Mathematical Analysis II*.

Whilst preparing the present book, we thought that we should allow for the possibility of choosing between two levels of presentation. For this reason, the contents are first discussed in an elementary way, and at a successive stage, they are examined from several, more penetrating, angles.

The agile organisation of the subject matter helps instructors to determine effortlessly which parts to present during lectures and where to stop.

In a course addressed to engineering or computer science students, for example, it may be appropriate to discuss multiple integrals from a more concrete point of view, starting from normal domains in the plane and in space. For mathematics and physics students, on the other hand, a more rigorous approach to Riemann's or Lebesgue's theory might be more fitting.

Another possibility is to leave the chapter on the Lebesgue integral for a third-year class. Similarly, we are convinced that the theory of regular surfaces in space should be enough for the majority of undergraduate degrees, while some lecturers might think it useful to teach, in an elementary way, k-dimensional manifolds and the generalisations of Stokes's theorem and of the divergence theorem to $\mathbb{R}^n$.

In the same line of thought, the various chapters' appendices provide further opportunities to go deeper into certain topics: among them, the Ascoli-Arzelà theorem, the regularity of convex functions in $\mathbb{R}^n$, the Gamma function or L^p spaces, all topics that are paramount in modern mathematical analysis. Other instances include the Weierstrass theorem on polynomial approximation of continuous functions or Peano's existence theorem (typically only existence, without uniqueness) for non-linear ODEs and systems under general assumptions.

The authors believe that any textbook can contribute to the success of a lecture course only to a point, and the choices made by lecturers are decisive in this respect. Nonetheless, they still hope that this book will be welcomed by students and colleagues.

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Sequences and Series of Functions

1

1.1 Sequences of Functions: Pointwise and Uniform Convergence

Let I be a set of real numbers and $f_k : I \to \mathbb{R}$ a sequence of real functions defined on I. One says that f_k *converges* to the function $f : I \to \mathbb{R}$ *pointwise on* I whenever

$$\lim_{k\to+\infty} f_k(x) = f(x)\,, \qquad \forall\, x \in I\,.$$

In other words, if for any $\varepsilon > 0$ and any $x \in I$ there exists $\nu_{\varepsilon,x} \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \varepsilon\,, \qquad \forall\, k > \nu_{\varepsilon,x}\,.$$

In general, given $\varepsilon > 0$, the number $\nu_{\varepsilon,x}$ depends on the point x; if, instead, this number is independent of x, one speaks of *uniform convergence*.

Precisely, we say that f_k *converges uniformly on* I to f if, for any $\varepsilon > 0$, there exists $\nu_\varepsilon \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \varepsilon\,, \qquad \forall\, k > \nu_\varepsilon, \quad \forall\, x \in I\,.$$

Equivalently, f_k *converges uniformly on* I to f if, for any $\varepsilon > 0$, there exists $\nu_\varepsilon \in \mathbb{N}$ such that

$$\sup\{|f_k(x) - f(x)| : x \in I\} < \varepsilon, \qquad \forall\, k > \nu_\varepsilon\,.$$

Another way to express the same is saying that f_k *converges uniformly on* I to f if the following condition on the limit of a numerical sequence holds

$$\lim_{k\to+\infty} \sup\{|f_k(x) - f(x)| : x \in I\} = 0\,.$$

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If a sequence of functions converges, pointwise or uniformly, on a set I, then it converges pointwise or uniformly on every subset $I' \subseteq I$, respectively.

Uniform convergence on I implies pointwise convergence, as one deduces from the inequality

$$|f_k(x) - f(x)| \leq \sup\{|f_k(x) - f(x)| : x \in I\},$$

for any $x \in I$. The converse is in general false, as the following examples show.

Example 1 Let a, b be two real numbers and f_k the sequence (see Fig. 1.1) defined on the interval $I = [0, 1]$ by

$$f_k(x) = \begin{cases} a & \text{if} \quad 0 < x \leq 1/k \\ b & \text{if} \quad 1/k < x \leq 1 \end{cases} \qquad \text{or if } x = 0\,.$$

Given $x \in (0, 1]$, for any $k > 1/x$ we have $1/k < x \leq 1$ and therefore $f_k(x) = b$; hence the sequence f_k converges to the function $f(x) = b$ pointwise.

Moreover, since

$$\sup\{|f_k(x) - f(x)| : x \in [0, 1]\} = |a - b|\,,$$

we have uniform convergence only when $a = b$.

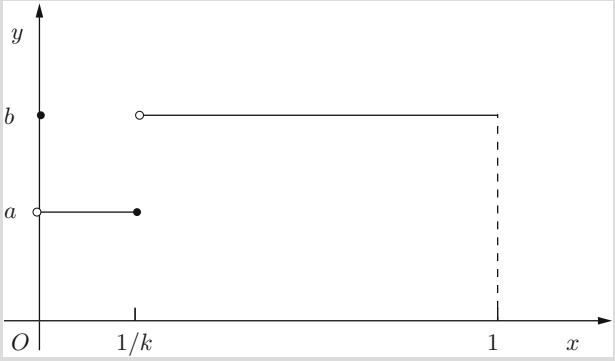


Fig. 1.1

Example 2 The sequence of functions defined on $I = [0, 1]$ by

$$f_k(x) = x^k\,, \tag{1.1}$$

(continued)

Example 2 (continued)
converges for every $x \in I$ to the function

$$f(x) = \begin{cases} 0 & \text{if} \quad x \in [0,1) \\ 1 & \text{if} \quad x = 1\,. \end{cases} \tag{1.2}$$

Furthermore,

$$\sup\{|f_k(x) - f(x)| \,:\, x \in [0,1]\} = \sup\{|f_k(x) - f(x)| \,:\, x \in [0,1)\} =$$

$$= \sup\left\{x^k \,:\, x \in [0,1)\right\} = 1\,.$$

Hence the sequence f_k does not converge uniformly on the interval $I = [0,1]$.

Example 3 Vice versa, it is easy to check that the sequence f_k defined in (1.1) converges uniformly on every interval of type $[0,b]$, with $0 < b < 1$.

In fact, on $[0,b]$ the pointwise limit $f(x)$ is identically zero, so

$$\sup\{|f_k(x) - f(x)| \,:\, x \in [0,b]\} = \sup\left\{x^k \,:\, x \in [0,b]\right\} = b^k,$$

and the least upper bound tends to zero as $k \to +\infty$.

Example 4 Consider the sequence of functions f_k (Fig. 1.2), defined on the real axis $I = \mathbb{R}$ by

$$f_k(x) = \frac{x^2}{k + x^2}\,, \qquad \forall\, k \in \mathbb{N}\,.$$

Figure 1.2 highlights the fact that the set $\{f_k(x) \le 1/2\} = [-\sqrt{k}, \sqrt{k}]$ increases as k increases, and the union over all $k \in \mathbb{N}$ coincides with $\mathbb{R}$. For any $x \in \mathbb{R}$ the pointwise limit vanishes

$$f(x) = \lim_{k \to +\infty} f_k(x) = 0\,,$$

yet the convergence of f_k to f is not uniform on $\mathbb{R}$; in fact

$$\sup\{|f_k(x) - f(x)| \,:\, x \in \mathbb{R}\} = \sup\left\{\frac{x^2}{k + x^2} \,:\, x \in \mathbb{R}\right\} = \lim_{x \to +\infty} \frac{x^2}{k + x^2} = 1.$$

(continued)

Example 4 (continued)

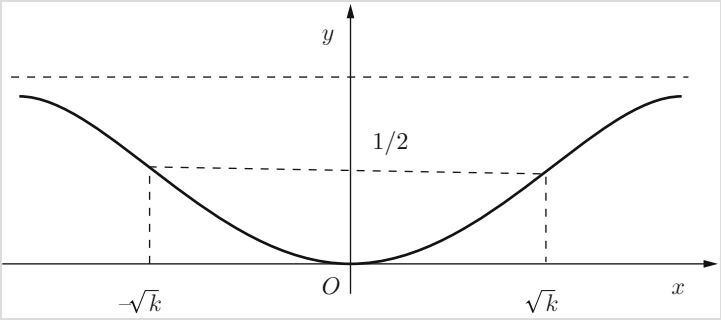


Fig. 1.2 $f_k(x) = \dfrac{x^2}{k + x^2}$

Example 5 The sequence f_k of the previous example converges uniformly to the function $f \equiv 0$ on every interval $[a, b]$, with $a, b \in \mathbb{R}$, $a < b$. In fact, since for any $k \in \mathbb{N}$ the function f_k is increasing for $x \geq 0$ and decreasing for $x \leq 0$, by setting $c = \max\{|a|, |b|\}$ we obtain

$$\sup\{|f_k(x) - f(x)| : x \in [a, b]\} = \sup\left\{\frac{x^2}{k + x^2} : x \in [a, b]\right\} =$$

$$= \max\left\{\frac{a^2}{k + a^2}, \frac{b^2}{k + b^2}\right\} = \frac{c^2}{k + c^2}$$

and the latter quantity tends to zero as $k \to +\infty$.

1.2 First Theorems on Uniform Convergence

Let us start by describing the continuity property of the uniform limit of continuous functions. Suppose $f_k : I \subseteq \mathbb{R} \to \mathbb{R}$ is a sequence of *continuous* functions on the subset I of $\mathbb{R}$, and assume that f_k converges *uniformly* on I to the function $f : I \to \mathbb{R}$; we shall prove that f is continuous on I. Observe that this result does not hold if we only assume that f_k converges to f pointwise. This is what happens, for instance, in Example 2 of the previous section, where the discontinuous function (1.2) is the pointwise limit of the sequence of continuous functions (1.1).

Theorem (Continuity of Limits) *Let $f_k : I \subseteq \mathbb{R} \to \mathbb{R}$ be a sequence of continuous functions that converges uniformly on I to the function f. Then f is continuous.*

Proof Let us verify that f is continuous at x_0, for any given $x_0 \in I$. By the uniform convergence hypothesis, given $\varepsilon > 0$, there exists ν such that

$$|f_k(x) - f(x)| < \varepsilon\,, \qquad \forall\, k > \nu\,, \quad \forall\, x \in I\,.$$

Let us choose $k_0 > \nu$; then clearly for any $x \in I$ we have

$$\begin{aligned} |f(x) - f(x_0)| &\le |f(x) - f_{k_0}(x)| + |f_{k_0}(x) - f_{k_0}(x_0)| + |f_{k_0}(x_0) - f(x_0)| \le \\ &< \varepsilon + |f_{k_0}(x) - f_{k_0}(x_0)| + \varepsilon\,. \end{aligned}$$

Because of the continuity of f_{k_0} it is possible to find $\delta > 0$ such that

$$x \in I, \quad |x - x_0| < \delta \quad \Rightarrow \quad |f_{k_0}(x) - f_{k_0}(x_0)| < \varepsilon$$

and so for $x \in I$, $|x - x_0| < \delta$, we obtain

$$|f(x) - f(x_0)| < 3\varepsilon\,.$$

More generally, we have the following result.

Theorem (Interchange of Limits) *Let $f_k : I \subseteq \mathbb{R} \to \mathbb{R}$ be a uniformly convergent sequence of functions on the set I. Suppose that the limit*

$$\lim_{x \to x_0} f_k(x)$$

exists for any $k \in \mathbb{N}$. Then the two limits

$$\lim_{k \to +\infty} \left(\lim_{x \to x_0} f_k(x) \right), \qquad \lim_{x \to x_0} \left(\lim_{k \to +\infty} f_k(x) \right) \tag{1.3}$$

exist and are equal.

Proof Set $l_k = \lim_{x \to x_0} f_k(x)$ and $f(x) = \lim_{k \to +\infty} f_k(x)$. Given $\varepsilon > 0$, by uniform convergence there exists $\nu \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \varepsilon\,, \qquad \forall\, k > \nu\,, \quad \forall\, x \in I\,.$$

Therefore

$$|f_k(x) - f_h(x)| \le |f_k(x) - f(x)| + |f(x) - f_h(x)| < 2\varepsilon \tag{1.4}$$

when $h, k > \nu$ and for any $x \in I$. Taking the limit in (1.4) as $x \to x_0$, we obtain

$$|l_k - l_h| \le 2\varepsilon, \qquad \forall\, h, k > \nu\,.$$

Then l_k is a Cauchy sequence, hence convergent to a real number, say l. In other words

$$\lim_{k\to+\infty} l_k = l\,.$$

This proves the existence of the first limit in (1.3). There remains to show that

$$\lim_{x\to x_0} f(x) = l\,. \tag{1.5}$$

For this we choose k_0 such that

$$|f_{k_0}(x) - f(x)| < \varepsilon\,, \qquad \forall\, x \in I \tag{1.6}$$

and such that

$$|l_{k_0} - l| < \varepsilon\,. \tag{1.7}$$

Corresponding to k_0 we then pick $\delta > 0$ such that

$$x \in I\,, \quad 0 < |x - x_0| < \delta \quad \Rightarrow \quad |f_{k_0}(x) - l_{k_0}| < \varepsilon\,. \tag{1.8}$$

From (1.6), (1.7), (1.8) we deduce

$$|f(x) - l| \le |f(x) - f_{k_0}(x)| + |f_{k_0}(x) - l_{k_0}| + |l_{k_0} - l| < 3\varepsilon$$

for any $x \in I$ such that $0 < |x - x_0| < \delta$. Put otherwise, (1.5) holds.

Notice that if we put

$$\lim_{x\to x_0} f_k(x) = f_k(x_0)\,, \qquad \forall\, k \in \mathbb{N}\,,$$

in the previous theorem's statement, that is if we assume the f_k are all continuous at x_0, by the equality of the limits in (1.3) we obtain

$$f(x_0) = \lim_{k\to+\infty} f_k(x_0) = \lim_{x\to x_0} f(x)\,.$$

Therefore f is continuous at x_0, which recovers the theorem on the continuity of the limit.

Uniformly convergent sequences of real functions $f_k : I \to \mathbb{R}$, just like sequences of real numbers, satisfy a property similar to Cauchy's criterion.

Uniform Cauchy Criterion *The sequence f_k converges uniformly on the set $I \subseteq \mathbb{R}$ to the function $f : I \to \mathbb{R}$ if and only if, for any $\varepsilon > 0$, there exists $\nu \in \mathbb{N}$ such that*

$$|f_k(x) - f_h(x)| < \varepsilon\,, \qquad \forall\, h, k > \nu\,, \quad \forall\, x \in I\,. \tag{1.9}$$

Proof If f_k converges uniformly on I to f, given $\varepsilon > 0$ there exists $\nu \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \varepsilon\,, \qquad \forall\, k > \nu\,, \quad \forall\, x \in I\,.$$

Hence for $h, k > \nu$ and any $x \in I$ we have

$$|f_k(x) - f_h(x)| \le |f_k(x) - f(x)| + |f(x) - f_h(x)| < 2\varepsilon\,.$$

Conversely, if Cauchy's condition (1.9) holds, for any given $x \in I$, $f_k(x)$ is in particular a Cauchy sequence of real numbers, which therefore converges to a real number that we may call $f(x)$.

Let us check that the sequence f_k converges uniformly to the function f. Given $\varepsilon > 0$, let $\nu \in \mathbb{N}$ be the index for which the Cauchy condition (1.9) holds. Taking the limit as $h \to +\infty$ in (1.9) we obtain the claim:

$$|f_k(x) - f(x)| \le \varepsilon\,, \qquad \forall\, k > \nu\,, \quad \forall\, x \in I\,.$$

1.3 Theorems on Interchanging Limits and Integrals or Derivatives

Let f_k be a sequence of integrable functions on a closed and bounded interval $I = [a, b]$, and suppose the f_k converge to some function f *pointwise* on I. We may ask whether f is integrable on $[a, b]$ and whether

$$\lim_{k\to+\infty} \int_a^b f_k(x)\,dx = \int_a^b f(x)\,dx\,. \tag{1.10}$$

The answer is typically no, as the following examples clarifies.

Example 1 For $x \in [0, 1]$ define (see the graphs in Fig. 1.3)

$$f_k(x) = k\,x\,e^{-kx^2}\,.$$

For any given $x \in [0, 1]$ the numerical sequence $f_k(x)$ tends to zero, hence the sequence f_k converges on $[0, 1]$ pointwise, and the pointwise limit is the function $f(x) = 0$ for every $x \in [0, 1]$. Let us compute the integral

$$\int_0^1 f_k(x)\,dx = \frac{1}{2}\left[-e^{-kx^2}\right]_0^1 = \frac{1}{2}(1 - e^{-k})\,,$$

(continued)

Example 1 (continued)
from which

$$\lim_{k\to+\infty}\int_0^1 f_k(x)\,dx = \frac{1}{2} \neq \int_0^1 f(x)\,dx = 0\,.$$

Note that the sequence f_k does *not* converge uniformly to f on the interval $[0, 1]$. In fact (the reader should do the computation)

$$\sup\{|f_k(x) - f(x)| : x \in [0, 1]\} = \max\{f_k(x) : x \in [0, 1]\} = f_k(x_k)$$

where x_k is the point in $[0, 1]$ at which the first derivative f_k' vanishes. As

$$x_k = \frac{1}{\sqrt{2k}}\,, \qquad f_k(x_k) = \sqrt{\frac{k}{2e}}\,,$$

$f_k(x_k)$ diverges to $+\infty$ when $k \to +\infty$.

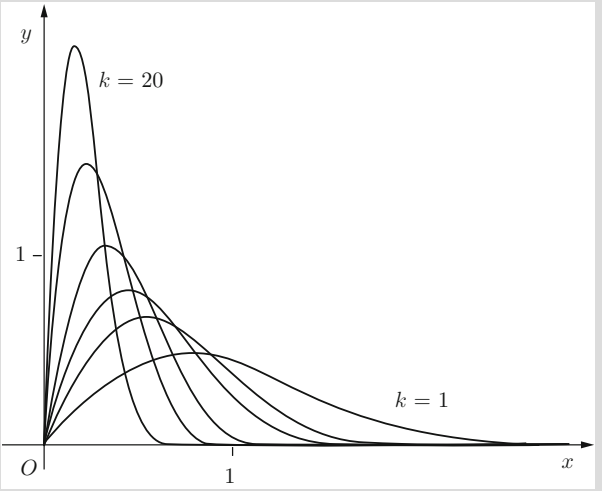


Fig. 1.3 $f_k(x)$ for $k = 1, 2, 3, 5, 10, 20$

If we drop the continuity of the functions $f_k(x)$, we may consider a further example:

Example 2 For $x \in [0, 1]$ define

$$f_k(x) = \begin{cases} k & \text{if} \quad 0 < x < 1/k \\ 0 & \text{if} \quad 1/k \le x \le 1 \quad \text{or if} \quad x = 0 \end{cases}$$

(continued)

Example 2 (continued)
Then for any $x \in [0, 1]$, $f_k(x)$ converges to $f(x) = 0$ (because given $x \in (0, 1]$, for any $k \geq 1/x$ we have $1/k \leq x \leq 1$ and then $f_k(x) = 0$). Furthermore,

$$\int_0^1 f_k(x)\,dx = 1\,, \qquad \int_0^1 f(x)\,dx = 0\,.$$

Also in this case the sequence f_k does *not* converge uniformly to f on the interval $[0, 1]$. In fact

$$\sup\{|f_k(x) - f(x)| : x \in [0, 1]\} = \max\{f_k(x) : x \in [0, 1]\} = k\,,$$

which diverges to $+\infty$ as $k \to +\infty$.

Hence pointwise convergence alone is not enough to warrant formula (1.10) for interchanging limits and integration. The following result proves that the uniform convergence of a sequence f_k to f implies (1.10).

Theorem (Interchange of Limits and Integrals) *If f_k is a sequence of continuous functions that converges uniformly to f on $[a, b]$, then*

$$\lim_{k\to+\infty} \int_a^b f_k(x)\,dx = \int_a^b f(x)\,dx\,.$$

Proof By the theorem on the continuity of limits, $f(x)$ is a continuous function on the closed and bounded interval $[a, b]$, and thus it is integrable there.

Moreover

$$\left|\int_a^b f_k(x)\,dx - \int_a^b f(x)\,dx\right| = \left|\int_a^b \{f_k(x) - f(x)\}dx\right| \leq$$

$$\leq \int_a^b |f_k(x)\,dx - f(x)|\,dx \leq (b-a)\cdot \max_{a\leq x\leq b} |f_k(x) - f(x)|\,,$$

so the claim follows when $k \to +\infty$.

We shall return to this important aspect in Chap. 9, where we shall discuss more general and complete results. Let us now study when limits can be interchanged with derivatives.

Let f_k be a sequence of differentiable functions on an interval $I \subseteq \mathbb{R}$ that converges on I *pointwise* to some function f. We may ask whether f, too, is differentiable on I and whether

$$\lim_{k\to+\infty} f_k'(x) = f'(x)\,, \qquad \forall\, x \in I\,.$$

The answer is still negative, as shown by the following examples.

Example 3 For $x \in I = \mathbb{R}$ (I could be an arbitrary interval of $\mathbb{R}$ with 0 in its interior) consider the sequence

$$f_k(x) = \sqrt{x^2 + \frac{1}{k}}\,.$$

It is easy to see that the f_k are continuous on I, and so are their first derivatives. Moreover, the sequence f_k converges uniformly on I to the function $f(x) = \sqrt{x^2} = |x|$, which is not differentiable at 0; in fact

$$|f_k(x) - f(x)| = \left|\sqrt{x^2 + \frac{1}{k}} - \sqrt{x^2}\right| = \frac{\left(x^2 + \frac{1}{k}\right) - x^2}{\sqrt{x^2 + \frac{1}{k}} + \sqrt{x^2}} =$$

$$= \frac{\frac{1}{k}}{\sqrt{x^2 + \frac{1}{k}} + \sqrt{x^2}} \le \frac{\frac{1}{k}}{\sqrt{\frac{1}{k}}} = \sqrt{\frac{1}{k}}\,,$$

since $\sqrt{x^2 + \frac{1}{k}} + \sqrt{x^2} \ge \sqrt{\frac{1}{k}}$ for any $x \in I$ and any $k \in \mathbb{N}$.

Let us now show that even if we assume that the limit $f(x)$ is differentiable, Eq. (1.3) might not still hold.

Example 4 On $I = \mathbb{R}$ (or any proper real interval I) define the sequence of functions

$$f_k(x) = \frac{1}{k}\,\sin kx\,, \qquad\qquad \forall\, k \in \mathbb{N}\,, \quad \forall\, x \in \mathbb{N}\,.$$

The sequence f_k converges uniformly to $f(x) = 0$ for any $x \in I$. On the other hand, since

$$f_k'(x) = \cos kx\,, \qquad \forall\, k \in \mathbb{N}\,, \quad \forall\, x \in I\,,$$

we have $f_k'(0) = 1$ for any $k \in \mathbb{N}$ and $f'(0) = 0$.

Theorem (Interchange of Limits and Derivatives) *Let f_k be a sequence of differentiable functions with continuous derivative on $[a, b]$. Suppose there exists $x_0 \in [a, b]$ such that the numerical sequence $f_k(x_0)$ converges on $\mathbb{R}$ and the sequence of derivatives f_k' converges uniformly on $[a, b]$. Then f_k converges uniformly on $[a, b]$ to a differentiable function f with continuous derivative on $[a, b]$, and furthermore*

$$\lim_{k \to +\infty} f_k'(x) = f'(x)\,. \tag{1.11}$$

Proof As f_k' is continuous, by the fundamental theorem of calculus we may write

$$f_k(x) = f_k(x_0) + \int_{x_0}^{x} f_k'(t)\,dt\,, \qquad \forall\, k \in \mathbb{N}\,, \quad \forall\, x \in [a,b]\,. \tag{1.12}$$

Moreover, calling $l \in \mathbb{R}$ the limit of the numerical sequence $f_k(x_0)$, and letting $g(x)$ denote the uniform limit of the sequence $f_k'(x)$ on $[a,b]$, by the theorem on the continuity of limits the function g is continuous on $[a,b]$, and hence also integrable on that interval. As $k \to +\infty$ in (1.12), we may interchange the limit and the integral, so that

$$\lim_{k\to+\infty} f_k(x) = l + \int_{x_0}^{x} g(t)\,dt\,, \qquad \forall\, x \in [a,b]\,.$$

Hence the sequence $f_k(x)$ converges on $[a,b]$ pointwise. If $f(x)$ denotes the limit, we have

$$f(x) = l + \int_{x_0}^{x} g(t)\,dt\,, \qquad \forall\, x \in [a,b]\,, \tag{1.13}$$

and from (1.12), (1.13) we deduce

$$\begin{aligned} |f_k(x) - f(x)| &\le |f_k(x_0) - l| + \left| \int_{x_0}^{x} \{f_k'(t) - g(t)\}\,dt \right| \le \\ &\le |f_k(x_0) - l| + (b-a)\cdot \sup\{|f_k'(t) - g(t)| \,:\, t \in [a,b]\}\,. \end{aligned}$$

But f_k' converges uniformly to g on $[a,b]$, so

$$\lim_{k\to+\infty} \sup\{|f_k(x) - f(x)| \,:\, x \in [a,b]\} = 0\,,$$

in other words $f_k(x)$ converges uniformly to $f(x)$ on $[a,b]$. Finally, from (1.13) we deduce, by the fundamental theorem of calculus, that f is differentiable and

$$f'(x) = g(x)\,, \qquad \forall\, x \in [a,b]\,.$$

Therefore $\lim_{k\to+\infty} f_k' = g = f'$ and (1.11) holds.

The theorem on interchanging limits and derivatives is still valid under less restrictive assumptions. Namely, it holds without assuming that the derivatives f_k' are continuous on $[a,b]$. Before we prove the more general statement, let us start with two lemmas, the first of which has independent interest.

Lemma 1 *Let f_k be a sequence of differentiable functions on a closed bounded interval $[a,b]$. Suppose there exists $x_0 \in [a,b]$ such that $f_k(x_0)$ converges, and that*

the sequence of derivatives f_k' converges uniformly on $[a, b]$. Then also f_k converges uniformly on $[a, b]$.

Notice (similar considerations hold for the other results in this section) that Lemma 1 immediately implies a similar result for sequences of functions defined on intervals that are not necessarily closed, nor bounded: *assuming f_k is a sequence of differentiable functions on some interval I (possibly not closed nor bounded) in $\mathbb{R}$, if there exists $x_0 \in I$ such that $f_k(x_0)$ converges, and provided the sequence of derivatives f_k' converges uniformly on every closed bounded interval contained in I, then f_k converges uniformly as well, on every interval $[a, b] \subseteq I$.*

Proof of Lemma 1 For any $x \in [a, b]$ and $h, k \in \mathbb{N}$

$$|f_k(x) - f_h(x)| \le |f_k(x_0) - f_h(x_0)| + |(f_k(x) - f_h(x)) - (f_k(x_0) - f_h(x_0))| . \quad (1.14)$$

Applying to the function $f_k - f_h$ the mean value theorem, there exists x_1 between x_0 and x such that

$$|(f_k(x) - f_h(x)) - (f_k(x_0) - f_h(x_0))| = |x - x_0| \cdot |f_k'(x_1) - f_h'(x_1)| . \quad (1.15)$$

From (1.14), (1.15) we deduce that for any $x \in [a, b]$

$$|f_k(x) - f_h(x)| \le |f_k(x_0) - f_h(x_0)| + (b - a) \cdot |f_k'(x_1) - f_h'(x_1)| .$$

By the uniform Cauchy criterion, given $\varepsilon > 0$ there exists $\nu_1 \in \mathbb{N}$ such that

$$|f_k'(x) - f_h'(x)| < \varepsilon , \qquad \forall\, h, k > \nu_1 , \quad \forall\, x \in [a, b] .$$

Moreover, Cauchy's criterion on numerical sequences guarantees the existence of $\nu_2 \in \mathbb{N}$ such that

$$|f_k(x_0) - f_h(x_0)| < \varepsilon , \qquad \forall\, h, k > \nu_2 .$$

Then for any $h, k > \nu = \max\{\nu_1, \nu_2\}$ we have

$$|f_k(x) - f_h(x)| < 2\varepsilon \qquad\qquad \forall\, x \in [a, b],$$

and the claim follows from the uniform Cauchy criterion.

Lemma 2 *Let f_k be a sequence of differentiable functions on a closed bounded interval $[a, b]$. Given $x_0 \in [a, b]$, set*

$$g_k(x) = \frac{f_k(x) - f_k(x_0)}{x - x_0} , \qquad \forall\, x \in [a, b] - \{x_0\} .$$

If the sequence of derivatives f_k' converges uniformly on $[a, b]$, then also the sequence g_k converges uniformly on $[a, b] - \{x_0\}$.

Proof For $h, k \in N$ and $x \in [a, b] - \{x_0\}$ we have

$$g_k(x) - g_h(x) = \frac{(f_k(x) - f_h(x)) - (f_k(x_0) - f_h(x_0))}{x - x_0}.$$

The mean value theorem applied to $f_k - f_h$ ensures that for any $x \in [a, b] - \{x_0\}$ there exists a point x_1, strictly lying between x_0 and x, such that

$$g_k(x) - g_h(x) = f_k'(x_1) - f_h'(x_1). \tag{1.16}$$

By the uniform convergence of f_k' on $[a, b]$ it follows that for any $\varepsilon > 0$ there exists $\nu \in \mathbb{N}$ such that

$$|f_k'(x) - f_h'(x)| < \varepsilon, \qquad \forall\, h, k > \nu, \quad \forall\, x \in [a, b].$$

Then (1.16) implies

$$|g_k(x) - g_h(x)| < \varepsilon, \qquad \forall\, h, k > \nu, \quad \forall\, x \in [a, b] - \{x_0\}.$$

The conclusion now follows from the uniform Cauchy criterion.

Theorem (Interchange of Limits and Derivatives—General Case) *Let f_k be a sequence of differentiable functions on $[a, b]$. Suppose there exists $x_0 \in [a, b]$ such that $f_k(x_0)$ converges, and that f_k' converges uniformly on $[a, b]$. Then f_k converges uniformly on $[a, b]$ to a differentiable function f on $[a, b]$, and*

$$f'(x) = \lim_{k \to +\infty} f_k'(x). \tag{1.17}$$

Proof By Lemma 1 the sequence f_k converges uniformly on $[a, b]$; let us call f the limit function. Given $x_0 \in [a, b]$, put

$$g_k(x) = \frac{f_k(x) - f_k(x_0)}{x - x_0}, \qquad \forall\, x \in [a, b] - \{x_0\},$$
$$g(x) = \frac{f(x) - f(x_0)}{x - x_0}, \qquad \forall\, x \in [a, b] - \{x_0\}.$$

Lemma 2 guarantees that $g_k(x)$ converges uniformly to $g(x)$ on $[a, b] - \{x_0\}$. The theorem on interchanging limits implies

$$\lim_{k \to +\infty} f_k'(x_0) = \lim_{k \to +\infty} \left(\lim_{x \to x_0} g_k(x) \right) = \lim_{x \to x_0} \left(\lim_{k \to +\infty} g_k(x) \right) = \lim_{x \to x_0} g(x) = f'(x_0)$$

and thus we recover (1.17), because x_0 is a generic point in $[a, b]$.

1.4 Uniform Convergence and Monotonicity

In this section we shall discuss two classical results regarding uniform convergence under a monotonicity hypothesis. The first theorem (by Dini) assumes monotonicity in the parameter k, the second one supposes monotonicity in the variable x.

Theorem 1 (Dini) *Let $I = [a, b]$ be a closed and bounded interval and consider a sequence $f_k : I \to \mathbb{R}$ of continuous functions, monotone in k (for instance, increasing: $f_k(x) \le f_{k+1}(x)$ for any $k \in \mathbb{N}$, $x \in I$), and pointwise convergent on $[a, b]$ to some continuous function f. Then f_k converges uniformly to f on $[a, b]$.*

Proof Consider, for example, an increasing sequence f_k, that is to say $f_k(x) \le f_{k+1}(x) \le f(x)$ for any $k \in \mathbb{N}$ and any $x \in I = [a, b]$.

Suppose, by contradiction, that f_k does not converge uniformly to f on $[a, b]$. This means there exists $\varepsilon_0 > 0$ such that for any $\nu \in \mathbb{N}$ we can find $k > \nu$ and $x \in [a, b]$ for which

$$|f_k(x) - f(x)| = f(x) - f_k(x) \ge \varepsilon_0 .$$

Hence for any $\nu = h \in \mathbb{N}$, there exist $k_h \to +\infty$ and $x_h \in [a, b]$ such that

$$f(x_h) - f_{k_h}(x_h) \ge \varepsilon_0 .$$

But the monotonicity of f_k in k forces $f_{k_h} \ge f_i$ when $k_h \ge i$. So we obtain

$$f(x_h) - f_i(x_h) \ge \varepsilon_0 , \qquad \forall\, h \in \mathbb{N} , \quad \forall\, i \le k_h .$$

The sequence x_h, being bounded, admits a subsequence x_{h_j} converging to a point x_0 of the interval $[a, b]$. Taking the limit as $j \to +\infty$ in

$$f(x_{h_j}) - f_i(x_{h_j}) \ge \varepsilon_0 , \qquad \forall\, h_j \in \mathbb{N} , \quad \forall\, i \le k_{h_j} ,$$

due to the continuity of f and f_i we have

$$f(x_0) - f_i(x_0) \ge \varepsilon_0 \qquad \forall\, i \in \mathbb{N} .$$

Taking the limit when $i \to +\infty$ we reach the contradiction $0 \ge \varepsilon_0$.

The following theorem gives another criterion, again in terms of monotonicity, for the uniform convergence of a sequence that converges pointwise on a closed and bounded interval I to some continuous function f on I.

Theorem 2 *Let $f_k : [a,b] \to \mathbb{R}$ be a sequence of increasing (or decreasing) functions in $x \in [a,b]$, which converges pointwise to the continuous function $f : [a,b] \to \mathbb{R}$. Then f is increasing (or respectively decreasing), and the convergence is uniform on $[a,b]$.*

Proof Divide the interval $[a,b]$ in $h \in \mathbb{N}$ parts (of the same length) using subdivision points

$$a = x_0 < x_1 < x_2 < \ldots < x_h = b\,.$$

Fix $\varepsilon > 0$. By the uniform continuity of f on $[a,b]$ it is possible to choose h large enough so that

$$|f(y) - f(x)| < \varepsilon\,, \qquad \forall\, x, y \in [x_i, x_{i+1}]\,,$$

for any $i = 0, 1, \ldots, h-1$. The pointwise convergence of f_k to f implies there exists $\nu \in \mathbb{N}$ such that

$$|f_k(x_i) - f(x_i)| < \varepsilon\,, \qquad \forall i = 0, 1, \ldots, h \quad \forall\, k > \nu\,.$$

Let us consider the case where the $f_k : [a,b] \to \mathbb{R}$ are increasing functions. If $x \in [x_i, x_{i+1}]$ for some $i \in \{0, 1, \ldots, h-1\}$, since $x_i \le x \le x_{i+1}$ we have

$$f_k(x_i) \le f_k(x) \le f_k(x_{i+1})\,, \qquad \forall\, k \in \mathbb{N}\,.$$

Therefore for $x \in [x_i, x_{i+1}]$ with i given, using (1.2) first and then (1.2), (1.2), we deduce

$$f_k(x) - f(x) = f_k(x) - f_k(x_{i+1}) + f_k(x_{i+1}) - f(x_{i+1}) + f(x_{i+1}) - f(x) \le$$
$$\le (f_k(x_{i+1}) - f(x_{i+1})) + (f(x_{i+1}) - f(x)) < 2\varepsilon\,,$$

for any $k > \nu$. Similarly

$$f_k(x) - f(x) = f_k(x) - f_k(x_i) + f_k(x_i) - f(x_i) + f(x_i) - f(x) \ge$$
$$\ge (f_k(x_i) - f(x_i)) + (f(x_i) - f(x)) > -2\varepsilon\,,$$

for any $k > \nu$. By (1.2), (1.2), f_k converges uniformly to f on $[a,b]$.

For a quick comparison here are the hypotheses and conclusions of Theorems 1 and 2.

Theorem 1 (Dini) *Supposing*

1. *$I = [a,b]$ is a closed and bounded interval;*
2. *$f_k : [a,b] \to \mathbb{R}$ is a continuous function for any $k \in \mathbb{N}$;*
3. *$f_k(x) \le f_{k+1}(x)$ for any $k \in \mathbb{N}$ and any $x \in I$;*

4. f_k *converges pointwise to a function* $f : [a, b] \to \mathbb{R}$*;*
5. f *is continuous on* $[a, b]$.

Then: f_k *converges uniformly to* f *on* I.

Theorem 2 *Supposing*

1. $I = [a, b]$ *is a closed and bounded interval;*
2. $f_k : [a, b] \to \mathbb{R}$ *is an increasing function for any* $k \in \mathbb{N}$*;*
3. f_k *converges pointwise to a function* $f : [a, b] \to \mathbb{R}$*;*
4. f *is continuous on* $[a, b]$.

Then: f_k *converges uniformly to* f *on* I *(and* f *is increasing on* $[a, b]$*).*

If we consider the examples in Sect. 1.1, the sequence of Example 2, $f_k = x^k$ for $x \in [0, 1]$, converges pointwise, but not uniformly, to the function f defined by $f(x) = 0$ when $x \in [0, 1)$, $f(1) = 1$. The latter does not satisfy the continuity assumption required in Theorems 1 and 2.

The same sequence of functions, restricted to the semi-open interval $[0, 1)$, converges pointwise, but not uniformly, to the zero function. In this case it is the first hypothesis in Theorems 1, 2 that does not hold over the corresponding interval.

The same sequence, restricted to an interval like $[0, b]$, with $0 < b < 1$ (as in Example 3, Sect. 1.1), fulfils all the assumptions of Theorems 1 and 2 (and the conclusion, too).

The sequence of Example 1, Sect. 1.1, is not covered by Theorems 1 and 2. The functions are not continuous (assumption 2 in Theorem 1 does not hold), nor are they monotone in $x \in [0, 1]$ (assumption 2 in Theorem 2 is false). If restricted to the semi-open interval $(0, 1]$, the sequence is monotone in x, but now the domain is no longer closed.

Example 4, Sect. 1.1, considers the convergence of the sequence f_k given by

$$f_k(x) = \frac{x^2}{k + x^2}, \qquad \forall\, k \in \mathbb{N},$$

on $\mathbb{R}$, which is not a bounded interval. For this reason Theorems 1 and 2 do not apply (and in fact the sequence of Example 4 does not converge uniformly).

Conversely, in Example 1, Sect. 1.1, we showed that this sequence converges uniformly on every interval $[a, b]$, $a, b \in \mathbb{R}$, $a < b$. Uniform convergence may be deduced using Dini's Theorem 1, since f_k is decreasing in k and all other hypotheses hold.

1.5 Series of Functions

If f_k is a sequence of real functions defined on the subset I of $\mathbb{R}$, we indicate by s_k the *sequence of partial sums*

$$\begin{aligned} s_1 &= f_1 \\ s_2 &= f_1 + f_2 \\ &\cdots\cdots\cdots\cdots \\ s_k &= f_1 + f_2 + \ldots + f_k \\ &\cdots\cdots\cdots\cdots \end{aligned}$$

The sequence of functions s_k is called *series* (*of functions*) with general term f_k, and we shall also use for it the expression

$$f_1 + f_2 + \ldots + f_k + \ldots \tag{1.18}$$

If, for any $x \in I$, the numerical series with general term $f_k(x)$

$$f_1(x) + f_2(x) + \ldots + f_k(x) + \ldots$$

is convergent, i.e. if the sequence $s_k(x)$ converges (it has finite limit) for every $x \in I$, one says that the series of functions (1.18) *converges pointwise on* I.

When the sequence of functions s_k converges uniformly on I, we say the series of functions (1.18) *converges uniformly on* I. In either case, the limit of s_k as $k \to +\infty$ is called *sum* of the series of general term f_k, and we denote it by

$$\sum_{k=1}^{\infty} f_k \,. \tag{1.19}$$

At times, (1.19) also indicates the series of general term f_k, apart from its sum. Often, as in the case of numerical series, one uses distinct summation indices for a series' general term and (for example) the sequence of partial sums of a convergent series:

$$\begin{aligned} s_k &= \sum_{i=1}^{k} f_i \,, \qquad \forall\, k \in \mathbb{N}\,; \\ f &= \lim_{k\to+\infty} s_k = \lim_{k\to+\infty} \sum_{i=1}^{k} f_i = \sum_{i=1}^{\infty} f_i \,. \end{aligned}$$

In analogy to numerical series, (1.18) is said to *converge absolutely* on I if the series of general term $|f_k|$ converges pointwise on I.

Cauchy's criteria, both pointwise and uniform, for sequences immediately generate the following criteria for series. It suffices to use the identity, valid for any k, $p \in N$,

$$s_{k+p}(x) - s_k(x) = f_{k+1}(x) + \ldots + f_{k+p}(x)\,.$$

Cauchy Criterion for Series *The series of general term f_k converges pointwise on I if, for any $\varepsilon > 0$ and any $x \in I$, there exists $\nu_{\varepsilon,x} \in \mathbb{N}$ such that*

$$|f_{k+1}(x) + \ldots + f_{k+p}(x)| < \varepsilon\,, \qquad \forall\, k > \nu_{\varepsilon,x}\,, \forall\, p \in \mathbb{N}\,.$$

Uniform Cauchy Criterion for Series *The series of general term f_k converges uniformly on I if, for any $\varepsilon > 0$, there exists $\nu_\varepsilon \in \mathbb{N}$ such that*

$$|f_{k+1}(x) + \ldots + f_{k+p}(x)| < \varepsilon\,, \qquad \forall\, k > \nu_\varepsilon\,, \quad \forall\, p \in \mathbb{N}\,, \quad \forall\, x \in I\,.$$

The series of functions (1.18) is called *totally convergent* on I if there exists a sequence of non-negative real numbers M_k such that

$$|f_k(x)| \le M_k\,, \qquad \forall\, x \in I\,, \forall\, k \in \mathbb{N}$$

and if the numerical series

$$M_1 + M_2 + \ldots + M_k + \ldots$$

converges. If a series of functions converges pointwise, uniformly or totally on some set I, it converges pointwise, uniformly or totally on every subset $I' \subseteq I$, respectively.

The next proposition is simple, but important to establish the type of convergence of a series of functions.

Proposition *The total convergence of a series of functions implies its uniform convergence. In other words (the last implication is obvious):*

$$\textit{total convergence} \Rightarrow \textit{uniform convergence} \Rightarrow \textit{pointwise convergence}.$$

Proof Let $f_1 + f_2 + \ldots + f_k + \ldots$ be a totally convergent series of functions on the set $I \subseteq \mathbb{R}$, and suppose

$$M_1 + M_2 + \ldots + M_k + \ldots$$

is a convergent numerical series with non-negative terms such that $|f_k(x)| \leq M_k$ for any $x \in I$ and any $k \in \mathbb{N}$.

By Cauchy's criterion for numerical series, for any $\varepsilon > 0$ there exists $\nu \in \mathbb{N}$ such that

$$M_{k+1} + \ldots + M_{k+p} < \varepsilon\,, \qquad \forall\, k > \nu\,, \forall\, p \in \mathbb{N}\,.$$

Hence for any $x \in I$, $k > \nu$ and $p \in \mathbb{N}$,

$$\begin{aligned} |f_{k+1}(x) + \ldots + f_{k+p}(x)| \leq |f_{k+1}(x)| + \ldots + |f_{k+p}(x)| \leq \\ \leq M_{k+1} + \ldots + M_{k+p} < \varepsilon\,. \end{aligned}$$

The series of general term f_k then satisfies the uniform Cauchy criterion for series (1.5), and therefore it converges uniformly on I.

In practice, in order to check whether a series of functions converges totally it may be convenient to choose the sequence M_k in the following optimal way

$$M_k = \sup\{|f_k(x)| \,:\, x \in I\}\,, \qquad \forall\, k \in \mathbb{N}\,.$$

Example 1 Consider, for any $x \in I = \mathbb{R}$, the series of functions

$$\sum_{k=1}^{\infty} f_k\,, \qquad \text{where} \qquad f_k(x) = \frac{x}{x^4 + 3k^4}\,, \qquad \forall\, k \in \mathbb{N}\,. \tag{1.20}$$

Given $k \in \mathbb{N}$, the derivative of $f_k(x)$,

$$f_k'(x) = \frac{3(k^4 - x^4)}{(x^4 + 3k^4)^2}\,, \qquad \forall\, k \in \mathbb{N}\,,$$

vanishes at $x = \pm k$. It is easy to see (Fig. 1.4) that the point $x = k$ is an absolute maximum point for $f_k(x)$, while $x = -k$ is an absolute minimum point. Furthermore

$$M_k = \sup\{|f_k(x)| \,:\, x \in \mathbb{R}\} = f_k(x) = \frac{1}{4k^3}\,, \qquad \forall\, k \in \mathbb{N}$$

and the numerical series of general term $1/k^3$ converges (it is a generalised harmonic series). Therefore the series of functions (1.20) is totally convergent everywhere on $\mathbb{R}$, and hence it converges on $\mathbb{R}$ uniformly and pointwise.

(continued)

Example 1 (continued)

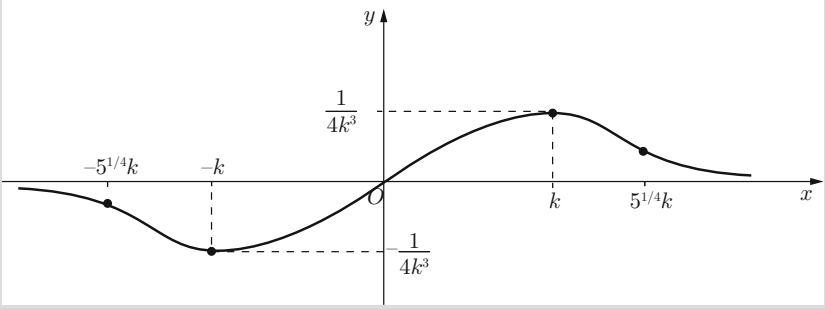


Fig. 1.4 $f_k(x) = \frac{x}{x^4+3k^4}$

Example 2 A very special series of functions

$$f_1 + f_2 + \ldots + f_k + \ldots$$

is the one where every f_k is constant on the set I considered. If so, the series' uniform convergence is equivalent to the pointwise convergence, whereas the total convergence is equivalent to the absolute convergence of the numerical series. In this way we can immediately build a series of (constant) functions that converges uniformly but not totally: it suffices to choose f_k constant on I, equal to $(-1)^k/k$, for any $k \in \mathbb{N}$.

The next example describes a series of functions with constant sign that converges uniformly but not totally on a set.

Example 3 For $x \in I = [1, +\infty)$ and for any $k \in \mathbb{N}$ set

$$f_k(x) = \begin{cases} 1/k & \text{if} \quad k \le x < k+1 \\ 0 & \text{otherwise} \end{cases}$$

The partial sum s_k is

$$s_k(x) = \sum_{i=1}^{k} f_i(x) = \begin{cases} 1/i & \text{if} \quad i \le x < i+1 \\ 0 & \text{if} \quad x \ge k+1 \end{cases} \qquad i = 1, 2, \ldots, k;$$

and it converges as $k \to +\infty$ to the function f defined by

$$f(x) = \frac{1}{i}, \qquad \text{if} \qquad i \le x < i+1 \qquad \forall\, i \in \mathbb{N}.$$

(continued)

Example 3 (continued)
The sequence s_k converges uniformly to f on the interval $I = [1, +\infty)$, because

$$\sup\{|s_k(x) - f(x)| : x \in I\} = \frac{1}{k+1}, \qquad \forall k \in \mathbb{N}.$$

However there is no convergent numerical series, say of general term M_k, such that

$$|f_k(x)| \le M_k, \qquad \forall x \in I, \quad \forall k \in \mathbb{N},$$

because $f_k(k) = 1/k$, and therefore the series of general term $M_k \ge 1/k$, cannot converge.

The theorem on the continuity of limits (Sect. 1.2) implies the following:

Theorem (Continuity of a Sum) *The sum of a uniformly convergent series of continuous functions is continuous.*

The results regarding the interchange of limits and integrals or derivatives (Sect. 1.3) imply analogous results for series of functions. For this consider the series

$$\sum_{k=1}^{\infty} f_k\,. \tag{1.21}$$

Theorem (Integration of Series) *Let f_k be a sequence of continuous functions on a closed bounded interval $[a, b]$, and suppose that the series (1.21) converges uniformly to f on $[a, b]$. Then*

$$\int_a^b f(x)\,dx = \sum_{k=1}^{\infty} \int_a^b f_k(x)\,dx\,. \tag{1.22}$$

Theorem (Differentiation of Series) *Let f_k be a sequence of differentiable functions with continuous derivatives on $[a, b]$. If the series (1.21) converges on $[a, b]$ to f, then f is differentiable with continuous derivative on $[a, b]$, and*

$$f'(x) = \sum_{k=1}^{\infty} f_k'(x)\,, \qquad \forall x \in [a, b]\,, \tag{1.23}$$

as long as the series of the derivatives converges uniformly on $[a, b]$.

We remark that the above result may be formulated in more generality, once we use the (general version of the) theorem for interchanging limits and derivatives of Sect. 1.3.

Sometimes formula (1.22) is expressed as follows

$$\int_a^b \sum_{k=1}^{\infty} f_k(x)\,dx = \sum_{k=1}^{\infty} \int_a^b f_k(x)\,dx$$

and one says that the series can be integrated term by term.

Analogously, formula (1.23) can be rewritten as

$$\left(\sum_{k=1}^{\infty} f_k(x)\right)' = \sum_{k=1}^{\infty} f_k'(x)\,dx\,, \qquad \forall\, x \in [a,b]\,,$$

and one says the series can be differentiated term by term.

1.6 Power Series

Let a_k, $k = 0, 1, 2, \ldots$, be a sequence of real numbers. The series of functions

$$a_0 + a_1 x + a_2 x^2 + \ldots + a_k x^k + \ldots \tag{1.24}$$

is called *power series* with coefficients $a_0, a_1, a_2, \ldots, a_k, \ldots$.

A power series satisfies one of the following:

(i) the series converges only at $x = 0$;
(ii) the series converges at any $x \in \mathbb{R}$;
(iii) there exists a real number $\varrho > 0$ such that the series converges for $|x| < \varrho$ and does not converge for $|x| > \varrho$.

In particular, the convergence set of the power series (1.24), i.e. the set of points $x \in \mathbb{R}$ at which (1.24) converges, is an interval centred at the origin, namely: just $\{0\}$ in case (i), the whole $\mathbb{R}$ in case (ii), and an interval between $-\varrho, \varrho$ in case (iii). To prove these claims let us begin with the following result.

Theorem 1 *If the power series (1.24) converges at some $\xi \neq 0$, it converges totally on any closed, bounded interval contained in $(-|\xi|, |\xi|)$.*

Proof The convergence of the numerical series

$$a_0 + a_1 \xi + a_2 \xi^2 + \ldots + a_k \xi^k + \ldots$$

implies that the sequence $a_k \xi^k$ is infinitesimal as $k \to +\infty$, and hence bounded. Put equivalently, there exists $M > 0$ such that

$$|a_k \xi^k| \leq M\,, \qquad \forall k \in \mathbb{N}\,.$$

So let η be a point in the open interval $(-|\xi|, |\xi|)$; we will show that (1.24) converges totally on $[-|\eta|, |\eta|]$, from which – suitably choosing η – the total convergence on any interval $[a, b] \subset (-|\xi|, |\xi|)$ will follow, because

$$[a, b] \subseteq [-|\eta|, |\eta|] \subset (-|\xi|, |\xi|)\,.$$

For any $x \in \mathbb{R}$ such that $|x| \leq |\eta|$, we have

$$|a_k x^k| = |a_k \xi^k| \cdot \left(\frac{|x|}{|\xi|}\right)^k \leq M \cdot \left(\frac{|\eta|}{|\xi|}\right)^k\,, \qquad \forall\, k \in \mathbb{N}\,,$$

and the numerical series of general term $M_k = M \cdot (|\eta|/|\xi|)^k$ is convergent, since it is a geometric series of ratio $(|\eta|/|\xi|) < 1$. Therefore the power series (1.24) converges totally on $[-|\eta|, |\eta|]$.

One calls *convergence radius* of the power series (1.24) the supremum $\varrho \in [0, +\infty]$ of the set X of real numbers x at which it converges:

$$\varrho = \sup\, X\,, \quad \text{where} \quad X = \left\{ x \in \mathbb{R} : \quad \sum_{k=0}^{\infty} a_k x^k \text{ converges} \right\}\,. \tag{1.25}$$

The convergence radius ϱ is zero if and only if (1.24) converges only at $x = 0$, while $\varrho = +\infty$ if and only if the series converges at any $x \in \mathbb{R}$. These properties are proved as in the proof of Theorem 2, which treats the case $0 < \varrho < +\infty$.

Theorem 2 *Let $0 < \varrho < +\infty$. The convergence radius of the power series (1.24) equals ϱ if and only if the series converges for $|x| < \varrho$ and does not converge for $|x| > \varrho$.*

Proof Let $\varrho \in (0, +\infty)$ be the convergence radius of (1.24), i.e. ϱ is the least upper bound defined in (1.25). We will show that (1.24) converges at any x such that $|x| < \varrho$. Fix a point x. By the supremum's properties, there exists $\xi \in X$ (ξ is such that the numerical series of general term $a_k \xi^k$ is convergent) for which $|x| < \xi \leq \varrho$. By Theorem 1 the series (1.24) converges at x.

Now we show that (1.24) does not converge at any x such that $|x| > \varrho$. In fact, if by contradiction (1.24) converged at some ξ such that $|\xi| > \varrho$, by Theorem 1 it would in particular converge at any point of the interval $[\varrho, |\xi|)$, in contrast to ϱ being the supremum (property (1.25)).

Conversely, suppose (1.24) converges for $|x| < \varrho_1$ and does not converge when $|x| > \varrho_1$. We have to prove that ϱ_1 is the supremum defined in (1.25). Since the power series converges on $|x| < \varrho_1$, X contains the interval $(-\varrho_1, \varrho_1)$ and therefore $\varrho_1 \leq \varrho$. But we cannot have $\varrho_1 < \varrho$, because otherwise the series would converge by Theorem 1 on the entire interval $[\varrho_1, \varrho)$, contradicting the fact that the series does not converge for $|x| > \varrho_1$.

When the convergence radius is non-zero and different from $+\infty$, nothing can be said in general on the power series' convergence at $x = -\varrho$ or $x = \varrho$, as explained by the following examples.

The (well-known geometric) series

$$1 + x + x^2 + \ldots + x^k + \ldots$$

has convergence radius $\varrho = 1$ (it converges at every point of $(-1, 1)$ and does not converge externally). At the endpoints of the convergence interval it does not converge, neither at $x = -1$ not at $x = 1$. By contrast, the power series

$$1 + x + \frac{x^2}{2^2} + \ldots + \frac{x^k}{k^2} + \ldots,$$

whose convergence radius is still 1, converges at both endpoints $x = -1$ and $x = 1$. Finally, the power series

$$1 + x + \frac{x^2}{2} + \ldots + \frac{x^k}{k} + \ldots,$$

converges at $x = -1$, but not at $x = 1$ (the reader should deduce, from this fact only, that the convergence radius of (1.6) is 1).

The next criterion is useful when we need to compute the convergence radius.

Theorem 3 (Cauchy-Hadamard) *Given the power series*

$$a_0 + a_1 x + a_2 x^2 + \ldots + a_k x^k + \ldots, \tag{1.26}$$

if the limit

$$l = \lim_{k \to +\infty} \sqrt[k]{|a_k|} \tag{1.27}$$

exists, then the convergence radius of (1.26) *is*

$$\varrho = \begin{cases} +\infty & \text{if} \quad l = 0 \\ 1/l & \text{if} \quad 0 < l < +\infty \\ 0 & \text{if} \quad l = +\infty \end{cases}.$$

Observe more generally that the same conclusion holds if we replace the existence of limit (1.27) with the condition that l is the *limit superior*

$$l = \lim_{k \to +\infty}'' \sqrt[k]{|a_k|}\,. \tag{1.28}$$

Proof For any $x \neq 0$ we have

$$\lim_{k\to+\infty} \sqrt[k]{|a_k x^k|} = l \cdot |x| .$$

If $l = 0$, then, the well-known root test for numerical series implies that series (1.26) converges absolutely at any $x \neq 0$, so the convergence radius is $+\infty$. When $l = +\infty$, the root test tells that (1.26) does not converge absolutely for any $x \neq 0$, and the convergence radius is $\varrho = 0$. If $0 < l < +\infty$, the test guarantees that (1.26) converges absolutely for $|x| < 1/l$ and does not converge absolutely for $|x| > 1/l$. Hence by Theorem 2 the convergence radius is $\varrho = 1/l$.

For example, using the Cauchy-Hadamard criterion we deduce that, for any $p \in \mathbb{R}$, the convergence radius of

$$1 + x + \frac{x^2}{2^p} + \ldots + \frac{x^k}{k^p} + \ldots ,$$

equals $\varrho = 1$.

Proceeding in analogy to Theorem 3, but now applying the ratio test instead of the root test, we arrive at the following convergence criterion.

Theorem 4 (d'Alembert) *Given the power series*

$$a_0 + a_1 x + a_2 x^2 + \ldots + a_k x^k + \ldots ,$$

where $a_k \neq 0$ *for all* $k \in \mathbb{N}$*, if the limit*

$$l = \lim_{k\to+\infty} \left| \frac{a_{k+1}}{a_k} \right|$$

exists then the convergence radius of (4) equals

$$\varrho = \begin{cases} +\infty & \text{if} \quad l = 0 \\ 1/l & \text{if} \quad 0 < l < +\infty \\ 0 & \text{if} \quad l = +\infty \end{cases} .$$

Consider the power series

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \ldots + a_k x^k + \ldots ; \tag{1.29}$$

the series obtained differentiating (1.29) term by term

$$\sum_{k=1}^{\infty} k\, a_k x^{k-1} = a_1 + 2a_2 x + 3a_3 x^2 + \ldots + k\, a_k x^{k-1} + \ldots \tag{1.30}$$

is called derivative series of the power series (1.29).

Theorem 5 (Convergence Radius of the Derivative Series) *A power series has the same convergence radius as its derivative series.*

Proof (First method): Indicate by ϱ and ϱ' the convergence radii of (1.29) and (1.30) respectively.

We will first show that if (1.29) converges at $x = x_0$, then (1.30) converges for $|x| < |x_0|$ (whence $\varrho \le \varrho'$). Suppose then that the numerical series

$$\sum_{k=0}^{\infty} a_k x_0^k$$

converges. The general term $a_k x_0^k$ converges to zero, and so there exists $L > 0$ such that $|a_k x_0^k| \le L$ for any $k \in \mathbb{N}$. The absolute convergence, for $|x| < |x_0|$, of (1.30) is a consequence of the comparison test and of the inequality

$$|k\, a_k x^{k-1}| = \frac{k|a_k x_0^k|}{|x_0|} \cdot \left|\frac{x}{x_0}\right|^{k-1} \le \frac{kL}{|x_0|} \cdot \left|\frac{x}{x_0}\right|^{k-1}, \qquad \forall\, k \in \mathbb{N},$$

since the series on the right is convergent. Therefore $\varrho \le \varrho'$.

Conversely, let us prove that if (1.30) converges at $x = x_1$ then (1.29) converges absolutely when $|x| < |x_1|$ (and then $\varrho \ge \varrho'$). Arguing as above, there exists $M > 0$ such that $|k\, a_k x_1^{k-1}| \le M$ for every $k \in \mathbb{N}$. The convergence on $|x| < |x_1|$ of (1.29) follows from the inequality

$$|a_k x^k| = \frac{1}{k}|k\, a_k x_1^{k-1}| \cdot |x_1| \cdot \left|\frac{x}{x_1}\right|^k \le \frac{M|x_1|}{k} \cdot \left|\frac{x}{x_1}\right|^k, \qquad \forall\, k \in \mathbb{N},$$

because the right-most series converges. All in all, $\varrho = \varrho'$.

Proof (Second method): This other argument is shorter than the previous one, but it requires Theorem 3 by Cauchy-Hadamard in its version (1.28), the one with the limit superior.

Instead of the derivative series (1.30) let us consider

$$\sum_{k=1}^{\infty} k\, a_k x^k = a_1 x + 2a_2 x^2 + \ldots + k a_k x^k + \ldots = x \cdot \sum_{k=1}^{\infty} k\, a_k x^{k-1}.$$

It has the same convergence radius as the derivative series, since it is obtained from the latter by multiplying by x, which does not depend on k. Let us then prove convergence on $x \neq 0$. By Theorem 3 (Cauchy-Hadamard) it will suffice to show

$$\lim_{k\to+\infty}{}'' \sqrt[k]{|a_k|} = \lim_{k\to+\infty}{}'' \sqrt[k]{k|a_k|}\,,$$

but this is straightforward from

$$\lim_{k\to+\infty} \sqrt[k]{k} = 1\,.$$

Theorem 6 (Differentiation and Integration of Power Series) *If the power series* (1.29) *has non-zero convergence radius and*

$$f(x) = \sum_{k=0}^{\infty} a_k x^k\,, \qquad \forall\, |x| < \varrho, \quad \textit{with} \quad \varrho > 0, \tag{1.31}$$

denotes the sum, then

$$f'(x) = \sum_{k=1}^{\infty} k\, a_k x^{k-1}\,, \qquad \forall\, |x| < \varrho\,. \tag{1.32}$$

$$\int_0^x f(t)\,dt = \sum_{k=0}^{\infty} \frac{a_k}{k+1}\, x^{k+1}\,, \qquad \forall\, |x| < \varrho\,. \tag{1.33}$$

Proof Theorem 5, on the convergence radius of the derivative series, implies that (1.33) has the same convergence radius as (1.31), and in turn the latter has the same radius as (1.32). Hence the three series converge uniformly on every closed and bounded subinterval of $(-\varrho, \varrho)$. By the theorems of differentiation and integration of general series the claim now follows.

To close the section let us address the more general situation of power series *centred at* $x_0 \in \mathbb{R}$ (possibly non-zero), meaning power series of type

$$\sum_{k=0}^{\infty} a_k (x - x_0)^k\,. \tag{1.34}$$

The study of such series is reduced to the case where $x_0 = 0$ once we define a new variable $y = x - x_0$ (subsequently the variable returns to be called x, as is customary). In particular, if $\varrho > 0$ is the convergence radius of

$$\sum_{k=0}^{\infty} a_k x^k\,,$$

centred at $x_0 = 0$ and associated with (1.34), then the power series (1.34) centred at $x_0 \in \mathbb{R}$ will converge absolutely on $|x - x_0| < \varrho$ and will not converge on $|x - x_0| > \varrho$.

1.7 Taylor Series

Let $f(x)$ be a real function defined on an interval (a, b) in $\mathbb{R}$ and let $x_0 \in (a, b)$ be a point. We seek to establish whether there exists a power series centred at x_0 that converges on (a, b) to f, which is usually phrased by saying that f can be *expanded in power series around* x_0 *on the interval* (a, b).

The first result in this direction goes as follows.

Theorem 1 *If the power series*

$$\sum_{k=0}^{\infty} a_k (x - x_0)^k \tag{1.35}$$

has convergence radius $\varrho > 0$*, its sum* $f(x)$ *is differentiable infinitely many times for* $|x - x_0| < \varrho$*, and for any* $m \in \mathbb{N}$ *the* m*th derivative equals*

$$f^{(m)}(x) = \sum_{k=m}^{\infty} k(k-1)\cdots(k-m+1) a_k (x - x_0)^{k-m} . \tag{1.36}$$

Furthermore, f *admits a series expansion of the form*

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k . \tag{1.37}$$

Proof Formula (1.36) arises by repeatedly applying Theorem 6 of the previous section (in particular, the theorem on differentiating power series). Putting $x = x_0$ in (1.36), all terms after the first vanish, and $f^{(m)}(x_0) = m!\, a_m$ for every $m \in \mathbb{N}$. Substituting $a_k = f^{(k)}(x_0)/k!$ in (1.35) gives (1.37).

By Theorem 1 we know that if f can be expanded in power series around x_0 on (a, b), then on some neighbourhood of x_0 inside (a, b), of the form $|x - x_0| < \varrho$, we necessarily have that

(i) f is differentiable infinitely many times when $|x - x_0| < \varrho$;

(ii) the power series takes the form:

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k = \\ = f(x_0) + f'(x_0)(x-x_0) + \ldots + \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + \ldots \tag{1.38}$$

Assuming that condition (i) holds then, (1.38) is called *Taylor series* of f, and the problem reduces to finding conditions under which

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \ldots + \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + \ldots \tag{1.39}$$

for $x \in (a,b)$. The common way to express this is to say *f can be expanded in Taylor series* on (a,b).

This example shows that condition i) is not enough, on its own, to warrant (1.39), nor does the convergence of the Taylor series (1.38) on (a,b), alone, imply that the sum should necessarily equal $f(x)$ for any $x \in (a,b)$. The function f defined for $x \in \mathbb{R}$ by

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if} \quad x \neq 0 \\ \\ 0 & \text{if} \quad x = 0 \end{cases}$$

is infinitely differentiable on $\mathbb{R}$ and

$$f(0) = 0, \quad f'(0) = 0, \quad \ldots, \quad f^{(k)}(0) = 0, \qquad \forall\, k \in \mathbb{N}.$$

Hence the Taylor series of f, around $x_0 = 0$, converges pointwise (and uniformly) to the zero function $g(x) = 0$ rather than to $f(x)$.

Theorem 1 (Criterion for Taylor Series Expansion) *If f admits derivatives of any order on (a,b) and there exist $M, L \geq 0$ such that*

$$\left| f^{(k)}(x) \right| \leq M L^k, \qquad \forall\, x \in (a,b) \tag{1.40}$$

(in particular, if the derivatives of f are uniformly bounded on (a,b)), then for any $x_0 \in (a,b)$ the function f admits a Taylor series expansion around x_0 on the interval (a,b).

Proof Let us examine the definition of nth remainder $R_n(x)$ in Taylor's formula:

$$R_n(x) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k .$$

We have to prove that, as $n \to +\infty$, $R_n(x)$ tends to zero for any $x \in (a, b)$. The Lagrange formula for $R_n(x)$ reads

$$R_n(x) = \frac{f^{(n+1)}(x_1)}{(n+1)!} (x - x_0)^{n+1} ,$$

where x_1 is a suitable value between x_0 and x. By assumption (1.40) we deduce

$$|R_n(x)| \le \frac{ML^{(n+1)}}{(n+1)!} |x - x_0|^{n+1} .$$

Now the theorem's proof reduces to showing that for any x

$$\lim_{n \to +\infty} \frac{(L \cdot |x - x_0|)^{n+1}}{(n+1)!} = 0 , \tag{1.41}$$

and this follows from the fact that the series

$$\sum_{n=0}^{\infty} \frac{(L \cdot |x - x_0|)^{n+1}}{(n+1)!}$$

is convergent by the ratio test, since

$$\lim_{n \to +\infty} \frac{(L \cdot |x - x_0|)^{n+1}}{(n+1)!} \cdot \frac{n!}{(L \cdot |x - x_0|)^n} = \lim_{n \to +\infty} \frac{L \cdot |x - x_0|}{n+1} = 0 ,$$

and the general term in (1.41) goes to zero.

In the special case where $x_0 = 0$, the Taylor series (1.39) reads

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(k)}(0)}{k!}x^k + \ldots$$

and is sometimes called *Maclaurin series* of f.

As an application of the previous Theorem 1 let us prove that the functions

$$e^x , \qquad \sin x , \qquad \cos x$$

admit a Maclaurin series. For $y = e^x$ we have

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^k}{k!} + \ldots$$

and since the derivatives of e^x are uniformly bounded on every interval of type $(-a, a)$, with $a > 0$, the exponential function in base e can be expanded in Maclaurin series on $(-a, a)$, by virtue of Theorem 1. But a is arbitrary, so

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^k}{k!} + \ldots \qquad \forall x \in \mathbb{R}\,.$$

Since

$$\left|D^k \sin x\right| \leq 1, \qquad \left|D^k \cos x\right| \leq 1, \qquad \forall x \in \mathbb{R}$$

for any $k \in \mathbb{N}$, in analogy to what we did for e^x it can be shown that $\sin x$ and $\cos x$ can be expanded on $\mathbb{R}$ to give the following series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \qquad \forall x \in \mathbb{R}\,.$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \qquad \forall x \in \mathbb{R}\,.$$

Figure 1.5 depicts the function $\sin x$ and its Taylor polynomials of orders 3, 5 and 7.

Here is a further useful theorem for obtaining series expansions.

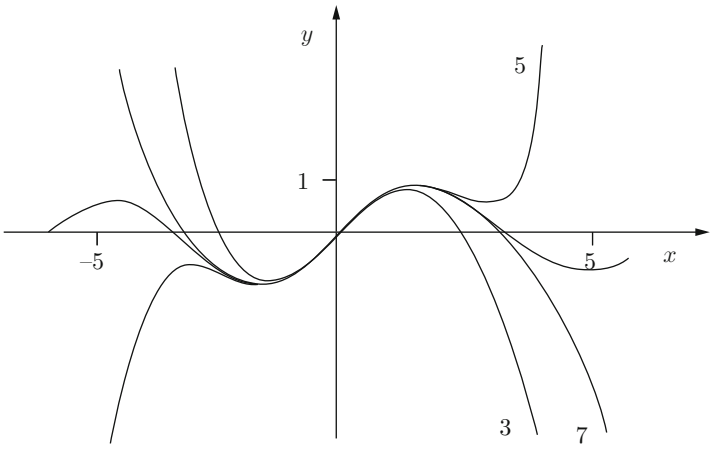


Fig. 1.5

Theorem 2 *Let f be a function with derivatives of any order on (a, b), and let $x_0 \in (a, b)$. If*

$$f'(x) = \sum_{k=1}^{\infty} \frac{f^{(k)}(x_0)}{(k-1)!}(x - x_0)^{k-1}, \qquad \forall\, x \in (a, b), \tag{1.42}$$

i.e. if the derivative series of the Taylor series of f has sum f', then f can be expanded in Taylor series around x_0 on the interval (a, b).

Proof Take an arbitrary closed subinterval I of (a, b), with $x_0 \in I$. The derivative series (1.42), as any convergent power series, converges uniformly on I. Interchanging the limit and the derivative we deduce that the Taylor series of f, being convergent at x_0, must converge on I, for whichever $I \subset (a, b)$ containing x_0, and its sum is a function g such that $g(x_0) = f(x_0)$ and $g'(x) = f'(x)$ for any $x \in (a, b)$. Integrating, it follows that $g(x) = f(x)$ for any $x \in (a, b)$.

Now we shall apply Theorem 2 to expand in Maclaurin series the functions

$$\log(1 + x), \qquad \operatorname{arctg} x.$$

The Maclaurin series of $y = \log(1 + x)$ is

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{k-1}\frac{x^k}{k} + \ldots$$

Its derivative series

$$1 - x + x^2 - \ldots + (-1)^{k-1}x^{k-1} + \ldots$$

is the geometric series of ratio $-x$ and starting at 1, which converges (pointwise) on the interval $(-1, 1)$. The latter's sum equals

$$\frac{1}{1+x} = 1 - x + x^2 - \ldots + (-1)^{k-1}x^{k-1} + \ldots, \qquad \forall\, x \in (-1, 1),$$

which is exactly the derivative of $y = \log(1 + x)$. By Theorem 2 therefore, $f(x) = \log(1 + x)$ can be expanded in Maclaurin series on $(-1, 1)$, and

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{k-1}\frac{x^k}{k} + \ldots, \qquad \forall\, x \in (-1, 1). \tag{1.43}$$

The Maclaurin series of $y = \operatorname{arctg} x$ is

$$x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots + (-1)^{k-1}\frac{x^{2k-1}}{2k-1} + \ldots$$

The derivative series

$$1 - x^2 + x^4 - \ldots + (-1)^{k-1} x^{2(k-1)} + \ldots$$

is the geometric series of ratio $-x^2$ and first term 1, which converges on $(-1, 1)$. The sum is the function $y = 1/(1+x^2)$, which equals the derivative of $y = \operatorname{arctg} x$. Again, Theorem 2 guarantees that $f(x) = \operatorname{arctg} x$ admits Maclaurin series on $(-1, 1)$ and

$$\operatorname{arctg} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots + (-1)^{k-1} \frac{x^{2k-1}}{2k-1} + \ldots, \qquad \forall\, x \in (-1, 1)\,. \tag{1.44}$$

As a matter of fact (1.44) holds on the entire closed interval $[-1, 1]$, by *Abel's theorem* (whose proof can be found in Sect. 1.12 in the Appendix). When $x = 1$, in particular, we recover the formula

$$\frac{\pi}{4} = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{2k-1}\,. \tag{1.45}$$

Applying the aforementioned Abel theorem, it is immediate to verify that (1.43) holds at $x = 1$, so

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \ldots + (-1)^{k-1} \frac{1}{k} + \ldots$$

Consider now the *binomial series*. Let us prove that for any $\alpha \in \mathbb{R}$ the function

$$f(x) = (1+x)^{\alpha}\,, \qquad\qquad x \in (-1, 1)\,,$$

admits Maclaurin series on $(-1, 1)$, and that

$$(1+x)^{\alpha} = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k\,, \qquad\qquad x \in (-1, 1)\,, \tag{1.46}$$

where the numbers $\binom{\alpha}{k}$, called *binomial coefficients*, are given by

$$\binom{\alpha}{k} = \begin{cases} 1 & \text{for} \quad k = 0 \\ \dfrac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!} & \text{for} \quad k \in \mathbb{N}\,. \end{cases} \tag{1.47}$$

For $k \in \mathbb{N}$ we have

$$f^{(k)}(x) = \alpha(\alpha-1)\cdots(\alpha-k+1)(1+x)^{\alpha-k},$$

and so at $x = 0$

$$f^{(k)}(0) = \alpha(\alpha-1)\cdots(\alpha-k+1)\,.$$

Hence the Maclaurin series of f reads

$$1 + \binom{\alpha}{1} x + \binom{\alpha}{2} x^2 + \ldots + \binom{\alpha}{k} x^k + \ldots \tag{1.48}$$

If α is a positive integer, (1.48) reduces to a polynomial, i.e. the sum of $\alpha+1$ term, since the binomial coefficients $\binom{\alpha}{k}$ vanish for $k > \alpha$. When $\alpha \notin \mathbb{N}$ we have

$$\frac{\left|\binom{\alpha}{k+1} x^{k+1}\right|}{\left|\binom{\alpha}{k} x^k\right|} = \left|\frac{\alpha-k}{k+1}\right| \cdot |x|$$

and by the ratio test the convergence radius is 1. Calling $g(x)$, with $x \in (-1,1)$, the sum of (1.48) and differentiating term by term we obtain

$$g'(x) = \sum_{k=1}^{\infty} k \binom{\alpha}{k} x^{k-1}\,, \qquad \forall\, x \in (-1,1)\,.$$

Now multiplying by $1+x$,

$$\begin{aligned}(1+x)\cdot g'(x) &= \sum_{k=1}^{\infty} k \binom{\alpha}{k} x^{k-1} + \sum_{k=1}^{\infty} k \binom{\alpha}{k} x^k = \\ &= \sum_{k=0}^{\infty} (k+1) \binom{\alpha}{k+1} x^k + \sum_{k=1}^{\infty} k \binom{\alpha}{k} x^k = \\ &= \alpha + \sum_{k=1}^{\infty} \left[(k+1)\binom{\alpha}{k+1} + \binom{\alpha}{k}\right] x^k\,.\end{aligned} \tag{1.49}$$

Keeping in account that

$$(k+1)\binom{\alpha}{k+1}+k\binom{\alpha}{k}=\frac{\alpha\cdot(\alpha-1)\cdot\cdots\cdot(\alpha-k+1)(\alpha-k)}{1\cdot 2\cdot\ldots\cdot(k-1)\cdot k}+$$
$$+\frac{\alpha\cdot(\alpha-1)\cdot\cdots\cdot(\alpha-k+1)k}{1\cdot 2\cdot\ldots\cdot(k-1)\cdot k}=$$
$$=\frac{\alpha\cdot(\alpha-1)\cdot\cdots\cdot(\alpha-k+1)}{1\cdot 2\cdot\ldots\cdot(k-1)\cdot k}\cdot\alpha=\binom{\alpha}{k}\alpha\,,$$

from (1.49) we infer

$$(1+x)\cdot g'(x)=\alpha+\sum_{k=1}^{\infty}k\binom{\alpha}{k}\alpha\,x^k=\alpha\sum_{k=0}^{\infty}\binom{\alpha}{k}x^k=\alpha\,g(x)\,.$$

Therefore

$$\frac{d}{dx}\left[g(x)(1+x)^{-\alpha}\right]=g'(x)(1+x)^{-\alpha}-g(x)\,\alpha(1+x)^{-\alpha-1}=$$
$$=\alpha\,g(x)(1+x)^{-\alpha-1}-g(x)\,\alpha(1+x)^{-\alpha-1}=0$$

and so the function $g(x)(1+x)^{-\alpha}$ is constant. But $g(0)=1$ (recall $g(x)$ is the sum of (1.48)), so we find

$$g(x)(1+x)^{-\alpha}=g(0)=1$$

and eventually

$$g(x)=(1+x)^{\alpha}=f(x)\,,\qquad\forall\,x\in(-1,1)\,.$$

Consider expansion (1.46) in the special case $\alpha=-1/2$. For any $k\in\mathbb{N}$, from (1.47) we then have

$$\binom{-1/2}{k}=(-1)^k\,\frac{1\cdot 3\cdot 5\cdot\ldots\cdot(2k-1)}{2^k\cdot k!}=$$
$$=(-1)^k\,\frac{1\cdot 3\cdot 5\cdot\ldots\cdot(2k-1)}{2\cdot 4\cdot 6\cdot\ldots\cdot 2k}=(-1)^k\,\frac{(2k-1)!!}{(2k)!!}\,,$$

where $n!!$ is the product of all odd numbers between 1 and n if n is odd, and the product of all even numbers from 2 to n if n is even. Hence

$$\frac{1}{\sqrt{1+x}}=1+\sum_{k=1}^{\infty}(-1)^{k-1}\,\frac{(2k-1)!!}{(2k)!!}\,x^k\,,\qquad x\in(-1,1)\,.$$

Actually, by Abel's theorem (see section "Abel's Theorem on Power Series" in Appendix) such expansion also holds at $x = 1$.

Similarly, if we take $\alpha = 1/2$ in (1.46) we obtain the noteworthy expansion

$$\sqrt{1+x} = 1 + \frac{x}{2} + \sum_{k=2}^{\infty} (-1)^k \frac{(2k-3)!!}{(2k)!!} x^k , \qquad x \in (-1, 1] . \tag{1.50}$$

Finally, observe that replacing x with $-x^2$ in (1.50) and integrating term by term, we arrive at the remarkable expansion

$$\arcsin x = x + \sum_{k=1}^{\infty} \frac{(2k-1)!!}{(2k)!!} \frac{x^{2k+1}}{2k+1} , \qquad x \in (-1, 1) ,$$

which converges at $x = \pm 1$ as well, as is easy to see.

1.8 Fourier Series

A function $f(x)$, defined at $x \in \mathbb{R}$, is called *periodic of period* T (or T*-periodic*), if

$$f(x+T) = f(x) , \qquad \forall\, x \in \mathbb{R} .$$

If a function is T-periodic, it is also periodic of period $2T, 3T, \ldots, kT, \ldots$

Let a, b be real numbers and $k \in \mathbb{N}$. The function

$$f(x) = a \cos kx + b \sin kx \tag{1.51}$$

is periodic of period 2π. More precisely, since

$$\cos kx = \cos(kx + 2\pi) = \cos\left(k\left(x + \frac{2\pi}{k}\right)\right) ,$$

and similarly for $\sin kx$, the function $f(x)$ in (1.51) satisfies the relationship

$$f(x) = f\left(x + \frac{2\pi}{k}\right) , \qquad \forall\, x \in \mathbb{R} ,$$

and is therefore $2\pi/k$-periodic.

Of particular interest, also in view of the applications to Physics, are the linear combinations of functions like (1.51),

$$s_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx) , \qquad \forall\, n \in \mathbb{N} ,$$

called *trigonometric polynomials*. The real numbers a_0 and a_k, b_k, for $k = 1, 2, \ldots, n$, are called *coefficients* of the trigonometric polynomial. Clearly, for any $n \in \mathbb{N}$, $s_n(x)$ is 2π-periodic.

Suppose the sequence (of partial sums) $s_n(x)$ in (1.8) converges for any $x \in \mathbb{R}$. In other terms, given $a_0 \in \mathbb{R}$ and a_k, b_k, for $k \in \mathbb{N}$, suppose that the series

$$\frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx), \tag{1.52}$$

called *trigonometric series* with coefficients a_0, a_k, b_k, is convergent for any $x \in \mathbb{R}$. Since $s_n(x)$ is periodic for any $n \in \mathbb{N}$, then also the sum $f(x)$ will be a periodic function of period 2π.

Conversely, given a 2π-periodic function $f(x)$ we may ask whether it admits a *trigonometric expansion*, i.e. whether we can find coefficients a_0, a_k, b_k, for $k \in \mathbb{N}$, such that series (1.52) converges for every $x \in \mathbb{R}$ with sum $f(x)$.

Assume $f(x)$ can be expanded in trigonometric series as in (1.52), i.e.

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx), \qquad \forall\, x \in \mathbb{R}, \tag{1.53}$$

and suppose that (1.52) is integrable term by term on an interval of width 2π (by the theorem of integration of series of Sect. 1.5, a sufficient condition for this is demanding that the series converges uniformly on the interval). Then we may follow Fourier's method to determine the coefficients a_0, a_k, b_k. To be precise, we consider (1.53), for example with $x \in [-\pi, \pi]$, and multiply both sides of (1.53) by $\cos mx$ or $\sin mx$, with $m \in \mathbb{N}$ given, then integrate between $-\pi$ and π. Thus we obtain

$$\int_{-\pi}^{\pi} f(x) \cos mx\, dx = \frac{a_0}{2} \int_{-\pi}^{\pi} \cos mx\, dx + \tag{1.54}$$
$$+ \sum_{k=1}^{n} \left(a_k \int_{-\pi}^{\pi} \cos kx \cdot \cos mx\, dx + b_k \int_{-\pi}^{\pi} \sin kx \cdot \cos mx\, dx \right);$$

$$\int_{-\pi}^{\pi} f(x) \sin mx\, dx = \frac{a_0}{2} \int_{-\pi}^{\pi} \sin mx\, dx + \tag{1.55}$$
$$+ \sum_{k=1}^{n} \left(a_k \int_{-\pi}^{\pi} \cos kx \cdot \sin mx\, dx + b_k \int_{-\pi}^{\pi} \sin kx \cdot \sin mx\, dx \right).$$

Noting that

$$\int_{-\pi}^{\pi} \cos kx \cdot \cos mx \, dx = \begin{cases} 0 & \text{if} \quad m \neq k \\ \pi & \text{if} \quad m = k \neq 0 \,, \end{cases} \tag{1.56}$$

$$\int_{-\pi}^{\pi} \cos kx \cdot \sin mx \, dx = 0 \,, \tag{1.57}$$

$$\int_{-\pi}^{\pi} \sin kx \cdot \sin mx \, dx = \begin{cases} 0 & \text{if} \quad m \neq k \\ \pi & \text{if} \quad m = k \neq 0 \,, \end{cases} \tag{1.58}$$

from (1.54) we find the relation

$$a_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, \cos mx \, dx \,, \qquad m = 0, 1, 2, \ldots, \tag{1.59}$$

(also for $m = 0$, due to the factor $1/2$ that multiplies a_0 in (1.53)), while from (1.55) we obtain

$$b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, \sin mx \, dx \,, \qquad m = 1, 2, 3, \ldots. \tag{1.60}$$

In any case, irrespective of the function $f(x)$, integrable between $-\pi$ and π, the constants defined by (1.59) for $m = 0, 1, 2, \ldots$ and by (1.60) for $m = 1, 2, \ldots$, are called the *Fourier coefficients* of $f(x)$. Series (1.52), when the coefficients are the Fourier coefficients (1.59), (1.60) of $f(x)$, is the *Fourier series* of $f(x)$.

Observe that if $f : [-\pi, \pi] \to \mathbb{R}$ is an *even* function, i.e. $f(x) = f(-x)$ for any $x \in [-\pi, \pi]$, the Fourier series of f has all coefficients b_k zero. And if f is *odd*, i.e. $f(x) = -f(-x)$ for any $x \in [-\pi, \pi]$, the Fourier series of f reduces to a series containing only sine functions.

In this and the following section, given a 2π-periodic, integrable function f between $-\pi$ and π, we shall examine the problem of expanding it in Fourier series, meaning when it might coincide with the sum of its Fourier series.

To that end let us introduce some notation. Let $f(x)$ be defined on $\mathbb{R}$. If for $x \in \mathbb{R}$ the limit

$$\lim_{h \to 0^+} \; f(x + h)$$

exists, we shall denote it by $f(x+)$. If

$$\lim_{h \to 0^-} \; f(x + h)$$

exists, we call it $f(x-)$.

We shall also say $f : [a, b] \to \mathbb{R}$ is *piecewise regular* on $[a, b]$ if there are finitely many points x_i, $i = 0, 1, \ldots, N$, with $a = x_0 < x_1 < \ldots < x_N = b$, such

that f is differentiable with continuous derivative on every interval (x_{i-1}, x_i) and the restriction of f' to (x_{i-1}, x_i) can be extended continuously to the closed interval $[x_{i-1}, x_i]$. We say $f : \mathbb{R} \to \mathbb{R}$ is *piecewise regular* on $\mathbb{R}$ when it is piecewise regular on every closed and bounded subinterval of $\mathbb{R}$.

The following fact, regarding the *pointwise convergence* of Fourier series, is proved in Sect. 1.9.

Theorem (Pointwise Convergence of Fourier Series) *Let f be a 2π-periodic, piecewise-regular function on $\mathbb{R}$. For any $x \in \mathbb{R}$ the Fourier series of f converges to*

$$\frac{1}{2}\left[f(x+) + f(x-)\right], \tag{1.61}$$

i.e. to the average of the right and the left limits of f at x. In particular, the Fourier series of f converges to $f(x)$ at all points x where it is continuous.

Example 1 Let f be the 2π-periodic function obtained by extending the function $x \in [-\pi, \pi] \to |x|$ periodically over $\mathbb{R}$ (Fig. 1.6).

As f is even, what we have said earlier ensures that $b_k = 0$ for every $k \in \mathbb{N}$, while an easy computation gives $a_0 = \pi$ and

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} |x| \cos kx\, dx = \frac{2}{\pi}\int_0^{\pi} x \cos kx\, dx =$$

$$= \frac{2}{\pi}\left\{\left[\frac{1}{k}\, x \sin kx\right]_0^{\pi} - \frac{1}{k}\int_0^{\pi} \sin kx\, dx\right\} = \frac{2}{\pi}\,\frac{(-1)^k - 1}{k^2}$$

for $k = 1, 2, \ldots$.

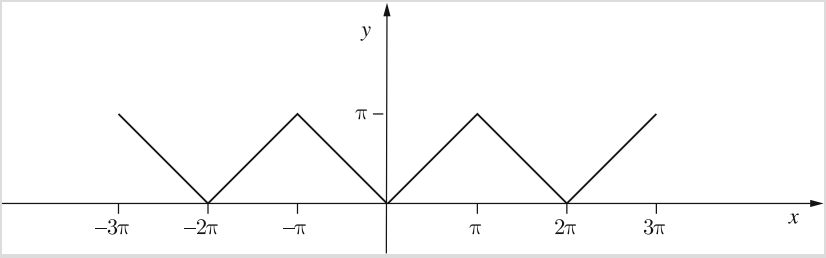


Fig. 1.6

The Fourier series of f is then

$$\frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos(2k+1)x}{(2k+1)^2} \,. \tag{1.62}$$

By the theorem on pointwise convergence of Fourier series, the latter converges to $f(x)$ for any $x \in \mathbb{R}$. As (1.62) is totally convergent, the Fourier series of f converges uniformly on $\mathbb{R}$ as well (also see Example 3 below). Putting $x = 0$ in (1.62) produces the remarkable identity

$$\frac{\pi^2}{8} = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \,. \tag{1.63}$$

Example 2 Let f be the 2π-periodic function obtained by extending periodically the function

$$g(x) = \begin{cases} 0 & \text{if} \quad -\pi < x \le 0 \\ 1 & \text{if} \quad 0 < x \le \pi \end{cases}$$

to $\mathbb{R}$. The Fourier coefficients of f are then

$$a_0 = 1, \qquad a_k = 0, \qquad b_k = \frac{1 - (-1)^k}{k\pi} \,, \qquad\qquad \forall\, k \in \mathbb{N} \,,$$

so the Fourier series of f is

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{\sin(2k+1)x}{2k+1} \,. \tag{1.64}$$

By the theorem on Fourier pointwise convergence, this series converges to $f(x)$ for any $x \in \mathbb{R}$ different from $k\pi$, $k \in \mathbb{Z}$. Note that for $x = 0$ the Fourier series of f has sum $1/2$, which is the average of the left and right limits of f at 0.

Taking $x = \pi/2$ in (1.64) produces the following expression for π:

$$\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \,,$$

which coincides with the Taylor series of $\operatorname{arctg} x$ at $x = 1$ (see (1.45)).

Let us now establish sufficient conditions for the *uniform convergence* of a Fourier series.

Also the following result will be proved in the next section.

Theorem (Uniform Convergence of Fourier Series) *Let f be a 2π-periodic, piecewise-regular continuous function on $\mathbb{R}$. The Fourier series of f converges totally, hence uniformly, to f on $\mathbb{R}$.*

Example 3 Consider again the function f of Example 1, i.e. the 2π-periodic extension of $x \in [-\pi, \pi] \to |x|$ to $\mathbb{R}$ (see Fig. 1.6).

We have already remarked that its Fourier series converges uniformly. The same conclusion may be reached using the above uniform convergence theorem.

We state, without proof, a useful generalisation of the theorem on uniform convergence for Fourier series.

Theorem *Let f be a 2π-periodic, piecewise-regular function. The Fourier series of f converges uniformly to f on any interval $[a, b]$ where f is continuous.*

Example 4 Let f be the 2π-periodic function

$$f(x) = \begin{cases} 0 & \text{if} \quad -\pi < x \le 0 \\ x & \text{if} \quad 0 < x \le \pi \,. \end{cases}$$

Its Fourier coefficients are

$$a_0 = \frac{\pi}{2}, \qquad a_k = \frac{(-1)^k - 1}{\pi k^2}, \qquad b_k = \frac{(-1)^{k+1}}{k},$$

so the Fourier series of f is

$$\frac{\pi}{4} - \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{\cos(2k+1)x}{(2k+1)^2} - \sum_{k=1}^{\infty} \frac{(-1)^k \sin kx}{k} \,. \tag{1.65}$$

Since f is continuous at all points of $\mathbb{R}$ different from $\pi + 2k\pi$, with $k \in \mathbb{Z}$, the previous theorem guarantees that (1.65) converges uniformly to f on every closed interval $[a, b] \subset (-\pi, \pi)$. Figure 1.7 shows the graphs of the first Fourier polynomials of (1.65).

(continued)

Example 4 (continued)

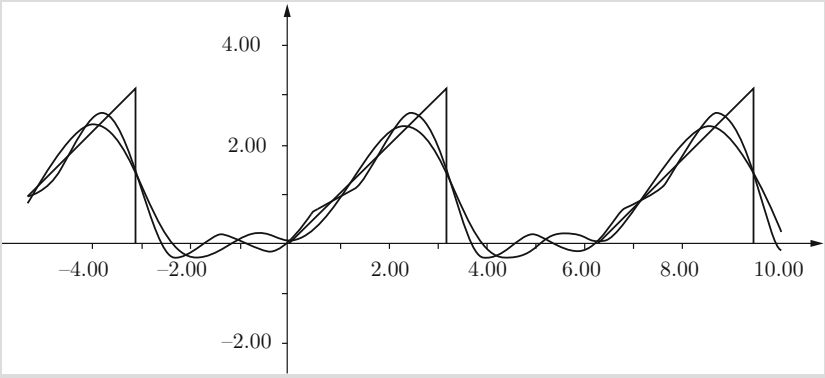


Fig. 1.7

Series (1.65), for $x = \pi$, reduces to

$$\frac{\pi}{4} + \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2};$$

by (1.63) this sum equals

$$\frac{\pi}{2} = \frac{1}{2} [f(\pi+) + f(\pi-)],$$

in agreement with the theorem on pointwise convergence and with (1.61).

We finish by pointing out that in certain cases, even if we do not know that the Fourier series of a piecewise-regular function f converges uniformly (or even pointwise) to f, we may still be able to integrate term by term.

The proof of the following theorem is postponed to the next section.

Theorem (Term-by-Term Integration of Fourier Series) *Let f be a piecewise-regular, 2π-periodic function. Given $x_0, x \in [-\pi, \pi]$, the following formula holds*

$$\int_{x_0}^{x} f(t)\, dt = \frac{a_0}{2} (x - x_0) + \sum_{k=1}^{\infty} \int_{x_0}^{x} (a_k \cos kt + b_k \sin kt)\, dt\,. \tag{1.66}$$

1.9 The Convergence of Fourier Series

This section is devoted to proving a number of results on the convergence of Fourier series that were stated in the previous section. In particular we shall prove the theorems on pointwise convergence, on uniform convergence and on term-by-term integration of Fourier series.

Let us start by proving *Bessel's inequality* (1.68), which expresses an important property of a Fourier series' coefficients.

Bessel Inequality *Let $f : [-\pi, \pi] \to \mathbb{R}$ be a bounded integrable function. If s_n denotes the nth partial sum of the Fourier series of f, we have*

$$\frac{1}{\pi}\int_{-\pi}^{\pi} |f(x) - s_n(x)|^2\, dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f^2(x)\, dx - \left[\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2)\right], \quad (1.67)$$

and moreover

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty}(a_k^2 + b_k^2) \le \frac{1}{\pi}\int_{-\pi}^{\pi} f^2(x)\, dx\,. \quad (1.68)$$

Proof First of all observe that (1.68) is a consequence of (1.67): as the left-hand side integral in (1.67) is non-negative, in fact, for any $n \in \mathbb{N}$ we have

$$\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2) \le \frac{1}{\pi}\int_{-\pi}^{\pi} f^2(x)\, dx\,,$$

and passing to the limit as $n \to +\infty$ produces (1.68).

To prove (1.67) let us note that

$$\begin{aligned}\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\, s_n(x)\, dx &= \frac{a_0}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx + \\ &+ \sum_{k=1}^{n}\left[\frac{a_k}{\pi}\int_{-\pi}^{\pi} f(x)\, \cos kx\, dx + \frac{b_k}{\pi}\int_{-\pi}^{\pi} f(x)\, \sin kx\, dx\right] = \\ &= \frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2)\,. \end{aligned} \quad (1.69)$$

Similarly, using (1.56), (1.57) and (1.58), we obtain

$$\frac{1}{\pi}\int_{-\pi}^{\pi} s_n^2(x)\, dx = \frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2)\,.$$

From this relation and (1.69) we deduce

$$\frac{1}{\pi}\int_{-\pi}^{\pi}|f(x)-s_n(x)|^2\,dx=$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}f^2(x)\,dx-\frac{2}{\pi}\int_{-\pi}^{\pi}f(x)s_n(x)\,dx+\frac{1}{\pi}\int_{-\pi}^{\pi}s_n^2(x)\,dx=$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}f^2(x)\,dx-\frac{a_0^2}{2}-\sum_{k=1}^{n}(a_k^2+b_k^2)\,,$$

and hence claim (1.67).

From (1.68) we recover in particular

$$\lim_{k\to+\infty}a_k=\lim_{k\to+\infty}b_k=0\,,$$

so we can state the following

Corollary *If $f:[-\pi,\pi]\to\mathbb{R}$ is a bounded integrable function,*

$$\lim_{k\to+\infty}\int_{-\pi}^{\pi}f(x)\,\sin kx\,dx=\lim_{k\to+\infty}\int_{-\pi}^{\pi}f(x)\,\cos kx\,dx=0\,. \tag{1.70}$$

In order to prove the theorem on the pointwise convergence of Fourier series we start with a technical lemma.

Lemma 1 *For any $n\in\mathbb{N}$ and any $x\in\mathbb{R}$ we have*

$$\frac{1}{2}+\cos x+\cos 2x+\ldots+\cos nx=\frac{\sin\left((n+\frac{1}{2})x\right)}{2\,\sin\frac{x}{2}}\,. \tag{1.71}$$

Furthermore, if f is 2π-periodic and integrable on $[-\pi,\pi]$ and s_n is the nth partial sum of its Fourier series, then

$$s_n(x)=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x+t)\,\frac{\sin\left(\left(n+\frac{1}{2}\right)t\right)}{2\,\sin\frac{t}{2}}\,dt\,. \tag{1.72}$$

Proof Formula (1.71) arises by summing over k from 1 to n the two sides of the elementary identity

$$\sin\left(\left(k+\frac{1}{2}\right)x\right)-\sin\left(\left(k-\frac{1}{2}\right)x\right)=2\,\sin\frac{x}{2}\,\cos kx\,.$$

To prove (1.72) we note that (1.71) implies

$$
\begin{aligned}
s_n(x) &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\left[\frac{1}{2}+\sum_{k=1}^{n}(\cos ky\,\cos kx+\sin ky\,\sin kx)\right]dy = \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\left[\frac{1}{2}+\sum_{k=1}^{n}\cos k(y-x)\right]dy = \\
&= \frac{1}{\pi}\int_{-\pi-x}^{\pi-x} f(x+t)\left[\frac{1}{2}+\sum_{k=1}^{n}\cos kt\right]dt = \\
&= \frac{1}{\pi}\int_{-\pi-x}^{\pi-x} f(x+t)\,\frac{\sin\left(\left(n+\frac{1}{2}\right)t\right)}{2\,\sin\frac{t}{2}}\,dt\,.
\end{aligned}
$$

Then (1.72) follows from the above once we recall that a T-periodic function g satisfies

$$
\int_a^{a+T} g(t)\,dt = \int_0^T g(t)\,dt
$$

for any $a \in \mathbb{R}$.

Proof of the Pointwise Convergence of Fourier Series Fix $x \in \mathbb{R}$ and $k \in \mathbb{N}$. Since (1.71) implies

$$
\frac{1}{\pi}\int_0^{\pi}\frac{\sin\left(\left(k+\frac{1}{2}\right)t\right)}{2\,\sin\frac{t}{2}}\,dt = \frac{1}{\pi}\int_{-\pi}^{0}\frac{\sin\left(\left(k+\frac{1}{2}\right)t\right)}{2\,\sin\frac{t}{2}}\,dt = \frac{1}{2}\,,
$$

by (1.72) we then find

$$
\begin{aligned}
&s_k(x)-\frac{1}{2}[f(x+)+f(x-)] = \\
&= \frac{1}{\pi}\int_0^{\pi}\frac{f(x+t)-f(x+)}{2\,\sin\frac{t}{2}}\,\sin\left(\left(k+\frac{1}{2}\right)t\right)dt + \\
&\quad+\frac{1}{\pi}\int_{-\pi}^{0}\frac{f(x+t)-f(x-)}{2\,\sin\frac{t}{2}}\,\sin\left(\left(k+\frac{1}{2}\right)t\right)dt\,.
\end{aligned}
\tag{1.73}
$$

Setting

$$F(t) = \begin{cases} \dfrac{f(x+t) - f(x+)}{2 \sin \dfrac{t}{2}} & 0 < t \le \pi \\ 0 & t = 0 \\ \dfrac{f(x+t) - f(x-)}{2 \sin \dfrac{t}{2}} & -\pi \le t < 0 \end{cases}, \tag{1.74}$$

since f is piecewise regular we deduce

$$\lim_{t \to 0^+} F(t) = f'_+(x), \qquad \lim_{t \to 0^-} F(t) = f'_-(x)$$

by de l'Hôpital's rule, where $f'_+(x)$, $f'_-(x)$ are the right and left derivatives of f at x. Then F is piecewise continuous on $[-\pi, \pi]$, and hence integrable and bounded.

Moreover, from (1.73) we have

$$s_k(x) - \frac{1}{2}[f(x+) + f(x-)] = \frac{1}{\pi} \int_{-\pi}^{\pi} F(t) \sin\left(\left(k + \frac{1}{2}\right) t\right) dt =$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} F(t) \cos\frac{t}{2} \sin kt \, dt + \frac{1}{\pi} \int_{-\pi}^{\pi} F(t) \sin\frac{t}{2} \cos kt \, dt \, .$$

The claim then follows by taking the limit as $k \to +\infty$ and applying the relationships in (1.70).

Due to the previous corollary, it is clear that the pointwise-convergence theorem is still valid if we replace the piecewise regularity of f with the weaker assumption that for any $x \in \mathbb{R}$ the function F, defined by (1.74), is bounded and integrable on $[-\pi, \pi]$.

Let us now present a theorem on the uniform convergence of Fourier series. First, though, a lemma.

Lemma 2 *Let f be 2π-periodic, continuous on $\mathbb{R}$ and piecewise regular. Then*

$$k a_k = -b'_k \, , \qquad k b_k = a'_k \, , \qquad \forall \, k \in \mathbb{N} \, , \tag{1.75}$$

where the Fourier coefficients a'_k and b'_k of the derivative f' are given by

$$a'_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f'(x) \cos kx \, dx \, , \qquad b'_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f'(x) \sin kx \, dx \, .$$

Proof As f is continuous and piecewise regular, integrating by parts we obtain

$$ka_k = \frac{k}{\pi}\int_{-\pi}^{\pi} f(x)\cos kx\,dx = \frac{1}{\pi}\left[f(x)\sin kx\right]_{-\pi}^{\pi} - \frac{1}{\pi}\int_{-\pi}^{\pi} f'(x)\sin kx\,dx =$$
$$= -\frac{1}{\pi}\int_{-\pi}^{\pi} f'(x)\sin kx\,dx = -b'_k\,.$$

The second relationship in (1.75) is similar.

Proof of the Uniform Convergence of Fourier Series To prove the claim it suffices to show that

$$\sum_{k=1}^{\infty}(|a_k| + |b_k|) < +\infty\,. \tag{1.76}$$

The elementary inequality $2ab \le a^2 + b^2$, valid for any $a, b \in \mathbb{R}$, implies

$$2|a_k| = 2k|a_k|\frac{1}{k} \le k^2 a_k^2 + \frac{1}{k^2}\,;$$

the term $2|b_k|$ can be bounded in a similar way. To attain (1.76) it is enough to check that

$$\sum_{k=1}^{\infty} k^2(a_k^2 + b_k^2) < +\infty\,.$$

But the above follows from (1.75) by applying Bessel's inequality to the Fourier coefficients of f', which is bounded and integrable on $[-\pi, \pi]$. In fact,

$$\sum_{k=1}^{\infty} k^2(a_k^2 + b_k^2) = \sum_{k=1}^{\infty}[(a'_k)^2 + (b'_k)^2] \le \frac{1}{\pi}\int_{-\pi}^{\pi}[f'(x)]^2\,dx < +\infty\,.$$

We close the section with the proof of the theorem on term-by-term integration of Fourier series. Call F the integral function defined on $[-\pi, \pi]$ as

$$F(x) = \int_{-\pi}^{x}\left(f(t) - \frac{a_0}{2}\right)dt\,, \qquad \forall\, x \in [-\pi, \pi]$$

and extended to $\mathbb{R}$ periodically with period 2π.

This F is continuous on $[-\pi, \pi]$. It vanishes at the interval's endpoints, since $F(-\pi) = 0$ and

$$F(\pi) = \int_{-\pi}^{\pi} f(t)\,dt - \pi a_0 = 0$$

by the definition of the Fourier coefficient a_0 in (1.59).

Therefore F is continuous on the entire $\mathbb{R}$. By the theorem on the uniform convergence of Fourier series, the Fourier series of F converges to F uniformly. For $k \in \mathbb{N}$, denote by

$$\alpha_k = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\cos kx\,dx\,, \qquad \beta_k = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\sin kx\,dx$$

the Fourier coefficients of F. As $F' = f$, by Lemma 2 we have

$$k\alpha_k = b_k\,, \qquad k\beta_k = a_k\,, \qquad \forall\, k \in \mathbb{N}\,.$$

Then given $x_0, x \in [-\pi, \pi]$,

$$\begin{aligned}
&\int_{x_0}^{x}\left(f(t) - \frac{a_0}{2}\right)dt = F(x) - F(x_0) = \\
&= \sum_{k=1}^{\infty}[\alpha_k(\cos kx - \cos kx_0) + \beta_k(\sin kx - \sin kx_0)] = \\
&= \sum_{k=1}^{\infty}\left[-b_k\frac{\cos kx - \cos kx_0}{k} + a_k\frac{\sin kx - \sin kx_0}{k}\right] = \\
&= \sum_{k=1}^{\infty}\left[b_k\int_{x_0}^{x}\sin kt\,dt + a_k\int_{x_0}^{x}\cos kt\,dt\right]
\end{aligned}$$

and (1.66) is proved.

Appendix to Chap. 1

1.10 The Ascoli-Arzelà Theorem

Let f_k be a sequence of continuous real functions on the closed and bounded interval $I = [a, b]$ of $\mathbb{R}$ (more generally, this section's discussion holds if we assume $I \subseteq \mathbb{R}$ is a *compact subset* of $\mathbb{R}$).

We say that the functions f_k are *uniformly bounded* (on $[a, b]$) if there exists $M > 0$ such that

$$|f_k(x)| \le M\,, \qquad \forall\, k \in \mathbb{N}\,, \qquad \forall\, x \in [a, b]\,.$$

We say that the functions f_k are *equicontinuous* (on $[a, b]$) if, for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x, y \in [a, b], \quad |x - y| < \delta \quad \Rightarrow \quad |f_k(x) - f_k(y)| < \varepsilon\,, \qquad \forall\, k \in \mathbb{N}\,. \tag{1.77}$$

In particular, when the functions f_k are *uniformly Lipschitz*, i.e. there exists $L > 0$ such that

$$|f_k(x) - f_k(y)| \le L|x - y|\,, \qquad \forall\, k \in \mathbb{N} \quad \forall\, x, y \in [a, b]\,,$$

then they are equicontinuous on $[a, b]$ as well.

Let us prove the remarkable *Ascoli-Arzelà theorem*.

Ascoli-Arzelà Theorem *Every sequence f_k of uniformly bounded and equicontinuous functions on the closed and bounded interval $[a, b] \subset \mathbb{R}$ admits a uniformly convergent subsequence.*

Proof Call $X = [a, b] \cap \mathbb{Q}$ the set of rational points of the interval $[a, b]$. As X is countable, we may represent it by a sequence x_j, with $j \in \mathbb{N}$. Let us split the proof in two steps.

Step 1 (*construction of a subsequence via Cantor's diagonal method*): by assumption the numerical sequence $f_k(x_1)$ is bounded on $\mathbb{R}$, so there exists a subsequence $f_k^{(1)}$ of f_k such that $f_k^{(1)}(x_1)$ converges to some real number y_1.

As the sequence $f_k^{(1)}(x_2)$ is bounded on $\mathbb{R}$, there exists a subsequence $f_k^{(2)}$ of $f_k^{(1)}$ such that $f_k^{(2)}(x_2)$ converges to some real number y_2. Iterating the process for every $h = 1, 2, \ldots$, we obtain a subsequence $f_k^{(h)}$ of f_k such that $f_k^{(h)}(x_h)$ converges to some $y_h \in \mathbb{R}$. Moreover, for any $h \in \mathbb{N}$, $f_k^{(h+1)}$ is a subsequence of $f_k^{(h)}$. So we have the following picture:

$$\begin{array}{llllll}
f_1^{(1)}(x_1), & f_2^{(1)}(x_1), & f_3^{(1)}(x_1), & \ldots & f_k^{(1)}(x_1), & \ldots \to y_1 \\
f_1^{(2)}(x_2), & f_2^{(2)}(x_2), & f_3^{(2)}(x_2), & \ldots & f_k^{(2)}(x_2), & \ldots \to y_2 \\
f_1^{(3)}(x_3), & f_2^{(3)}(x_3), & f_3^{(3)}(x_3), & \ldots & f_k^{(3)}(x_3), & \ldots \to y_3 \\
\ldots & \ldots & \ldots & \ldots & \ldots & \ldots\ldots\ldots \\
f_1^{(h)}(x_h), & f_2^{(h)}(x_h), & f_3^{(h)}(x_h), & \ldots & f_k^{(h)}(x_h), & \ldots \to y_h \\
\ldots & \ldots & \ldots & \ldots & \ldots & \ldots\ldots\ldots
\end{array} \tag{1.78}$$

Now consider the diagonal sequence

$$g_k(x) = f_k^{(k)}(x)\,, \qquad \forall\, k \in \mathbb{N}\,.$$

For every j, the sequence $g_k(x_j)$, with $k \ge j$, is extracted from the jth sequence in (1.78), so when $k \to +\infty$ we have

$$g_k(x_j) \to y_j\,, \qquad \forall\, j = 1, 2, 3, \ldots\,. \tag{1.79}$$

Step 2 (*Cauchy sequence*): given $\varepsilon > 0$, pick $\delta > 0$ so that the equicontinuity condition (1.77) holds. Divide the bounded interval $[a, b]$ in a finite number of

subintervals of the same length, say into $s \in \mathbb{N}$ parts of length less than δ (so it is enough that $s > (b-a)/\delta$). In each of these parts we choose a point of X, say $x_{j_1}, x_{j_2}, x_{j_3}, \ldots, x_{j_s}$. Then for any $x \in [a,b]$ there exists x_{j_r}, for some $r \in \{1, 2, 3, \ldots, s\}$, such that

$$|x - x_{j_r}| < \delta\,. \tag{1.80}$$

By (1.79), for any $r = 1, 2, 3, \ldots, s$ the sequence $g_k(x_{j_r})$ is convergent as $k \to +\infty$, and hence it is a Cauchy sequence. Then there exists $\nu \in \mathbb{N}$ such that

$$|g_k(x_{j_r}) - g_h(x_{j_r})| < \varepsilon\,, \tag{1.81}$$

for any $h, k > \nu$, and any $r = 1, 2, 3, \ldots, s$.

Let us use the equicontinuity (1.77) and, for any $x \in [a,b]$, consider the point $x_{j_r} \in X$ for which (1.80) holds. From (1.81) it follows that

$$|g_k(x) - g_h(x)| \le |g_k(x) - g_k(x_{j_r})| + |g_k(x_{j_r}) - g_h(x_{j_r})| +$$

$$+ |g_h(x_{j_r}) - g_h(x)| < 3\varepsilon\,,$$

for any $h, k > \nu$. Therefore g_k is a Cauchy sequence for the uniform convergence. The uniform Cauchy criterion of Sect. 1.2 allows to conclude.

1.11 The Weierstrass Approximation Theorem

A frequent problem in Mathematical Analysis is the approximation of a given function $f(x)$ by functions of a special kind.

When $f(x)$ is a continuous function on a closed and bounded interval $[a,b]$, a classical theorem due to Weierstrass provides an exhaustive answer: the approximating functions are polynomials, as long as we interpret the word approximation in the sense of uniform convergence.

Weierstrass Approximation Theorem *Any continuous function $f(x)$ on $[a,b]$ is a uniform limit of a sequence of polynomials.*

Proof The variable change $x \to (x-a)/(b-a)$ maps polynomials to polynomials, and reduces us to consider the interval $[0,1]$.

We may also assume $f(0) = f(1) = 0$; in fact, once the theorem is proved in that case, setting

$$g(x) = f(x) - f(0) - x[f(1) - f(0)]\,, \qquad x \in [0,1]\,,$$

we have $g(0) = g(1) = 0$, and the function $f(x) - g(x)$ is a polynomial. So if $g(x)$ is a uniform limit of polynomials, the same is true for $f(x)$.

So assume $f(0) = f(1) = 0$ and let us extend f to $\mathbb{R}$, by defining it equal to 0 outside $[0, 1]$, and thus obtaining a uniformly continuous function on $\mathbb{R}$. Define the sequence of polynomials

$$q_k(x) = c_k(1 - x^2)^k\,, \qquad k \in \mathbb{N}\,,$$

where c_k, for any $k \in \mathbb{N}$, is a constant chosen so that

$$\int_{-1}^{1} q_k(x)\,dx = c_k \int_{-1}^{1} (1 - x^2)^k\,dx = 1\,. \tag{1.82}$$

Using Bernoulli's inequality $(1 - x^2)^k \geq 1 - kx^2$ (true for any x such that $|x| \leq 1$ and any $k \in \mathbb{N}$), we have

$$\begin{aligned} \int_{-1}^{1} (1 - x^2)^k\,dx = 2\int_{0}^{1} (1 - x^2)^k\,dx \geq 2\int_{0}^{1/\sqrt{k}} (1 - x^2)^k\,dx \geq \\ \geq 2\int_{0}^{1/\sqrt{k}} (1 - kx^2)\,dx = \frac{4}{3\sqrt{k}} > \frac{1}{\sqrt{k}}\,. \end{aligned} \tag{1.83}$$

From (1.82), (1.83), then,

$$c_k < \sqrt{k}\,, \qquad \forall\, k \in \mathbb{N}\,.$$

Therefore, for any $\delta \in (0, 1)$ and any x such that $\delta \leq |x| \leq 1$,

$$0 \leq q_k(x) = c_k(1 - x^2)^k \leq \sqrt{k}(1 - \delta^2)^k\,, \qquad \forall\, k \in \mathbb{N}\,. \tag{1.84}$$

Therefore the polynomial sequence $q_k(x)$ converges to zero uniformly when $\delta \leq |x| \leq 1$. Let us introduce another sequence

$$p_k(x) = \int_{-1}^{1} f(x + t)q_k(t)\,dt \qquad \forall\, k \in \mathbb{N}\,, x \in [0, 1]\,. \tag{1.85}$$

Just like the sequence q_k, this too is made of polynomials, because a variable change in the integral gives

$$p_k(x) = \int_{0}^{1} f(t)q_k(t - x)\,dt\,.$$

By (1.82) we also have

$$\int_{-1}^{1} f(x)q_k(t)\,dt = f(x)\int_{-1}^{1} q_k(t)\,dt = f(x), \qquad \forall\, k \in \mathbb{N}\,, \forall\, x \in [0, 1]\,. \tag{1.86}$$

Given $\varepsilon > 0$, choose $\delta > 0$ so that

$$|x - y| < \delta \qquad \Rightarrow \qquad |f(x) - f(y)| < \frac{\varepsilon}{2}\,, \tag{1.87}$$

and let

$$M = \max\{|f(x)| : 0 \le x \le 1\}\,.$$

We use (1.85), (1.86) first, then (1.87), (1.84), and finally (1.82) to obtain

$$\begin{aligned} |p_k(x) - f(x)| &\le \int_{-1}^{1} |f(x+t) - f(x)| q_k(t)\,dt \le \\ &\le 2M \int_{-1}^{-\delta} q_k(t)\,dt + \frac{\varepsilon}{2} \int_{-\delta}^{\delta} q_k(t)\,dt + \\ &\quad + 2M \int_{\delta}^{1} q_k(t)\,dt \le 4M\sqrt{k}(1-\delta^2)^k + \frac{\varepsilon}{2}\,. \end{aligned}$$

The last quantity is less than ε when k is large enough. The claim now follows.

1.12 Abel's Theorem on Power Series

Consider a power series

$$a_0 + a_1 x + a_2 x^2 + \ldots + a_k x^k + \ldots \tag{1.88}$$

whose convergence radius ϱ is a positive real number, and suppose the series also converges at $x = \varrho$ (a similar reasoning holds at $x = -\varrho$). *Abel's theorem* implies that the sum $f(x)$ of (1.88) is continuous not just on $(-\varrho, \varrho)$, but at $x = \varrho$ as well. Hence the series' sum at $x = \varrho$ can be computed using the values of f in the limit $x \to \varrho^-$.

We shall discuss two versions of Abel's theorem. They differ in the proof and in the conclusion. The thesis of Theorem 2 regards the series' uniform convergence, which implies the sum's continuity of Theorem 1.

To simplify the notation let us consider the case $\varrho = 1$.

This does not affect the result's generality. In fact, if (1.88) converges at $x = \varrho$, the series of general term $b_k = a_k \varrho^k$ clearly converges.

Since

$$\sum_{k=0}^{\infty} b_k \in \mathbb{R}\,,$$

Abel's theorem for the case $\varrho = 1$ talks about the properties of the power series

$$\sum_{k=0}^{\infty} b_k x^k\,, \qquad -1 < x \le 1\,,$$

i.e.

$$\sum_{k=0}^{\infty} a_k \varrho^k x^k\,, \qquad -1 < x \le 1\,.$$

Putting $y = \varrho x$, we deduce similar properties for

$$\sum_{k=0}^{\infty} a_k y^k\,, \qquad -\varrho < y \le \varrho\,.$$

Theorem 1 (Abel) *Let* $\sum_{k=0}^{\infty} a_k$ *be a convergent numerical series and set*

$$f(x) = \sum_{k=0}^{\infty} a_k x^k\,, \qquad -1 < x < 1\,.$$

Then as $x \to 1^-$ *the function* $f(x)$ *converges to* $\sum_{k=0}^{\infty} a_k$.

Proof Take the sequence s_n of partial sums of the given numerical series

$$s_n = \sum_{k=0}^{n} a_k\,, \qquad n = 0, 1, 2, \ldots,$$

and define s_n also for $n = -1$ by $s_{-1} = 0$. Given that $s_n - s_{n-1} = a_n$ for any $n = 0, 1, 2, \ldots$, if $|x| < 1$ we have

$$\sum_{k=0}^{n} a_k x^k = \sum_{k=0}^{n} (s_k - s_{k-1}) x^k =$$

$$= s_n x^n + \sum_{k=0}^{n-1} s_k x^k - x \cdot \sum_{k=0}^{n} s_{k-1} x^{k-1} =$$

$$= s_n x^n + (1 - x) \cdot \sum_{k=0}^{n-1} s_k x^k\,, \qquad -1 < x < 1\,.$$

In the limit as $n \to +\infty$ we obtain

$$f(x) = \lim_{n\to+\infty} \sum_{k=0}^{n} a_k x^k = (1-x) \cdot \sum_{k=0}^{\infty} s_k x^k \,. \tag{1.89}$$

Put $s = \lim_{n\to+\infty} s_n = \sum_{k=0}^{\infty} a_k$. Given $\varepsilon > 0$, let $\nu \in \mathbb{N}$ be such that

$$|s_n - s| < \frac{\varepsilon}{2}, \qquad \forall\, n > \nu.$$

Using the geometric series' sum in the form

$$(1-x) \cdot \sum_{k=0}^{\infty} x^k = 1\,, \qquad \forall\, x \in \mathbb{R} : -1 < x < 1\,,$$

from (1.89) we obtain, for $0 < x < 1$,

$$\begin{aligned} |f(x) - s| &= \left| (1-x) \cdot \sum_{k=0}^{\infty} (s_k - s) x^k \right| \leq \\ &\leq (1-x) \cdot \sum_{k=0}^{\nu} |s_k - s| + (1-x) \cdot \sum_{k=\nu+1}^{\infty} |s_k - s| \cdot x^k < (1-x) \cdot \sum_{k=0}^{\nu} |s_k - s| + \frac{\varepsilon}{2}\,. \end{aligned} \tag{1.90}$$

Finally, since the finite sum $(1-x) \cdot \sum_{k=0}^{\nu} |s_k - s|$ tends to zero when $x \to 1$, there exists $\delta > 0$ such that

$$(1-x) \cdot \sum_{k=0}^{\nu} |s_k - s| < \frac{\varepsilon}{2} \qquad \forall\, x \in (1-\delta, 1)\,. \tag{1.91}$$

From (1.90), (1.91) it follows that $|f(x) - s| < \varepsilon$ for $1 - \delta < x < 1$. In other words,

$$\lim_{x\to 1^-} f(x) = s = \sum_{k=0}^{\infty} a_k\,.$$

Theorem 2 (Abel) *Let* $\sum_{k=0}^{\infty} a_k$ *be a convergent numerical series. Then the power series*

$$\sum_{k=0}^{\infty} a_k x^k \tag{1.92}$$

converges uniformly on the interval $[0, 1]$. *Consequently, the sum* $f(x)$ *of* (1.92) *is a continuous function on* $[0, 1]$.

Since by assumption the power series (1.92) converges at $x = 1$, by Theorem 1, Sect. 1.6, it must converge totally on every closed and bounded subinterval of $(-1, 1)$. Hence we might equivalently conclude, in Theorem 2, that (1.92) converges uniformly on any closed and bounded subinterval of $(-1, 1]$.

Proof of Theorem 2 By the convergence of $\sum_{k=0}^{\infty} a_k$ and Cauchy's criterion for numerical series, for any $\varepsilon > 0$ there exists $\nu \in \mathbb{N}$ such that

$$\left| \sum_{k=n}^{n+p} a_k \right| < \varepsilon\,, \qquad \forall\, n > \nu\,, \qquad \forall\, p = 0, 1, 2, \ldots\,. \tag{1.93}$$

We claim that the uniform Cauchy criterion for series (1.92) holds on $[0, 1]$. To that end, define

$$\begin{aligned} b_0 &= a_n \\ b_1 &= a_n + a_{n+1} \\ b_2 &= a_n + a_{n+1} + a_{n+2} \\ &\ldots \\ &\ldots \\ b_p &= a_n + a_{n+1} + a_{n+2} + \ldots + a_{n+p}. \end{aligned}$$

By (1.93)

$$|b_k| < \varepsilon\,, \qquad \forall\, k = 0, 1, 2, \ldots, p\,. \tag{1.94}$$

As

$$a_n = b_0,\ a_{n+1} = b_1 - b_0,\ a_{n+2} = b_2 - b_1, \ldots,\ a_{n+p} = b_p - b_{p-1}\,,$$

moreover, we have

$$\begin{aligned} \sum_{k=n}^{n+p} a_k x^k &= a_n x^n + a_{n+1} x^{n+1} + \ldots + a_{n+p} x^{n+p} = \\ &= b_0 x^n + (b_1 - b_0) x^{n+1} + \ldots + (b_p - b_{p-1}) x^{n+p} = \\ &= b_0 (x^n - x^{n+1}) + b_1 (x^{n+1} - x^{n+2}) + \ldots + \\ &\quad + b_{p-1} (x^{n+p-1} - x^{n+p}) + b_p x^{n+p}\,. \end{aligned}$$

But $0 \le x \le 1$, so $(x^n - x^{n+1}) \ge 0$, $(x^{n+1} - x^{n+2}) \ge 0$, and so on. Hence from (1.94) we obtain

$$\begin{aligned}\left|\sum_{k=n}^{n+p} a_k x^k\right| &\le |b_0|(x^n - x^{n+1}) + |b_1|(x^{n+1} - x^{n+2}) + \ldots + \\ &+ |b_{p-1}|(x^{n+p-1} - x^{n+p}) + |b_p| x^{n+p} \le \\ &\le \varepsilon[(x^n - x^{n+1}) + (x^{n+1} - x^{n+2}) + \ldots + \\ &+ (x^{n+p-1} - x^{n+p}) + x^{n+p}] = \varepsilon x^n \,.\end{aligned}$$

Again, $0 \le x \le 1$ implies

$$\left|\sum_{k=n}^{n+p} a_k x^k\right| < \varepsilon \,, \qquad \forall\, n > \nu \,, \quad \forall\, p = 0, 1, 2, \ldots$$

for any $x \in [0, 1]$. Therefore the uniform Cauchy criterion ensures that (1.92) converges uniformly on $[0, 1]$. Consequently, by the theorem on the sum's continuity (Sect. 1.5), the sum $f(x)$ of (1.92) is a continuous function on $[0, 1]$.

We finish with an application of Abel's theorem to the computation of the limit of the integral of a power series' sum. In the following statement ϱ is the convergence radius of given series. Note how, in principle, the theorem applies even if the absolute value of ϱ is less than the convergence radius. In such a case, though, the result is not new: it descends from the theorem on integration by series, due to the uniform convergence on the interval.

Theorem 3 *Let* $f(x) = \sum_{k=0}^{\infty} a_k x^k$. *Then*

$$\lim_{\eta \to \varrho^-} \int_0^{\eta} f(x)\, dx = \sum_{k=0}^{\infty} \frac{a_k}{k+1} \varrho^{k+1} \,, \tag{1.95}$$

provided the numerical series on the right is convergent.

Proof We begin by proving the theorem for $\varrho = 1$, i.e.

$$\lim_{\eta \to 1^-} \int_0^{\eta} f(x)\, dx = \sum_{k=0}^{\infty} \frac{a_k}{k+1} \,, \tag{1.96}$$

only assuming the right-hand-side numerical series in ((1.96)) is convergent. By Theorem 1 in Sect. 1.6 the power series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$$

converges for any $|x| < 1$. By Theorem 5, Sect. 1.6, the derivative series

$$\sum_{k=1}^{\infty} a_k x^k$$

is absolutely convergent for $|x| < 1$, with uniform convergence on the interval $[0, \eta]$ for any $0 < \eta < 1$.

Consider $f(x) = \sum_{k=0}^{\infty} a_k x^k$ for any $|x| < 1$, and fix $0 < \eta < 1$. Integrating between 0 and η we have

$$\int_0^{\eta} f(x)\, dx = \sum_{k=0}^{\infty} \frac{a_k}{k+1}\, \eta^{k+1}\,. \tag{1.97}$$

We have assumed that the numerical series on the right in (1.96) is convergent, so Abel's theorem implies

$$\lim_{\eta \to 1^-} \sum_{k=0}^{\infty} \frac{a_k}{k+1}\, \eta^{k+1} = \sum_{k=0}^{\infty} \frac{a_k}{k+1}\,,$$

and by (1.97)

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} = \lim_{\eta \to 1^-} \sum_{k=0}^{\infty} \frac{a_k}{k+1}\, \eta^{k+1} = \lim_{\eta \to 1^-} \int_0^{\eta} f(x)\, dx\,. \tag{1.98}$$

Hence from (1.98) we deduce (1.96), i.e. the claim when $\varrho = 1$.

For the general case we set $b_k = a_k \varrho^{k+1}$, and in the integral we substitute $x = \varrho \cdot t$. This gives

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1}\, \varrho^{k+1} = \sum_{k=0}^{\infty} \frac{b_k}{k+1}\,, \tag{1.99}$$

$$\lim_{\eta \to \varrho^-} \int_0^{\eta} f(x)\, dx = \lim_{\eta \to \varrho^-} \int_0^{\eta} \sum_{k=0}^{\infty} a_k x^k\, dx = \tag{1.100}$$

$$= \lim_{\eta \to 1^-} \int_0^{\eta} \sum_{k=0}^{\infty} a_k (\varrho \cdot t)^k\, \varrho \cdot dt = \lim_{\eta \to 1^-} \int_0^{\eta} \sum_{k=0}^{\infty} b_k t^k\, dt\,.$$

The equality of (1.99) and (1.100) follows from (1.96), once we replace a_k with b_k, under the hypothesis that the numerical series of (1.99) are convergent. This concludes the proof of (1.95).



Metric Spaces and Banach Spaces

2

2.1 Introduction

This chapter collects the main notions from *topology* regarding metric spaces and Banach spaces. Although this material might already be known in particular cases (at least for the standard topology of $\mathbb{R}$) we shall recall it in order to keep the text self-contained. Certain sections might be postponed for a subsequent study.

The starting point is the observation that the notions of *convergent sequence* of real numbers and *continuous real function* of one real variable actually depend on certain features of the absolute value. More specifically, the crucial properties are the following:

$$|x-y|=0 \qquad \text{if and only if} \quad x=y\,, \qquad \forall\, x,\, y\,\in\mathbb{R}; \tag{2.1}$$

$$|x-y|=|y-x|, \qquad \forall\, x,\, y\,\in\mathbb{R}; \tag{2.2}$$

$$|x-y|\le|x-z|+|z-y|, \qquad \forall\, x,\, y,\, z\,\in\mathbb{R}. \tag{2.3}$$

The quantity $d(x,\,y)=|x-y|$ measures the *distance* between the numbers $x,\,y\,\in\mathbb{R}$. Throughout the chapter we shall see that once we axiomatise the idea of distance of two points in a set X, by introducing a function $d(x,\,y)$ satisfying properties similar to (2.1), (2.2), and (2.3), many definitions and theorems of Mathematical Analysis become simpler and more natural. At the same time, in this way we shall acquire the tools necessary for treating more general aspects.

2.2 Metric Spaces

Let X be a set and $d : X\times X\to[0,+\infty)$ a function that associates with each pair $(x,\,y)$ of points of X a real number $d(x,\,y)\ge 0$. One says that d is a *distance* (or

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metric) on X, provided the following conditions are satisfied:

$$d(x, y) = 0 \qquad \text{if and only if} \quad x = y\,, \qquad \forall\, x, y \in X; \tag{2.4}$$

$$d(x, y) = d(y, x), \qquad \forall\, x, y \in X; \tag{2.5}$$

$$d(x, y) \le d(x, z) + d(z, y), \qquad \forall\, x, y, z \in X. \tag{2.6}$$

Condition (2.6) is called *triangle inequality*. Given $x, y \in X$, the number $d(x, y)$ is called the *distance between x and y*.

If d is a distance on the set X, we say (X, d) is a *metric space*. More concisely, we shall say X is a *metric space* when no confusion arises.

Let us see how to apply the properties that define a distance d to prove the following inequality, called *Lipschitz inequality of the distance*:

$$|d(x, z) - d(y, z)| \le d(x, y)\,, \qquad \forall\, x, y, z \in X\,. \tag{2.7}$$

From the triangle inequality (2.6) we have

$$d(x, z) \le d(x, y) + d(y, z)\,,$$

or equivalently,

$$d(x, z) - d(y, z) \le d(x, y)\,. \tag{2.8}$$

Using again the triangle inequality gives

$$d(y, z) \le d(y, x) + d(x, z)\,,$$

and so

$$-d(y, x) \le d(x, z) - d(y, z)\,. \tag{2.9}$$

From (2.8), (2.9), and keeping in account property (2.5), namely that $d(x, y) = d(y, x)$, we eventually obtain

$$-d(x, y) \le d(x, z) - d(y, z) \le d(x, y)\,,$$

which is (2.7).

For any $x_0 \in X$ and any $r > 0$, the *open ball* with centre x_0 and radius r is the set

$$B_r(x_0) = \{x \in X : d(x, x_0) < r\}\,.$$

A set $A \subseteq X$ is called *open* if any one of its points is the centre of an open ball contained in A. This means that for every $x_0 \in A$ there exists $r > 0$ such that

$B_r(x_0) \subseteq A$. The empty set $\emptyset$ is assumed open. A set $C \subseteq X$ is said to be *closed* if its complement $A = X - C$ is open. The collection of all open sets of a metric space (X, d) is called *topology induced by the metric* d.

With the next result we list a few properties of open sets.

Proposition 1 *In a metric space every open ball is an open set, every union of open sets is an open set, the intersection of two open sets is an open set.*

Proof Let us start by the first statement. To show that the open ball $B_r(x_0)$ is open we must prove that for any $x \in B_r(x_0)$ there exists $s > 0$ such that $B_s(x) \subset B_r(x_0)$. For that let $s = r - d(x, x_0)$, and then we have

$$d(y, x) < s \quad \Rightarrow \quad d(y, x_0) \leq d(y, x) + d(x, x_0) < r \, .$$

This implies $y \in B_s(x) \Rightarrow y \in B_r(x_0)$.

Next, consider a family of open sets $(A_i)_{i \in I}$ with $x \in \bigcup_{i \in I} A_i$. Then there exists $i \in I$ such that $x \in A_i$. As the A_i are open, there exists $r > 0$ such that

$$B_r(x) \subseteq A_i \subseteq \bigcup_{i \in I} A_i \, .$$

and so $\bigcup_{i \in I} A_i$ is an open set.

For the last claim consider two open sets A_1, A_2 and let $x \in A_1 \cap A_2$. Then there exist $r_1, r_2 > 0$ such that $B_{r_1}(x) \subseteq A_1$ and $B_{r_2}(x) \subseteq A_2$. Putting $r = \min\{r_1, r_2\}$, we have $B_r(x) \subseteq A_1 \cap A_2$.

From Proposition 1, using *De Morgan's law*

$$\bigcap_{i \in I} (X - A_i) = X - \bigcup_{i \in I} A_i \, ,$$

we immediately obtain the following

Proposition 2 *In a metric space the intersection of closed sets is a closed set, and the union of two closed sets is a closed set.*

Let us consider some examples of metric spaces.

Example 1 Let X be a set and d the function

$$d(x, y) = \begin{cases} 0 & \text{if} \quad x = y \\ 1 & \text{if} \quad x \neq y \end{cases}$$

for $x, y \in X$. Immediately d is a metric, called *discrete metric*, and any subset of X is open for this metric.

Example 2 The function $d(x, y) = |x - y|$ is a metric on the set $\mathbb{R}$ of real numbers, called *Euclidean metric*. It induces the usual topology on the real line, in which a set is open if and only if it is a union of open intervals.

*Example 3 (*Euclidean Metric on $\mathbb{R}^n$*)* Given $n \in \mathbb{N}$ and points $x, y \in \mathbb{R}^n$ of coordinates $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n)$, put

$$d_n(x, y) = \left[\sum_{i=1}^{n}(x_i - y_i)^2\right]^{1/2}, \tag{2.10}$$

called *Euclidean distance*, or *standard metric*, of $\mathbb{R}^n$. In the subsequent sections we shall also use the symbol $|x - y|$ to denote the standard distance $d_n(x, y)$, without further distinguishing the case $n = 1$ from the general situation $n \geq 1$.

It is straightforward to check that d_n satisfies (2.4) and (2.5) in the definition of metric. Let us prove the triangle inequality (2.6):

$$d_n(x, y) \leq d_n(x, z) + d_n(z, y)\,.$$

Setting $a = x - z$, $b = z - y$, or in coordinates

$$a_i = x_i - z_i\,, \qquad b_i = z_i - y_i\,, \qquad \forall\, i = 1, 2, \ldots, n\,,$$

it will suffice to prove that

$$\sum_{i=1}^{n}(a_i + b_i)^2 \leq \left\{\left(\sum_{i=1}^{n} a_i^2\right)^{1/2} + \left(\sum_{i=1}^{n} b_i^2\right)^{1/2}\right\}^2 .$$

This is the same as (computing the squares on either side)

$$\sum_{i=1}^{n} a_i \cdot b_i \leq \left(\sum_{i=1}^{n} a_i^2\right)^{1/2} \cdot \left(\sum_{i=1}^{n} b_i^2\right)^{1/2} . \tag{2.11}$$

For this, observe that for any $t \in \mathbb{R}$

$$0 \leq \sum_{i=1}^{n}(a_i + t b_i)^2 = \left(\sum_{i=1}^{n} b_i^2\right) t^2 + 2\left(\sum_{i=1}^{n} a_i b_i\right) t + \left(\sum_{i=1}^{n} a_i^2\right) = \alpha t^2 + 2\beta t + \gamma\,.$$

(continued)

Example 3 (continued)
Therefore (if α does not vanish, i.e. if the b_i are not all zero, for otherwise (2.11) is obvious) choosing $t = -\beta/\alpha$ we obtain $\gamma - \beta^2/\alpha \geq 0$. Hence

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right) \cdot \left(\sum_{i=1}^{n} b_i^2\right), \tag{2.12}$$

which is equivalent to the *Cauchy-Schwarz inequality*:

$$\left|\sum_{i=1}^{n} a_i b_i\right| \leq \left(\sum_{i=1}^{n} a_i^2\right)^{1/2} \cdot \left(\sum_{i=1}^{n} b_i^2\right)^{1/2}, \tag{2.13}$$

which clearly implies (2.11).

Therefore the function $d_n : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined in (2.10) is a metric. The open balls centred at x_0 are, for $n = 2$, the familiar open discs of the plane (without the boundary circle); for $n = 3$ they are solid open balls in space (without the spherical surface).

Regarding n-dimensional *Euclidean space* $(\mathbb{R}^n, d_n)$, let us introduce a few concepts that will be useful later. Given $a = (a_1, a_2, \ldots, a_n)$ and $b = (b_1, b_2, \ldots, b_n)$ in $\mathbb{R}^n$, the set

$$(a, b) = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : a_i < x_i < b_i, \forall\, i = 1, 2, \ldots, n\}$$

will be called *open interval* in $\mathbb{R}^n$, of endpoints a and b. The point $(a + b)/2$ is the *centre* of the interval (a, b).

It is easy to verify that any open interval in $\mathbb{R}^n$ is contained in an open ball with the same centre, and every open ball in $\mathbb{R}^n$ is contained in some open interval of the same centre.

Example 4 Let $C^0([a, b])$ denote the set of continuous functions on the interval $[a, b]$ on $\mathbb{R}$. For $f, g \in C^0([a, b])$ define

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in [a, b]\}.$$

As (2.4) and (2.5) in the definition of metric are clearly valid, we shall just prove the triangle inequality (2.6):

$$d(f, g) \leq d(f, h) + d(h, g)\,. \tag{2.14}$$

For any $x \in [a, b]$

$$|f(x) - g(x)| \leq |f(x) - h(x)| + |h(x) - g(x)| \leq d(f, h) + d(h, g)\,.$$

(continued)

Example 4 (continued)
Taking the supremum over $x \in [a, b]$ we obtain (2.14). Figure 2.1 shows an open ball for $f \in C^0([a, b])$.

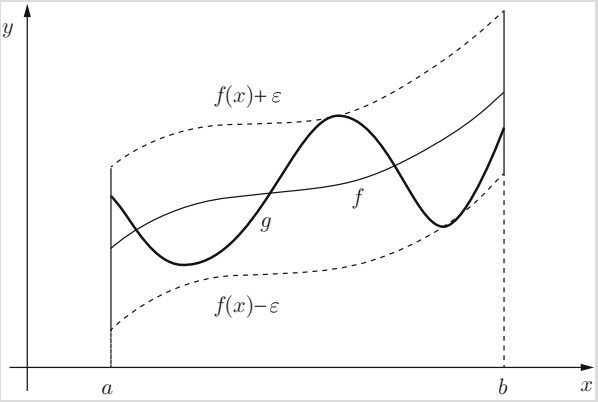


Fig. 2.1 $f, g \in C^0([a, b]) : d(f, g) < \varepsilon$

Let us briefly examine, in the section's second part, some definitions and properties on a metric space (X, d).

A subset I of X is a *neighbourhood* of the point x if there exists an open set $A \subseteq I$ such that $x \in A$. A point x is called *internal* to $Y \subseteq X$ if Y is a neighbourhood of x.

The (possibly empty) set of internal points of Y is called the *interior of* Y, indicated by $\overset{\circ}{Y}$. As a point x belongs to $\overset{\circ}{Y}$ if and only if there is an open set A contained in Y such that $x \in A$, the *interior of* Y is the *union of all open sets contained in* Y, and hence it is the largest open set contained in Y. Therefore a *set* Y *is open if and only if* $Y = \overset{\circ}{Y}$.

A point x is *external* to $Y \subseteq X$ if it is internal to the complement $X - Y$.

A point x is a *limit point* of Y if for any neighbourhood I of x we have $I \cap (Y - \{x\}) \neq \emptyset$, i.e. if every neighbourhood of x contains points of Y other than x itself. A point $x \in Y$ is called *isolated* when it is not a limit point of Y, in other words there exists a neighbourhood I of x such that $I \cap Y = \{x\}$.

The union of $Y \subseteq X$ with the set of its limit points is the *closure* of Y, and is denoted by $\overline{Y}$. It is immediate to show that *a set is closed if and only if it contains its limit points*; consequently a set Y is closed if and only if $Y = \overline{Y}$.

A point x is called a *boundary point* of Y if any neighbourhood of x contains both points of Y and points of the complement $X - Y$. The set ∂Y of boundary

points of Y therefore consists of points common to the closures of Y and of $X - Y$:

$$\partial Y = \overline{Y} \cap \overline{X - Y}\,.$$

Furthermore

$$\overline{Y} = Y \cup \partial Y\,,$$

so Y is *closed* if and only if, as already said, it contains its limit points, in other words *if and only if it contains its boundary points*.

We shall call a set Y a *domain* if it is the closure of an open set.

For instance, on $\mathbb{R}^n$ the closure $\overline{B_r(x_0)}$ of an open ball $B_r(x_0)$ is a domain. Its boundary $\partial B_r(x_0) = \overline{B_r(x_0)} - B_r(x_0)$ is not a domain.

We end the section with the definition of bounded sets. A subset A of the metric space (X, d) is *bounded* if it is contained in an open ball, i.e. there exist $x_0 \in X$ and $r > 0$ such that

$$A \subseteq B_r(x_0)\,. \tag{2.15}$$

Note that if (2.15) holds for some $x_0 \in X$, then it holds for any other centre $x_1 \in X$, as long as we replace r with $r_1 = r + d(x_0, x_1)$; in fact, due to

$$d(x, x_0) < r \quad \Rightarrow \quad d(x, x_1) \le d(x, x_0) + d(x_0, x_1) < r_1\,,$$

we have $B_r(x_0) \subset B_{r_1}(x_1)$.

Moreover, the *diameter* $\delta(A)$ of a bounded set A is

$$\delta(A) = \sup\{d(x, y) : x \in A, y \in A\}\,.$$

The diameter is increasing with respect to the inclusion of sets: if $A \subseteq B$ then $\delta(A) \le \delta(B)$.

2.3 Sequences in a Metric Space: Continuous Functions

Let (X, d) be a metric space and x_k a sequence of points of X, i.e. a function from $\mathbb{N}$ to X. We say that x_k *tends*, or *converges* to a point $x \in X$ if, for any $\varepsilon > 0$, there exists $\nu \in \mathbb{N}$ such that

$$d(x_k, x) < \varepsilon \qquad \forall\, k > \nu\,.$$

This means $d(x_k, x) \to 0$ as $k \to +\infty$ (note that $d(x_k, x)$, for $k \in \mathbb{N}$, is a sequence of real numbers, and so we can apply the well-known definition of limit of a real sequence).

Just as in $\mathbb{R}$, it can be proved more generally that in metric spaces *limits are unique*, i.e. *a convergent sequence cannot have two distinct limits*.

In fact if we assume, by contradiction, that x and y are limits of the sequence x_k, with $x \neq y$, the triangle inequality forces

$$d(x, y) \leq d(x, x_k) + d(x_k, y)\,.$$

Taking the limit as $k \to +\infty$ gives $d(x, y) = 0$ and so $x = y$, which is absurd.

As we do in $\mathbb{R}$, we may characterise closed sets using sequences. It can be easily proved in fact that *a subset C of the metric space (X, d) is closed if and only if the limit of every convergent sequence in C belongs to C*.

Let us go back to Example 3 of the previous section on *Euclidean n-space* $(\mathbb{R}^n, d_n)$, and consider a sequence $x_k = (x_{k,1}, x_{k,2}, \ldots, x_{k,n}) \in \mathbb{R}^n$. The following result characterises the convergence of x_k to $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ in terms of the convergence in $\mathbb{R}$ of the coordinates $x_{k,i}$ to x_i, for every $i = 1, 2, \ldots, n$. Recall (Example 3, Sect. 2.2) that the *Euclidean distance* d_n on $\mathbb{R}^n$ is defined by

$$d_n(x, y) = \left[\sum_{i=1}^{n} (x_i - y_i)^2\right]^{1/2}\,. \tag{2.16}$$

Proposition 1 *Let d_n be the Euclidean distance on $\mathbb{R}^n$, $x_k = (x_{k,1}, x_{k,2}, \ldots, x_{k,n})$ $(k \in \mathbb{N})$ a sequence in $\mathbb{R}^n$, and let $x = (x_1, x_2, \ldots, x_n)$ be a point in $\mathbb{R}^n$. Then*

$$d_n(x_k, x) \to 0\,, \qquad as \qquad k \to +\infty\,,$$

if and only if, for any $i = 1, 2, \ldots, n$,

$$x_{k,i} \to x_i\,, \qquad as \qquad k \to +\infty\,.$$

Proof First observe that if $a_1, a_2, \ldots, a_n$ are non-negative real numbers,

$$\max\{a_1, a_2, \ldots, a_n\} \leq \sqrt{\sum_{i=1}^{n} a_i^2} \leq \sum_{i=1}^{n} a_i \leq n \cdot \max\{a_1, a_2, \ldots, a_n\}\,, \tag{2.17}$$

where the second inequality is equivalent to $\displaystyle\sum_{i=1}^{n} a_i^2 \leq \left(\sum_{i=1}^{n} a_i\right)^2$, which is easy to prove.

Recalling definition (2.16) for the Euclidean distance d_n, from (2.17) we find

$$|x_{k,i} - x_i| \leq d_n(x_k, x) \leq n \cdot \max\{|x_{k,j} - x_j| \,:\, 1 \leq j \leq n\}\,,$$

for any $i = 1, 2, \ldots, n$, and the claim follows.

Inspired by (2.17) let us define the general notion of equivalent distances. Two distances d_1 and d_2 on a set X are *equivalent* if there exist positive constants m, M such that

$$m\, d_2(x, y) \le d_1(x, y) \le M\, d_2(x, y), \qquad \forall\, x, y \in X\,. \tag{2.18}$$

Clearly the role of d_1,d_2 is symmetrical, so that

$$\frac{1}{M}\, d_1(x, y) \le d_2(x, y) \le \frac{1}{m}\, d_1(x, y), \qquad \forall\, x, y \in X\,.$$

If two metrics d_1,d_2 on a set X are equivalent, a sequence x_k converges to some point $x \in X$ for the metric d_1 if and only if x_k converges to $x \in X$ for the metric d_2.

This is because $d_1(x_k, x) \to 0$ as $k \to +\infty$ if and only if $d_2(x_k, x) \to 0$ as $k \to +\infty$, a direct consequence of definition (2.18).

Let us now pass to another important definition. Let (X, d_X), (Y, d_Y) be metric spaces. A function $f : X \to Y$ is said to be (*sequentially*) *continuous* at $x_0 \in X$ if, for any sequence x_k in X that converges to x_0, we have $f(x_k) \to f(x_0)$. That is to say,

$$d_X(x_k, x_0) \to 0 \qquad \Rightarrow \qquad d_Y(f(x_k), f(x_0)) \to 0\,.$$

In analogy to the case of real functions of one real variable one can prove (see the following Proposition 2) that f is *sequentially continuous* at x_0 if and only if it is *continuous* at x_0, i.e. for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in X : d_X(x, x_0) < \delta \qquad \Rightarrow \qquad d_Y(f(x), f(x_0)) < \varepsilon\,.$$

Moreover, f is called *continuous on X* when it is continuous at every point $x_0 \in X$.

Proposition 2 *Let $f : X \to Y$ be a function from X to Y and $x_0 \in X$ a point. The following conditions are equivalent:*

$$x_k \in X\ \ \forall k \in \mathbb{N}, \qquad d_X(x_k, x_0) \to 0 \ \ \Rightarrow\ \ d_Y(f(x_k), f(x_0)) \to 0\,; \tag{2.19}$$

$$\forall \varepsilon > 0\ \ \exists \delta > 0 : x \in X, \ \ d_X(x, x_0) < \delta \ \ \Rightarrow\ \ d_Y(f(x), f(x_0)) < \varepsilon. \tag{2.20}$$

Proof Start with (2.19) $\Rightarrow$ (2.20). By contradiction, if (2.20) did not hold, then

$$\exists\, \varepsilon_0 > 0 : \ \forall\, \delta > 0\,, \ \exists\, x \in X : d_X(x, x_0) < \delta, \ \ d_Y(f(x), f(x_0)) \ge \varepsilon_0\,. \tag{2.21}$$

Set $\delta = 1/k$ with $k \in \mathbb{N}$ and call $x = x_k$ the value appearing in (2.21) depending on the choice of $\delta = 1/k$:

$$\exists\, \varepsilon_0 > 0 : \ \forall\, k \in \mathbb{N}\,, \ \exists\, x_k \in X : d_X(x_k, x_0) < \frac{1}{k}, \ \ d_Y(f(x_k), f(x_0)) \ge \varepsilon_0\,.$$

Hence x_k would converge to x_0 in X, but the sequence $f(x_k)$ does not converge to $f(x_0)$ in Y, i.e (2.19) does not hold.

Now to (2.20) $\Rightarrow$ (2.19). Given $\varepsilon > 0$, let $\delta > 0$ be a real number for which (2.20) holds. As $d_X(x_k, x_0) \to 0$, there exists $\nu \in \mathbb{N}$ such that $d_X(x_k, x_0) < \delta$ for any $k > \nu$. From (2.20), then,

$$d_Y(f(x_k), f(x_0)) < \varepsilon\,, \qquad \forall\, k > \nu\,,$$

i.e. $d_Y(f(x_k), f(x_0)) \to 0$ as $k \to +\infty$.

Proposition 3 *Let (X, d) be a metric space and x_0 a given point in X. Then the function*

$$x \in X \;\to\; d(x, x_0)$$

is continuous from X to $\mathbb{R}$ (as usual, $\mathbb{R}$ has the Euclidean topology).

Proof Applying the distance's *Lipschitz inequality*, i.e (2.7), for $x, y \in A$ we have

$$|d(y, x_0) - d(x, x_0)| \le d(x, y)\,.$$

Then if x_k is a sequence of points tending to x,

$$|d(x_k, x_0) - d(x, x_0)| \le d(x_k, x)$$

and therefore $d(x_k, x_0) \to d(x, x_0)$ as $k \to +\infty$.

More generally we have the following

Proposition 4 *Let (X, d) be a metric space and A a subset of X. For $x \in X$, the distance of x to A, defined as*

$$d(x, A) = \inf_{a \in A} d(x, a)\,,$$

satisfies the Lipschitz inequality

$$|d(y, A) - d(x, A)| \le d(x, y)\,, \tag{2.22}$$

for any $x, y \in X$. In particular

$$x \in X \to d(x, A)$$

is a continuous function on X.

Proof Fix x in X. For any $\varepsilon > 0$ there exists $a \in A$ such that $d(x, a) < d(x, A) + \varepsilon$. Hence for any $y \in X$,

$$d(y, a) \leq d(y, x) + d(x, a) < d(y, x) + d(x, A) + \varepsilon \,.$$

But clearly $d(y, A) \leq d(y, a)$, so

$$d(y, A) < d(y, x) + d(x, A) + \varepsilon \,,$$

and by the arbitrariness of ε,

$$d(y, A) \leq d(y, x) + d(x, A) \,, \qquad \forall\, x, y \in X \,. \tag{2.23}$$

Interchanging x and y,

$$d(x, A) \leq d(x, y) + d(y, A) \,, \qquad \forall\, x, y \in X \,. \tag{2.24}$$

Now (2.23) and (2.24) imply the claim (2.22).

As an application of Proposition 4 we shall prove a separation-type result for closed sets under a continuous function.

Separation Theorem *Let A and B be non-empty closed and disjoint sets in a metric space (X, d). There exists a continuous function $f : X \to [0, 1]$ such that*

$$f(x) = \begin{cases} 0 & \text{if} \quad x \in A \\ 1 & \text{if} \quad x \in B \,. \end{cases}$$

Proof Since $A \bigcap B = \emptyset$ we have

$$d(x, A) + d(x, B) > 0 \,, \qquad \forall\, x \in X \,,$$

and therefore the function from X to $[0, 1]$

$$\frac{d(x, A)}{d(x, A) + d(x, B)}$$

is continuous, it vanishes on A and it equals 1 on B.

2.4 Vector Spaces: Linear Maps

In this and the subsequent section we shall recall some results from Linear Algebra which we assume are well known to the reader. For this reason statements will not be proved.

Let V denote a *vector space*, i.e. a set equipped with an operation, called *sum* and denoted by $+$, and an operation called *scalar multiplication*, sometimes denoted by $\cdot$, so that the following axioms holds (with $x, y \in V, \lambda, \mu \in \mathbb{R}$):

$$\begin{aligned}
&\textit{the sum is associative and commutative}; && (2.25)\\
&\textit{there exists a unique element in } V, \textit{ called neutral element } 0,\\
&\textit{such that } x + 0 = x, \textit{ for any } x \in V;\\
&\textit{for any } x \in V \textit{ there exists a unique element, called opposite to } x\\
&\textit{and written } -x, \textit{ such that } x + (-x) = 0;\\
&(\lambda + \mu)x = \lambda x + \mu x, \qquad (\lambda\mu)x = \lambda(\mu x);\\
&\lambda(x + y) = \lambda x + \lambda y;\\
&0 \cdot x = 0, \qquad 1 \cdot x = x\,. && (2.26)
\end{aligned}$$

*Example 1 (*The Space $\mathbb{R}^n$*)* If $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ denote two ordered n-tuples of real numbers, i.e. x, y are elements of $\mathbb{R}^n$, we shall call *sum* of x and y the element in $\mathbb{R}^n$

$$x + y = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)\,.$$

If $\lambda \in \mathbb{R}$, $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, the *product* of λ by x is the element in $\mathbb{R}^n$

$$\lambda \cdot x = (\lambda x_1, \lambda x_2, \ldots, \lambda x_n)\,.$$

Properties (2.25)–(2.26) are easy as long as we take $0 = (0, 0, \ldots, 0)$, called *zero vector*, as neutral element (by abuse of notation we have indicated with 0 both the real number zero and the neutral element of $\mathbb{R}^n$).

We may then regard $\mathbb{R}^n$ as a *vector space*, in which case its elements are called *vectors*. In the next section we shall examine in detail some properties of this vector space.

*Example 2 (*The Space $\mathbb{R}^X$ of Real Functions Defined on $X \subseteq \mathbb{R}$*)* If f and g are real functions defined on the subset X of $\mathbb{R}$, we call *sum* of f and g the function

$$f + g : x \in X \to f(x) + g(x) \in \mathbb{R}\,.$$

(continued)

Example 2 (continued)
If $\lambda \in \mathbb{R}$ and $f : X \to \mathbb{R}$, the *product* λf is the function

$$\lambda f : x \in X \to \lambda f(x) \in \mathbb{R} .$$

Easily, (2.25)–(2.26) hold if the neutral element is the zero function. Thus $\mathbb{R}^X$ is a vector space. The special case $X = \mathbb{N}$ produces the vector space of real sequences. In the next section we will study the important *subspaces* of $\mathbb{R}^X$ called C^0, C^1, C^k.

A *linear combination* of elements $x_1, x_2, \ldots, x_k$ in a vector space V is a vector of the form

$$\lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_k x_k ,$$

where $\lambda_1, \lambda_2, \ldots, \lambda_k$ are real numbers called the linear combination's *coefficients*. The vectors $x_1, x_2, \ldots, x_k$ are said to be *linearly dependent* when there exists a linear combination with coefficients $\lambda_1, \lambda_2, \ldots, \lambda_k$, not all zero, that equals the zero vector, i.e.

$$\lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_k x_k = 0 .$$

If $x_1, x_2, \ldots, x_k$ are not linearly dependent we call them *linearly independent*. In this case the only linear combination equal to 0 is the combination whose coefficients $\lambda_1, \lambda_2, \ldots, \lambda_k$ all vanish.

The vector space V is said to have *dimension* n if there exist linearly independent vectors $\{x_1, x_2, \ldots, x_n\}$, but it is impossible to find $n + 1$ linearly independent vectors in V. In case V reduces to the zero vector only we say it has dimension zero.

When V has dimension n a set $\{x_1, x_2, \ldots, x_n\}$ of linearly independent vectors is called a *basis* of V. The next result characterises bases in vector spaces.

Proposition *The vectors $\{x_1, x_2, \ldots, x_n\}$ form a basis for V if and only if any vector x of V can be written in a unique way as linear combination of $x_1, x_2, \ldots, x_n$:*

$$x = \lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_n x_n .$$

If V and W are vector spaces, a map $L : V \to W$ is *linear* if

$$L(\lambda x + \mu y) = \lambda L(x) + \mu L(y) , \tag{2.27}$$

for any $x, y \in V$ and $\lambda, \mu \in \mathbb{R}$.

Note how from definition (2.27) it easily follows that the *range* $L(V)$ of a linear map L,

$$L(V) = \{y \in W : \ y = L(x), \ \text{with } x \in V\},$$

is a vector space contained in W, i.e. a *vector subspace* of W. Similarly, the *kernel* $N(L)$ of L, given by

$$N(L) = \{x \in V : \ L(x) = 0\},$$

is a vector subspace of V.

In the special situation where L is a linear map from V to $\mathbb{R}$ (the target space W is $\mathbb{R}$), L is called a *linear functional*, or *linear form*, on V.

2.5 The Vector Space $\mathbb{R}^n$ and Its Dual

The definition of the vector space $\mathbb{R}^n$, well known to readers, was recalled in the previous section's Example 1. Apart from the *sum of elements* of $\mathbb{R}^n$ and the *product of a real number and an element* of $\mathbb{R}^n$, it is important to introduce here the *standard scalar* (or *inner*) *product* of elements $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ in $\mathbb{R}^n$. It is denoted by (x, y) and defined by

$$(x, y) = \sum_{i=1}^{n} x_i \, y_i \, .$$

The scalar product satisfies the following properties (for $x, y, z \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$):

$$\begin{aligned}
&(x, y) = (y, x)\,; \\
&(x + y, z) = (x, z) + (y, z)\,; \\
&\lambda(x, y) = (\lambda x, y)\,; \\
&(x, x) \geq 0\,; \\
&(x, x) = 0 \qquad \Leftrightarrow \qquad x = 0\,.
\end{aligned}$$

We call *modulus*, or (*Euclidean*) *norm*, of a vector $x \in \mathbb{R}^n$ the quantity

$$|x| = (x, x)^{1/2} = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} .$$

In this notation the *Cauchy-Schwarz inequality* in (2.13) reads

$$|(x, y)| \leq |x| \cdot |y|\,, \qquad \forall\, x, y \in \mathbb{R}^n\,.$$

Due to the Cauchy-Schwarz inequality, the ratio $(x, y)/(|x| \cdot |y|)$ is a real number in $[-1, 1]$. Hence the following notion is well defined: if x, y are non-zero vectors in $\mathbb{R}^n$, the *angle* between them is the real number $\alpha \in [0, \pi]$ such that

$$\cos \alpha = \frac{(x, y)}{|x| \cdot |y|} .$$

The vectors

$$\begin{aligned} e_1 &= (1, 0, \ldots, 0) , \\ e_2 &= (0, 1, \ldots, 0) , \\ &\ldots\ldots\ldots\ldots\ldots \\ e_n &= (0, 0, \ldots, 1) \end{aligned}$$

form a basis of $\mathbb{R}^n$. In fact if the vector h has components $(h_1, h_2, \ldots, h_n)$ then

$$h = h_1 e_1 + h_2 e_2 + \ldots h_n e_n , \tag{2.28}$$

and such a representation is *unique*. The basis $\{e_1, e_2, \ldots, e_n\}$ is called *canonical basis* of $\mathbb{R}^n$.

As already said in the previous section regarding a general vector space V, a *linear functional* on $\mathbb{R}^n$ is a linear map L, i.e.

$$L(\lambda x + \mu y) = \lambda L(x) + \mu L(y) , \tag{2.29}$$

for any $x, y \in V$ and $\lambda, \mu \in \mathbb{R}$.

The collection of linear functionals from $\mathbb{R}^n$ to $\mathbb{R}$ is the *dual space* of $\mathbb{R}^n$, and is denoted by $(\mathbb{R}^n)^*$.

Observe that $(\mathbb{R}^n)^*$, with the usual *sum of functions* and *product of a function by a real number*, is a vector space of dimension n.

Indeed, let us show how to construct a basis in $(\mathbb{R}^n)^*$. For any $i = 1, 2, \ldots, n$ call e^i the linear functional on $\mathbb{R}^n$ that associates with any vector $h \in \mathbb{R}^n$ its ith component

$$e^i(h) = (h, e_i) = h_i . \tag{2.30}$$

If $L : \mathbb{R}^n \to \mathbb{R}$ is a generic linear functional, setting $a_i = L(e_i)$ for any $i = 1, 2, \ldots, n$, by (2.28), (2.29), and (2.30) we have

$$\begin{aligned} L(h) &= L(h_1 e_1 + h_2 e_2 + \ldots + h_n e_n) = \\ &= h_1 a_1 + h_2 a_2 + \ldots + h_n a_n = a_1 e^1(h) + a_2 e^2(h) + \ldots + a_n e^n(h) , \end{aligned} \tag{2.31}$$

for any $h \in \mathbb{R}^n$. Hence

$$L = a_1 e^1 + a_2 e^2 + \ldots + a_n e^n \,, \tag{2.32}$$

meaning that L is a linear combination of the functionals $e^1, e^2, \ldots, e^n$. Moreover, it is straightforward to check that (2.32) is a unique representation. By the previous section's proposition, then, $\{e^1, e^2, \ldots, e^n\}$ is a basis of $(\mathbb{R}^n)^*$, called *canonical dual basis*.

The following identities express the relationship between the canonical bases of $\mathbb{R}^n$ and its dual:

$$e^i(e_j) = \delta_{ij} \,, \qquad \forall\, i, j = 1, 2, \ldots, n \,.$$

The elements of the canonical dual basis of $(\mathbb{R}^n)^*$ are also written $dx_1, dx_2, \ldots, dx_n$, so for any $i = 1, 2, \ldots, n$ and $h \in \mathbb{R}^n$,

$$dx_i(h) = e^i(h) = h_i \,.$$

We associate with any *linear map* $L : \mathbb{R}^n \to \mathbb{R}^m$ the $m \times n$ matrix A whose columns are the vectors $L(e_1), L(e_2), \ldots, L(e_n) \in \mathbb{R}^m$, where $\{e_1, e_2, \ldots, e_n\}$ is the canonical basis of $\mathbb{R}^n$. Calling a_{ij}, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$ the generic element of A, we then have

$$a_{ij} = (\varepsilon_i, L(e_j)) \,, \tag{2.33}$$

where $\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_m\}$ is the canonical basis of $\mathbb{R}^m$. From (2.33) we easily deduce the following result.

Proposition 1 *If $L : \mathbb{R}^n \to \mathbb{R}^m$ is a linear map and A its matrix, then*

$$L(h) = A \cdot h \tag{2.34}$$

for any $h \in \mathbb{R}^n$, where $A \cdot h$ is the product of A by the column vector h:

$$A \cdot h = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} h_1 \\ h_2 \\ \ldots \\ h_n \end{pmatrix} =$$

$$A \cdot h = \begin{pmatrix} a_{11}h_1 + a_{12}h_2 + \ldots + a_{1n}h_n \\ a_{21}h_1 + a_{22}h_2 + \ldots + a_{2n}h_n \\ \ldots\ldots\ldots\ldots \\ a_{m1}h_1 + a_{m2}h_2 + \ldots + a_{mn}h_n \end{pmatrix} .$$

We point out explicitly that, conversely, given an $m \times n$ matrix A, (2.34) defines a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$, whose associated matrix is precisely A. We then conclude that (2.34) defines a *1-1 correspondence* between *the set of linear maps* from $\mathbb{R}^n$ to $\mathbb{R}^m$ and *the set of* $m \times n$ *matrices*.

If $L : \mathbb{R}^n \to \mathbb{R}$ is a linear functional, the matrix associated with L reduces to the row vector (or *covector*)

$$a = (a_1, a_2, \ldots, a_n) ,$$

where $a_i = L(e_i)$, $i = 1, 2, \ldots, n$. Then (2.31) implies, for any $h \in \mathbb{R}^n$,

$$L(h) = (a, h) . \tag{2.35}$$

From what we have observed above, we conclude that a map $L : \mathbb{R}^n \to \mathbb{R}$ is linear if and only if there exists a vector $a \in \mathbb{R}^n$ satisfying (2.35). The representation is, moreover, unique.

Let us recall the following.

Proposition 2 *Let $L : \mathbb{R}^n \to \mathbb{R}^m$, $M : \mathbb{R}^m \to \mathbb{R}^k$ be linear maps, and A, B the associated matrices. The composite linear map $M \circ L : \mathbb{R}^n \to \mathbb{R}^k$ is represented by the matrix $C = B \cdot A$, the product of A and B.*

We wrap up this Linear Algebra summary by recalling the definition and essential properties of the *rank* of a matrix. If A is an $m \times n$ matrix, the rank $r(A)$ is defined as the *largest size of non-zero minors* in A. In particular, if A is a square $n \times n$ matrix, its rank is n if and only if the determinant of A is different from zero.

It is known that the rank also coincides with the maximum number of linearly independent rows (and columns). More generally, one could prove the following fact.

Theorem 1 *Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be a linear map and A the $m \times n$ matrix representing it. Then the rank $r(A)$ of A equals the dimension of the range $L(\mathbb{R}^n)$ of L, while $n - r(A)$ is equal to the dimension of the kernel $N(L)$ of L.*

In particular, for $L : \mathbb{R}^n \to \mathbb{R}^n$, i.e. $m = n$, if the determinant of the associated matrix A is non-zero, the previous theorem implies $L(\mathbb{R}^n) = \mathbb{R}^n$, and the kernel of L reduces to the zero vector only. These facts immediately force the map L to be invertible, and the inverse L^{-1} is still a linear map. Hence we may state the following theorem.

Theorem 2 *Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map with associated $n \times n$ matrix A. L is invertible if and only if the determinant of A is non-zero. If so, the inverse map $L^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is linear and is represented by the inverse matrix A^{-1}.*

The properties of *determinants* of square matrices are well known to the reader. We just recall the following important property.

Theorem 3 *Let A, B be $n \times n$ matrices. The determinant of the product of A and B equals the product of the determinants of A and B:*

$$\det A \cdot B = \det A \cdot \det B \,.$$

The reader should note that in the above formula the "$\cdot$" on the left denotes the matrix product, whereas the one on the right is the product in $\mathbb{R}$.

2.6 Normed Vector Spaces

A *norm* on a vector space V is a function that assigns to every vector $x \in V$ a real number $\|x\| \geq 0$ and satisfies the following conditions

$$\|x\| = 0 \qquad \text{if and only if } \; x = 0\,, \quad \forall\, x \in V; \tag{2.36}$$

$$\|\lambda x\| = |\lambda| \cdot \|x\|, \qquad \forall\, \lambda \in \mathbb{R}\,, \quad \forall\, x \in V; \tag{2.37}$$

$$\|x + y\| \leq \|x\| + \|y\|\,, \qquad \forall\, x, y \in V. \tag{2.38}$$

Relationship (2.38) is called *triangle inequality*. When $\|\cdot\|$ is a norm on V, one says V is a *normed (vector) space*. Sometimes, to remind us of the vector space on which we are operating, the notation $\|\cdot\|_V$ is used for the norm of V.

The simplest instance of a normed vector space is the set of real numbers equipped with the norm given by the absolute value.
Any norm on a vector space V defines a special metric on V. Precisely, if $\|\cdot\|_V$ is a norm on the vector space V, setting

$$d(x, y) = \|x - y\|\,, \qquad \forall\, x, y \in V\,, \tag{2.39}$$

defines a distance on V.

Furthermore, the distance d of (2.39) enjoys the following properties:

$$d(x + z, y + z) = d(x, y)\,, \qquad \forall\, x, y, z \in V\,,$$

(d is *translation-invariant*), and

$$d(\lambda x, \lambda y) = |\lambda| \cdot d(x, y)\,, \qquad \forall\, \lambda \in \mathbb{R}\,,\ \forall\, x, y \in V\,.$$

As any normed vector space is in particular a metric space, we have the notion of *convergent sequences*. Precisely, a sequence x_k has limit $x \in V$ if and only if

$$\lim_{k \to +\infty} \|x_k - x\|_V = 0 .$$

The definition, given for metric spaces, of *equivalent distances* translates into the notion of *equivalent norms*. Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on V are *equivalent* when there exist positive constants m, M such that

$$m\|x\|_2 \leq \|x\|_1 \leq M\|x\|_2 , \qquad \forall\, x \in V . \tag{2.40}$$

If *two norms* $\|\cdot\|_1$, $\|\cdot\|_2$ *on* V *are equivalent, a sequence* x_k *converges to* $x \in V$ *in the norm* $\|\cdot\|_1$ *if and only if* x_k *converges to* $x \in V$ *in the norm* $\|\cdot\|_2$.

In fact $\|x_k - x\|_1 \to 0$ as $k \to +\infty$ if and only if $\|x_k - x\|_2 \to 0$ as $k \to +\infty$, because of definition (2.40).

Example 4, Sect. 2.8, will discuss a space that carries two non-equivalent norms. If W is another normed vector space, with norm $\|\cdot\|_W$, a function $f : V \to W$ from V to W is said to be *continuous* at a point $x_0 \in V$ if

$$\|x_k - x_0\|_V \to 0 \quad \Rightarrow \quad \|f(x_k) - f(x_0)\|_W \to 0$$

as $k \to +\infty$. Equivalently, for any sequence x_k converging in V to x_0, the corresponding sequence $f(x_k)$ converges to $f(x_0)$ in W.

The following functions are continuous on any normed vector space V: the *translation by a vector* $y \in V$, i.e. the function

$$x \in V \to x + y \in V ,$$

and the *homothety by the factor* $\lambda \in \mathbb{R}$, i.e.

$$x \in V \to \lambda x \in V .$$

To see this it suffices to note that for any sequence x_k in V we have

$$\|(x_k + y) - (x_0 + y)\| = \|x_k - x_0\|, \qquad \forall\, k \in \mathbb{N};$$
$$\|\lambda x_k - \lambda x_0\| = |\lambda| \cdot \|x_k - x_0\|, \qquad \forall\, k \in \mathbb{N} .$$

Two important instances of normed space are $\mathbb{R}^n$, dealt with extensively in the next section, and $C^0([a, b])$. The latter is the set of continuous real functions on the closed and bounded interval $[a, b]$ endowed with the norm

$$\|f\|_{C^0} = \sup\{|f(x)| : x \in [a, b]\} . \tag{2.41}$$

Notice that by the continuity of f the supremum in (2.41) is indeed the maximum. This norm on $C^0([a,b])$ is sometimes denoted by $\|f\|_\infty$.
In Example 4, Sect. 2.2, we introduced the distance

$$d(f,g) = \sup\{|f(x) - g(x)| : x \in [a,b]\}.$$

In agreement with (2.41), we have

$$d(f,g) = \|f - g\|_{C^0}, \qquad \forall\, f, g \in C^0([a,b]). \tag{2.42}$$

Arguing as in Example 4, Sect. 2.2, it is easy to see that (2.42) defines a norm, called *supremum norm* (or *sup norm* for short).

The convergence of a sequence $f_k \in C^0([a,b])$, $k \in \mathbb{N}$, under the supremum norm coincides with the uniform convergence of the sequence f_k (see Sect. 1.1 in Chap. 1).

Further properties of $C^0([a,b])$, and its subspaces $C^1([a,b])$ and $C^k([a,b])$, $k \in \mathbb{N}$, will be addressed in the next sections.

2.7 The Normed Vector Space $\mathbb{R}^n$

The *Euclidean norm* of an element $x = (x_1, x_2, \ldots, x_n)$ of $\mathbb{R}^n$, also called just *norm* or *modulus* of x, is defined by

$$|x| = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}.$$

We recall that, using the symbols for distances on metric spaces, in Sect. 2.2 the *Euclidean norm* $|x|$ of x was written $d_n(x,0)$, which is the *Euclidean distance* of the point x to 0 in $\mathbb{R}^n$.

More generally, for any $p \in [1, +\infty)$ we define the function $\|\cdot\|_p : \mathbb{R}^n \to \mathbb{R}$ (which we shall prove is a norm on $\mathbb{R}^n$)

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}.$$

So we have, in particular, $\|x\|_p = |x|$ when $p = 2$. Now let us define, for $p = +\infty$,

$$\|x\|_\infty = \max\{|x_i|\,;\, i = 1, 2, \ldots, n\}. \tag{2.43}$$

Figure 2.2 shows the open balls in the plane centred at the origin with respect to some of the norms introduced.

One reason for the symbol $\|\cdot\|_\infty$ in (2.43) is given by the property stated in Proposition 1.

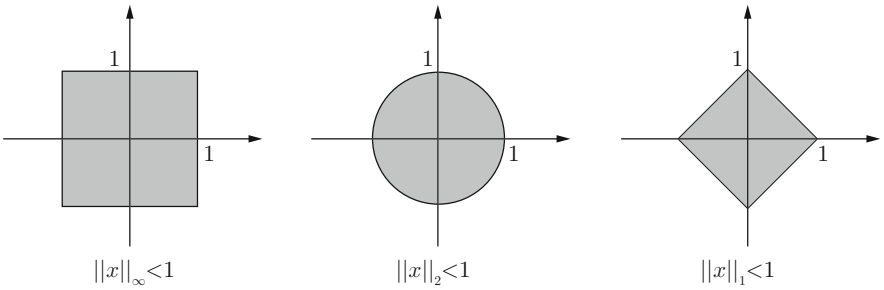


Fig. 2.2

Proposition 1 *The following limit condition holds:*

$$\lim_{p\to+\infty} \|x\|_p = \|x\|_\infty\,, \qquad \forall\, p \in \mathbb{R}^n\,. \tag{2.44}$$

Proof Given $x \in \mathbb{R}^n$, we have

$$\|x\|_\infty = \max\{|x_i|\,;\, i = 1, 2, \ldots, n\} = |x_{i_0}|\,,$$

for some $i_0 \in \{1, 2, \ldots, n\}$. For any $p \geq 1$, therefore,

$$\|x\|_\infty^p = |x_{i_0}|^p \leq \sum_{i=1}^n |x_i|^p \leq n|x_{i_0}|^p = n\|x\|_\infty^p\,,$$

and so

$$\|x\|_\infty \leq \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} = \|x\|_p \leq n^{1/p}\|x\|_\infty\,. \tag{2.45}$$

As $p \to +\infty$, $n^{1/p} \to 1$ and (2.44) follows.

Theorem *For any $p \in [1, +\infty]$ the function $\|\cdot\|_p : \mathbb{R}^n \to \mathbb{R}$ is a norm on $\mathbb{R}^n$.*

Proof Properties (2.36) and (2.37) are straightforward. For the triangle inequality (2.38) we examine the cases $p = 1$, $p = +\infty$ and $1 < p < +\infty$ separately.

If $p = 1$, for any $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ in $\mathbb{R}^n$ we have

$$\begin{aligned} \|x + y\|_1 &= \sum_{i=1}^n |x_i + y_i| \leq \sum_{i=1}^n (|x_i| + |y_i|) = \\ &= \sum_{i=1}^n |x_i| + \sum_{i=1}^n |y_i| = \|x\|_1 + \|y\|_1\,. \end{aligned} \tag{2.46}$$

When $p = +\infty$, for any $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ in $\mathbb{R}^n$

$$\begin{aligned} \|x + y\|_\infty = \max_{1\le i\le n} |x_i + y_i| \le \max_{1\le i\le n} (|x_i| + |y_i|) \le \\ \le \max_{1\le i\le n} |x_i| + \max_{1\le i\le n} |y_i| = \|x\|_\infty + \|y\|_\infty \,. \end{aligned} \tag{2.47}$$

The case $1 < p < \infty$ is more complicated. Hence we shall divide the proof in three steps, all of independent interest, that go under the names of Young inequality, Hölder inequality and Minkowski inequality.

Young's Inequality *Fix $p > 1$ and let p' be the conjugated exponent, defined as*

$$\frac{1}{p} + \frac{1}{p'} = 1 \qquad \textit{i.e.} \qquad p' = \frac{p}{p-1} \,.$$

Then for any pair of non-negative real numbers a, b,

$$ab \le \frac{a^p}{p} + \frac{b^{p'}}{p'},$$

with equality if and only if $a^p = b^{p'}$.

Proof Fix $b > 0$ (when $b = 0$ the claim is trivial) and consider the function $f : [0, +\infty) \to [0, +\infty)$ defined by

$$f(t) = \frac{t^p}{p} + \frac{b^{p'}}{p'} - tb \,.$$

We must prove $f(a) \ge 0$, or equivalently $f(t) \ge 0$ for any $t \ge 0$.

The derivative of f is $f'(t) = t^{p-1} - b$ for any $t \ge 0$. As $f'(t) < 0$ if and only if $t < b^{1/(p-1)}$, the function f reaches its minimum at $t = b^{1/(p-1)}$. Hence

$$\begin{aligned} f(t) \ge f(b^{1/(p-1)}) = \frac{b^{p/(p-1)}}{p} + \frac{b^{p'}}{p'} - b^{1/(p-1)+1} = \\ = \left(\frac{1}{p} + \frac{1}{p'} - 1\right) b^{p'} = 0 \,, \end{aligned}$$

with equality occurring only at the minimum point $t = a = b^{1/(p-1)}$, i.e. for $a^p = b^{p'}$.

Hölder's Inequality *Fix $p > 1$ and let $p' = p/(p-1)$ be the conjugate exponent. Then the scalar product of $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$*

$$(x, y) = \sum_{i=1}^{n} x_i \, y_i$$

satisfies the inequality

$$|(x, y)| \le \|x\|_p \cdot \|y\|_{p'} . \tag{2.48}$$

Since $p = 2$ implies $p' = 2$, (2.48) coincides with the *Cauchy-Schwarz inequality*

$$|(x, y)| \le |x| \cdot |y| , \qquad \forall\, x, y \in \mathbb{R}^n .$$

Now, (2.48) holds for $p = 1$, too. In such a case the convention is that $p' = +\infty$, for in fact

$$|(x, y)| \le \sum_{i=1}^{n} |x_i\, y_i| \le \max_{1\le i\le n} |y_i| \cdot \sum_{i=1}^{n} |x_i| = \|y\|_\infty \cdot \|x\|_1 .$$

Proof of Hölder's Inequality if x or y are zero the inequality is trivial. Suppose then $x \neq 0$ and $y \neq 0$. Set

$$a_i = \frac{|x_i|}{\|x\|_p} , \qquad b_i = \frac{|y_i|}{\|y\|_{p'}} , \qquad \forall\, i = 1, 2, \ldots, n , \tag{2.49}$$

and use, for any $i = 1, 2, \ldots, n$ Young's inequality

$$a_i b_i \le \frac{1}{p} \frac{|x_i|^p}{\|x\|_p^p} + \frac{1}{p'} \frac{|y_i|^{p'}}{\|y\|_{p'}^{p'}} .$$

Summing over $i = 1, 2, \ldots, n$,

$$\sum_{i=1}^{n} a_i\, b_i \le \frac{1}{p} \frac{\|x\|_p^p}{\|x\|_p^p} + \frac{1}{p'} \frac{\|y\|_{p'}^{p'}}{\|y\|_{p'}^{p'}} = \frac{1}{p} + \frac{1}{p'} = 1 . \tag{2.50}$$

Recalling (2.49), from (2.50) we have

$$\sum_{i=1}^{n} |x_i\, y_i| \le \|x\|_p \cdot \|y\|_{p'}$$

and hence (2.48) follows.

Finally, Minkowski's inequality will conclude the proof of the previous theorem, thus establishing that $\|\cdot\|_p : \mathbb{R}^n \to \mathbb{R}$ is a norm on $\mathbb{R}^n$ for any $p \in [1, +\infty]$.

Minkowski's Inequality *For any $p \in [1, +\infty]$ and $x, y \in \mathbb{R}^n$*

$$\|x + y\|_p \le \|x\|_p + \|y\|_p . \tag{2.51}$$

Proof Minkowski's inequality has already been proved for $p = 1$ in (2.46) and $p = +\infty$ in (2.47), so we only need to consider $1 < p < +\infty$.

Let $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$. For any $i \in \{1, 2, \ldots, n\}$

$$|x_i + y_i|^p = |x_i + y_i|^{p-1} \cdot |x_i + y_i| \leq |x_i + y_i|^{p-1} \cdot (|x_i| + |y_i|)\,.$$

By Hölder's inequality it follows that

$$\sum_{i=1}^{n} |x_i + y_i|^p \leq \sum_{i=1}^{n} |x_i + y_i|^{p-1} \cdot |x_i| + \sum_{i=1}^{n} |x_i + y_i|^{p-1} \cdot |y_i| \leq$$

$$\leq \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)\cdot p'} \right)^{1/p'} \cdot \|x\|_p + \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)\cdot p'} \right)^{1/p'} \cdot \|y\|_p,$$

and since $(p-1) \cdot p' = p$,

$$\|x + y\|_p^p = \sum_{i=1}^{n} |x_i + y_i|^p \leq \|x + y\|_p^{p/p'} (\|x\|_p + \|y\|_p)\,. \tag{2.52}$$

Now observe that Minkowski's inequality (2.51) is trivial when $\|x + y\|_p = 0$. If this is not the case, we divide (2.52) by $\|x + y\|_p^{p/p'}$, and from $p - p/p' = 1$ the claim follows.

Proposition 2 *The norms $\|\cdot\|_p$ on $\mathbb{R}^n$, for any $1 \leq p \leq +\infty$, are all equivalent.*

We point out that we shall prove in Sect. 2.10 that any norm on $\mathbb{R}^n$ is equivalent to any other.

Proof From (2.45)we have:

$$\|x\|_\infty \leq \|x\|_p \leq n^{1/p} \|x\|_\infty\,, \qquad \forall\, x \in \mathbb{R}^n\,.$$

Therefore, whatever $p \geq 1$ is, the norm $\|\cdot\|_p$ is equivalent to $\|\cdot\|_\infty$, and the claim is proved.

Example 1 (The Space l^p) We obtain a generalisation of the normed vector space $\mathbb{R}^n$ by looking at, for any $p \in [1, +\infty]$, the space l^p of sequences $x = (x_i)_{i \in \mathbb{N}}$ such that

$$\|x\|_p = \left(\sum_{i=1}^{+\infty} |x_i|^p \right)^{1/p} < +\infty\,. \tag{2.53}$$

(continued)

Example 1 (continued)
We want to prove that (2.53) defines a norm on l^p, and we shall only show the triangle inequality. By Minkowski's inequality, if $x = (x_i)_{i\in\mathbb{N}} \in l^p$ and $y = (y_i)_{i\in\mathbb{N}} \in l^p$, for any $n \in \mathbb{N}$ we have

$$\left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/p} \leq \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p} \leq$$
$$\leq \sup_{n\in\mathbb{N}} \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \sup_{n\in\mathbb{N}} \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p} = \|x\|_p + \|y\|_p \,. \tag{2.54}$$

Hence the sequence of partial sums on the right-hand side in (2.54) is bounded from above, and when $n \to +\infty$ we recover the triangle inequality

$$\|x + y\|_p = \lim_{n\to+\infty} \left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/p} \leq \|x\|_p + \|y\|_p \,, \qquad \forall\, x, y \in l^p \,.$$

This implies $x + y \in l^p$ and that the triangle inequality holds. Therefore l^p, for any $p \in [1, +\infty]$, is a normed vector space.

2.8 Complete Metric Spaces: Banach Spaces

Let (X, d) be a metric space. A sequence x_k is a *Cauchy sequence* in X if, for any $\varepsilon > 0$, there exists $\nu \in \mathbb{N}$ such that

$$d(x_k, x_h) < \varepsilon \,, \qquad \forall\, h, k > \nu \,.$$

The metric space (X, d) is called *complete* if every Cauchy sequence converges.

Since any Cauchy sequence of real numbers is convergent, the Euclidean space $\mathbb{R}$ with the distance $d(x, y) = |x - y|$ is complete. On the other hand the set $\mathbb{Q}$ of rationals, endowed with the same metric, is not complete, as is well known.
Suppose V is a normed vector space with norm $\|\cdot\|$. If it is complete as a metric space for the distance

$$d(x, y) = \|x - y\|_V$$

induced by $\|\cdot\|$, we call it a *Banach space*.
Below we provide a few important examples of Banach spaces.

Example 1 Let $C^0([a,b])$ denote the *space of continuous functions* on the closed and bounded interval $[a,b] \subset \mathbb{R}$. We claim that $C^0([a,b])$ is a Banach space for the *supremum norm*

$$\|f\|_{C^0} = \sup\{|f(x)| : x \in [a,b]\} \tag{2.55}$$

introduced in Sect. 2.2. Since the convergence, under norm (2.55), of a sequence $f_k \in C^0([a,b])$, $k \in \mathbb{N}$, coincides with the uniform convergence of the f_k, the uniform Cauchy criterion (Sect. 1.2) ensures that $C^0([a,b])$ is complete.

Example 2 Let $C^1([a,b])$ denote the *space of continuous functions f with continuous derivative* on $[a,b]$, with norm

$$\|f\|_{C^1} = \sup_{x\in[a,b]} |f(x)| + \sup_{x\in[a,b]} |f'(x)| \,. \tag{2.56}$$

We claim that $C^1([a,b])$ is a Banach space. The proof that (2.56) defines a norm goes exactly as for $C^0([a,b])$ (see also Example 4 in Sect. 2.2). Let us then show $C^1([a,b])$ is complete. Take a Cauchy sequence f_k in $C^1([a,b])$ for the norm (2.56). Then both the sequence f_k and the sequence of derivatives f_k' are Cauchy for the sup norm of $C^0([a,b])$. As the latter space is complete, there exist continuous functions f and g on $[a,b]$ such that

$$\|f_k - f\|_{C^0} \to 0 \,, \qquad \|f_k' - g\|_{C^0} \to 0 \,.$$

But $f_k \to f$ and $f_k' \to g$ uniformly on $[a,b]$ by interchanging the limit and the derivative (Sect. 1.3), so $f' = g$. Therefore $C^1([a,b])$, with norm (2.56), is complete and hence a Banach space.

Example 3 In a similar way one can prove that *the space $C^k([a,b])$ of differentiable functions up to order $k \in \mathbb{N}$ with continuous k^{th} derivative on* $[a,b]$, equipped with norm

$$\|f\|_{C^k} = \sum_{i=0}^{k} \sup\{|f^{(i)}(x)| : x \in [a,b]\} \,,$$

where $f^{(0)} = f$, is a Banach space.

Example 4 Consider the space $C^1([a,b])$ of continuous functions f with continuous derivative on $[a,b]$, and put on it the *supremum norm* (2.55) that we have on $C^0([a,b])$. Thus we obtain a normed vector space (because (2.55) is a norm) that is *not* complete. For

(continued)

Example 4 (continued)
example, as shown in Example 3, Sect. 1.3, the sequence

$$f_k = \sqrt{x^2 + \frac{1}{k}},$$

converges uniformly on the interval $[-1, 1]$ to $f(x) = |x|$, which is continuous but not everywhere differentiable on the interval. Hence f_k is a sequence in $C^1([-1, 1])$, actually a Cauchy sequence for the norm of $C^0([-1, 1])$, but it does not admit limit in $C^1([-1, 1])$. If there existed a function $g \in C^1([-1, 1])$, limit of f_k for the norm of $C^0([-1, 1])$, we would have $f_k(x) \to g(x)$ for any $x \in [-1, 1]$, and therefore $g(x) = |x|$, although the latter does not belong to $C^1([-1, 1])$.

The above considerations also imply that the C^0 and C^1 norms *are not equivalent* on $C^1([a, b])$. For the C^0 norm the space $C^1([a, b])$ is not complete, whereas it is Banach for the natural norm (2.56).

Example 5 Given $f \in C^0([a, b])$ put

$$\|f\|_p = \left(\int_a^b |f(x)|^p \, dx \right)^{1/p}, \tag{2.57}$$

where $p \in [1, +\infty)$ is a given real number. In Sect. 9.9, Chap. 9, we will prove that (2.57) defines a norm under more general hypotheses. Here we remark that $C^0([a, b])$, with that norm, is not Banach. For instance, for any $k \in \mathbb{N}$ let f_k be the continuous function on $[-1, 1]$ defined by

$$f_k(x) = \begin{cases} -1 & \text{if} \quad -1 \le x \le -1/k \\ kx & \text{if} \quad -1/k \le x \le 1/k \\ 1 & \text{if} \quad 1/k \le x \le 1 \end{cases}.$$

When $h, k \in \mathbb{N}, h > k$, the function $f_k(x) - f_h(x)$ is zero outside $[-1/k, 1/k]$. As $|f_k(x) - f_h(x)| \le 2$ for every $x \in [-1, 1]$,

$$\|f_k - f_h\|_p^p = \int_{-1/k}^{1/k} |f_k(x) - f_h(x)|^p \, dx \le \frac{2^{p+1}}{k},$$

hence f_k is a Cauchy sequence in $C^0([-1, 1])$.

If f_k converged to some $f \in C^0([-1, 1])$ in norm (2.57), we would have

$$\begin{aligned} 0 = \lim_{k \to +\infty} \|f_k - f\|_p^p &= \lim_{k \to +\infty} \int_{-1/k}^{1/k} |f_k(x) - f(x)|^p \, dx + \\ &+ \lim_{k \to +\infty} \left(\int_{-1}^{-1/k} |-1 - f(x)|^p \, dx + \int_{1/k}^{1} |1 - f(x)|^p \, dx \right) = \\ &= \int_{-1}^{0} |-1 - f(x)|^p \, dx + \int_{0}^{1} |1 - f(x)|^p \, dx, \end{aligned}$$

(continued)

Example 5 (continued)
and therefore $f(x) = -1$ for $x \in [-1, 0)$, $f(x) = 1$ for $x \in (0, 1]$, in contrast to the continuity of f.

In Sect. 9.9 we will drop the continuity of f and so achieve a larger class of functions, called L^p: this will be a Banach space under norm (2.57).

The following Propositions 1 and 2 are useful in several situations.

Proposition 1 *If a Cauchy sequence x_k in a metric space X admits a subsequence with limit $x \in X$, then the whole sequence converges to x.*

Proof given $\varepsilon > 0$, let $\nu \in \mathbb{N}$ be such that

$$d(x_k, x_h) < \varepsilon\,, \qquad \forall\, h, k > \nu\,.$$

Write x_{k_r} for the subsequence of x_k that converges to $x \in X$. Choose $\overline{r} > \nu$ such that

$$d(x_{k_{\overline{r}}}, x) < \varepsilon\,.$$

As $k_r \geq r$, for any $r \in \mathbb{N}$, in particular we have $k_{\overline{r}} > \nu$ and so

$$d(x_k, x) \leq d(x_k, x_{k_{\overline{r}}}) + d(x_{k_{\overline{r}}}, x) < 2\varepsilon\,, \qquad \forall\, k > \nu\,.$$

Proposition 2 *Let (X, d) be a complete metric space and Y a closed subset in X. Then the metric space (Y, d) is complete.*

Proof Let x_k be a Cauchy sequence in Y. As X is complete there exists $x \in X$ such that $x_k \to x$. But Y being closed forces $x \in Y$.

2.9 Lipschitz Functions: The Contraction Theorem

Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. One says that f is *Lipschitz* if there exists a constant L such that

$$d_Y(f(x), f(y)) \leq L\, d_X(x, y)\,, \qquad \forall\, x, y \in X\,.$$

Clearly a Lipschitz function $f : X \to Y$ is continuous on X.

When (X, d_X) and (Y, d_Y) coincide (let d denote the common distance), a Lipschitz function $f : X \to X$ *with Lipschitz constant* $L < 1$, i.e. such that

$$d(f(x), f(y)) \leq L\, d(x, y)\,, \qquad \forall\, x, y \in X\,,$$

is often called a *contraction* on the metric space (X, d).

The next theorem, also known as *Banach-Caccioppoli fixed-point theorem*, has many applications in Analysis and will be invoked several times in the sequel.

Contraction Theorem *Let (X, d) be a complete metric space and $f : X \to X$ a contraction. Then there exists a unique point $x \in X$ such that $f(x) = x$.*

A point $x \in X$ such that $f(x) = x$ is called a *fixed point* of f. The theorem then asserts that a contraction of X possesses a unique fixed point.

Proof Pick an arbitrary point $x_0 \in X$ and define recursively the sequence x_k as follows:

$$x_1 = f(x_0)\,, \quad x_2 = f(x_1)\,, \quad \ldots \quad , x_{k+1} = f(x_k)\,, \quad \ldots$$

More compactly, $x_{k+1} = f(x_k)$ for any $k = 0, 1, 2, \ldots$

Let us prove that x_k is a Cauchy sequence. For that observe

$$\begin{aligned}
&d(x_1, x_2) = d(f(x_0), f(x_1)) \leq Ld(x_0, x_1) = Ld(x_0, f(x_0))\,,\\
&d(x_2, x_3) = d(f(x_1), f(x_2)) \leq Ld(x_1, x_2) \leq L^2 d(x_0, f(x_0))\,,\\
&\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\\
&d(x_k, x_{k+1}) = d(f(x_{k-1}), f(x_k)) \leq Ld(x_{k-1}, x_k) \leq L^k d(x_0, f(x_0))\,,
\end{aligned}$$

and so forth. Moreover, by the triangle inequality

$$\begin{aligned}
d(x_k, x_{k+p}) &\leq d(x_k, x_{k+1}) + d(x_{k+1}, x_{k+2}) + \ldots + d(x_{k+p-1}, x_{k+p}) \leq\\
&\leq (L^k + L^{k+1} + \ldots + L^{k+p-1})\, d(x_0, f(x_0)) =\\
&= \frac{L^k - L^{k+p}}{1 - L} d(x_0, f(x_0))\,.
\end{aligned}$$

As $0 \leq L < 1$ we then have

$$d(x_k, x_{k+p}) \leq \frac{L^k}{1 - L} d(x_0, f(x_0))\,. \qquad \forall\, k, p\,,$$

Since $L^k \to 0$ as $k \to +\infty$, the right side (independent of $p \in \mathbb{N}$) can be made smaller than any given quantity $\varepsilon > 0$, as long as we choose k large enough. Hence x_k is a Cauchy sequence.

As (X, d) is complete, there exists $x \in X$ such that $x_k \to x$. Taking the limit of $x_{k+1} = f(x_k)$ (valid for any $k = 0, 1, 2, \ldots$) as $k \to +\infty$, by the continuity of f we obtain

$$x = f(x)\,,$$

showing that x is a fixed point.

Regarding the uniqueness, suppose there is a point $y \in X$ such that $f(y) = y$. Then

$$d(x, y) = d(f(x), f(y)) \le Ld(x, y)\,,$$

which is absurd if $d(x, y) \neq 0$, because $L < 1$. Eventually $d(x, y) = 0$, and so $x = y$.

Now we shall prove a useful theorem on the extension of Lipschitz functions $f : X \to \mathbb{R}$.

Mcshane Lemma (Extension of Lipschitz Functions) *Let (X, d) be a metric space and A a subset of X. Any Lipschitz function $f : A \to \mathbb{R}$ with constant L can be extended to the entire space X to a Lipschitz function $g : X \to \mathbb{R}$ with the same constant L.*

Proof Define

$$g(x) = \inf\{f(y) + Ld(x, y) : y \in A\}\,, \qquad \forall x \in X\,, \tag{2.58}$$

and let us prove that

$$g(x) = f(x)\,, \qquad \forall x \in A\,, \tag{2.59}$$

$$|g(x) - g(y)| \le Ld(x, y)\,, \qquad \forall x, y \in X\,. \tag{2.60}$$

To that end observe that from definition (2.58), taking $y = x$, we have $g(x) \le f(x)$ for any $x \in A$. Furthermore, if $x, y \in A$ then

$$f(x) = f(x) - f(y) + f(y) \le Ld(x, y) + f(y)\,,$$

and taking the infimum over all $y \in A$ we attain the opposite inequality $f(x) \le g(x)$ for any $x \in A$. This proves (2.59).

On the other hand, if $x \in X$ then definition (2.58) gives

$$g(x) \le f(a) + Ld(x, a)\,, \qquad \forall a \in A\,,$$

and for any $y \in X$,

$$g(x) \leq f(a) + Ld(x, y) + Ld(y, a)\,, \qquad \forall\, a \in A$$

using the triangle inequality. Now take the least upper bound over $a \in A$ to obtain

$$g(x) \leq g(y) + Ld(x, y)\,.$$

This implies (2.60) since x and y may be interchanged.

2.10 Compact Sets: Continuous Functions on Compact Sets

A subset K in a metric space (X, d) is *compact* if every sequence x_k of points of K admits a subsequence that converges to a point $x \in K$.
If $K \subseteq X$ is a compact set according to the above definition, it is usually called *sequentially compact* in order to distinguish the above from the definition in the Appendix (see Sect. 2.12), which uses *open covers*. At any rate, in Sect. 2.12 we shall prove that the metric structure of the space X guarantees that the two notions are equivalent.

Proposition *If (X, d) is a metric space and $K \subseteq X$ is compact, then K is closed.*

Proof Take a sequence x_k of points in K with limit $x \in X$. We claim $x \in K$. As K is compact, there exists a subsequence x_{k_h} converging to some $x_0 \in K$. Necessarily $x = x_0$, and so $x \in K$.
In case $K = X$ is compact, we call (X, d) a *compact metric space*.

By relying on the Bolzano-Weierstrass theorem for $\mathbb{R}$, the following result is easy to prove.

Characterisation of Compact Subsets of $\mathbb{R}^n$ (Heine-Borel Theorem) *A subset $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof Suppose K is a closed and bounded subset, and let us take a sequence $x_k = (x_{k,1}, x_{k,2}, \ldots, x_{k,n})$ in K. The sequences $x_{k,i}$, for $i = 1, 2, \ldots, n$, are bounded sequences of real numbers. The Bolzano-Weierstrass theorem tells that we may extract from x_k a subsequence whose first coordinate converges. From the latter we may extract a subsequence whose second coordinate converges, and so on. This produces a strictly increasing sequence k_h of natural numbers such that $x_{k_h,i} \to x_i$ for any $i = 1, 2, \ldots, n$. Therefore $x_{k_h} \to x = (x_1, x_2, \ldots, x_n)$. But K is closed, so $x \in K$ and K is compact.

Vice versa, if K is compact it is closed by the previous proposition. If it were unbounded, there would exist a sequence x_k in K such that

$$\lim_{k\to+\infty} |x_k| = +\infty\,,$$

from which we could extract no convergent sequence, violating the compactness of K.

Continuous functions defined on compact sets enjoy similar properties to those of continuous functions of one real variable defined on closed and bounded intervals. The reader will also note that the proofs of these properties in metric spaces are natural generalisations of those for functions of one real variable.

Weierstrass Theorem *Let K be a compact subset in the metric space (X, d). Any continuous function $f : K \to \mathbb{R}$ admits maximum and minimum on K.*

Proof Set

$$M = \sup\{f(x) : x \in K\}\,,$$

and let us show that there exists a sequence x_k in K such that

$$\lim_{k\to+\infty} f(x_k) = M\,. \tag{2.61}$$

If $M = +\infty$, by the supremum's properties, for any $k \in N$ there exists $x_k \in \mathbb{N}$ such that $f(x_k) > k$, so $f(x_k) \to M$. If, instead, $M < +\infty$, for any $k \in \mathbb{N}$ there exists $x_k \in K$ such that

$$M - \frac{1}{k} < f(x_k) \le M\,,$$

and again $f(x_k) \to M$.

As K is compact, there is a subsequence x_{k_h} of x_k such that $x_{k_h} \to x_0 \in K$. Since f is continuous,

$$\lim_{h\to+\infty} f(x_{k_h}) = f(x_0)\,.$$

Then by (2.61)

$$M = \lim_{k\to+\infty} f(x_k) = \lim_{h\to+\infty} f(x_{k_h}) = f(x_0)\,.$$

Therefore $M < +\infty$ and $f(x_0) = M = \max\{f(x) : x \in K\}$. The reasoning is analogous for establishing the existence of a minimum point.

The following is a consequence of the Weierstrass theorem.

Equivalence of Norms on $\mathbb{R}^n$ *All norms on $\mathbb{R}^n$ are equivalent.*

Proof It is enough to show that any norm $\|\cdot\|$ on $\mathbb{R}^n$ is equivalent to the standard norm $|\cdot|$. For any $x \in \mathbb{R}^n$ we have

$$\|x\| = \left\| \sum_{i=1}^{n} x_i e_i \right\| \leq \sum_{i=1}^{n} |x_i| \cdot \|e_i\| \leq \max\{\|e_i\| : i = 1, 2, \ldots, n\} \sum_{i=1}^{n} |x_i| \leq \tag{2.62}$$
$$\leq \sqrt{n}\, \max\{\|e_i\| : i = 1, 2, \ldots, n\}\, |x| = M|x|\,.$$

In particular, for $x, y \in \mathbb{R}^n$ it follows that $\|x - y\| \leq M|x - y|$, so the function $x \to \|x\|$ is continuous on $\mathbb{R}^n$ in the Euclidean metric. As the set $K = \{x \in \mathbb{R}^n : |x| = 1\}$ is compact, there exists $m = \min\{\|x\| : x \in K\}$. This minimum m is certainly positive, because it is the value of the norm $\|\cdot\|$ at some point $x \neq 0$. If $x \in \mathbb{R}^n - \{0\}$ then

$$\|x\| = |x| \cdot \left\| \frac{x}{|x|} \right\| \geq m|x|\,.$$

From this inequality together with (2.62) the claim follows.

Consider two metric spaces (X, d_X), (Y, d_Y), and a subset $K \subseteq X$. A function $f : K \to Y$ is *uniformly continuous* on K if, for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x, x' \in K : d_X(x, x') < \delta \quad \Rightarrow \quad d_Y(f(x), f(x')) < \varepsilon\,.$$

For instance, a Lipschitz function $f : K \to Y$, and in particular a contraction, is uniformly continuous on K.

The next theorem provides us with a sufficient condition for a continuous function to be *uniformly continuous.*

Theorem (Cantor) *Let K be a compact subset in the metric space (X, d_X). Then a continuous function $f : K \to X$ is uniformly continuous on K.*

Proof Suppose by contradiction that f is not uniformly continuous on K. Then there exist $\varepsilon_0 > 0$ and also two sequences $x_k,\ x'_k$ in K such that

$$d_Y(f(x_k), f(x'_k)) \geq \varepsilon_0\,, \tag{2.63}$$

but at the same time

$$x_k, x'_k \in K : d_X(x_k, x'_k) < \frac{1}{k}\,, \tag{2.64}$$

for any $k \in \mathbb{N}$. The compactness of K forces the existence of a subsequence x_{k_h} of x_k and a point $x_0 \in K$ such that $x_{k_h} \to x_0$.

From (2.64) the corresponding subsequence x'_{k_h} converges to x_0, since

$$d_X(x'_{k_h}, x_0) \le d_X(x'_{k_h}, x_{k_h}) + d_X(x_{k_h}, x_0) < \frac{1}{k_h} + d_X(x_{k_h}, x_0)\,.$$

But f is continuous, so $f(x_{k_h}) \to f(x_0)$, $f(x'_{k_h}) \to f(x_0)$, i.e.

$$d_Y(f(x_{k_h}), f(x_0)) \to 0\,, \qquad d_Y(f(x'_{k_h}), f(x_0)) \to 0\,,$$

and then

$$d_Y(f(x_{k_h}), f(x'_{k_h})) \to 0\,,$$

contradicting (2.63).

2.11 Connected Open Subsets of $\mathbb{R}^n$

An open subset $A \subseteq \mathbb{R}^n$ is *connected* when there is no partition $\{A_1, A_2\}$ of A into non-empty open sets. Equivalently, an open set A is *connected* if and only if

$$A_1, A_2 \quad \text{open in} \quad \mathbb{R}^n, \qquad A_1 \cap A_2 = \emptyset\,, \qquad A_1 \cup A_2 = A$$

imply that one of the two sets A_1 and A_2 is empty.

A *segment* $[x_1, x_2]$ in $\mathbb{R}^n$ with endpoints x_1 and x_2 is the set of points $x(t)$ in $\mathbb{R}^n$ such that

$$x(t) = x_1 + t(x_2 - x_1)\,, \qquad \forall\, t \in [0, 1]\,.$$

Let $x_1, x_2, \ldots, x_k$ be k points of $\mathbb{R}^n$ ($k \ge 2$), with $x_i \ne x_{i+1}$ for any $i = 1, 2, \ldots, k-1$. A *polygonal path* in $\mathbb{R}^n$ with vertices $x_1, x_2, \ldots, x_k$ is the union of the line segments $[x_i, x_{i+1}]$, for $i = 1, 2, \ldots, k-1$. The points x_1, x_k are the *path's endpoints*.

Theorem on Connected Open Subsets of $\mathbb{R}^n$ *If $A \subseteq \mathbb{R}^n$ is a connected open set, any pair of points of A are endpoints of a polygonal path entirely contained in A.*

Proof Pick $x_1 \in A$ and call A_1 the set of endpoints $x \in A$ of polygonal paths starting at x_1 and all contained in A. We claim $A_1 = A$.

The set A_1 is open, because if $x \in A_1$ then x belongs to the open set A and then there is an open ball $I_\delta(x)$, centred at x with radius δ, contained in A. If $y \in I_\delta(x)$, $y \ne x$, the line segment $[x, y]$ is part of a radial segment in the ball $I_\delta(x)$, and is clearly contained in $I_\delta(x)$. Hence joining the previous path between x_1 and x with the line segment $[x, y]$ produces a polygonal path from x_1 to y, still contained in A. Therefore $y \in A_1$ by definition, and then $I_\delta(x) \subseteq A_1$, making A_1 open.

Now set $A_2 = A - A_1$ and let us show that A_2, too, is open. If A_2 is empty there is nothing to prove. So suppose $x \in A_2$. Then there exists an open ball $I_\delta(x)$, of centre x and radius δ, contained in A. If $y \in I_\delta(x)$, $y \neq x$, the line segment $[x, y]$ is contained in $I_\delta(x)$. If y were in A_1, i.e. if there existed a polygonal path in A joining x_1 and y, attaching this path to $[y, x]$ would produce a polygonal path from x_1 to x contained in A, resulting in $x \in A_1$. This would contradict $x \in A_2 = A - A_1$. Eventually then $y \notin A_1$, so $y \in A_2$ and therefore $I_\delta(x) \subseteq A_2$, showing A_2 is open.

We have found two open sets A_1, A_2, with $A_1 \cap A_2 = \emptyset$, $A_1 \cup A_2 = A$. By assumption A is connected, and A_1 is non-empty since $x_1 \in A_1$, so necessarily $A_2 = A - A_1 = \emptyset$, i.e. $A_1 = A$.

Let us point out explicitly that if A is an open subset in $\mathbb{R}^n$ such that any pair of points of A are endpoints of a path inside A, we say A is *path connected*. The theorem we have just proved may then be rephrased as follows: *any connected open set is path connected*. Interestingly, the converse is true as well.

More generally, we shall now define *connected domains* in $\mathbb{R}^n$. Recall that a *domain* $D \subseteq \mathbb{R}^n$ is the closure of an open set. We shall say D is a *connected domain* if it is the closure of a connected open subset of $\mathbb{R}^n$.

The following result extends to $\mathbb{R}^n$ the intermediate value theorem for functions of one real variable.

Intermediate Value Theorem *Let A be a connected open set (or domain) in $\mathbb{R}^n$, and $f : A \to \mathbb{R}$ a continuous function on A. Then f assumes every value between the infimum and the supremum of f on A.*

Proof Fix $l \in (\inf_A f, \sup_A f)$. There exist x and y internal to A such that $f(x) < l < f(y)$ (obvious if A is open, and an easy consequence of the continuity of f if A is a domain). Hence there is a polygonal path, of vertices $x_1 = x, x_2, \ldots, x_k = y$, completely contained in the interior of A. As the restriction of f to every line segment $[x_i, x_{i+1}]$, $i = 1, 2, \ldots, k-1$, is a continuous function of one real variable, namely

$$t \to f(x(t)) = f(x_i + t(x_{i-1} - x_i)), \qquad \forall\, t \in [0, 1],$$

the image of each interval $[x_i, x_{i+1}]$ is a closed interval in $\mathbb{R}$.

Furthermore, the images of two consecutive intervals $[x_{i-1}, x_i]$, $[x_i, x_{i+1}]$ overlap because the value $f(x_i)$ belongs to both. Therefore the images' union is an interval. Iterating the process, the image of the polygonal path of vertices $x_1 = x, x_2, \ldots, x_k = y$ is an interval, which contains $f(x)$, $f(y)$ and hence it also contains l.

Appendix to Chap. 2

2.12 Further Compactness Theorems: Generalised Weierstrass Theorem

The main aim of the section is characterising compact sets in a metric space and show a few interesting applications. First we need some definitions.

Consider a metric space (X, d) and a subset Y. A family $\mathcal{R}$ of subsets of X is a *cover* of Y if Y is contained in the union of the elements of $\mathcal{R}$.

If $\mathcal{R}' \subseteq \mathcal{R}$ is a subfamily of $\mathcal{R}$ that still covers Y, we say $\mathcal{R}'$ is a *subcover* of Y extracted from $\mathcal{R}$. A cover $\mathcal{R}$ is called *open* when it consists of open subsets of X, and *finite* if it is made of finitely many subsets.

Finally, a subset $Y \subset X$ is *totally bounded* if for any $r > 0$ there exists a finite cover of Y consisting of open balls of radius r

$$B_r(x_1), \quad B_r(x_2), \quad \ldots, B_r(x_N),$$

with $x_1, x_2, \ldots, x_N \in Y$.

It is easy to see that *a totally bounded set is bounded*.

Theorem (Characterisation of Compact Sets) *Let (X, d) be a metric space and K a subset of X. The following three properties are equivalent:*

(a) K is compact*;*
(b) K is totally bounded, and any Cauchy sequence in K converges to some point of K*;*
(c) every open cover of K admits a finite subcover.

Condition (c) is often taken as the general definition of compactness. The above theorem allows to conclude that for metric spaces this notion of compactness is equivalent to that of Sect. 2.10, whereby *a set K in a metric space (X, d) is called compact if every sequence x_k in K admits a subsequence that converges to a point of $x \in K$*.

Proof (a) $\Rightarrow$ (b). Let x_k be a Cauchy sequence in K. As K is compact, we can extract a subsequence with limit $x \in K$, so by Proposition 1, Sect. 2.8, x_k must converge to x. This proves the second part of (b).

By contradiction, suppose K is not totally bounded. Then there exists $r_0 > 0$ such that K cannot be covered by finitely many open balls of radius r_0 and centred in K. In particular, given $x_1 \in K$, the ball $B_{r_0}(x_1)$ cannot contain K and so there exists $x_2 \in K$ such that $d(x_1, x_2) \geq r_0$. As $B_{r_0}(x_1)$, $B_{r_0}(x_2)$ cannot cover K, there exists $x_3 \in K$ such that $d(x_1, x_3) \geq r_0$ and $d(x_2, x_3) \geq r_0$. Continuing in this way

we build a sequence x_k in K for which

$$d(x_h, x_k) \geq r_0\,, \qquad \forall\, h, k \in \mathbb{N},\ h \neq k\,.$$

But then x_k has no convergent subsequence, which cannot happen since K is compact.

(b) $\Rightarrow$ (c). Suppose by contradiction there is an open cover $\mathcal{R}$ of K without finite subcovers. We will show that it is possible to find a sequence x_k in K such that, setting

$$B_k = B_{2^{-k}}(x_k)\,,$$

we have

(i) $B_k \cap B_{k+1} \neq \emptyset$ for any $k \in \mathbb{N}$;
(ii) for any k there is no finite subfamily of $\mathcal{R}$ covering $B_k \cap K$.

As K is totally bounded, it can be covered by a finite number of balls centred at points in it, all with radius $1/2$. By the assumption on $\mathcal{R}$, at least one such neighbourhood, say $B_1 = B_{2^{-1}}(x_1)$, with $x_1 \in K$, is such that there is no finite subfamily of $\mathcal{R}$ covering $B_1 \cap K$.

Next we consider a finite cover of K made of balls centred at points in K, all with radius $1/4$. By the choice of B_1, at least one such neighbourhood $B_2 = B_{2^{-2}}(x_2)$, with $x_2 \in K$, will have a non-empty intersection with $B_1 \cap K$ and will be such that no finite subfamily of $\mathcal{R}$ covers $B_2 \cap K$. In particular $B_1 \cap B_2 \neq \emptyset$. In the same way we can find a neighbourhood $B_3 = B_{2^{-3}}(x_3)$ with $x_3 \in K$ and $B_2 \cap B_3 \neq \emptyset$, such that no finite subfamily of $\mathcal{R}$ covers $B_3 \cap K$. By iterating the construction we build a sequence x_k of points of K verifying (i) and (ii).

Note that x_k is a Cauchy sequence: from (i), given $k \in \mathbb{N}$ there exists $z \in B_k \cap B_{k+1}$ and so

$$d(x_k, x_{k+1}) \leq d(x_k, z) + d(z, x_{k+1}) \leq \frac{1}{2^k} + \frac{1}{2^{k+1}} < \frac{1}{2^{k-1}}\,.$$

Fix a positive integer k_0. If $h > k > k_0$ then

$$\begin{aligned} d(x_k, x_h) &\leq d(x_k, x_{k+1}) + \ldots + d(x_{h-1}, x_h) \leq \\ &\leq \frac{1}{2^{k-1}} + \ldots + \frac{1}{2^{h-2}} < \frac{1}{2^{k_0-1}}\,. \end{aligned}$$

Hence x_k is Cauchy, and by (b) it converges to some $x \in K$.

Let $A \in \mathcal{R}$ be an open set of the cover containing x. There is a ball $B_r(x)$ around x contained in A. As $x_k \to x$, there exists an integer k such that $d(x, x_k) < r/2$ and

$2^{-k} < r/2$. Therefore

$$B_k = B_{2^{-k}}(x_k) \subseteq B_r(x) \subseteq A$$

and, by (ii), we reach a contradiction.

(c) $\Rightarrow$ (a). Let us first show that K is closed. Given $x_0 \notin K$, for any $x \in K$ there exists $B_{r_x}(x)$ around x such that

$$x_0 \notin \overline{B_{r_x}(x)}\,. \tag{2.65}$$

As x varies in K, the neighbourhoods $B_{r_x}(x)$ generate an open cover of K, so we may select a finite number of them whose union contains K. Since, by (2.65),

$$x_0 \notin \bigcup_{i=1}^{N} \overline{B_i} = C$$

and C is closed, there exists a neighbourhood of x_0 disjoint from C, and hence disjoint from K. Therefore x_0 is external to K. In conclusion, every point not in K is external to K, meaning K is closed.

To prove the compactness of K it suffices to show that any sequence x_k in K admits a convergent subsequence. Suppose by contradiction that x_k has no convergent subsequence. Then for any $x \in K$ there exists a ball around x containing only a finite number of terms x_k. But K can be covered by finitely many such neighbourhoods, so the sequence x_k must be finite, which is absurd.

This theorem, together with the Ascoli-Arzelà theorem of Sect. 1.10, Chap. 1, easily implies the next compactness criterion in $C^0([a,b])$.

Compactness Criterion in $C^0([a,b])$ *Let Y be a subset of the metric space $C^0([a,b])$. The following properties are equivalent:*

(a) *every sequence f_k in Y admits a convergent subsequence;*
(b) *Y consists of functions that are uniformly bounded, i.e. there exists $M > 0$ such that*

$$|f(x)| \le M\,, \qquad \forall\, x \in [a,b]\,, \forall\, f \in Y\,, \tag{2.66}$$

and equicontinuous: for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon\,, \qquad \forall\, x, y \in [a,b] : |x - y| < \delta,\ \forall\, f \in Y\,. \tag{2.67}$$

Observe that (a) is equivalent to demanding that the closure $\overline{Y}$ is a compact subset of $C^0([a,b])$.

Proof The implication (b) $\Rightarrow$ (a) is precisely the Ascoli-Arzelà theorem, since the convergence in $C^0([a,b])$ is the uniform convergence.

(a) $\Rightarrow$ (b). As noted above, (a) is the same as saying $\overline{Y}$ is compact. In particular $\overline{Y}$ is bounded and hence Y is bounded, proving (2.66).

Let us prove (2.67). As $\overline{Y}$ is totally bounded, given $\varepsilon > 0$ there exist $f_1, f_2, \ldots, f_N$ in $\overline{Y}$ such that $\overline{Y}$ is contained in the union over $i \in \{1, 2, \ldots, N\}$ of the balls $B_i = B_{\varepsilon/4}(f_i)$.

Take $f \in Y$. Then there exists $i \in \{1, 2, \ldots, N\}$ such that

$$d(f, f_i) = \sup\{|f(x) - f_i(x)| : x \in [a,b]\} < \frac{\varepsilon}{4}.$$

Moreover, the f_i, $i = 1, 2, \ldots, N$, are uniformly continuous, so there exists $\delta > 0$ such that

$$|x - y| < \delta \quad \Rightarrow \quad |f_i(x) - f_i(y)| < \frac{\varepsilon}{2}, \qquad \forall\, i = 1, 2, \ldots, N.$$

Hence for $x, y \in [a,b]$ and $|x - y| < \delta$ we have

$$|f(x) - f(y)| \le |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| < \\ < \frac{\varepsilon}{4} + \frac{\varepsilon}{2} + \frac{\varepsilon}{4} = \varepsilon$$

and (2.67) is proven.

In Sect. 2.3 we defined the *continuity* of a function $f : X \to Y$ between metric spaces. We showed (see Proposition 2, Sect. 2.3) that the following are equivalent:

$$\forall\, x_k \quad \textit{sequence in} \quad X : d_X(x_k, x_0) \to 0 \quad \Rightarrow \quad d_Y(f(x_k), f(x_0)) \to 0,$$
$$\forall\, \varepsilon > 0\, \exists\, \delta > 0 : x \in X, d_X(x, x_0) < \delta \quad \Rightarrow \quad d_Y(f(x), f(x_0)) < \varepsilon.$$

Either can be taken as definition of continuity of f at x_0. As always, we say that f is continuous on X if it is continuous at every point $x_0 \in X$.

Proposition (Characterisation of Continuity) *A function f between metric spaces (X, d_X) and (Y, d_Y) is continuous on X if and only if, for any open set $A \subseteq Y$, the set*

$$f^{-1}(A) = \{x \in X : f(x) \in A\}$$

is open in X.

By the previous proposition $f : X \to Y$ is continuous on X if and only if the pre-image $f^{-1}(C)$ of any closed set $C \subseteq Y$ is closed in X.

Proof Let f be continuous on X and A an open set in Y. We claim $f^{-1}(A)$ is open in X. Take a generic point $x_0 \in f^{-1}(A)$ and let us show there exists an open ball $I_\delta(x_0)$ around x_0 contained in $f^{-1}(A)$.

As A is open in Y and $f(x_0) \in A$, there exists $\varepsilon > 0$ such that

$$d_Y(y, f(x_0)) < \varepsilon \quad \Rightarrow \quad y \in A\,. \tag{2.68}$$

By the continuity of f at x_0, there exists $\delta > 0$ such that

$$d_X(x, x_0) < \delta \quad \Rightarrow \quad d_Y(f(x), f(x_0)) < \varepsilon\,. \tag{2.69}$$

From (2.68) and (2.69) we have

$$d_X(x, x_0) < \delta \quad \Rightarrow \quad f(x) \in A \quad \Rightarrow \quad x \in f^{-1}(A)\,.$$

Hence the open ball $I_\delta(x_0) = \{x \in X : d_X(x, x_0) < \delta\}$ is contained in $f^{-1}(A)$, as required.

Conversely, suppose $f^{-1}(A)$ is open, for any open set $A \subset Y$. Given $x_0 \in X$ and $\varepsilon > 0$, put

$$A = \{y \in Y : d_Y(y, f(x_0)) < \varepsilon\}\,.$$

This is open in Y so $f^{-1}(A)$ is open in X, and therefore there exists $\delta > 0$ such that

$$d_X(x, x_0) < \delta \quad \Rightarrow \quad x \in f^{-1}(A) \quad \Rightarrow \quad f(x) \in A\,.$$

By definition of A we obtain

$$d_X(x, x_0) < \delta \quad \Rightarrow \quad d_Y(f(x), f(x_0)) < \varepsilon\,.$$

An important application of the above characterisation of metric compactness is a generalisation of the Weierstrass theorem.

Generalised Weierstrass Theorem *Let f be a continuous function between metric spaces (X, d_X) and (Y, d_Y). If K is compact in X, the image*

$$f(K) = \{y \in Y : \exists\, x \in X :\ y = f(x)\}\,.$$

is compact in Y.

Proof Let $\mathcal{R} = \{A_i\}$ be an open cover of $f(K)$. Then $\{f^{-1}(A_i)\}$ is a cover of K, and by continuity the previous proposition (characterising continuity) implies that $\{f^{-1}(A_i)\}$ is an open cover of K.

As K is compact in X, there is a finite number of indices $i_1, i_2, \ldots, i_N$ such that

$$K \subset f^{-1}(A_{i_1}) \cup f^{-1}(A_{i_2}) \cup \ldots \cup f^{-1}(A_{i_N})\,;$$

and so

$$f(K) \subset A_{i_1} \cup A_{i_2} \cup \ldots \cup A_{i_N}\,.$$

Therefore K is compact in Y.

We close the section by observing that the Weierstrass theorem of Sect. 2.10 is a corollary of the above. When f is real-valued, in fact (i.e. $Y = \mathbb{R}$), to say $f(K)$ is compact means $f(K)$ is closed and bounded, and as such it has maximum and minimum due to the characterisation of compact subsets of $\mathbb{R}$ (Heine-Borel theorem, Sect. 2.10).

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Functions of Several Variables

3

In this chapter we begin the systematic study of the properties of functions of several real variables. We shall consider functions $f : \mathbb{R}^n \to \mathbb{R}^m$, with $n, m \geq 1$, starting from the case $f : \mathbb{R}^n \to \mathbb{R}$, i.e. $m = 1$ (*scalar functions*), and then pass in Sect. 3.14 to $m > 1$ (*vector-valued functions*).

3.1 Round-up of Topology in $\mathbb{R}^n$

The topological properties of $\mathbb{R}^n$ which we briefly recall here have been already introduced and studied in Chap. 2, in relationship to general metric spaces.

For any element $x = (x_1, x_2, \ldots, x_n)$ in $\mathbb{R}^n$ we write $|x|$ for the *modulus* (also called *norm* or *distance to the origin*) of x, defined as

$$|x| = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} .$$

Let x_0 be a given element of $\mathbb{R}^n$. The *open ball* around x_0 of radius $\delta > 0$ is the non-empty subset of $\mathbb{R}^n$

$$I_\delta = \{x \in \mathbb{R}^n \, : \, |x - x_0| < \delta\} \tag{3.1}$$

consisting of points whose distance to x_0 is less than δ. Sometimes it becomes important to denote explicitly that x_0 is the ball's centre, so we shall write $I_\delta(x_0)$ to denote (3.1).

Let A be a subset in $\mathbb{R}^n$. An element x_0 of $\mathbb{R}^n$ is called *interior* to A if there exists an open ball around x_0 contained in A, *exterior* to A if there is an open ball around x_0 contained in the complement of A (i.e. x_0 is interior to the complement of A), and finally x_0 is a *boundary point* of A if every open ball centred at x_0 contains points of both A and its complement (i.e. x_0 is neither interior nor exterior to A).

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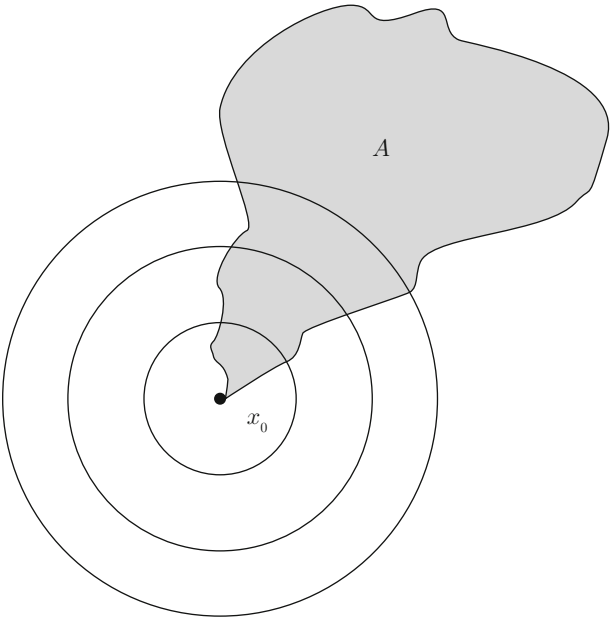


Fig. 3.1

The point x_0 is a *limit point* of $A \subseteq \mathbb{R}^n$ if every open ball around x_0 contains a point of A distinct from x_0 (as in Fig. 3.1).

Immediately, all interior points of a set A are limit points of A, and no exterior point is a limit point of A. Boundary points may or not be limit points: if a boundary point of A is not a limit point, it is called an *isolated point* of A.

A set $A \subseteq \mathbb{R}^n$ is *open* if for any $x_0 \in A$ there is an open ball around x_0 contained in A (every x_0 is interior to A). A set $C \subseteq \mathbb{R}^n$ is *closed* if the complement $A = \mathbb{R}^n - C$ is open.

Obviously there are subsets of $\mathbb{R}^n$ that are neither open nor closed. The empty set $\emptyset$ and the entire $\mathbb{R}^n$ are the only subsets of $\mathbb{R}^n$ that are simultaneously open and closed.

The *closure* of a set A, written $\overline{A}$, is the subset of $\mathbb{R}^n$ arising from the union of A and its limit points. The closure $\overline{A}$ is a closed set, and more precisely it coincides with the intersection of all closed sets containing A. One can also prove that $\overline{A}$ is the union of A and its boundary points.

A *domain* in $\mathbb{R}^n$ is the closure of an open (non-empty) set. Hence a domain is closed, and consists of the union of an open set and its boundary.

For instance, a closed ball in $\mathbb{R}^n$ is a domain, whilst the union of a closed ball and an isolated point is not.

A set $A \subseteq \mathbb{R}^n$ is *bounded* if it is contained in an open ball $I_M(0)$ centred at the origin, i.e. there exists $M > 0$ such that

$$|x| < M\,, \qquad \forall\, x \in A\,.$$

A set $K \subset \mathbb{R}^n$ is *compact* if every sequence of elements of K admits a subsequence that converges to some element of K.

The Heine-Borel theorem, proved in Sect. 2.10, states that *a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded*. A further characterisation proved in Sect. 2.12 is that *$K \subset \mathbb{R}^n$ is compact if and only if every collection of open sets whose union contains K possesses a finite subcollection whose union still contains K*.

An open set $A \subseteq \mathbb{R}^n$ is *connected* if there do not exist two non-empty disjoint open sets whose union is A. In formulas, there are no open sets $A_1, A_2 \subseteq \mathbb{R}^n$, such that

$$\begin{cases} A_1 \neq \emptyset\,, & A_2 \neq \emptyset \\ A_1 \cap A_2 = \emptyset\,, & A_1 \cup A_2 = A\,. \end{cases}$$

Equivalently, if we can find open sets $A_1, A_2 \subseteq \mathbb{R}^n$ such that

$$A_1 \cap A_2 = \emptyset\,, \qquad A_1 \cup A_2 = A\,,$$

then one of A_1, A_2 must be empty.

Finally, a *domain* is *connected* if it is the closure of a connected open set.

3.2 Limits and Continuity

The definition of limit for functions of one real variable is easily extended to several variables. To do that we consider a subset A in $\mathbb{R}^n$ and a function $f : A \to \mathbb{R}$, so f is a real function defined on $A \subseteq \mathbb{R}^n$. Let x_0 be a limit point of A.

We say $f(x)$ tends to $\ell \in \mathbb{R} \cup \{\pm\infty\}$ as $x \to x_0$, written

$$\lim_{x \to x_0} f(x) = \ell \quad (\text{or} \quad f(x) \to \ell \quad \text{as } x \to x_0),$$

if, for any neighbourhood $U \subseteq \mathbb{R}$ of ℓ, there exists a neighbourhood $I_\delta(x_0) \subseteq \mathbb{R}^n$ of x_0 such that

$$f(x) \in U\,, \qquad \forall\, x \in I_\delta(x_0) \cap A - \{x_0\}\,.$$

As for the one-variable case, the definition can be rephrased *in terms of* ε, δ. For example, if $\ell \in \mathbb{R}$ (i.e. the limit is finite), $f(x)$ tends (or converges) to ℓ as $x \to x_0$

if, for any given $\varepsilon > 0$, there is a number $\delta > 0$ such that

$$|f(x) - \ell| < \varepsilon\,, \qquad \forall\, x \in A - \{x_0\} : \ |x - x_0| < \delta\,. \tag{3.2}$$

The reader should reformulate the cases $\ell = \pm\infty$ in a similar fashion.

For the sake of an example, consider the function f of two real variables x, y

$$f(x) = \frac{x^2}{\sqrt{x^2 + y^2}}\,.$$

Its domain of definition is the open set $A = \mathbb{R}^2 - \{(0,0)\}$ of points different from the origin (the punctured plane). The point $(0, 0)$ is a limit point of A.

Let us show that

$$\lim_{(x,y)\to(0,0)} \frac{x^2}{\sqrt{x^2 + y^2}} = 0\,.$$

By the inequalities

$$0 \le \frac{x^2}{\sqrt{x^2 + y^2}} \le \frac{x^2 + y^2}{\sqrt{x^2 + y^2}} = \sqrt{x^2 + y^2}\,,$$

we deduce that, for any $\varepsilon > 0$,

$$0 \le f(x, y) < \varepsilon \quad \text{(and so } |f(x, y)| < \varepsilon\text{)}$$

for any $(x, y) \neq (0, 0)$ such that $\sqrt{x^2 + y^2} < \varepsilon$. In other words (3.2) holds for $\delta = \varepsilon$.

Now we will show that

$$\lim_{(x,y)\to(0,0)} \frac{x}{\sqrt{x^2 + y^2}} \quad \textit{does not exist}.$$

Setting $g(x, y) = x/\sqrt{x^2 + y^2}$, for any neighbourhood I_δ of $(0, 0)$ there both exist points $(0, y)$ on the y-axis where

$$g(0, y) = 0\,,$$

and also points on the x-axis where

$$g(x, 0) = \frac{x}{|x|} = \begin{cases} 1 & \text{if} \quad x > 0 \\ -1 & \text{if} \quad x < 0\,. \end{cases} \tag{3.3}$$

Therefore (and irrespective of the fact that the function in (3.3) assumes distinct values), the function $g(x, y)$ does not admit limit $\ell \in \mathbb{R} \cup \{\pm\infty\}$ as $(x, y) \to (0, 0)$.

Let $f : A \to \mathbb{R}$ be a function defined on $A \subseteq \mathbb{R}^n$ and $x_0 \in A$. If x_0 is not isolated for A (i.e. if x_0 is a limit point of A), the function is said *continuous* at x_0 when

$$\lim_{x \to x_0} f(x) = f(x_0) .$$

Conventionally, f is continuous at every isolated point. We also say f is *continuous on the set* A if it is continuous at every point $x_0 \in A$.

In the rest of the section we will quickly recall a few important theorems on continuous functions on $\mathbb{R}^n$, which generalise the similar results for real functions of one real variable. The latter were proved in Sects. 2.10 and 2.11 of Chap. 2. Let us begin with two results on continuous functions defined on a compact set.

Weierstrass' Theorem *A continuous real function on a compact set K in $\mathbb{R}^n$ assumes its minimum and maximum values on K.*

Cantor's Theorem *A continuous real function on a compact set $K \subseteq \mathbb{R}^n$ is uniformly continuous on K: i.e., for any $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$x_1, x_2 \in K,\ |x_1 - x_2| < \delta \quad \Rightarrow \quad |f(x_1) - f(x_2)| < \varepsilon .$$

At last, the following result on continuous functions on connected domains is useful.

Intermediate Value Theorem *A continuous real function on a bounded connected domain D in $\mathbb{R}^n$ assumes all values between its minimum and maximum values.*

3.3 Partial Derivatives

Let A be an open set in $\mathbb{R}^n$, $f : A \to \mathbb{R}$ a function defined on A, $x = (x_1, x_2, \ldots, x_n)$ a point of A and $i \in \{1, 2, \ldots, n\}$ a given index. The *partial derivative* of f in the variable x_i at the point x is the limit

$$\lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h} , \tag{3.4}$$

whenever such limit exists and is finite. The partial derivative is denoted by any one for the following symbols

$$\frac{\partial f}{\partial x_i} , \quad \frac{\partial}{\partial x_i} f , \quad f_{x_i} , \quad f_{x_i}(x) , \quad D_i f , \quad D_i f(x) , \quad D_{x_i} f .$$

So by definition the partial derivative of $f(x) = f(x_1, \ldots, x_i, \ldots, x_n)$ in the variable x_i is computed by considering the remaining variables

$$x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n$$

fixed, and differentiating with respect to x_i using the usual rules for real functions of *one* real variable.

For example $f : \mathbb{R}^2 \to \mathbb{R}$ is a function in the two real variables x, y, defined by

$$f(x, y) = x^2 \cos y \,,$$

and its partial derivatives f_x, f_y with respect to x, y, respectively, are

$$f_x = 2x \cos y \,, \qquad f_y = -x^2 \sin y \,.$$

Consider instead $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = \sin(xy) \,.$$

The partial derivatives are

$$f_x = y \cos(xy) \,, \qquad f_y = x \cos(xy) \,.$$

Now take the function in n real variables $x = (x_1, x_2, \ldots, x_n)$:

$$f(x) = |x|^2 = \sum_{i=1}^{n} x_i^2 \,.$$

The partial derivative of f in x_i reads

$$f_{x_i} = 2x_i \,, \qquad \forall\, i \in \{1, 2, \ldots, n\} \,,$$

i.e. $f_{x_1} = 2x_1$, $f_{x_2} = 2x_2$ etc.

Finally, if $f : \mathbb{R}^n \to \mathbb{R}$ is given by

$$f(x) = \sin |x|^2 \,,$$

by the chain rule (in *one* real variable!) we obtain

$$f_{x_i} = 2x_i \cos |x|^2 \,, \qquad \forall\, i \in \{1, 2, \ldots, n\} \,.$$

If, at a point x, all partial derivatives f_{x_1}, f_{x_2}, $\ldots$, f_{x_n} exist, the function is said to *admit partial derivatives at* x. When f admits partial derivatives at any x in the open set A, we say it *admits partial derivatives on* A. Warning: the reader

should not confuse the existence of the derivatives with the notion of differentiability introduced in Sect. 3.5.

As for one real variable, also functions in several real variables may be defined on an open set A and not admit partial derivatives at some point in A. For instance, the function of two real variables defined on $\mathbb{R}^2$ by

$$f(x, y) = \sqrt{x^2 + y^2}\,, \qquad \forall\, (x, y) \in \mathbb{R}^2\,,$$

admits partial derivatives at any $(x, y) \neq (0, 0)$, given by

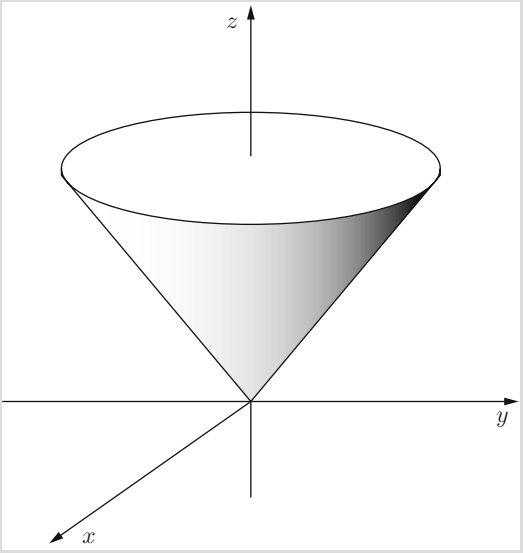
$$f_x = \frac{x}{\sqrt{x^2 + y^2}}\,, \qquad f_y = \frac{y}{\sqrt{x^2 + y^2}}\,,$$

but it *does not* admit partial derivatives at $(0, 0)$, because at that point the partial derivatives are not defined. In fact, the difference quotient with respect to x is

$$\frac{f(0 + h, 0) - f(0, 0)}{h} = \frac{\sqrt{h^2}}{h} = \frac{|h|}{h},$$

which does *not* admit limit as $h \to 0$. The singularity at $(0, 0)$ is highlighted in the graph of Fig. 3.2, a half cone with vertex at the origin.

Fig. 3.2
$f(x, y) = \sqrt{x^2 + y^2}$



Sometimes, to work with partial derivatives, it is useful to use a vector notation. Consider, for any index $i \in \{1, 2, \ldots, n\}$, the vector

$$e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$$

whose components are all zero except for the ith one that equals 1. The partial derivative $f_{x_i}(x)$, at a point x in the open subset $A \subseteq \mathbb{R}^n$, is the limit

$$f_{x_i}(x) = \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h} \tag{3.5}$$

(if such exists and is finite). Clearly this limit coincides with the limit in definition (3.4).

According to what we have said, the partial derivatives of a function $f : A \to \mathbb{R}$ are well defined whenever the limit of the difference quotient exists and is finite at some x in an *open* set $A \subseteq \mathbb{R}^n$. More generally, if D is not open, we may consider the partial derivative as limit of the difference quotient at a point x *interior* to $D \subseteq \mathbb{R}^n$. We cannot, *in general*, consider the difference quotient at boundary points, in analogy to what happens at the endpoints of an interval for functions of one variable, when one defines one-sided derivatives. For example, consider the *closed cone* D in $\mathbb{R}^2$ defined by (Fig. 3.3)

$$D = \left\{ (x, y) \in \mathbb{R}^2 : \frac{1}{2}|x| \leq |y| \leq 2|x| \right\}.$$

At the origin the difference quotient cannot be defined, since we cannot compute the values of a function f at $(0, 0)$ or $(h, 0)$ when $h \neq 0$ in order to define the partial derivative f_x, nor can we compute the values of f at $(0, 0)$ or $(0, k)$ for $k \neq 0$, to define f_y.

Therefore on the boundary of D the difference quotient is not used, and one rather proceeds as follows. Suppose D is a *domain* in $\mathbb{R}^n$, i.e. the closure of an open set

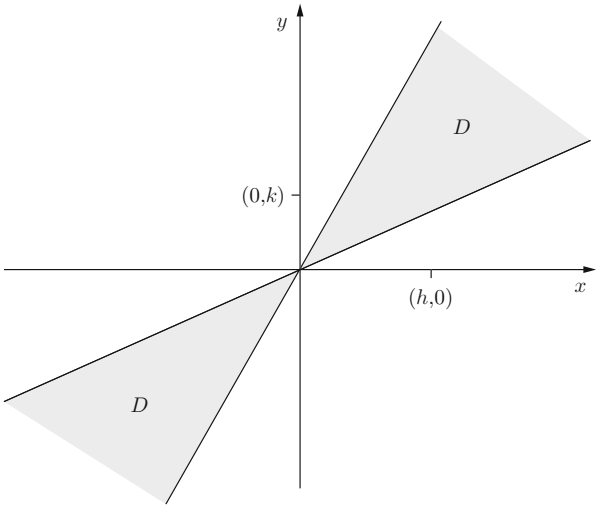


Fig. 3.3

(like the one in Fig. 3.3). Assume that for $i \in \{1, 2, \ldots, n\}$ the partial derivative $f_{x_i}(x)$ exists at any x interior to D, and that the derivative f_{x_i} is a continuous function on the interior of D. At any point x_0 of D (either interior or boundary point) it makes sense to consider the limit

$$\lim_{x \to x_0} f_{x_i}(x)\,. \tag{3.6}$$

If x_0 is *interior* to D, by the continuity of f_{x_i} the limit is equal to $f_{x_i}(x_0)$. In other words if limit (3.6) exists and is finite, we say it is, by definition, the partial derivative of f in x_i at the boundary point x_0.

Thus the partial derivative f_{x_i} at the boundary of a domain is defined as the *continuous extension* of $f_{x_i}(x)$ from the domain's interior, under the assumption that such extension does exist. When, instead, at a boundary point x_0 of D the limit does not exist, or it exists but is not finite, one says f does not admit partial derivative in x_i at x_0.

In case limit (3.6) exists at any x_0 on the boundary of D, the partial derivative $f_{x_i}(x)$ is well defined as *continuous extension* from the interior to the entire domain D, and it can be shown that this produces a continuous function on D.

3.4 Higher Derivatives. Schwarz's Theorem

Let $f(x) = f(x_1, \ldots, x_i, \ldots, x_n)$ be a function of n real variables that *admits partial derivatives* on an open set $A \subseteq \mathbb{R}^n$:

$$f_{x_1}(x),\ f_{x_2}(x),\ \ldots,\ f_{x_n}(x) \qquad \forall\, x \in A\,.$$

Every partial derivative $f_{x_i}(x)$, $i \in \{1, 2, \ldots, n\}$ is well defined on A. If, for some $i \in \{1, 2, \ldots, n\}$, $f_{x_i}(x)$ itself admits partial derivatives, its partial derivatives

$$\frac{\partial}{\partial x_j} f_{x_i}\,, \qquad \forall\, j = 1, 2, \ldots, n\,, \tag{3.7}$$

are called *second (partial) derivatives* of f, and are variously denoted by

$$\frac{\partial^2 f}{\partial x_i \partial x_j}\,,\ f_{x_i x_j}\,,\ f_{x_i x_j}(x)\,,\ D_{ij} f\,,\ D_{ij} f(x)\,,\ D_{x_i x_j} f\,.$$

All second derivatives of f, if they exist, are obtained as in (3.7), as the indices $j = 1, 2, \ldots, n$ and $i = 1, 2, \ldots, n$ vary. Thus we have a *matrix* of second derivatives, called *Hessian matrix* and denoted by $D^2 f$. The Hessian matrix is a square $n \times n$ matrix, and its entries are the second derivatives $f_{x_i x_j}$. In matrix notation

$$D^2 f = (f_{x_i x_j})\,, \quad \text{or} \quad D^2 f(x) = (f_{x_i x_j}(x))\,,$$

or explicitly

$$D^2 f = \begin{pmatrix} f_{x_1x_1} & f_{x_1x_2} & \cdots & f_{x_1x_n} \\ f_{x_2x_1} & f_{x_2x_2} & \cdots & f_{x_2x_n} \\ \cdots & \cdots & \cdots & \cdots \\ f_{x_nx_1} & f_{x_nx_2} & \cdots & f_{x_nx_n} \end{pmatrix} .$$

When all second derivatives $f_{x_ix_j}$, $i, j = 1, 2, \ldots, n$, of f exist at some x, then the Hessian matrix $D^2 f(x)$ is defined at x. If this is true for any x in an open set $A \subseteq \mathbb{R}^n$, one says that *f admits second partial derivatives on A*.

The diagonal elements $f_{x_1x_1}, f_{x_2x_2}, \ldots, f_{x_nx_n}$ in $D^2 f$ are sometimes called *pure second derivatives*, to distinguish them from the other entries called *mixed second derivatives*.

In general the mixed derivative $f_{x_ix_j}$ $(i \neq j)$ is different from the derivative $f_{x_jx_i}$, obtained from the former by interchanging the order of the variables; an example will be discussed in the section's second part. The following theorem (due to Schwarz) establishes natural sufficient conditions for the mixed derivatives $f_{x_ix_j}$ and $f_{x_jx_i}$ $(i \neq j)$ to be equal: *if $f_{x_ix_j}(x)$ and $f_{x_jx_i}(x)$ are continuous functions, they coincide*. Going back to the matrix, by Schwarz's theorem if the entries of $D^2 f(x)$ are continuous functions on some open set A, then $D^2 f(x)$ is a *symmetric* matrix for any $x \in A$.

Schwarz's Theorem *Let A be an open set in $\mathbb{R}^n$, $x_0 \in A$ and f a real function that admits second partial derivatives on A. If $f_{x_ix_j}$ and $f_{x_jx_i}$, with $i \neq j$, are continuous functions at x_0, then*

$$f_{x_ix_j}(x_0) = f_{x_jx_i}(x_0) .$$

Proof Suppose preliminarily $n = 2$, i.e. f is a function of two real variables denoted by (x, y). Pick $(x_0, y_0) \in A \subseteq \mathbb{R}^2$ and take (x, y) generic in A, with $x \neq x_0$ and $y \neq y_0$, as in Fig. 3.4.

Evaluating f at the points in Fig. 3.4 we define the following two functions of one real variable

$$F(x) = f(x, y) - f(x, y_0) \qquad \text{(with } y \text{ fixed)},$$
$$G(y) = f(x, y) - f(x_0, y) \qquad \text{(with } x \text{ fixed)}.$$

Applying the mean value theorem to $F(x)$ on the interval between x_0 and x, there exists a point x_1 such that

$$F(x) - F(x_0) = F'(x_1)(x - x_0) = [f_x(x_1, y) - f_x(x_1, y_0)](x - x_0) .$$

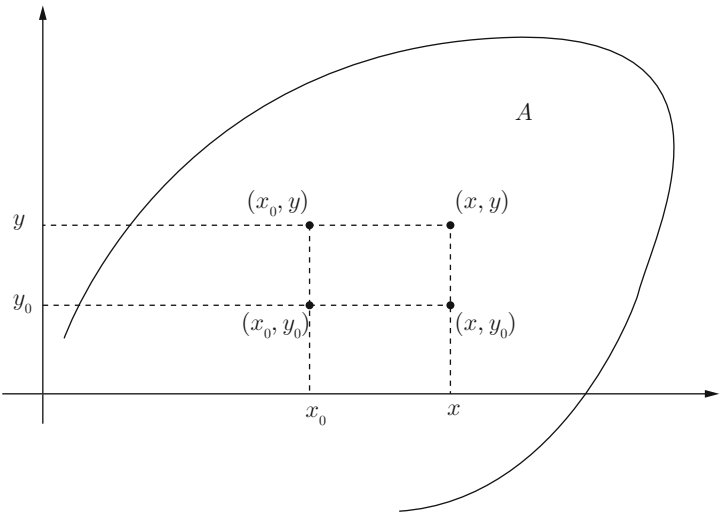


Fig. 3.4

Now apply the mean value theorem to the function $f_x(x_1, y)$ (in the variable y) on the interval between y_0, y, to obtain a point y_1 in that interval such that

$$\begin{aligned} F(x) - F(x_0) &= [f_x(x_1, y) - f_x(x_1, y_0)](x - x_0) = \\ &= f_{xy}(x_1, y_1)(x - x_0)(y - y_0)\,. \end{aligned} \tag{3.8}$$

Proceeding similarly for $G(y)$, we find y_2 (in the interval between y_0, y) and x_2 (in the interval between x_0, x) such that

$$\begin{aligned} G(y) - G(y_0) = G'(y_2)(y - y_0) &= [f_y(x, y_2) - f_y(x_0, y_2)](y - y_0) = \\ &= f_{yx}(x_2, y_2)(x - x_0)(y - y_0)\,. \end{aligned} \tag{3.9}$$

A direct computation shows

$$F(x) - F(x_0) = G(y) - G(y_0)\,, \tag{3.10}$$

because

$$\begin{aligned} F(x) - F(x_0) &= f(x, y) - f(x, y_0) - [f(x_0, y) - f(x_0, y_0)]\,, \\ G(y) - G(y_0) &= f(x, y) - f(x_0, y) - [f(x, y_0) - f(x_0, y_0)]\,. \end{aligned}$$

Comparing (3.8), (3.9) and (3.10) we arrive at the condition

$$f_{xy}(x_1, y_1)(x - x_0)(y - y_0) = f_{yx}(x_2, y_2)(x - x_0)(y - y_0)\,,$$

and since $x \neq x_0$ and $y \neq y_0$,

$$f_{xy}(x_1, y_1) = f_{yx}(x_2, y_2)\,.$$

The points (x_1, y_1), (x_2, y_2) lie inside the dashed rectangle of Fig. 3.4, so they satisfy (if $x > x_0$ and $y > y_0$)

$$x_0 < x_1\,, x_2 < x\,, \qquad y_0 < y_1\,, y_2 < y\,.$$

In the limit as $(x, y) \to (x_0, y_0)$, the points (x_1, y_1), (x_2, y_2) tend to (x_0, y_0). By the continuity, of f_{xy} and f_{yx} at (x_0, y_0) we obtain the claim

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)\,.$$

The general case $n \geq 2$ is immediate from $n = 2$ if we consider the function $g : \mathbb{R}^2 \to \mathbb{R}$, in the variables x_i, x_j, $i \neq j$, defined by

$$g(x_i, x_j) = f(x_1, \ldots, x_i, \ldots, x_j, \ldots, x_n)$$

as (x_i, x_j) varies, where the other coordinates are kept fixed, as if they were parameters. It is also clear, to conclude, that it suffices to assume that the second derivatives $f_{x_i x_j}$, $f_{x_j x_i}$ are continuous only in the pair (x_i, x_j).

Let us remark that the mixed derivatives' mere existence at a point does not guarantee that the order of differentiation can be interchanged. For instance, look at the function $f(x, y)$ of two variables

$$f(x, y) = \begin{cases} 0 & \text{if} \quad (x, y) = (0, 0) \\ \dfrac{x^3 y - x y^3}{x^2 + y^2} & \text{if} \quad (x, y) \neq (0, 0)\,. \end{cases}$$

The reader can check that $f(x, y)$ is continuous also at $(x, y) = (0, 0)$; even more, the first derivatives $f_x(x, y)$ and $f_y(x, y)$ are continuous at $(0, 0)$. To compute the mixed derivatives at $(0, 0)$ we start by noting that

$$f_x(x, y) = \frac{(3x^2 y - y^3)(x^2 + y^2) - 2x(x^3 y - x y^3)}{(x^2 + y^2)^2} = \frac{x^4 y + 4x^2 y^3 - y^5}{(x^2 + y^2)^2}$$

for any $(x, y) \neq (0, 0)$, whereas at $(0, 0)$:

$$f_x(0, 0) = \lim_{h \to 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \to 0} 0 = 0\,.$$

(continued)

Furthermore

$$f_{xy}(0,0) = \lim_{k\to 0} \frac{f_x(0,k) - f_x(0,0)}{k} = \lim_{k\to 0} \frac{-k^5}{k^5} = -1\,.$$

When we interchange x and y the function f changes sign (i.e. $f(x,y) = -f(y,x)$), so

$$f_{yx}(0,0) = -f_{xy}(0,0) = 1\,.$$

Hence at $(0,0)$ we have $f_{yx} \neq f_{xy}$. By Schwarz's theorem, the mixed derivatives are clearly both (by the aforementioned symmetry) discontinuous at $(0,0)$. (The reader should check this fact).

Schwarz's theorem and the other considerations of this section extend seamlessly to partial derivatives of order $k \in \mathbb{N}$. We write

$$\frac{\partial^k f}{\partial x_1^{k_1}\,\partial x_2^{k_2}\ldots\,\partial x_n^{k_n}}\,, \qquad \text{where} \quad k = k_1 + k_2 + \ldots + k_n\,, \tag{3.11}$$

to denote the kth partial derivative of $f = f(x_1, x_2, \ldots, x_n)$, k_1 times with respect to x_1, k_2 times in x_2 and so on. This notation clearly does not emphasise the *order* in which derivatives are computed. The symbol (3.11) is consistent in case the kth derivative is continuous, because if so, the result does not depend on the order of differentiation by Schwarz's theorem.

So for instance the kth derivatives of a function of two variables $f(x,y)$, if continuous, are the following $k+1$ functions

$$\frac{\partial^k f}{\partial x^k}\,, \quad \frac{\partial^k f}{\partial x^{k-1}\partial y}\,, \quad \frac{\partial^k f}{\partial x^{k-2}\partial y^2}\,, \quad \ldots \quad , \frac{\partial^k f}{\partial x \partial y^{k-1}}\,, \quad \frac{\partial^k f}{\partial y^k}\,.$$

3.5 Gradient. Differentiability

Suppose $f = f(x) = f(x_1, x_2, \ldots, x_n)$ *admits first partial derivatives* at a point x in an open set $A \subseteq \mathbb{R}^n$ on which f is defined. The *gradient* of f at x is by definition the vector Df whose components are the partial derivatives of f. The gradient is denoted by

$$Df(x)\,,\ \nabla f\,,\ \nabla f(x)\,,\ \operatorname{grad} f\,,\ \operatorname{grad} f(x)\,,$$

and for given x it is the vector in $\mathbb{R}^n$ of components

$$Df(x) = (f_{x_1}(x),\, f_{x_2}(x), \ldots, f_{x_n}(x))\,.$$

Saying that a function *admits (first) partial derivatives* at $x \in \mathbb{R}^n$ is the same as saying that f *has a gradient*. A stronger condition is requiring that at $x \in \mathbb{R}^n$ the function be *differentiable*, in the following sense.

Let A be an open set in $\mathbb{R}^n$. One says a function $f : A \to \mathbb{R}$ is *differentiable* at $x \in A$ if it admits partial derivatives at x (i.e., the gradient $Df(x)$ is defined) and

$$\lim_{h\to 0} \frac{f(x+h) - f(x) - (Df(x), h)}{|h|} = 0\,. \tag{3.12}$$

Note that in (3.12), h is a variable in $\mathbb{R}^n$, $|h|$ is its norm and $(Df(x), h)$ is the inner product of h and the gradient of f at x.

A function f is said *differentiable* on A if it is differentiable at any $x \in A$. Given x, the *linear* map defined on $\mathbb{R}^n$ by $h \to (Df(x), h)$ is called the *differential* of f at the point x, and is denoted by $df(x)$. Therefore $df(x)$ is the linear map (or *linear functional* – the word *functional* is employed here to denote a linear map from $\mathbb{R}^n$ to $\mathbb{R}$) in the variable $h \in \mathbb{R}^n$ defined by

$$df(x)(h) = (Df(x), h), \qquad \forall\, h \in \mathbb{R}^n\,. \tag{3.13}$$

At first sight the following definition of differentiability might seem more general: f is *differentiable* at x if there exists a *linear functional* $L : \mathbb{R}^n \to \mathbb{R}$ such that

$$\lim_{h\to 0} \frac{f(x+h) - f(x) - L(h)}{|h|} = 0\,. \tag{3.14}$$

Recall that every linear functional on $\mathbb{R}^n$ is represented by an inner product (see Sect. 2.5), i.e. for any linear map L there exists a vector $\ell \in \mathbb{R}^n$ such that

$$L(h) = (\ell, h) \qquad \forall\, h \in \mathbb{R}^n\,.$$

Then (3.14) may be rephrased as

$$\lim_{h\to 0} \frac{f(x+h) - f(x) - (\ell, h)}{|h|} = 0\,.$$

If ℓ has components $\ell = (\ell_i)$, for any $i \in \{1, 2, \dots, n\}$ we set $h = te_i = (0, \dots, 0, t, 0, \dots, 0)$ with $t \in \mathbb{R}$. Then

$$\lim_{t\to 0} \frac{f(x+te_i) - f(x) - t\ell_i}{|t|} = 0$$

or equivalently, computing the limits $t \to 0^+$, $t \to 0^-$ separately,

$$\lim_{t\to 0} \frac{f(x+te_i) - f(x) - t\ell_i}{t} = 0\,.$$

The immediate consequence is

$$\lim_{t\to 0} \frac{f(x+te_i)-f(x)}{t} = \ell_i\,,$$

so f admits partial derivative at x, which equals $f_{x_i} = \ell_i$. Repeating the argument for any $i \in \{1, 2, \ldots, n\}$ we obtain $Df = \ell$, and also

$$(Df(x), h) = (\ell, h) = L(h)\,, \qquad \forall\, h \in \mathbb{R}^n\,.$$

All in all, definitions (3.12) and (3.14) are equivalent.

A second remark: take (3.13) and drop the differential's dependence on x, so

$$df(h) = (Df, h)\,, \qquad \forall\, h \in \mathbb{R}^n\,. \tag{3.15}$$

We remind that the symbol dx_i, for any $i \in \{1, 2, \ldots, n\}$, denotes the linear functional $dx_i : \mathbb{R}^n \to \mathbb{R}$ such that $dx_i(h) = h_i$ for any $h \in \mathbb{R}^n$ (see (2.5)). In this notation, (3.15) reads

$$df = \sum_{i=1}^{n} f_{x_i} dx_i\,, \tag{3.16}$$

which should be understood as an equality between the linear functional df and the linear combination of the linear functionals $dx_1, dx_2, \ldots, dx_n$ on the right.

Now, recall that a function $f(x)$ is "*little o*" of $g(x)$ as $x \to x_0$, expressed by $f(x) = o(g(x))$, if $g(x)$ is an infinitesimal function as $x \to x_0$ and the ratio $f(x)/g(x)$ tends to zero. Then a function $f(x)$, defined on an open set A, is differentiable at $x \in A$ if it admits partial derivatives at x and

$$f(x+h) = f(x) + (Df(x), h) + o(|h|)\,, \qquad as \qquad h \to 0\,. \tag{3.17}$$

Especially in view of the application to physical sciences, it is useful to observe that, using (3.13), we may rephrase condition (3.17) as follows

$$\triangle f = f(x+\triangle x) - f(x) = df(x)(\triangle x) + o(|\triangle x|)\,, \qquad as \qquad \triangle x \to 0\,.$$

This means that when we pass from x to $x + \triangle x$, the increment of the function $\triangle f$ differs from the differential df by an infinitesimal quantity of order higher than the increment $\triangle x$. In the notation of (3.16), the above formula is often expressed as

$$\triangle f \simeq df = \sum_{i=1}^{n} f_{x_i} dx_i$$

in the applications.

Geometrically speaking, the differentiability at a point is related to the existence of the *tangent plane* to the function's graph at the point. Namely, if f is differentiable at $x_0 \in A$, putting $x = x_0$ and $h = x - x_0$ in (2.8) gives

$$f(x) = f(x_0) + (Df(x_0), x - x_0) + o(|x - x_0|)$$

and the *linear* function defined on $\mathbb{R}^n$ by

$$x \in \mathbb{R}^n \to f(x_0) + (Df(x_0), x - x_0) \tag{3.18}$$

approximates $f(x)$ at $x = x_0$ up to infinitesimals of order higher than the distance $|x - x_0|$ to the contact point. The graph of the linear map (3.18) (to be accurate, this function is *affine*) is the *tangent plane* to the graph of $f(x)$ at x_0, where f is differentiable.

One of the reasons for considering differentiability is that the latter forces continuity (generalising the similar condition for one real variable). If we suppose that f is *differentiable at* $x \in A$, then (3.17) immediately implies that f *is continuous at that point*, because

$$|(Df(x), h)| \le |Df(x)| \cdot |h| \to 0 \qquad \text{as} \qquad h \to 0$$

by the Cauchy-Schwarz inequality, and so

$$\lim_{h\to 0} f(x + h) = \lim_{h\to 0} [f(x) + (Df(x), h) + o(|h|)] = f(x)\,.$$

Observe that while continuity, is a consequence of differentiability, it does not follow from, the existence of the first partial derivatives, as illustrated by this example: the function

$$f(x, y) = \begin{cases} 0 & \text{if} \quad (x, y) = (0, 0) \\ \dfrac{xy}{x^2 + y^2} & \text{if} \quad (x, y) \neq (0, 0)\,, \end{cases} \tag{3.19}$$

defined on $\mathbb{R}^2$ is not continuous at the origin $(0, 0)$: for instance, for $x = y$

$$\lim_{x\to 0} f(x, x) = \lim_{x\to 0} \frac{x^2}{x^2 + x^2} = \frac{1}{2},$$

which differs from the value $f(0, 0) = 0$ (more generally $f(x, mx) = 1/(1 + m^2)$ is constant along the straight line $y = mx$ through the origin, hence $f(x, y)$ cannot admit limit as $(x, y) \to (0, 0)$). On the other hand both partial derivatives exist at $(0, 0)$ (and at any other point in $\mathbb{R}^2$), because f is null along the axes, where $x = 0$ or $y = 0$, so

$$f_x(0, 0) = 0\,, \qquad f_y(0, 0) = 0\,.$$

(continued)

Obviously the function f in (3.19) is not differentiable at $(0, 0)$, since the limit as $(h, k) \to (0, 0)$ of

$$\frac{f(h,k)}{\sqrt{h^2+k^2}} = \frac{hk}{(h^2+k^2)^{3/2}}$$

does not exist. The non-differentiability of f at the origin can also be deduced from the fact that f is not continuous at that point. In Fig. 3.5 we have represented the graph of $f(x, y)$ defined in (3.19), as (x, y) varies in the square $[-1, 1] \times [-1, 1]$. The picture clearly shows the singularity in the middle, at the axes' crossing, and that the function is constant along the line $y = x$, for instance.

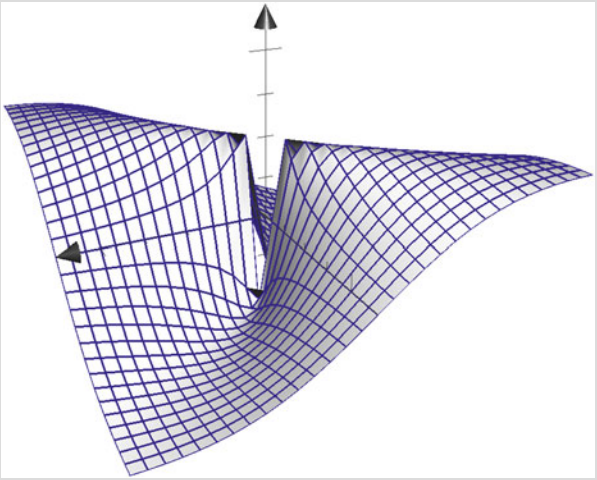


Fig. 3.5 The function $f(x, y)$ defined in (3.19)

The first derivatives' continuity, is a sufficient condition for differentiability. We have in fact the following important result.

Theorem (Differentiability Criterion) *Suppose $f : A \to \mathbb{R}$ admits continuous partial derivatives $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ at a point $x \in A$. Then f is differentiable at x.*

Proof For simplicity we restrict to the case $n = 2$ and examine a function of two variables $f = f(x, y)$, $(x, y) \in A \subseteq \mathbb{R}^2$. Consider the quantity

$$f(x+h, y+k) - f(x, y) = [f(x+h, y+k) - f(x, y+k)] + \\ + [f(x, y+k) - f(x, y)] .$$

Applying the mean value theorem twice, there exist points x_1 (between x and $x+h$) and y_1 (between y and $y+k$) such that

$$f(x+h, y+k) - f(x, y) = f_x(x_1, y+k) \cdot h + f_y(x, y_1) \cdot k\,.$$

Using this, let us estimate the quotient appearing in definition (3.12):

$$\left| \frac{f(x+h, y+k) - f(x, y) - f_x(x, y) \cdot h - f_y(x, y) \cdot k}{\sqrt{h^2+k^2}} \right| \leq \tag{3.20}$$

$$\leq |f_x(x_1, y+k) - f_x(x, y)| \cdot \frac{|h|}{\sqrt{h^2+k^2}} + |f_y(x, y_1) - f_y(x, y)| \cdot \frac{|k|}{\sqrt{h^2+k^2}} \leq$$

$$\leq |f_x(x_1, y+k) - f_x(x, y)| + |f_y(x, y_1) - f_y(x, y)|.$$

When $(h, k) \to (0, 0)$, x_1 tends to x and y_1 to y, i.e. $(x_1, y_1) \to (x, y)$. As the partial derivatives f_x, f_y are continuous at (x, y), (3.20) implies

$$\lim_{(h,k)\to(0,0)} \frac{f(x+h, y+k) - f(x, y) - f_x(x, y) \cdot h - f_y(x, y) \cdot k}{\sqrt{h^2+k^2}} = 0\,,$$

and so f is differentiable at (x, y).

A continuous function on an open set $A \subseteq \mathbb{R}^n$ is said to be *of class* C^0 *on* A, or simply a C^0 function, written $f \in C^0(A)$. If f admits continuous first partial derivatives on A, f is said to be *of class* C^1 *on* A, written $f \in C^1(A)$.

In general, if f admits continuous partial derivatives of order $k \in \mathbb{N}$ on A we call it *of class* C^k *on* A, written $f \in C^k(A)$. With this notation, the differentiability criterion, and the implication 'differentiable $\Rightarrow$ continuous', now read:

$$f \in C^1(A) \quad \Rightarrow \quad f \quad \textit{differentiable on} \quad A \quad \Rightarrow \quad f \in C^0(A)\,.$$

Similarly, applying the differentiability criterion to kth order partial derivatives produces

$$f \in C^k(A) \quad \Rightarrow \quad f \in C^{k-1}(A)\,, \qquad \forall\, k \in \mathbb{N}\,.$$

3.6 Composite Functions

Let $x_1(t), x_2(t), \ldots, x_n(t)$ be n real functions defined on an interval $I \subset \mathbb{R}$. Call $x : I \to \mathbb{R}^n$ the map from I to $\mathbb{R}^n$ whose components are the $x_i(t)$, $i = 1, 2, \ldots, n$. Then $x(t)$ is the vector in $\mathbb{R}^n$, dependent on the variable $t \in I$, whose components

are

$$x(t) = (x_1(t), x_2(t), \dots, x_n(t)) , \qquad \forall\, t \in I .$$

In the terminology of Chapter 6 we say the map $x : I \to \mathbb{R}^n$ is a *curve* in $\mathbb{R}^n$.

Let A be an open set in $\mathbb{R}^n$ that contains the range $x(I)$ of the map $x : I \to \mathbb{R}^n$, i.e. suppose $x(t) \in A$ for any $t \in I$. If $f : A \to \mathbb{R}$ is a real function of n real variables defined on A, the composite function

$$F(t) = f(x(t)) = f \circ x(t) , \qquad \forall\, t \in I$$

is well defined on I. The composite function $F : I \to \mathbb{R}$ is a real function of one real variable. The next result establishes when $F = f \circ x$ is differentiable, and gives a formula for differentiating it. As usual, if $t \in I$ is an endpoint of the interval I, we consider the right derivative for the first endpoint or the left derivative for the second endpoint.

Theorem (Chain Rule) *Suppose that the vector $x(t)$ is differentiable at $t \in I$ (i.e. the n component functions $x_1(t), x_2(t), \dots, x_n(t)$ admit derivatives at $t \in I$) and that f is differentiable at the point $x(t)$. Then the composite function $F(t) = f(x(t))$ is differentiable at $t \in I$, with derivative*

$$F'(t) = (Df(x(t)), x'(t)) = \sum_{i=1}^{n} f_{x_i}(x(t)) \cdot x_i'(t) . \tag{3.21}$$

Proof Start from the differentiability hypothesis for f, which we express as follows

$$f(y) = f(x) + (Df(x), y - x) + o(|y - x|) , \quad \text{as} \quad y \to x .$$

Put $x = x(t)$ and $y = x(t + h)$ (with $t, t + h \in I$), to obtain

$$f(x(t + h)) = f(x(t)) + (Df(x(t)), x(t + h) - x(t)) + o(|x(t + h) - x(t)|) .$$

Then the difference quotient of the composite $F(t) = f(x(t))$ reads

$$\begin{aligned} \frac{F(t + h) - F(t)}{h} &= \frac{f(x(t + h)) - f(x(t))}{h} = \\ &= \left(Df(x(t)), \frac{x(t + h) - x(t)}{h} \right) + \frac{o(|x(t + h) - x(t)|)}{h} . \end{aligned} \tag{3.22}$$

Let us now compute separately the limits in the right-most side of (3.22) when $h \to 0$. Given the similarity of the terms, we shall only compute the second one. Suppose $|x(t+h)-x(t)| \neq 0$ when $h \neq 0$. If that is not the case we proceed exactly

as in the proof of the chain rule in *one* real variable. We have

$$\begin{aligned} &\lim_{h\to 0}\left|\frac{o(|x(t+h)-x(t)|)}{h}\right| = \\ &= \lim_{h\to 0}\frac{|o(|x(t+h)-x(t)|)|}{|x(t+h)-x(t)|}\cdot\lim_{h\to 0}\frac{|x(t+h)-x(t)|}{|h|} = 0\,. \end{aligned} \tag{3.23}$$

The limit in (3.23) vanishes since it is the product of a quantity that tends to zero (by definition of "*little o*") and a quantity with finite limit:

$$\begin{aligned} \lim_{h\to 0}\frac{|x(t+h)-x(t)|}{|h|} &= \lim_{h\to 0}\left(\sum_{i=1}^{n}\left(\frac{x_i(t+h)-x_i(t)}{h}\right)^2\right)^{1/2} = \\ &= \left(\sum_{i=1}^{n}(x_i'(t))^2\right)^{1/2} = |x'(t)|\,. \end{aligned}$$

As $h\to 0$, from (3.22) and (3.23) we obtain

$$\lim_{h\to 0}\frac{F(t+h)-F(t)}{h} = (Df(x(t)),x'(t))\,.$$

Consider for example the function of two variables

$$f(x,y) = x^2 - y^2\,.$$

Compose it with the function from $\mathbb{R}$ to $\mathbb{R}^2$ defined by

$$x(t) = -t^2\,, \qquad y(t) = t\,. \tag{3.24}$$

This is a curve in the xy-plane: precisely, since $x(t) = -(y(t))^2$, it is a parabola of equation $x = -y^2$. Using the composite $F(t) = f(x(t),y(t))$ we then find the space curve of equations

$$\begin{cases} x = x(t) \\ y = y(t) \\ z = F(t) = f(x(t),y(t)). \end{cases}$$

The image of the curve lies on the graph for the surface of equation $z = f(x,y)$ (Fig. 3.6).

To find the local maximum and minimum points of f lying on curve (3.24), we can first solve the equation $F'(t) = 0$. By the chain rule

$$\begin{aligned} F'(t) = f_x\cdot x' + f_y\cdot y' = 2x\cdot x' - 2y\cdot y' = \\ = -2t^2\cdot(-2t) - 2t\cdot 1 = 2t(2t^2-1)\,, \end{aligned} \tag{3.25}$$

(continued)

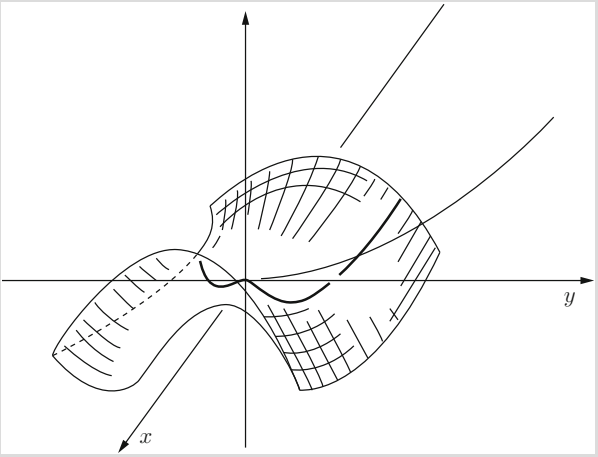


Fig. 3.6

which vanishes at $t = 0$ and $t = \pm\sqrt{2}/2$. At these points, the second derivative's sign

$$F''(0) = -2 < 0\,, \qquad F''\left(\pm\frac{\sqrt{2}}{2}\right) = 4 > 0$$

tells that $t = 0$ is a local maximum point, while $t = \pm\sqrt{2}/2$ give two (absolute) minima.

The reader can check the result obtained in (3.25) by differentiating $F(t) = f(x(t), y(t)) = (-t^2)^2 - t^2$ directly. More generally, the reader can verify by direct inspection, using the differentiation rules in one variable, that, formula (3.21) holds, in the particular case in which $f(x) = f(x_1, x_2, \ldots, x_n) = \sum_{i=1}^{n} f_i(x_i)$ is a sum of functions f_i, each depending on the variable x_i only.

As the partial derivative of a function of several variables is calculated with respect to one of the real variables and considering the others as fixed parameters, there is a similar chain rule also when the components themselves depend upon several real variables. Precisely, we have the following result, a direct consequence of the chain rule shown above.

Chain Rule *Indicate by $t = (t_1, t_2, \ldots, t_k)$ a variable in $\mathbb{R}^k$ and let $x_1(t)$, $x_2(t)$, $\ldots$, $x_n(t)$ be n functions defined on an open set $B \subset \mathbb{R}^k$ and admitting partial derivatives in t_j, for some $j \in \{1, 2, \ldots, k\}$, at a point $t \in B$.*

Let A be an open subset of $\mathbb{R}^n$ containing the range $x(B)$ of the map $x : B \to \mathbb{R}^n$, and $f : A \to \mathbb{R}$ a differentiable function at the point $x(t)$. Then the composite

function $F(t) = f(x(t)) = f \circ x(t)$ is defined on B, it admits partial derivatives in t_j at t, and the partial derivatives equal

$$\frac{\partial F}{\partial t_j}(t) = \left(Df(x(t)), \frac{\partial x}{\partial t_j}(t) \right) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x(t)) \cdot \frac{\partial x_i}{\partial t_j}(t) \,. \tag{3.26}$$

If, furthermore, f and $x_1(t), x_2(t), \ldots, x_n(t)$ are C^1 functions, then the composite $F = f \circ x$ is of class C^1 as well, and hence differentiable on A.

Due to its importance, we stress once more that, from formula (3.26), if the functions $x : B \subseteq \mathbb{R}^k \to \mathbb{R}^n$ and $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ are C^1 (we say $x : B \to \mathbb{R}^n$ is of class C^1 if all its components x_i, $1 = 1, 2, \ldots, n$, are of class C^1), then the composite function $f \circ x$, too, is of class C^1. In such a case, by the differentiability criterion $f \circ x$ is differentiable and the gradient $DF = D(f \circ x)$ has components $\partial F / \partial t_j$, $j \in \{1, 2, \ldots, k\}$, expressed by formula (3.26).

3.7 Directional Derivatives

We call a unit vector in $\mathbb{R}^n$ a *direction*.

Let A be an open set in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ a function defined on A. Fix $x \in A$. Given a direction λ in $\mathbb{R}^n$ (so $\lambda \in \mathbb{R}^n$, $|\lambda| = 1$), the *directional derivative* of f along λ at the point x is

$$\lim_{h \to 0} \frac{f(x + h\lambda) - f(x)}{h} \,,$$

provided such limit exists and is finite. The directional derivative is denoted by

$$\frac{\partial f}{\partial \lambda}, \quad \frac{\partial f}{\partial \lambda}(x), \quad D_\lambda, \quad D_\lambda f, \quad D_\lambda f(x) \,.$$

In particular, suppose λ is the direction of a coordinate axis, i.e. for some index $i \in \{1, 2, \ldots, n\}$

$$\lambda = e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$$

with all components zero except the ith one, equal to 1. Then the directional derivative $\partial f / \partial \lambda$ coincides with the partial derivative $\partial f / \partial x_i = f_{x_i}$ (compare to formula (3.5)).

The following criterion is useful for computing directional derivatives.

Directional Derivative of a Differentiable Function *Let A be an open set in $\mathbb{R}^n$ and $x \in A$ a point. If f is differentiable at x, it admits at x directional derivative*

along any direction $\lambda \in \mathbb{R}^n$*, and the latter equals*

$$\frac{\partial f}{\partial \lambda}(x) = (Df(x), \lambda) = \sum_{i=1}^{n} f_{x_i}(x) \cdot \lambda_i \,. \tag{3.27}$$

Proof The directional derivative $\partial f/\partial\lambda$ is the derivative in t, evaluated at $t = 0$, of $t \to f(x + t\lambda)$. By the chain rule (previous section)

$$\frac{\partial f}{\partial \lambda}(x) = \left[\frac{d}{dt} f(x + t\lambda)\right]_{t=0} = (Df(x), \lambda) = \sum_{i=1}^{n} f_{x_i}(x) \cdot \lambda_i \,.$$

If a function admits all derivatives at some point but is *not* differentiable there, the representation formula (3.27) might not be valid. Consider for example the function of two variables

$$f(x, y) = \frac{x^2 y}{x^2 + y^2} \quad \text{if} \quad (x, y) \neq (0, 0), \qquad \text{and} \qquad f(0, 0) = 0 \,. \tag{3.28}$$

The directional derivative at $(0, 0)$ along $\lambda = (\lambda_1, \lambda_2)$ is the limit as $h \to 0$ of the ratio

$$\frac{f(0 + h\lambda) - f(0)}{h} = \frac{\lambda_1^2 \lambda_2}{\lambda_1^2 + \lambda_2^2} \,, \tag{3.29}$$

so the directional derivative at $(0, 0)$ exists for any $\lambda = (\lambda_1, \lambda_2)$ and equals the value (3.29). At $(0, 0)$, in particular, the partial derivatives (the directional derivatives along $(\lambda_1, \lambda_2) = (1, 0)$ and $(\lambda_1, \lambda_2) = (0, 1)$) are both zero. In this case formula (3.27), i.e.

$$\frac{\partial f}{\partial \lambda} = f_x \lambda_1 + f_y \lambda_2 \,,$$

does not hold, since the right-hand side is identically zero for any $\lambda = (\lambda_1, \lambda_2)$. The reader may check that (3.28) is not differentiable at $(0, 0)$.

Let f be differentiable at some point x in an open set A in $\mathbb{R}^n$. Fix x and consider the directional derivative at x as the direction $\lambda \in \mathbb{R}^n$ varies ($|\lambda| = 1$). Formula (3.27) expresses the directional derivative as the inner product of the gradient Df and the direction λ:

$$\frac{\partial f}{\partial \lambda} = (Df, \lambda) \,.$$

By the Cauchy-Schwarz inequality in $\mathbb{R}^n$ we have

$$\left|\frac{\partial f}{\partial \lambda}\right| \leq |Df| \cdot |\lambda| = |Df| \,,$$

with equality occurring if and only if the vectors Df and λ are parallel. Hence in general

$$-|Df| \le \frac{\partial f}{\partial \lambda} \le |Df|\,.$$

Moreover, if $Df(x) \neq 0$,

$$\frac{\partial f}{\partial \lambda} = |Df|$$

if and only if $\lambda = Df/|Df|$ (λ is the normalised gradient, and has *the same orientation* as Df). On the other hand

$$\frac{\partial f}{\partial \lambda} = -|Df|$$

if and only if λ *is the gradient's direction but with opposite orientation* ($\lambda = -Df/|Df|$).

Therefore the gradient Df, if non-zero, provides the oriented direction along which the directional derivative is largest, while $-Df$ defines the oriented direction along which the directional derivative is smallest. In a nutshell, *the gradient gives the direction of maximum increase* (if non-zero).

Fix distinct points x_1, x_2 in $\mathbb{R}^n$ and consider the function

$$f(x) = c|x - x_1|^2 \cdot |x - x_2|^2\,, \qquad (c > 0)\,.$$

It clearly reaches a minimum at $x = x_1$ and at $x = x_2$. Let us check this by examining the gradient lines, i.e. the curves of steepest descent, in the special case $n = 2$. To simplify notations let us fix the two points along one axis, with the origin as midpoint. Consider for instance the function of two variables

$$f(x, y) = \frac{1}{4}\,[(x-1)^2 + y^2] \cdot [(x+1)^2 + y^2]\,.$$

whose gradient $Df = (f_x, f_y)$ is

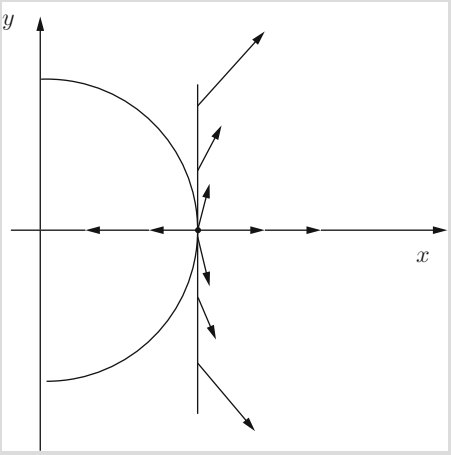
$$f_x = x(x^2 + y^2 - 1)\,, \qquad f_y = y(x^2 + y^2 + 1)\,.$$

The partial derivative f_x vanishes at $x = 0$ (y-axis) and for $x^2 + y^2 = 1$ (unit circle centred at the origin). The partial derivative f_y vanishes only for $y = 0$ (x-axis). Hence the gradient is zero on the intersection of these sets, i.e. at the three points $(-1, 0)$, $(0, 0)$, $(1, 0)$.

For example, on a neighbourhood of $(1, 0)$ the picture is that of Fig. 3.7: at $(1, 0)$ the gradient is zero, along the vertical line $x = 1$ we have $f_x \ge 0$

(continued)

Fig. 3.7



and f_y is positive for $y > 0$ and negative for $y < 0$. Along this line the gradient then grows larger as we move away from the point. Similarly, along the x-axis the gradient vector is horizontal and becomes larger as we move away from $(1, 0)$. The point $(1, 0)$ is therefore a minimum point for the function, because on a neighbourhood the gradient points away in every direction.

Near $(-1, 0)$ the situation is similar. On the other hand, around $(0, 0)$ the picture is that of Fig. 3.8: along the y-axis the gradient is vertical and always points away from the origin. Along the x-axis the gradient is horizontal and always points towards the origin.

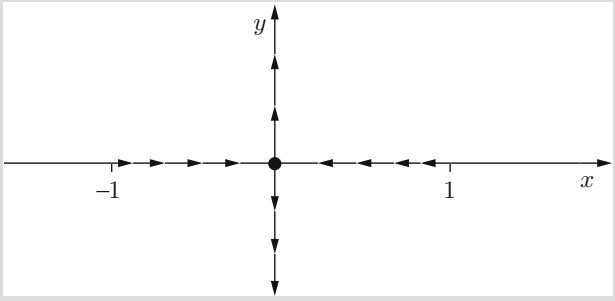


Fig. 3.8

This means that the function $f(x, 0)$, in the variable x only, is largest at $x = 0$ (the derivative f_x is positive for $x < 0$ and negative for $x > 0$). Instead, the function $f(0, y)$ is smallest at $y = 0$ (f_y is negative before $y = 0$ and positive afterwards). Therefore $(0, 0)$ is neither a maximum nor a minimum point for $f(x, y)$.

In Fig. 3.9 we plotted the gradient of $f(x, y)$ for other values (x, y). Note in particular how the gradient is vertical along the unit circle. For reference, the gradient along $y = x$ is also shown.

(continued)

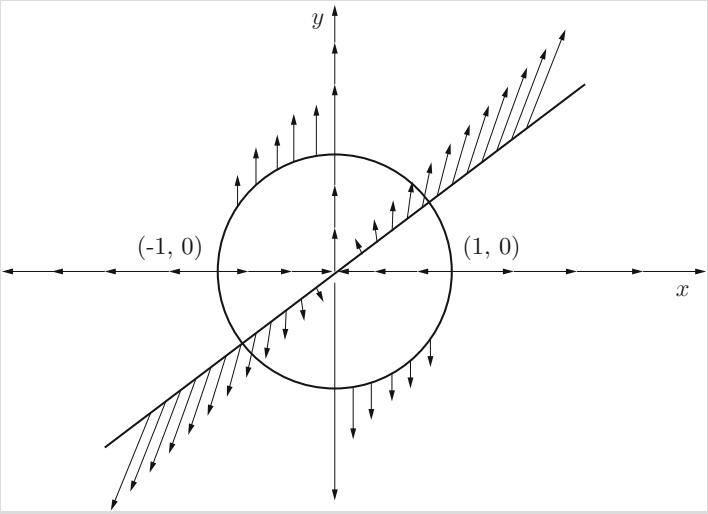
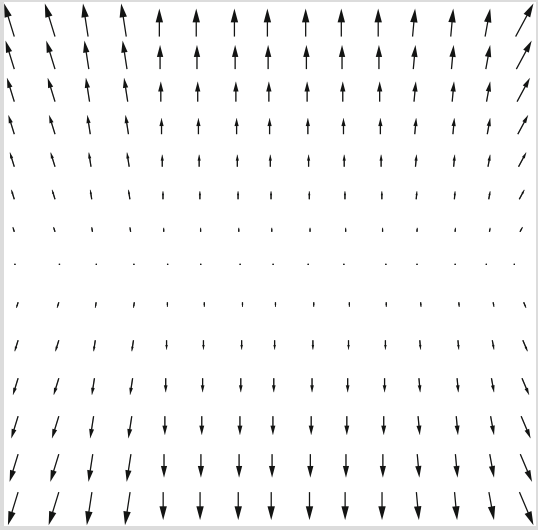


Fig. 3.9

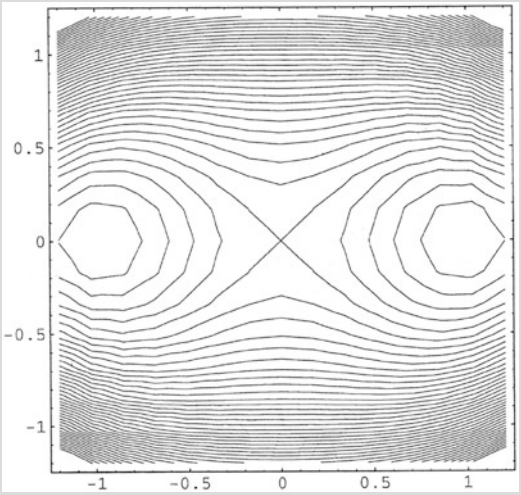
Finally, Fig. 3.10 shows the gradient field of the function $f(x, y)$, while Fig. 3.11 is a plot of the level curves wrapping around the minimum points $(\pm 1, 0)$.

Fig. 3.10



(continued)

Fig. 3.11



3.8 Functions with Vanishing Gradient on Connected Sets

Recall that an open set $A \subseteq \mathbb{R}^n$ is *connected* if

$$A_1, A_2 \text{ open subsets of } \mathbb{R}^n, \quad A_1 \cap A_2 = \emptyset, \quad A_1 \cup A_2 = A$$

force one of A_1, A_2 to be empty.

The following result is about functions with null partial derivatives on a connected subset of $\mathbb{R}^n$, and generalises the analogous simple fact in one variable, where a derivative is zero on some interval.

Functions with Zero Gradient *If a function f has zero gradient at all points in a connected open set $A \subseteq \mathbb{R}^n$, then f is constant on A.*

Proof By assumption all partial derivatives are zero on A, so the derivatives are continuous and f is differentiable on A. Fix $x_0 \in A$ and define the set

$$A_1 = \{x \in A : \quad f(x) = f(x_0)\}.$$

Clearly $A = A_1 \cup A_2$, where

$$A_2 = \{x \in A : \quad f(x) \neq f(x_0)\}.$$

Being differentiable, f is continuous on A, so the set A_2 is open. Let us show A_1 is open as well. Take $x_1 \in A_1$ (so $f(x_1) = f(x_0)$) and let I_δ be an open ball, centred at

x_1 with radius δ, contained in A. We claim that $I_\delta \subseteq A_1$, i.e. $f(x) = f(x_0) = f(x_1)$ for any $x \in I_\delta$.

If $x = x_1$ there is nothing to prove. If $x \neq x_1$ and $x \in I_\delta$ the one-variable function $\varphi : [0, 1] \to \mathbb{R}$ defined by

$$\varphi(t) = f(x_1 + t(x - x_1)), \qquad \forall\, t \in [0, 1],$$

assumes values $f(x_1)$ (at $t = 0$) and $f(x)$ (at $t = 1$). Its derivative, computed with the chain rule, is

$$\varphi'(t) = (Df(x_1 + t(x - x_1)),\ x - x_1) = 0, \qquad \forall\, t \in [0, 1].$$

The function $\varphi : [0, 1] \to \mathbb{R}$ has zero derivative on $[0, 1]$ and hence is constant. In particular, $\varphi(0) = \varphi(1)$, i.e. $f(x) = f(x_1)$. This shows A_1 is open.

So now $A = A_1 \cup A_2$, $A_1 \cap A_2 = \emptyset$ with A_1, A_2 open in $\mathbb{R}^n$ and $A_1 \neq \emptyset$. As A is connected in $\mathbb{R}^n$, we have $A_2 = \emptyset$ and so $A_1 = A$, i.e.

$$f(x) = f(x_0) \qquad \forall\, x \in A.$$

Let us apply the above theorem to prove the identity

$$\operatorname{arctg} \frac{y}{x} + \operatorname{arctg} \frac{x}{y} = \frac{\pi}{2}, \qquad \forall\, x > 0,\ \forall\, y > 0. \tag{3.30}$$

The reader should check that the function of two variables

$$f(x, y) = \operatorname{arctg} \frac{y}{x} + \operatorname{arctg} \frac{x}{y},$$

has vanishing partial derivatives away from the axes. As $A = \{(x, y) \in \mathbb{R}^2 : x > 0,\ y > 0\}$ is open and connected in $\mathbb{R}^2$,

$$f(x, y) = \textit{constant} \qquad \forall\, (x, y) \in A.$$

The value of the constant can be determined by evaluating $f(x, y)$ at any point of A, say $(x, y) = (t, t)$, $t > 0$. Then $f(t, t) = \pi/2$, and (3.30) is proved.

With the same technique the reader may prove the identity

$$\operatorname{arctg} \frac{y}{x} + \operatorname{arctg} \frac{x}{y} = -\frac{\pi}{2}, \qquad \forall\, x > 0,\ \forall\, y < 0.$$

We now propose a second proof of the previous theorem. The argument is probably simpler, but it shifts the crucial point to a result concerning open connected subsets of $\mathbb{R}^n$, which anyhow has independent interest.

Start by recalling that a *line segment* $[x_1, x_2]$ in $\mathbb{R}^n$ of endpoints $x_1 \neq x_2$ (also known as an *interval* in $\mathbb{R}^n$) is the set of points $x(t) \in \mathbb{R}^n$ such that

$$x(t) = x_1 + t(x_2 - x_1)\,, \qquad \forall\, t \in [0,1]\,.$$

Consider points $x_1, x_2, \ldots, x_k$, k $(k \geq 2)$ in $\mathbb{R}^n$, with $x_i \neq x_{i+1}$ for any $i = 1, 2, \ldots, k-1$. A *polygonal path* in $\mathbb{R}^n$ of vertices $x_1, x_2, \ldots, x_k$ is the union of the line segments $[x_i, x_{i+1}]$, $i = 1, 2, \ldots, k-1$. The vertices x_1 and x_k are the path's *endpoints*.

In Sect. 2.11 we proved the following fact on connected open set in $\mathbb{R}^n$: *if $A \subseteq \mathbb{R}^n$ is connected and open, any two points of A are the endpoints of a polygonal path entirely contained in A.*

Using this characterisation we can prove in a different way the result saying that *if a function's gradient vanishes at every point of a connected open subset of $\mathbb{R}^n$, the function is constant on the subset.*

Suppose f is a function with zero gradient on an open, connected set $A \subset \mathbb{R}^n$. We fix $x_1 \in A$ and shall prove $f(x) = f(x_1)$ for any $x \in A$. If $x \neq x_1$ (otherwise there is nothing to prove), by the aforementioned result on connected open subsets there exists a polygonal path from x_1 to x contained in A. Call $x_1, x_2, \ldots, x_k = x$ the path's vertices. It suffices to show that f is constant on each interval $[x_i, x_{i+1}]$, $i = 1, 2, \ldots, k-1$, because at the intervals' endpoints we have

$$f(x_1) = f(x_2), \quad f(x_2) = f(x_3), \quad \ldots \quad , f(x_{k-1}) = f(x_k)\,,$$

and so $f(x_1) = f(x_k)$, i.e. $f(x_1) = f(x)$. So we may just prove that f is constant on every interval contained in A.

Let $[x_1, x_2]$ be an interval in $\mathbb{R}^n$ contained in A, and consider the one-variable function $\varphi : [0,1] \to \mathbb{R}$ defined by

$$\varphi(t) = f(x_1 + t(x - x_1))\,, \qquad t \in [0,1]\,.$$

Using the chain rule, its derivative

$$\varphi'(t) = (Df(x_1 + t(x - x_1)),\, x - x_1)\,, \qquad \forall\, t \in [0,1]\,,$$

vanishes on $[0,1]$ because $Df = 0$ on A. Then $\varphi : [0,1] \to \mathbb{R}$ is constant on $[0,1]$, i.e. f is constant on the line segment $[x_1, x_2]$.

3.9 Homogeneous Functions

A set $A \subseteq \mathbb{R}^n$ is called a *cone* in $\mathbb{R}^n$ if it satisfies

$$x \in A \quad \Rightarrow \quad tx \in A\,, \quad \forall\, t > 0\,.$$

For example, $\mathbb{R}^n$ itself is a cone, as is $\mathbb{R}^n - \{0\}$. A hyperplane through the origin is a cone. In $\mathbb{R}^2$ the first quadrant (either closed or open) is a cone, and so is the union of the first and third quadrants. Also $\mathbb{R}^n$ minus a ray emanating from the origin is a cone.

Let A be a cone in $\mathbb{R}^n$ and α a real number. A function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is said *homogeneous of degree* α on A if

$$f(tx) = t^\alpha f(x) \qquad \forall\, x \in A\,, \quad \forall\, t > 0\,.$$

(One might say *positively homogeneous* to emphasise that the scaling factor is positive).

For example, the norm $f(x) = |x|$ is a homogeneous function of degree 1 on $\mathbb{R}^n$. More generally, $f(x) = |x|^p$, $p \in \mathbb{R}$, is homogeneous of degree p on $\mathbb{R}^n$ (on $\mathbb{R}^n - \{0\}$ if $p < 0$).

If $A = (a_{ij})$ is an $n \times n$ matrix, the associated quadratic form $f(x) = \sum_{i,j=1}^{n} a_{ij} x_i x_j$ is homogeneous of degree 2 on $\mathbb{R}^n$. The functions

$$f(x, y) = \frac{x}{y}\,, \quad g(x, y) = \sin\frac{x}{y}\,, \quad h(x, y) = e^{x/y}\,,$$

are homogeneous of degree zero on the cone $A = \{(x, y) \in \mathbb{R}^2 : y \neq 0\}$. The function

$$f(x, y) = x^3 y \cos\frac{xy}{x^2 + y^2}$$

is homogeneous of degree 4 on $\mathbb{R}^2 - \{(0, 0)\}$.

The following theorem by Euler characterises homogeneous functions of degree α.

Euler's Theorem for Homogeneous Functions *Let $f : A \to \mathbb{R}$ be a differentiable function on an open cone $A \subseteq \mathbb{R}^n$. Then f is homogeneous of degree α on A if and only if the Euler identity*

$$(Df(x), x) = \alpha f(x)\,, \qquad \forall\, x \in A \tag{3.31}$$

holds.

Proof For given $x \in A$, consider the function $F : (0, +\infty) \to \mathbb{R}$ defined by

$$F(t) = \frac{f(tx)}{t^\alpha}\,, \qquad \forall\, t > 0\,.$$

The function f is homogeneous of degree α in x, i.e. $f(tx) = t^\alpha f(x)$, if and only if F is constant, equal to $f(x)$, for any $t \in (0, +\infty)$. This happens precisely when the derivative $F'(t)$ is identically zero on $(0, +\infty)$. By assumption f is differentiable

on A, so by the chain rule

$$F'(t) = \frac{1}{t^{2\alpha}}\left[\sum_{i=1}^{n} f_{x_i}(tx)\cdot x_i \cdot t^{\alpha} - \alpha t^{\alpha-1} f(tx)\right] =$$

$$= \frac{1}{t^{\alpha+1}}\left[\sum_{i=1}^{n} f_{x_i}(tx)\cdot x_i \cdot t - \alpha f(tx)\right].$$

Hence $F'(t) = 0$ for any $t \in (0, +\infty)$ if and only if

$$\sum_{i=1}^{n} f_{x_i}(tx)\cdot x_i \cdot t = \alpha f(tx)\,, \qquad \forall\, x \in A\,, \quad \forall\, t > 0\,. \tag{3.32}$$

Since x belongs to A if and only if $tx \in A$ for any given $t > 0$, if we replace x with tx identity (3.32) assumes the form

$$\sum_{i=1}^{n} f_{x_i}(x) x_i = \alpha f(x)\,, \qquad \forall\, x \in A\,,$$

which is just the, Euler identity (3.31).

Here is a second feature of homogeneous, functions of degree α.

Homogeneity of the Gradient of a Homogeneous Function *Let $f : A \to \mathbb{R}$ be a function having partial derivatives on an open cone $A \subseteq \mathbb{R}^n$. If f is homogeneous of degree α on A, for any $i \in \{1, 2, \ldots, n\}$ the partial derivative f_{x_i} is homogeneous of degree $\alpha - 1$ on A.*

Proof By assumption

$$f(tx) - t^{\alpha} f(x) = 0\,, \qquad \forall\, x \in A\,, \quad \forall\, t > 0\,.$$

Given $t > 0$, then, the partial derivative in x_i of the left-hand side is zero, namely

$$f_{x_i}(tx)\cdot t - t^{\alpha} f_{x_i}(x) = 0\,, \qquad \forall\, x \in A\,, \quad \forall\, t > 0\,. \tag{3.33}$$

(Note that, in this case, we did not use the chain rule in n variables, but in one; this is the reason for not demanding that f be differentiable). Dividing equation (3.33) by t shows that f_{x_i} is homogeneous of degree $\alpha - 1$ on A.

3.10 Functions Defined by Integrals

Occasionally it becomes necessary to establish whether an integral function admits partial derivatives, or is differentiable (or simply, if it is continuous). The cases of

present concern are functions $\Phi : A \to \mathbb{R}$ defined on open sets $A \subseteq \mathbb{R}^n$ by means of an integral expression of the type

$$\Phi(x) = \int_{\alpha(x)}^{\beta(x)} f(x,t)\,dt\,, \qquad x \in A \subseteq \mathbb{R}^n\,, \tag{3.34}$$

where $f : A \times \mathbb{R} \to \mathbb{R}$ is a continuous function in $n+1$ real variables. Let us start with a result about continuity.

Continuity of Integral Functions *Let $f : A \times \mathbb{R} \to \mathbb{R}$ be a continuous function of $n+1$ real variables and α, β two continuous functions on A. Then the integral function $\Phi : A \to \mathbb{R}$ in (3.34) is continuous on A.*

Proof Consider first the function $F : A \times \mathbb{R}^2 \to \mathbb{R}$, of $n+2$ real variables, defined by

$$F(x,y,z) = \int_y^z f(x,t)\,dt\,, \qquad x \in A \subseteq \mathbb{R}^n\,, y \in \mathbb{R}\,, z \in \mathbb{R}\,,$$

and let us show it is continuous on $A \times \mathbb{R} \times \mathbb{R}$. From this it will follow that, the $\Phi : A \to \mathbb{R}$ in (3.34), i.e. $\Phi(x) = F(x, \alpha(x), \beta(x))$, is continuous on A as composite of continuous functions.

Let (x_0, y_0, z_0) be a given point of $A \times \mathbb{R} \times \mathbb{R}$. Let $K \times [a,b] \times [a,b]$ be a compact set contained in $A \times \mathbb{R} \times \mathbb{R}$ and containing a neighbourhood of (x_0, y_0, z_0).

By the Weierstrass theorem there exists $M > 0$ such that $|f(x,t)| \le M$ for any $(x,t) \in K \times [a,b]$. For any $(x,y,z) \in K \times [a,b] \times [a,b]$ we have

$$\begin{aligned}
&|F(x,y,z) - F(x_0,y_0,z_0)| \le \\
&\le \left| \int_y^{y_0} |f(x,t)|\,dt \right| + \left| \int_{y_0}^{z_0} |f(x,t) - f(x_0,t)|\,dt \right| + \left| \int_{z_0}^{z} |f(x,t)|\,dt \right| \le \\
&\le M|y - y_0| + \left| \int_{y_0}^{z_0} |f(x,t) - f(x_0,t)|\,dt \right| + M|z - z_0|\,.
\end{aligned}$$

Since $g(x,t) = |f(x,t) - f(x_0,t)|$ is continuous when $(x,y) \in K \times [a,b]$, by Cantor's theorem it is uniformly continuous on that set. Hence if we fix $\varepsilon > 0$, there exists $\delta > 0$ such that, for any pair (x_1,t_1), $(x_2,t_2) \in K \times [a,b]$ with distance less than δ from one another, we have $|g(x_1,t_1) - g(x_2,t_2)| < \varepsilon$. In particular, for $(x_1,t_1) = (x,t)$ and $(x_2,t_2) = (x_0,t_0)$,

$$|g(x,t)| = |f(x,t) - f(x_0,t)| < \varepsilon\,, \qquad |x - x_0| < \delta\,,$$

since $g(x_2, t_2) = g(x_0, t_0) = 0$. If moreover $|y - y_0| < \varepsilon$, $|z - z_0| < \varepsilon$, we eventually obtain

$$|F(x, y, z) - F(x_0, y_0, z_0)| \leq M\varepsilon + \varepsilon|z_0 - y_0| + M\varepsilon \leq (2M + b - a)\varepsilon\,.$$

Now suppose $f : A \times \mathbb{R} \to \mathbb{R}$ is a continuous function of $n + 1$ real variables, of class C^1 in the variable $x \in A$ (in the sense that the partial derivatives f_{x_i}, $i = 1, 2, \ldots, n$ are continuous on $A \times \mathbb{R}$) and that $\alpha, \beta : A \to \mathbb{R}$ are of class $C^1(A)$. Then the following theorem holds.

Differentiability of Integral Functions *Under the previous hypotheses, the integral function $\Phi : A \to \mathbb{R}$ is of class $C^1(A)$ (in particular, differentiable on A) and the components Φ_{x_i} of the gradient $D\Phi$ are*

$$\frac{\partial \Phi}{\partial x_i} = f(x, \beta(x))\,\frac{\partial \beta}{\partial x_i} - f(x, \alpha(x))\,\frac{\partial \alpha}{\partial x_i} + \int_{\alpha(x)}^{\beta(x)} \frac{\partial f}{\partial x_i}\,(x, t)\,dt \tag{3.35}$$

for any $i \in \{1, 2, \ldots, n\}$.

Proof As before let $F : A \times \mathbb{R}^2 \to \mathbb{R}$ denote the function of $n + 2$ real variables

$$F(x, y, z) = \int_y^z f(x, t)\,dt\,, \qquad x \in A \subseteq \mathbb{R}^n\,,\ y \in \mathbb{R}\,,\ z \in \mathbb{R}\,.$$

Given $x \in A$, $y \in \mathbb{R}$, the function

$$G(z) = \int_y^z f(x, t)\,dt$$

is differentiable by the fundamental theorem of calculus, with derivative

$$G'(z) = \frac{\partial F}{\partial z}\,(x, y, z) = f(x, z)\,.$$

Fix $x \in A$, $z \in \mathbb{R}$. For the same reason as above, since

$$F(x, y, z) = -\int_z^y f(x, t)\,dt\,,$$

the partial derivative in y exists and equals

$$\frac{\partial F}{\partial y}\,(x, y, z) = -f(x, y)\,.$$

Now let us prove that the partial derivative f_{x_i}, for given $i \in \{1, 2, \ldots, n\}$, is

$$\frac{\partial F}{\partial x_i} = \int_y^z \frac{\partial f}{\partial x_i}\,(x, t)\,dt\,. \tag{3.36}$$

To that end consider $x \in A$. As A is open, there exists $\delta > 0$ such that $x + he_i \in A$ for any $h \in [-\delta, \delta]$, where $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ has ith component equal to 1 and the others null. Then

$$\frac{F(x + he_i, y, z) - F(x, y, z)}{h} = \int_y^z \frac{f(x + he_i, t) - f(x, t)}{h}\, dt\,. \tag{3.37}$$

By the mean value theorem (in one real variable) there exists $\vartheta \in (0, 1)$ such that

$$\frac{f(x + he_i, t) - f(x, t)}{h} = f_{x_i}(x + \vartheta he_i, t)\,. \tag{3.38}$$

By assumption the derivative f_{x_i} is continuous on $A \times \mathbb{R}$. In particular it is continuous on the set (when $y < z$, for instance)

$$\{(x, t) \in [x - \delta e_i\,,\ x + \delta e_i] \times [y, z]\}\,,$$

which is compact in $\mathbb{R}^{n+1}$. Therefore f_{x_i} is uniformly continuous there, and for any $\varepsilon > 0$ there exists $\delta_1 \le \delta$ such that

$$|f_{x_i}(x + \vartheta he_i, t) - f_{x_i}(x, t)| < \varepsilon$$

for any $\vartheta \in (0, 1)$ and $t \in [y, z]$, provided $|h| \le \delta_1$.

The previous estimate together with (3.37), (3.38) imply

$$\left| \frac{F(x + he_i, y, z) - F(x, y, z)}{h} - \int_y^z \frac{\partial f}{\partial x_i}(x, t)\, dt \right| \le$$

$$\le \int_y^z |f_{x_i}(x + \vartheta he_i, t) - f_{x_i}(x, t)| \le \varepsilon |z - y|\,,$$

for any $|h| \le \delta_1$, and then (3.36) follows.

By the integral function's continuity, F is then of class $C^1(A \times \mathbb{R}^2)$.

Let us now go back to the integral function Φ, which may be represented via F as

$$\Phi(x) = F(x, \alpha(x), \beta(x)) = \int_{\alpha(x)}^{\beta(x)} f(x, t)\, dt\,, \qquad x \in A \subseteq \mathbb{R}^n\,.$$

Since, by assumption, $\alpha(x)$ and $\beta(x)$ are of class $C^1(A)$, the chain rule guarantees that Φ is $C^1(A)$ and that (3.35) holds.

3.11 Taylor Formula and Higher-Order Differentials

Let A be an open set in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ a function of class $C^k(A)$ for some $k \in \mathbb{N}$. Consider points $x, x+h$ in A, with $h \neq 0$, with the property that the segment $[x, x+h]$ is contained in A. The points $x(t)$ of the segment are of the form

$$x(t) = x + th\,, \qquad \forall\, t \in [0,1]\,.$$

Consider the composite function

$$F(t) = f(x(t)) = f(x+th)$$

for $t \in [0,1]$. Its first derivative is by definition the directional derivative of f along h, in case h is a direction, i.e. $|h| = 1$. In any case, since f is of class $C^1(A)$ and so differentiable on A, by the chain rule the composite function F is differentiable on $[0,1]$ with derivative

$$F'(t) = (Df(x+th), h) = \sum_{i=1}^{n} f_{x_i}(x+th)h_i\,, \tag{3.39}$$

where h_i, $i = 1, 2, \ldots, n$, are the components of h. Applying the chain rule to (3.39) we see that F' is differentiable and $(F')'$ is

$$\begin{aligned} F''(t) &= \sum_{j=1}^{n} \frac{\partial}{\partial x_j} \left(\sum_{i=1}^{n} f_{x_i}(x+th)h_i \right) h_j = \\ &= \sum_{i,j=1}^{n} f_{x_i x_j}(x+th)h_i h_j\,. \end{aligned} \tag{3.40}$$

More generally, $F : [0,1] \to \mathbb{R}$ is differentiable k times and the derivative of order k equals

$$F^{(k)}(t) = \frac{d^k F}{dt^k}(t) = \sum_{i_1, i_2, \ldots, i_k = 1}^{n} f_{x_{i_1} x_{i_2} \ldots x_{i_k}}(x+th)h_{i_1} h_{i_2} \ldots h_{i_k}\,. \tag{3.41}$$

The kth derivative $F^{(k)}$ is a homogeneous polynomial in the components of the vector h, and the coefficients are the partial derivatives of f of order k.

Using the Taylor expansion with Lagrange remainder for one-variable functions, there exists a real number $\vartheta \in (0,1)$ such that

$$F(1) = F(0) + F'(0) + \frac{F''(0)}{2} + \ldots + \frac{F^{(k-1)}(0)}{(k-1)!} + \frac{F^{(k)}(\vartheta)}{k!}\,. \tag{3.42}$$

Note that $\vartheta \in (0, 1)$ depends on x and h. Substituting (3.39), (3.40), (3.41) in (3.42) gives the *Taylor formula with Lagrange remainder for functions of n real variables*.

Let us make formula (3.42) explicit, distinguishing the cases $k = 1$, $k = 2$ and $k > 2$.

In analogy to the case $n = 1$, for one variable, when $k = 1$ the Taylor formula with Lagrange remainder is equivalent to the mean value theorem.

Mean Value Theorem *Let f be a function of class $C^1(A)$ and x, $x + h$ points in $A \subseteq \mathbb{R}^n$ ($h \neq 0$) such that the segment $[x, x + h]$ is contained in A. There exists a real number $\vartheta \in (0, 1)$, depending on x and h, such that*

$$f(x + h) = f(x) + \sum_{i=1}^{n} f_{x_i}(x + \vartheta h)h_i = f(x) + (Df(x + \vartheta h), h)\,.$$

Proof It suffices to write (3.42) with $k = 1$ (which is the mean value theorem in one variable)

$$F(1) = F(0) + F'(\vartheta)$$

and represent the derivative F' using (3.39).

For $k = 2$ we obtain:

Second-Order Taylor Formula with Lagrange Remainder *Let f be of class $C^2(A)$ and x, $x + h$ ($h \neq 0$) points in $A \subseteq \mathbb{R}^n$ such that the segment $[x, x + h]$ is contained in A. There is a real number $\vartheta \in (0, 1)$, depending on x and h, such that*

$$f(x + h) = f(x) + \sum_{i=1}^{n} f_{x_i}(x)h_i + \frac{1}{2} \sum_{i,j=1}^{n} f_{x_i x_j}(x + \vartheta h)h_i h_j =$$

$$= f(x) + (Df(x), h) + \frac{1}{2}\,(D^2 f(x + \vartheta h) \cdot h, h)\,.$$

As usual Df denotes the *gradient* and $D^2 f$ the *Hessian matrix* of second partial derivatives of f. Moreover, (Df, h) is the Euclidean inner product of the vectors Df and h, while $D^2 f \cdot h$ is the matrix product between the $n \times n$ matrix $D^2 f$ and the $n \times 1$ matrix h (a column vector).

Proof From the second-order Taylor formula (3.42) for $F : [0, 1] \to \mathbb{R}$,

$$F(1) = F(0) + F'(0) + \frac{1}{2}\,F''(\vartheta)$$

and the expressions (3.39) and (3.40) for F' and F'', we obtain

$$f(x+h) = f(x) + \sum_{i=1}^{n} f_{x_i}(x)h_i + \frac{1}{2}\sum_{i,j=1}^{n} f_{x_i x_j}(x+\vartheta h)h_i h_j \tag{3.43}$$

for a suitable $\vartheta \in (0,1)$. Clearly $\sum_{i=1}^{n} f_{x_i}(x)h_i$ equals the inner product $(Df(x), h)$. There remains to examine the last summand in (3.43), which equals $\frac{1}{2}(D^2 f(x+\vartheta h)\cdot h, h)$ since

$$\begin{aligned} D^2 f \cdot h &= \begin{pmatrix} f_{x_1x_1} & f_{x_1x_2} & \cdots & f_{x_1x_n} \\ f_{x_2x_1} & f_{x_2x_2} & \cdots & f_{x_2x_n} \\ \cdots & \cdots & \cdots & \cdots \\ f_{x_nx_1} & f_{x_nx_2} & \cdots & f_{x_nx_n} \end{pmatrix} \cdot \begin{pmatrix} h_1 \\ h_2 \\ \cdots \\ h_n \end{pmatrix} = \\ &= \begin{pmatrix} f_{x_1x_1}h_1 + f_{x_1x_2}h_2 + \ldots + f_{x_1x_n}h_n \\ f_{x_2x_1}h_1 + f_{x_2x_2}h_2 + \ldots + f_{x_2x_n}h_n \\ \cdots\cdots\cdots\cdots \\ f_{x_nx_1}h_1 + f_{x_nx_2}h_2 + \ldots + f_{x_nx_n}h_n \end{pmatrix} \end{aligned} \tag{3.44}$$

and the inner product of $D^2 f \cdot h$ and h is

$$\begin{aligned} &(f_{x_1x_1}h_1 + f_{x_1x_2}h_2 + \ldots + f_{x_1x_n}h_n)\cdot h_1 + \\ &+ (f_{x_2x_1}h_1 + f_{x_2x_2}h_2 + \ldots + f_{x_2x_n}h_n)\cdot h_2 + \ldots \\ &\ldots + (f_{x_nx_1}h_1 + f_{x_nx_2}h_2 + \ldots + f_{x_nx_n}h_n)\cdot h_n = \\ &= \sum_{i=1}^{n}\left(\sum_{j=1}^{n} f_{x_ix_j}h_j\right)h_i = \sum_{i,j=1}^{n} f_{x_ix_j}h_i h_j\,. \end{aligned} \tag{3.45}$$

In the next section we shall use the version of Taylor's formula of order two in n variables that involves Peano's remainder.

Second-Order Taylor Formula with Peano Remainder *Let f be a C^2 function on the open set A in $\mathbb{R}^n$. Take $x, x+h \in A$ with $h \neq 0$ such that the segment $[x, x+h]$ is contained in A. Then*

$$f(x+h) = f(x) + (Df(x), h) + \frac{1}{2}(D^2 f(x)\cdot h, h) + o(|h|^2)\,.$$

Before we prove it, let us see a useful property.

Lemma *For any $m \times n$ matrix $A = (a_{ij})$ and any vector $h \in \mathbb{R}^n$, the inequality*

$$|A \cdot h| \le |A| \cdot |h| \tag{3.46}$$

holds, where

$$|A| = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2}\,.$$

denotes the matrix norm.

Proof We have

$$A \cdot h = \begin{pmatrix} a_{11}h_1 + a_{12}h_2 + \ldots + a_{1n}h_n \\ a_{21}h_1 + a_{22}h_2 + \ldots + a_{2n}h_n \\ \cdots \\ a_{m1}h_1 + a_{m2}h_2 + \ldots + a_{mn}h_n \end{pmatrix},$$

and by the Cauchy-Schwarz inequality in $\mathbb{R}^n$ (see (2.12))

$$|A \cdot h| = \sqrt{\sum_{i=1}^{m} (a_{i1}h_1 + a_{i2}h_2 + \ldots + a_{in}h_n)^2} \le$$

$$\le \sqrt{\sum_{i=1}^{m} \left(\sum_{j=1}^{n} (a_{ij})^2 \right) |h|^2} = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2} \cdot |h|\,,$$

in other words (3.46).

Proof of the Second-Order Taylor Formula with Peano Remainder Starting from the analogous formula with Lagrange remainder there remains to show

$$\frac{1}{2}(D^2 f(x + \vartheta h) \cdot h, h) = \frac{1}{2}(D^2 f(x) \cdot h, h) + o(|h|^2)\,,$$

i.e.

$$\lim_{h \to 0} \frac{(D^2 f(x + \vartheta h) \cdot h, h) - (D^2 f(x) \cdot h, h)}{|h|^2} = 0\,. \tag{3.47}$$

The Cauchy-Schwarz inequality and (3.46) imply

$$\begin{aligned}&\left|\frac{(D^2 f(x+\vartheta h)\cdot h,h)-(D^2 f(x)\cdot h,h)}{|h|^2}\right|=\\&=\frac{|([D^2 f(x+\vartheta h)-D^2 f(x)]\cdot h,h)|}{|h|^2}\le\\&\le\frac{|[D^2 f(x+\vartheta h)-D^2 f(x)]\cdot h|}{|h|}\le\\&\le|D^2 f(x+\vartheta,h)-D^2 f(x)|\,.\end{aligned}\tag{3.48}$$

As the second partial derivatives of f are continuous, when $h\to 0$ inequality (3.48) yields (3.47).

To address the Taylor formula of order $k>2$ we introduce the *higher-order differentials* of a function $f:A\to\mathbb{R}$. Let f be of class $C^k(A)$ and write df for the differential of f at x with increment h:

$$df(x)=(Df(x),h)=\sum_{i=1}^{n} f_{x_i}(x)h_i\,.\tag{3.49}$$

The differential df, sometimes called *first-order differential* of f, is a linear functional in the variable $h\in\mathbb{R}^n$, i.e. a homogeneous polynomial of degree 1 in the components of h.

If $k\ge 2$, the partial derivatives $f_{x_i}(x)$ are of class $C^1(A)$ and so differentiable. Fixing $h\in\mathbb{R}^n$, the differential of $df(x)$ in (3.49), written $d^2 f(x)$, is called *second-order differential* of f at $x\in A$, and it equals

$$\begin{aligned}d^2 f(x)=d(df(x))=d\left(\sum_{i=1}^{n} f_{x_i}(x)h_i\right)=\\=\sum_{j=1}^{n}\frac{\partial}{\partial x_j}\left(\sum_{i=1}^{n} f_{x_i}(x)h_i\right)h_j=\sum_{i,j=1}^{n} f_{x_i x_j}(x)h_i h_j\,.\end{aligned}$$

The second-order differential is a quadratic form in $h\in\mathbb{R}^n$, i.e. a homogeneous polynomial of degree two in the components of h.

Analogously, if $k\ge 3$ the *third-order differential* $d^3 f(x)$ of f is the differential of $d^2 f(x)$ with h constant:

$$d^3 f(x)=\sum_{i_1,i_2,i_3=1}^{n} f_{x_{i_1}x_{i_2}x_{i_3}}(x)h_{i_1}h_{i_2}h_{i_3}.$$

Iterating the process, the *differential of order k* is

$$d^k f(x) = \sum_{i_1, i_2, \ldots, i_k = 1}^{n} f_{x_{i_1} x_{i_2} \ldots x_{i_k}}(x) h_{i_1} h_{i_2} \ldots h_{i_k}, \tag{3.50}$$

a homogeneous polynomial of degree $k \in \mathbb{N}$ in the components of $h \in \mathbb{R}^n$.

Comparing expression (3.50) with (3.41) and (3.42) gives back the Taylor formula for arbitrary k.

***k*th Order Taylor Formula with Lagrange Remainder** *Let f be a function of class $C^k(A)$ with $k \in \mathbb{N}$. Given $x, x+h \in A \subseteq \mathbb{R}^n$, $h \neq 0$, such that $[x, x+h] \subset A$, there exists a real number $\vartheta \in (0,1)$, depending on x and h, for which*

$$f(x+h) = f(x) + df(x) + \frac{1}{2} d^2 f(x) + \ldots + \frac{1}{(k-1)!} d^{k-1} f(x) + \frac{1}{k!} d^k f(x+\vartheta h)\,.$$

3.12 Quadratic Forms. Definite, Semi-definite and Indefinite Square Matrices

With the aim of formulating, in the next section, useful criteria for the study of *local maximum and minimum points* of a function of several variables, we shall introduce some definitions for square matrices $A = (a_{ij})$, $i, j = 1, 2, \ldots, n$.

Associated with any *square matrix* $A = (a_{ij})$, $i, j = 1, 2, \ldots, n$, is the function $F : \mathbb{R}^n \to \mathbb{R}$ defined by

$$F(\lambda) = (A \cdot \lambda, \lambda), \qquad \lambda \in \mathbb{R}^n\,,$$

where: λ is the variable in $\mathbb{R}^n$, $A \cdot \lambda$ is the matrix product of the $n \times n$ matrix A by the $n \times 1$ column vector λ, and $(A \cdot \lambda, \lambda)$ is the inner product of the column vectors $A \cdot \lambda$ and λ. It can be proved, as in the previous section (see (3.44) and (3.45)) that

$$F(\lambda) = (A \cdot \lambda, \lambda) = \sum_{i,j=1}^{n} a_{ij} \lambda_i \lambda_j, \qquad \forall\, \lambda \in \mathbb{R}^n\,.$$

Hence $F(\lambda)$ is a homogeneous quadratic polynomial in the components of λ, i.e. $F(\lambda)$ is a *quadratic form* in the variable $\lambda \in \mathbb{R}^n$ called the *quadratic form associated with the matrix* $A = (a_{ij})$.

A square matrix $A = (a_{ij})$ is said to be *positive definite* if the associated quadratic form is positive on $\mathbb{R}^n - \{0\}$, i.e.

$$\sum_{i,j=1}^{n} a_{ij} \lambda_i \lambda_j > 0\,, \qquad \forall\, \lambda \in \mathbb{R}^n\,,\ \lambda \neq 0.$$

The matrix is *positive semi-definite* if the associated quadratic form is non-negative on $\mathbb{R}^n$:

$$\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \geq 0\,, \qquad \forall\,\lambda \in \mathbb{R}^n\,.$$

Similarly, if $\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j < 0$ for any non-zero $\lambda \in \mathbb{R}^n$, the matrix $A = (a_{ij})$ is *negative definite*, and *negative semi-definite* when $\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \leq 0$ for any $\lambda \in \mathbb{R}^n$.

Finally, $A = (a_{ij})$ is *indefinite* if none of the above hold, i.e. if the associated quadratic form changes sign on $\mathbb{R}^n$. Equivalently, there exist $\lambda, \mu \in \mathbb{R}^n$ such that $\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j > 0$, $\sum_{i,j=1}^{n} a_{ij}\mu_i\mu_j < 0$.

Example 1 Consider the 2×2 matrices

$$(a_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad (b_{ij}) = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}, \qquad (c_{ij}) = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

The first is positive definite, since the associated quadratic form is

$$\sum_{i,j=1}^{2} a_{ij}\lambda_i\lambda_j = a_{11}\lambda_1^2 + 2a_{12}\lambda_1\lambda_2 + a_{22}\lambda_2^2 = \lambda_1^2 + 2\lambda_2^2 > 0\,,$$

for any $\lambda \in \mathbb{R}^2$, $\lambda \neq 0$. The second one is indefinite, as the associated quadratic form

$$\sum_{i,j=1}^{2} b_{ij}\lambda_i\lambda_j = b_{11}\lambda_1^2 + b_{12}\lambda_1\lambda_2 + b_{21}\lambda_2\lambda_1 + b_{22}\lambda_2^2 = \lambda_1^2 + 2\lambda_1\lambda_2$$

is positive when $\lambda = (\lambda_1, \lambda_2) = (1, 0)$, while it is negative for $\lambda = (\lambda_1, \lambda_2) = (1, -1)$.

The third matrix is positive semi-definite, because the quadratic form

$$\sum_{i,j=1}^{2} c_{ij}\lambda_i\lambda_j = \lambda_1^2 + 4\lambda_1\lambda_2 + 4\lambda_2^2 = (\lambda_1 + 2\lambda_2)^2$$

is greater than or equal to zero for any $\lambda = (\lambda_1, \lambda_2) \in \mathbb{R}^2$, but vanishes for $\lambda_1 = -2\lambda_2$. Note that the determinant of c_{ij} is zero, a typical property of (positive or negative) semi-definite matrices that are not definite.

Now let us examine a property of $n \times n$ (positive or negative) semi-definite matrices, and then we will look at the 2×2 case in more detail. Such properties will be employed in the next section. We refer to Sect. 3.16 in the Appendix, on complementary material about quadratic forms, for more properties.

Characterisation of Definite Matrices *A matrix $A = (a_{ij})$, $i, j = 1, 2, \ldots, n$, is positive definite if and only if there exists a constant $m > 0$ such that*

$$\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \geq m|\lambda|^2\,, \qquad \forall\,\lambda \in \mathbb{R}^n\,. \tag{3.51}$$

A matrix $A = (a_{ij})$ is negative definite if and only if there exists a constant $m > 0$ such that $\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \leq -m|\lambda|^2$ for any $\lambda \in \mathbb{R}^n$.

Proof If (3.51) holds, evidently $\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j > 0$ for any non-zero $\lambda \in \mathbb{R}^n$. Hence $A = (a_{ij})$ is positive definite. Vice versa, suppose $A = (a_{ij})$ is positive definite. Consider the quadratic form F associated with A

$$F(\lambda) = (A \cdot \lambda, \lambda) = \sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j\,,$$

as $\lambda \in \mathbb{R}^n$ varies in

$$K = \{\lambda \in \mathbb{R}^n : |\lambda| = 1\}\,.$$

The set K is compact in $\mathbb{R}^n$ and F is continuous (on $\mathbb{R}^n$), so by the Weierstrass theorem it assumes a minimum on K. Let λ_0 be a minimum point ($|\lambda_0| = 1$), so

$$\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \geq F(\lambda_0)\,, \qquad \forall\,\lambda \in \mathbb{R}^n\,, |\lambda| = 1\,.$$

By assumption $A = (a_{ij})$ is positive definite, and since $\lambda_0 \neq 0$, we have $F(\lambda_0) = (A \cdot \lambda_0, \lambda_0) > 0$. Set $m = F(\lambda_0)$, so

$$\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \geq m\,, \qquad \forall\,\lambda \in \mathbb{R}^n\,, |\lambda| = 1\,. \tag{3.52}$$

Consider now a generic vector $\lambda \in \mathbb{R}^n - \{0\}$. Call $\mu = \lambda/|\lambda|$, so $|\mu| = 1$ and from (3.52),

$$m \leq \sum_{i,j=1}^{n} a_{ij}\mu_i\mu_j = \sum_{i,j=1}^{n} a_{ij}\frac{\lambda_i}{|\lambda|}\cdot\frac{\lambda_j}{|\lambda|} = \frac{1}{|\lambda|^2}\cdot\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j\,,$$

whence (3.51) follows, for any $\lambda \in \mathbb{R}^n - \{0\}$. Finally, when $\lambda = 0$ the claim (3.51) is obvious.

The characterisation of negative definite matrices is similar.

Let us now pass to 2×2 matrices. We shall suppose the matrices are *symmetric*, just like the *Hessian matrix* $D^2 f = (f_{x_i x_j})$ of second derivatives of a C^2 function f. Note that when studying definite, semi-definite or indefinite matrices $A = (a_{ij})$ we can always reduce to the symmetric case, provided we replace a_{ij} $(i \neq j)$ with the average $(a_{ij} + a_{ij})/2$, because this does not alter the nature of the quadratic form associated with A. For details see Sect. 3.16 in the Appendix, about further material on quadratic forms.

Characterisation of 2×2 **Matrices** *Let*

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$$

be a 2×2 *symmetric matrix, and write as usual* $\det A = a_{11}a_{22} - a_{12}^2$ *for its determinant. Then*

$$\begin{aligned} \det A > 0 \quad &\Rightarrow \quad A \text{ is definite}; \\ \det A = 0 \quad &\Rightarrow \quad A \text{ is semi-definite}; \\ \det A < 0 \quad &\Rightarrow \quad A \text{ is indefinite}. \end{aligned}$$

In particular, when $\det A > 0$*, A is positive definite if* $a_{11} > 0$*, and negative definite if* $a_{11} < 0$*. When* $\det A = 0$*, A is positive semi-definite if* $a_{11}, a_{22} \geq 0$*, while negative semi-definite if* $a_{11}, a_{22} \leq 0$*.*

Proof The quadratic form associated with A is

$$F(\lambda_1, \lambda_2) = a_{11}\lambda_1^2 + 2a_{12}\lambda_1\lambda_2 + a_{22}\lambda_2^2, \qquad \forall\, \lambda = (\lambda_1, \lambda_2) \in \mathbb{R}^2.$$

To begin with suppose $a_{11} \neq 0$. Consider $\lambda = (\lambda_1, \lambda_2) \in \mathbb{R}^2$ with $\lambda_2 \neq 0$, and set $t = \lambda_1/\lambda_2$. The polynomial

$$\varphi(t) = \frac{F(\lambda_1, \lambda_2)}{\lambda_2^2} = a_{11}t^2 + 2a_{12}t + a_{22}$$

is quadratic in $t \in \mathbb{R}$, and its graph is a parabola with vertex at $t = -a_{12}/a_{11}$, where the derivative $\varphi'(t)$ vanishes. The minimum value (if $a_{11} > 0$) or maximum value $(a_{11} < 0)$ of φ is

$$\varphi\left(-\frac{a_{12}}{a_{11}}\right) = a_{22} - \frac{a_{12}^2}{a_{11}} = \frac{\det A}{a_{11}}. \tag{3.53}$$

Hence if $\det A > 0$ and $a_{11} > 0$, the convex parabola $\varphi(t)$ assumes positive values for any $t \in \mathbb{R}$. Therefore $F(\lambda_1, \lambda_2) > 0$ for any $(\lambda_1, \lambda_2) \in \mathbb{R}^2$ with $\lambda_2 \neq 0$. When

$\lambda_2 = 0$, $F(\lambda_1, 0) = a_{11}\lambda_1^2 > 0$ for any $\lambda_1 \neq 0$. This means A is positive definite. If $\det A > 0$ and $a_{11} < 0$ one can prove in a similar manner that A is negative definite.

We wrap up the case $\det A = a_{11}a_{22} - a_{12}^2 > 0$ by noting that $a_{11} \neq 0$ is automatic, for otherwise $\det A = -a_{12}^2 \leq 0$.

If $\det A = 0$ and $a_{11} > 0$ ($\det A = a_{11}a_{22} - a_{12}^2 = 0$ implies $a_{22} \geq 0$) we proceed as above to find $F(\lambda_1, \lambda_2) \geq 0$, and $F(\lambda_1, \lambda_2) = 0$ for $\lambda_1/\lambda_2 = t = -a_{12}/a_{11}$. Hence A is positive semi-definite. If $a_{11} = 0$ and $a_{22} > 0$ we go about similarly, interchanging λ_1 and λ_2. Finally if $a_{11} = a_{22} = 0$, necessarily $a_{12}^2 = a_{11} \cdot a_{22} = 0$ and so $F(\lambda_1, \lambda_2) = 0$ for any $(\lambda_1, \lambda_2) \in \mathbb{R}^2$. In any case F is positive semi-definite. The case $\det A = 0$, $a_{11} \leq 0$, $a_{22} \leq 0$ is analogous and gives a negative semi-definite quadratic form.

At last, when $\det A < 0$ and $a_{11} \neq 0$, from (3.53) we deduce that if $a_{11} > 0$ then $\varphi : \mathbb{R} \to [\det A/a_{11}, +\infty)$, while if $a_{11} < 0$ then $\varphi : \mathbb{R} \to (-\infty, \det A/a_{11}]$. In either case φ assumes both positive and negative values, so $F(\lambda_1, \lambda_2)$ is indefinite. If $a_{11} = 0$ and $a_{22} \neq 0$ we just interchange λ_1, λ_2. If $a_{11} = a_{22} = 0$ then $a_{12} \neq 0$ and clearly $F(\lambda_1, \lambda_2) = 2a_{12}\lambda_1\lambda_2$ is an indefinite quadratic form.

3.13 Local Maxima and Minima

Let f be defined on a set $A \subseteq \mathbb{R}^n$. A point $x_0 \in A$ is a *local maximum point* of f on A if there exists a ball $I_\delta(x_0)$, centred at x_0 with radius δ, such that

$$f(x_0) \geq f(x) \qquad \forall\, x \in I_\delta(x_0) \cap A\,.$$

Similarly, $x_0 \in A$ is a *local minimum point* of f on A if there exists a ball $I_\delta(x_0)$, of centre x_0 and radius δ, for which

$$f(x_0) \leq f(x) \qquad \forall\, x \in I_\delta(x_0) \cap A\,.$$

The following criterion is useful when searching for local extrema in the interior of a differentiable function's domain.

First-Order Necessary Condition *If $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ has partial derivatives at a local maximum or minimum point x_0 in the interior of A, then $Df(x_0) = 0$.*

Proof To fix ideas suppose x_0 is a local maximum point. Moreover x_0 is interior to A and f is differentiable at x_0.

Given an index $i \in \{1, 2, \ldots, n\}$, as usual we write e_i for the vector

$$e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$$

whose components are all zero except the ith one, equal to 1.

As x_0 is an interior maximum point, the one-variable function $F(t) = f(x_0+te_i)$ is defined on a neighbourhood of $t = 0$, and has a local maximum at $t = 0$. But F is differentiable and by definition its derivative F' is the partial derivative $f_{x_i}(x_0)$. Hence $f_{x_i}(x_0) = F'(0) = 0$, as requested.

According to the above condition, local extrema lying in the interior of the domain of a function f admitting partial derivatives should be seeked among the points at which all partial derivatives f_{x_i} vanish simultaneously. Such points are called *critical* or *stationary* for f. Therefore *the gradient vanishes at a critical point.*

The function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = x^2 + y^2$ has gradient $Df = (2x, 2y)$, which is zero at $(x, y) = (0, 0)$. Clearly $(0, 0)$ is a minimum point for f on $\mathbb{R}^2$, since $f(x, y) = x^2 + y^2 \geq 0 = f(0, 0)$ for any $(x, y) \in \mathbb{R}^2$. The graph of f is shown in Fig. 3.12.

Also $g : \mathbb{R}^2 \to \mathbb{R}$, $g(x, y) = x^2 - y^2$, has vanishing gradient $Dg = (2x, -2y)$ at $(x, y) = (0, 0)$. The latter is then a critical point. But it is neither a maximum nor a minimum point because, for instance, the sign of g along the axes is

$$g(x, 0) > 0\,, \qquad \forall\, x \neq 0\,, \qquad\qquad g(0, y) < 0 \qquad \forall\, y \neq 0\,.$$

Hence g assumes, on any neighbourhood I_δ of the origin, values both larger and smaller than $g(0, 0) = 0$. We say $(0, 0)$ is a *saddle point* of g (see Fig. 3.13).

Figures 3.14 and 3.15 show the level curves of $f(x, y)$ and $g(x, y)$ as (x, y) varies in the square $[-1, 1] \times [-1, 1]$. In Fig. 3.14 the level curves are closed and surround the minimum point $(0, 0)$ of $f(x, y)$. In Fig. 3.15, on the contrary, level curves do not close around the saddle point $(0, 0)$. Note the curious fact that the computer produced Fig. 3.15 without plotting the line $y = x$, which is a level curve of $g(x, y)$.

Another necessary condition (but of order two) is phrased in terms of the *Hessian matrix* $D^2 f = (f_{x_i x_j})$, and uses the properties of definite, semi-definite and indefinite matrices introduced in the previous section.

Second-Order Necessary Condition *If $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is a C^2 function on a neighbourhood of an interior local minimum (or maximum) point $x_0 \in A$, the Hessian matrix $D^2 f(x_0)$ at x_0 is positive semi-definite (or negative semi-definite, respectively).*

Proof Let $x_0 \in A$ be a local minimum point for f, interior to A. Let λ be a given vector in $\mathbb{R}^n$. The function

$$F(t) = f(x_0 + t\lambda)$$

of the real variable t is defined on a neighbourhood of $t = 0$, and has a local minimum at $t = 0$. The second derivative F'', if it exists, satisfies $F''(0) \geq 0$.

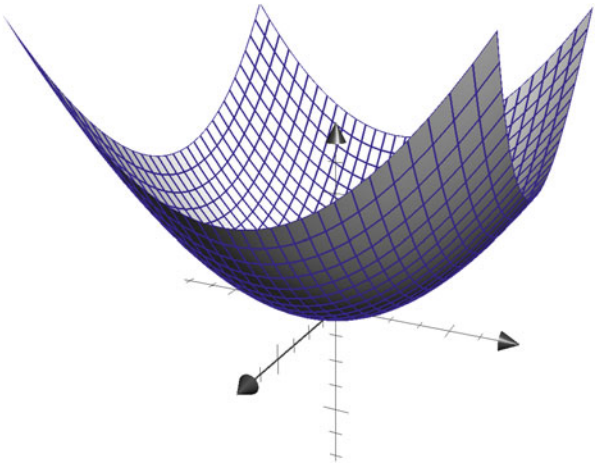


Fig. 3.12 $f(x, y) = x^2 + y^2$

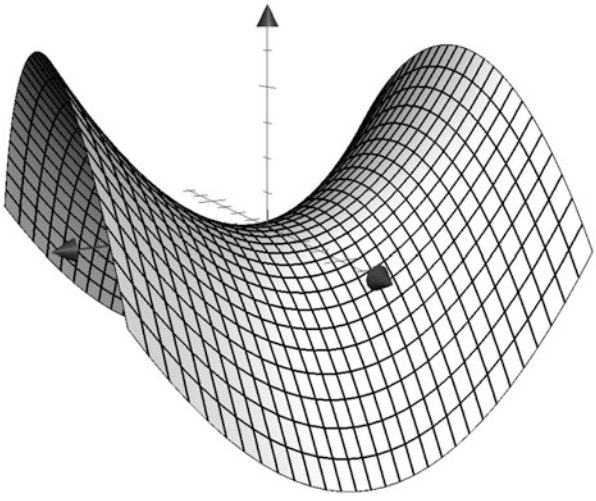


Fig. 3.13 $g(x, y) = x^2 - y^2$

Fig. 3.14

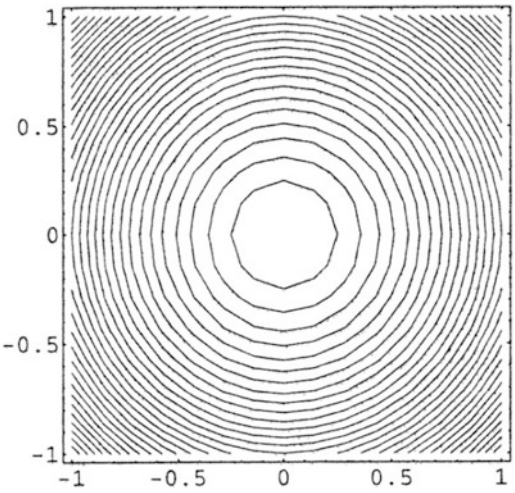
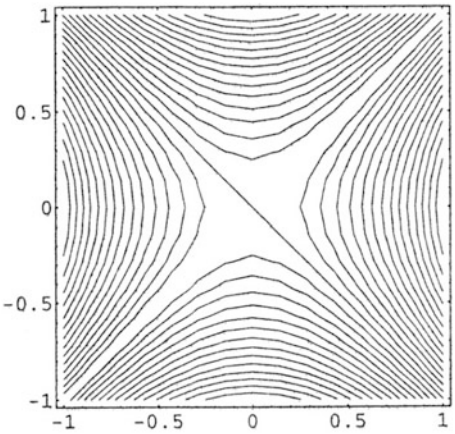


Fig. 3.15



As $f \in C^2$, the chain rule gives

$$F''(t) = \frac{d}{dt}\left(\frac{d}{dt} f(x_0 + t\lambda)\right) = \frac{d}{dt}\left(\sum_{i=1}^{n} f_{x_i}(x_0 + t\lambda)\lambda_i\right) =$$

$$= \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\left(\sum_{i=1}^{n} f_{x_i}(x_0 + t\lambda)\lambda_i\right)\lambda_j = \sum_{i,j=1}^{n} f_{x_i x_j}(x_0 + t\lambda)\lambda_i\lambda_j$$

whence

$$F''(0) = \sum_{i,j=1}^{n} f_{x_i x_j}(x_0)\lambda_i\lambda_j \geq 0\,, \qquad \forall\, \lambda \in \mathbb{R}^n\,.$$

Therefore $D^2 f(x_0)$ is a positive semi-definite matrix. The case where x_0 is an (interior) local maximum point is treated similarly.

Remark The second-order condition implies in particular that *if* $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ *is a* C^2 *function on a neighbourhood of an interior local minimum (or maximum)* x_0, *then*

$$f_{x_i x_i}(x_0) \geq 0 \qquad (f_{x_i x_i}(x_0) \leq 0)\,,$$

for any $i = 1, 2, \ldots, n$. In fact, since the Hessian matrix $D^2 f(x_0)$ at the minimum point x_0 is positive semi-definite (if x_0 is a maximum the argument is similar), then

$$\sum_{i,j=1}^{n} f_{x_i x_j}(x_0)\lambda_i \lambda_j \geq 0\,, \qquad \forall\, \lambda \in \mathbb{R}^n\,.$$

Given $i \in \{1, 2, \ldots, n\}$ and taking $\lambda = e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$, we find $f_{x_i x_j}(x_0) \geq 0$.

Now we shall present a sufficient condition for a critical point to be a local extremum. We consider first the general n variable case, and then pass to functions in 2 variables.

Sufficient Condition in n **Variables** *Let* $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ *be a* C^2 *function on a neighbourhood of an interior critical point* x_0. *If the Hessian matrix* $D^2 f(x_0)$ *is positive definite,* x_0 *is a local minimum point. If* $D^2 f(x_0)$ *is negative definite,* x_0 *is a local maximum point. If* $D^2 f(x_0)$ *is indefinite,* x_0 *is neither a local maximum nor a local minimum.*

Proof Consider first the case where the Hessian matrix $D^2 f(x_0)$ is positive definite. By the characterisation of definite matrices (see (3.51) in particular), there exists a constant $m > 0$ such that

$$(D^2 f(x_0) \cdot \lambda, \lambda) = \sum_{i,j=1}^{n} f_{x_i x_j}(x_0)\lambda_i \lambda_j \geq m|\lambda|^2\,, \qquad \forall\, \lambda \in \mathbb{R}^n\,.$$

By assumption x_0 is critical for f, so $Df(x_0) = 0$. Using Taylor's formula of order two with Peano remainder we find

$$f(x_0 + \lambda) - f(x_0) = (Df(x_0), \lambda) + \frac{1}{2}(D^2 f(x_0) \cdot \lambda, \lambda) + o(|\lambda|^2) \geq$$

$$\geq \frac{m}{2}|\lambda|^2 + o(|\lambda|^2) = |\lambda|^2 \left\{ \frac{m}{2} + \frac{o(|\lambda|^2)}{|\lambda|^2} \right\}.$$

Since

$$\lim_{\lambda\to 0}\frac{o(|\lambda|^2)}{|\lambda|^2}=0\,,$$

there exists $\delta>0$ such that

$$\frac{|o(|\lambda|^2)|}{|\lambda|^2}<\frac{m}{4}\,,\qquad \forall\,\lambda\in\mathbb{R}^n-\{0\},\ |\lambda|<\delta\,.$$

Hence

$$\begin{aligned} f(x_0+\lambda)-f(x_0)&\geq|\lambda|^2\left\{\frac{m}{2}+\frac{o(|\lambda|^2)}{|\lambda|^2}\right\}\geq\\ &\geq|\lambda|^2\left\{\frac{m}{2}-\frac{m}{4}\right\}=\frac{m}{4}|\lambda|^2\,,\qquad\forall\,\lambda\in\mathbb{R}^n,\ |\lambda|<\delta\,. \end{aligned}$$

In the end, x_0 is a minimum point for f on the ball $I_\delta(x_0)$ around x_0 with radius δ.

The proof for $D^2f(x_0)$ negative definite is analogous. When instead $D^2f(x_0)$ is indefinite, the second-order necessary condition shows that x_0 cannot be a local minimum point for f in A, otherwise $D^2f(x_0)$ should be positive semi-definite; nor can it be a local maximum point, in which case $D^2f(x_0)$ would be negative semi-definite.

The Case of Semi-definite Hessian Is Not Covered by the Sufficient Condition If in fact the Hessian matrix $D^2f(x_0)$ is positive definite at a critical point x_0, then x_0 is a local minimum point. If the Hessian is negative definite, x_0 is a local maximum point. If the matrix is indefinite the point is none of the above. If the Hessian matrix at x_0 is semi-definite we cannot say anything unless we study the function on a neighbourhood of x_0 using other means.

For example, the functions in three variables

$$f(x,y,z)=x^4+y^4+z^4\,,\qquad g(x,y,z)=x^4+y^4-z^4\,,$$

have a critical point at the origin $(0,0,0)$, and in both cases the Hessian is zero at the point. Hence we have a critical point with semi-definite Hessian matrix. In the first case $(0,0,0)$ is clearly a local (absolute) minimum point in $\mathbb{R}^3$, since

$$f(x,y,z)\geq 0=f(0,0,0)\,,\qquad\forall\,(x,y,z)\in\mathbb{R}^3\,.$$

For g the origin is neither a maximum nor a minimum, because

$$\begin{aligned} &g(x,y,0)=x^4+y^4>0=g(0,0,0)\,,\qquad\forall\,(x,y)\neq(0,0)\,,\\ &g(0,0,z)=-z^4<0=g(0,0,0)\,,\qquad\forall\,z\ \neq 0\,. \end{aligned}$$

We wrap up this section with a sufficient condition in the two-dimensional case. For a C^2 function f of two real variables we denote with $Hf(x, y)$ the *Hessian determinant*, i.e. the determinant of the Hessian matrix $D^2 f(x, y)$:

$$Hf(x, y) = \det D^2 f(x, y) = \begin{vmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{vmatrix} = f_{xx} \cdot f_{yy} - (f_{xy})^2 .$$

Sufficient Condition in 2 Variables *Let $f : A \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a function of class C^2 on a neighbourhood of an interior point (x_0, y_0) of A. If*

$$\begin{cases} f_x(x_0, y_0) = f_y(x_0, y_0) = 0 \\ Hf(x_0, y_0) > 0\,; \qquad f_{xx}(x_0, y_0) > 0 \end{cases} \tag{3.54}$$

then (x_0, y_0) is local minimum point. If

$$\begin{cases} f_x(x_0, y_0) = f_y(x_0, y_0) = 0 \\ Hf(x_0, y_0) > 0\,; \qquad f_{xx}(x_0, y_0) < 0 \end{cases} \tag{3.55}$$

then (x_0, y_0) is a local maximum point for. Finally if

$$\begin{cases} f_x(x_0, y_0) = f_y(x_0, y_0) = 0 \\ Hf(x_0, y_0) < 0 \end{cases} \tag{3.56}$$

then (x_0, y_0) is critical, but neither a maximum nor a minimum point.

Proof Under (3.54), or (3.55) or (3.56), the point (x_0, y_0) is always critical for f. Keeping in account the characterisation of 2×2 matrices, if the Hessian determinant $Hf(x_0, y_0)$ is positive the matrix $D^2 f(x_0, y_0)$ is positive definite in case (3.54) (where $f_{xx}(x_0, y_0) > 0$), and negative definite in case (3.55) (where $f_{xx}(x_0, y_0) < 0$). In case (3.56), the determinant $Hf(x_0, y_0)$ is negative and so the Hessian matrix $D^2 f(x_0, y_0)$ is indefinite.

The conclusion follows directly from the general sufficient condition in n variables.

3.14 Vector-Valued Functions

Let n, m be natural numbers and A a subset of $\mathbb{R}^n$. A map associating with every point in A a unique point in $\mathbb{R}^m$ is *a function* from $A \subseteq \mathbb{R}^n$ to $\mathbb{R}^m$. We write it as $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and denote by

$$(f_1, f_2, \ldots, f_m)$$

its components. More explicitly, since the variable x has components $(x_1, x_2, \ldots, x_n)$, we shall use the notation

$$(f_1(x_1, x_2, \ldots, x_n),\ f_2(x_1, x_2, \ldots, x_n), \ldots,\ f_m(x_1, x_2, \ldots, x_n))$$

or, more succinctly,

$$f = (f_\alpha(x))_{\alpha=1,2,\ldots,m} \qquad \forall\, x = (x_i)_{i=1,2,\ldots,n} \in A\,.$$

Evidently every component $f_\alpha(x)$, $\alpha = 1, 2, \ldots, m$, is a *scalar* function, defined on A with values in $\mathbb{R}$, i.e. $f_\alpha : A \subseteq \mathbb{R}^n \to \mathbb{R}$, for any $\alpha = 1, 2, \ldots, m$. Therefore we can apply to f_α all of the properties, e.g., about continuity, and differentiability, examined in previous sections.

If $n = m > 1$, i.e. if $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ with $n \geq 2$, we call $f = (f_1, f_2, \ldots, f_n)$ a *vector field* on $\mathbb{R}^n$. The geometric interpretation is to associate with every point $x \in A \subseteq \mathbb{R}^n$ the vector $f(x) = (f_1(x), f_2(x), \ldots, f_n(x))$ of $\mathbb{R}^n$.

The physical sciences consider in particular vector fields $v : A \subseteq \mathbb{R}^3 \to \mathbb{R}^3$ in space (or on $\mathbb{R}^n$), of components (v_1, v_2, v_3). Explicitly,

$$v(x, y, z) = (v_1(x, y, z),\ v_2(x, y, z),\ v_3(x, y, z))\ , \tag{3.57}$$

where (x, y, z) is a generic point in $\mathbb{R}^3$. Using the canonical basis e_1, e_2, e_3 of $\mathbb{R}^3$, (3.57) reads

$$v(x, y, z) = v_1(x, y, z)e_1 + v_2(x, y, z)e_2 + v_3(x, y, z)e_3\,.$$

In Physics a customary notation is

$$v(x, y, z) = v_1(x, y, z)\mathbf{i} + v_2(x, y, z)\mathbf{j} + v_3(x, y, z)\mathbf{k}\,,$$

where $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are the canonical basis of $\mathbb{R}^3$, with components $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$.

A particularly relevant *continuous* vector field on $\mathbb{R}^n$, i.e. a continuous map $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$, is given by the gradient $Dg : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ of a function g of class $C^1(A)$, i.e. setting $f = Dg$. Vector fields on $\mathbb{R}^n$ are studied in detail in Chap. 7 in relationship to differential 1-forms. If f is a vector field that equals the gradient $f = Dg$ of some function $g \in C^1$, we say the vector field is *conservative* and that $g : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is a *potential* of f.

To highlight the differences, at times crucial, between the target space being $\mathbb{R}$ or $\mathbb{R}^m$, $m > 1$, we will say that f is a scalar function if $m = 1$, i.e. $f : \mathbb{R}^n \to \mathbb{R}$, while if $m > 1$, $f : \mathbb{R}^n \to \mathbb{R}^m$ will be referred to as a *vector-valued function*.

For the sake of an example, let us consider a few differences between the scalar and vector-valued cases. For vector-valued functions the mean value theorem does not usually hold, nor does more generally the Taylor formula with Lagrange remainder. To see this let us

(continued)

consider the simplest *vector-valued* case, where $n = 1$ and $m = 2$. Consider the function $f : \mathbb{R} \to \mathbb{R}^2$,

$$f(t) = (r\,\cos t,\ r\,\sin t)\,, \tag{3.58}$$

where r is a given positive parameter. In the language of Chapter 6, f is a *curve* in $\mathbb{R}^2$. Its image is the circle with centre $(0, 0)$ and radius r on the plane.

As $f(0) = f(2\pi) = (r, 0)$, we have $f(2\pi) - f(0) = (0, 0)$. Yet the derivative $f'(t)$ does not vanish at any intermediate point in $[0, 2\pi]$. On the contrary, the modulus of $f'(t)$

$$|f'(t)| = |(-r\,\sin t,\ r\,\cos t)| = r$$

is constant ($\neq 0$). Therefore there is no $\vartheta \in [0, 2\pi]$ such that $f(2\pi) - f(0) = 2\pi \cdot f'(\vartheta)$.

Another problem is related to extrema. It does not make sense to talk about local maxima or minima for a vector-valued function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$, because $\mathbb{R}^m$, for $m > 1$, is not an ordered set. It is, on the other hand, reasonable to speak of the local maxima or minima of the components f_α of f (albeit not so useful), just like it makes sense to consider the local extrema of the modulus $|f|$ of f.

As for scalar functions, though, it makes sense to speak of *critical points* of a vector-valued function. For example the function $f : \mathbb{R} \to \mathbb{R}^2$ defined in (3.58) has no critical points (although its modulus $|f|$ is constant, equal to r, on the entire $\mathbb{R}$).

Let x_0 be a limit point of the set $A \subseteq \mathbb{R}^n$, $f : A \to \mathbb{R}^m$ a function and $\ell \in \mathbb{R}^m$ a given point. We say $f(x)$ *tends (or converges) to* ℓ *as* $x \to x_0$, written

$$\lim_{x \to x_0} f(x) = \ell \quad (or: \quad f(x) \to \ell \quad as \quad x \to x_0)\,,$$

if, for any neighbourhood $I_\varepsilon(\ell) \subseteq \mathbb{R}^m$ of ℓ, there exists a neighbourhood $I_\delta(x_0) \subseteq \mathbb{R}^n$ of x_0 such that

$$f(x) \in I_\varepsilon(\ell)\,, \qquad \forall\, x \in I_\delta(x_0) \cap A - \{x_0\}\,.$$

In terms of inequalities, $f(x)$ tends (or converges) to ℓ as $x \to x_0$ if, for any given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - \ell| < \varepsilon\,, \qquad \forall\, x \in A - \{x_0\}: \quad |x - x_0| < \delta\,.$$

The definition of limit is completely analogous to the known case of scalar functions. In the present vectorial case it is good to note that $|f(x) - \ell|$ denotes a norm on $\mathbb{R}^m$, while $|x - x_0|$ is a norm on $\mathbb{R}^n$.

Let $\ell = (\ell_\alpha)_{\alpha=1,2,\dots,m}$. From the inequality

$$|f_\alpha(x) - \ell_\alpha| \le |f(x) - \ell|\,, \qquad \forall\, \alpha = 1, 2, \dots, m\,,$$

we immediately have that if $f(x)$ converges to ℓ as $x \to x_0$ then every component $f_\alpha(x)$ converges to ℓ_α as $x \to x_0$. The converse holds too. In fact, if $|f_\alpha(x) - \ell_\alpha| < \varepsilon$ for any $x \in A - \{x_0\}$ such that $|x - x_0| < \delta$ and for any $\alpha = 1, 2, \ldots, m$, then

$$|f(x) - \ell| = \sqrt{\sum_{\alpha=1}^{m} (f_\alpha(x) - \ell_\alpha)^2} < \varepsilon\sqrt{m}$$

as well. Hence *a necessary and sufficient condition* for $f = (f_\alpha) : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ to tend to $\ell = (\ell_\alpha) \in \mathbb{R}^m$ as $x \to x_0$ is that every component f_α, $\alpha = 1, 2, \ldots, m$, must tend to ℓ_α as $x \to x_0$.

There is a notion of limit also when $|x| \to +\infty$, or for divergent functions $|f| \to +\infty$. For example, if x_0 is a limit point of $A \subseteq \mathbb{R}^n$, we say $|f(x)| \to +\infty$ as $x \to x_0$ if for any $M > 0$ there exists $\delta > 0$ such that

$$|f(x)| > M\,, \qquad \forall\, x \in A - \{x_0\} : \qquad |x - x_0| < \delta\,.$$

We leave it to the reader to formulate the other cases.

We have continuity at x_0, as usual, whenever $f(x) \to f(x_0)$ as $x \to x_0$, which is the same as the continuity at x_0 of each component f_α of f. Hence a function $f = (f_\alpha) : A \to \mathbb{R}^m$ is of class C^0 on $A \subseteq \mathbb{R}^n$, written

$$f \in C^0(A; \mathbb{R}^m)\,,$$

if and only if each component f_α, $\alpha = 1, 2, \ldots, m$ is of class $C^0(A)$.

Let A be an *open* set in $\mathbb{R}^n$, $f : A \to \mathbb{R}^m$ a function with components f_α, $\alpha = 1, 2, \ldots, m$, and fix a point x in A. The function f *admits partial derivatives* at x if each component f_α of f admits partial derivatives at x, and f admits partial derivatives on A if it admits partial derivatives at any $x \in A$. In these circumstances the $m \times n$ *matrix of partial derivatives*

$$Df = \left(\frac{\partial f_\alpha}{\partial x_i}\right)_{\substack{\alpha=1,2,\ldots,m \\ i=1,2,\ldots,n}} = \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1} & \dfrac{\partial f_1}{\partial x_2} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \dfrac{\partial f_2}{\partial x_1} & \dfrac{\partial f_2}{\partial x_2} & \cdots & \dfrac{\partial f_2}{\partial x_n} \\ \cdots & \cdots & \cdots & \cdots \\ \dfrac{\partial f_m}{\partial x_1} & \dfrac{\partial f_m}{\partial x_2} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{pmatrix}.$$

is well defined. This is called the *Jacobian matrix* of the vector-valued function f. It is also denoted with the symbol

$$Df = \frac{\partial(f_1, f_2, \ldots, f_m)}{\partial(x_1, x_2, \ldots, x_n)}\,,$$

highlighting the components f_α of f ($\alpha = 1, 2, \ldots, m$) and the components x_i of x ($i = 1, 2, \ldots, n$). Let A be an open set of $\mathbb{R}^n$. We say $f : A \to \mathbb{R}^m$ is *differentiable* at $x \in A$ if it admits partial derivatives at x (i.e. the Jacobian matrix exists) and

$$\lim_{h\to 0} \frac{f(x+h) - f(x) - Df(x)\cdot h}{|h|} = 0\,. \tag{3.59}$$

Observe that $Df(x)\cdot h$ is the product of the $m \times n$ Jacobian matrix $Df(x)$ and the vector h, seen as an $n \times 1$ matrix. The product $Df(x)\cdot h$ is then an $m \times 1$ matrix, which is coherent with $f(x+h) - f(x)$, dimension-wise.

In the $m = 1$ (scalar) case, $Df(x)\cdot h \in \mathbb{R}$, since the product

$$Df(x)\cdot h = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n}\right)\cdot \begin{pmatrix} h_1 \\ h_2 \\ \ldots \\ h_n \end{pmatrix} = \left(\frac{\partial f}{\partial x_1}h_1 + \frac{\partial f}{\partial x_2}h_2 + \ldots + \frac{\partial f}{\partial x_n}h_n\right)$$

boils down to the inner product $(Df(x), h)$ between the vectors $Df(x)$ and h, one recovers the usual differentiability definition for scalar functions.

The function f is *differentiable on A* if it is differentiable at any $x \in A$. Given x, the *linear* map $\mathbb{R}^n \to \mathbb{R}^m$ defined by $h \in \mathbb{R}^n \to (Df(x), h) \in \mathbb{R}^m$ is called the *differential* of f at x, and we denote it by $df(x)$. Hence $df(x) : \mathbb{R}^n \to \mathbb{R}^m$ is the linear map (or *linear functional*) in the variable $h \in \mathbb{R}^n$ defined by

$$df(x)(h) = Df(x)\cdot h\,, \qquad \forall\, h \in \mathbb{R}^n\,. \tag{3.60}$$

Using the "*little o*" notation, a function $f(x)$ defined on the open set A is differentiable at $x \in A$ if it has partial derivatives at x and

$$\begin{aligned} f(x+h) &= f(x) + Df(x)\cdot h + o(|h|)\,, \\ &= f(x) + df(x)(h) + o(|h|)\,, \end{aligned}$$

as $h \to 0$. The differentiability of a vector-valued function is then reduced to the differentiability of its components. We have in fact the following result.

Differentiability of Vector-Valued Functions *The vector-valued function $f : A \to \mathbb{R}^m$, of components f_α, $\alpha = 1, 2, \ldots, m$, is differentiable at x in the open subset $A \subset \mathbb{R}^n$ if and only if the scalar components $f_\alpha : A \to \mathbb{R}$, for any $\alpha \in \{1, 2, \ldots, m\}$ are differentiable at x. Moreover, if f is differentiable on A, the differential $df(x)$, evaluated at $h \in \mathbb{R}^n$, is the vector of $\mathbb{R}^m$*

$$df(x)(h) = (df_1(x)(h), df_2(x)(h), \ldots, df_m(x)(h)) \tag{3.61}$$

of components $df_\alpha(x)$, $\alpha = 1, 2, \ldots, m$.

Proof It is enough to observe that the limit in (3.59) reads, component-wise,

$$\lim_{h\to 0} \frac{f_\alpha(x+h) - f_\alpha(x) - \sum_{i=1}^n \dfrac{\partial f_\alpha}{\partial x_i} h_i}{|h|} =$$
$$= \lim_{h\to 0} \frac{f_\alpha(x+h) - f_\alpha(x) - (Df_\alpha(x), h)}{|h|} = 0\,,$$

for any $\alpha = 1, 2, \ldots, m$, recalling that the limits of vector-valued functions may be computed through the limits of their scalar components.

Formula (3.61) follows from (3.60), which expresses the differential $df(x) \cdot h = Df(x) \cdot h$ of the vector-valued function f, because the product $Df(x) \cdot h$ between the $m \times n$ Jacobian matrix and the column vector h equals the $m \times 1$ matrix of components

$$\sum_{i=1}^n \frac{\partial f_\alpha}{\partial x_i} h_i = (Df_\alpha(x), h) = df_\alpha(x)(h)\,, \qquad \forall\, \alpha = 1, 2, \ldots, m\,.$$

With the above theorem we may transfer to the vectorial case the main results about differentiating scalar functions, in practice by applying them to the scalar components f_α of f. In this way we find that *if $f : A \to \mathbb{R}^m$ is differentiable at $x \in A \subseteq \mathbb{R}^n$, then f is also continuous at x.*

We also have the differentiability criterion: *if $f : A \to \mathbb{R}^m$ is defined on the open set $A \subseteq \mathbb{R}^n$ and the Jacobian matrix Df is continuous at $x \in A$ (i.e. all elements $\partial f_\alpha / \partial x_i$ of Df are continuous at x) then f is differentiable at x.*

If the Jacobian matrix Df is continuous on an open subset $A \subset \mathbb{R}^n$, we say $f : A \to \mathbb{R}^m$ is of class C^1 on A and write

$$f \in C^1(A; \mathbb{R}^m)\,.$$

By the previous properties

$$f \in C^1(A; \mathbb{R}^m) \;\Rightarrow\; f \quad \textit{is differentiable on} \quad A \;\Rightarrow\; f \in C^0(A; \mathbb{R}^m)\,.$$

Of particular significance is the formula for the Jacobian of the composite of vector-valued functions. Consider a vector-valued function $f : A \to \mathbb{R}^m$, defined on an open set A in $\mathbb{R}^n$, and let $f = f(x)$ with $x \in A$. Suppose $x = g(t)$ itself is a vector-valued function, with $g : B \to \mathbb{R}^n$, B open in $\mathbb{R}^k$ ($k \in \mathbb{N}$) with range $g(B) \subseteq A$. Then the *composite function* $F : B \to \mathbb{R}^m$ is well defined, as

$$F = f \circ g\,,$$

also written $F(t) = f(g(t))$, for $t \in \mathbb{R}^k$. The Jacobian of F is computed by the next theorem.

Chain Rule for Vector-Valued Functions *Let $g : B \subseteq \mathbb{R}^k \to g(B) \subseteq A \subseteq \mathbb{R}^n$ and $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be C^1 functions. The composite $F = f \circ g$ is of class C^1 (in*

particular, differentiable) and its Jacobian DF equals the matrix product

$$DF = Df \cdot Dg\,. \tag{3.62}$$

Proof Consider the functions $f = f(x)$, $x = g(t)$ and the composite $F(t) = f \circ g(t) = f(g(t))$. Let $F_\alpha(t) = f_\alpha(g(t))$, $\alpha = 1, 2, \ldots, m$, denote the components of F. By the chain rule (3.26) for scalar functions, applied to the components $F_\alpha(t)$, we have

$$\frac{\partial F_\alpha}{\partial t_j}(t) = \sum_{i=1}^{n} \frac{\partial f_\alpha}{\partial x_i} \frac{\partial g_i}{\partial t_j}\,, \tag{3.63}$$

for any $\alpha = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, k$. This is the same as (3.62) in terms of the components. In fact (3.62) says the Jacobian matrix DF, of components

$$DF = \left(\frac{\partial F_\alpha}{\partial t_j}\right)\,,$$

with $\alpha = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, k$, is the product of the $m \times n$ matrix Df of components $\partial f_\alpha / \partial x_i$, times the $n \times k$ matrix Dg of components $\partial g_i / \partial t_j$. Furthermore, since $f_\alpha(x)$ and $x_i(t)$ are C^1 functions for any $\alpha = 1, 2, \ldots, m$ and $i = 1, 2, \ldots, n$, from (3.63) we deduce that F_α is of class C^1, whence $F = (F_\alpha)$ is of class C^1 too. Therefore F is differentiable.

The above result gives a way to compute the Jacobian of a composite vector-valued function; in the same way, the next result expresses the composite function's differential.

Chain Rule for the Differential of a Composite Map *Let $g : B \subseteq \mathbb{R}^k \to g(B) \subseteq A \subseteq \mathbb{R}^n$ and $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be C^1 functions. Then the differential dF of the composite $F = f \circ g$ is the composition*

$$dF = df \circ dg\,. \tag{3.64}$$

Proof In order to prove (3.64) we compute

$$df \circ dg = df(dg) = (df_\alpha(dg))\,, \tag{3.65}$$

where $f_\alpha, \alpha = 1, 2, \ldots, m$, are the components of f. Recalling that the differential of a vector-valued function g has as components the differentials of the components g_i (see (3.61)), we set $dg = (dg_i)$, $i = 1, 2, \ldots, n$, to obtain

$$df_\alpha(dg) = \sum_{i=1}^{n} \frac{\partial f_\alpha}{\partial x_i}\, dg_i\,. \tag{3.66}$$

Moreover, calling $h = (h_j)$, $j = 1, 2, \ldots, k$, the independent variable in $B \subseteq \mathbb{R}^k$, we have

$$dg_i(h) = \sum_{j=1}^{k} \frac{\partial g_i}{\partial t_j} h_j \,. \tag{3.67}$$

From (3.66) and (3.67) we then find

$$df_\alpha(dg) = \sum_{i=1}^{n} \sum_{j=1}^{k} \frac{\partial f_\alpha}{\partial x_i} \frac{\partial g_i}{\partial t_j} h_j = \sum_{j=1}^{k} \left(\sum_{i=1}^{n} \frac{\partial f_\alpha}{\partial x_i} \frac{\partial g_i}{\partial t_j} \right) h_j$$

and using the chain rule (3.63),

$$df_\alpha(dg) = \sum_{j=1}^{k} \frac{\partial F_\alpha}{\partial t_j} h_j = dF_\alpha \,. \tag{3.68}$$

Comparing (3.65) and (3.68) we end up with $dF = (dF_\alpha) = df \circ dg$, i.e. the claim.

Now consider the case $m = n$. If $f : A \to \mathbb{R}^n$ is defined on an open set $A \subseteq \mathbb{R}^n$ and differentiable at point $x_0 \in A$, either one of the symbols

$$J(x_0) \,, \quad J_f(x_0) \,, \quad \det \frac{\partial(f_1, f_2, \ldots, f_n)}{\partial(x_1, x_2, \ldots, x_n)}(x_0)$$

denotes the Jacobian determinant, i.e. the determinant of the $n \times n$ matrix $Df(x_0)$. Recalling that the determinant of a product is the product of the factors' determinants, from (3.62) we deduce the next result.

Jacobian of Composite Functions *Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ and $g : B \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be functions of class C^1 on open sets A, B, with $g(B) \subseteq A$. Setting $F = f \circ g$, we have*

$$J_F(t) = J_f(g(t)) \cdot J_g(t) \,, \tag{3.69}$$

As an application of formulas (3.62) and (3.69) we have

Jacobian of Inverse Functions *Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a C^1 function with C^1 inverse $f^{-1} : B \to A$. Then*

$$Df^{-1}(y) = [Df(f^{-1}(y))]^{-1} \,, \qquad \forall\, y \in B \,; \tag{3.70}$$

$$J_{f^{-1}}(y) = \frac{1}{J_f(f^{-1}(y))} \,, \qquad \forall\, y \in B \,. \tag{3.71}$$

Appendix to Chap. 3

3.15 Convex Functions

Given distinct points $x_1, x_2 \in \mathbb{R}^n$, the *(line) segment* with endpoints x_1, x_2 is the set of points in $\mathbb{R}^n$ of the form $tx_1 + (1-t)x_2$, where t is a real parameter in $[0, 1]$. One says $tx_1 + (1-t)x_2$, $t \in [0, 1]$, is *convex combination* of x_1 and x_2.

A set $A \subseteq \mathbb{R}^n$ is *convex* if the segment between any two points x_1, x_2 of A is contained in A. In symbols

$$x_1, x_2 \in A \quad \Rightarrow \quad tx_1 + (1-t)x_2 \in A\,, \qquad \forall\, t \in [0, 1]\,.$$

Let A be a convex subset of $\mathbb{R}^n$. A function $f : A \to \mathbb{R}$ is *convex* if

$$f(tx_1 + (1-t)x_2) \le tf(x_1) + (1-t)f(x_2)\,,$$

for any $x_1, x_2 \in A$ and $t \in [0, 1]$.

The definition extends to functions of n real variables the analogous notion for functions defined on a real interval. The special case of one-variable functions is well-suited to construct examples of convex functions on intervals, which are certainly well known to the reader.

An example of convex function on $\mathbb{R}^n$, with $n \ge 1$, is the norm

$$f(x) = |x|\,, \qquad\qquad x \in \mathbb{R}^n\,.$$

By the triangle inequality,

$$f(tx_1 + (1-t)x_2) = |tx_1 + (1-t)x_2| \le |tx_1| + |(1-t)x_2| = t|x_1| + (1-t)|x_2| = \\ = tf(x_1) + (1-t)f(x_2)$$

for any $x_1, x_2 \in A$ and any $t \in [0, 1]$. Also $f(x) = |x|^2$ and $f(x) = |x|^p$, $p \ge 1$, are convex on $\mathbb{R}^n$. More generally,

$$f(x) = g(|x|)\,, \qquad\qquad x \in \mathbb{R}^n\,,$$

with $g : [0, +\infty) \to \mathbb{R}$ convex and increasing. In fact

$$f(tx_1 + (1-t)x_2) = g\left(|tx_1 + (1-t)x_2|\right) \le g(t|x_1| + (1-t)|x_2|) \le \\ \le tg(|x_1|) + (1-t)g(|x_2|) = tf(x_1) + (1-t)f(x_2)\,.$$

Fig. 3.16

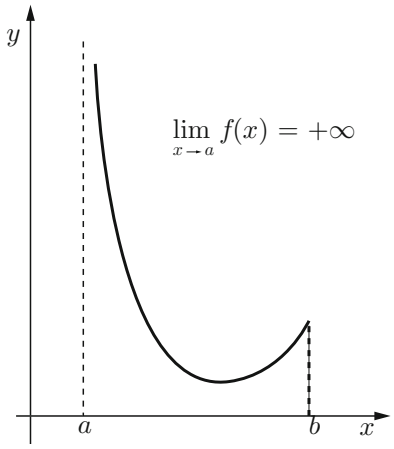
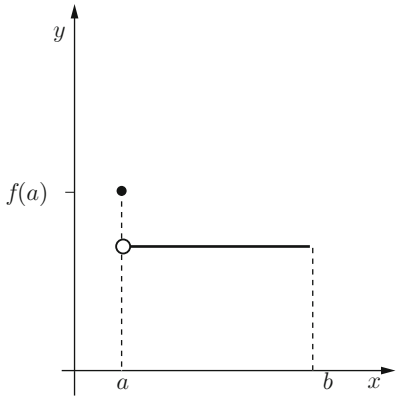


Fig. 3.17



As in the one-dimensional case (Fig. 3.16), a convex function $f : A \to \mathbb{R}$ of n variables may be unbounded on the neighbourhood of a boundary point of the set A; it may also be discontinuous on the boundary of A (Fig. 3.17).

On the other hand, on the interior of A f is necessarily continuous. To prove that, we need to define locally bounded and locally Lipschitz functions.

A function $f : A \to \mathbb{R}$, defined on a subset A of $\mathbb{R}^n$ with non-empty interior, is *locally upper bounded* (or *locally bounded from above*) on A if for any interior point x_0 of A, there exists a ball $B_r(x_0)$, of centre x_0 and radius $r > 0$, contained in A, and there exists a constant $M = M(x_0, r)$ such that

$$f(x) \leq M\,, \qquad \forall\, x \in B_r(x_0)\,.$$

Similarly, $f : A \to \mathbb{R}$ is *locally lower bounded* (or *locally bounded from below*) on A if for any interior point x_0 there exist a ball $B_r(x_0)$ contained in A and a constant

$m = m(x_0, r)$ such that

$$f(x) \geq m\,, \qquad \forall\, x \in B_r(x_0)\,.$$

Finally, $f : A \to \mathbb{R}$ is *locally bounded* on A if it is locally bounded on A both from below and from above.

A function $f : A \to \mathbb{R}$, defined on $A \subset \mathbb{R}^n$ with non-empty interior, is *locally Lipschitz* on A if, for any interior point x_0 of A, there exist a ball $B_r(x_0) \subset A$ and a constant $L = L(x_0, r)$ such that

$$|f(x_1) - f(x_2)| \leq L|x_1 - x_2|\,, \qquad \forall\, x_1, x_2 \in B_r(x_0)\,. \tag{3.72}$$

An immediate consequence of definition (3.72) is that every locally Lipschitz function is continuous at any interior point of A.

In the sequel we shall always consider functions on sets $A \subseteq \mathbb{R}^n$ with non-empty interior. For these the following theorem holds.

Local Lipschitz Continuity of Convex Functions *Every convex function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz on A.*

The proof is divided in several steps,: first we will establish local upper boundedness, then continuity on the interior of A, which implies local boundedness, and finally we will prove that the function is locally Lipschitz.

Let us begin from the following relationship, called *discrete Jensen inequality*, to distinguish it from the *integral Jensen inequality* discussed at the end of Chap. 8 on multiple integration.

(Discrete) Jensen Inequality *Let $f : A \to \mathbb{R}$ be a convex function on the convex set $A \subset \mathbb{R}^n$. If $x_1, x_2, \ldots, x_k$ are k points in A, $k \geq 2$, and $t_1, t_2, \ldots, t_k$ are non-negative real numbers, then*

$$\frac{\sum_{i=1}^{k} t_i x_i}{\sum_{i=1}^{k} t_i} \in A\,, \tag{3.73}$$

$$f\left(\frac{\sum_{i=1}^{k} t_i x_i}{\sum_{i=1}^{k} t_i}\right) \leq \frac{\sum_{i=1}^{k} t_i f(x_i)}{\sum_{i=1}^{k} t_i}\,. \tag{3.74}$$

Proof We use induction on $k \geq 2$. When $k = 2$ condition (3.73) reads

$$\frac{t_1 x_1 + t_2 x_2}{t_1 + t_2} = \frac{t_1}{t_1 + t_2} x_1 + \frac{t_2}{t_1 + t_2} x_2 \in A$$

and (3.74) is

$$f\left(\frac{t_1 x_1 + t_2 x_2}{t_1 + t_2}\right) \leq \frac{t_1}{t_1 + t_2} f(x_1) + \frac{t_2}{t_1 + t_2} f(x_2).$$

Now we recast the above relationships in the form of convexity inequalities

$$t x_1 + (1 - t) x_2 \in A\,, \qquad f(t x_1 + (1 - t) x_2) \leq t f(x_1) + (1 - t) f(x_2)\,,$$

by setting

$$t = \frac{t_1}{t_1 + t_2}\,, \qquad 1 - t = 1 - \frac{t_1}{t_1 + t_2} = \frac{t_2}{t_1 + t_2}\,.$$

Suppose (3.73) and (3.74) hold for some $k \geq 2$ and let us prove

$$\frac{\sum_{i=1}^{k+1} t_i x_i}{\sum_{i=1}^{k+1} t_i} \in A\,, \tag{3.75}$$

$$f\left(\frac{\sum_{i=1}^{k+1} t_i x_i}{\sum_{i=1}^{k+1} t_i}\right) \leq \frac{\sum_{i=1}^{k+1} t_i f(x_i)}{\sum_{i=1}^{k+1} t_i}\,. \tag{3.76}$$

Put

$$t = \frac{\sum_{i=1}^{k} t_i}{\sum_{i=1}^{k+1} t_i}\,, \quad \text{from which} \quad 1 - t = \frac{t_{k+1}}{\sum_{i=1}^{k+1} t_i}\,.$$

Then

$$\frac{\sum_{i=1}^{k+1} t_i x_i}{\sum_{i=1}^{k+1} t_i} = \frac{\sum_{i=1}^{k} t_i x_i}{\sum_{i=1}^{k+1} t_i} + \frac{t_{k+1} x_{k+1}}{\sum_{i=1}^{k+1} t_i} = t \frac{\sum_{i=1}^{k} t_i x_i}{\sum_{i=1}^{k} t_i} + (1 - t) x_{k+1} = t x + (1 - t) x_{k+1}\,,$$

where we have set

$$x = \frac{\sum_{i=1}^{k} t_i x_i}{\sum_{i=1}^{k} t_i}$$

and observing that $x \in A$ by the inductive hypothesis (3.73). As A is convex, we have $tx + (1-t)x_{k+1} \in A$ and (3.75) holds. Furthermore, the convexity of f and the inductive hypothesis (3.74) give

$$f\left(\frac{\sum_{i=1}^{k+1} t_i x_i}{\sum_{i=1}^{k+1} t_i}\right) = f(tx + (1-t)x_{k+1}) \le tf(x) + (1-t)f(x_{k+1}) =$$

$$= tf\left(\frac{\sum_{i=1}^{k} t_i x_i}{\sum_{i=1}^{k} t_i}\right) + (1-t)f(x_{k+1}) \le t\frac{\sum_{i=1}^{k} t_i f(x_i)}{\sum_{i=1}^{k} t_i} + (1-t)f(x_{k+1}) =$$

$$= \frac{\sum_{i=1}^{k} t_i f(x_i)}{\sum_{i=1}^{k+1} t_i} + \frac{t_{k+1} f(x_{k+1})}{\sum_{i=1}^{k+1} t_i},$$

i.e. (3.76).

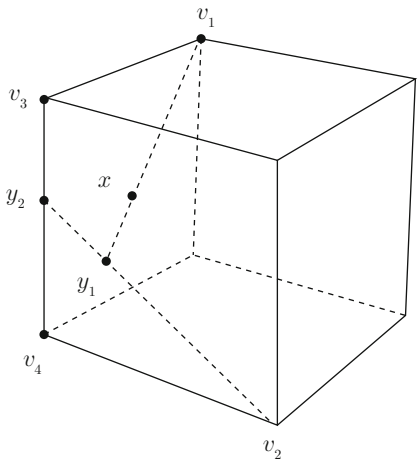
Local Upper Boundedness of Convex Functions *Every convex function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is locally bounded from above on A.*

Proof For any point x_0 in the interior of the convex set $A \subseteq \mathbb{R}^n$ consider an n-dimensional cube Q of centre x_0 (x_0 is the intersection of the cube's diagonals) whose closure is contained in A. We shall prove f is upper bounded on Q. This will imply, f is bounded from above on every ball $B(r, x_0)$ contained in Q.

Write x_i, $i = 1, 2, \ldots, 2^n$, for the vertices of Q. The first thing we prove is the geometrically intuitive property that any point in the cube's interior is a convex combination of the vertices. That is, for any interior x there are non-negative reals $t_1, t_2, \ldots, t_{2^n}$ such that

$$\sum_{i=1}^{2^n} t_i = 1\,, \qquad x = \sum_{i=1}^{2^n} t_i x_i\,. \tag{3.77}$$

Fig. 3.18



Actually we can prove that, more precisely, any x interior to Q is a convex combination of at most $n+1$ vertices of Q. This means some the numbers $t_1, t_2, \ldots, t_{2^n}$ are zero.

The proof is by recurrence on n. The reader can start by understanding the case $n = 3$ using Fig. 3.18. We denote the chosen vertices of Q by v_i, $i = 1, 2, \ldots, n+1$. In particular, v_1 is any one vertex of Q, say $v_1 = x_1$. Consider the straight line through v_1 and x, by assumption distinct from v_1 (being interior to Q). This line meets the face opposite to v_1 at some point y_1, as in Fig. 3.18. The points v_1, x and y_1 are collinear by construction, so x is a convex combination of v_1 and y_1, i.e. there exists $t_1 \in (0, 1)$ such that

$$x = t_1 v_1 + (1 - t_1) y_1 \,. \tag{3.78}$$

Let v_2 be the vertex opposite to v_1. If $y_1 = v_2$ the proof ends and in (3.77) it suffices to choose $t_2 = 1 - t_1$. Otherwise, consider the line through v_2 and y_1, which intersects the cube's boundary (an edge if $n = 3$, as in Fig. 3.18) at some y_2. The point y_1 is a convex combination of v_2 and y_2, so there is $s \in (0, 1)$ such that

$$y_1 = s v_2 + (1 - s) y_2 \,.$$

Combining the above with (3.78) gives

$$x = t_1 v_1 + (1 - t_1)[s v_2 + (1 - s) y_2] \,,$$

whence, setting $t_2 = (1 - t_1)s$,

$$(1 - t_1)(1 - s) = (1 - t_1) - (1 - t_1)s = 1 - t_1 - t_2 \,,$$

and then

$$x = t_1 v_1 + t_2 v_2 + (1 - t_1 - t_2) y_2 \,.$$

Iterating the construction $n-1$ times we find y_{n-1} on an edge, which is then a convex combination of the vertices v_n and v_{n+1}. The process stops after n steps, giving

$$x = t_1 v_1 + t_2 v_2 + \ldots + t_n v_n + \left(1 - \sum_{i=1}^{n} t_i\right) v_n \,.$$

So if we set $t_{n+1} = 1 - \sum_{i=1}^{n} t_i$, we obtain (3.77). Because of (3.77), and since $\sum_{i=1}^{n+1} t_i = 1$, Jensen's inequality implies

$$f(x) = f\left(\sum_{i}^{n+1} t_i x_i\right) \le \sum_{i}^{n+1} t_i f(x_i) \le M \sum_{i}^{n+1} t_i = M \,,$$

where $M = \max_i \{f(x_i)\}$. This proves f is bounded on Q.

Continuity of Convex Functions *Every convex function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is continuous on the interior of A. Precisely, for any interior point x_0 of A, there exists a ball $B(r, x_0)$, with centre x_0, radius $r > 0$ and closure contained in A, for which the estimate*

$$|f(x) - f(x_0)| \le \frac{M - f(x_0)}{r} |x - x_0| \,, \quad \forall\, x \in B_r(x_0) \,, \tag{3.79}$$

holds with $M = \sup\{f(x) : x \in B_r(x_0)\}$.

Proof Observe that the local upper boundedness of convex functions implies that the upper bound M is finite if $B_r(x_0)$ has closure contained in A. It is clear that estimate (3.79) forces f to be continuous at x_0, so we may just prove (3.79).

First let us show (3.79) for $x_0 = 0$. Assume $x_0 = 0$ is interior to A and let $B_r(0)$ be a ball of centre 0, radius $r > 0$, with closure contained in A and for which

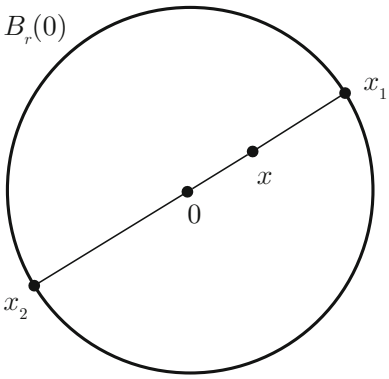
$$\sup\{f(x) : x \in B_r(0)\} = M < +\infty \,.$$

We claim

$$|f(x) - f(0)| \le \frac{M - f(0)}{r} |x| \,, \qquad \forall\, x \in B_r(0) \,. \tag{3.80}$$

Condition (3.80) is obvious if $x = 0$. If $x \neq 0$, call x_1, x_2 the two intersections of the boundary of $B_r(0)$ with the line through x, $x_0 = 0$, as in Fig. 3.19.

Fig. 3.19



Let x_1 be the point with the direction $x/|x|$ of x, and x_2 the antipodal point. As $|x_1| = |x_2| = r$,

$$x_1 = r\frac{x}{|x|}\,, \qquad\qquad x_2 = -r\frac{x}{|x|}\,. \tag{3.81}$$

It is easy to write x as convex combination of x_1 and 0: in fact if

$$x = tx_1 + (1-t)0 = tx_1\,,$$

we have $x = tx_1 = tr\dfrac{x}{|x|}$ and so $\dfrac{tr}{|x|} = 1$, i.e. $t = \dfrac{|x|}{r}$. Therefore

$$x = \frac{|x|}{r}x_1 + \left(1 - \frac{|x|}{r}\right)0$$

and by convexity

$$f(x) \le \frac{|x|}{r}f(x_1) + \left(1 - \frac{|x|}{r}\right)f(0),$$

so

$$f(x) - f(0) \le \frac{|x|}{r}[f(x_1) - f(0)] \le \frac{M - f(0)}{r}|x|, \tag{3.82}$$

which proves one part of (3.80). Referring to Fig. 3.19, write $x_0 = 0$ as convex combination of x_2 and x. Recalling the expression for x_2 in (3.81) we obtain

$$0 = tx_2 + (1-t)x = \left(-\frac{tr}{|x|} + 1 - t\right)x$$

so $-\dfrac{tr}{|x|}+1-t=0$, i.e. $t=\frac{|x|}{r+|x|}$. Therefore

$$0=\frac{|x|}{r+|x|}x_2+\frac{r}{r+|x|}x,$$

and by convexity

$$f(0)\le\frac{|x|}{r+|x|}f(x_2)+\frac{r}{r+|x|}f(x),$$

so

$$\begin{aligned}(r+|x|)f(0)&\le|x|f(x_2)+rf(x),\\ r[f(0)-f(x)]&\le|x|[f(x_2)-f(0)],\\ f(0)-f(x)&\le\frac{|x|}{r}[f(x_2)-f(0)]\le\frac{M-f(0)}{r}|x|.\end{aligned}\tag{3.83}$$

Now (3.82) and (3.83) yield (3.80). Finally, if x_0 is a generic interior point, apply (3.80) to $g(x)=f(x+x_0)$, observing that $g(x)$ is defined on a neighbourhood of $x=0$. This gives

$$|g(x)-g(0)|\le\frac{M-g(0)}{r}|x|\,,\qquad\forall\,x\in B_r(0),$$

i.e.

$$\big|f(x+x_0)-f(x_0)\big|\le\frac{M-f(x_0)}{r}|x|\,,\qquad\forall\,x\in B_r(0)\,,$$

leading to (3.79) by a change of notation (replacing $x-x_0$ with x).

From the continuity we have just discussed one proves the theorem mentioned earlier:

Local Lipschitz Continuity of Convex Functions *Every convex function $f:A\subseteq\mathbb{R}^n\to\mathbb{R}$ is locally Lipschitz on A. Precisely, let x_0 be interior to A and $B_{r_0}(x_0)$ a ball with centre x_0, radius r_0 and whose closure $\overline{B_{r_0}(x_0)}$ consists of interior points of A. Then for any ball $B_r(x_0)$ centred at x_0 with radius $r<r_0$, the estimate*

$$\big|f(x)-f(y)\big|\le\frac{M-m}{r_0-r}|x-y|,\qquad\forall\,x,y\in B_r(x_0)\,,\tag{3.84}$$

holds with $M=\max\big\{f(x):x\in\overline{B_{r_0}(x_0)}\big\}$, $m=\min\big\{f(x):x\in\overline{B_{r_0}(x_0)}\big\}$.

Proof We shall refer to Fig. 3.20. Consider a ball $B_{r_0}(x_0)$ whose closure $\overline{B_{r_0}(x_0)}$ is made of interior points of A and let $B_r(x_0)$ be a concentric neighbourhood of radius $r<r_0$. As f is continuous, it assumes maximum and minimum on $\overline{B_{r_0}(x_0)}$. For any

Fig. 3.20

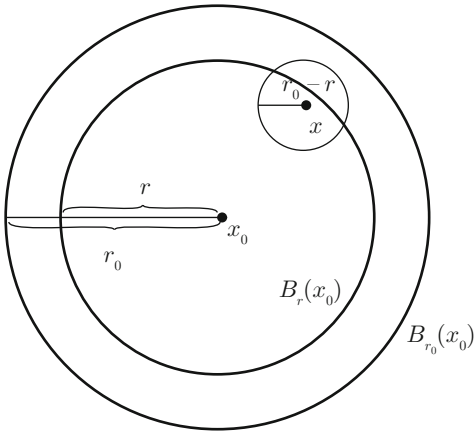
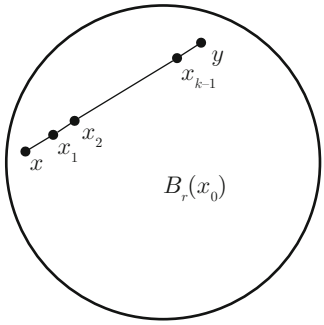


Fig. 3.21



$x \in B_r(x_0)$, the ball $B_{r_0-r}(x)$ of centre x and radius $r_0 - r$ is contained in $B_{r_0}(x_0)$ (see Fig. 3.20). By continuity at x (Eq. (3.79)), for any $x \in B_r(x_0)$ and any $y \in B_{r_0-r}(x)$ we have

$$\big|f(y) - f(x)\big| \le \frac{M - f(x)}{r_0 - r}|y - x| \le \frac{M - m}{r_0 - r}\big|y - x|. \tag{3.85}$$

This proves (3.84) for a generic x in $B_r(x_0)$ but with $y \in B_{r_0-r}(x)$. Let us now address the case in the Lipschitz theorem, with $x, y \in B_r(x_0)$. Divide the line segment $[x, y]$ in k parts of equal length $|x - y|/k$ as in Fig. 3.21, using points x_j, $j = 0, 1, \ldots, k$

$$x_0 = x, \quad x_1, \quad x_2, \ \ldots \ , x_k = y$$

defined as

$$x_j = x + \frac{j}{k}(y - x), \qquad j = 0, 1, \ldots, k\,.$$

Choose $k \in \mathbb{N}$ large, i.e. $k > \dfrac{|x-y|}{r_0 - r}$, so that

$$|x_{j+1} - x_j| = \left|\frac{1}{k}(y-x)\right| = \frac{1}{k}|x-y| < r_0 - r\,, \qquad \forall\, j = 0, 1, \ldots, k-1\,.$$

Thus $x_{j+1} \in B_{r_0-r}(x_j)$ and we may apply (3.85) with $x = x_j \in B_r(x_0)$ and $y = x_{j+1} \in B_{r_0-r}(x_j)$:

$$|f(x_{j+1}) - f(x_j)| \le \frac{M-m}{r_0-r}|x_{j+1} - x_j| = \frac{M-m}{r_0-r} \cdot \frac{|x-y|}{k}\,.$$

Adding the above relations for all $j = 0, 1, \ldots, k-1$ produces

$$|f(x) - f(y)| = \left|\sum_{j=0}^{k-1}[f(x_{j+1}) - f(x_j)]\right| \le \sum_{j=0}^{k-1}|f(x_{j+1}) - f(x_j)| \le$$

$$\le \frac{M-m}{r_0-r} \cdot \frac{|x-y|}{k} \cdot \sum_{j=0}^{k-1} 1 = \frac{M-m}{r_0-r}|x-y|\,.$$

The following criterion has independent theoretical interest, besides being useful to establish whether a given function $f : A \to \mathbb{R}$ is convex on some open set A in $\mathbb{R}^n$.

Convexity Criterion for Differentiable Functions *Let A be an open and convex set in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ a differentiable function on A. Then f is convex on A if and only if*

$$f(x) \ge f(x_0) + (Df(x_0), x - x_0)\,, \qquad \forall\, x, x_0 \in A\,, \tag{3.86}$$

or if and only if

$$(Df(y) - Df(x), y - x) \ge 0\,, \qquad \forall\, x, y \in A\,. \tag{3.87}$$

Condition (3.86) is interpreted geometrically by saying that the graph of f lies above the tangent plane, in correspondence to any point $x \in A$ and for any tangent point $x_0 \in A$. In view of (3.87) we say that the Jacobian $Df : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ of a convex function is a *monotone operator* on $\mathbb{R}^n$, thus extending the monotonicity, well known for $n = 1$, of the derivatives of convex functions in one real variable.

Proof If f is convex, by definition

$$f(tx + (1-t)x_0) \le tf(x) + (1-t)f(x_0)$$

for any $x, x_0 \in A$ and $t \in [0, 1]$. We rewrite this as

$$f(x_0 + t(x - x_0)) \leq f(x_0) + t\big[f(x) - f(x_0)\big],$$

whence

$$\frac{f(x_0 + t(x - x_0)) - f(x_0)}{t} \leq f(x) - f(x_0)\,.$$

The limit as $t \to 0^+$ of the left-hand side is the derivative of $\varphi(t) = f(x_0 + t(x - x_0))$ at $t = 0$. By the chain rule, when $t \to 0^+$ we have

$$(Df(x_0), x - x_0) \leq f(x) - f(x_0)\,.$$

Conversely, if (3.86) holds, given $x,\ x_0 \in A$ we consider the convex combination $x_0 + t\,(x - x_0)$. Then (3.86) implies the two inequalities

$$f(x) \geq f(x_0 + t\,(x - x_0)) + (Df(x_0 + t\,(x - x_0)), (1 - t)\,(x - x_0)) \quad (3.88)$$

$$f(x_0) \geq f(x_0 + t\,(x - x_0)) + (Df(x_0 + t\,(x - x_0)), -t\,(x - x_0))\,. \quad (3.89)$$

Multiplying (3.88) by t and (3.89) by $(1 - t)$, then adding, gives the convexity of f:

$$tf(x) + (1 - t)f(x_0) \;\geq\; f(x_0 + t\,(x - x_0)) \;=\; f(tx + (1 - t)x_0)\,.$$

We claim that from (3.86) we can obtain (3.87). Consider, besides

$$f(x) \;\geq\; f(x_0) + (Df(x_0), x - x_0)\,, \qquad \forall\, x, x_0 \in A$$

(this is (3.86)), also the relationship coming from (3.86) when we interchange x and x_0:

$$f(x_0) \geq f(x) + (Df(x), x_0 - x)\,, \qquad \forall\, x, x_0 \in A.$$

Adding the two we find

$$0 \geq (Df(x_0) - Df(x), x - x_0)\,, \qquad \forall\, x, x_0 \in A,$$

i.e. (3.87). Vice versa, start from (3.87) and apply the mean value theorem to $\varphi : [0, 1] \to \mathbb{R}$,

$$\varphi(t) = f(x_0 + t\,(x - x_0))\,.$$

There exists $\vartheta \in (0,1)$ such that $\varphi(1) - \varphi(0) = \varphi'(\vartheta)$, i.e.

$$f(x) - f(x_0) = (Df(x_0 + \vartheta\,(x - x_0)), x - x_0)\,.$$

By (3.87)

$$\begin{aligned}&(Df(x_0 + \vartheta\,(x - x_0)) - Df(x_0), \vartheta\,(x - x_0)) = \\ &= \vartheta\;(Df(x_0 + \vartheta\,(x - x_0)) - Df(x_0), x - x_0)\;\geq\;0\,,\end{aligned}$$

whence (3.86)

$$f(x) - f(x_0) = (Df(x_0 + \vartheta\,(x - x_0)), x - x_0) \geq (Df(x_0), x - x_0)\,.$$

Finally, let us discuss a convexity criterion relying on the properties of the Hessian matrix $D^2 f$ of the second derivatives of f.

Convexity Criterion for C^2 Functions *Let A be an open convex set of $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ a function of class $C^2(A)$. Then f is convex on A if and only if the Hessian matrix $D^2 f(x)$ is positive semi-definite for any $x \in A$.*

Proof Suppose $D^2 f(x)$ is positive semi-definite for any $x \in A$. Given $x, x_0 \in A$, Taylor's formula of order two with Lagrange remainder ensures that there is a real number $\vartheta \in (0,1)$ such that

$$f(x) = f(x_0) + (Df(x_0), x - x_0) + \frac{1}{2}\left(D^2 f(x_0 + \vartheta(x - x_0)) \cdot (x - x_0), x - x_0\right)$$

As the Hessian matrix is positive semi-definite at every point in the convex set $A \subset \mathbb{R}^n$, and since

$$x_0 + \vartheta(x - x_0) = \vartheta x + (1 - \vartheta)\, x_0$$

is a convex combination of $x, x_0 \in A$, and hence belongs to A, it follows that

$$\begin{aligned}&f(x) - f(x_0) - (Df(x_0), x - x_0) = \\ &= \frac{1}{2}\left(D^2 f(x_0 + \vartheta\,(x - x_0)) \cdot (x - x_0), x - x_0\right) \geq 0\,.\end{aligned}$$

By the convexity criterion (3.86) f is convex on A.

Conversely, let us suppose f is convex on A. As A is open, for any $x_0 \in A$ and any $\lambda \in \mathbb{R}^n$ the function $\varphi : [0,1] \to \mathbb{R}$ given by

$$\varphi(t) = f(x_0 + t\,\lambda)$$

is defined on an entire neighbourhood of $t = 0$, say $t \in (-\delta, \delta)$ with $\delta > 0$. For such t, since

$$x_0 = \frac{1}{2}(x_0 + t\,\lambda) + \frac{1}{2}(x_0 - t\,\lambda)\,,$$

the convexity of f implies

$$f(x_0) \le \frac{1}{2} f(x_0 + t\lambda) + \frac{1}{2} f(x_0 - t\,\lambda)$$

and so, equivalently,

$$\frac{f(x_0 + t\lambda) + f(x_0 - t\,\lambda) - 2\,f(x_0)}{t^2} \ge 0\,,$$

for any $t \in (-\delta, \delta)$, $t \neq 0$. We can calculate the limit as $t \to 0$ of the above left-hand side applying twice de l'Hôpital's rule and the chain rule. This gives

$$\begin{aligned}
0 \le \lim_{t\to 0} \frac{f(x_0 + t\,\lambda) + f(x_0 - t\lambda) - 2\,f(x_0)}{t^2} = \\
= \lim_{t\to 0} \frac{(Df(x_0 + t\,\lambda), \lambda) - (Df(x_0 - t\,\lambda), \lambda)}{2\,t} = \\
= \lim_{t\to 0} \frac{1}{2} \left\{ \left(D^2 f(x_0 + t\,\lambda) \cdot \lambda, \lambda \right) + \left(D^2 f(x_0 - t\,\lambda) \cdot \lambda, \lambda \right) \right\} = \\
= \left(D^2 f(x_0) \cdot \lambda, \lambda \right)\,.
\end{aligned}$$

In the final limit we have used the assumption $f \in C^2(A)$. Therefore $D^2 f(x_0)$ is positive semi-definite.

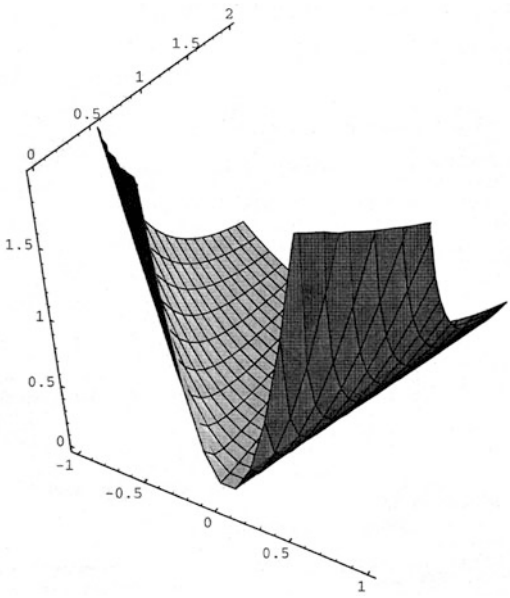
Remark When $n = 2$, using the characterisation of semi-definite 2×2 matrices of Sect. 3.12, we can make the convexity criterion for C^2 functions $f : A \subseteq \mathbb{R}^2 \to \mathbb{R}$ more precise. For that let us denote by $Hf(x, y) = \det D^2 f(x, y)$ the Hessian determinant

$$Hf(x, y) = \begin{vmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{vmatrix} = f_{xx} \cdot f_{yy} - \left(f_{xy}\right)^2\,.$$

Then the function $f = f(x, y)$ is convex on the open set $A \subset \mathbb{R}^2$ if and only if

$$\begin{cases} Hf(x, y) = f_{xx} \cdot f_{yy} - \left(f_{xy}\right)^2 \ge 0 \\ f_{xx}(x, y) \ge 0 \quad f_{yy}(x, y) \ge 0\,, \end{cases} \qquad \forall\,(x, y) \in A.$$

Fig. 3.22 $f(x, y) = \frac{x^2}{y}$



Example 1 The function of two real variables

$$f(x, y) = \frac{x^2}{y}$$

is convex on the open set $A = \{(x, y) \in \mathbb{R}^2 : y > 0\}$. In fact we can apply the convexity criterion of C^2 functions. The Hessian matrix is

$$D^2 f(x, y) = \begin{pmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & 2\frac{x^2}{y^3} \end{pmatrix}$$

and its determinant $Hf(x, y) = \det D^2 f(x, y)$ is identically zero. Since $f_{xx}(x, y) > 0$ and $f_{yy}(x, y) \geq 0$ for any $(x, y) \in A$, the function is convex on A. The graph of $f(x, y)$, when (x, y) is in the square $[-1, 1] \times [0, 2]$, is shown in Fig. 3.22.

Example 2 Generalising the previous example, verify that the function of two real variables

$$f(x, y) = x^p \cdot y^q$$

with $p, q \in \mathbb{R}$, is convex on the open convex subset of $\mathbb{R}^2$

$$A = \{(x, y) \in \mathbb{R}^2 : \, x > 0, y > 0\}$$

if and only if

$$p\,(p-1) \geq 0, \qquad q\,(q-1) \geq 0, \qquad pq(1-p-q) \geq 0\,.$$

3.16 Complements on Quadratic Forms

We continue the study of *quadratic forms* on $\mathbb{R}^n$ started in Sect. 3.12.

Given a *square* $n \times n$ *matrix* $A = (a_{ij})$, $i, j = 1, 2, \ldots, n$, consider the homogeneous polynomial $F(\lambda)$ of degree two in the components of $\lambda \in \mathbb{R}^n$ with coefficients a_{ij} defined by

$$F(\lambda) = (A \cdot \lambda, \lambda) = \sum_{i,j=1}^{n} a_{ij} \lambda_i \lambda_j \,, \qquad \lambda \in \mathbb{R}^n, \tag{3.90}$$

($A \cdot \lambda$ is the matrix product of A by the column vector λ). We call $F(\lambda)$ the *quadratic form associated with the matrix* $A = (a_{ij})$.

In expression (3.90) we may rewrite the ij terms with $i \neq j$ in the form

$$a_{ij}\lambda_i\lambda_j + a_{ji}\lambda_j\lambda_i = (a_{ij} + a_{ji})\,\lambda_i\lambda_j = \frac{a_{ij} + a_{ji}}{2}\,\lambda_i\lambda_j + \frac{a_{ij} + a_{ji}}{2}\,\lambda_j\lambda_i \,.$$

Replacing a_{ij} with $(a_{ij} + a_{ji})/2$ (the quadratic form does not change), we can always suppose the matrix $A = (a_{ij})$ is *symmetric*, i.e.

$$a_{ij} = a_{ji} \qquad \forall\, i, j = 1, 2, \ldots, n.$$

An *eigenvalue* of the symmetric matrix A is by definition a number (in general a *complex number*) α such that the linear system

$$A\,x = \alpha x \tag{3.91}$$

admits at least one non-zero solution x, called an *eigenvector* of A associated with the eigenvalue α.

System (3.91) may be recast in the equivalent form

$$(A - \alpha I)\,x = 0\,, \tag{3.92}$$

where I is the identity matrix. By Cramer's rule, (3.91) or equivalently (3.92) has a non-zero solution x if and only if the determinant of the coefficient matrix is zero:

$$\det(A - \alpha I) = \begin{vmatrix} a_{11} - \alpha & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} - \alpha & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{n1} & a_{n2} & \ldots & a_{nn} - \alpha \end{vmatrix} = 0 \tag{3.93}$$

Expanding the determinant shows that the left-hand side of (3.93) is a polynomial of degree n in α, called the *characteristic polynomial* of A, with the following form

$$\det(A - \alpha I) = (-1)^n \alpha^n + (-1)^{n-1}(a_{11} + a_{22} + \ldots + a_{nn})\alpha^{n-1} + \ldots + \det A \,. \tag{3.94}$$

The eigenvalues α of A are exactly the roots of the characteristic polynomial (3.94) of A, i.e. the (complex) solutions to equation (3.93).

By the fundamental theorem of algebra the characteristic polynomial (3.94) admits n roots, real or complex, with multiplicity. Given the characteristic polynomial's special structure, and since A is symmetric, it can be proved that its roots, i.e. the eigenvalues of A, are all real.

Theorem 1 *The characteristic polynomial's roots, i.e. the eigenvalues of the symmetric matrix A, are all real.*

Proof Let α be an eigenvalue of A and x ($x \neq 0$) a corresponding eigenvector, i.e. α is a root ($\alpha \in \mathbb{C}$ in general) of (3.93) and x is a complex solution ($x \in \mathbb{C}^n$, a vector with n complex components) of system (3.91). Take the inner product of either side of (3.91) with $\overline{x}$, the complex conjugate of x (whose components are the complex conjugates of the components of x):

$$(A\,x, \overline{x}) = \alpha\,(x, \overline{x}) \;.$$

Set $x = x_k + i\,y_k$, where i is the imaginary unit ($i^2 = -1$). On the right-hand side we have

$$(x, \overline{x}) = \sum_{k=1}^{n} (x_k + i\,y_k)\,(x_k - i\,y_k) = \sum_{k=1}^{n} (x_k^2 + y_k^2) = |x|^2 \;.$$

On the left, due to the symmetry of A,

$$\begin{aligned}
(A\,x, \overline{x}) &= \sum_{h,k=1}^{n} a_{hk}(x_h + i\,y_h)\,(x_k - i\,y_k) = \\
&= \sum_{h,k=1}^{n} a_{hk}\,\left(x_h x_k + i\,(y_h x_k - y_k x_h) + y_h y_k\right) = \\
&= \sum_{h,k=1}^{n} a_{kh}\,(x_k x_h - i\,(y_k x_h - y_h x_k) + y_k y_h) = \overline{(A\,x, \overline{x})} \;,
\end{aligned}$$

i.e. $(A\,x, \overline{x})$ coincides with its conjugate $\overline{(A\,x, \overline{x})}$. Hence $(A\,x, \overline{x})$ is a real number and so is

$$\alpha = \frac{(A\,x, \overline{x})}{|x|^2}.$$

By Theorem 1 the n eigenvalues of a symmetric matrix $A = (a_{ij})$, counted with multiplicity, are real. We denote them by $\alpha_1, \alpha_2, \ldots, \alpha_n$ in increasing order. Call m the smallest among the α_i, $i = 1, 2, \ldots, n$, and M the largest one. Then

$$m = \alpha_1 \leq \alpha_2 \leq \ \ldots \ \leq \alpha_n = M\,.$$

Theorem 2 *If m, M are the smallest and largest eigenvalues of a symmetric matrix A, then*

$$m\ |\lambda|^2 \leq (A \cdot \lambda, \lambda) \leq M\,|\lambda|^2\ , \qquad \forall\, \lambda \in \mathbb{R}^n. \tag{3.95}$$

Proof Consider the quadratic form

$$F(\lambda) = (A \cdot \lambda, \lambda) = \sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j\ \ ,$$

for λ in the compact set $K = \{\lambda \in \mathbb{R}^n : |\lambda| = 1\} \subset \mathbb{R}^n$. By the Weierstrass theorem the continuous function F assumes maximum and minimum on K. In other words there exist λ_1, λ_2, with $|\lambda_1| = |\lambda_2| = 1$, such that

$$F(\lambda_1) \leq \sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j\ \leq\ F(\lambda_2)\ , \qquad \forall\, \lambda \in \mathbb{R}^n,\ |\lambda| = 1. \tag{3.96}$$

Set $m_1 = F(\lambda_1)$, $m_2 = F(\lambda_2)$. For any $\lambda \in \mathbb{R}^n - \{0\}$, define $\mu = \lambda/\,|\lambda|$. As $|\mu| = 1$, from (3.96) we have

$$m_1 \leq \sum_{i,j=1}^{n} a_{ij}\mu_i\mu_j = \sum_{i,j=1}^{n} a_{ij}\,\frac{\lambda_i}{|\lambda|}\cdot\frac{\lambda_j}{|\lambda|} = \frac{1}{|\lambda|^2}\cdot\sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j \leq m_2\,.$$

Now set

$$G(\lambda) = \frac{1}{|\lambda|^2}\ \cdot \sum_{i,j=1}^{n} a_{ij}\lambda_i\lambda_j\ ,$$

giving $|\lambda_1| = |\lambda_2| = 1$, $G(\lambda_1) = m_1$, $G(\lambda_2) = m_2$ and

$$m_1 \leq G(\lambda) \leq m_2\ , \qquad \forall\, \lambda \in \mathbb{R}^n - \{0\}. \tag{3.97}$$

Hence λ_1 is a minimum point (local and absolute) for G on $\{\lambda \in \mathbb{R}^n : \lambda \neq 0\}$, whilst λ_2 is a maximum point. If at these points the gradient of G exists, it must vanish. Now,

$$\frac{\partial G}{\partial \lambda_i} = \frac{1}{|\lambda|^4} \cdot \left[\frac{\partial}{\partial \lambda_i} (A \cdot \lambda, \lambda) \cdot |\lambda|^2 - (A \cdot \lambda, \lambda) \cdot \frac{\partial}{\partial \lambda_i} |\lambda|^2 \right].$$

Clearly

$$\frac{\partial}{\partial \lambda_i} |\lambda|^2 = \frac{\partial}{\partial \lambda_i} \left(\lambda_1^2 + \lambda_2^2 + \ldots + \lambda_n^2 \right) = 2\lambda_i$$

and

$$\begin{aligned} \frac{\partial}{\partial \lambda_i} (A \cdot \lambda, \lambda) = \frac{\partial}{\partial \lambda_i} \big(& a_{11}\lambda_1^2 + 2a_{12}\lambda_1\lambda_2 + \ldots + 2\,a_{1i}\lambda_1\lambda_i + \ldots + 2\,a_{2i}\lambda_2\lambda_i + \ldots \\ & + \ldots + a_{ii}\lambda_i^2 + \ldots + a_{nn}\lambda_n^2 \big) = \\ = 2\big(& a_{1i}\lambda_1 + a_{2i}\lambda_2 + \ldots + a_{ni}\lambda_n \big) = 2 \sum_{j=1}^{n} a_{ji}\lambda_j \,. \end{aligned}$$

Therefore

$$\frac{\partial G}{\partial \lambda_i} = \frac{2}{|\lambda|^2} \left[\sum_{j=1}^{n} a_{ji}\lambda_j - \frac{(A \cdot \lambda, \lambda)}{|\lambda|^2} \lambda_i \right];$$

$$DG = \frac{2}{|\lambda|^2} \left[A \cdot \lambda - G(\lambda)\lambda \right].$$

When $\lambda = \lambda_1$ and $\lambda = \lambda_2$ the gradient DG is zero, i.e.

$$A \cdot \lambda - G(\lambda)\lambda = 0\,, \qquad \text{for } \lambda = \lambda_1 \text{ and } \lambda = \lambda_2\,.$$

As λ_1 and λ_2 are non-zero vectors, necessarily $G(\lambda_1) = m_1$ and $G(\lambda_2) = m_2$ are eigenvalues of A, with respective eigenvectors λ_1, λ_2.

We claim m_1 is the smallest eigenvalue of A and m_2 is the largest. If α is another eigenvalue of A with (non-zero) eigenvector λ_0, then $A\lambda_0 = \alpha\,\lambda_0$, so $(A\,\lambda_0, \lambda_0) = \alpha\,|\lambda_0|^2$. From (3.97)

$$m_1 \leq G(\lambda_0) = \frac{(A\,\lambda_0, \lambda_0)}{|\lambda_0|^2} = \alpha \leq m_2\,.$$

In the theorem's notation we then have $\alpha_1 = m = m_1$, $\alpha_n = M = m_2$, and (3.97) implies

$$m\,|\lambda|^2 \leq (A \cdot \lambda, \lambda) \leq M\,|\lambda|^2\,, \qquad \forall\, \lambda \in \mathbb{R}^n - \{0\}\,.$$

As the relationship is trivial when $\lambda = 0$, the theorem is proved.

Theorem 3 *A symmetric matrix $A = (a_{ij})$ is positive definite (negative definite) if and only if all eigenvalues are positive (negative). The matrix A is positive semi-definite (negative semi-definite) if and only if the eigenvalues are non-negative (non-positive). At last, A is indefinite if and only if there are eigenvalues of opposite sign.*

Proof Suppose the eigenvalues of A are positive. Then clearly the smallest one is positive too, and (3.95) implies $(A \cdot \lambda, \lambda) > 0$ for any $\lambda \in \mathbb{R}^n - \{0\}$. Hence A is positive definite. Similarly, if $m \geq 0$ then $(A \cdot \lambda, \lambda) \geq 0$ for any $\lambda \in \mathbb{R}^n$. The cases $M < 0$ and $M \leq 0$ are treated similarly.

If instead there exist two eigenvalues of opposite sign $\alpha_1 < 0 < \alpha_2$, calling λ_1, λ_2 the corresponding (non-zero) eigenvectors, we find $A\lambda_1 = \alpha_1\lambda_1$, $A\lambda_2 = \alpha_2\lambda_2$, and so

$$(A\lambda_1, \lambda_1) = \alpha_1 |\lambda_1|^2 < 0, \qquad (A\,\lambda_2, \lambda_2) = \alpha_2\, |\lambda_2|^2 > 0\,.$$

Hence the associated quadratic form $(A \cdot \lambda, \lambda)$ assumes both signs and A is indefinite.

The converse is immediate using (3.95) and (some of the steps in the proof of) Theorem 2. In fact

$$m \leq \frac{(A \cdot \lambda, \lambda)}{|\lambda|^2} \leq M\,, \qquad \forall\, \lambda \in \mathbb{R}^n - \{0\},$$

where m, M are the minimum and maximum eigenvalues of A. These, moreover, are values attained by $G(\lambda) = (A \cdot \lambda, \lambda)/\,|\lambda|^2$ at certain (non-zero) values of λ.

Note that from Theorems 2 and 3 we can deduce the characterisation of positive definite (symmetric) matrices from Sect. 3.12, whereby *a symmetric matrix $A = (a_{ij})$ is positive definite if and only if there exists a constant $m > 0$ such that*

$$(A \cdot \lambda, \lambda) \geq m\, |\lambda|^2\,, \qquad \forall\, \lambda \in \mathbb{R}^n.$$

We state without proof a characterisation for definite $n \times n$ symmetric matrices $A = (a_{ij})$. Consider the n *submatrices* $A_1, A_2, \ldots, A_n$ constructed starting from the top left corner of A and enlarging along the main diagonal, as follows:

$$A_1 = (a_{11})\,, \quad A_2 = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\,, \qquad A_3 = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}\,, \ \ldots\,,$$

$$\ldots, A_{n-1} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1,n-1} \\ a_{21} & a_{22} & \ldots & a_{2,n-1} \\ \ldots & \ldots & \ldots & \ldots \\ a_{n-1,1} & a_{n-1,2} & \ldots & a_{n-1,n-1} \end{pmatrix}\,, \qquad A_n = A\,.$$

Theorem 4 *A symmetric matrix $A = (a_{ij})$ is positive definite if and only if*

$$\det A_k > 0\,, \qquad \forall\, k = 1, 2, \ldots, n\,,$$

and negative definite if and only if

$$\det A_1 = a_{11} < 0\,, \quad \det A_2 > 0\,, \quad \det A_3 < 0,\ \ldots\ , (-1)^n \det A_n = (-1)^n \det A > 0\,.$$

We end the section by studying the convexity of quadratic forms.

Characterisation of Convex Quadratic Forms *Let $A = (a_{ij})$ be a symmetric matrix. The associated quadratic form $F(\lambda) = (A \cdot \lambda, \lambda)$ is convex for $\lambda \in \mathbb{R}^n$ if and only if A is positive semi-definite.*

Proof (First Method) Easily, the function F is of class $C^2(\mathbb{R}^n)$. Let us compute the Hessian matrix D^2F, starting from the gradient DF :

$$\begin{aligned}
\frac{\partial}{\partial \lambda_i} F\,(\lambda) &= \frac{\partial}{\partial \lambda_i}\,(A \cdot \lambda, \lambda) = \\
&= \frac{\partial}{\partial \lambda_i} \Big(a_{11}\lambda_1^2 + 2a_{12}\lambda_1\lambda_2 + \ldots + 2a_{1i}\lambda_1\lambda_i + \ldots \\
&\quad \ldots + 2\,a_{2i}\lambda_2\lambda_i + \ldots + a_{ii}\lambda_i^{\,2} + \ldots + a_{nn}\lambda_n^2 \Big) = \\
&= 2\,(a_{1i}\lambda_1 + a_{2i}\lambda_2 + \ldots + a_{ni}\lambda_n) = 2\sum_{j=1}^{n} a_{ji}\lambda_j\,,
\end{aligned}$$

so

$$\frac{\partial^2}{\partial \lambda_i \partial \lambda_j} F\,(\lambda) = \frac{\partial^2}{\partial \lambda_i \partial \lambda_j}\,(A \cdot \lambda, \lambda) = \frac{\partial}{\partial \lambda_j} 2 \sum_{j=1}^{n} a_{ji}\lambda_j = 2a_{ji}\,.$$

As A is symmetric, we have $a_{ji} = a_{ij}$ for any $i,\ j = 1, 2, \ldots, n$ and the Hessian matrix D^2F, constant on $\mathbb{R}^n$, is equal to $2A$, the matrix obtained doubling all the entries of A. Hence for the associated quadratic forms:

$$(D^2F \cdot \lambda, \lambda) = 2\,(A \cdot \lambda, \lambda)\ ,$$

and D^2F is positive semi-definite if and only if A is positive semi-definite. By the convexity criterion for C^2 functions (see previous section), F is convex on $\mathbb{R}^n$ if and only if A is positive semi-definite.

Proof (Second Method) This argument is directly based on the definition of convexity and the *Cauchy-Schwarz inequality of quadratic forms*. We state immediately the inequality and justify it at the end of the present proof. If the matrix A is positive semi-definite, for any $\lambda, \mu \in \mathbb{R}^n$

$$|(A \cdot \lambda, \mu)| \leq (A \cdot \lambda, \lambda)^{1/2} \cdot (A \cdot \mu, \mu)^{1/2} \,. \tag{3.98}$$

Suppose A is positive semi-definite and let us show that the quadratic form $F(\lambda) = (A \cdot \lambda, \lambda)$ is convex on $\mathbb{R}^n$. For any $\lambda, \mu \in \mathbb{R}^n$ and any $t \in [0, 1]$

$$\begin{aligned} &F(t\lambda + (1-t)\mu) = \\ &= (A \cdot (t\lambda + (1-t)\mu), t\lambda + (1-t)\mu) = \\ &= t^2 (A \cdot \lambda, \lambda) + t(1-t)(A \cdot \lambda, \mu) + t(1-t)(A \cdot \mu, \lambda) + (1-t)^2 (A \cdot \mu, \mu) \,. \end{aligned}$$

By the symmetry of A we have $(A \cdot \lambda, \mu) = (A \cdot \mu, \lambda)$. Furthermore, using the Cauchy-Schwarz inequality (3.98),

$$\begin{aligned} &F(t\lambda + (1-t)\mu) = \\ &= t^2 (A \cdot \lambda, \lambda) + 2t(1-t)(A \cdot \lambda, \mu) + (1-t)^2 (A \cdot \mu, \mu) \leq \\ &\leq t^2 (A \cdot \lambda, \lambda) + \\ &\quad + 2t(1-t)(A \cdot \lambda, \lambda)^{1/2} \cdot (A \cdot \mu, \mu)^{1/2} + (1-t)^2 (A \cdot \mu, \mu) = \\ &= \left[t (A \cdot \lambda, \lambda)^{1/2} + (1-t)(A \cdot \mu, \mu)^{1/2} \right]^2 , \end{aligned}$$

and so

$$(A \cdot (t\lambda + (1-t)\mu), t\lambda + (1-t)\mu)^{1/2} \leq t (A \cdot \lambda, \lambda)^{1/2} + (1-t)(A \cdot \mu, \mu)^{1/2} \,,$$

i.e. $\lambda \in \mathbb{R}^n \to G(\lambda) = (A \cdot \lambda, \lambda)^{1/2}$ is convex. But then $F(\lambda) = [G(\lambda)]^2$ is convex on $\mathbb{R}^n$ as well. In fact, by the monotonicity and convexity of $s \to s^2$ when $s \geq 0$, we have

$$F(t\lambda + (1-t)\mu) = [G(t\lambda + (1-t)\mu)]^2 \leq [tG(\lambda) + (1-t)G(\mu)]^2 \leq$$

$$\leq t\,[G(\lambda)]^2 + (1-t)\,[G(\mu)]^2 = tF(\lambda) + (1-t)F(\mu)\,.$$

Vice versa, if $F(\lambda) = (A \cdot \lambda, \lambda)$ is convex on $\mathbb{R}^n$, as $0 \in \mathbb{R}^n$ is a convex combination of λ and $-\lambda$,

$$0 = \frac{1}{2}\lambda + \frac{1}{2}(-\lambda)\,,$$

we see A is positive semi-definite, because $(A \cdot (-\lambda), (-\lambda)) = (A \cdot \lambda, \lambda)$, and

$$(A \cdot \lambda, \lambda) = \frac{1}{2}\,(A \cdot \lambda, \lambda) + \frac{1}{2}\,(A \cdot (-\lambda), (-\lambda)) \geq$$

$$\geq \left(A \cdot \left(\frac{1}{2}\lambda + \frac{1}{2}(-\lambda)\right), \frac{1}{2}\lambda + \frac{1}{2}(-\lambda)\right) = (A \cdot 0, 0) = 0\,.$$

In the above proof we have used the following:

Cauchy-Schwarz Inequality for Quadratic Forms *Let A be a positive semi-definite symmetric matrix. Then for any $\lambda, \mu \in \mathbb{R}^n$*

$$|(A \cdot \lambda, \mu)| \leq (A \cdot \lambda, \lambda)^{1/2} \cdot (A \cdot \mu, \mu)^{1/2}\,. \tag{3.99}$$

Proof For any $\lambda, \mu \in \mathbb{R}^n$, the symmetry of A gives $(A \cdot \lambda, \mu) = (A \cdot \mu, \lambda)$. Hence for any $\lambda, \mu \in \mathbb{R}^n$ and $t \in \mathbb{R}$

$$(A \cdot (t\lambda + \mu), t\lambda + \mu) = t^2\,(A \cdot \lambda, \lambda) + t\,(A \cdot \lambda\mu) + t\,(A \cdot \mu, \lambda) + (A \cdot \mu, \mu) =$$

$$= t^2\,(A \cdot \lambda, \lambda) + 2t\,(A \cdot \lambda, \mu) + (A \cdot \mu, \mu)\,.$$

The quadratic polynomial $t^2\,(A \cdot \lambda, \lambda) + 2t\ (A \cdot \lambda, \mu) + (A \cdot \mu, \mu)$ is non-negative for any $t \in \mathbb{R}$ because A is positive semi-definite. Hence its discriminant Δ is less than or equal to zero:

$$\Delta = 4\left\{[(A \cdot \lambda, \mu)]^2 - (A \cdot \lambda, \lambda) \cdot (A \cdot \mu, \mu)\right\} \leq 0\,,$$

which is claim (3.99).

3.17 The Maximum Principle for Harmonic Functions

Let Ω be an open subset of $\mathbb{R}^n$. A C^2 function $u : \Omega \to \mathbb{R}$ is said to be *harmonic* on Ω if it solves the linear, second-order partial differential equation

$$\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \ldots + \frac{\partial^2 u}{\partial x_n^2} = 0\,, \qquad x \in \Omega\,.$$

The above PDE is called *Laplace equation* and is written more compactly as $\Delta u = 0$, i.e.

$$\Delta u = \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = 0\,, \qquad x \in \Omega\,.$$

In dimension one, $n = 1$, where $\Omega = (a, b)$ is a real open interval, a harmonic function $u : (a, b) \to \mathbb{R}$ satisfies $u'' = 0$ for any $x \in (a, b)$. Integrating twice it is straightforward to see that any function of that type is affine on (a, b), i.e. of the form $u(x) = m\,x + q$ with $m, q \in \mathbb{R}$.

If $n \geq 2$, there exist many harmonic maps besides affine functions

$$u(x) = q + \sum_{i=1}^{n} m_i x_i\,, \qquad x = (x_i) \in \Omega\,,$$

$m_i, q \in \mathbb{R}$. For instance, when $n = 2$, the following are all harmonic on their domains of definition (the reader should check this):

$$u(x, y) = x^2 - y^2\,, \qquad u(x, y) = \log\sqrt{x^2 + y^2}\,,$$

$$u(x, y) = e^x \cos\, y\,, \qquad u(x, y) = \operatorname{arctg}\frac{x}{y}\,.$$

For $n \geq 3$, the function

$$u(x) = |x|^{2-n}\,, \qquad x \in \Omega = \mathbb{R}^n - \{0\}$$

is harmonic. The function $u(x, y) = x^3 - 3xy^2$ is harmonic on $\mathbb{R}^2$; its graph, when (x, y) is in the square $[-1, 1] \times [-1, 1]$, is plotted in Fig. 3.23. The gradient is zero only at $(x, y) = (0, 0)$, but this is not a maximum nor a minimum point of u (the reader should check this analytically). This is a general feature: no harmonic function has local extrema on the interior of its domain of definition. The maximum principle, discussed below, is based on this fact.

The reader should observe the level curves of $u(x, y) = x^3 - 3xy^2$ in Fig. 3.24. They are not closed (because the function does not have interior local extrema) and therefore the curves all reach the boundary.

(continued)

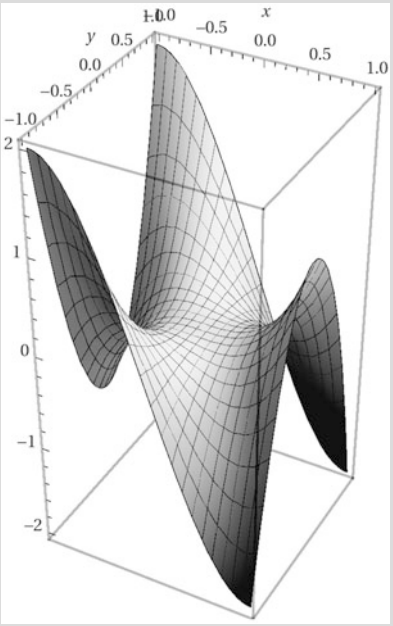


Fig. 3.23 Graph of $u(x, y) = x^3 - 3xy^2$

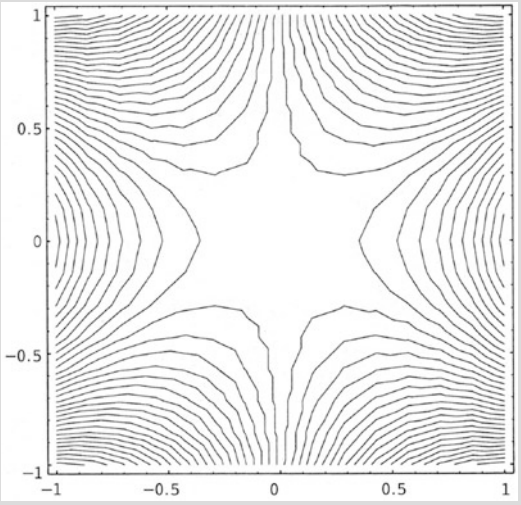


Fig. 3.24 Level curves of $u(x, y) = x^3 - 3xy^2$

Let Ω be an open *bounded* subset of $\mathbb{R}^n$ and u a function defined on $\overline{\Omega}$ and continuous there, i.e. $u \in C^0(\overline{\Omega})$. Denote by m, M the minimum and maximum of u on the boundary $\partial\Omega$ of Ω (these exist by the Weierstrass theorem, since $\partial\Omega$ is compact in $\mathbb{R}^n$):

$$m = \min\{u(x) : x \in \partial\Omega\}, \qquad M = \max\{u(x) : x \in \partial\Omega\}. \tag{3.100}$$

If $u \in C^2(\Omega)$ is harmonic on Ω, moreover, the maximum and minimum of u *on the entire* $\overline{\Omega}$ are equal to M, m respectively. We have in fact:

Maximum Principle *Let Ω be an open bounded set in $\mathbb{R}^n$ and $u \in C^0(\overline{\Omega}) \cap C^2(\Omega)$ a harmonic function on Ω. Call m, M the minimum and maximum of u on the boundary $\partial\Omega$, as in* (3.100)*. Then*

$$m \le u(x) \le M, \qquad \forall x \in \overline{\Omega}. \tag{3.101}$$

A consequence of the maximum principle is that a harmonic function u cannot have an absolute maximum or minimum point that is *interior* and *strict*. For instance, for maxima, there is no $x_0 \in \Omega$ such that $u(x_0) > u(x)$ for any $x \in \overline{\Omega} - \{x_0\}$.

For example, $u(x) =$ constant satisfies the maximum principle's assumptions (and the conclusion!), and every point in an open and bounded subset $\Omega \subset \mathbb{R}^n$ is a maximum and a minimum point for u interior to $\overline{\Omega}$, but not strict.

The hypothesis that the open set Ω is *bounded* is essential. For example, for $n = 2$ the function $u(x, y) = e^x \cos y$ is harmonic on the unbounded set

$$\Omega = \left\{(x, y) \in \mathbb{R}^2 : -\frac{\pi}{2} < y < \frac{\pi}{2}\right\}$$

and vanishes along the boundary $\partial\Omega = \left\{(x, y) \in \mathbb{R}^2 : |y| = \pi/2\right\}$. Therefore the numbers m, M in (3.100) are both zero and (3.101) does not hold on Ω. The reader can verify that, on the other hand, estimate (3.101) holds on the bounded set $\Omega \cap \{(x, y) \in \mathbb{R}^2 : |x| < r\}$, whichever $r > 0$ is chosen.

Proof of the Maximum Principle For any $\varepsilon > 0$ define

$$v_\varepsilon(x) = u(x) + \varepsilon |x|^2, \qquad \forall x \in \overline{\Omega}.$$

The function v_ε is of class $C^0(\overline{\Omega}) \cap C^2(\Omega)$, and for any $x \in \Omega$

$$\Delta v_\varepsilon(x) = \Delta u(x) + \varepsilon \Delta |x|^2 =$$
$$= \varepsilon \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2} \left(x_1^2 + x_2^2 + \ldots + x_n^2 \right) = 2n\varepsilon > 0 .$$

The second-order necessary conditions for local extrema guarantee that v_ε cannot have local maxima in Ω, because if x_0 were one such then

$$\Delta v_\varepsilon(x_0) = \sum_{i=1}^{n} \frac{\partial^2 v_\varepsilon}{\partial x_i^2}(x_0) \leq 0 .$$

Hence the absolute maximum of v_ε on $\overline{\Omega}$, which exists by the Weierstrass theorem, is reached on the boundary $\partial\Omega$, and

$$v_\varepsilon(x) \leq \max \{ v_\varepsilon(x) : x \in \partial\Omega \} = \max \left\{ u(x) + \varepsilon |x|^2 : x \in \partial\Omega \right\} \qquad \forall x \in \overline{\Omega}.$$

As Ω is bounded, there exists $R > 0$ such that $|x| \leq R$ for any $x \in \Omega$, and even for any $x \in \overline{\Omega}$. Then, for such x,

$$v_\varepsilon(x) \leq \max \left\{ u(x) + \varepsilon |x|^2 : x \in \partial\Omega \right\} \leq$$
$$\leq \max \left\{ u(x) + \varepsilon R^2 : x \in \partial\Omega \right\} = M + \varepsilon R^2 ,$$

i.e.

$$u(x) + \varepsilon |x|^2 \leq M + \varepsilon R^2 , \qquad \forall x \in \overline{\Omega} .$$

Taking the limit as $\varepsilon \to 0$ gives $u(x) \leq M$ for any $x \in \overline{\Omega}$.

The other inequality is proved similarly, considering the local minima of $w_\varepsilon(x) = u(x) - \varepsilon |x|^2$.

Many applications often require to solve the following differential problem, called *Dirichlet problem*. Consider an open bounded set $\Omega \subseteq \mathbb{R}^n$ and two functions $f, g : \overline{\Omega} \to \mathbb{R}$ that are continuous on $\overline{\Omega}$. The problem asks to determine a function $u \in C^0(\overline{\Omega}) \cap C^2(\Omega)$ that solves the differential equation

$$\Delta u(x) = f(x) , \qquad x \in \Omega , \tag{3.102}$$

and satisfies

$$u(x) = g(x) , \qquad \forall x \in \partial\Omega \tag{3.103}$$

on the boundary of Ω. It can be proved that, under suitable hypotheses on f and $\partial\Omega$, such a solution u exists. Here we just prove that the Dirichlet problem (3.102) and (3.103) has one solution at most.

Solution's Uniqueness for Dirichlet Problems *The Dirichlet problem* (3.102) *and* (3.103) *admits one solution at most.*

Proof Suppose u_1, u_2 are of class $C^0(\overline{\Omega}) \cap C^2(\Omega)$ and satisfy the differential equation (3.102) and the boundary condition (3.103). The difference $u = u_1 - u_2$ solves the Laplace equation

$$\Delta u(x) = \Delta u_1(x) - \Delta u_2(x) = f(x) - f(x) = 0\,, \qquad x \in \Omega$$

and the boundary condition

$$u(x) = u_1(x) - u_2(x) = g(x) - g(x) = 0\,, \qquad \forall\, x \in \partial\Omega\,.$$

By the maximum principle

$$m \le u(x) \le M, \qquad \forall\, x \in \overline{\Omega}\,,$$

where $m = \min\{u(x) : x \in \partial\Omega\} = 0\,,\ M = \max\{u(x) :\ x \in \partial\Omega\} = 0\,$. Therefore $u = 0$ on $\overline{\Omega}$, meaning $u_1 = u_2$ on $\overline{\Omega}$.



Ordinary Differential Equations

4

4.1 Introduction: The Initial Value Problem

Let $[a, b]$ be a real interval and $g(x)$ a continuous function on $[a, b]$. The problem of finding a *primitive* of $g(x)$ on $[a, b]$, i.e. a function $y(x)$ such that

$$y'(x) = g(x)\,, \qquad \forall\, x \in [a, b]\,, \tag{4.1}$$

is solved by the *fundamental theorem of calculus*. Namely any primitive of g, i.e. a solution to Eq. (4.1), has the form

$$y(x) = y_0 + \int_{x_0}^{x} g(t)\,dt\,, \qquad \forall\, x \in [a, b]\,, \tag{4.2}$$

where x_0 is a given point in $[a, b]$ and y_0 is an arbitrary real number.

Equation (4.1) is a very special example of *differential equation*. The solution $y(x)$ represented in (4.2) satisfies not only Eq. (4.1), but also the *initial condition*

$$y(x_0) = y_0 + \int_{x_0}^{x_0} g(t)\,dt = y_0\,.$$

One says $y(x)$ in (4.2) solves the *initial value problem*, or *Cauchy problem*,

$$\begin{cases} y'(x) = g(x) \\ y(x_0) = y_0. \end{cases}$$

We also say the differential equation (4.1) is of *order one*, because the highest derivative that appears is the first derivative.

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N. Fusco et al., *Mathematical Analysis*, La Matematica per il 3+2 137,
https://doi.org/10.1007/978-3-031-04151-8_4

Another example of differential equation of order one is

$$y'(x) = \lambda\, y(x)\,, \qquad \forall\, x \in \mathbb{R}\,, \tag{4.3}$$

where λ is a given real parameter. A solution to (4.3) is a function $y = y(x)$, differentiable on $\mathbb{R}$, such that $y'(x) = \lambda y(x)$, for any $x \in \mathbb{R}$.

Immediately, a solution to Eq. (4.3) is the function

$$y(x) = ce^{\lambda x}\,, \qquad \forall\, x \in \mathbb{R}\,, \tag{4.4}$$

where c is an arbitrary constant in $\mathbb{R}$. In fact, differentiating (4.4) gives $y'(x) = c\lambda e^{\lambda x} = \lambda y(x)$.

An interesting problem is establishing whether all possible solutions of (4.3) are of the form (4.4), as the parameter c varies.

The answer is yes: suppose $y(x)$ solves (4.3) and put

$$z(x) = y(x) \cdot e^{-\lambda x}\,, \qquad \forall\, x \in \mathbb{R}\,.$$

Since, by assumption, $y'(x) - \lambda y(x) = 0$, we have

$$z'(x) = y'(x) \cdot e^{-\lambda x} + y(x) \cdot (-\lambda) \cdot e^{-\lambda x} =$$

$$= e^{-\lambda x} \cdot (y'(x) - \lambda y(x)) = 0\,, \qquad \forall\, x \in \mathbb{R}\,.$$

Hence $z(x) = c =$ constant on $\mathbb{R}$ and $y(x) \cdot e^{-\lambda x} = c$, i.e. formula (4.4) holds.

The *Cauchy problem* associated with the differential equation (4.3) is

$$\begin{cases} y'(x) = \lambda y(x) \\ y(x_0) = y_0\,. \end{cases} \tag{4.5}$$

with given x_0, y_0 in $\mathbb{R}$. As every solution to the differential equation is of the form (4.4), $y(x) = ce^{\lambda x}$, to solve the initial value problem (4.5) we need to find, if possible, the constant c so to satisfy the *initial condition* $y(x_0) = y_0$. We have

$$y(x_0) = ce^{\lambda x_0} = y_0\,,$$

whence $c = y_0 e^{-\lambda x_0}$. Therefore the Cauchy problem (4.5) is solvable, and the unique solution is

$$y(x) = y_0 e^{\lambda(x - x_0)}\,.$$

An example of *second-order* differential equation is

$$y''(x) = \lambda y'(x)\,, \qquad \forall\, x \in \mathbb{R}\,, \tag{4.6}$$

where λ is a given parameter in $\mathbb{R}$.

The substitution $z(x) = y'(x)$ transforms (4.6) into

$$z'(x) = \lambda z(x)\,,$$

which was solved earlier and has as solutions

$$z(x) = ce^{\lambda x}\,, \qquad \forall\, x \in \mathbb{R}\,.$$

As $y'(x) = z(x) = ce^{\lambda x}$, the solutions $y(x)$ to the second-order equation (4.6) are, if $\lambda \neq 0$,

$$y(x) = \int ce^{\lambda x}\,dx = c_1 + \frac{c}{\lambda}e^{\lambda x} = c_1 + c_2 e^{\lambda x}\,,$$

where c_1, c_2 are real constants (in particular, we have set $c_2 = c/\lambda$, $\lambda \neq 0$; as c is an arbitrary constant, so is c_2). When $\lambda = 0$, instead,

$$y(x) = \int c\,dx = c_1 + cx\,, \tag{4.7}$$

with c_1, c arbitrary constants. The *initial conditions* to be imposed in order to uniquely determine the function $y(x)$ among all possible solutions to the second-order equation (4.6) are

$$y(x_0) = y_0\,, \qquad y'(x_0) = y_0'\,, \tag{4.8}$$

where y_0, y_0' are given real numbers. In the case at hand, if $\lambda \neq 0$ we obtain

$$y(x_0) = c_1 + c_2 e^{\lambda x_0} = y_0\,, \qquad y'(x_0) = c_2 \lambda e^{\lambda x_0} = y_0'\,,$$

and solving for c_1, c_2 we can write the solution $y(x)$ as

$$y(x) = y_0 + \frac{y_0'}{\lambda}(e^{\lambda(x-x_0)} - 1)\,.$$

If $\lambda = 0$, starting from (4.7) we find the solution

$$y(x) = y_0 + y_0'(x - x_0)\,. \tag{4.9}$$

To verify the result just found, the reader should note that, for $\lambda = 0$, the second-order equation (4.6) reduces to $y''(x) = 0$ for any $x \in \mathbb{R}$. The latter is satisfied precisely by the functions whose graph is a straight line. Imposing conditions (4.8) determines a unique line, expressed analytically by (4.9).

Summarising the computations of example (4.6), we have proved that the following *initial value problem*

$$\begin{cases} y''(x) = \lambda y'(x) \\ y(x_0) = y_0 \\ y'(x_0) = y_0' \,, \end{cases}$$

admits one, and only one, solution, whatever the values of the real parameters λ, x_0, y_0, y_0' are.

Another differential equation of second order, relevant for the applications, is the *equation of harmonic motion*

$$y'' + \omega^2 y = 0 \,, \tag{4.10}$$

where $\omega \in \mathbb{R}$, $\omega \neq 0$ (note that for $\omega = 0$ the differential equation (4.10) coincides with (4.6) with $\lambda = 0$, treated earlier). A family of solutions is

$$y(x) = c_1 \cos \omega x + c_2 \, \sin \omega x \,, \tag{4.11}$$

with c_1, c_2 constants in $\mathbb{R}$ (the reader should check this).

Let us show that every solution to Eq. (4.10) is of the form (4.11). For this we consider a generic solution $y(x)$ to (4.10). Define $z_1(x)$, $z_2(x)$ to be solutions of the linear system in 2 equations and 2 unknowns

$$\begin{cases} y(x) = z_1(x) \cos \omega x + z_2(x) \, \sin \omega x \\ \dfrac{y'(x)}{\omega} = -z_1(x) \, \sin \omega x + z_2(x) \cos \omega x \,. \end{cases} \tag{4.12}$$

The coefficients' determinant is non-zero, since

$$\begin{vmatrix} \cos \omega x & \sin \omega x \\ -\sin \omega x & \cos \omega x \end{vmatrix} = 1 \,, \qquad \forall \, x \in \mathbb{R} \,,$$

hence system (4.12) admits unique solutions for any $x \in \mathbb{R}$, and $z_1(x)$, $z_2(x)$ are well defined (moreover, computing the values explicitly, one sees they are differentiable on $\mathbb{R}$). Differentiating the first equation in (4.12) and using the second one we find

$$\begin{aligned} y' &= (z_1' \cos \omega x + z_2' \sin \omega x) + \omega(-z_1 \sin \omega x + z_2 \cos \omega x) = \\ &= (z_1' \cos \omega x + z_2' \sin \omega x) + y' \,, \end{aligned}$$

so

$$z_1' \cos \omega x + z_2' \sin \omega x = 0 \,. \tag{4.13}$$

Differentiating the second equation of (4.12) and using the first,

$$\frac{y''}{\omega} = (-z_1' \sin \omega x + z_2' \cos \omega x) + \omega(-z_1 \cos \omega x - z_2 \sin \omega x) =$$
$$= (-z_1' \sin \omega x + z_2' \cos \omega x) - \omega y \,,$$

so from $y'' + \omega^2 y = 0$ it follows

$$-z_1' \sin \omega x + z_2' \cos \omega x = 0 \,. \tag{4.14}$$

By (4.13) and (4.14), the derivatives z_1', z_2' solve the 2×2 homogeneous linear system

$$\begin{cases} z_1' \cos \omega x + z_2' \sin \omega x = 0 \\ -z_1' \sin \omega x + z_2' \cos \omega x = 0 \,. \end{cases}$$

As the coefficients' determinant is identically 1, there is a unique solution, given by the pair $(0, 0)$, i.e.

$$z_1'(x) = z_2'(x) = 0 \,, \qquad \forall\, x \in \mathbb{R} \,.$$

Therefore $z_1(x) = \text{constant} = c_1$, $z_2(x) = \text{constant} = c_2$, and (4.11) holds.

So all solutions to the equation of *harmonic motion* (4.10) are represented by formula (4.11). Consequently, whichever the values of the real parameters $\omega \neq 0$, y_0, y_0', the *initial value problem*

$$\begin{cases} y''(x) + \omega^2 y(x) = 0 \\ y(0) = y_0 \\ y'(0) = y_0' \,, \end{cases}$$

(just for simplicity we chose $x_0 = 0$ as initial point) admits a unique solution, given by

$$y(x) = y_0 \cos \omega x + \frac{y_0'}{\omega} \sin \omega x \,.$$

The latter arises from (4.11) by choosing the constants c_1, c_2 in order to satisfy the initial conditions $y(0) = y_0$ and $y'(0) = y_0'$.

Given $n \in \mathbb{N}$, the simplest *differential equation of order n* is

$$y^{(n)}(x) = g(x) \,, \qquad \forall\, x \in [a, b] \,, \tag{4.15}$$

where $g : [a, b] \to \mathbb{R}$ is a given continuous function. For $n = 1$ we recover the problem of finding the primitives of $g(x)$, as of (4.1).

A solution to (4.15) is a function $y(x)$ defined on $[a, b]$, differentiable n times on that interval and whose n^{th} derivative is equal to g on $[a, b]$. We immediately see that any such solution has *continuous* n^{th} derivative on $[a, b]$, since the latter coincides with $g(x)$. Therefore we can use Taylor's formula of order n with integral remainder: given $x_0 \in [a, b]$, for any $x \in [a, b]$ we have

$$\begin{aligned} y(x) = y(x_0) + y'(x_0)(x - x_0) + \frac{y''(x_0)}{2}(x - x_0)^2 + \ldots + \\ + \frac{y^{(n-1)}(x_0)}{(n-1)!}(x - x_0)^{n-1} + \int_{x_0}^{x} \frac{(x-t)^{n-1}}{(n-1)!}\, y^{(n)}(t)\, dt\,. \end{aligned} \tag{4.16}$$

Due to $y^{(n)}(x) = g(x)$, for any $x \in [a, b]$, we substitute this value in the integral of (4.16), and it is easy to verify that the function

$$\begin{aligned} y(x) = y_0 + y_0'(x - x_0) + \frac{y_0''}{2}(x - x_0)^2 + \ldots + \\ + \frac{y_0^{(n-1)}}{(n-1)!}(x - x_0)^{n-1} + \int_{x_0}^{x} \frac{(x-t)^{n-1}}{(n-1)!}\, g(t)\, dt\,, \end{aligned} \tag{4.17}$$

solves the *initial value problem* of order n

$$\begin{cases} y^{(n)} = g(x) \\ y(x_0) = y_0 \\ y'(x_0) = y_0' \\ \ldots\ldots\ldots\ldots \\ y^{(n-1)}(x_0) = y_0^{(n-1)} \end{cases}$$

for equation (4.15), where y_0, y_0', …, $y_0^{(n-1)}$ are n given real numbers. Notice that formula (4.17) generalises to $n \geq 1$ expression (4.2), corresponding to $n = 1$.

The differential equations considered thus far, of order one, two, $n \in \mathbb{N}$, have a common feature: they are *linear* differential equations, because the relationship between the derivatives of the unknown function $y(x)$ is linear. An example of initial value problem for a non-linear equation of order one is

$$\begin{cases} y' = y^2 \\ y(0) = 1\,. \end{cases} \tag{4.18}$$

Let us show (without thinking about uniqueness, for the time being) that

$$y(x) = \frac{1}{1 - x} \tag{4.19}$$

solves the initial value problem (4.18): $y(0) = 1$ and

$$y'(x) = \frac{1}{(1-x)^2} = y^2 .$$

We discover a novel phenomenon compared to the earlier linear equations. Before, in fact, all equations and associated initial value problems could be solved on the interval of definition ($[a, b]$ or $\mathbb{R}$) of the problem's data (we will show this is the case for all *linear* differential equations). In case (4.18), on the contrary, equation $y'(x) = [y(x)]^2$ is a priori defined for any $x \in \mathbb{R}$, while the solution (4.19) is defined on a proper subset, the interval $(-\infty, 1)$: this is a neighbourhood of $x_0 = 0$, at which the initial condition $y(0) = 1$ is given.

The reader will have noticed that the function $y(x)$ in (4.19) is also defined on $(1, +\infty)$, which does not contain the initial point $x_0 = 0$. Since $y(x)$ is not differentiable everywhere on $(-\infty, +\infty)$, it cannot be a global solution of the differential equation. Hence the interval $(1, +\infty)$ cannot be considered, i.e., the function $y(x)$ in (4.19) is solution of the initial value problem (4.18) on the interval $(-\infty, 1)$.

The previous discussion also says that, in general, we can consider at least two ways of solving an initial value problem: (1) *globally*, by which we determine a solution *on the entire interval, which is given beforehand*, where the equation is defined; (2) *locally*, whereby a solution is defined *on a neighbourhood of the point* x_0 in the interval of concern.

The most general differential equation of order n is

$$F(x, y, y', y'', \dots, y^{(n)}) = 0 , \tag{4.20}$$

where F is a real function in $n+2$ real variables defined on an open subset of $\mathbb{R}^{n+2}$.

This is referred to as an *ordinary differential equation* (ODE), to distinguish it from a *partial differential equation* (PDE), which, as the name says, involves the partial derivatives of the unknown function. For example the *Laplace equation*

$$\Delta u = \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = 0 , \qquad x \in \Omega ,$$

seen in Sect. 3.17, is a linear PDE of order two. The unknown $u : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ is a function in several real variables.

The possibility of solving (4.20) for the highest-order derivative $y^{(n)}$ is a different problem from that of actually solving the equation, and will be addressed in Chapter 11, on *implicit functions*. In the sequel we shall consider differential equations of order $n \in \mathbb{N}$ that are in so-called *normal* form

$$y^{(n)} = f(x, y, y', y'', \dots, y^{(n-1)}) , \tag{4.21}$$

where f is a function of $n+1$ variables defined on an open set in $\mathbb{R}^{n+1}$.

For example, the first-order *differential* equation $(y')^2+1=0$ has no real solutions because the *algebraic* equation $t^2+1=0$ cannot be solved for $t\in\mathbb{R}$.

The first-order ODE $(y')^2+x^3=0$, instead, is equivalent to the pair of *normal* differential equations

$$y'=(-x)^{3/2}=|x|^{3/2}\,,\qquad x\in(-\infty,0]\,,$$
$$y'=-(-x)^{3/2}=-|x|^{3/2}\,,\qquad x\in(-\infty,0]\,,$$

or more succinctly,

$$y'(x)=\pm|x|^{3/2}\,,\qquad x\in(-\infty,0]\,.$$

Fix a point $(x_0,y_0,y_0',y_0'',\dots,y_0^{(n-1)})$ in $\mathbb{R}^{n+1}$. Given a function f in $n+1$ variables defined on a neighbourhood of the point, we may consider the *Cauchy problem*

$$\begin{cases} y^{(n)}=f(x,y,y',y'',\dots,y^{(n-1)})\\ y(x_0)=y_0\\ y'(x_0)=y_0'\\ \dots\dots\dots\dots\dots\dots\dots\\ y^{(n-1)}(x_0)=y_0^{(n-1)}\,, \end{cases}$$

relative to the normal ODE (4.21) of order n (in general non-linear).

In even more generality we can consider a Cauchy problem relative to a *system* of ODEs in normal form. The unknown is now a *vector-valued function*

$$y(x)=(y_1(x),y_2(x),\dots,y_k(x))\,,$$

for a given k in $\mathbb{N}$. The system of differential equations can be written as

$$\begin{cases} y_1^{(n_1)}=f_1(x,y_1,y_2,\dots,y_k,\dots,y_1^{(n_1-1)},y_2^{(n_2-1)},\dots,y_k^{(n_k-1)})\\ y_2^{(n_2)}=f_2(x,y_1,y_2,\dots,y_k,\dots,y_1^{(n_1-1)},y_2^{(n_2-1)},\dots,y_k^{(n_k-1)})\\ \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\\ y_k^{(n_k)}=f_k(x,y_1,y_2,\dots,y_k,\dots,y_1^{(n_1-1)},y_2^{(n_2-1)},\dots,y_k^{(n_k-1)})\,. \end{cases}\tag{4.22}$$

The system is said to have order n_1 in the unknown y_1, order n_2 in the unknown y_2 etc. For short, the ODE system (4.22) is of order $n=\max\{n_1,n_2,\dots,n_k\}$.

A *differential equation of order n* in normal form

$$y^{(n)}(x)=f(x,y,y',y'',\dots,y^{(n-1)})\,,\tag{4.23}$$

like (4.21), can always be recast as a *system of ODEs* in normal form of *order one*, by substituting

$$y = y_1, \quad y' = y_2, \quad y'' = y_3, \ldots, \quad y^{(n-1)} = y_n\,;$$

Thus Eq. (4.23) becomes equivalent to the ODE system

$$\begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ \ldots\ldots\ldots\ldots\ldots \\ y_{n-1}' = y_n \\ y_n' = f(x, y_1, y_2, \ldots, y_n). \end{cases}$$

Clearly, a substitution of this sort will turn *an ODE system of order larger than* 1, like (4.22), *into a first-order system*.

Hence a generic system of normal ODEs, of arbitrary order, can be equivalently expressed as a system of order-one normal ODEs of the type

$$\begin{cases} y_1' = f_1(x, y_1, y_2, \ldots, y_n) \\ y_2' = f_2(x, y_1, y_2, \ldots, y_n) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_n' = f_n(x, y_1, y_2, \ldots, y_n). \end{cases} \tag{4.24}$$

The *vector-valued function* $y(x) = (y_1(x), y_2(x), \ldots, y_n(x))$ is the system's unknown. Introducing the vector-valued function $f = (f_1, f_2, \ldots, f_n)$, system (4.24) can be written as

$$y' = f(x, y)\,. \tag{4.25}$$

The reader should bear in mind that in (4.25), x belongs in $\mathbb{R}$ while y and f are vector-valued functions. If, for instance, system (4.25) is defined for any x and y, then $y : \mathbb{R} \to \mathbb{R}^n$ and $f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$.

Fix $x_0 \in \mathbb{R}$ and $y_0 \in \mathbb{R}^n$ so that the point $(x_0, y_0) \in \mathbb{R} \times \mathbb{R}^n$ falls in the domain of f. The corresponding *initial value problem* is written

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0\,. \end{cases} \tag{4.26}$$

In the next section we shall discuss sufficient conditions for the *local solvability*, i.e. on neighbourhoods of x_0, of the initial value problem (4.26). Then, in Sect. 4.4, we shall consider certain assumptions on f meant to ensure the existence of a *global solution* on some interval given a priori.

4.2 Cauchy's Local Existence and Uniqueness Theorem

Here we will prove *Cauchy's theorem* on the local existence and uniqueness of solutions to first-order ODE systems in normal form of the type

$$y' = f(x, y)\,.$$

Fix $x_0 \in \mathbb{R}$ and $y_0 \in \mathbb{R}^n$ and suppose the function f is defined on a neighbourhood

$$I \times J = \{(x, y) \in \mathbb{R} \times \mathbb{R}^n : |x - x_0| \le a\,,\ |y - y_0| \le b\} \tag{4.27}$$

of the point $(x_0, y_0) \in \mathbb{R} \times \mathbb{R}^n$ $(a, b > 0)$.

As usual, $|x - x_0|$ is the *absolute value* of $x - x_0 \in \mathbb{R}$ whereas $|y - y_0|$ is the *norm* (or *modulus*) of the vector $y - y_0 \in \mathbb{R}^n$. Moreover the function f is $\mathbb{R}^n$-valued, i.e. $f : I \times J \to \mathbb{R}^n$.

Assume furthermore that

$$f(x, y) \quad \text{is } \textit{continuous} \text{ on} \quad I \times J \tag{4.28}$$

and *Lipschitz* in the variable y uniformly in $x \in I$, meaning there is a constant $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \le L|y_1 - y_2|\,, \qquad \forall\, x \in I\,,\ y_1, y_2 \in J\,. \tag{4.29}$$

Cauchy Theorem (Local Existence and Uniqueness) *Under hypotheses* (4.28) *and* (4.29)*, there exist a real number $\delta > 0$ and a unique differentiable function $y = y(x)$, $y : [x_0 - \delta, x_0 + \delta] \to \mathbb{R}^n$, which solves the initial value problem*

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases} \tag{4.30}$$

on $[x_0 - \delta, x_0 + \delta]$.

If $f(x, y)$ verifies the hypotheses of Cauchy's theorem (f continuous and Lipschitz in y) only when x belongs in a right neighbourhood $[x_0, x_0 + a]$ of x_0, the same proof shows that there exist a number $\delta > 0$ and a unique function $y = y(x)$, $y : [x_0, x_0 + \delta] \to \mathbb{R}^n$, defined and differentiable on $[x_0, x_0 + \delta]$, that solves (4.30) on the right neighbourhood. Clearly a similar statement holds for left neighbourhoods. Moreover, analogous considerations can be made for the ensuing existence results.

The number $\delta > 0$ in Cauchy's statement can be estimated explicitly. Below we offer two arguments: the first, of *functional nature* (it uses the fixed-point theorem

in *metric spaces*, in particular *spaces of functions*), provides the following estimate:

$$\delta < \min\left\{a;\ \frac{b}{M};\ \frac{1}{L}\right\},$$

with a, b coming from (4.27), L given in (4.29) and

$$M = \max\{|f(x, y)| :\ (x, y) \in I \times J\}. \tag{4.31}$$

Observe that such M must exist (finite) by the Weierstrass theorem.

The second argument, given below, relies on *real analysis* (it depends on properties of definite integrals and the uniform convergence of series and sequences of functions) and estimates δ more accurately:

$$\delta = \min\left\{a;\ \frac{b}{M}\right\}.$$

Anyway, Cauchy's theorem ensures the local existence and uniqueness of a solution to the initial value problem (4.30), i.e. of a solution defined on a neighbourhood $[x_0-\delta, x_0+\delta]$ of x_0 which is typically smaller than the initial interval $[x_0-a, x_0+a]$.

As already mentioned, we will prove the result in two ways. It may be instructive for the reader to read both, in order to spot the different methods, analyse the varying level of abstraction and understand the similarities. At a successive reading the two arguments usually become more similar than what might seem at first.

In both proofs of Cauchy's theorem, given below, the first step consists in formulating the initial value problem (4.30) in an equivalent *integral form*. The quantities x_0, y_0 and δ have the same meaning as before.

Initial Value Problem in Integral Form *Fix $\delta > 0$. The following statements are equivalent:*

(i) *there exists a differentiable function $y = y(x)$ on $[x_0 - \delta, x_0 + \delta]$, for which $y'(x) = f(x, y(x))$ for any $x \in [x_0 - \delta, x_0 + \delta]$, and such that $y(x_0) = y_0$;*
(ii) *there exists a continuous function $y = y(x)$ on $[x_0 - \delta, x_0 + \delta]$, such that*

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt\,, \qquad \forall\, x \in [x_0 - \delta, x_0 + \delta]\,.$$

Proof (i) $\Rightarrow$ (ii). By hypothesis there is a function $y = y(x)$, $y : [x_0-\delta, x_0+\delta] \to \mathbb{R}^n$, differentiable on the interval $[x_0 - \delta, x_0 + \delta]$ and such that

$$y'(x) = f(x, y(x)) \tag{4.32}$$

for any x in the interval. If $x \in [x_0 - \delta, x_0 + \delta]$, integrating component-wise both sides of (4.32) between x_0 and x, we obtain

$$y(x) - y(x_0) = \int_{x_0}^{x} y'(t)\, dt = \int_{x_0}^{x} f(t, y(t))\, dt\,,$$

and (ii) follows since $y(x_0) = y_0$ by assumption.
(ii) $\Rightarrow$ (i). Now $y(x)$ is continuous and such that

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt\,, \qquad \forall\, x \in [x_0 - \delta, x_0 + \delta]\,. \tag{4.33}$$

Putting $x = x_0$ in (4.33) gives $y(x_0) = y_0$. By the fundamental theorem of calculus (applied to each component of the vector-valued function $y = y(x)$, $y : [x_0 - \delta, x_0 + \delta] \to \mathbb{R}^n$), $y(x)$ is differentiable with derivative

$$y'(x) = f(x, y(x))\,, \qquad \forall\, x \in [x_0 - \delta, x_0 + \delta]\,.$$

Hence (i) holds.

It will be useful, in the proof of Cauchy's theorem (but also elsewhere, e.g. in Sect. 6.3, to establish the formula for the length of a regular curve in $\mathbb{R}^n$), to know the following result, that compares the integral of the modulus with the modulus of the integral. It is a generalisation to $n > 1$ of a property of scalar functions that the reader will certainly know.

Lemma *Let $g : [a, b] \to \mathbb{R}^n$ be a continuous function. Then*

$$\left| \int_a^b g(t)\, dt \right| \le \int_a^b |g(t)|\, dt\,. \tag{4.34}$$

If g has components $(g_1, g_2, \ldots, g_n)$, the left-hand-side integral denotes the vector with components

$$\int_a^b g(t)\, dt = \left(\int_a^b g_1(t)\, dt, \int_a^b g_2(t)\, dt, \ldots, \int_a^b g_n(t)\, dt \right), \tag{4.35}$$

while as customary $|g(t)|$ is the Euclidean norm (or modulus) of $g(t)$:

$$|g(t)| = \sqrt{g_1^2(t) + g_2^2(t) + \ldots + g_n^2(t)}\,.$$

Proof of the Lemma Call v the vector in (4.35):

$$v = (v_1, v_2, \ldots, v_n) = \int_a^b g(t) dt\,. \tag{4.36}$$

The modulus squared $|v|^2$ is the inner product of v with itself (as for any vector in $\mathbb{R}^n$), so

$$|v|^2 = (v, v) = v_1 \cdot v_1 + v_2 \cdot v_2 + \ldots + v_n \cdot v_n = \\ = v_1 \int_a^b g_1(t)dt + v_2 \int_a^b g_2(t) + \ldots + v_n \int_a^b g_n(t)dt \,.$$

But v is independent of $t \in [a, b]$, i.e. each component v_i is constant in t, so

$$|v|^2 = \int_a^b \{v_1 g_1(t) + v_2 g_2(t) + \ldots + v_n g_n(t)\}dt \leq \\ \leq \int_a^b |v_1 g_1(t) + v_2 g_2(t) + \ldots + v_n g_n(t)|dt \,. \tag{4.37}$$

The *Cauchy-Schwarz inequality* (2.13) here reads

$$|(v, g(t))| = \left|\sum_{i=1}^n v_i g_i(t)\right| \leq \left(\sum_{i=1}^n v_i^2\right)^{1/2} \cdot \left(\sum_{i=1}^n g_i^2(t)\right)^{1/2} = |v| \cdot |g(t)| \,. \tag{4.38}$$

From (4.37) and (4.38) we then have

$$|v|^2 \leq \int_a^b |v| \cdot |g(t)|dt = |v| \cdot \int_a^b |g(t)|dt \,.$$

When $|v| \neq 0$, dividing by $|v|$ yields

$$|v| \leq \int_a^b |g(t)|dt \,,$$

which is (4.34) once we recall the meaning of $v = \int_a^b g(t)dt$ defined in (4.36). When $|v| = 0$, $\left|\int_a^b g(t)dt\right| = 0$: the left-hand side of (4.34) is zero, and the relation is trivially true.

Proof of Cauchy's Theorem (First Method) We shall prove the integral form (ii). Consider a real number $\delta > 0$ satisfying constraints (4.2). More precisely, suppose $\delta > 0$ satisfies the inequalities

$$\delta \leq \min\left\{a;\ \frac{b}{M}\right\}, \qquad \delta < \frac{1}{L}, \tag{4.39}$$

where a and b are as in (4.27), M is the maximum of $|f|$ defined in (4.31), and L is the Lipschitz constant in (4.29).

Let $I_\delta = [x_0 - \delta, x_0 + \delta]$ be the closed neighbourhood of x_0 of half-width δ. Consider the metric space $C^0(I_\delta)$ of continuous functions $y : I_\delta \to \mathbb{R}^n$ on I_δ with the supremum norm. Then take the set $B \subset C^0(I_\delta)$ of continuous functions on I_δ such that

$$\sup\{|y(x) - y_0| : \quad x \in I_\delta\} \leq b\,.$$

In other words,

$$B = \{y \in C^0(I_\delta) : \|y - y_0\|_{C^0} \leq b\}$$

is a closed ball in $C^0(I_\delta)$, with centre y_0 and radius b.

By Proposition 1 in Sect. 2.8 the set B, being closed in the complete metric space $C^0(I_\delta)$, is a complete metric space as well.

Define $F : B \to B$ by

$$F(y) = z\,, \tag{4.40}$$

where z is the vector-valued function

$$z(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt\,, \qquad x \in I_\delta\,. \tag{4.41}$$

We claim F is a map from B to B, i.e. that for $y \in B$ the above function z belongs to the ball $B \subset C^0(I_\delta)$. In fact, $z \in C^0(I_\delta)$. Moreover, by (4.34),

$$|z(x) - y_0| = \left|\int_{x_0}^{x} f(t, y(t))\, dt\right| \leq \left|\int_{x_0}^{x} |f(t, y(t))|\, dt\right|\,,$$

for any $x \in I_\delta$.

By assumption $|y(t) - y_0| \leq b$ for any $t \in I_\delta$, so $(t, y(t)) \in I \times J$ and then $|f(t, y(t))| \leq M$. Hence, given that $M\delta \leq b$,

$$|z(x) - y_0| \leq \left|\int_{x_0}^{x} |f(t, y(t))|\, dt\right| \leq M|x - x_0| \leq M\delta \leq b\,.$$

Therefore, as said, F maps the metric space B to itself. Now we will show F is a contraction on B. By (4.34) we have

$$\begin{aligned} |F(y_1) - F(y_2)| &= \left|\int_{x_0}^{x} \{f(t, y_1(t)) - f(t, y_2(t))\}\, dt\right| \leq \\ &\leq \left|\int_{x_0}^{x} |f(t, y_1(t)) - f(t, y_2(t))|\, dt\right|\,. \end{aligned}$$

Using the Lipschitz hypothesis (4.29) of f with respect to y we obtain

$$|F(y_1)(x) - F(y_2)(x)| \le \left| \int_{x_0}^{x} |f(t, y_1(t)) - f(t, y_2(t))|\, dt \right| \le$$
$$\le \left| \int_{x_0}^{x} L|y_1(t) - y_2(t)|\, dt \right| \le L\delta \|y_1 - y_2\|_{C^0} ,$$

and taking the supremum over $x \in I_\delta$,

$$\|F(y_1) - F(y_2)\|_{C^0} \le L\delta \|y_1 - y_2\|_{C^0} .$$

Hence $F : B \to B$ is a Lipschitz map. But by (4.39) the Lipschitz constant $L\delta$ is less than 1, so F is a contraction on B.

By the fixed-point theorem (Sect. 2.9) there exists exactly one function $y \in B$ such that $y = F(y)$. Recalling definition (4.40), (4.41) of F, we then have

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt , \qquad \forall\, x \in [x_0 - \delta, x_0 + \delta] ,$$

in other words (ii).

***Proof of Cauchy's Theorem* (*Second Method*)** We will use the notation introduced at the beginning, in particular the numbers a, b (the half-length of the interval I and the radius of the ball J), M (the maximum of $|f(x, y)|$ when $(x, y) \in I \times J$ as in (4.31)) and L (the Lipschitz constant of (4.29)). Consider the positive real number

$$\delta = \min \left\{ a;\ \frac{b}{M} \right\} .$$

Call I_δ the interval $[x_0 - \delta, x_0 + \delta]$ and define the sequence $y_k : I_\delta \to \mathbb{R}^n$, $k = 0, 1, 2, \ldots$, recursively:

$$\begin{cases} y_0(x) = y_0 \\ y_{k+1}(x) = y_0 + \displaystyle\int_{x_0}^{x} f(t, y_k(t))\, dt . \end{cases} \tag{4.42}$$

First, we need to show the sequence y_k is well defined. This means, more precisely, that the integrand $f(t, y_k(t))$ is well defined, that is,

$$y_k(t) \in J = \{y \in \mathbb{R}^n : |y - y_0| \le b\} , \qquad \forall\, t \in I_\delta . \tag{4.43}$$

We shall prove (4.43) by induction on $k = 0, 1, 2, \ldots$. When $k = 0$ evidently $y_0 \in J$. Now suppose $y_k(t) \in J$ for any $t \in I_\delta$, so $(t, y_k(t)) \in I_\delta \times J$ for any $t \in I_\delta$. Hence

$$|f(t, y_k(t))| \le M , \qquad \forall\, t \in I_\delta .$$

By (4.34), furthermore, for $x \in I_\delta$ we have

$$|y_{k+1}(x) - y_0| = \left| \int_{x_0}^{x} f(t, y_k(t))\, dt \right| \le \left| \int_{x_0}^{x} |f(t, y_k(t))|\, dt \right| \le$$
$$\le M|x - x_0| \le M\delta \le b\,,$$

because $\delta \le b/M$ by definition of δ. Hence $y_{k+1}(t) \in J$ for any $t \in I_\delta$, and (4.42) is well defined.

Let us go back to the (well-defined) sequence y_k in (4.42) and proceed recursively, by proving the following estimates. For any $x \in I_\delta$

$$|y_1(x) - y_0| = \left| \int_{x_0}^{x} f(t, y_k(t))\, dt \right| \le M|x - x_0|\,, \tag{4.44}$$

since $f(t, y_k(t)) \le M$ for any $t \in I_\delta$. The Lipschitz assumption in y tells, from (4.44), that

$$\begin{aligned} |y_2(x) - y_1(x)| &= \left| \int_{x_0}^{x} \{f(t, y_1(t)) - f(t, y_0)\}\, dt \right| \le \\ &\le \left| \int_{x_0}^{x} |f(t, y_1(t)) - f(t, y_0)|\, dt \right| \le \left| \int_{x_0}^{x} L|y_1(t) - y_0|\, dt \right| \le \\ &\le ML \left| \int_{x_0}^{x} |t - x_0|\, dt \right| = \frac{ML}{2} |x - x_0|^2\,. \end{aligned} \tag{4.45}$$

Again the Lipschitz condition in y applied to (4.45) gives

$$\begin{aligned} |y_3(x) - y_2(x)| &= \left| \int_{x_0}^{x} \{f(t, y_2(t)) - f(t, y_1(t))\}\, dt \right| \le \\ &\le \left| \int_{x_0}^{x} |f(t, y_2(t)) - f(t, y_1(t))|\, dt \right| \le \\ &\le \left| \int_{x_0}^{x} L|y_2(t) - y_1(t)|\, dt \right| \le \\ &\le \frac{ML^2}{2} \left| \int_{x_0}^{x} |t - x_0|^2\, dt \right| = \frac{ML^2}{3!} |x - x_0|^3\,. \end{aligned}$$

Iterating (or by induction) we arrive at the estimate

$$|y_{k+1}(x) - y_k(x)| \le \frac{ML^k}{(k+1)!} |x - x_0|^{k+1}\,, \qquad \forall\, x \in I_\delta\,. \tag{4.46}$$

Relation (4.46) implies that the sequence $y_k(x)$ converges uniformly on the interval I_δ. In fact, by

$$y_{k+1} = y_0 + (y_1 - y_0) + (y_2 - y_1) + \ldots + (y_{k+1} - y_k)\,,$$

the function y_{k+1} is the $(k+1)^{\text{st}}$ partial sum of the series

$$y_0 + \sum_{k=0}^{\infty} (y_{k+1}(x) - y_k(x))\,, \tag{4.47}$$

which converges on I_δ totally, by estimate (4.46). In fact the generic term of the series in (4.47) is bounded from above as follows

$$|y_{k+1}(x) - y_k(x)| \le \frac{ML^k}{(k+1)!}\,|x - x_0|^{k+1} \le \frac{M}{L}\,\frac{(L\delta)^{k+1}}{(k+1)!}$$

for $k = 0, 1, 2, \ldots$ and any $x \in I_\delta$. The corresponding numerical series is the convergent exponential series

$$\frac{M}{L} \sum_{k=0}^{\infty} \frac{(L\delta)^{k+1}}{(k+1)!} = \frac{M}{L}\,(e^{L\delta} - 1)\,.$$

In this way we have proved that the sequence $y_k(x)$ converges uniformly on I_δ. Let $y(x)$ be the limit, and let us take the limit in (4.42) as $k \to +\infty$. Since f is Lipschitz in y, the composite function $f(t, y_k(t))$ converges uniformly to $f(t, y(t))$ on I_δ. Therefore we can interchange the limit and the integral to obtain

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\,dt\,, \qquad \forall\, x \in I_\delta,$$

which means that $y(x)$ solves the Cauchy problem in the integral form (ii).

That this solution is unique follows again by the integral form (ii). Indeed, suppose $z(x)$ solves the initial value problem on some interval $I_{\delta_1} = [x_0 - \delta_1, x_0 + \delta_1]$. By the integral characterisation $z(x)$ satisfies the equation

$$z(x) = y_0 + \int_{x_0}^{x} f(t, z(t))\,dt\,, \qquad \forall\, x \in I_{\delta_1}.$$

We claim that on the interval I_{δ_2}, $z(x)$ coincides with the earlier solution $y(x)$, where $\delta_2 = \min\{\delta; \delta_1\}$. In fact, by the same method used for the sequence $y_k(x)$, we find

$$|z(x) - y_0| = \left| \int_{x_0}^{x} f(t, z(t))\,dt \right| \le M|x - x_0|\,, \tag{4.48}$$

for any $x \in I_{\delta_2}$. Recalling the definition of $y_1(x)$ in (4.42), as f is Lipschitz in y, (4.48) gives

$$
\begin{aligned}
|z(x) - y_1(x)| &= \left| \int_{x_0}^{x} \{f(t, z(t)) - f(t, y_0)\}\, dt \right| \leq \\
&\leq \left| \int_{x_0}^{x} |f(t, z(t)) - f(t, y_0)|\, dt \right| \leq \left| \int_{x_0}^{x} L|z(t) - y_0|\, dt \right| \leq \\
&\leq ML \left| \int_{x_0}^{x} |t - x_0|\, dt \right| = \frac{ML}{2} |x - x_0|^2 ,
\end{aligned}
$$

and iterating (by induction),

$$
|z(x) - y_k(x)| \leq \frac{ML^k}{(k+1)!} |x - x_0|^{k+1} , \qquad \forall\, x \in I_{\delta_2} .
$$

Since the right-hand side tends to zero as $k \to +\infty$, the function $z(x)$ is equal to $y(x) = \lim_{k \to +\infty} y_k(x)$ for any $x \in I_{\delta_2}$, as required.

The two proofs of Cauchy's theorem (the 'functional' one that uses the fixed-point theorem for metric spaces, and the second one of real-analytical flavour) are based on the *method of successive approximation* of (4.42). In the functional-like proof, the method of successive approximation is part of the proof of the fixed-point theorem. That means that the initial value problem's solution is *constructed* using *successive approximations*.

Let us exemplify the method by solving the following initial value problem

$$
\begin{cases}
y' = z \\
z' = -y \\
y(0) = 0 \\
z(0) = 1
\end{cases}
\tag{4.49}
$$

in the unknowns $(y(x), z(x))$, with $(y, z) : \mathbb{R} \to \mathbb{R}^2$. The vector-valued function associated with the initial value problem (4.49) is

$$
f = (f_1, f_2) = (z, -y) .
$$

Consider a neighbourhood $I_\delta = [-\delta, +\delta]$ of the point $x_0 = 0$ and let us define the sequence $(y_k(x), z_k(x))$, where $(y_k, z_k) : I_\delta \to \mathbb{R}^2$, $k = 0, 1, 2, \ldots$, starting from the initial datum

$$
(y_0(x), z_0(x)) = (0, 1) ,
$$

(continued)

and then recursively

$$\begin{cases} y_{k+1}(x) = y_0 + \displaystyle\int_0^x f_1(y_k(t), z_k(t))\,dt = y_0 + \int_0^x z_k(t)\,dt \\ z_{k+1}(x) = z_0 + \displaystyle\int_0^x f_2(y_k(t), z_k(t))\,dt = z_0 - \int_0^x y_k(t)\,dt\,. \end{cases}$$

We then obtain

$$\begin{cases} y_1(x) = 0 + \displaystyle\int_0^x 1\,dt = x \\ z_1(x) = 1 - \displaystyle\int_0^x 0\,dt = 1\,, \end{cases}$$

$$\begin{cases} y_2(x) = 0 + \displaystyle\int_0^x 1\,dt = x \\ z_2(x) = 1 - \displaystyle\int_0^x t\,dt = 1 - \frac{x^2}{2}\,, \end{cases}$$

$$\begin{cases} y_3(x) = 0 + \displaystyle\int_0^x \left(1 - \frac{t^2}{2}\right) dt = x - \frac{x^3}{3!} \\ z_3(x) = 1 - \displaystyle\int_0^x t\,dt = 1 - \frac{x^2}{2}\,, \end{cases}$$

$$\begin{cases} y_4(x) = 0 + \displaystyle\int_0^x \left(1 - \frac{t^2}{2}\right) dt = x - \frac{x^3}{3!} \\ z_4(x) = 1 - \displaystyle\int_0^x \left(t - \frac{t^3}{3!}\right) dt = 1 - \frac{x^2}{2} + \frac{x^4}{4!}\,. \end{cases}$$

The pairs $(y_k(x), z_k(x))$, for $k = 0, 1, 2, \ldots$, form a sequence of *successive approximations* of the solution $(y(x), z(x))$, in the sense that $(y_k(x), z_k(x))$ converges uniformly on I_δ (in this particular example, whichever δ we pick) to the solution $(y(x), z(x))$. In the limit as $k \to +\infty$

$$\begin{cases} y(x) = \displaystyle\lim_{k\to+\infty} y_k(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \\ z(x) = \displaystyle\lim_{k\to+\infty} z_k(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \end{cases}$$

so $y(x) = \sin x$, $z(x) = \cos x$. Checking that these two functions solve the initial value problem (4.49) is immediate.

Using the method described above, the reader should show that the initial value problem

$$\begin{cases} y' = z \\ z' = y \\ y(0) = 0 \\ z(0) = 1 \end{cases}$$

is solved by $y(x) = (e^x - e^{-x})/2 = \sinh x, z(x) = (e^x + e^{-x})/2 = \cosh x$.

4.3 First Consequences of Cauchy's Theorem

It may be complicated to verify directly the Cauchy theorem's hypotheses, especially the Lipschitz property of $f(x, y)$ in y. The following Corollary 1 gives us an easy way to do just that.

As in the previous section, given $x_0 \in \mathbb{R}$ and $y_0 \in \mathbb{R}^n$, consider a function $f(x, y)$ defined on a neighbourhood $I \times J$ of $(x_0, y_0) \in \mathbb{R} \times \mathbb{R}^n$, of the form

$$I \times J = \{(x, y) \in \mathbb{R} \times \mathbb{R}^n : |x - x_0| \leq a\,,\ |y - y_0| \leq b\}\,,$$

with $a, b > 0$.

Corollary 1 *Let $f = f(x, y)$, with $x \in \mathbb{R}$, $y \in \mathbb{R}^n$, be a function defined on $I \times J$ with values in $\mathbb{R}^n$. Suppose $f(x, y)$ and its partial derivatives*

$$\frac{\partial f}{\partial y_i}(x, y)\,, \qquad\qquad i = 1, 2, \ldots, n\,,$$

are continuous on $I \times J$. Then there exist a positive number δ and a unique function $y : [x_0 - \delta, x_0 + \delta] \to \mathbb{R}^n$ solving the initial value problem

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0\,. \end{cases} \tag{4.50}$$

on $[x_0 - \delta, x_0 + \delta]$.

Proof We aim to show that under the given hypotheses $f(x, y)$ is Lipschitz in $y \in J$, uniformly in $x \in I$. Then it suffices to invoke Cauchy's local existence and uniqueness theorem.

Call $(f_\alpha)_{\alpha=1,\dots,n}$ the components of $f : I \times J \to \mathbb{R}^n$. As the partial derivatives $\partial f_\alpha / \partial y_i$, $i, \alpha = 1, 2, \dots, n$, are continuous, so is

$$(x, y) \in I \times J \quad \to \quad \left(\sum_{i=1}^{n} \left(\frac{\partial f_\alpha}{\partial y_i}(x, y) \right)^2 \right)^{1/2}, \qquad \alpha = 1, 2, \dots, n, \tag{4.51}$$

on $I \times J$. By the Weierstrass theorem the function in (4.51) has a maximum on $I \times J$, say L_α.

The function in (4.51) is just the norm of the gradient of $f_\alpha(x, y)$ with respect to y, viewing $x \in I$ as a fixed parameter. Let $D_y f_\alpha(x, y)$ be the gradient of f_α in y, so

$$|D_y f_\alpha(x, y)| = \left(\sum_{i=1}^{n} \left(\frac{\partial f_\alpha}{\partial y_i}(x, y) \right)^2 \right)^{1/2} \le L_\alpha, \qquad \forall\, (x, y) \in I \times J. \tag{4.52}$$

Next we apply to $y \in J \to f_\alpha(x, y) \in \mathbb{R}$ the mean value theorem (Sect. 3.11): given $x \in I$, $y_1, y_2 \in J$, there is a real number ϑ_α between 0 and 1, such that

$$f_\alpha(x, y_1) - f_\alpha(x, y_2) = (D_y f_\alpha(x, y_2 + \vartheta_\alpha(y_1 - y_2)),\ y_1 - y_2)$$

(on the right, as usual, $(\cdot, \cdot)$ is the inner product in $\mathbb{R}^n$). The point

$$(x, y_2 + \vartheta_\alpha(y_1 - y_2)) = \vartheta_\alpha(x, y_1) + (1 - \vartheta_\alpha)(x, y_2)$$

is a convex combination of (x, y_1) and (x, y_2) in $I \times J$, so it belongs to $I \times J$ as well. The Cauchy-Schwarz inequality and (4.52) give

$$\begin{aligned} |f(x, y_1) - f(x, y_2)| &\le \sum_{\alpha=1}^{n} |f_\alpha(x, y_1) - f_\alpha(x, y_2)| = \\ &= \sum_{\alpha=1}^{n} |D_y f_\alpha(x, y_2 + \vartheta_\alpha(y_1 - y_2))| \cdot |y_1 - y_2| \le \\ &\le \sum_{\alpha=1}^{n} L_\alpha \cdot |y_1 - y_2| = L \cdot |y_1 - y_2|, \end{aligned}$$

for any $x \in I$, $y_1, y_2 \in J$, where $L = \sum_{\alpha=1}^{n} L_\alpha$, as required.

Next we will show that the solution $y(x)$ to (4.50) is actually more regular than just differentiable on I_δ.

Regularity Theorem *In the Cauchy theorem's notations, if $f = f(x, y)$ is of class $C^k(I \times J)$ for some $k \in \{0, 1, 2, \ldots\}$, the solution to the initial value problem is a C^{k+1} function $y : I_\delta \to \mathbb{R}^n$. Therefore if f is C^∞ also the solution y is C^∞.*

Proof We use induction, starting from $k = 0$. If f is $C^0(I \times J)$, the initial value problem's solution $y(x)$ is differentiable and solves the ODE

$$y'(x) = f(x, y(x)), \qquad \forall\, x \in I_\delta.$$

The derivative $y'(x)$ then equals a continuous function, hence is continuous on I_δ. Therefore y is C^1.

Suppose now that the statement is true for some $k \in \{0,1,2,\ldots\}$ and let us prove that $f \in C^{k+1}(I \times J)$ implies y is C^{k+2}.

By induction, since in particular $f \in C^k(I \times J)$, the solution y is of class C^{k+1}. But $f(x, y(x))$ is the composite of f, of class C^{k+1}, and y, of class C^{k+1}, so $y'(x) = f(x, y(x))$ is C^{k+1} on I_δ, i.e. y is C^{k+2}.

Cauchy's local existence and uniqueness theorem for first-order systems also has an effect on systems of arbitrarily high order. As we explained in the introduction to this chapter, *an ODE system of any order can be reduced to an ODE system of order one*. In the sequel, for simplicity, we shall only consider ordinary differential *equations*, in normal form and of order n.

Fix a point $(x_0, y_0, y_0', y_0'', \ldots, y_0^{n-1})$ in $\mathbb{R} \times \mathbb{R}^n$ and numbers $a, b \in \mathbb{R}$. As before we let I be the interval $[x_0 - a, x_0 + a] \subset \mathbb{R}$. Call Y_0 the point in $\mathbb{R}^n$ of coordinates $(y_0, y_0', y_0'', \ldots, y_0^{n-1})$ and $J = \{Y \in \mathbb{R}^n : |Y - Y_0| \le b\}$ the ball centred at $Y_0 \in \mathbb{R}^n$.

Corollary 2 *Let $f = f(x, Y)$, with $x \in I \subset \mathbb{R}$, $Y \in J \subset \mathbb{R}^n$, be a real function that is continuous on $I \times J$ and Lipschitz in $Y \in J$, uniformly in $x \in I$. There exist a positive number $\delta \le a$ and a unique function $y : [x_0 - \delta, x_0 + \delta] \to \mathbb{R}$, differentiable n times on $[x_0 - \delta, x_0 + \delta]$, that solves on that interval the Cauchy problem*

$$\begin{cases} y^{(n)} = f(x, y, y', y'', \ldots, y^{(n-1)}) \\ y(x_0) = y_0 \\ y'(x_0) = y_0' \\ \ldots\ldots\ldots\ldots \\ y^{(n-1)}(x_0) = y_0^{(n-1)}. \end{cases} \tag{4.53}$$

Proof Denoting

$$y = y_1, \quad y' = y_2, \quad y'' = y_3, \quad \ldots, \quad y^{(n-1)} = y_n,$$

Eq. (4.53) is equivalent to the first-order system

$$\begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ \cdots\cdots\cdots\cdots \\ y_{n-1}' = y_n \\ y_n' = f(x, y_1, y_2, \ldots, y_n)\,, \end{cases}$$

which we may write as

$$y' = F(x, y)\,,$$

with $y = y(x)$ vector-valued of components $(y_1(x), y_2(x), \ldots, y_n(x))$ and $F(x, y)$ vector-valued of components

$$F(x, y) = (y_2, y_3, \ldots, f(x, y_1, y_2, \ldots, y_n))\,.$$

Clearly $F(x, y)$ is continuous on $I \times J$ and Lipschitz in $y \in J$, uniformly in $x \in I$. The claim is then a direct consequence of Cauchy's theorem.

Combining Corollaries 2 and 1 we obtain an existence and uniqueness result for ODEs in normal form, under the hypothesis that $f = f(x, Y)$ admits continuous partial derivatives in Y. For brevity we shall not state this explicitly, also in view of the fact that it is easily deducible from Corollaries 1 and 2.

Cauchy's theorem extends to functions $f(x, y)$ that are not Lipschitz in y, but continuous in the pair of variables (x, y). As regards existence, we refer to Peano's theorem, proved in the Appendix (Sect. 4.9)). That said, the uniqueness stated in Cauchy's theorem may not hold if $f(x, y)$ is not Lipschitz in y.

For example, consider the following initial value problem of order one

$$\begin{cases} y' = 2\sqrt{|y|} \\ y(0) = 0 \end{cases} \tag{4.54}$$

where $y : \mathbb{R} \to \mathbb{R}$. The function $f(y) = 2\sqrt{|y|}$ (constant in x) is continuous on $\mathbb{R}$ but not Lipschitz.

The initial value problem (4.54) has infinitely many solutions defined on $\mathbb{R}$. The reader may check that the following two are solutions

$$y(x) = 0\,, \qquad \forall\, x \in \mathbb{R}\,;$$

$$y(x) = x|x| = \begin{cases} x^2 & \text{if} \quad x \geq 0 \\ -x^2 & \text{if} \quad x < 0\,. \end{cases}$$

Note in particular that $y(x) = x|x|$ is differentiable also at $x = 0$, at which point the derivative is $y'(0) = 0$.

Among other solutions to (4.54), we have those of the form

$$y(x) = \begin{cases} 0 & \text{if} \quad x < k \\ (x-k)^2 & \text{if} \quad x \geq k\,. \end{cases} \tag{4.55}$$

for any given $k \geq 0$. As before, the function $y(x)$ in (4.55) is differentiable at $x = k$, too, and it satisfies the differential equation at this point. The reader should now construct other solutions (the full answer is given in Sect. 4.9).

4.4 The Global Existence and Uniqueness Theorem: Extension of Solutions

Some applications require that we establish the existence of the solution to a Cauchy problem *on a given interval*. The 'local' Cauchy theorem gave us sufficient conditions for the *local resolution*, by providing a solution *on a neighbourhood* I_δ of the initial point x_0. Here, instead, we shall give sufficient conditions for the *global resolution*, by establishing the existence of a solution *on the entire interval, fixed in advance*, on which the ODE is defined. In the proof of the main result we will show that the solution to the initial value problem, which *a priori* is defined on just a neighbourhood of the initial point x_0, can actually be *extended* to the whole interval.

The assumptions we make on $f = f(x, y)$ (more restrictive than the Cauchy theorem's ones) are the following: given two real numbers α, β ($\alpha < \beta$), the function f is defined on the set

$$[\alpha, \beta] \times \mathbb{R}^n = \{(x, y) \in \mathbb{R} \times \mathbb{R}^n : x \in [\alpha, \beta]\}\,.$$

Suppose

$$f(x, y) \quad \text{is } \textit{continuous} \text{ on} \quad [\alpha, \beta] \times \mathbb{R}^n \tag{4.56}$$

and *locally Lipschitz* in $y \in \mathbb{R}^n$, uniformly in $x \in [\alpha, \beta]$, in the sense that for any M, there is a constant $L > 0$ (typically dependent on M) such that

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|\,, \tag{4.57}$$

for any $x \in [\alpha, \beta]$, $y_1, y_2 \in \mathbb{R}^n$, with $|y_1|, |y_2| \leq M$. Suppose finally that there are two constants L_1, L_2 such that

$$|f(x, y)| \leq L_1 + L_2|y|\,, \tag{4.58}$$

for any $x \in [\alpha, \beta]$, $y \in \mathbb{R}^n$.

Notice that (4.58) is in particular satisfied when f is *bounded* on $[\alpha, \beta] \times \mathbb{R}^n$ (taking $L_2 = 0$), or when $f(x, 0)$ is bounded and $f(x, y)$ is *globally Lipschitz* on $[\alpha, \beta] \times \mathbb{R}^n$ (not just *locally*), meaning that (4.57) holds with L independent of M. In the latter case, in fact, putting $y_1 = y$ and $y_2 = 0$ in (4.57) we obtain

$$|f(x, y)| \leq |f(x, 0)| + L|y| \leq L_1 + L|y|\,,$$

where $L_1 = \sup\{|f(x, 0)| : x \in [\alpha, \beta]\}$.

Hence there are two remarkable cases in which the above hypotheses holds: $f(x, y)$ globally Lipschitz on $[\alpha, \beta] \times \mathbb{R}^n$, or $f(x, y)$ bounded and locally Lipschitz on $[\alpha, \beta] \times \mathbb{R}^n$, for instance when f is bounded and C^1.

Global Existence and Uniqueness Theorem *If f satisfies* (4.56), (4.57) *and* (4.58), *then for any $x_0 \in [\alpha, \beta]$, $y_0 \in \mathbb{R}^n$ there exists a unique function $y = y(x)$, $y : [\alpha, \beta] \to \mathbb{R}^n$, that solves on the entire interval $[\alpha, \beta]$ the initial value problem*

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0\,. \end{cases} \tag{4.59}$$

Example 1 The technique for solving the following initial value problem (relative to a *separable ODE*) will be explained in the next section. Here we shall just verify that the solution to

$$\begin{cases} y' = 1 + y^2 \\ y(0) = 0 \end{cases}$$

is $y(x) = \operatorname{tg} x$. In fact, $y(0) = 0$, $y' = 1/\cos^2 x$ and

$$1 + y^2 = 1 + \frac{\operatorname{sen}^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = y'\,.$$

It is clear that the solution is not defined on $\mathbb{R}$ but only on the neighbourhood $(-\pi/2, \pi/2)$ of $x = 0$. The conclusion of the global existence and uniqueness theorem does not hold. More precisely, the theorem is not applicable on an arbitrary interval $[\alpha, \beta]$ fixed in advance. The reason is that the function $f(y) = 1 + y^2$ (constant in x) does not satisfy (4.58).

Example 2 In contrast to the previous example, the initial value problem

$$\begin{cases} y' = \dfrac{1 + x^2}{1 + y^2} \\ y(1) = 1\,, \end{cases}$$

(continued)

Example 2 (continued)
satisfies all the hypotheses of the global existence and uniqueness theorem. In particular (4.58) holds, since $f(x, y) = (1 + x^2)/(1 + y^2)$ is bounded as x varies in a bounded interval $[\alpha, \beta]$ and $y \in \mathbb{R}$.

It is easy to see $y(x) = x$ is a solution defined on the entire $\mathbb{R}$.

Proof of the Global Existence and Uniqueness Theorem Consider $\alpha \leq x_0 < \beta$ and let us show the initial value problem (4.59) admits a (unique) solution to the right of x_0, on the interval $[x_0, \beta]$. A similar argument applies to the case $[\alpha, x_0]$, $\alpha \neq x_0$.

Cauchy's local existence and uniqueness theorem warrants that there exist a number $\delta_0 > 0$ and a (unique) solution $y : I_{\delta_0} \to \mathbb{R}^n$ to (4.59). In particular $y(x)$ solves the initial value problem on $[x_0, x_0 + \delta_0]$.

Define $(x_1, y_1) \in [x_0, \beta] \times \mathbb{R}^n$ by $x_1 = x_0 + \delta_0$ and $y_1 = y(x_1)$. Let us solve the new initial value problem

$$\begin{cases} y' = f(x, y) \\ y(x_1) = y_1 \,. \end{cases} \tag{4.60}$$

By Cauchy's theorem there exist $\delta_1 > 0$ and $z : I_{\delta_1} \to \mathbb{R}^n$, with $I_{\delta_1} = [x_1 - \delta_1, x_1 + \delta_1]$, solving (4.60). As $y(x_1) = y_1$, also $y(x)$ solves (4.60) on $[x_1 - \delta_1, x_1] \subseteq [x_0, x_1]$. By uniqueness, $z(x)$ coincides with $y(x)$ to the left of x_1, and is therefore a right-sided extension of $y(x)$. Changing notation we may call this function $y(x)$ and consider it a solution to the Cauchy problem (both (4.59) and (4.60)) on the larger interval

$$[x_0, x_1 + \delta_1] = [x_0, x_0 + \delta_0 + \delta_1] \,.$$

Now we repeat the argument starting from the initial datum $(x_2, y_2) \in [x_0, \beta] \times \mathbb{R}^n$, with $x_2 = x_1 + \delta_1$ and $y_2 = y(x_2)$. We extend the solution $y(x)$ to an interval of type

$$[x_0, x_2 + \delta_2] = [x_0, x_0 + \delta_0 + \delta_1 + \delta_2] \,,$$

with $\delta_2 > 0$. Iterating, we end up extending the solution to

$$[x_0, x_0 + \delta_0 + \delta_1 + \ldots + \delta_k] \,.$$

We claim that after finitely many iterations (i.e. for k finite), we will have

$$x_{k+1} = x_0 + \delta_0 + \delta_1 + \ldots + \delta_k \geq \beta, \tag{4.61}$$

so that $y(x)$ solves the initial value problem on the entire interval $[x_0, \beta]$.

Assumption (4.58) is the main difference between the local and global versions of Cauchy's theorem, and is employed at this point to prove (4.61). For this let us return to Cauchy's statement, applied repeatedly with initial values (x_k, y_k), $k = 0, 1, 2, \ldots$, and with $a_k = \beta - x_k$ and $b_k > 0$ chosen at a later stage. Cauchy's theorem gives for δ_k the following estimate

$$\delta_k = \min\left\{a_k; \frac{b_k}{M_k}\right\} = \min\left\{\beta - x_k; \frac{b_k}{M_k}\right\},$$

where

$$M_k = \max\{|f(x, y)| : x \in [x_k, \beta],\ |y - y_k| \le b_k\}.$$

From (4.58) we deduce

$$|f(x, y)| \le L_1 + L_2|y| \le L_1 + L_2(|y_k| + b_k),$$

for any $x \in [\alpha, \beta]$, $|y - y_k| \le b_k$. Put $b_k = L_1 + L_2|y_k|$, so

$$\begin{aligned} \frac{M_k}{b_k} &= \frac{1}{b_k} \max\{|f(x, y)| : x \in [x_k, \beta],\ |y - y_k| \le b_k\} \le \\ &\le \frac{1}{b_k}(L_1 + L_2|y_k| + L_2 b_k) = 1 + L_2. \end{aligned} \tag{4.62}$$

Now, suppose by contradiction $x_k < \beta$ for any $k \in \mathbb{N}$. Then we could not have

$$\delta_k = \frac{b_k}{M_k}, \qquad \forall\, k \in \mathbb{N},$$

because in that case (4.62) would imply $\delta_k \ge 1/(1 + L_2)$ for any k, hence $x_k \to +\infty$. Therefore there is an integer k such that $\delta_k = \beta - x_k$, and $x_{k+1} = x_k + \delta_k = \beta$.

Having proved that there exists $k \in \mathbb{N}$ such that $x_k \ge \beta$, the function $y(x)$ solves problem (4.59) on the entire interval $[x_0, \beta]$.

In the proof of the global existence and uniqueness we have shown that the solution to the ODE

$$y' = f(x, y), \tag{4.63}$$

defined *locally* on a neighbourhood of the initial point, is actually *extendable* to the entire given interval. In case the hypotheses of the global theorem are not satisfied the following considerations might be of use.

Let $y = y(x)$ be a solution to (4.63) on some interval (a, b) (hereafter the endpoints are allowed to be infinite). We say that a function $y_1(x)$ is an *extension* of $y(x)$ if $y_1(x)$ solves the same ODE (4.63) on an interval $(a_1, b_1) \supseteq (a, b)$ and $y_1(x) = y(x)$ for any $x \in (a, b)$.

An extension $y_1(x)$ of $y(x)$ to (a_1, b_1) is called *maximal* when any extension y_2 of y to an interval (a_2, b_2) implies that (a_1, b_1) contains (a_2, b_2).

Existence of the Maximal Extension *If $f = f(x, y)$ satisfies (4.28), (4.29) in Cauchy's theorem, every solution $y = y(x)$ to the ODE (4.63) has a maximal extension.*

Proof We divide the argument in two steps.

Step one: consider two solutions y_1, y_2 of (4.63), respectively defined on (a_1, b_1), (a_2, b_2), that coincide at some point $x_0 \in I = (a_1, b_1) \cap (a_2, b_2)$. We claim $y_1(x) = y_2(x)$ for any $x \in I$.

We restrict to points $x \in I$ such that $x \geq x_0$, since the other case is similar. Set

$$\overline{x} = \inf\{x \in I : x \geq x_0,\ y_1(x) \neq y_2(x)\}\,. \tag{4.64}$$

Then $y_1(\overline{x}) = y_2(\overline{x})$. In fact, if $\overline{x} = x_0$, by assumption $y_1(x_0) = y_2(x_0)$. If $\overline{x} > x_0$, assuming $y_1(\overline{x}) \neq y_2(\overline{x})$ would give $y_1(x) \neq y_2(x)$ for any x in some neighbourhood of $\overline{x}$ (by the continuity of y_1, y_2). But $\overline{x}$ is the infimum.

Hence y_1, y_2 solve on I the same ODE, and they satisfy $y_1(\overline{x}) = y_2(\overline{x})$. Cauchy's theorem with initial point $\overline{x}$ implies that $y_1(x)$ and $y_2(x)$ coincide on a neighbourhood of $\overline{x}$, violating the definition of $\overline{x}$ in (4.64). Therefore the set in (4.64) is empty, and $y_1(x) = y_2(x)$ for any $x \in I$.

Second step: let (a, b) be the open interval where $y(x)$ solves (4.63). Consider the set $\mathcal{P}$ of all solutions of (4.63) that extend y. Saying $y_1 \in \mathcal{P}$ means that y_1 solves (4.63) on some interval $(a_1, b_1) \supseteq (a, b)$, and that y_1 coincides with y on (a, b).

To define the maximal extension of y, start by introducing the endpoints of the maximal interval (a_m, b_m):

$$a_m = \inf\{a_1 : \exists\, y_1 \in \mathcal{P} \textit{ solution to } (4.63) \textit{ on } (a_1, b_1)\}, \tag{4.65}$$

$$b_m = \sup\{b_1 : \exists\, y_1 \in \mathcal{P} \textit{ solution to } (4.63) \textit{ on } (a_1, b_1)\}. \tag{4.66}$$

As $y \in \mathcal{P}$, we have $a_m \leq a < b \leq b_m$, so (a_m, b_m) is not empty. By definitions (4.65), (4.66) we infer that, for any $x \in (a_m, b_m)$, there is a solution $y_1 \in \mathcal{P}$ to (4.63) on (a_1, b_1), where $x \in (a_1, b_1)$. Now define y_m at x by

$$y_m(x) = y_1(x)\,. \tag{4.67}$$

This does not depend on the particular solution y_1 chosen: if y_2 were another solution to (4.63) on some (a_2, b_2), with $x \in (a_2, b_2)$, by step one we would have $y_1(x) = y_2(x)$ because $x \in (a_1, b_1) \cap (a_2, b_2)$. Therefore $y_m(x)$ in (4.67) is well defined.

In particular $y_m(x) = y(x)$ for any $x \in (a, b)$, and so y_m is an extension of y. Furthermore, y_m is a maximal extension, because by definitions (4.65) and (4.66), we have $(a_m, b_m) \supseteq (a_1, b_1)$ for any $y_1 \in \mathcal{P}$ defined on (a_1, b_1).

Finally, let us show y_m solves (4.63) on (a_m, b_m). Fix $x \in (a_m, b_m)$, and in definition (4.67) take $y_1 \in \mathcal{P}$ solution to (4.63) on $(a_1, b_1) \subseteq (a_m, b_m)$, with $x \in (a_1, b_1)$. Since $y_m(x) = y_1(x)$ for any $x \in (a_1, b_1)$, on this interval $y_m'(x) = y_1'(x)$, and so y_m solves (4.63) on (a_m, b_m).

Consider an open subset A in $\mathbb{R} \times \mathbb{R}^n$ and let $f = f(x, y)$ be a function defined on A that satisfies the conditions ensuring the local existence of a solution to the initial value problem

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$$

for any point $(x_0, y_0) \in A$. For example, f might be *of class* C^1, in which case Corollary 1 in Sect. 4.3 applies. More simply, it suffices to take f *continuous* on A, due to Peano's theorem (see the Appendix, Sect. 4.9).

Below we state, without proof, a result on extending solutions to maximal intervals only under a continuity assumption.

Extension Theorem *Let $f : A \subseteq \mathbb{R} \times \mathbb{R}^n$ be continuous and $y : (a, b) \to \mathbb{R}^n$ a solution to the ODE (4.63). There exists a maximal solution y_m of (4.63), defined on a maximal interval $(a_m, b_m) \supseteq (a, b)$ with $a_m, b_m \in \mathbb{R} \cup \{\pm\infty\}$, that extends $y(x)$, i.e. such that $y_m(x) = y(x)$ for any $x \in (a, b)$.*

Moreover, if $x \to b_m^-$, then $y_m(x)$ "tends" to the boundary of A, meaning that for any compact set $K \subset A$, there exists $\delta > 0$ such that $(x, y_m(x)) \notin K$ if $x \in (b_m - \delta, b_m)$. The behaviour is analogous for $x \to a_m^+$.

Let us emphasise that the extension theorem does not claim that the maximal solution $y_m(x)$ converges to a point on the boundary of A as x tends to one endpoint of the maximal interval. It says that $(x, y_m(x)) \notin K$ for any compact subset $K \subset A$, when x is sufficiently close to the interval's endpoint.

For example, the real function

$$y(x) = e^{\cos \frac{1}{x}} \tag{4.68}$$

has derivative

$$y'(x) = e^{\cos \frac{1}{x}} \cdot \frac{1}{x^2} \sin \frac{1}{x} = y(x) \cdot \frac{1}{x^2} \sin \frac{1}{x}.$$

(continued)

Hence $y = y(x)$ solves the ODE $y' = f(x, y)$ where

$$f(x, y) = y \cdot \frac{1}{x^2} \sin \frac{1}{x} .$$

The function f is continuous in x and locally Lipschitz in y, for instance on $A = \{(x, y) \in \mathbb{R}^2 : x > 0\}$. Now, $y = y(x)$ in (4.68) solves $y' = f(x, y)$, it tends to ∂A as $x \to 0^+$ in the sense of the theorem, but does not admit limit as $x \to 0^+$.

If we seek a globally Lipschitz example in y, on the same set $A = \{(x, y) \in \mathbb{R}^2 : x > 0\}$, we may consider the function (constant in y)

$$f(x) = \frac{1}{x^2} \sin \frac{1}{x} .$$

Now $y(x) = \cos(1/x)$ solves $y' = f(x)$.

4.5 Solving First-Order ODEs in Normal Form

In this section and the next we shall examine ODEs of type

$$y' = f(x, y) , \tag{4.69}$$

where $f(x, y)$ is a real-valued function. Earlier we saw under which conditions on f equation (4.69) admits a solution satisfying the initial condition $y(x_0) = y_0$ on the neighbourhood of a point x_0.

Hence under the Cauchy theorem's assumptions, given x_0 we determine a solution to (4.69) for any real number y_0 (in some interval where f is defined and satisfies the conditions). We then say that the set of solutions depends on a real parameter (called y_0 above, but typically denoted with c, for *constant*). Such a solution set is called *general integral*.

Each element in this family of functions, obtained fixing the parameter c, is a *particular solution* of the differential equation (4.69). Conversely, a solution might not be a particular integral, and these further solutions are called *singular integrals*. We shall examine, especially in the next section, differential equations admitting singular integrals. Here we will consider special examples of functions $f(x, y)$ for which we are able to compute (more or less explicitly) the general integral of the corresponding equation (4.69).

If the differential equation is given in the form

$$y' = f(x) \cdot g(y) , \tag{4.70}$$

with $f(x)$ and $g(y)$ continuous, the equation is called *separable* (this refers to the fact that the variables are separable, as explained below). Supposing $g(y) \neq 0$ for any y, dividing (4.70) by $g(y)$ and integrating in x we obtain

$$\int \frac{y'(x)}{g(y(x))}\,dx = \int f(x)\,dx\,.$$

This is an expression of the form

$$G(y(x)) = F(x) + c$$

where F is a primitive of f and G a primitive of $1/g$.

When G is invertible, we obtain a family of explicit solutions represented by $y = y(x, c) = G^{-1}(F(x) + c)$.

Under the variable change $y = y(x)$ in the left-hand side the integral reduces (4.70) to

$$\int \frac{dy}{g(y)} = \int f(x)\,dx\,.$$

Example 1 Determine the general integral of the ODE

$$y' = x \cdot y^3\,. \tag{4.71}$$

Separating the variables and integrating, we obtain

$$\int \frac{dy}{y^3} = \int x\,dx\,,$$

whence

$$-\frac{1}{2y^2} = \frac{x^2}{2} + c\,.$$

If we denote by c the arbitrary constant $-2c$, we then have

$$y = \pm\frac{1}{\sqrt{c - x^2}}\,. \tag{4.72}$$

Note that (4.71), apart from the solutions in (4.72) plotted in Fig. 4.1, also admits the singular integral $y \equiv 0$.

(continued)

Example 1 (continued)

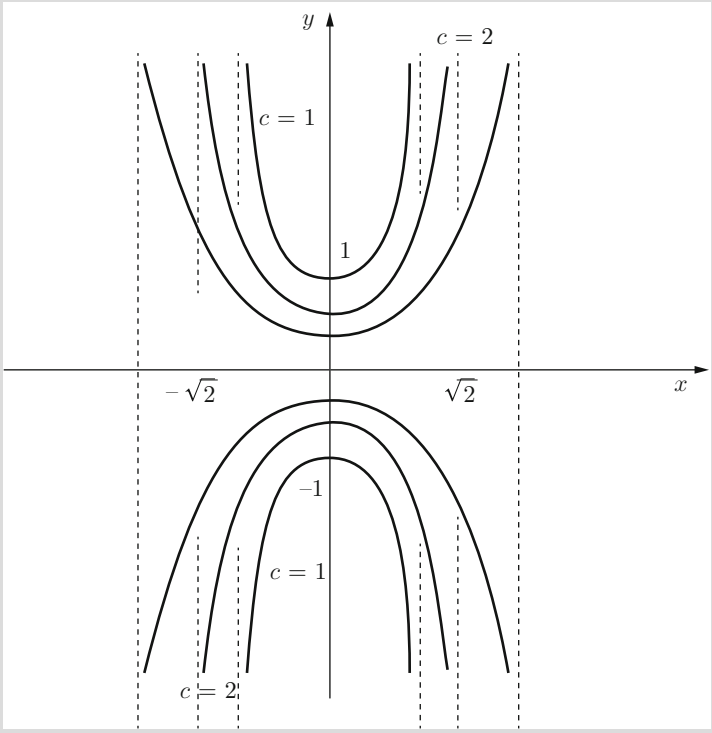


Fig. 4.1 $y = y(x, c)$ given in (4.72)

Equations of type

$$y' = g(ax + by)\,, \tag{4.73}$$

with a, b non-zero, are immediately reducible to the separable case. For this put $z = z(x) = ax + by$, so $z' = a + by'$ and then (4.73) becomes

$$z' = a + bg(z)\,,$$

which is a separable equation, and can be solved with the above method.

The differential equation

$$y' = g\left(\frac{y}{x}\right)\,, \tag{4.74}$$

called *homogeneous*, is solved using the substitution $z = \dfrac{y}{x}$. This turns (4.74) into

$$xz' + z = g(z)\,,$$

which is separable.

Example 2 Determine the solutions of the homogeneous ODE

$$y' = \frac{2xy}{x^2 + y^2}\,. \tag{4.75}$$

Dividing numerator and denominator by x^2 we obtain

$$y' = \frac{2\dfrac{y}{x}}{1 + \left(\dfrac{y}{x}\right)^2},$$

and putting $z = \dfrac{y}{x}$ we easily obtain the separable equation

$$xz' = \frac{z - z^3}{1 + z^2}\,,$$

whose solution is the general integral

$$\frac{z}{z^2 - 1} = cx\,.$$

Recalling that $z = \dfrac{y}{x}$, we then obtain

$$\frac{y}{y^2 - x^2} = c.$$

Again, let c replace the constant $1/c$, so the general integral of (4.75) reads

$$y^2 - x^2 = cy\,,$$

representing a family of hyperbolas. Also note that $y \equiv 0$ is a particular integral of (4.75).

All equations of the form

$$y' = g\left(\frac{ax + by + c}{a'x + b'y' + c'}\right)\,, \tag{4.76}$$

with a, b, a', b' not all zero, are reducible to homogeneous equations.

Call r and r' the straight lines of equations $ax+by+c = 0$ and $a'x+b'y'+c' = 0$. If r and r' are parallel, i.e. $a' = ka$ and $b' = kb$ for some real number k, then (4.76)

reduces to an equation of type (4.73), solved with the substitution $z = ax + by$. If r and r' are not parallel, so that the coefficients a, a' and b, b' are not proportional, the lines meet at a point (x_0, y_0). Put

$$\xi = x - x_0\,, \qquad \eta = y - y_0\,,$$

so $ax_0 + by_0 + c = 0$ implies $ax + by + c = a\xi + b\eta$, and similarly $a'x + b'y + c' = a'\xi + b'\eta$. Since we also have

$$\frac{d\eta}{d\xi} = \frac{dy}{dx}\,,$$

Eq. (4.76) now reads

$$\eta' = g\left(\frac{a\xi + b\eta}{a'\xi + b'\eta}\right) = g\left(\frac{a + b\frac{\eta}{\xi}}{a' + b'\frac{\eta}{\xi}}\right)\,,$$

which is homogeneous.

Example 3 Consider

$$y' = \frac{2x - 6}{y - x + 2}\,. \tag{4.77}$$

The lines $2x - 6 = 0$, $y - x + 2 = 0$ meet at $(3, 1)$. We put $\xi = x - 3$, $\eta = y - 1$ so that (4.77) transforms into

$$\eta' = \frac{2\xi}{\eta - \xi} = \frac{2}{(\eta/\xi) - 1}\,.$$

The substitution $z = z(\xi) = \eta/\xi$ gives the separable equation

$$\xi z' = -\frac{z^2 - z - 2}{z - 1}.$$

An easy computation then furnishes the general integral

$$(z + 1)^2 \cdot (z - 2) = \frac{c}{\xi^3}\,.$$

Recalling that $z = \eta/\xi$ and $\xi = x - 3$, $\eta = y - 1$, we find the general integral of (4.77)

$$(y + x - 4)^2 \cdot (y - 2x + 5) = c\,.$$

We close the section considering a case that can be handled with differential forms. The reader might come back here after studying the topic in Chap. 7.

If the differential equation is presented as

$$y' = -\frac{a(x, y)}{b(x, y)}, \tag{4.78}$$

or equivalently

$$\omega = a(x, y)dx + b(x, y)dy = 0, \tag{4.79}$$

the solutions to (4.78) arise from examining the differential form ω. In fact, if the form is exact (see Sect. 7.3, Chap. 7) and if $F(x, y)$ denotes a primitive of ω, the solutions of (4.79) are precisely those satisfying the implicit equation

$$F(x, y) = c. \tag{4.80}$$

Example 4 Given

$$y' = \frac{e^{2x} - xy^2}{x^2 y},$$

observe that the differential form

$$\omega = (xy^2 - e^{2x})dx + x^2 y\, dy, \tag{4.81}$$

is exact, since it is defined on the entire plane and

$$\frac{\partial}{\partial y}(xy^2 - e^{2x}) = 2xy = \frac{\partial}{\partial x}(x^2 y)$$

(see Theorem 2, Sect. 7.4).

Easily, a primitive of the form (4.81) is

$$F(x, y) = \frac{1}{2}(x^2 y^2 - e^{2x}).$$

It follows from (4.80) that the general integral of the given ODE is given by the expression $x^2 y^2 - e^{2x} = c$.

4.6 Solving First-Order ODEs Not in Normal Form

Consider the differential equation, not in normal form,

$$y = xy' + g(y'), \tag{4.82}$$

where g is differentiable. It is called *Clairaut equation*. If $y(x)$ is a solution of (4.82) admitting second derivative, by differentiating (4.82) in x we find

$$y''(x + g'(y')) = 0\,,$$

and so either

$$y'' = 0 \tag{4.83}$$

or

$$x + g'(y') = 0\,. \tag{4.84}$$

For (4.83) we have $y' = c$, and because of (4.82), necessarily

$$y = xc + g(c)\,. \tag{4.85}$$

Another solution to (4.82) comes from (4.84). Set $y' = t$, so (4.84) and (4.82) give parametric equations

$$\begin{cases} x = -g'(t) \\ y = -tg'(t) + g(t)\,. \end{cases} \tag{4.86}$$

The solution to (4.86) is called a *singular integral* of the Clairaut equation. It can be proved that this curve is the *envelope* of the family of lines (4.85), i.e. at any point it is tangent to one line in the family.

Example 1 Consider the Clairaut equation

$$y = xy' - \frac{1}{3}\,(y')^3\,.$$

From (4.85) we obtain solutions

$$y = cx - \frac{1}{3}\,c^3\,, \tag{4.87}$$

while (4.86) produces the singular integral

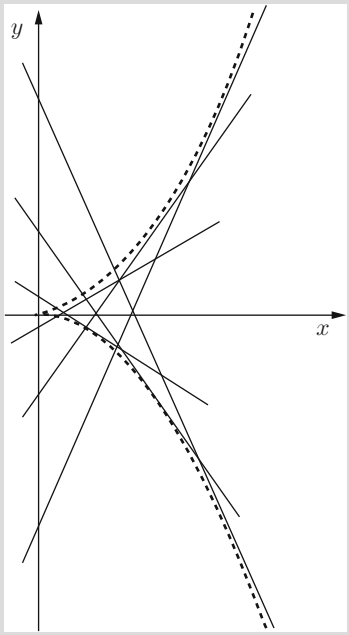
$$\begin{cases} x = t^2 \\ y = \dfrac{2}{3}\,t^3\,, \end{cases} \tag{4.88}$$

the curve obtained as union of the graphs of the two functions of equations $y = \pm(2/3)\,x^{3/2}$. The reader should check that the latter envelops the family of lines (4.87). Put otherwise, (4.87) is the *equation of all tangent lines* to the curve (4.88) (see Fig. 4.2).

(continued)

Example 1 (continued)

Fig. 4.2



Consider now the non-normal equation

$$x = g(y')\,, \tag{4.89}$$

where g is differentiable. Let us take y' as independent variable and put $t = y'$. From (4.89) we have

$$\frac{dy}{dt} = \frac{dy}{dx}\,\frac{dx}{dt} = tg'(t)\,,$$

and integrating by parts

$$y(t) = \int tg'(t)\,dt = tg(t) - G(t) + c\,,$$

with G a primitive of g. From this relation and (4.89) we determine the general integral

$$\begin{cases} x = g(t) \\ y = tg(t) - G(t) + c\,. \end{cases} \tag{4.90}$$

Example 2 To solve

$$x = \frac{1}{1 + (y')^2}\,, \tag{4.91}$$

we can certainly use (4.90), where

$$g(t) = \frac{1}{1+t^2}\,, \qquad G(t) = \operatorname{arctg} t\,.$$

Then

$$\begin{cases} x = \dfrac{1}{1+t^2} \\ y = \dfrac{t}{1+t^2} - \operatorname{arctg} t + c\,. \end{cases}$$

Solving for the parameter t in terms of x, we obtain $t = \pm\sqrt{(1-x)/x}$, and the general integral of (4.91) is explicitly given by

$$y = \pm\left(\sqrt{x - x^2} - \operatorname{arctg}\sqrt{\frac{1-x}{x}}\right) + c\,.$$

The reasoning employed for tackling

$$y = g(y')\,. \tag{4.92}$$

is similar. Putting $y' = t$, by (4.92) we have

$$\frac{dx}{dt} = \frac{dx}{dy}\,\frac{dy}{dt} = \frac{g'(t)}{t}\,.$$

Then the general integral of (4.92) is, in parametric form,

$$\begin{cases} x = \displaystyle\int \frac{g'(t)}{t}\,dt \\ y = g(t)\,. \end{cases} \tag{4.93}$$

4.7 Solving Higher-Order Equations

Consider a differential equation of second order where y does not appear:

$$F(x, y', y'') = 0\,. \tag{4.94}$$

The substitution $z(x) = y'(x)$ clearly turns (4.94) into the first-order equation

$$F(x, z, z') = 0 \tag{4.95}$$

in the unknown z. If $z = z(x, c)$ is the general integral of (4.95), the general integral of (4.94) is then

$$y(x) = c_1 + \int_{x_0}^{x} z(t, c)\, dt\,.$$

In the same way, an equation of order $n \geq 2$

$$F(x, y^{(k)}, y^{(k+1)}, \ldots, y^{(n)}) = 0\,, \tag{4.96}$$

where the variables $y, y', \ldots, y^{(k-1)}$ $(1 \leq k < n)$ do not appear explicitly, can be transformed into an equation of order $n - k$

$$F(x, z, z', \ldots, z^{(n-k)}) = 0 \tag{4.97}$$

by the substitution $z(x) = y^{(k)}(x)$. If $z = z(x, c_1, c_2, \ldots, c_{n-k})$ is the general integral of (4.97), the solutions to (4.96) arise from solving the ODE of order k

$$y^{(k)} = z(x, c_1, c_2, \ldots, c_{n-k})\,.$$

The latter may be solved for instance as we did for (4.15) in the introduction to this chapter (Sect. 4.1).

Now consider a second-order equation that does not explicitly depend on x

$$F(y, y', y'') = 0\,. \tag{4.98}$$

We may think of y as an independent variable, and putting $z(y) = y'$ we find

$$y'' = \frac{dy'}{dx} = \frac{dz}{dy}\frac{dy}{dx} = z'z\,.$$

Thus (4.98) becomes of order one:

$$F(y, z, z'z) = 0\,.$$

If $z = z(y) = z(y, c)$ is a general integral of the latter equation, the solutions of (4.98) are obtained by solving the separable equation

$$y' = z(y)\,.$$

Example 1 Let us solve the initial value problem

$$\begin{cases} \dfrac{y''}{(1+(y')^2)^{3/2}} = \dfrac{1}{y^2} \\ y(0) = 1\,, \quad y'(0) = 0\,. \end{cases} \tag{4.99}$$

Put $z(y) = y'$, so $y'' = z'z$ and the ODE in (4.99) reads

$$\frac{z'z}{(1+z^2)^{3/2}} = \frac{1}{y^2}\,,$$

which is separable. The latter's general integral is

$$\frac{1}{\sqrt{1+z^2}} = \frac{1}{y} + c\,. \tag{4.100}$$

The Cauchy conditions $y(0) = 1$, $y'(0) = 0$ imply $z(1) = 0$, imposing which on (4.100) produces $c = 0$.

Therefore, recalling that $z = y'$, if y solves the given initial value problem, y also solves

$$\begin{cases} \sqrt{1+(y')^2} = y \\ y(0) = 1\,. \end{cases} \tag{4.101}$$

The ODE in (4.101) is of the type $y = g(y')$. Using (4.93) from the previous section, the general integral, in parametric form, is

$$\begin{cases} x = \displaystyle\int \frac{1}{\sqrt{1+t^2}}\,dt = \log\left(t + \sqrt{1+t^2}\right) + c \\ y = \sqrt{1+t^2}\,. \end{cases}$$

Solving the first equation for the parameter t gives $t = \sinh(x-c)$. Substituting in the second equation produces $y = \cosh(x-c)$. Imposing the initial condition $y(0) = 1$ finally gives the solution to (4.99):

$$y = \cosh x\,.$$

4.8 Qualitative Study of Solutions

In this section we will show how, sometimes, one can infer properties of solutions, in particular those allowing to plot graphs, starting only from the analytical expression of the given ODE, and without solving the equation explicitly – which by the way may not be solvable in closed form. These methods are referred to as the *qualitative study of the solutions*.

Let us consider some examples.

Example 1 The initial value problem

$$\begin{cases} y' = \dfrac{1}{y} - x \\ y(1) = 1\,. \end{cases} \tag{4.102}$$

admits a solution $y(x)$ defined on a neighbourhood of $x_0 = 1$, because the function $f(x, y) = 1/y - x$ is C^1 on a neighbourhood of the point $(x_0, y_0) = (1, 1)$, to be precise on the open set $A = \{(x, y) \in \mathbb{R}^2 : y > 0\}$.

The sign of the derivative y' speaks about the monotonicity of $y(x)$. In fact, from $y' = 1/y - x$ we have $y' \le 0$ whenever $1/y - x \le 0$, i.e. on A, restricted to $y > 0$ when $y > 1/x$. In the xy-plane the region in question lies above the hyperbola $y = 1/x$, see Fig. 4.3.

Since the point corresponding to the initial datum $(x_0, y_0) = (1, 1)$ lies on the hyperbola, for $x_0 = 1$ we have $y'(x_0) = 0$. The solution's graph, with horizontal tangent at $x_0 = 1$, crosses the hyperbola. Restricting to a neighbourhood of x_0, we then have $y'(x) > 0$ for $x < x_0$ and $y'(x) < 0$ for $x > x_0$. Therefore x_0 is a maximum point of $y(x)$.

On a right neighbourhood of x_0 the function $y(x)$ is strictly decreasing. We may ask whether the graph of $y(x)$ intersects the hyperbola in Fig. 4.3, and the answer is no. If that happened at some point x_1, the solution at x_1 would be strictly decreasing, in contrast to the fact that $y'(x) \ge 0$ on a right neighbourhood of x_1 (corresponding to points on the graph below the hyperbola).

So for $x > x_0$ the solution's graph stays above the hyperbola $y = 1/x$, $y > 0$. In particular, it stays inside $A = \{(x, y) \in \mathbb{R}^2 : y > 0\}$, the domain of definition of $f(x, y) = 1/y - x$. The solution is therefore decreasing and defined for any $x > x_0$.

Let ℓ be the limit of $y(x)$ as $x \to +\infty$, which exists due to the monotonicity of $y(x)$.

By general principles $0 \le \ell < y(1) = 1$. We claim that, actually, $\ell = 0$. By contradiction, if $0 < \ell < 1$, the ODE $y'(x) = f(x, y(x))$ says

$$\lim_{x\to+\infty} y'(x) = \lim_{x\to+\infty} f(x, y(x)) = \lim_{x\to+\infty} \frac{1}{y(x)} - x = -\infty\,,$$

and integrating $y'(x)$ we would also have $y(x) \to -\infty$.

Hence $\ell = 0$. Since $y(x) \to 0$ when $x \to +\infty$, the solution has the x-axis as horizontal asymptote.

Differentiating the ODE in (4.102) we obtain

$$y'' = -\frac{1}{y^2}\, y' - 1\,.$$

In particular, for $x < x_0 = 1$ (and $y > 0$) we have $y'' < 0$ (since $y' > 0$) so $y(x)$ is concave. Moreover, as $y(x)$ is strictly increasing and concave for $x < 1$, it must tend to zero as x approaches a certain finite value a.

Therefore the solution is defined on the interval $(a, +\infty)$, and using the differential equation again we find

$$\lim_{x\to a^+} y(x) = 0\,, \qquad \lim_{x\to a^+} y'(x) = +\infty\,,$$

(continued)

Example 1 (continued)

showing that $y(x)$ approaches the x-axis as $x \to a^+$, and in the limit the tangent line is vertical. The solution's graph, based on the information gathered, is represented in Fig. 4.4.

Fig. 4.3

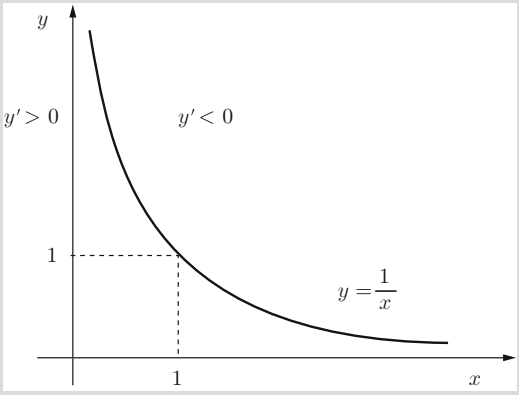
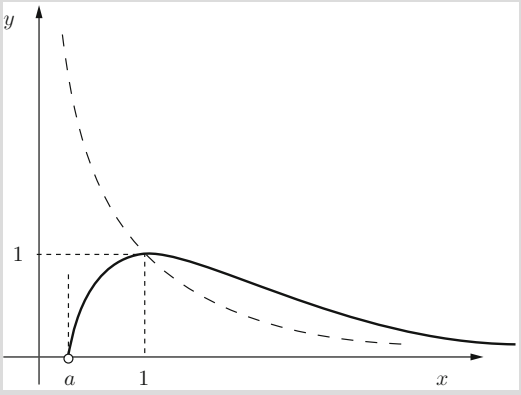


Fig. 4.4



In the following two examples it is possible to solve the ODE using the methods seen in Sect. 4.5. The equation of Example 2 is separable and in Example 3 it is homogeneous. However, in both cases the solution is implicit, and it is not an easy task to deduce qualitative properties from the analytical expressions.

Example 2 Let us study the initial value problem

$$\begin{cases} y' = y^6 - y^3 - 2 \\ y(0) = 1\,. \end{cases} \tag{4.103}$$

(continued)

Example 2 (continued)
The solution $y(x)$, which exists by Cauchy's theorem, is strictly decreasing on a neighbourhood of the initial point $x_0 = 0$. In fact the derivative $y' = y^6 - y^3 - 2$ is negative inside the interval between the roots of the right-hand-side polynomial, i.e. for $-1 < y < 2^{1/3}$, and the initial value $y(0) = 1$ satisfies that.

The constant functions $y \equiv -1$ and $y \equiv 2^{1/3}$ solve the ODE. By Cauchy's uniqueness, no other solution can, for any $x = x_1$, take the values $y(x_1) = -1$ or $y(x_1) = 2^{1/3}$, because otherwise we would have two distinct solutions of the same equation, with the same initial value at x_1.

Hence the graph of $y(x)$ cannot intersect the horizontal lines $y = -1$ and $y = 2^{1/3}$. Being continuous, $y(x)$ satisfies the bounds $-1 < y(x) < 2^{1/3}$ for any x in its domain.

In the strip $-1 < y < 2^{1/3}$ the function $f(x, y) = y^6 - y^3 - 2$ (constant in x) is bounded and Lipschitz in y, so the global existence and uniqueness theorem says $y(x)$ is defined for any $x \in \mathbb{R}$.

Call ℓ the limit of $y(x)$ as $x \to +\infty$ (ℓ exists since $y(x)$ is monotone, and similarly for $x \to -\infty$). Then necessarily $-1 \le \ell < 2^{1/3}$. Now we shall prove that $\ell = -1$.

As $y(x)$ has finite limit when $x \to +\infty$, also $y'(x)$ admits limit as $x \to +\infty$. Moreover, by de L'Hôpital's rule this limit must be zero:

$$0 = \left(\frac{\ell}{+\infty} =\right) \lim_{x\to+\infty} \frac{y(x)}{x} = \lim_{x\to+\infty} y'(x)\,. \tag{4.104}$$

Recall that de L'Hôpital's rule applies not only to the indeterminate forms $0/0$ and ∞/∞, but also when only the denominator diverges, as in (4.104). Furthermore, one of the assumptions of de L'Hôpital's theorem is the existence of the limit of $y'(x)$ on the right. This follows from the ODE $y'(x) = f(x, y(x))$, because $y(x)$ has finite limit when $x \to +\infty$,

$$\begin{aligned} \lim_{x\to+\infty} y'(x) &= \lim_{x\to+\infty} f(x, y(x)) = \\ &= \lim_{x\to+\infty} [y(x)]^6 - [y(x)]^3 - 2 = \ell^6 - \ell^3 - 2\,. \end{aligned} \tag{4.105}$$

Hence $y'(x)$ admits limit when $x \to +\infty$, and by (4.104) the limit is 0.

Here is a second method, that does not rely on de L'Hôpital's theorem, for proving that $\lim_{x\to+\infty} y'(x) = 0$. Using $y'(x) = f(x, y(x))$, the derivative $y'(x)$ admits limit as $x \to +\infty$ as in (4.105):

$$\lim_{x\to+\infty} y'(x) = \ell^6 - \ell^3 - 2.$$

This quantity is a number $\ell' \le 0$ because $-1 \le \ell < 2^{1/3}$. If ℓ' were negative, we would have $y(x) \to -\infty$ as $x \to +\infty$, against the boundedness of $y(x)$. Therefore $\ell' = \lim_{x\to+\infty} y'(x) = 0$.

From (4.105) it follows that $\ell^6 - \ell^3 - 2 = 0$, so $\ell = -1$ or $\ell = 2^{1/3}$. The latter possibility must be discarded since $y(x)$ is decreasing, with $-1 < y(x) < 2^{1/3}$ for any $x \in \mathbb{R}$. All in all, $y(x) \to -1$ as $x \to +\infty$ (and similarly, $y(x) \to 2^{1/3}$ when $x \to -\infty$).

(continued)

Example 2 (continued)

Differentiating in x the ODE in (4.103) allows to establish the sign of the second derivative. In fact

$$y'' = \frac{d}{dx}\,(y^6 - y^3 - 2) = 3y^2(2y^3 - 1)\cdot y'\,,$$

and $y'(x) < 0$ for any $x \in \mathbb{R}$ implies $y'' > 0$ when $y < 2^{-1/3}$, $y \neq 0$. So, the solution $y(x)$ is convex when y is less than $2^{-1/3}$ and concave otherwise. At $y = 2^{-1/3}$ the function $y(x)$ has an inflection.

Figure 4.5 represents the solution's graph based on the above features.

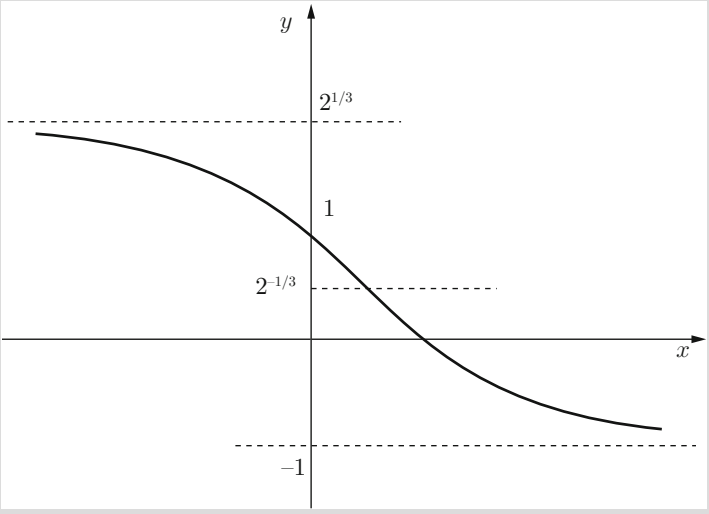


Fig. 4.5

Example 3 Let us study the solutions of

$$y' = \frac{y+x}{y-x}\,. \tag{4.106}$$

The first derivative's sign is determined by the sign of

$$f(x,y) = \frac{y+x}{y-x}$$

(continued)

Example 3 (continued)
and is represented in Fig. 4.6. Along the line $y = x$ (where the ODE's right-hand side is not well defined) the first derivative diverges to infinity, whilst along $y = -x$ the first derivative vanishes. Moreover, $(0, 0)$ is a singular point for the ODE.

The sign of the second derivative can be found by differentiating the equation. Easy calculations produce

$$y'' = 2\,\frac{y - xy'}{(y - x)^2}\,,$$

and keeping in account $y' = (y + x)/(y - x)$, we have

$$y'' = 2\,\frac{y^2 - 2xy - x^2}{(y - x)^3}\,. \tag{4.107}$$

Then $y'' = 0$ when the numerator of (4.107) vanishes, i.e. along the straight lines $y/x = 1 \pm \sqrt{2}$.

By direct inspection the two linear functions $y = (1 \pm \sqrt{2})x$ solve (4.106). The graph of any other solution, by Cauchy's uniqueness, cannot meet these lines.

The qualitative graph of the general integral of (4.106) is shown in Fig. 4.7.

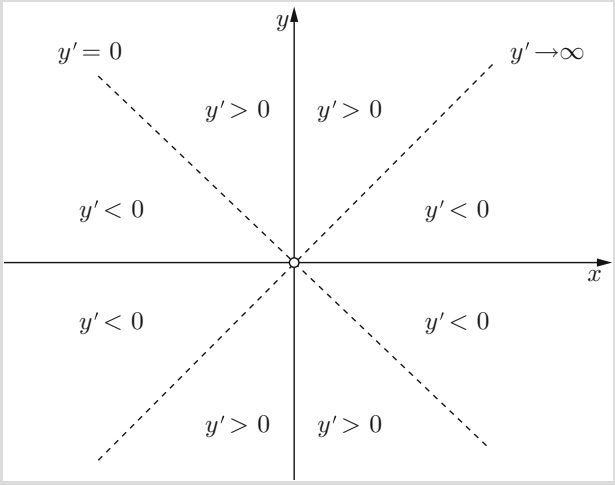


Fig. 4.6

(continued)

Example 3 (continued)

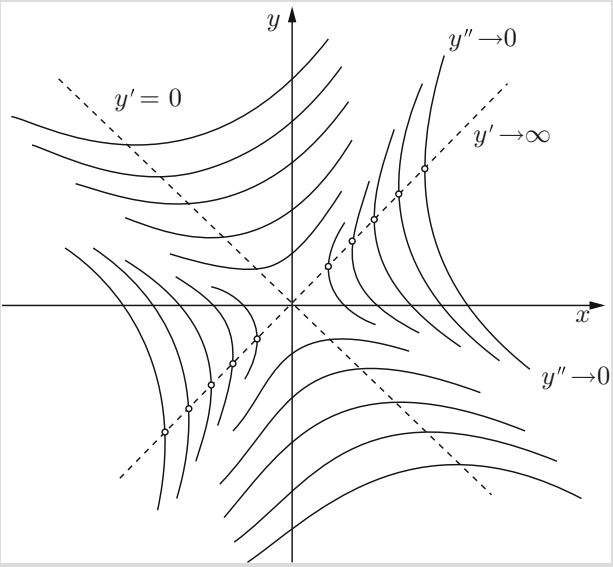


Fig. 4.7

Appendix to Chap. 4

4.9 Peano's Theorem

Peano's theorem, which will be proved in this section, is a result on local existence for Cauchy problems relative to systems of order one in normal form

$$y' = f(x, y)\,.$$

The theorem holds under the sole assumption that the function f is continuous.

Precisely, given $x_0 \in \mathbb{R}$ and $y_0 \in \mathbb{R}^n$, $a, b > 0$, suppose f is *continuous* on a set $I \times J \subset \mathbb{R} \times \mathbb{R}^n$ of the form

$$I \times J = \{(x, y) \in \mathbb{R} \times \mathbb{R}^n : x_0 \le x \le x_0 + a,\ |y - y_0| \le b\}\,.$$

We could consider a left neighbourhood $x_0 - a \le x \le x_0$ of the point x_0, instead of a right neighbourhood, and the conclusion of Peano's theorem would be valid there. Recall that $|y - y_0|$ is the *norm* (or *modulus*) in $\mathbb{R}^n$ of the vector $y - y_0$.

Peano's Local Existence Theorem *Assuming f is continuous, there exist a real number $\delta > 0$ and a function $y = y(x)$, $y : [x_0, x_0 + \delta] \to \mathbb{R}^n$, defined and differentiable on $[x_0, x_0 + \delta]$, that solves on that interval the initial value problem*

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \,. \end{cases} \tag{4.108}$$

Under the hypotheses of Peano's theorem there is no uniqueness. For example, as observed in Sect. 4.3, the following initial value problem of order one

$$\begin{cases} y' = 2\sqrt{|y|} \\ y(0) = 0 \,, \end{cases}$$

with $y : \mathbb{R} \to \mathbb{R}$, admits infinitely many solutions on $\mathbb{R}$. One family of solutions, depending on two real parameters h, k, $h \le 0 \le k$, is

$$y(x) = \begin{cases} -(x-h)^2 & \text{if} \quad x \le h \\ 0 & \text{if} \quad h < x < k \\ (x-k)^2 & \text{if} \quad x \ge k \,. \end{cases}$$

Proof of Peano's Theorem As in the Cauchy theorem, let M be the maximum

$$M = \max\{|f(x, y)| : (x, y) \in I \times J\}$$

and $\delta > 0$ the number

$$\delta = \min\left\{a; \frac{b}{M}\right\} .$$

Call I_δ the interval $[x_0, x_0 + \delta]$ and let us define a sequence of functions $y_k : I_\delta \to \mathbb{R}^n$, $k \in \mathbb{N}$. We begin with setting $y_k(x) \equiv y_0$ when $x < x_0$. Such extension of y_k outside I_δ will come in handy later (for $i = 1$).

Now we define the element y_k in the sequence. For this we divide I_δ in k closed subintervals of the same length

$$\left[x_0, x_0 + \frac{\delta}{k}\right], \quad \left[x_0 + \frac{\delta}{k}, x_0 + \frac{2\delta}{k}\right], \ \ldots \ , \left[x_0 + \frac{(k-1)\delta}{k}, x_0 + \delta\right] ;$$

and then define $y_k(x)$ on each piece recursively, first for $x < x_0$ (as we did above), then on $[x_0, x_0 + \delta/k]$, and so forth.

Precisely, on the generic interval

$$I_{k,i} = \left[x_0 + \frac{(i-1)\delta}{k}, x_0 + \frac{i\delta}{k} \right], \qquad i = 1, 2, \ldots, k,$$

we define $y_k(x)$ by the formula

$$y_k(x) = y_0 + \int_{x_0}^{x} f(t, y_k(t - \delta/k))\, dt\,. \tag{4.109}$$

Hence y_k is given on $I_{k,i}$ using the values that y_k assumes on the preceding interval $I_{k,i-1}$. Consider $I_{k,0}$ (the symbol should be clear) and recall y_k is constant and equal to y_0 on such interval.

We claim the sequence y_k is well defined. In other words, the integrand $f(t, y_k(t - \delta/k))$ is well defined, or equivalently,

$$y_k(x) \in J = \{y \in \mathbb{R}^n : |y - y_0| \le b\} \qquad \forall\, x \in I_{k,i-1} \tag{4.110}$$

where $t - \delta/k$ is a generic point in $I_{k,i-1}$. We shall prove (4.110) by induction on $i = 1, 2, \ldots, k$. When $i = 1$ recall that $y_k(x) \equiv y_0$ for $x < x_0$. Hence $y_k \equiv y_0 \in J$ for any $x \in I_{k,0}$. For generic i, suppose $y_k(x) \in J$ for any $x \in I_{k,i-1}$. For these points x we then have $(x, y_k(x)) \in I \times J$, and so

$$|f(x, y_k(x)| \le M, \qquad \forall\, x \in I_{k,i-1}\,. \tag{4.111}$$

For any $x \in I_{k,i}$, from definition (4.109)

$$\begin{aligned} |y_k(x) - y_0| &= \left| \int_{x_0}^{x} f(t, y_k(t - \delta/k))\, dt \right| \le \\ &\le \left| \int_{x_0}^{x} |f(t, y_k(t - \delta/k)|\, dt \right| \le M|x - x_0| \le M\delta \le b \end{aligned}$$

since $\delta \le b/M$. Therefore $y_k(x) \in J$ for any $x \in I_{k,i}$ and (4.109) is well defined.

Now, it is a good idea to think of expression (4.109) as a global definition on the interval $[x_0, x_0 + \delta]$ (more precisely, on $[x_0 - 1/k, x_0 + \delta]$, after setting $y_k(x) \equiv y_0$ when $x < x_0$):

$$y_k(x) = y_0 + \int_{x_0}^{x} f(t, y_k(t - \delta/k))\, dt\,, \qquad \forall\, x \in [x_0, x_0 + \delta]\,. \tag{4.112}$$

From (4.110) we have

$$|y_k(x) - y_0| \le b, \qquad \forall\, x \in [x_0, x_0 + \delta]\,. \tag{4.113}$$

Moreover, (4.111) implies that if $x_1, x_2 \in [x_0, x_0 + \delta]$ then

$$\begin{aligned} |y_k(x_2) - y_k(x_1)| &= \left| \int_{x_1}^{x_2} f(t, y_k(t - \delta/k))\, dt \right| \leq \\ &\leq \left| \int_{x_1}^{x_2} |f(t, y_k(t - \delta/k))|\, dt \right| \leq M|x_2 - x_1|\,. \end{aligned} \tag{4.114}$$

We shall retain the definitions and results of Sect. 1.10 regarding the Ascoli-Arzelà theorem. By (4.113), (4.114) the functions $y_k(x)$ are uniformly bounded and equicontinuous on the compact interval $[x_0, x_0 + \delta]$. The Ascoli-Arzelà theorem then guarantees the existence of a subsequence $y_{k_h}(x)$ of $y_k(x)$ that converges uniformly on $[x_0, x_0 + \delta]$ to a continuous function $y(x)$.

The uniform convergence then implies $y_{k_h}(x - \delta/k_h)$ converges, as $h \to +\infty$, to $y(x)$. Since f is uniformly continuous on $I \times J$, the composite map $f(x, y_{k_h}(x - \delta/k_h))$ converges uniformly to $f(x, y(x))$ on $[x_0, x_0 + \delta]$.

Next, consider the integral relation (4.112) for the indices k_h. As the convergence is uniform, we may take the limit inside the integral in (4.112) as $h \to +\infty$, to obtain

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt\,, \qquad \forall\, x \in [x_0, x_0 + \delta]\,.$$

By the integral form of Sect. 4.2, the function $y(x)$ is a solution to the initial value problem (4.108).

Check for updates

Linear Differential Equations

5

5.1 General Properties

Consider the ODE of order n in normal form

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \ldots + a_1(x)y' + a_0(x)y = g(x)\,, \tag{5.1}$$

where the functions $a_0(x), a_1(x), \ldots, a_{n-1}(x)$, called *coefficients* of the equation, and $g(x)$ are continuous on the interval $[a, b] \subset \mathbb{R}$. The function $g(x)$ will be referred to as the *right-hand side* of Eq. (5.1).

Equation (5.1) is a *linear differential equation*, because the map L that associates with any function $u(x)$ of class $C^n([a, b])$ the function

$$L(u) = u^{(n)} + a_{n-1}u^{(n-1)} + \ldots + a_1(x)u' + a_0(x)u\,,$$

is linear, i.e. it satisfies

$$L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)\,, \tag{5.2}$$

for any pair u, v of $C^n([a, b])$-functions and any $\alpha, \beta \in \mathbb{R}$.

From that, a few properties of the *solutions* (also known as *integrals*) to (51.1) follow. For example, if u and v are solutions

$$L(u) = g\,, \qquad L(v) = g\,,$$

then the difference $\omega = u - v$ is a solution of the *homogeneous equation*

$$L(\omega) = 0\,.$$

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N. Fusco et al., *Mathematical Analysis*, La Matematica per il 3+2 137,
https://doi.org/10.1007/978-3-031-04151-8_5

Under the given hypotheses, we may apply to (5.1) the theorem on global existence and uniqueness for the associated *Cauchy problem* (see Sect. 4.4), which takes the form

Existence and Uniqueness Theorem *If the coefficients and the right-hand side of* (5.1) *are continuous functions on* $[a, b]$*, then for any* $x_0 \in [a, b]$ *and any* $(y_0, y_0', \ldots, y_0^{(n-1)}) \in \mathbb{R}^n$ *there exists a unique solution to equation* (5.1)*, defined on* $[a, b]$ *and satisfying the initial conditions*

$$\begin{cases} y(x_0) = y_0 \\ y'(x_0) = y_0' \\ \cdots\cdots\cdots \\ y^{(n-1)}(x_0) = y_0^{(n-1)} . \end{cases}$$

In fact, Eq. (5.1) can be put in normal form

$$y^{(n)} = f(x, y, y', y'', \ldots, y^{(n-1)}) ,$$

(see Sect. 4.1) provided we choose

$$f(x, y, y', y'', \ldots, y^{(n-1)}) = g(x) - [a_{(n-1)}(x) y^{(n-1)} + \ldots + a_1(x) y' + a_0(x) y] ,$$

which is continuous on $[a, b] \times \mathbb{R}^n$ and Lipschitz in all variables except possibly the first one.

Let us begin by studying the homogeneous equation

$$L(y) = y^{(n)} + a_{(n-1)}(x) y^{(n-1)} + \ldots + a_1(x) y' + a_0(x) y = 0 . \qquad (5.3)$$

Proposition 1 *For any* $k \in \mathbb{N}$*, if* $y_1, y_2, \ldots, y_k$ *are solutions (integrals) of* (5.3)*, any linear combination*

$$c_1 y_1 + c_2 y_2 + \ldots + c_k y_k$$

of them with real coefficients $c_1, c_2, \ldots, c_k$ *is a solution (an integral) of the same equation.*

Proof This is a simple consequence of linearity, i.e. (5.2).

Proposition 2 *For any $x_0 \in [a, b]$ the zero function u is the only particular integral of (5.3) that satisfies the initial conditions*

$$u(x_0) = 0\,, \quad u'(x_0) = 0\,, \quad \ldots \quad , u^{(n-1)}(x_0) = 0\,. \tag{5.4}$$

Proof Consequence of the existence and uniqueness theorem (in particular, of the uniqueness), since the zero function solves the ODE (5.3) with initial conditions (5.4).

The *general integral* of a linear ODE of order n is the set of all solutions to the equation. For determining such *general integral* it is useful to recall some concepts.

Let $u_1, u_2, \ldots, u_k$ be real functions defined on an interval I. We shall say $u_1, u_2, \ldots, u_k$ are *linearly dependent* if there are k real numbers $c_1, c_2, \ldots, c_k$, not all zero, such that

$$c_1 u_1(x) + c_2 u_2(x) + \ldots + c_k u_k(x) = 0\,, \qquad \forall\, x \in I\,. \tag{5.5}$$

The $u_1, u_2, \ldots, u_k$ are said *linearly independent* if they are not linearly dependent, i.e. if the only linear combination $c_1 u_1 + c_2 u_2 + \ldots + c_k u_k$ fulfilling (5.5) is the one where $c_1 = c_2 = \ldots = c_k = 0$.

Now let $y_1, y_2, \ldots, y_n$ be n particular integrals on $[a, b]$ of the homogeneous ODE (5.3). In order to decide whether they are linearly independent or not we introduce the *Wronskian determinant* of $y_1, y_2, \ldots, y_n$. This is the function $W(x)$ defined for $x \in [a, b]$ by the determinant

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) & \ldots & y_n(x) \\ y_1'(x) & y_2'(x) & \ldots & y_n'(x) \\ \ldots & \ldots & \ldots & \ldots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \ldots & y_n^{(n-1)}(x) \end{vmatrix}.$$

Wronskian Theorem *If $y_1, y_2, \ldots, y_n$ are particular integrals of the homogeneous equation (5.3), then*

(i) *there is a point $x_0 \in [a, b]$ such that $W(x_0) = 0$ if and only if $y_1, y_2, \ldots, y_n$ are linearly dependent;*

(ii) *there is a point $x_1 \in [a, b]$ such that $W(x_1) \neq 0$ if and only if $y_1, y_2, \ldots, y_n$ are linearly independent.*

Proof Let us prove (i) first. Suppose $W(x_0) = 0$, so the linear system in the unknowns $\xi_1, \xi_2, \ldots, \xi_n$

$$\begin{cases} y_1(x_0)\xi_1 + y_2(x_0)\xi_2 + \ldots + y_n(x_0)\xi_n = 0 \\ y_1'(x_0)\xi_1 + y_2'(x_0)\xi_2 + \ldots + y_n'(x_0)\xi_n = 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_1^{(n-1)}(x_0)\xi_1 + y_2^{(n-1)}(x_0)\xi_2 + \ldots + y_n^{(n-1)}(x_0)\xi_n = 0 \end{cases}$$

has a solution $(c_1, c_2, \ldots, c_n)$ different from $(0, 0, \ldots, 0)$. Hence the function $u(x) = c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x)$ is a particular integral of (5.3), that vanishes at x_0 together with its $n-1$ derivatives. By Proposition 2, $u(x) = 0$ for any $x \in [a, b]$, and then the integrals $y_1, y_2, \ldots, y_n$ are linearly dependent.

Conversely, let $c_1, c_2, \ldots, c_n$ be n real numbers, not all zero, such that

$$c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x) = 0\,, \qquad \forall\, x \in [a, b]\,. \tag{5.6}$$

Differentiating, for any $x \in [a, b]$ we obtain $n-1$ equations

$$\begin{cases} c_1 y_1'(x) + c_2 y_2'(x) + \ldots + c_n y_n'(x) = 0 \\ c_1 y_1''(x) + c_2 y_2''(x) + \ldots + c_n y_n''(x) = 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ c_1 y_1^{(n-1)}(x) + c_2 y_2^{(n-1)}(x) + \ldots + c_n y_n^{(n-1)}(x) = 0. \end{cases} \tag{5.7}$$

Therefore the linear system of n Eqs. (5.6) and (5.7), for any $x \in [a, b]$, has a solution $(c_1, c_2, \ldots, c_n)$ different from the trivial solution $(0, 0, \ldots, 0)$. Hence the determinant $W(x)$ is zero for any $x \in [a, b]$.

To prove (ii) suppose $W(x_1) \neq 0$. The functions $y_1, y_2, \ldots, y_n$ must be linearly independent, otherwise what we have seen above would imply $W(x) = 0$ for any $x \in [a, b]$. Vice versa, if $y_1, y_2, \ldots, y_n$ are linearly independent, by i) we have $W(x) \neq 0$ for any $x \in [a, b]$.

From the above result we deduce the following proposition

Proposition 3 *If $y_1, y_2, \ldots, y_n$ are particular integrals of the homogeneous equation (5.3), their Wronskian $W(x)$ is either identically zero (if the particular integrals are dependent), or different from zero for any $x \in [a, b]$ (if the particular integrals are independent).*

The determinant's properties allow to prove the following result

Proposition 4 *The Wronskian of n particular integrals of* (5.3) *is a particular integral of the linear homogeneous equation of order one*

$$y' = -a_{n-1}(x)y\,.$$

Proof Recall that the derivative of a square matrix' determinant equals the sum of the determinants obtained differentiating each single row. From this, the Wronskian derivative $W'(x)$ is the sum of n determinants, of which the first $n-1$ are zero because they have two identical rows, and the last one is

$$\begin{vmatrix} y_1(x) & y_2(x) & \dots & y_n(x) \\ y_1'(x) & y_2'(x) & \dots & y_n'(x) \\ \dots & \dots & \dots & \dots \\ y_1^{(n-2)}(x) & y_2^{(n-2)}(x) & \dots & y_n^{(n-2)}(x) \\ y_1^{(n)}(x) & y_2^{(n)}(x) & \dots & y_n^{(n)}(x) \end{vmatrix}\,. \tag{5.8}$$

As, for any $i = 1, 2, \dots, n$,

$$y_i^{(n)} = -\left(a_{n-1}(x)y_i^{(n-1)}(x) + \dots + a_0(x)y_i(x)\right)\,,$$

(5.8) is the sum of n determinants: one is $-a_{n-1}(x)\cdot W(x)$, while the others vanish because they have two proportional rows. All in all we obtain

$$W'(x) = -a_{n-1}(x)W(x)\,, \qquad \forall\, x \in [a,b]\,.$$

5.2 General Integral of Linear ODEs

Let us begin with the case of a homogeneous equation of order n

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = 0\,, \tag{5.9}$$

with continuous coefficients on the interval $[a,b] \subset \mathbb{R}$. We observed in the previous section that if $y_1, y_2, \dots, y_n$ are n particular integrals of (5.9), any linear combination $y_0(x)$ with real coefficients

$$y_0(x) = c_1 y_1(x) + c_2 y_2(x) + \dots + c_n y_n(x) \tag{5.10}$$

is still a particular integral of (5.9)

The aim of this section is proving that:

(a) a homogeneous equation of order n always admits n linearly independent integrals $y_1, y_2, \ldots, y_n$;
(b) if $y_1, y_2, \ldots, y_n$ are n linearly independent integrals, any other particular integral $y_0(x)$ of (5.9) can be written as in (5.10).

In other words, the solution set to (5.9), i.e. the *general integral*, is a vector subspace of $C^n([a, b])$, of which the n solutions $y_1, y_2, \ldots, y_n$ are a basis.

Theorem (Existence of n Linearly Independent Integrals) *The homogeneous ODE (5.9), with continuous coefficients on $[a, b]$, always admits n linearly independent integrals.*

Proof Pick x_0 in $[a, b]$ and let I denote the $n \times n$ identity matrix

$$I = \begin{pmatrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \ldots & 1 \end{pmatrix}.$$

By the existence and uniqueness theorem, for any $k = 1, 2, \ldots, n$, there is a particular integral of (5.9) satisfying the initial conditions

$$\begin{cases} y_k(x_0) = \delta_{1k} \\ y_k'(x_0) = \delta_{2k} \\ \ldots\ldots\ldots \\ y_k^{(n-1)}(x_0) = \delta_{nk} \end{cases}$$

where $\delta_{hk} = 1$ if $h = k$ and $\delta_{hk} = 0$ if $h \neq k$. Then the Wronskian of $y_1, y_2, \ldots, y_n$ satisfies

$$W(x_0) = \det I = 1\,.$$

By the Wronskian theorem, $W(x) \neq 0$ for any x, so $y_1, y_2, \ldots, y_n$ are linearly independent.

Theorem (General Integral of Homogeneous Equations) *Let $y_1, y_2, \ldots, y_n$ be n linearly independent particular integrals of (5.9). The general integral of (5.9) is given by linear combinations*

$$c_1 y_1 + c_2 y_2 + \ldots + c_n y_n\,, \qquad (c_1, c_2, \ldots, c_n) \in \mathbb{R}^n\,.$$

Proof We must show that for any particular solution $y(x)$ of (5.9) there is a unique n-tuple $(c_1, c_2, \ldots, c_n)$ of real numbers such that

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x)\,,$$

for any $x \in [a, b]$. For that, fix x_0 in $[a, b]$ and set

$$y(x_0) = y_0\,,\ y'(x_0) = y_0'\,,\ \ \ldots\ \ ,\ y^{(n-1)}(x_0) = y_0^{(n-1)}\,.$$

Consider the $n \times n$ linear system in the unknowns $\xi_1, \xi_2, \ldots, \xi_n$

$$\begin{cases} y_1(x_0)\xi_1 + y_2(x_0)\xi_2 + \ldots + y_n(x_0)\xi_n = y_0 \\ y_1'(x_0)\xi_1 + y_2'(x_0)\xi_2 + \ldots + y_n'(x_0)\xi_n = y_0' \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_1^{(n-1)}(x_0)\xi_1 + y_2^{(n-1)}(x_0)\xi_2 + \ldots + y_n^{(n-1)}(x_0)\xi_n = y_0^{(n-1)}\,. \end{cases} \tag{5.11}$$

As the Wronskian of $y_1, y_2, \ldots, y_n$ is non-zero at x_0, system (5.11) has a unique solution $(c_1, c_2, \ldots, c_n)$. The function

$$c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x)$$

is an integral of (5.9), which assumes at x_0 the same initial values of the solution $y(x)$. By uniqueness we have

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x)\,, \qquad \forall\, x \in [a, b]\,,$$

and the claim follows.

Remark If we call V the set of solutions of the homogeneous equation (5.9), we have in practice proved that V is a *vector space of dimension n*, because there are n linearly independent elements of V that generate any other element as a linear combination.

Let us now pass to *non-homogeneous* linear equations

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \ldots + a_1(x)y' + a_0(x)y = g(x)\,, \tag{5.12}$$

where the coefficients and the right-hand side are continuous on $[a, b]$. The homogeneous equation with the same coefficients

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \ldots + a_1(x)y' + a_0(x)y = 0\,,$$

is the *homogeneous equation associated* with (5.12).

We begin observing that if v is a particular integral of (5.12) and u is an integral of the associated homogeneous equation, then $u + v$ is another particular integral of (5.12).

We set out to prove that *to solve a non-homogeneous linear ODE it suffices to know one particular integral and the general integral of the associated homogeneous equation*. Namely, the following result holds.

Theorem (General Integral of Non-homogeneous Linear Equations) *Let v_0 be a particular integral of (5.12) and suppose $y_1, y_2, \ldots, y_n$ are n linearly independent integrals of the associated homogeneous equation. Then the general integral of (5.12) is*

$$c_1 y_1 + c_2 y_2 + \ldots + c_n y_n + v_0\,, \qquad (c_1, c_2, \ldots, c_n) \in \mathbb{R}^n\,.$$

Proof We have to show that for any particular solution $y(x)$ to (5.12), there is a unique vector $(c_1, c_2, \ldots, c_n)$ of real numbers such that

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x) + v_0(x)\,, \qquad \forall\, x \in [a, b]. \tag{5.13}$$

Let x_0 be a point di $[a, b]$ and set

$$y(x_0) = y_0\,,\ y'(x_0) = y_0'\,,\ \ \ldots\ \ ,\ y^{(n-1)}(x_0) = y_0^{(n-1)}\,.$$

The linear system in the unknowns $\xi_1, \xi_2, \ldots, \xi_n$

$$\begin{cases} y_1(x_0)\xi_1 + y_2(x_0)\xi_2 + \ldots + y_n(x_0)\xi_n = y_0 - v_0(x_0) \\ y_1'(x_0)\xi_1 + y_2'(x_0)\xi_2 + \ldots + y_n'(x_0)\xi_n = y_0' - v_0'(x_0) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_1^{(n-1)}(x_0)\xi_1 + y_2^{(n-1)}(x_0)\xi_2 + \ldots + y_n^{(n-1)}(x_0)\xi_n = y_0^{(n-1)} - v_0^{(n-1)}(x_0) \end{cases}$$

has a unique solution $(c_1, c_2, \ldots, c_n)$, since its determinant equals the value at x_0 of the Wronskian of $y_1, y_2, \ldots, y_n$, which is non-zero by the linear independence of $y_1, y_2, \ldots, y_n$.

The function

$$c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x) + v_0(x)$$

is an integral of (5.12), which assumes at x_0 the same value as the solution $y(x)$. By the uniqueness theorem, then, (5.13) holds.

Now we find the general integral of the first-order linear equation

$$y' + a_0(x)y = g(x)\,, \tag{5.14}$$

where $a_0(x)$ is continuous on the interval $[a, b]$. We fix $x_0 \in [a, b]$ and denote by $u(x)$ the integral of the associated homogeneous ODE, which at x_0 equals 1, i.e. the solution to the Cauchy problem

$$\begin{cases} u' + a_0(x)u = 0 \\ u(x_0) = 1\,. \end{cases}$$

By Proposition 2 in the previous section, $u(x) \neq 0$ for any $x \in [a, b]$ (since if $u(x)$ were zero somewhere in $[a, b]$ it would have to vanish everywhere on $[a, b]$). The ODE is separable, so *separating* to the left-hand side the unknown variable u gives

$$\frac{u'(x)}{u(x)} = -a_0(x)\,, \qquad \forall\, x \in [a, b].$$

Integrating both sides, for any $x \in [a, b]$ we have

$$\int_{x_0}^{x} \frac{u'(t)}{u(t)}\,dt = \log \frac{u(x)}{u(x_0)} = -\int_{x_0}^{x} a_0(t)\,dt$$

and since $u(x_0) = 1$,

$$u(x) = e^{-\int_{x_0}^{x} a_0(t)\,dt}\,. \tag{5.15}$$

Hence the homogeneous equation's general integral is

$$ce^{-\int_{x_0}^{x} a_0(t)\,dt}\,, \qquad c \in \mathbb{R}\,.$$

By the theorem on non-homogeneous linear ODEs, the general integral of (5.14) is the function

$$ce^{-\int_{x_0}^{x} a_0(t)\,dt} + y_0(x) \qquad c \in \mathbb{R}\,, \tag{5.16}$$

where y_0 is a particular integral of (5.14).

To determine y_0 we shall use *Lagrange's method of variation of parameters*. If $u(x)$ is a (non-zero) particular integral of the associated homogeneous equation, we seek $y_0(x)$ of the form

$$y_0(x) = \varphi(x)u(x)\,,$$

where $\varphi(x)$ is to be determined. We impose that $y_0(x)$ solves the complete Eq. (5.14). Then $y_0' = \varphi' u + \varphi u'$, and demanding that $y_0' + a_0(x)y_0 = g(x)$ be

satisfied, we deduce

$$\varphi' u + \varphi u' + a_0(x)\varphi u = g(x)\,. \tag{5.17}$$

By assumption $\varphi u' + a_0(x)\,\varphi u = \varphi(u' + a_0(x)\,u) = \varphi \cdot 0 = 0$, so from (5.17) we obtain $\varphi' u = g(x)$ i.e. $\varphi' = g(x)/u$. If we choose, for example, $\varphi(x)$ so that $\varphi(x_0) = 0$, we find

$$\varphi(x) = \int_{x_0}^{x} \frac{g(t)}{u(t)}\,dt\,,$$

whence

$$y_0(x) = \varphi(x)u(x) = u(x)\int_{x_0}^{x} \frac{g(t)}{u(t)}\,dt$$

is the integral of (5.14) that vanishes at x_0.

Remembering (5.15) and (5.16), eventually the general integral of (5.14) is

$$e^{-\int_{x_0}^{x} a_0(t)\,dt}\left(\int_{x_0}^{x} g(t)\,e^{\int_{x_0}^{t} a_0(s)\,ds}\,dt + c\right) \qquad c \in \mathbb{R}\,.$$

Example 1 Let us solve an ODE of type

$$y' = \frac{y}{\operatorname{tg} x} + h(x)\sin x\,,$$

where $h(x)$ is continuous on $(0, \pi)$. The associated homogeneous equation

$$y' = \frac{y}{\operatorname{tg} x}$$

is linear and separable, and admits the general integral $y = c\sin x$. We seek a particular solution of the full equation of the form $y_0 = \varphi(x)\sin x$. Then

$$\varphi'(x)\sin x = h(x)\sin x\,,$$

and so

$$y_0(x) = \sin x\int_{x_0}^{x} h(t)\,dt\,,$$

where x_0 is an (arbitrary) given point in $(0, \pi)$. The general integral therefore reads

$$c \in \mathbb{R} \to \sin x\left(c + \int_{x_0}^{x} h(t)\,dt\right)\,.$$

5.3 The Method of Variation of Parameters

The previous section should stress the importance of finding a particular integral for a linear equation of order n, once we know n linearly independent integrals y_1, y_2, …, y_n for the associated homogeneous equation. This is enough to obtain the general solution of the ODE.

In this section we shall describe a technique, due to Lagrange, that allows to determine a particular integral starting from n linearly independent integrals y_1, y_2, …, y_n. We begin by looking at a linear equation of *second order*

$$y'' + a_1(x)\, y' + a_0(x) y = g(x)\,, \tag{5.18}$$

where $g(x)$ is continuous on $[a, b]$. Let $y_1(x)$, $y_2(x)$ be two solutions on $[a, b]$ of the associated homogeneous equation, i.e.

$$y_i'' + a_1(x) y_i' + a_0(x)\, y_i = 0\,, \qquad i = 1, 2 \tag{5.19}$$

on $[a, b]$. Suppose $y_1(x)$, $y_2(x)$ are linearly independent. We look for a solution to (5.18) of the form

$$y = \varphi_1(x) y_1 + \varphi_2(x) y_2\,, \tag{5.20}$$

imposing on $\varphi_1(x)$, $\varphi_2(x)$ the condition that $y = y(x)$ *may be differentiated as if* $\varphi_i(x)$ *were constants*, i.e.

$$y' = \varphi_1(x) y_1' + \varphi_2(x) y_2'\,. \tag{5.21}$$

This is the same as demanding

$$\varphi_1'(x) y_1 + \varphi_2'(x) y_2 = 0\,. \tag{5.22}$$

Differentiating (5.21) in x produces

$$y'' = \varphi_1(x) y_1'' + \varphi_2(x) y_2'' + \varphi_1'(x) y_1' + \varphi_2'(x) y_2'\,.$$

Hence the function y given by (5.20) solves Eq. (5.18) if and only if

$$\varphi_1\big(y_1'' + a_1(x)\, y_1' + a_0(x)\, y_1\big) + \varphi_2\left(y_2'' + a_1(x)\, y_2' + a_0(x)\, y_2\right) + \\ + \varphi_1'(x)\, y_1' + \varphi_2'(x)\, y_2' = g(x)\,.$$

By (5.19), this happens precisely when

$$\varphi_1'(x) y_1' + \varphi_2'(x) y_2' = g(x)\,. \tag{5.23}$$

The system made of (5.22) and (5.23)

$$\begin{cases} \varphi_1'(x)y_1 + \varphi_2'(x)y_2 = 0 \\ \varphi_1'(x)\,y_1' + \varphi_2'(x)y_2' = g(x) \end{cases} \tag{5.24}$$

in the unknowns $\varphi_1'(x)$, $\varphi_2'(x)$, has a unique solution because its determinant is the Wronskian $W(x)$ of y_1, y_2, and the latter is non-zero by the Wronskian theorem (Sect. 5.1). By Cramer's rule the solutions then are

$$\varphi_1'(x) = \frac{\begin{vmatrix} 0 & y_2(x) \\ g(x) & y_2'(x) \end{vmatrix}}{W(x)}\,,$$

$$\varphi_2'(x) = \frac{\begin{vmatrix} y_1(x) & 0 \\ y_1'(x) & g(x) \end{vmatrix}}{W(x)}\,.$$

Let us now discuss the above method in general, for ODEs of any order.

Method of Variation of Parameters (Lagrange) *Let y_1, y_2, …, y_n, for $x \in [a, b]$, be n linearly independent integrals of the homogeneous equation associated with the ODE*

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \ldots + a_1(x)y' + a_0(x)y = g(x)\,. \tag{5.25}$$

Suppose the derivatives of the n-tuple $(\varphi_1, \varphi_2, \ldots, \varphi_n)$ solve on $[a, b]$ the linear system in the unknowns $\xi_1, \xi_2, \ldots, \xi_n$:

$$\begin{cases} y_1(x)\xi_1 + y_2(x)\xi_2 + \ldots + y_n(x)\xi_n = 0 \\ y_1'(x)\xi_1 + y_2'(x)\xi_2 + \ldots + y_n'(x)\xi_n = 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_1^{(n-1)}(x)\xi_1 + y_2^{(n-1)}(x)\xi_2 + \ldots + y_n^{(n-1)}(x)\xi_n = g(x) \end{cases}\,. \tag{5.26}$$

Then the linear combination

$$v(x) = \sum_{i=1}^{n} y_i(x)\varphi_i(x)\,, \qquad x \in [a, b]\,,$$

is a particular integral of the linear ODE (5.25).

Proof The linear system (5.26) has (only) one solution because the coefficient matrix' determinant, i.e. the Wronskian $W(x)$ of the solutions y_i , $i = 1, 2, \ldots, n$, is non-zero by the Wronskian theorem (Sect. 5.1). Let $\psi_i = \psi_i(x)$, $i = 1, 2, \ldots, n$, be the solution. Then $(\varphi_1, \varphi_2, \ldots, \varphi_n)$ arises from integrating $(\psi_1, \psi_2, \ldots, \psi_n)$:

$$\varphi_i(x) = \int_{x_0}^{x} \psi_i(t)\, dt\,, \qquad \forall\, i = 1, 2, \ldots, n$$

for a given $x_0 \in [a, b]$.

The derivatives $\xi_i = \varphi_i'$ of $\varphi_i(x)$, $i = 1, 2, \ldots, n$, satisfy system (5.26) by assumption. Therefore for any $x \in [a, b]$,

$$\begin{aligned}
v'(x) &= \sum_{i=1}^{n} y_i'(x)\varphi_i(x)\,,\\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\
v^{(n-1)}(x) &= \sum_{i=1}^{n} y_i^{(n-1)}(x)\varphi_i(x)\,,\\
v^{(n)}(x) &= \sum_{i=1}^{n} y_i^{(n)}(x)\,\varphi_i(x) + g(x)\,.
\end{aligned}$$

As y_i, for any $i \in \{1, 2, \ldots, n\}$, solves the associated homogeneous equation, the claim follows:

$$\begin{aligned}
&v^{(n)} + a_{n-1}(x)v^{(n-1)} + \ldots + a_1(x)v' + a_0(x)v =\\
&= g(x) + \sum_{i=1}^{n} \Big(y_i^{(n)} + a_{n-1}(x)y_i^{(n-1)} + \ldots +\\
&\qquad + a_1(x)y_i' + a_0(x)y_i \Big)\varphi_i(x) = g(x)\,, \qquad \forall\, x \in [a, b].
\end{aligned}$$

Example 1 Let us apply the method to the non-homogeneous linear ODE

$$y'' + y = \frac{1}{\sin x} \tag{5.27}$$

of order two. The associated homogeneous equation has two linearly independent integrals

$$y_1(x) = \cos x\,, \qquad y_2(x) = \sin x\,,$$

so we look for a particular solution of (5.27) of the form

$$y = \varphi_1(x)\cos x + \varphi_2(x)\sin x\,,$$

(continued)

Example 1 (continued)
so that the derivatives $\varphi_1'(x)$ and $\varphi_2'(x)$ of the unknown φ_1, φ_2 satisfy

$$\begin{cases} \varphi_1'(x)\cos x + \varphi_2'(x)\sin x = 0 \\ -\varphi_1'(x)\sin x + \varphi_2'(x)\cos x = 1/\sin x\,. \end{cases}$$

Using Cramer's rule,

$$\varphi_1'(x) = -1\,, \qquad \varphi_2'(x) = \frac{\cos x}{\sin x}\,,$$

and integrating, $\varphi_1(x) = -x$, $\varphi_2(x) = \log|\sin x|$. Hence

$$y(x) = -x\cos x + \sin x \log|\sin x|$$

is a particular integral of (5.27).

5.4 Bernoulli Equations

A Bernoulli equation is an ODE of order one of type

$$y' = a(x)y + b(x)y^\alpha\,, \tag{5.28}$$

with $a(x)$, $b(x)$ continuous on $[a, b] \subset \mathbb{R}$ and α a real number different from 0 and 1.

This (non-linear) ODE is in normal form $y' = f(x, y)$. The function

$$f(x, y) = a(x)y + b(x)y^\alpha\,,$$

is continuous for $(x, y) \in [a, b] \times (0, +\infty)$, but not globally Lipschitz in y. Nevertheless, restricting $f(x, y)$ to any compact set $[a, b] \times [c, d]$ (with $c > 0$), the function is continuous and Lipschitz in y. By Cauchy's existence and uniqueness theorem (Sect. 4.4), for any $(x_0, y_0) \in [a, b] \times [c, d]$ there is a unique integral curve of (5.28), defined on a suitable neighbourhood of x_0, passing through (x_0, y_0).

Let us show that a Bernoulli equation can be transformed into another linear ODE by changing the unknown function. We divide (5.28) by y^α (thus, for $\alpha > 0$, we are neglecting the zero solution). We have

$$\frac{y'}{y^\alpha} = a(x)y^{1-\alpha} + b(x)\,. \tag{5.29}$$

Setting $z(x) = [y(x)]^{1-\alpha}$ we obtain

$$z'(x) = \frac{d}{dx}[y(x)]^{1-\alpha} = (1-\alpha)\frac{y'}{y^\alpha}$$

so (5.29) turns into the linear equation in $z(x)$

$$z' = (1-\alpha)\, a(x)\, z + (1-\alpha)\, b(x)\,,$$

which is solved in the way shown earlier. Once we have $z(x)$, we will compute $y = z^{1/(1-\alpha)}$, solution to (5.28).

Example 1 Let us find two solutions, if possible defined on the entire $(0, +\infty)$, of the Bernoulli equation

$$x\, y' = -y^2 \log x - 2y\,. \tag{5.30}$$

Divide (5.30) by xy^2,

$$\frac{y'}{y^2} = -\frac{2}{xy} - \frac{\log x}{x} \tag{5.31}$$

and set $z(x) = 1/y$. In this way (5.31) becomes

$$z' = \frac{2z}{x} + \frac{\log x}{x}\,. \tag{5.32}$$

This is linear and of order one. The general integral of the homogeneous equation associated with (5.32) is $z = cx^2$. If we want a particular solution of (5.31) of type $\varphi(x)x^2$, we find for φ the constraint

$$\varphi'(x)x^2 = \frac{\log x}{x}\,,$$

and so

$$\varphi(x) = \int \frac{\log x}{x^3}\, dx = -\frac{\log x}{2x^2} - \frac{1}{4x^2} + c\,.$$

Eventually the general integral of (5.31) is

$$z(x) = cx^2 - \frac{\log x}{2} - \frac{1}{4}\,, \tag{5.33}$$

so the general integral of (5.30) is

$$y(x) = \left(cx^2 - \frac{\log x}{2} - \frac{1}{4}\right)^{-1}. \tag{5.34}$$

In order for $y = y(x) = y(x, c)$ to be defined on $(0, +\infty)$ it is necessary and sufficient that the $z(x)$ in (5.33) is positive for any $x > 0$. As $z(x) \to +\infty$ when $x \to 0^+$, we need

$$z(x) = cx^2 - \frac{\log x}{2} - \frac{1}{4} > 0\,, \qquad \forall x > 0. \tag{5.35}$$

(continued)

Example 1 (continued)
The necessary condition for having (5.35) is that the constant c be positive. If that is the case, $z(x)$ has a minimum for $x > 0$ at $x_0 = 1/(2\sqrt{c})$, with minimum value $\log(2\sqrt{c})/2$. Therefore the solutions to the Bernoulli ODE (5.30) that are defined on the whole $(0, +\infty)$ are the functions in (5.34) for which $c > 1/4$.

5.5 Homogeneous Equations with Constant Coefficients

Consider the linear homogeneous equation of order n

$$L(y) = y^{(n)} + a_{n-1}y^{(n-1)} + \ldots + a_1 y' + a_0 y = 0\,, \tag{5.36}$$

with *constant coefficients* $a_0, a_1, \ldots, a_{n-1} \in \mathbb{R}$. We set out to show that to determine n linearly independent integrals of (5.36) it suffices to know the roots of the algebraic equation of degree n

$$p(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \ldots + a_1\lambda + a_0 = 0\,, \tag{5.37}$$

called *characteristic equation* of (5.36) ($p(\lambda)$ is the *characteristic polynomial* of the linear ODE (5.36)).

Here we should introduce the notion of *complex solution* $u_1(x) + iu_2(x)$ to equation (5.36). If $u_1(x)$, $u_2(x)$ are real-valued functions defined on the interval I of $\mathbb{R}$, the *derivative* in x of the complex-valued function $u(x) = u_1(x) + iu_2(x)$ is the function $u'(x) = u_1'(x) + iu_2'(x)$. We shall say $u(x) = u_1(x) + iu_2(x)$ is a *particular integral* of (5.36) if, for any $x \in I$, the differential identity

$$L(u) = u^{(n)} + a_{n-1}u^{(n-1)} + \ldots + a_1 u' + a_0 u = 0$$

holds over $\mathbb{C}$. When $u(x) = u_1(x) + iu_2(x)$ is a complex solution to (5.36), the functions $u_1(x), u_2(x)$ are real solutions of (5.36). In fact $L(u) = L(u_1) + iL(u_2)$, so if u is a complex solution then $L(u_1) + iL(u_2)$ is the complex number zero, i.e. $L(u_1) = L(u_2) = 0$.

We recall that for any complex number $z = \alpha + i\beta$, the complex number e^z is defined as

$$e^z = e^{\alpha+i\beta} = e^\alpha\,(\cos\beta + i\sin\beta)\,.$$

It is easy to verify that, given $z \in \mathbb{C}$, the function $f(x) = e^{xz}$ is differentiable in $x \in \mathbb{R}$, and

$$f'(x) = z \cdot e^{xz}\,.$$

Finally, since for any $x \in \mathbb{R}$

$$e^{ix} = \cos x + i \sin x\,, \qquad e^{-ix} = \cos x - i \sin x\,,$$

we have the *Euler formulas*

$$\sin x = \frac{e^{ix} - e^{-ix}}{2}\,, \qquad \cos x = \frac{e^{ix} + e^{-ix}}{2}\,, \qquad \forall\, x \in \mathbb{R}.$$

Proposition 1 *For any real (or complex) number λ, the function*

$$x \in \mathbb{R} \;\rightarrow\; e^{\lambda x}$$

is a real (or complex) solution of (5.36) *if and only if λ is a root of the characteristic equation* (5.37)*.*

Proof It is enough to note

$$L(e^{\lambda x}) = e^{\lambda x}\; p(\lambda)\,,$$

with $L(y)$ and $p(\lambda)$ as in (5.36) and (5.37).

Proposition 2 *If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are n distinct roots of the characteristic equation* (5.37)*, the n integrals*

$$e^{\lambda_1 x}, \quad e^{\lambda_2 x}, \ldots, e^{\lambda_n x}, \qquad x \in \mathbb{R},$$

are linearly independent.

Proof By contradiction, suppose there exist n constants, not all zero, $c_1, c_2, \ldots, c_n$ such that

$$c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} + \ldots + c_n e^{\lambda_n x} = 0\,, \tag{5.38}$$

for any $x \in \mathbb{R}$. Multiplying (5.38) by $e^{-\lambda_n x}$ and differentiating in x gives

$$\begin{aligned} &c_1(\lambda_1 - \lambda_n)e^{(\lambda_1-\lambda_n)x} + c_2(\lambda_2 - \lambda_n)e^{(\lambda_2-\lambda_n)x} + \\ &+ \ldots + c_{n-1}(\lambda_{n-1} - \lambda_n)e^{(\lambda_{n-1}-\lambda_n)x} = 0\,, \qquad \forall\, x \in \mathbb{R}. \end{aligned} \tag{5.39}$$

Multiplying (5.39) by $e^{-(\lambda_{n-1}-\lambda_n)x}$ and differentiating, we find

$$c_1(\lambda_1-\lambda_n)(\lambda_1-\lambda_{n-1})\,e^{(\lambda_1-\lambda_{n-1})\,x}+c_2(\lambda_2-\lambda_n)(\lambda_2-\lambda_{n-1})\,e^{(\lambda_2-\lambda_{n-1})x}+$$
$$+\ldots+c_{n-2}(\lambda_{n-2}-\lambda_n)(\lambda_{n-2}-\lambda_{n-1})\,e^{(\lambda_{n-2}-\lambda_{n-1})\,x}=0,\qquad \forall\, x\in\mathbb{R}.$$

Proceeding in the same way, we obtain the relation

$$c_1(\lambda_1-\lambda_n)(\lambda_1-\lambda_{n-1})\cdot\ldots\cdot(\lambda_1-\lambda_2)e^{(\lambda_1-\lambda_2)x}=0\,,\qquad \forall\, x\in\mathbb{R},$$

which forces $c_1=0$, since the roots λ_i are distinct. Similarly one sees that $c_2=c_3=\ldots=c_n=0$, which is absurd.

By the above proposition, the general integral of a homogeneous linear ODE of order n with constant coefficients, whose characteristic equation has distinct roots $\lambda_1,\lambda_2,\ldots,\lambda_n$, is then

$$y=c_1\,e^{\lambda_1 x}+c_2e^{\lambda_2 x}+\ldots+c_n\,e^{\lambda_n x}\,.$$

As regards multiple characteristic roots, the following proposition holds.

Proposition 3 *For any real (or complex) root λ of multiplicity $r\in\mathbb{N}$ of the characteristic equation, the function*

$$u(x)=x^k e^{\lambda x}\,,\qquad x\in\mathbb{R},$$

$k=0,1,\ldots,r-1$ is a real (complex) solution to (5.36)*.*

Proof The Leibniz formula applied m times gives

$$D^{(m)}\left(x^k e^{\lambda x}\right)=\sum_{h=0}^{m}\binom{m}{h}\lambda^{m-h}e^{\lambda x}D^{(h)}x^k=$$
$$=e^{\lambda x}\sum_{h=0}^{m}\binom{m}{h}\lambda^{m-h}D^{(h)}x^k$$

and so

$$\begin{aligned} L(u) &= u^{(n)} + a_{n-1}u^{(n-1)} + \ldots + a_1 u' + a_0 u = \\ &= e^{\lambda x}\left\{ \sum_{h=0}^{n} \binom{n}{h} \lambda^{n-h} D^{(h)} x^k + a_{n-1} \sum_{h=0}^{n-1} \binom{n-1}{h} \lambda^{n-h-1} D^{(h)} x^k + \right. \\ &\quad \left. + \ldots + a_1 \sum_{h=0}^{1} \binom{1}{h} \lambda^{1-h} D^{(h)} x^k + a_0 x^k \right\} = \\ &= e^{\lambda x}\left\{ p^{(k)}(\lambda) + \binom{k}{k-1} x p^{(k-1)}(\lambda) + \ldots + \binom{k}{1} x^{k-1} p'(\lambda) + x^k p(\lambda) \right\}. \end{aligned} \tag{5.40}$$

But λ is a characteristic root of multiplicity r, so

$$p(\lambda) = p'(\lambda) = \ldots = p^{(r-1)}(\lambda)\,.$$

Then, as $0 \le k \le r-1$, the last term in (5.40) is zero, and the claim follows.

The next proposition is a straightforward consequence of the above one.

Proposition 4 *Let $\lambda = \alpha + i\beta$ be a complex root of multiplicity $r \in \mathbb{N}$ for the characteristic equation. The real functions*

$$x \in \mathbb{R} \to x^k e^{\alpha x} \cos \beta x\,, \qquad x \in \mathbb{R} \to x^k e^{\alpha x} \sin \beta x\,,$$

for any $k = 0, 1, \ldots, r-1$, are particular integrals of the homogeneous ODE (5.36).

Proof By Proposition 3 the complex function

$$x \in \mathbb{R} \to x^k e^{\lambda x}\,,$$

for any $k = 0, 1, \ldots, r-1$, is a complex solution to (5.36). Since

$$x^k e^{\lambda x} = x^k e^{\alpha x}(\cos \beta x + i \sin \beta x)\,,$$

we have

$$0 = L(x^k e^{\lambda x}) = L(x^k e^{\alpha x} \cos \beta x) + i\, L(x^k e^{\alpha x} \sin \beta x)\,,$$

and so $L(x^k e^{\alpha x} \cos \beta x) + i\, L(x^k e^{\alpha x} \sin \beta x)$ is the complex number zero, i.e.

$$L(x^k e^{\alpha x} \cos \beta x) = L(x^k e^{\alpha x} \sin \beta x) = 0\,.$$

We are now in a position to generalise Proposition 2 to the case of multiple characteristic roots.

General Integral of Homogeneous Equations with Constant Coefficients *If the characteristic equation* (5.37) *of ODE* (5.36) *admits p distinct roots $\lambda_1, \lambda_2, \ldots, \lambda_p$ of multiplicities $r_1, r_2, \ldots, r_p$, the general integral of* (5.36) *is*

$$y(x) = \left(c_{1,1} + c_{1,2}x + \ldots + c_{1,r_1}x^{r_1-1}\right) e^{\lambda_1 x} + \\ + \ldots + \left(c_{p,1} + c_{p,2}x + \ldots + c_{p,r_p}x^{r_p-1}\right) e^{\lambda_p x} .$$

Proof From $r_1 + r_2 + \ldots + r_p = n$, we have then n integrals

$$\begin{array}{l} e^{\lambda_1 x}\,,\ \ x\,e^{\lambda_1 x},\ \ \ldots\ ,\ \ x^{r_1-1}\,e^{\lambda_1 x}, \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ e^{\lambda_p x}\,,\ \ x\,e^{\lambda_p x},\ \ \ldots\ ,\ \ x^{r_p-1}\,e^{\lambda_p x}. \end{array} \tag{5.41}$$

The claim is proven once we show they are linearly independent over $\mathbb{R}$. Suppose by contradiction that they are not. Then there exist p polynomials $P_1(x)$, $P_2(x)$, ..., $P_p(x)$, with $\deg P_k(x) \le r_k - 1$ for any $k = 1, 2, \ldots, p$, at least one of which is not zero, so that

$$P_1(x)e^{\lambda_1 x} + P_2(x)e^{\lambda_2 x} + \ldots + P_p(x)\,e^{\lambda_p x} = 0\,, \qquad \forall\, x \in \mathbb{R}. \tag{5.42}$$

Assume, to fix ideas, $P_1(x)$ is not the zero polynomial, and call k $(0 \le k \le r_p)$ the degree of $P_p(x)$. Multiply (5.42) by $e^{-\lambda_p x}$ and then differentiate $k+1$ times. In this way we arrive at a relation like

$$Q_1(x)e^{(\lambda_1-\lambda_p)x} + Q_2(x)\,e^{(\lambda_2-\lambda_p)x} + \ldots + Q_{p-1}(x)e^{(\lambda_{p-1}-\lambda_p)x} = 0\,,$$

for any $x \in \mathbb{R}$, where the polynomial $Q_1(x)$ has the same degree of $P_1(x)$. Repeating the argument, as was done in Proposition 2, we obtain a relation of type

$$Z(x)e^{(\lambda_1-\lambda_2)x} = 0\,, \tag{5.43}$$

with $Z(x)$ a polynomial of the same degree as $P_1(x)$. Using (5.43), $Z(x)$ must be identically zero, violating the assumption that $P_1(x)$ is not the zero polynomial. Therefore $P_1(x) = 0$ for any $x \in \mathbb{R}$, and the same can be shown for the other $P_k(x)$, for any $k = 2, \ldots, p$. The claim then follows.

The n integrals (5.41) in general might contain imaginary terms. If we want to keep with real solutions, we will show next that from (5.41) we can determine n real, linearly independent integrals.

The characteristic equation has real coefficients, so if it possesses a complex root $\lambda = \alpha + i\beta$ with multiplicity r, it will also have the complex-conjugate number $\overline{\lambda} = \alpha - i\beta$ as root, with the same multiplicity r. It can be shown (see Proposition 4) that equation (5.36) admits the $2r$ real integrals

$$x^k e^{\alpha x} \cos \beta x \,, \qquad x^k e^{\alpha x} \sin \beta x \,,$$

for $k \in \{0, 1, \ldots, r-1\}$, and these may take the place of the $2r$ complex integrals coming from the roots $\lambda, \overline{\lambda}$ of (5.37). The new solution set is still made of n linearly independent integrals.

5.6 Equations with Constant Coefficients and Special Right-Hand Side

Let us consider the non-homogeneous equation with constant coefficients

$$y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x) \,. \tag{5.44}$$

For certain choices of the right-hand side $g(x)$, to find a particular integral it is not necessary to know n independent integrals of the homogeneous equation. Start with

Proposition 1 *If $a_0 \neq 0$ and $g(x)$ is a polynomial of degree k, there is a polynomial of degree k that is a particular integral of* (5.44).

Proof Assume

$$g(x) = b_0 + b_1 x + b_2 x^2 + \ldots + b_k x^k \,.$$

By the principle of identity of polynomials, the polynomial

$$p(x) = c_0 + c_1 x + c_2 x^2 + \ldots + c_k x^k$$

is an integral of (5.44) if and only if the monomial coefficients of equal degree are the same in $g(x)$ and in

$$L(p) = p^{(n)} + a_{n-1} p^{(n-1)} + \ldots + a_1 p' + a_0 p \,.$$

That is,

$$c_k a_0 = b_k \,,$$

$$c_{k-1} a_0 + k c_k a_1 = b_{k-1} \,,$$

$$c_{k-2} a_0 + (k-1)\, c_{k-1} a_1 + k(k-1)\, c_k a_2 = b_{k-2} \,,$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$c_0 a_0 + c_1 a_1 + 2\, c_2 a_2 + \ldots + (k-1)! c_{k-1} a_{k-1} + k! c_k a_k = b_0 \,,$$

where it is understood that $a_n = 1$ and $a_h = 0$ for $h > n$. These relationships determine the coefficients of $p(x)$ uniquely. Note that the previous proof also provides us with a simple method to find the polynomial solution.

Similar arguments allow to prove

Proposition 2 *Suppose the coefficients of* (5.44) *satisfy*

$$a_0 = a_1 = \ldots = a_{m-1} = 0 \,, \quad a_m \neq 0 \,, \qquad (0 < m < n)$$

and the right-hand side $g(x)$ is a polynomial of degree k. Then there is a polynomial of degree $m + k$ of type

$$x^m \left(c_m + c_{m+1} x + c_{m+2} x^2 + \ldots + c_{m+k} x^k \right) \,,$$

that is a particular integral of (5.44).

More generally, the following result holds.

Proposition 3 *Suppose the right-hand side $g(x)$ is of the form*

$$g(x) = e^{\lambda x}\, p_k(x) \,,$$

where λ is a real or complex number and $p_k(x)$ a polynomial of degree k. Then there is a polynomial $p(x)$ such that the function

$$e^{\lambda x}\, p(x)$$

is an integral of (5.44). *Furthermore, $p(x)$ has degree k if λ is not a root of the characteristic equation, while it is a polynomial of degree $m + k$ of type*

$$x^m \left(c_m + c_{m+1} x + c_{m+2} x^2 + \ldots + c_{m+k} x^k \right)$$

if λ is root of multiplicity m of the characteristic equation.

Proof Write

$$g(x) = e^{\lambda x}\left(b_0 + b_1 x + b_2 x^2 + \ldots + b_k x^k\right) \tag{5.45}$$

and let $p(x)$ be a polynomial. Consider the function $v(x) = e^{\lambda x} p(x)$. By the Leibniz rule, for any $i = 1, 2, \ldots, n$

$$v^{(i)} = D^{(i)}\left(e^{\lambda x} p(x)\right) = \sum_{h=0}^{i} \binom{i}{h} \lambda^{i-h} e^{\lambda x} p^{(h)}(x) = e^{\lambda x} \sum_{h=0}^{i} \binom{i}{h} \lambda^{i-h} p^{(h)}(x)\,,$$

and so

$$\begin{aligned}
L(v) = v^{(n)} + a_{n-1} v^{(n-1)} + \ldots + a_1 v' + a_0 v = \\
= e^{\lambda x}\left\{ \sum_{h=0}^{n} \binom{n}{h} \lambda^{n-h} p^{(h)}(x) + a_{n-1} \sum_{h=0}^{n-1} \binom{n-1}{h} \lambda^{n-h-1} p^{(h)}(x) + \right. \\
\left. + \ldots + a_1 \sum_{h=0}^{1} \binom{1}{h} \lambda^{1-h} p^{(h)}(x) + a_0 p(x) \right\} = \\
= e^{\lambda x}\left\{ \binom{n}{n} p^{(n)}(x) + \ldots + \right. \\
+ \left[a_i + \binom{i+1}{i} a_{i+1}\lambda + \ldots + \binom{n-1}{i} a_{n-1}\lambda^{n-i-1} + \binom{n}{i}\lambda^{n-i} \right] p^{(i)}(x) + \\
+ \ldots + \left[a_1 + \binom{2}{1} a_2 \lambda + \ldots + \binom{n-1}{1} a_{n-1}\lambda^{n-2} + \binom{n}{1}\lambda^{n-1} \right] p'(x) + \\
\left. + \left[a_0 + a_1\lambda + \ldots + a_{n-1}\lambda^{n-1} + \lambda^n \right] p(x) \right\}.
\end{aligned}$$

For $i = 0, 1, \ldots, n-1$ let

$$\alpha_i = a_i + \binom{i+1}{i} a_{i+1}\lambda + \ldots + \binom{n-1}{i} a_{n-1}\lambda^{n-i-1} + \binom{n}{i}\lambda^{n-i}\,, \tag{5.46}$$

so that

$$\begin{aligned}
L(v) = v^{(n)} + a_{n-1} v^{(n-1)} + \ldots + a_1 v' + a_0 v = \\
= e^{\lambda x}\left\{ p^{(n)} + \alpha_{n-1} p^{(n-1)} + \ldots + \alpha_1 p' + \alpha_0 p \right\}.
\end{aligned}$$

Hence, keeping in account the expression of $g(x)$ in (5.45), the function $v(x) = e^{\lambda x} p(x)$ solves the equation if and only if $p(x)$ solves the linear equation

$$p^{(n)} + \alpha_{n-1} p^{(n-1)} + \ldots + \alpha_1 p' + \alpha_0 p = b_0 + b_1 x + b_2 x^2 + \ldots + b_k x^k .$$

Note that if λ is not a characteristic root of (5.44), the coefficient α_0 in (5.46) is non-zero. On the other hand, if λ is a root of multiplicity m $(0 < m < n)$,

$$\alpha_0 = \alpha_1 = \ldots = \alpha_{m-1} = 0 , \qquad \alpha_m \neq 0 .$$

Now the claim follows from Propositions 1 and 2.

Finally, suppose $g(x)$ has the form

$$g(x) = e^{\lambda x} \left(p_1(x) \cos \mu x + p_2(x) \sin \mu x \right) , \tag{5.47}$$

for some polynomials $p_1(x)$, $p_2(x)$, a complex number λ and $\mu \in \mathbb{R}$. One can then prove the following result.

Proposition 4 *If $g(x)$ has the form (5.47), there exist two real polynomials $q_1(x), q_2(x)$ such that*

$$v(x) = e^{\lambda x} \left(q_1(x) \cos \mu x + q_2(x) \sin \mu x \right)$$

is a particular solution of (5.44). Let moreover k be the degree of $p_1(x) + p_2(x)$. Then the polynomial $q_1(x) + q_2(x)$ has degree k if $\lambda \pm i\mu$ is not a characteristic root, and it has degree $m + k$ if $\lambda \pm i\mu$ is a root of multiplicity m.

5.7 Linear Euler Equations

We call *Euler equation* a linear ODE of the type

$$x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \ldots + a_1 x\, y' + a_0 y = g(x) , \tag{5.48}$$

where $a_0, a_1, \ldots, a_{n-1} \in \mathbb{R}$. When considered on the intervals $(0, +\infty)$ or $(-\infty, 0)$, the equation reduces to an equation with constant coefficients using the substitution $x = e^t$.

Supposing first $x > 0$, we perform the variable change $x = e^t$, i.e. $t = \log x$. Set $z(t) = y(e^t)$, so that $y(x) = z(\log x)$, and then

$$\frac{dy}{dx} = \frac{dz}{dt}\frac{1}{x};$$

$$\frac{d^2y}{dx^2} = \frac{d^2z}{dt^2}\frac{1}{x^2} - \frac{dz}{dt}\frac{1}{x^2} = \left(\frac{d^2z}{dt^2} - \frac{dz}{dt}\right)\frac{1}{x^2};$$

$$\frac{d^3y}{dx^3} = \left(\frac{d^3z}{dt^3} - \frac{d^2z}{d^2t}\right)\frac{1}{x^3} + \left(\frac{d^2z}{dt^2} - \frac{dz}{dt}\right)\frac{-2}{x^3} = \left(\frac{d^3z}{dt^3} - 3\frac{d^2z}{d^2t} + 2\frac{dz}{dt}\right)\frac{1}{x^3};$$

. . . . . .

The equation now has constant coefficients:

$$z^{(n)} + b_{n-1}z^{(n-1)} + \ldots + b_1 z' + b_0 z = g(e^t)\,. \tag{5.49}$$

If $x < 0$, we use the substitution $x = -e^t$, so (5.48) still becomes an ODE of type (5.49).

In the special case $g = 0$, the (homogeneous) ODE (5.49) has as solution $z(t) = e^{\alpha t}$, if α solves the corresponding characteristic equation. For $x > 0$ we therefore find

$$y(x) = z(\log x) = e^{\alpha \log x} = x^\alpha.$$

The fact that (5.48) for $x > 0$ admits particular integrals like $y = x^\alpha$ may be deduced also by substituting $y = x^\alpha$ in (5.48) directly, with α to be determined.

For example, for the second-order homogeneous Euler equation

$$x^2 y'' + a_1 x\, y' + a_0 y = 0\,,$$

if we set $y = x^\alpha$ then

$$y' = \alpha\, x^{\alpha-1} \qquad y'' = \alpha(\alpha-1)x^{\alpha-2}\,,$$

and substituting in (5.48) we obtain

$$\alpha(\alpha-1)x^\alpha + a_1\alpha x^\alpha + a_0 x^\alpha = 0\,, \qquad \forall x > 0.$$

In particular, for $x = 1$,

$$\alpha(\alpha-1) + a_1\alpha + a_0 = 0\,,$$

from which we find the (real or complex) values of α.

Example 1 Consider, for $x > 0$, the Euler ODE of order two

$$x^2 y'' + xy' - y = \frac{2 - \log^2 x}{\log^3 x}\,. \tag{5.50}$$

Let us first find the general integral of the associated homogeneous equation

$$x^2 y'' + xy' - y = 0\,, \tag{5.51}$$

by seeking solutions $y(x) = x^\alpha$. As $y' = \alpha x^{\alpha-1}$, $y'' = \alpha(\alpha-1)x^{\alpha-2}$, (5.51) yields

$$\alpha(\alpha-1)x^\alpha + \alpha x^\alpha - x^\alpha = 0\,, \qquad \forall x > 0,$$

and then $\alpha(\alpha-1) + \alpha - 1 = \alpha^2 - 1 = 0$, i.e. $\alpha = \pm 1$. So the general integral of the homogeneous equation (5.51) is

$$c_1\, x + \frac{c_2}{x}\,, \qquad x > 0. \tag{5.52}$$

Just for completeness, we recover the same with the substitution $x = e^t$. Putting $z(t) = y(e^t)$ gives $y(x) = z(\log x)$ and then

$$\frac{dy}{dx} = \frac{dz}{dt}\frac{1}{x}\,, \qquad \frac{d^2y}{dx^2} = \frac{d^2z}{dt^2}\frac{1}{x^2} - \frac{dz}{dt}\frac{1}{x^2} = \left(\frac{d^2z}{dt^2} - \frac{dz}{dt}\right)\frac{1}{x^2}\,.$$

Then (5.50) becomes

$$z'' - z = \frac{2 - t^2}{t^3}\,, \tag{5.53}$$

and the associated homogeneous equation $z'' - z = 0$ has general integral

$$z = c_1 e^t + c_2\, e^{-t}\,. \tag{5.54}$$

Recalling that $x = e^t$, we find once again the solution (5.52), exactly as with the method that used $y(x) = x^\alpha$.

To find a particular solution of (5.50) we will vary the parameters. Consider (5.53), and keeping (5.54) into account, a particular solution will have the form

$$y = \varphi_1(t)e^t + \varphi_2(t)e^{-t}\,, \tag{5.55}$$

where (see (5.24))

$$\begin{cases} \varphi_1'(t)e^t + \varphi_2'(t)e^{-t} = 0 \\ \varphi_1'(t)e^t - \varphi_2'(t)e^{-t} = (2 - t^2)/t^3\,. \end{cases}$$

(continued)

Example 1 (continued)
Hence

$$\varphi_1'(t) = \frac{1}{2}\frac{2-t^2}{t^3}e^{-t}\,, \qquad \varphi_2'(t) = \frac{1}{2}\frac{t^2-2}{t^3}e^{t}\,,$$

integrating which gives

$$\varphi_1(t) = \frac{1}{2}\left(\frac{-1}{t^2}+\frac{1}{t}\right)e^{-t}\,, \qquad \varphi_2(t) = \frac{1}{2}\left(\frac{1}{t^2}+\frac{1}{t}\right)e^{t}\,.$$

From (5.54) and (5.55) and the above relations it follows that the general integral of (5.53) is

$$z = c_1 e^t + c_2 e^{-t} + \frac{1}{t}.$$

In conclusion, the general solution of (5.50) is

$$y = c_1 x + \frac{c_2}{x} + \frac{1}{\log x}\,, \qquad x > 0.$$

Appendix to Chap. 5

5.8 Boundary Value Problems

Apart from *initial value problems* (or *Cauchy problems*) one can pose other types of problems for ODEs of order n, especially in view of the applications.

For example, given a second-order linear equation

$$y'' + a_1(x)y' + a_0(x)y = g(x) \tag{5.56}$$

with continuous coefficients $a_0(x)$, $a_1(x)$ and continuous right-hand side $g(x)$ on $[a, b]$, a *boundary value problem* consists in finding a solution that fulfils the *boundary conditions* ($A, B \in \mathbb{R}$)

$$y(a) = A\,, \qquad y(b) = B\,. \tag{5.57}$$

The *homogeneous problem associated with the boundary value problem* (5.56) and (5.57) is the problem relative to the associated homogeneous equation

$$y'' + a_1(x)y' + a_0(x)y = 0\,, \tag{5.58}$$

together with the *homogeneous* boundary conditions

$$y(a) = 0\,, \qquad y(b) = 0\,. \tag{5.59}$$

The general integral of Eq. (5.56) is

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + v(x)\,, \tag{5.60}$$

where $y_1(x)$, $y_2(x)$ is a system of linearly independent integrals for the homogeneous equation (5.58), and $v(x)$ is a particular integral of (5.56).

To find the constants c_1, c_2 so that the y in (5.60) meets the initial conditions in (5.57), we impose $y(a) = A$, $y(b) = B$, leading to the linear system in the unknowns c_1, c_2

$$\begin{cases} y_1(a)c_1 + y_2(a)c_2 = A - v(a) \\ y_1(b)c_1 + y_2(b)c_2 = B - v(b)\,. \end{cases} \tag{5.61}$$

If the coefficients' determinant

$$\Delta = \begin{vmatrix} y_1(a) & y_2(a) \\ y_1(b) & y_2(b) \end{vmatrix} \tag{5.62}$$

is different from zero, the linear system can be solved uniquely, and there is one pair of real numbers c_1, c_2 for which $y(x)$ in (5.60) satisfies the initial conditions $y(a) = A$, $y(b) = B$.

So if $\Delta \neq 0$, the starting boundary value problem (5.56) and (5.57) admits one, and only one, solution.

The associated homogeneous problem (5.58) and (5.59) certainly admits the solution $y(x) \equiv 0$. This will be the only solution if the determinant Δ in (5.62) is non-zero (since, as we said, irrespective of $g(x)$ and the initial values A, B, the boundary value problem has exactly one solution if $\Delta \neq 0$).

But when $\Delta = 0$, the homogeneous problem (5.58) and (5.59) will have non-zero solutions as well, called *eigensolutions* (or *eigenfunctions*). In fact, the general integral of (5.58) is the set of

$$y(x) = c_1 y_1(x) + c_2 y_2(x)\,,$$

as c_1, c_2 vary in $\mathbb{R}$. Such functions $y(x)$ satisfy the boundary conditions $y(a) = 0$, $y(b) = 0$ if and only if the constants c_1, c_2 solve the linear system

$$\begin{cases} y_1(a)c_1 + y_2(a)c_2 = 0 \\ y_1(b)c_1 + y_2(b)c_2 = 0\,. \end{cases} \tag{5.63}$$

Since the coefficients' matrix has determinant Δ equal to zero, the homogeneous system (5.63) has infinitely many solutions c_1, c_2.

For the same reason, if $\Delta = 0$ the boundary value problem (5.56) and (5.57) will have either infinitely many solutions or none, depending on whether the right-hand side term $g(x)$ and the constants A, B verify the Rouché-Capelli theorem for system (5.61).

For example, given $\ell > 0$ consider the homogeneous boundary value problem

$$\begin{cases} y'' + \lambda y = 0 \\ y(0) = y(\ell) = 0\,. \end{cases} \tag{5.64}$$

where λ is a real parameter. The values of λ for which there are *eigensolutions*, i.e. *non-zero* solutions to (5.64), are called *eigenvalues*.

From the above general discussion, the eigenvalues are the solutions of

$$\Delta(\lambda) = \begin{vmatrix} y_1(0,\lambda) & y_2(0,\lambda) \\ y_1(\ell,\lambda) & y_2(\ell,\lambda) \end{vmatrix} = 0\,, \tag{5.65}$$

where $y_1(x,\lambda)$, $y_2(x,\lambda)$, are linearly independent solutions of the ODE in (5.64).

Let us first show the eigenvalues are positive real numbers. Let λ be an eigenvalue with eigenfunction $y = y(x,\lambda)$. Multiplying the ODE in (5.64) by y and integrating from 0 to ℓ we obtain

$$\int_0^\ell yy''\,dx + \lambda \int_0^\ell y^2\,dx = 0\,.$$

Integrating by parts the first term, and keeping into account the boundary conditions we find

$$-\int_0^\ell (y')^2\,dx + \lambda \int_0^\ell y^2\,dx = 0\,.$$

But y is an eigenfunction, so it cannot vanish identically on $[0,\ell]$. Being continuous on that interval, if $x_0 \in [0,\ell]$ is such that $y(x_0) \neq 0$, then $y(x)$ cannot vanish near x_0, and so the integral of $[y(x)]^2$ over $[0,\ell]$ is strictly positive. Hence

$$\lambda = \frac{\displaystyle\int_0^\ell (y')^2\,dx}{\displaystyle\int_0^\ell y^2\,dx} > 0\,.$$

Relatively to real numbers $\lambda > 0$, the general integral of the equation in (5.64) is

$$y(x) = c_1 \cos\sqrt{\lambda}x + c_2 \sin\sqrt{\lambda}x\,, \tag{5.66}$$

(continued)

and the determinant $\Delta(\lambda)$ in (5.65) equals

$$\Delta(\lambda) = \begin{vmatrix} 1 & 0 \\ \cos\sqrt{\lambda}\ell & \sin\sqrt{\lambda}\ell \end{vmatrix} = \sin\sqrt{\lambda}\ell\,.$$

Therefore the eigenvalues are the positive solutions of $\sin\sqrt{\lambda}\ell = 0$, i.e.

$$\lambda_k = \frac{k^2\pi^2}{\ell^2}\,, \qquad k \in \mathbb{N}.$$

Solving system (5.63) we find the eigensolutions

$$y_k(x) = c_2 \sin\frac{k\pi}{\ell}x\,, \qquad k \in \mathbb{N}.$$

Let us now pass to the non-homogeneous boundary value problem

$$\begin{cases} y'' + \lambda y = g(x) \\ y(0) = y(\ell) = 0\,, \end{cases} \tag{5.67}$$

and let us find a necessary condition on $g(x)$ so that it admits a solution, when λ is an eigenvalue. Call y_k an eigenfunction with eigenvalue $\lambda = \lambda_k$, multiply the ODE in (5.67) by y_k and integrate between 0 and ℓ:

$$\int_0^\ell y'' y_k\,dx + \lambda_k \int_0^\ell y y_k dx = \int_0^\ell g(x) y_k\,dx\,.$$

Integrating by parts, and remembering that y and y_k satisfy the initial conditions, gives

$$\int_0^\ell y\left(y_k'' + \lambda_k y_k\right)\,dx = \int_0^\ell g(x) y_k\,dx\,.$$

Since $y_k'' + \lambda_k y_k = 0$, we deduce that, necessarily,

$$\int_0^\ell g(x) y_k\,dx = 0\,.$$

It could actually be proved that the above is also a sufficient condition, but we will stop here for the sake of brevity.

The boundary conditions (5.57), which prescribe the solution's value at the endpoints of the interval $[a, b]$, are not the only conditions appearing in the applications. Sometimes the problem assigns the value of the solution at one endpoint, and the

value of the first derivative at the other. More generally, one could consider boundary conditions of type

$$\begin{cases} \alpha_1 y(a) + \beta_1 y(b) + \gamma_1 y'(a) + \delta_1 y'(b) = A \\ \alpha_2 y(a) + \beta_2 y(b) + \gamma_2 y'(a) + \delta_2 y'(b) = B\,. \end{cases}$$

For special values of the constants, these incorporate the initial value conditions in (5.57).

Consider for instance the homogeneous problem

$$\begin{cases} y'' + \lambda y = 0 \\ y(0) = y(\pi) \\ y'(0) = y'(\pi)\,, \end{cases} \tag{5.68}$$

for $\lambda > 0$, where the boundary conditions are *periodic*.

The general integral of the homogeneous equation is (5.66). Imposing the periodicity conditions in (5.68) produces the homogeneous system

$$\begin{cases} (1 - \cos\sqrt{\lambda}\pi)c_1 - (\sin\sqrt{\lambda}\pi)c_2 = 0 \\ (\sin\sqrt{\lambda}\pi)c_1 + (1 - \cos\sqrt{\lambda}\pi)c_2 = 0\,, \end{cases} \tag{5.69}$$

whose determinant is $\Delta(\lambda) = 2(1 - \cos\sqrt{\lambda}\pi)$, vanishing for $\sqrt{\lambda} = 2k$, $k \in \mathbb{N}$. Hence the problem's *eigenvalues*, i.e. the values of λ for which (5.67) has *non-zero* solutions, are the numbers

$$\lambda_k = 4k^2\,, \qquad k \in \mathbb{N}.$$

If $\lambda = \lambda_k$ for some $k \in \mathbb{N}$, system (5.69) is true for any c_1, c_2. The corresponding linearly independent and 2π-periodic eigenfunctions are

$$y_{1,k}(x) = \cos 2kx\,,$$
$$y_{2,k}(x) = \sin 2kx\,,$$

as well as their linear combinations

$$c_1 y_{1,k}(x) + c_2 y_{2,k}(x) = c_1 \cos 2kx + c_2 \sin 2kx\,.$$

5.9 Linear Systems

A system of n differential equations of the form

$$\begin{cases} y_1' = a_{11}(x)y_1 + a_{12}(x)y_2 + \ldots + a_{1n}(x)y_n + g_1(x) \\ y_2' = a_{21}(x)y_1 + a_{22}(x)y_2 + \ldots + a_{2n}(x)y_n + g_2(x) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_n' = a_{n1}(x)y_1 + a_{n2}(x)y_2 + \ldots + a_{nn}(x)y_n + g_n(x) \end{cases} \tag{5.70}$$

is a *linear ODE system of order one*. The functions $a_{ij}(x)$ are the system's *coefficients* while the $g_i(x)$ are the *right-hand-side terms*. A solution is an n-tuple of differentiable functions on some interval $[a, b] \subset \mathbb{R}$ that simultaneously satisfy the n equations for any $x \in [a, b]$.

System (5.70) may be written more concisely as

$$y_i' = \sum_{j=1}^{n} a_{ij}(x)y_j + g_i(x)\,, \qquad i = 1, 2, \ldots, n,$$

or, using the square matrix $A(x) = \left[a_{ij}(x)\right]$ and column vectors

$$Y(x) = \begin{pmatrix} y_1(x) \\ y_2(x) \\ \vdots \\ y_n(x) \end{pmatrix}, \qquad G(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \\ \vdots \\ g_n(x) \end{pmatrix}$$

as

$$Y'(x) = A(x)Y(x) + G(x)\,. \tag{5.71}$$

If the coefficients and right-hand sides are continuous, we can apply to (5.70) the global existence and uniqueness theorem for *Cauchy problems* (see Sect. 4.4), which now is recast as in the following result.

Existence and Uniqueness Theorem *If the coefficients $a_{ij}(x)$ and the right-hand sides $g_i(x)$ of system (5.70) are continuous functions on $[a, b]$, then for any $x_0 \in [a, b]$ and any $(y_1^0, y_2^0, \ldots, y_n^0) \in \mathbb{R}^n$ there exists a unique solution to (5.70) that satisfies the initial conditions*

$$y_1(x_0) = y_1^0, \quad y_2(x_0) = y_2^0, \quad \ldots, \quad y_n(x_0) = y_n^0\,.$$

Consider first the *homogeneous* linear system

$$Y'(x) = A(x)Y(x) \tag{5.72}$$

and let

$$Y_1(x) = \begin{pmatrix} y_{11}(x) \\ y_{21}(x) \\ \vdots \\ y_{n1}(x) \end{pmatrix}, \quad Y_2(x) = \begin{pmatrix} y_{12}(x) \\ y_{22}(x) \\ \vdots \\ y_{n2}(x) \end{pmatrix}, \quad \dots, \quad Y_n(x) = \begin{pmatrix} y_{1n}(x) \\ y_{2n}(x) \\ \vdots \\ y_{nn}(x) \end{pmatrix},$$

denote n solutions. The function

$$V(x) = \begin{vmatrix} y_{11}(x) & y_{12}(x) & \dots & y_{1n}(x) \\ y_{21}(x) & y_{22}(x) & \dots & y_{2n}(x) \\ \dots & \dots & \dots & \dots \\ y_{n1}(x) & y_{n2}(x) & \dots & y_{nn}(x) \end{vmatrix}$$

is called *determinant* of these n solutions. As in the case of the Wronskian determinant (Sect. 5.1), it can be proved that

$$V'(x) = \left(\sum_{i=1}^{n} a_{ii}(x) \right) V(x), \qquad \forall\, x \in [a, b],$$

so $V(x)$ solves a linear ODE of order one. Eventually

$$V(x) = V(x_0) \cdot e^{\int_{x_0}^{x} \sum_{i=1}^{n} a_{ii}(t)\, dt},$$

with $x_0 \in [a, b]$. We then deduce the following result.

Proposition 1 *Assume $Y_1(x), Y_2(x), \dots, Y_n(x)$ are solutions to the homogeneous system (5.72). A necessary and sufficient condition for the vector-valued functions $Y_1, Y_2, \dots, Y_n$ to be linearly independent is that their values at an arbitrary point $x_0 \in [a, b]$ be linearly independent.*

To find n linearly independent integrals one goes about in the same way as for homogeneous linear equations of order n, considering the solutions of the Cauchy problems at x_0 with initial values

$$\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The general integral of system (5.72) descends from the following proposition.

Proposition 2 *If $Y_1(x), Y_2(x), \ldots, Y_n(x)$ are n linearly independent solutions of system (5.72), every other solution arises as linear combination of the Y_i.*

Proof The system's linearity ensures that any linear combination

$$\sum_{i=1}^{n} c_i Y_i(x)$$

will solve. If $Y(x)$ is a solution, imposing

$$\sum_{i=1}^{n} c_i Y_i(x_0) = Y(x_0)$$

for a given x_0 in $[a, b]$, we obtain a linear system in the unknowns $c_1, c_2, \ldots, c_n$, which has a unique solution since the coefficients' determinant is $V(x_0) \neq 0$. Therefore

$$\sum_{i=1}^{n} c_i Y_i(x) = Y(x)\,, \qquad \forall\, x \in [a, b].$$

Example 1 As in Sect. 4.2 (see (4.49)) we consider the system

$$\begin{cases} y_1' = y_2 \\ y_2' = -y_1 \end{cases}.$$

Choosing at $x_0 = 0$ initial value $y_1(x_0) = y_1^0 = 0$, $y_2(x_0) = y_2^0 = 1$, we find the solution

$$Y_1(x) = \begin{pmatrix} y_{11}(x) \\ y_{21}(x) \end{pmatrix} = \begin{pmatrix} \sin x \\ \cos x \end{pmatrix}.$$

If we choose at $x_0 = 0$ initial value $y_1(x_0) = 1$, $y_2(x_0) = 0$, then

$$Y_2(x) = \begin{pmatrix} y_{12}(x) \\ y_{22}(x) \end{pmatrix} = \begin{pmatrix} \cos x \\ -\sin x \end{pmatrix}.$$

The general integral is then

$$\begin{cases} y_1 = c_1 \sin x + c_2 \cos x \\ y_2 = c_1 \cos x - c_2 \sin x\,. \end{cases}$$

Let us pass to the general case for the full system (5.70), which we may write as (5.71). Observe first that if $\overline{Y}(x)$ is an integral of the associated homogeneous

system (5.72) and $Z(x)$ is an integral of (5.71), the sum

$$Y(x) = \overline{Y}(x) + Z(x)$$

is still a solution of the full system.

Proposition 3 *If $Z(x)$ is a solution of (5.71) and $Y_1(x), Y_2(x), \ldots, Y_n(x)$ are n linearly independent integrals of the associated homogeneous system (5.72), every solution of the full system has the form*

$$Y(x) = \sum_{i=1}^{n} c_i Y_i(x) + Z(x)\,.$$

Proof It is enough to apply Proposition 2 to $Y(x) - Z(x)$, noting that

$$(Y(x) - Z(x))' = Y'(x) - Z'(x) =$$

$$= (A(x)Y(x) + G(x)) - (A(x)Z(x) + G(x)) = A(x)\,(Y(x) - Z(x))\ .$$

Then $Y(x) - Z(x)$ is an integral of the homogeneous system.

Now we will prove that from n linearly independent integrals of the homogeneous system, it is possible to compute a particular integral

$$\begin{pmatrix} \overline{y}_1(x) \\ \overline{y}_2(x) \\ \vdots \\ \overline{y}_n(x) \end{pmatrix},$$

of the full system. To fix ideas, we seek the integral of (5.70) that satisfies the initial conditions

$$\overline{y}_1(x_0) = \overline{y}_2(x_0) = \ldots = \overline{y}_n(x_0) = 0\,. \tag{5.73}$$

As the determinant $V(x)$ is always non-zero, there are n continuous functions

$$\varphi_1(x)\,, \quad \varphi_2(x)\,, \quad \ldots, \quad \varphi_n(x)$$

with continuous first derivatives on $[a, b]$, such that

$$\begin{cases} y_{11}(x)\varphi_1(x) + y_{12}(x)\varphi_2(x) + \ldots + y_{1n}(x)\varphi_n(x) = \overline{y}_1(x) \\ y_{21}(x)\varphi_1(x) + y_{22}(x)\varphi_2(x) + \ldots + y_{2n}(x)\varphi_n(x) = \overline{y}_2(x) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_{n1}(x)\varphi_1(x) + y_{n2}(x)\varphi_2(x) + \ldots + y_{nn}(x)\varphi_n(x) = \overline{y}_n(x)\,. \end{cases} \tag{5.74}$$

Substituting $\overline{y}_k(x)$, $k = 1, 2, \ldots, n$, in (5.70) and using the fact that

$$Y_k(x) = \begin{pmatrix} y_{1k}(x) \\ y_{2k}(x) \\ \vdots \\ y_{nk}(x) \end{pmatrix}$$

solves (5.70) for any k, we deduce

$$\begin{cases} y_{11}(x)\varphi_1'(x) + y_{12}(x)\varphi_2'(x) + \ldots + y_{1n}(x)\varphi_n'(x) = g_1(x) \\ y_{21}(x)\varphi_1'(x) + y_{22}(x)\varphi_2'(x) + \ldots + y_{2n}(x)\varphi_n'(x) = g_2(x) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_{n1}(x)\varphi_1'(x) + y_{n2}(x)\varphi_2'(x) + \ldots + y_{nn}(x)\varphi_n'(x) = g_n(x)\,. \end{cases}$$

By Cramer's rule, we have

$$\varphi_k'(x) = \frac{1}{V(x)} \sum_{i=1}^{n} g_i(x) V_{ik}(x)\,, \qquad k = 1, 2, \ldots, n,$$

where $V_{ik}(x)$ is the cofactor of $y_{ik}(x)$ in $V(x)$. Now set $x = x_0$ in (5.74). Recalling (5.73), we have

$$\overline{\varphi}_1(x_0) = \overline{\varphi}_2(x_0) = \ldots = \overline{\varphi}_n(x_0) = 0\,.$$

So now

$$\varphi_k(x) = \int_{x_0}^{x} \sum_{i=1}^{n} g_i(t) \frac{V_{ik}(t)}{V(t)}\, dt\,, \qquad k = 1, 2, \ldots, n,$$

and therefore

$$\overline{y}_h(x) = \int_{x_0}^{x} \sum_{i,k=1}^{n} g_i(t) y_{hk}(x) \frac{V_{ik}(t)}{V(t)}\, dt\,, \qquad h = 1, 2, \ldots, n.$$

Note the analogy with the method of variation of parameters introduced in Sect. 5.3.



Curves and Integrals Along Curves

6

6.1 Regular Curves

Consider a point particle moving in three-dimensional space in the time interval $[t_0, t_1]$. In a given frame system, for any $t \in [t_0, t_1]$ we let $\{x(t), y(t), z(t)\}$ denote the particle's coordinates at time t. The function $\varphi : [t_0, t_1] \to \mathbb{R}^3$, of scalar components $x(t), y(t), z(t)$, is a *curve* describing the motion under examination. Writing φ in vector notation we have

$$\varphi(t) = x(t)\, e_1 + y(t)\, e_2 + z(t)\, e_3 ,$$

where e_1, e_2, e_3 are the unit vectors of the given frame.

Suppose $\varphi(t)$ can be differentiated twice. In Physics the vectors

$$\varphi'(t) = x'(t)\, e_1 + y'(t)\, e_2 + z'(t)\, e_3 ,$$
$$\varphi''(t) = x''(t)\, e_1 + y''(t)\, e_2 + z''(t)\, e_3 ,$$

are respectively called *velocity* and *acceleration* of the point particle at time t, and are pivotal in the motion's description.

Figure 6.1 represents the path of a particle whose motion has velocity of constant modulus, and shows the velocity and acceleration vectors at two points. We will show in this chapter that the velocity vector is *tangent* to the particle's trajectory, while the acceleration is *orthogonal* (if, as in the picture, the velocity's modulus is constant).

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N. Fusco et al., *Mathematical Analysis*, La Matematica per il 3+2 137,
https://doi.org/10.1007/978-3-031-04151-8_6

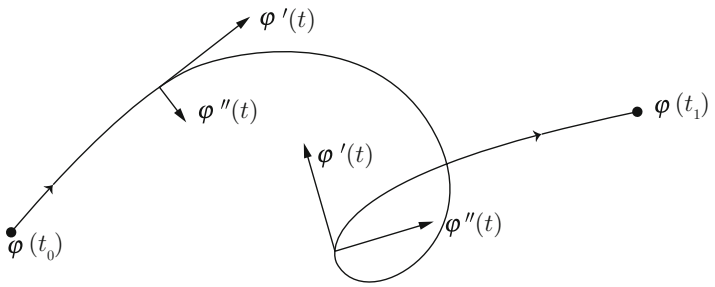


Fig. 6.1

In general, a *curve* is a continuous map $\varphi : I \to \mathbb{R}^n$, where I is a real interval. The relations

$$\begin{cases} x_1 = \varphi_1(t) \\ x_2 = \varphi_2(t) \\ \dots\dots\dots\dots \\ x_n = \varphi_n(t) \end{cases} \qquad t \in I\,,$$

expressing the components of φ, are the curve's *parametric equations* in the parameter t. Using the familiar notation for vectors in $\mathbb{R}^n$, we may write more compactly $\varphi(t) = (\varphi_i(t))_{i=1,\dots,n}$. The *image* $\varphi(I)$ of φ is called the *support* of the curve, and should not be confused with the curve itself, which is a function.

Example 1 The curves $\varphi : [0, 2\pi] \to \mathbb{R}^2$, $\psi : [0, 4\pi] \to \mathbb{R}^2$, of parametric equations

$$\begin{cases} \varphi_1(t) = \cos t \\ \varphi_2(t) = \sin t\,, \end{cases} \quad \forall\, t \in [0, 2\pi]\,, \qquad \begin{cases} \psi_1(t) = \cos t \\ \psi_2(t) = \sin t\,, \end{cases} \quad \forall\, t \in [0, 4\pi]\,, \tag{6.1}$$

have the same support (the unit *circle* centred at the origin), but are distinct curves.

Example 2 The curve $\varphi : \mathbb{R} \to \mathbb{R}^3$ of parametric equations

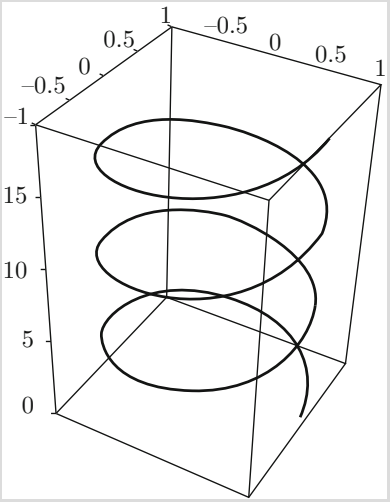
$$\varphi(t) = (a\cos t,\ a\sin t,\ b\,t)\,, \qquad \forall\, t \in \mathbb{R}, \tag{6.2}$$

with $a > 0$, $b \neq 0$, is a *circular helix* (see Fig. 6.2). The number $2\pi b$ measures how much a point moving along the curve goes up (or down, if $b < 0$) after a complete turn around the cylinder, and is called *pitch* of the helix.

(continued)

Example 2 (continued)

Fig. 6.2



A curve $\varphi : I \to \mathbb{R}^n$ is called *simple* if for any pair of distinct points t_1, t_2 in I, of which at least one is *internal* to I, we have $\varphi(t_1) \neq \varphi(t_2)$. A curve $\varphi : I \to \mathbb{R}^n$, defined on a closed and bounded interval $I = [a, b]$, is *closed* if $\varphi(a) = \varphi(b)$.

Example 3 The circle $\varphi = (\varphi_1, \varphi_2)$ defined in (6.1) is a simple closed curve. The circle $\psi = (\psi_1, \psi_2)$ defined in (6.1) is closed but not simple. The helix defined by (6.2) is simple but not closed.

The curve (*strophoid*) in $\mathbb{R}^2$ of parametric equations

$$\varphi(t) = \left(t^3 - t,\, t^2 - 1\right), \qquad \forall\, t \in \mathbb{R},$$

whose support is plotted in Fig. 6.3, is not simple, because $\varphi(1) = \varphi(-1) = (0, 0)$. Its restriction to $I = [-1, 1]$, instead, i.e. the curve ψ (*strophoid noose*) defined by

$$\psi(t) = \left(t^3 - t,\, t^2 - 1\right), \qquad \forall\, t \in [-1, 1],$$

is simple and closed.

(continued)

Example 3 (continued)

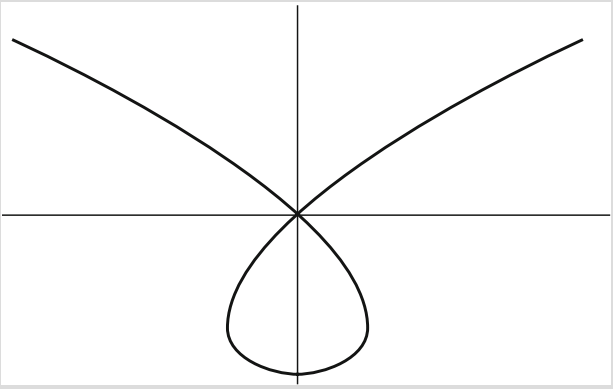


Fig. 6.3

It will turn out to be convenient to impose a few regularity assumptions on the curves we shall work with.

Let I be a closed and bounded interval $[a, b]$. A curve $\varphi : I \to \mathbb{R}^n$ is said *regular* if the map φ is C^1 on $I = [a, b]$ and for any $t \in (a, b)$ the vector $\varphi'(t)$ is non-zero, i.e. the derivatives $\varphi_1'(t), \ldots, \varphi_n'(t)$ do not vanish simultaneously.

Given distinct time instants $t_0, t_1 \in [a, b]$, consider the straight line in $\mathbb{R}^n$ passing through $\varphi(t_0)$ and $\varphi(t_1)$. It has parametric equation

$$x_i = x_i(t) = \varphi_i(t_0) + (t - t_0)\frac{\varphi_i(t_1) - \varphi_i(t_0)}{t_1 - t_0}, \qquad \forall\, i = 1, 2, \ldots, n\,.$$

If φ is a regular curve, the limit of the above expression, as $t_1 \to t_0$, gives the line of equations

$$x_i = x_i(t) = \varphi_i(t_0) + (t - t_0)\varphi_i'(t_0), \qquad \forall\, i = 1, 2, \ldots, n\,,$$

(see Fig. 6.4), called the *tangent line* to the curve φ at the point $\varphi(t_0)$. The vector $\varphi'(t_0)$ of coordinates $\big(\varphi_i'(t_0)\big)_{i=1,\ldots,n}$ is the *tangent vector* to φ at $\varphi(t_0)$, while

$$T(t_0) = \frac{\varphi'(t_0)}{|\varphi'(t_0)|} \tag{6.3}$$

is the *unit tangent vector*. The regularity condition $\varphi'(t_0) \neq 0$ for any $t_0 \in (a, b)$ guarantees the existence of a unique unit tangent vector to the curve at $\varphi(t_0)$. Hence the curve has no cusps or sharp corners.

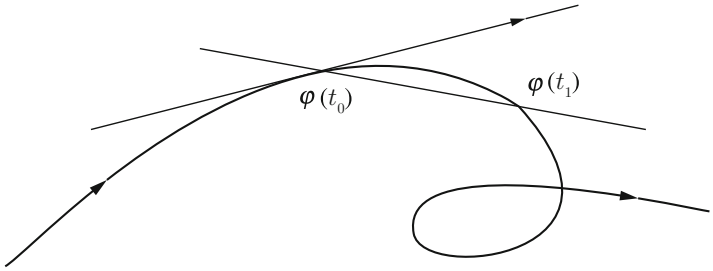


Fig. 6.4

Example 4 The curve in $\mathbb{R}^2$ of parametric equations

$$\varphi(t) = \left(t^3, t^2\right), \qquad \forall\, t \in [-1, 1], \tag{6.4}$$

is not regular, since $\varphi'(0) = (0, 0)$. To represent (6.4) on the xy-plane it is useful to observe that we can solve for t the first equation in

$$x = \varphi_1(t) = t^3, \qquad y = \varphi_2(t) = t^2, \qquad \forall\, t \in [-1, 1],$$

to obtain $t = x^{1/3}$. Substituting in the other relation gives

$$y = x^{2/3}, \qquad \forall\, x \in [-1, 1].$$

Hence the function's graph (Fig. 6.5) has a cusp at $x = 0$.

The same curve (6.4) may be viewed as union of two regular curves φ_+ (with support in the first quadrant) and φ_- (supported in the second quadrant) of parametric equations

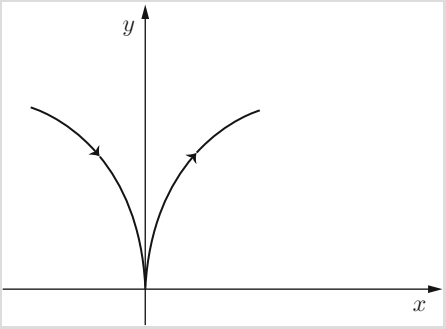
$$\varphi_+(t) = \left(t^3, t^2\right), \quad \forall\, t \in [0, 1]; \quad \varphi_-(t) = \left(t^3, t^2\right), \quad \forall\, t \in [-1, 0].$$

These two curves, that meet at the origin, are regular. For this reason we say that the curve φ in (6.4) is *piecewise regular*.

(continued)

Example 4 (continued)

Fig. 6.5



When $f : [a, b] \to \mathbb{R}$ is a C^1 function in one variable, the *plane curve* of parametric equations $\varphi(t) = (t, f(t))$ has as support the graph of f and is regular. In this case $y = f(x)$ is the *Cartesian equation* of the curve. Note that the tangent line to the curve at $\varphi(t_0)$ coincides with the usual tangent to the graph of f at $(t_0, f(t_0))$.

It is sometimes convenient to express a curve's *equation* in a suitable coordinate system. For example, if the equation of a *plane curve* in *polar coordinates* is

$$\varrho = \varrho(\vartheta)\,, \qquad \vartheta_0 \le \vartheta \le \vartheta_1\,,$$

called *polar equation*, its equations in Cartesian coordinates read

$$x = \varrho(\vartheta)\cos\vartheta\,, \qquad y = \varrho(\vartheta)\sin\vartheta\,, \qquad \vartheta_0 \le \vartheta \le \vartheta_1\,, \tag{6.5}$$

so $\vartheta \in [\vartheta_0, \vartheta_1]$ is the parameter and the plane curve is the map $\varphi : \mathbb{R} \to \mathbb{R}^2$ defined by $\varphi(\vartheta) = (\varrho(\vartheta)\cos\vartheta, \varrho(\vartheta)\sin\vartheta)$. Furthermore, when $\varrho(\vartheta)$ is a C^1 function, it is easy to see that the curve is regular if and only if

$$\big[\varrho(\vartheta)\big]^2 + \big[\varrho'(\vartheta)\big]^2 > 0, \qquad \forall\,\vartheta \in (\vartheta_0, \vartheta_1). \tag{6.6}$$

Example 5 The curve (*cardioid*) in $\mathbb{R}^2$, of polar equation

$$\varrho = a\,(1 + \cos\vartheta)\,, \qquad 0 \le \vartheta \le 2\pi\,,$$

with $a > 0$, is not regular since $\varrho(\pi) = \varrho'(\pi) = 0$. The point corresponding to $\vartheta = \pi$ in (6.5) is the origin. There, the cardioid has a cusp (Fig. 6.6).

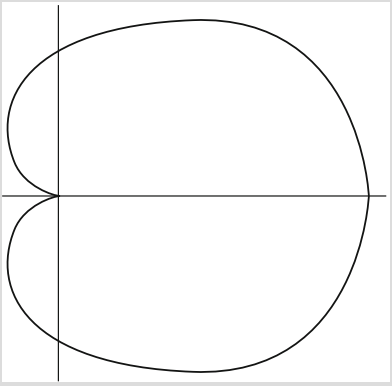
(continued)

Example 5 (continued)
Note however that the representation

$$\varrho = a\ (1 + \cos\vartheta)\,, \qquad -\pi\ \leq \vartheta \leq\ \pi\,, \tag{6.7}$$

defines a curve, still called *cardioid*, with the same support of Fig. 6.6, but now the map $\varphi : [-\pi, \pi] \to \mathbb{R}^2$, $\varphi(\vartheta) = (\varrho(\vartheta)\cos\vartheta,\ \varrho(\vartheta)\sin\vartheta)$, is regular, because for $\vartheta \in (-\pi, \pi)$ the regularity condition (6.6) holds.

Fig. 6.6



6.2 Oriented Curves

Consider the parametric curves φ, ψ in $\mathbb{R}^2$

$$\varphi(t) = (\cos\ t, \sin\ t)\,, \quad \forall\, t \in [0, 2\pi]; \qquad \psi(s) = (\cos\ 2s, \sin\ 2s)\,, \quad \forall\, s \in [0, \pi].$$

Both (simple, closed and regular) have as support the *unit circle* centred at the origin. Exactly as for the curve $\varphi(t)$, a point moving on the plane under the law of motion $\psi(s)$ travels around the unit circle counter-clockwise once. The modulus of the velocity $\left|\psi'(s)\right|$ of the point, therefore, is twice that of a point moving under $\varphi(t)$. Moreover, considering the map $g : t \in [0, 2\pi] \to s \in [0, \pi]$, $s = g(t) = t/2$, we have

$$\varphi(t) = \psi \circ g(t) = \psi(g(t)).$$

Conversely, if $g^{-1} : s \in [0, \pi] \to t \in [0, 2\pi]$ is the map $t = g^{-1}(s) = 2s$, then

$$\psi(s) = \varphi \circ g^{-1}(s) = \varphi(g^{-1}(s))\,.$$

By an invertible transformation of the parameter, therefore, we can pass from the parametric representation of φ to the parametric representation of ψ.
In general, two curves $\varphi : I \to \mathbb{R}^n$, $\psi : J \to \mathbb{R}^n$ are *equivalent* if there is a *reparametrisation*, i.e. a C^1 map $g : I \to J$ with $g'(t) \neq 0$ for any $t \in I$ (hence invertible), such that

$$\varphi(t) = \psi \circ g(t).$$

If so, the inverse map $g^{-1} : J \to I$ of g will be of class C^1, $(g^{-1})'(s) \neq 0$ for any $s \in J$ and

$$\psi(s) = \varphi \circ g^{-1}(s).$$

If φ and ψ are *equivalent* in the above sense we will write

$$\varphi \sim \psi\,.$$

The map g allowing to pass from the parametric representation of φ to that of ψ is a *diffeomorphism*.

Easily, $\sim$ is an *equivalence relation*. It induces a decomposition of the family of regular curves into mutually disjoint equivalence classes. In the sequel we shall use the word *curve* to indicate both an equivalence class $\gamma = [\varphi]$, and an arbitrary parametric representation of $\varphi(t)$. The context should clarify which sense is correct, unless we make it explicit.

For instance, the curves $\varphi = (\varphi_1, \varphi_2)$ and $\psi = (\psi_1, \psi_2)$ in $\mathbb{R}^2$ of parametric equations

$$\begin{cases} \varphi_1(t) = \cos t \\ \varphi_2(t) = \sin t, \end{cases} \quad t \in [0, 2\pi]\,, \qquad \begin{cases} \psi_1(s) = \sin 2\,s \\ \psi_2(s) = \cos 2\,s\,, \end{cases} \quad s \in \left[\frac{\pi}{4}, \frac{5\,\pi}{4}\right], \tag{6.8}$$

are equivalent. Namely, $\psi(s) = \varphi \circ g^{-1}(s)$, where $g^{-1} : \left[\frac{\pi}{4}, \frac{5\pi}{4}\right] \to [0, 2\pi]$ is the map $g^{-1}(s) = (5\pi/2) - 2\,s$. That is, g^{-1} is the (decreasing) *linear* map that sends $s = 5\pi/4$ to $g^{-1}(s) = 0$ and $s = \pi/4$ to $g^{-1}(s) = 2\pi$.

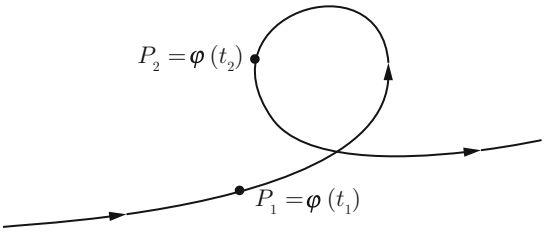
To any curve $\varphi : I \to \mathbb{R}^n$ we can assign an *orientation*, induced by the chosen parametric representation. We say a point $P_1 = \varphi(t_1)$ *precedes* $P_2 = \varphi(t_2)$ *in the chosen orientation* if $t_1 < t_2$ (see Fig. 6.7).

We can then define another (sharper) equivalence relation for regular curves, which we will denote by $\overset{\circ}{\sim}$.

We will say the regular curves $\varphi : I \to \mathbb{R}^n$, $\psi : J \to \mathbb{R}^n$ *are* $\overset{\circ}{\sim}$*-equivalent* if $\varphi \sim \psi$ and φ and ψ have the same orientation.

It is easy to see that $\varphi : I \to \mathbb{R}^n$, $\psi : J \to \mathbb{R}^n$ are $\overset{\circ}{\sim}$-equivalent if and only if the reparametrisation $s = g(t)$, with $g : I \to J$, always has positive derivative.

Fig. 6.7



For example the curves $\varphi = (\varphi_1, \varphi_2)$ and $\psi = (\psi_1, \psi_2)$ defined in (6.8) are not $\overset{\circ}{\sim}$-equivalent.

The equivalence class γ for $\sim$ therefore breaks into two equivalence classes for $\overset{\circ}{\sim}$, called *oriented curves*. Once we have chosen one orientation, arbitrarily, called *positive orientation*, the two equivalence classes are indicated by γ^+, γ^-.

6.3 The Length of a Curve

Consider a continuous curve $\varphi : [a, b] \to \mathbb{R}^n$ defined over a closed and bounded interval $[a, b]$. Any *partition*

$$a = t_0 < t_1 < \ldots < t_N = b$$

of $[a, b]$ determines a polygonal path $\mathcal{P}$ of vertices $\varphi(a), \varphi(t_1), \ldots, \varphi(t_{N-1}), \varphi(b)$, and conversely.

The *length* of this path is

$$\ell(\mathcal{P}) = \sum_{i=1}^{N} |\varphi(t_i) - \varphi(t_{i-1})| .$$

Intuitively, this number is an approximation from below of the "length" of the curve φ. The smaller the lengths of the segments, the better the approximation (Fig. 6.8).

Hence the following definition is natural. The *length of the continuous curve* $\varphi : [a, b] \to \mathbb{R}^n$ is the number

$$L(\varphi) = \sup \ell(\mathcal{P}), \tag{6.9}$$

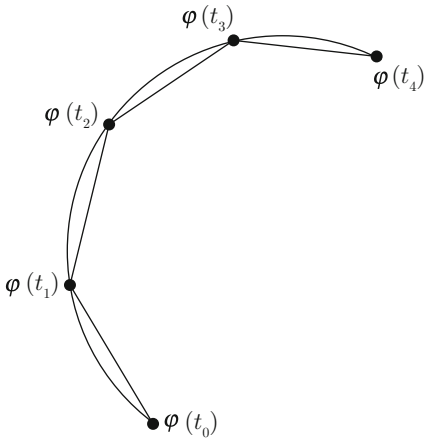
as $\mathcal{P}$ varies among all polygonal paths with vertices on the curve's support. If the supremum is finite, the curve φ is said to be *rectifiable*.

We start by showing that all $C^1[a, b]$ curves are rectifiable.

Rectifiability of C^1 Curves *A C^1 curve $\varphi : [a, b] \to \mathbb{R}^n$ is rectifiable, and its length $L(\varphi)$ is the integral*

$$L(\varphi) = \int_a^b \left|\varphi'(t)\right| dt. \tag{6.10}$$

Fig. 6.8



Proof Let us first prove that

$$\ell(\mathcal{P}) \le \int_a^b \left|\varphi'(t)\right| dt, \tag{6.11}$$

for any polygonal path $\mathcal{P}$ on the curve φ determined by a partition $a = t_0 < t_1 < \ldots < t_N = b$ of $[a, b]$. The lemma of Sect. 4.2 (whereby the integral of the norm bounds the norm of the integral) and the fundamental theorem of calculus on the components of φ tell us that

$$\ell(\mathcal{P}) = \sum_{i=1}^{N} |\varphi(t_i) - \varphi(t_{i-1})| = \sum_{i=1}^{N} \left| \int_{t_{i-1}}^{t_i} \varphi'(t) dt \right| \le$$

$$\le \sum_{i=1}^{N} \int_{i-1}^{t_i} \left|\varphi'(t)\right| dt = \int_a^b \left|\varphi'(t)\right| dt.$$

Hence (6.11) implies

$$L(\varphi) = \sup_{\mathcal{P}} \ell(\mathcal{P}) \le \int_a^b \left|\varphi'(t)\right| dt. \tag{6.12}$$

Conversely, as φ' is uniformly continuous on $[a, b]$, given $\varepsilon > 0$ there is $\delta > 0$ such that

$$s, t \in [a, b], \quad |t - s| < \delta \quad \Rightarrow \quad \left|\varphi'(t) - \varphi'(s)\right| < \varepsilon\,. \tag{6.13}$$

Consider a partition $a = t_0 < t_1 < \ldots < t_N = b$ of $[a, b]$ by intervals of length less than δ, and let $\mathcal{P}$ be the induced polygonal path. Fix the interval $[t_{i-1}, t_i]$ and

for any $s \in [t_{i-1}, t_i]$ compute

$$\varphi(t_i) - \varphi(t_{i-1}) = \int_{t_{i-1}}^{t_i} \varphi'(t)dt \; =$$

$$= \int_{t_{i-1}}^{t_i} \left\{\varphi'(t) - \varphi'(s)\right\} dt + \varphi'(s)\ (t_i\ -\ t_{i-1})\,.$$

Taking norms and remembering the uniform continuity in (6.13),

$$\left|\varphi'(s)\right| (t_i\ -\ t_{i-1}) \leq$$

$$\leq |\varphi(t_i)\ -\ \varphi(t_{i-1})| + \left| \int_{t_{i-1}}^{t_i} \left\{\varphi'(t)\ -\ \varphi'(s)\right\} dt \right| \leq$$

$$\leq |\varphi(t_i) - \varphi(t_{i-1})| + \varepsilon\ (t_i\ -\ t_{i-1})\,.$$

Then

$$\left|\varphi'(s)\right| \leq \frac{|\varphi(t_i)\ -\ \varphi(t_{i-1})|}{t_i\ -\ t_{i-1}} + \varepsilon,$$

and integrating over $s \in [t_{i-1}, t_i]$,

$$\int_{t_{i-1}}^{t_i} \left|\varphi'(s)\right| ds \leq |\varphi(t_i)\ -\ \varphi(t_{i-1})| + \varepsilon\ (t_i\ -\ t_{i-1})\ .$$

Finally, summing for $i = 1, 2, \ldots, N$,

$$\int_a^b \left|\varphi'(s)\right| ds \leq \ell(\mathcal{P}) + \varepsilon\,(b-a) \leq L(\varphi) + \varepsilon\,(b-a)\ . \tag{6.14}$$

Letting ε go to zero, from (6.14) and (6.12) we obtain claim (6.10).

Example 1 Let $y = f(x)$ ($f : [a, b] \to \mathbb{R}$) be a C^1 function and $\varphi : [a, b] \to \mathbb{R}^2$ the curve of components $(t, f(t))$, i.e. of parametric equations

$$\begin{cases} x = t \\ y = f(t)\,, \end{cases} \qquad \forall\, t \in [a, b].$$

The support of φ is the graph of f. Clearly φ is a regular curve, and its length, by formula (6.10), is

$$L(\varphi) = \int_a^b \sqrt{1 + (f'(x))^2}dx\,.$$

Example 2 If a plane curve $\varphi : [\vartheta_0, \vartheta_1] \to \mathbb{R}^2$ has polar equation $\varrho = \varrho(\vartheta)$, $\vartheta_0 \le \vartheta \le \vartheta_1$, with $\varrho(\vartheta)$ of class C^1, the length of φ equals

$$L(\varphi) = \int_a^b \sqrt{(\varrho'(\vartheta))^2 + (\varrho(\vartheta))^2}\, d\vartheta.$$

Example 3 Consider the continuous function $f : [0, 1] \to \mathbb{R}$, not of class C^1, defined by

$$f(x) = \begin{cases} 0 & \text{if} \quad x = 0 \\ x \sin \dfrac{\pi}{2x} & \text{if} \quad 0 < x \le 1\,. \end{cases}$$

We claim that $\varphi(t) = (t, f(t))$ is not rectifiable. Call $\mathcal{P}_N$, for any positive integer N, the polygonal path induced by the partition of $[0, 1]$

$$0 = t_N < t_{N-1} = \frac{1}{2N-1} < \ldots < t_i = \frac{1}{2i+1} < \ldots < t_0 = 1\,,$$

so $t_N = 0$ and $t_i = 1/(2i + 1)$, $i = 0, 1, 2, \ldots, N - 1$. Its vertices are $(0, 0)$ and the points $(1/(2i + 1)\,, (-1)^i/(2i + 1))$. Then

$$\ell(\mathcal{P}_N) > \sum_{i=1}^{N-1} |\varphi(t_i) - \varphi(t_{i-1})| = \sum_{i=1}^{N-1} \frac{\sqrt{4 + 16\,i^2}}{4\,i^2 - 1} > \sum_{i=1}^{N-1} \frac{1}{i}$$

and so

$$\lim_{N \to +\infty} \ell(\mathcal{P}_N) = +\infty.$$

Unless $\varphi : [a, b] \to \mathbb{R}^n$ is one-to-one, (6.10) will not give the measure of the support of φ. For example, the two curves $\varphi(t) = (\cos t, \sin t)$, for $t \in [0, 2\pi]$, and $\psi(t) = (\cos 2t, \sin 2t)$, for $t \in [0, 3\pi/2]$, have lengths $L(\varphi) = 2\pi$, $L(\psi) = 3\pi$ despite having the same support.

On the other hand the length of a regular curve γ is an intrinsic quantity, that does not depend on the particular parametric representation we have chosen. Let $\varphi : [a, b] \to \mathbb{R}^n$, $\psi : [\alpha, \beta] \to \mathbb{R}^n$ be arbitrary parametric representations of γ, and suppose $g : [a, b] \to [\alpha, \beta]$ is a *diffeomorphism* such that $\varphi(t) = \psi(g(t))$. Denoting by $t = g^{-1}(s)$ the inverse map of $s = g(t)$, we then have

$$\int_a^b \left|\varphi'(t)\right| dt = \int_a^b \left|\frac{d\psi}{ds}(g(t))\right| \cdot \left|g'(t)\right| dt =$$

$$= \int_{g(a)}^{g(b)} \left|\psi'(s)\right| \cdot \left|g'(g^{-1}(s))\right| \cdot \frac{dg^{-1}(s)}{ds}\, ds.$$

But $\left|g'(g^{-1}(s))\right| \cdot \dfrac{dg^{-1}(s)}{ds}$ is identically equal to 1 or -1, depending on whether $g(t)$ is strictly increasing or decreasing. In either case

$$\int_a^b \left|\varphi'(t)\right| dt = \int_\alpha^\beta \left|\psi'(s)\right| ds.$$

From the definition of length of a curve $\varphi : [a, b] \to \mathbb{R}^n$ we deduce that if φ_1 and φ_2 are restrictions of φ to the intervals $[a, c]$ and $[c, b]$, then

$$L(\varphi) = L(\varphi_1) + L(\varphi_2).$$

In general, (6.10) is still valid for piecewise-regular curves. We shall say a curve $\varphi : [a, b] \to \mathbb{R}^n$ is *piecewise regular* if there is a partition $\{a = a_0 < a_1 < \ldots < a_N = b\}$ of $[a, b]$ such that the curve $\varphi_i = \varphi|_{[a_{i-1}, a_i]}$ is regular, for any $i = 1, 2, \ldots, N$.

Now it is easy to show that the length of a piecewise-regular curve $\varphi : [a, b] \to \mathbb{R}^n$ equals

$$L(\varphi) = \sum_{i=1}^N L(\varphi_i) = \sum_{i=1}^N \int_{a_{i-1}}^{b_i} \left|\varphi_i'(t)\right| dt = \int_a^b \left|\varphi'(t)\right| dt.$$

Hence, as already said, (6.10) holds for piecewise-regular curves. In particular, for polygonal paths $\mathcal{P}$ we have $L(\mathcal{P}) = \ell(\mathcal{P})$.

Among all possible parametrisations of a regular curve, the one defined by the *arclength* is especially relevant from a geometric point of view. To fix ideas, think of a rectifiable curve with support $\Gamma \subset \mathbb{R}^n$ and fix on it a point P_0 and an orientation, as in Fig. 6.9. Associate with any point P of Γ the length $s(P)$ of the arc joining P_0 and P if P follows P_0, or minus that length if P precedes P_0. Thus we set up a 1-1 correspondence between the points of Γ and the points of some interval $[a, b]$, in such a way that if $s(P_1) = s_1$ and $s(P_2) = s_2$, the length of the arc joining P_1 and P_2 is exactly $|s_2 - s_1|$.

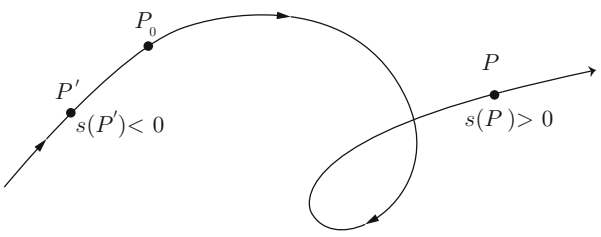


Fig. 6.9

Fix a parametrisation $\varphi : [a, b] \to \mathbb{R}^n$ of the regular curve γ and a point $t_0 \in [a, b]$. The function

$$s(t) = \int_{t_0}^{t} \left|\varphi'(\tau)\right| d\tau, \qquad \forall\, t \in [a, b]$$

is strictly increasing, differentiable, and such that $\dfrac{ds}{dt} > 0$ for any t. Hence $s = s(t)$ is a reparametrisation $s : [a, b] \to [s(a), s(b)]$. The parameter s is called *arclength*. In this way if $t = t(s)$ is the inverse of $s = s(t)$ and $\gamma(s) = \varphi(t(s))$, for any value of the parameter s we have

$$\gamma'(s) = \frac{d\varphi}{dt}\,(t(s)) \cdot \frac{dt}{ds}\,(s) = \frac{\varphi'(t(s))}{|\varphi'(t(s))|}\,,$$

and therefore

$$\left|\gamma'(s)\right| = 1, \qquad \forall\, s \in [s(a), s(b)]\,. \tag{6.15}$$

We deduce that $\gamma'(s)$ is the *unit tangent vector* to the curve at the generic point of arclength s. From (6.15) it follows that if s is an arclength parameter, to any subinterval $[s_1, s_2] \subseteq [s(a), s(b)]$ there corresponds an arc with length $s_2 - s_1$.

6.4 The Integral of a Function Along a Curve

Let γ be a regular curve and $\varphi : [a, b] \to \mathbb{R}^n$ a parametrisation. If f is a real function of n real variables, defined on the curve's support $\Gamma = \varphi([a, b])$, and if f is continuous on this support, it makes sense to consider the integral

$$\int_a^b f(\varphi(t)) \left|\varphi'(t)\right| dt$$

in the real variable t. When $\psi : [\alpha, \beta] \to \mathbb{R}^n$ is another parametrisation of γ, one can prove

$$\int_a^b f(\varphi(t)) \left|\varphi'(t)\right| dt = \int_\alpha^\beta f(\psi(s)) \left|\psi'(s)\right| ds. \tag{6.16}$$

In fact, if $g : [a, b] \to [\alpha, \beta]$ is a reparametrisation of equation $s = g(t)$, and $t = g^{-1}(s)$ is the inverse map,

$$\int_a^b f(\varphi(t)) \left|\varphi'(t)\right| dt =$$

$$= \int_{g(a)}^{g(b)} f\left(\varphi(g^{-1}(s))\right) \cdot \left|\frac{d\varphi}{dt}\,(g^{-1}(s))\right| \cdot \frac{dg^{-1}(s)}{ds}\, ds.$$

As $\varphi(t) = \psi(g(t))$, or $\psi(s) = \varphi(g^{-1}(s))$, we have

$$\psi'(s) = \frac{d\varphi(g^{-1}(s))}{ds} = \frac{d\varphi}{dt}(g^{-1}(s)) \cdot \frac{dg^{-1}(s)}{ds}$$

and so

$$\int_a^b f(\varphi(t)) \left|\varphi'(t)\right| dt = \pm \int_{g(a)}^{g(b)} f(\varphi(s)) \cdot \left|\psi'(s)\right| ds, \tag{6.17}$$

where we choose the $+$ sign if $\dfrac{dg^{-1}}{ds} > 0$, and the $-$ sign otherwise (recall g and g^{-1} are strictly monotone). Using (6.17) we obtain (6.16), since having $\dfrac{dg^{-1}}{ds} > 0$ for any s forces $g(a) = \alpha$ and $g(b) = \beta$, while $\dfrac{dg^{-1}}{ds} < 0$ for any s implies $g(a) = \beta$ and $g(b) = \alpha$.

We may then define the following notion. If γ is a regular curve in $\mathbb{R}^n$ with parametrisation $\varphi : [a, b] \to \mathbb{R}^n$ (also known as a *path*), and $f : \Gamma \to \mathbb{R}$ is a continuous function on the curve's support $\Gamma \subset \mathbb{R}^n$, the integral

$$\int_a^b f(\varphi(t)) \left|\varphi'(t)\right| dt$$

does not depend on the chosen parametrisation nor on the orientation induced on γ. This quantity is called *integral of f along the curve γ*, and one customarily indicates it by

$$\int_\gamma f ds. \tag{6.18}$$

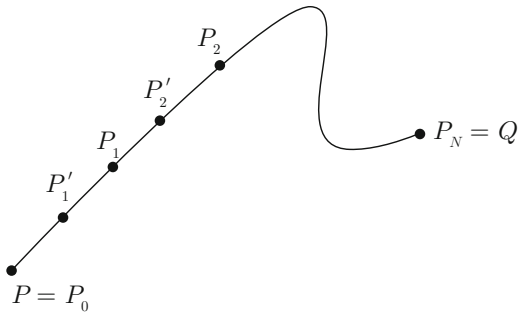
Observe that if $\gamma : [0, L] \to \mathbb{R}^n$ is an arclength parametrisation of γ (the parameter is the arclength s), since $\left|\gamma'(s)\right|$ is equal to 1 the integral (6.18) is

$$\int_0^L f(\gamma(s))\, ds.$$

The integral can be obtained in a more intrinsic way using a similar argument to the one employed for the integral of a continuous function on a segment $[a, b]$.

In fact, suppose $P = \gamma(0)$, $Q = \gamma(L)$ are the endpoints of the curve γ and let $P_0 = P$, $P_1, \ldots, P_N = Q$ be $N + 1$ points on the support Γ of γ (see Fig. 6.10) chosen so that P_{i-1} precedes P_i in the orientation given by the parameter s. Then every P_i defines an arclength value $s(P_i) = s_i$, and $s_{i-1} < s_i$ for any $i = 1, 2, \ldots, N$. On

Fig. 6.10



each arc γ_i between P_{i-1} and P_i let us pick a point P_i' with arclength s_i'. If $L(\gamma_i)$ is the length of γ_i, the sum

$$\sum_{i=1}^{N} f(P_i')L(\gamma_i) \tag{6.19}$$

equals

$$\sum_{i=1}^{N} f(\gamma(s_i'))(s_i - s_{i-1})\,. \tag{6.20}$$

As the function $f(\gamma(s))$ is continuous on $[0, L]$, (6.20) converges to

$$\int_0^L f\,(\gamma(s))\,ds,$$

as $\max\{s_i - s_{i-1} : 1 \le i \le N\} \to 0$. We therefore conclude that the sums (6.19) converge to the integral

$$\int_\gamma f ds,$$

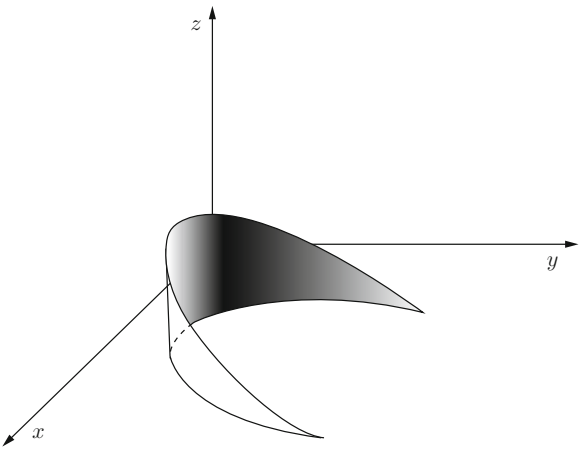
as $\max\{s_i - s_{i-1} : 1 \le i \le N\}$ goes to zero.

As for the integrals of continuous functions of one real variable, if γ is a plane curve with support Γ and $f : \Gamma \to [0, +\infty)$ is a continuous function, we may think of $\int_\gamma f ds$ as a measure of the area of the surface between Γ and the graph of f over Γ (see Fig. 6.11).

By the definition of integral along a curve, if γ is a regular curve with support Γ and $f, g : \Gamma \to \mathbb{R}$ are continuous,

$$\int_\gamma (\alpha f + \beta g)\,ds = \alpha \int_\gamma ds + \beta \int_\gamma g\,ds, \qquad \forall\, \alpha, \beta \in \mathbb{R}\,; \tag{6.21}$$

Fig. 6.11



$$\int_\gamma f\,ds \le \int_\gamma g\,ds, \qquad \text{if} \quad f \le g \quad \text{on} \quad \Gamma;$$

$$\left|\int_\gamma f\,ds\right| \le \int_\gamma |f|\,ds \le \max_\Gamma |f| \cdot L(\gamma). \tag{6.22}$$

Moreover, if $\gamma : [a,b] \to \mathbb{R}^n$ is the union of regular curves $\gamma_1 : [a,c] \to \mathbb{R}^n$, $\gamma_2 : [c,b] \to \mathbb{R}^n$, $a < c < b$, then

$$\int_\gamma f\,ds = \int_{\gamma_1} f\,ds + \int_{\gamma_2} f\,ds. \tag{6.23}$$

This decomposition, in analogy to what was observed about a regular curve's length, allows us to extend the notion of integral along a curve to piecewise-regular curves γ in the obvious way.

Now we introduce what is commonly known as the *centroid* of a curve. Suppose $\gamma : [a,b] \to \mathbb{R}^n$ is a piecewise-regular simple curve with support Γ and length $L(\gamma)$. The point $x_0 \in \mathbb{R}^n$ of coordinates

$$x_{0i} = \frac{1}{L(\gamma)} \int_\gamma x_i\,ds, \qquad \forall\, i = 1, 2, \ldots, n,$$

where x_i , $i = 1, 2, \ldots, n$, are the coordinates of the generic point $x \in \mathbb{R}^n$, is the *centroid* of the set Γ.

6.5 The Curvature of a Plane Curve

Consider a regular curve $\gamma : [a, b] \to \mathbb{R}^2$ parametrised by arclength s. We wish to define the notion of *curvature*, a scalar quantity describing (up to sign) the geometric features of the curve, irrespective of possible reparametrisations or coordinate changes.

Let us recall the *formula for differentiating the inner product in* $\mathbb{R}^n$. Suppose that $u(t), v(t) : I \to \mathbb{R}^n$ are differentiable functions on an interval $I \subseteq \mathbb{R}$, of respective components $(u_i(t))$, $(v_i(t))$. Then

$$\begin{aligned} \frac{d}{dt}\,(u(t), v(t)) &= \frac{d}{dt}\,\sum_{i=1}^{n} u_i(t)\, v_i(t) = \\ &= \sum_{i=1}^{n} u_i'(t)\, v_i(t) \; + \sum_{i=1}^{n} u_i(t)\, v_i'(t) = \\ &= \big(u'(t), v(t)\big) + \big(u(t), v'(t)\big) \; . \end{aligned} \tag{6.24}$$

Note in particular the following property of vector-valued functions of constant norm, a consequence of (6.24): *if* $w(t) : I \to \mathbb{R}^n$ *is differentiable on the interval* $I \subseteq \mathbb{R}$ *and* $|w(t)| = 1$ *for any* $t \in I$, *the vector* $w(t)$ *is orthogonal to* $w'(t)$ *at any point* $t \in I$:

$$\big(w'(t), w(t)\big) = 0\,, \qquad \forall\, t \in I\,. \tag{6.25}$$

The proof is an application of formula (6.24):

$$\big(w'(t), w(t)\big) = \frac{1}{2}\frac{d}{dt}\,(w(t), w(t)) = \frac{1}{2}\frac{d}{dt}\,|w(t)|^2 = 0\,, \qquad \forall\, t \in I\,.$$

Going back to the curve $\gamma : I = [a, b] \to \mathbb{R}^2$, in this section we will assume $\gamma(s)$ is of class C^2. As in (6.3), we call $T(s)$ the *unit tangent vector* to γ at the point $\gamma(s)$, i.e. the vector-valued function

$$T(s) = \frac{\gamma'(s)}{|\gamma'(s)|} = \gamma'(s)\,, \qquad s \in I\,,$$

since $\big|\gamma'(s)\big|$ is identically 1. Consider now the derivative $\dfrac{d}{ds}T(s) = T'(s)$. As $T(s)$ has length 1, from (6.25) we have

$$\big(T'(s), T(s)\big) = \big(\gamma''(s), \gamma'(s)\big) = 0\,, \qquad \forall\, s \in I\,, \tag{6.26}$$

and so the vector $\gamma''(s)$ is orthogonal to the unit tangent vector $\gamma'(s)$, for any $s \in I$.

We call *unit normal* to the curve the unit vector $N(s)$ obtained rotating the unit tangent vector $T(s)$ by $\pi/2$ in the same way as e_2 rotates over to e_1. Since (6.26) implies

$$\big(T'(s), T(s)\big) = 0\,, \qquad \forall\, s \in I\,,$$

the vector $T'(s)$, orthogonal to $T(s)$, is parallel to the unit normal $N(s)$. Therefore there exists a real number $k(s)$ such that

$$T'(s) = -k(s)\, N(s)\,, \qquad \forall\, s \in I\,, \tag{6.27}$$

called the *curvature* of the plane curve γ at $\gamma(s)$. As it stands, the curvature can be positive or negative (see Fig. 6.12).

Warning: sometimes the curvature is defined by the formula $T'(s) = k(s)\,\overline{N}(s)$, where $\overline{N}(s)$ is the opposite to the unit normal we have defined. The two curvature definitions clearly coincide, for $\overline{N}(s) = -N(s)$.

The circle centred on the negative normal semi-axis (that of $-N(s)$) if $k(s) > 0$, or on the positive semi-axis if $k(s) < 0$, passing through $\gamma(s)$ and with radius $1/|k(s)|$, is the *osculating circle* to the curve γ at $\gamma(s)$. The reciprocal $1/|k(s)|$ of the curvature's absolute value $|k(s)|$ is called *curvature radius*.

It is easy to show that the osculating circle to the curve at $\gamma(s)$ has, at that point, the same tangent line and same normal axis as the curve γ (Fig. 6.13).

Fig. 6.12

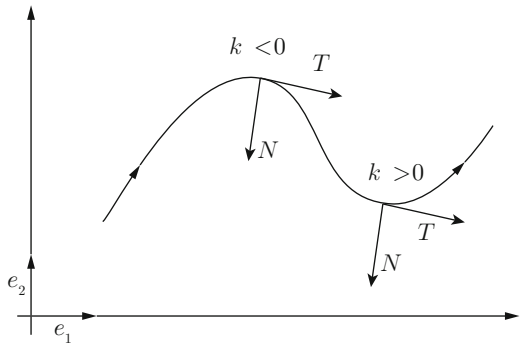
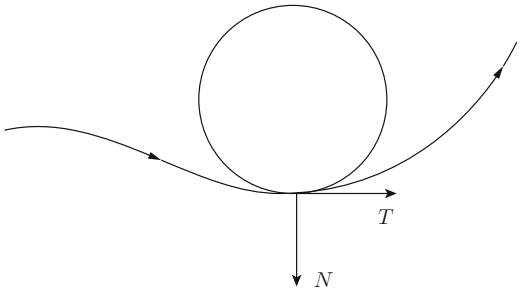


Fig. 6.13



Now we will prove that a *plane curve* of equations $\varphi(t) = (x(t), y(t))$ has *curvature*

$$k(t) = \frac{x'(t)\, y''(t) - x''(t)\, y'(t)}{\left[(x'(t))^2 + (y'(t))^2\right]^{3/2}} \tag{6.28}$$

at the point $\varphi(t)$.

Suppose $\gamma(s) = (x(s), y(s))$ is the parametric equation of the curve, with arclength s. Equation (6.27) reads

$$\left(x''(s), y''(s)\right) = -k(s)\left(y'(s), -x'(s)\right), \tag{6.29}$$

from which, in particular, supposing for instance $x'(s) \neq 0$,

$$k(s) = \frac{y''(s)}{x'(s)}. \tag{6.30}$$

Consider a diffeomorphism $t = t(s)$, $t : J \to I$ such that $\gamma(s) = \varphi(t(s))$ and $dt/ds > 0$ just to fix ideas. Then

$$\gamma'(s) = \varphi'(t(s))\frac{dt}{ds},$$

and since $\left|\gamma'\right| = 1$,

$$\frac{dt}{ds} = \frac{1}{|\varphi'|}, \qquad x'(s) = \frac{x'(t)}{|\varphi'|}, \qquad y'(s) = \frac{y'(t)}{|\varphi'|}.$$

Differentiating the last equation and recalling that

$$\left|\varphi'\right| = \left[\left(x'(t)\right)^2 + \left(y'(t)\right)^2\right]^{1/2} \tag{6.31}$$

we obtain

$$\begin{aligned} y''(s) &= \frac{d}{dt}\left(\frac{y'(t)}{|\varphi'|}\right)\frac{dt}{ds} = \\ &= \left[\frac{y''(t)}{|\varphi'|} - \frac{y'(t)\left(x'(t)\, x''(t) + y'(t)\, y''(t)\right)}{|\varphi'|^3}\right] \cdot \frac{1}{|\varphi'|}. \end{aligned}$$

From this relation and (6.31) we then infer

$$\frac{y''(s)}{x'(s)} = \frac{\left[\dfrac{y''(t)}{|\varphi'|} - \dfrac{y'(t)\left(x'(t)\,x''(t) + y'(t)\,y''(t)\right)}{|\varphi'|^3}\right]}{x'(t)} =$$

$$= \frac{y''(t)\,\left|\varphi'\right|^2 - y'(t)\left(x'(t)\,x''(t) + y'(t)y''(t)\right)}{x'(t)\,|\varphi'|^3} =$$

$$= \frac{y''(t)\,\left(x'(t)\right)^2 - y'(t)\,x'(t)x''(t)}{x'(t)\,|\varphi'|^3} = \frac{y''(t)\,x'(t) - y'(t)x''(t)}{|\varphi'|^3}\,.$$

Eventually, (6.28) follows from the above together with (6.30).

The outcome is the same in case $dt/ds < 0$, or if $x'(s) = 0$. In the latter case, instead of (6.30) we have, by (6.29),

$$k(s) = -\frac{x''(s)}{y'(s)},$$

with computations similar to the previous ones.

Example 1 A straight line clearly has zero curvature at every point. Suppose γ is an *ellipse* of equations $\varphi(t) = (x(t), y(t)) = (a\cos t, b\,\sin t)\,, a, b \in \mathbb{R}, t \in I \subseteq \mathbb{R}$, and let us determine its curvature at $\varphi(t)$. As

$$x'(t) = -a\sin t, \quad x''(t) = -a\cos t, \quad y'(t) = b\cos t, \quad y''(t) = -b\sin t,$$

from (6.28) an easy computation produces

$$k(t) = \frac{a\,b}{\left[a^2\sin^2 t + b^2\cos^2 t\right]^{3/2}}, \qquad \forall\, t \in I\,. \tag{6.32}$$

If $a > b$, the reader can show that the curvature (6.32) of the ellipse is largest at the vertices $(\pm a, 0)$ and smallest at the vertices $(0, \pm b)$. If $a = b$ the ellipse is a circle of radius a, which, as is to be expected, has constant curvature $1/a$.

In particular, if the curve γ is the graph of a function $f : I \to \mathbb{R}$ of class C^2, with the parametrisation $\varphi(t) = (t, f(t))\,, t \in I$, from (6.28) we obtain the following expression for the *curvature* of γ

$$k(t) = \frac{f''(t)}{\left[1 + (f'(t))^2\right]^{3/2}}, \qquad \forall\, t \in I.$$

6.6 The Cross Product in $\mathbb{R}^3$

The notion of *cross product* of two vectors in $\mathbb{R}^3$, which we introduce here, will be used in particular in the next two sections.

Let $u = (u_1, u_2, u_3)$, $v = (v_1, v_2, v_3)$ be vectors in $\mathbb{R}^3$. The *cross product* of u and v, written $u \wedge v$, is the vector of $\mathbb{R}^3$

$$u \wedge v = \begin{vmatrix} u_2 & v_2 \\ u_3 & v_3 \end{vmatrix} e_1 + \begin{vmatrix} u_3 & v_3 \\ u_1 & v_1 \end{vmatrix} e_2 + \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix} e_3 \,, \tag{6.33}$$

where e_1 , e_2 , e_3 are the elements of the canonical basis of $\mathbb{R}^3$ ($e_1 = (1, 0, 0)$ etc.). The components of $u \wedge v$ are the 2×2 minors of the matrix with columns u and v

$$\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \\ u_3 & v_3 \end{pmatrix}, \tag{6.34}$$

taken with alternating signs $+, -, +$. Operationally, (6.33) is computable as the formal expansion, along the first column, of the 3×3 determinant

$$\begin{vmatrix} e_1 & u_1 & v_1 \\ e_2 & u_2 & v_2 \\ e_3 & u_3 & v_3 \end{vmatrix} = e_1 \begin{vmatrix} u_2 & v_2 \\ u_3 & v_3 \end{vmatrix} - e_2 \begin{vmatrix} u_1 & v_1 \\ u_3 & v_3 \end{vmatrix} + e_3 \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix}.$$

Below we list the cross product's main properties. Let (u_1, u_2, u_3), (v_1, v_2, v_3) and (w_1, w_2, w_3) be the components of $u, v, w \in \mathbb{R}^3$.

Proposition 1 *For any $u, v, w \in \mathbb{R}^3$ and $\lambda, \mu \in \mathbb{R}$,*

$$u \wedge v = -v \wedge u \,; \tag{6.35}$$

$$(\lambda u + \mu w) \wedge v = \lambda(u \wedge v) + \mu(w \wedge v) \,; \tag{6.36}$$

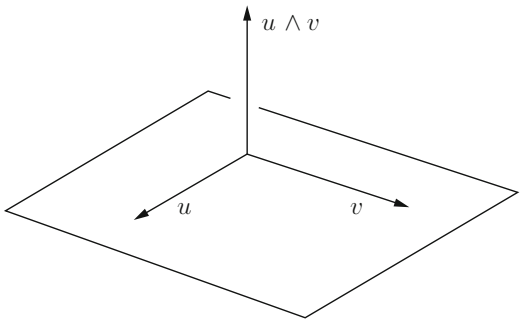
$$u \wedge v = 0 \quad \textit{if and only if } u \textit{ and } v \textit{ are linearly dependent}; \tag{6.37}$$

$$(u \wedge v, w) = \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} \,; \tag{6.38}$$

$$(u \wedge v, u) = 0, \qquad (u \wedge v, v) = 0 \,. \tag{6.39}$$

Proof Equation (6.35) follows from the fact that interchanging the two columns in (6.34) changes the sign of all minors. The linearity property (6.36) is straightforward from definition (6.33). Now, (6.37) arises by recalling that u and v are linearly independent if and only if matrix (6.34) has rank 2.

Fig. 6.14



Expanding along the last column the determinant in (6.38), by the definitions of cross product and inner product we have

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = w_1 \begin{vmatrix} u_2 & v_2 \\ u_3 & v_3 \end{vmatrix} - w_2 \begin{vmatrix} u_1 & v_1 \\ u_3 & v_3 \end{vmatrix} + w_3 \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix} =$$

$$= (w, u \wedge v) = (u \wedge v, w)\,.$$

At last, (6.39) is immediate from (6.38).

From (6.39) we also obtain the following result. See Fig. 6.14.

Proposition 2 *If two vectors u, v are linearly independent, their cross product $u \wedge v$ is orthogonal to the plane spanned by u, v. In particular the vectors u, v, $u \wedge v$, form a basis of $\mathbb{R}^3$.*

Note how each canonical basis element of $\mathbb{R}^3$ is the cross product of the other two:

$$e_1 = e_2 \wedge e_3\,, \qquad e_2 = e_3 \wedge e_1\,, \qquad e_3 = e_1 \wedge e_2\,. \tag{6.40}$$

Fix linearly independent vectors $\{u, v, w\}$ in $\mathbb{R}^3$ in a given order. We say that they are a *positively oriented basis*, or that they define the *positive orientation* on $\mathbb{R}^3$, if the determinant

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} \tag{6.41}$$

is positive. If, on the contrary, (6.41) is negative, $\{u, v, w\}$ define the *negative orientation* on $\mathbb{R}^3$.

For example the ordered basis $\{e_1, e_2, e_3\}$ is positively oriented, while $\{e_2, e_1, e_3\}$ is negatively oriented.

Choosing, in (6.38), $w = u \wedge v$, gives

Proposition 3 *If $u, v \in \mathbb{R}^3$ are linearly independent, the ordered triple $\{u, v, u \wedge v\}$ is a positively oriented basis.*

Example 1 The vectors $\{(1, 2, 3), (0, 1, 1), (-1, 1, 1)\}$ endow $\mathbb{R}^3$ with the positive orientation, because

$$\begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{vmatrix} = 1 .$$

The next result is meant to characterise geometrically the cross product.

Proposition 4 *For any $u, v, w \in \mathbb{R}^3$,*

$$(u \wedge v) \wedge w = (u, w)v - (v, w)\, u . \tag{6.42}$$

For any $u, v, x, y \in \mathbb{R}^3$,

$$(u \wedge v\,,\ x \wedge y) = \begin{vmatrix} (u, x) & (v, x) \\ (u, y) & (v, y) \end{vmatrix} . \tag{6.43}$$

Proof In case u, v, w coincide with one among e_1, e_2, e_3 in the canonical basis, by (6.39) and (6.40) it is easy to verify (6.42). In other words, (6.42) is proved by assessing it on the canonical basis. The general case follows by the linearity of the cross product and the inner product, because the expressions in (6.42) are linear in u, v and w. Formula (6.43) is proved along the same lines.

Choosing $x = u$, $y = v$ in (6.43) yields

$$|u \wedge v|^2 = (u \wedge v\,,\ u \wedge v) = \begin{vmatrix} (u, u) & (v, u) \\ (u, v) & (v, v) \end{vmatrix} =$$

$$= |u|^2\ |v|^2 - (u, v)^2 = |u|^2\ |v|^2\,(1 - \cos^2 \alpha) = |u|^2\ |v|^2 \sin^2 \alpha\,,$$

where α is the *angle* between the vectors u, v. From this relation and Propositions 2 and 3 one obtains

Proposition 5 *If $u, v \in \mathbb{R}^3$ are linearly independent, their cross product $u \wedge v$ is orthogonal to the plane spanned by u and v, it has norm $|u| \cdot |v| \sin \alpha$ and is oriented so that $\{u, v, u \wedge v\}$ is a positively oriented basis.*

From (6.42) we finally deduce a result generalising (6.40), which we shall use later.

Proposition 6 *If $u, v \in \mathbb{R}^3$ are orthogonal unit vectors, $|u| = |v| = 1$ and $(u, v) = 0$, then*

$$(u \wedge v) \wedge u = v, \qquad (u \wedge v) \wedge v = -u\,. \tag{6.44}$$

6.7 Biregular Curves in $\mathbb{R}^3$: Curvature

In this section, and in the next, $\gamma : [a, b] \to \mathbb{R}^3$ will indicate a regular curve in $\mathbb{R}^3$ parametrised by arclength s. We want to introduce two characteristic quantities of a curve, the *curvature* and the *torsion*. They not only describe the curve's geometric features, irrespective of coordinate changes or reparametrisations. As we will see, they completely determine the curve itself up to rigid motions.

First of all we remind the *differentiation rules* for the *inner product* and the *cross product* in $\mathbb{R}^3$. Suppose $u(t), v(t) : I \to \mathbb{R}^3$ are differentiable functions on the interval I, of respective components $(u_i(t))$, $(v_i(t))$. For the *inner product* we showed in (6.24) that

$$\frac{d}{dt}\,(u(t), v(t)) = \big(u'(t), v(t)\big) + \big(u(t), v'(t)\big)\,.$$

We recall in particular that *if $w(t) : I \to \mathbb{R}^3$ is differentiable on the interval $I \subseteq \mathbb{R}$ and $|w(t)| = 1$ for any $t \in I$, the vector $w(t)$ is at any $t \in I$ orthogonal to the vector $w'(t)$*:

$$\big(w'(t), w(t)\big) = 0\,, \qquad \forall\, t \in I\,. \tag{6.45}$$

The *derivative of the cross product* equals

$$\frac{d}{dt} u(t) \wedge v(t) = u'(t) \wedge v(t) + u(t) \wedge v'(t)\,. \tag{6.46}$$

Differentiating for instance the first component of $u(t) \wedge v(t)$ (see the definition of the *cross product* in the previous section) gives

$$\begin{aligned}\frac{d}{dt}\begin{vmatrix} u_2(t) & v_2(t) \\ u_3(t) & v_3(t) \end{vmatrix} &= \frac{d}{dt}\,(u_2(t)v_3(t) - u_3(t)v_2(t)) = \\ &= \big(u_2'(t)v_3(t) - u_3'(t)v_2(t)\big) + \big(u_2(t)v_3'(t) - u_3(t)v_2'(t)\big)\,,\end{aligned}$$

which is exactly the first component of $u'(t) \wedge v(t) + u(t) \wedge v'(t)$. Similarly, the other components of $d(u \wedge v)/dt$ and $u' \wedge v + u \wedge v'$ coincide.

Returning to the curve $\gamma : [a, b] \to \mathbb{R}^3$, in this section we shall assume that the function $\gamma(s)$ is C^2. We shall take a regular curve $\gamma : I = [a, b] \to \mathbb{R}^3$ of class $C^2(I)$ parametrised by arclength s.

If $T(s)$ is the *unit tangent vector* to γ at $s \in I$, since $|\gamma'(s)|$ is identically equal to 1 we have

$$T(s) = \frac{\gamma'(s)}{|\gamma'(s)|} = \gamma'(s)\,, \qquad s \in I\,. \tag{6.47}$$

Consider now the derivative $\dfrac{d}{ds}T(s) = T'(s)$ of the unit tangent vector $T(s)$. As $T(s)$ is a unit vector, (6.45) implies

$$\big(T'(s), T(s)\big) = \big(\gamma''(s), \gamma'(s)\big) = 0\,, \qquad \forall\, s \in I\,,$$

and so $\gamma''(s)$ is orthogonal to the unit tangent vector $\gamma'(s)$ for any $s \in I$. When $\gamma''(s)$ is non-zero everywhere on I, we say γ is *biregular*.

The *norm* of $\dfrac{d}{ds}T(s)$ is called the *curvature* $k(s)$ (if $k(s)$ is positive everywhere on I, γ is *biregular*). The curvature

$$k(s) = \left|\frac{d}{ds}T(s)\right| = |\gamma''(s)|$$

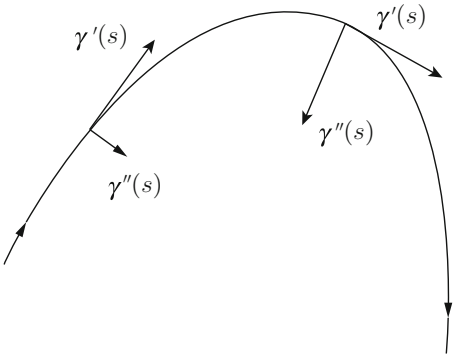
is a scalar quantity and has a major geometric meaning. It measures the variation of the tangent, i.e. how fast the curve bends away at points near $\gamma(s)$ from the tangent line at $\gamma(s)$. Hence the curvature is large when the curve, nearby $\gamma(s)$, deviates a lot from being a straight line (see Fig. 6.15).

When the space curve is actually contained in a plane, the definition of curvature for curves in $\mathbb{R}^3$ does not coincide with the one for plane curves. In space, in fact, it is not possible to define a sign for the curvature (since the normal to a plane curve, with given unit tangent vector, is determined up to sign, whereas in space it is not). Nonetheless, the two notions of curvature agree up to sign.

If γ is biregular, the unit vector

$$N(s) = \frac{\gamma''(s)}{|\gamma''(s)|} = \frac{\gamma''(s)}{k(s)} \qquad s \in I\,, \tag{6.48}$$

Fig. 6.15



called *unit normal* at $\gamma(s)$, is defined at every point. Since $T(s) = \gamma'(s)$, we have

$$T'(s) = k(s)\, N(s)\,, \qquad \forall\, s \in I \tag{6.49}$$

(recall that the curvature $k(s)$ is a number), i.e. *the vector $N(s)$ is everywhere* proportional to the derivative $T'(s)$ of the unit tangent vector $T(s)$, and by (6.45), it is *orthogonal to the unit tangent vector $T(s)$*.

The plane through $\gamma(s)$ spanned by the unit tangent and normal vectors is called *osculating plane*. The circle on the osculating plane with centre on the positive normal semi-axis (the one of $N(s)$), passing through $\gamma(s)$ and with radius $1/k(s)$ is the *osculating circle* to γ at $\gamma(s)$. The reciprocal $1/k(s)$ of the curvature $k(s)$ is the *curvature radius*.

Given a regular curve γ in space and any parametrisation $\varphi : I = [a, b] \to \mathbb{R}^3$, the curvature $k(t)$ at $\gamma(t)$ is

$$k(t) = \frac{\left|\varphi'(t) \wedge \varphi''(t)\right|}{|\varphi'(t)|^3}, \qquad t \in I\,. \tag{6.50}$$

In fact, suppose $\gamma : J \to \mathbb{R}^3$ is an arclength parametrisation of γ, and $t = t(s)$, $t : J \to I$, is a reparametrisation (a diffeomorphism) with the arclength $s = s(t)$ as inverse. Defining $\varphi(t) = \gamma(s(t))$, we have

$$\varphi'(t) = \gamma'(s(t))\frac{ds}{dt}\,, \quad \text{and so} \quad \left|\varphi'(t)\right| = \left|\frac{ds}{dt}\right|\,,$$

$$\varphi''(t) = \gamma''(s(t))\left(\frac{ds}{dt}\right)^2 + \gamma'(s(t))\frac{d^2s}{dt^2}\,.$$

As $\gamma' \wedge \gamma' = 0$, from the above and (6.47), (6.48) it follows

$$\varphi'(t) \wedge \varphi''(t) = \left(\frac{ds}{dt}\right)^3 \gamma' \wedge \gamma'' =$$

$$= \pm\left|\varphi'(t)\right|^3 k(s(t))\,[T(s) \wedge N(s)]$$

and so (6.50).

Suppose γ is a *cylindrical helix* of equations $\varphi(t) = (a\cos t, a\,\sin t, b\,t)$, with $a > 0$, $b \in \mathbb{R}$, $t \in I \subseteq \mathbb{R}$. Let us compute its curvature at $\varphi(t)$. Since

$$\varphi'(t) = (-a\sin t, a\cos t, b)\,, \qquad \varphi''(t) = (-a\cos t, -a\sin t, 0) \tag{6.51}$$

$$\varphi'(t) \wedge \varphi''(t) = \left(ab\sin t, -ab\cos t, a^2\right) \tag{6.52}$$

(continued)

we have $|\varphi'(t)| = \sqrt{a^2+b^2}$, $|\varphi' \wedge \varphi''| = a\sqrt{a^2+b^2}$. Using formula (6.50),

$$k(t) = \frac{|\varphi'(t) \wedge \varphi''(t)|}{|\varphi'(t)|^3} = \frac{a}{a^2+b^2}, \qquad \forall\, t \in I\,.$$

The curvature of this helix is therefore constant. If $b = 0$ the helix collapses to a circle of radius a, whose curvature is $1/a$, as we saw in Sect. 6.5 for plane curves.

Appendix to Chap. 6

6.8 Curves in $\mathbb{R}^3$: Torsion, Frenet Frame

Let $\gamma : I = [a, b] \to \mathbb{R}^3$ be a biregular curve (the definition of *biregular curve* is in the previous section) of class C^3 and let s be the arclength. We wish to define the concept of *torsion*. For that, retaining the previous section's notations for the unit tangent vector $T(s)$ and unit normal $N(s)$, we define the *unit binormal vector* $B(s)$

$$B(s) = T(s) \wedge N(s)\,. \tag{6.53}$$

By definition $B(s)$ is normal to both $T(s)$ and $N(s)$, making it orthogonal to the osculating plane at $\gamma(s)$. If we differentiate $B(s)$, the quantity $|B'(s)|$ is a scalar function measuring the binormal's variation, i.e. how far away the curve is from the osculating plane in the proximity of $\gamma(s)$. The larger $|B'(s)|$ is, the more the curve deviates from being a plane curve. Differentiating (6.53) and recalling (6.46) for the derivative of the cross product, we find

$$B'(s) = T'(s) \wedge N(s) + T(s) \wedge N'(s) = T(s) \wedge N'(s)\,,$$

since the first summand is zero because $T'(s)$ is parallel to $N(s)$. Hence B', apart from being orthogonal to B, is also orthogonal to T, and hence parallel to the unit normal N. Therefore there is, for any $s \in I$, a real number $\tau(s)$ such that

$$B'(s) = \tau(s)\, N(s)\,, \tag{6.54}$$

called *torsion*.

Note that if the curve lies entirely on a plane, $B(s)$ is always orthogonal to that plane, so the torsion is everywhere zero. Conversely, if $\tau(s)$ is identically null, then $B'(s) = 0$ for any $s \in I$ and hence $B(s)$ is a *constant* unit vector B_0. By (6.53) then,

$$\frac{d}{ds}\,(\gamma(s), B_0) = \big(\gamma'(s), B_0\big) = (T(s), T(s) \wedge N(s)) = 0\,.$$

Then there is a real number d_0 such that

$$(\gamma(s),\, B_0) = d_0\,, \qquad \forall s \in I.$$

In other words, the curve γ lies on the plane orthogonal to B_0, of equation $a_0x + b_0y + c_0z = d_0$, where (a_0, b_0, c_0) are the coordinates of B_0.

If γ is given in terms of an arbitrary parametric representation $\varphi(t)$, arguing as we did to prove (6.50) we can show that

$$\tau(t) = \frac{-\left(\varphi'(t) \wedge \varphi''(t)\,,\, \varphi'''(t)\right)}{|\varphi'(t) \wedge \varphi''(t)|^2}.$$

With this formula we can compute the *torsion of the cylindrical helix* of equations $\varphi(t) = (a\cos t,\ a\sin t,\ b\,t)$, with $a > 0$, $b \in \mathbb{R}$ and $t \in I$. From (6.51) and (6.52)

$$\varphi''(t) = (-a\cos t, -a\sin t, 0)\,,\ \varphi'''(t) = (a\sin t, -a\cos t, 0)\,,$$

$$\varphi'(t) \wedge \varphi''(t) = \left(ab\sin t, -ab\cos t, a^2\right),$$

and so

$$\tau(t) = \frac{-\left(\varphi'(t) \wedge \varphi''(t)\,,\, \varphi'''(t)\right)}{|\varphi'(t) \wedge \varphi''(t)|^2} = \frac{-b}{a^2 + b^2} \qquad \forall\, t \in I.$$

The torsion of the cylindrical helix is therefore constant. As expected, $\tau = 0$ if and only if $b = 0$, i.e. if the curve is a circle. Furthermore, $\tau > 0$ when $b < 0$, and conversely.

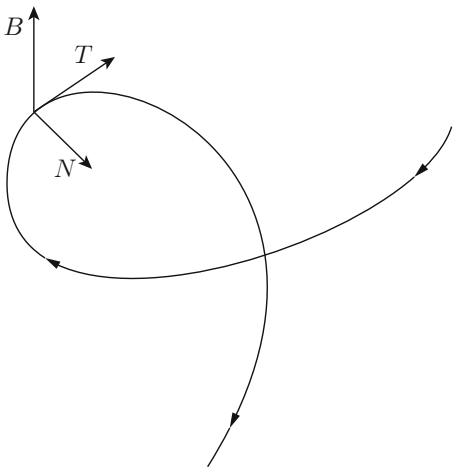
At last, at any point on the curve $\gamma(s)$ we can define a frame system with origin $\gamma(s)$ and axes given by $T(s)$, $N(s)$, $B(s)$ (Fig. 6.16). This is called *Frenet frame*. If the three unit vectors defining the axes of the Frenet frame are known at each point, formulas (6.49) and (6.54) allow to find the curvature and torsion of the curve. For completeness, let us compute the derivative of $N(s)$. As (6.53) and (6.44) imply

$$B(s) \wedge T(s) = (T(s) \wedge N(s)) \wedge T(s) = N(s)\,,$$

by (6.49) and (6.54) we find

$$N'(s) = \frac{d}{dt}B(s) \wedge T(s) = B'(s) \wedge T(s) + B(s) \wedge T'(s) =$$
$$= \tau(s)N \wedge T + k(s)B \wedge N = -\tau(s)B(s) - k(s)T(s).$$

Fig. 6.16



The relationships expressing $T'(s)$, $N'(s)$ and $B'(s)$ in terms of the Frenet frame at $\gamma(s)$ form a *system of differential equations* known as *Frenet formulas*:

$$\begin{cases} T' = k\,N \\ N' = -k\,T\, - \tau\,B \\ B' = \tau\,N\,. \end{cases} \tag{6.55}$$

The coordinate planes of the Frenet frame, apart from the osculating plane (containing T and N), are the *normal plane* (the one containing N and B) and the *rectifying plane* (containing T and B).

We stress that knowing curvature and torsion allows to recover a space curve completely. There is in fact the following result, whose proof we omit for the sake of brevity.

Theorem *Given two C^1 functions $k : I \to \mathbb{R}$, $\tau : I \to \mathbb{R}$ on an interval $I \subseteq \mathbb{R}$, with $k(s) > 0$ for any $s \in I$, there exists a regular curve $\gamma = \gamma(s)$, $s \in I$, of class C^3, for which s is the arclength function, $k(s)$ the curvature and $\tau(s)$ the torsion. Furthermore $\gamma(s)$ is unique up to rigid motions. That is, for any other curve $\overline{\gamma}(s)$ with curvature $k(s)$ and torsion $\tau(s)$, there exist an orthogonal transformation $L : \mathbb{R}^3 \to \mathbb{R}^3$ with positive determinant and a vector $\xi \in \mathbb{R}^3$, such that*

$$\overline{\gamma}(s) = L\,(\gamma(s)) + \xi.$$

The result is in practice a *theorem of existence and uniqueness* for the ODE system (6.55). In case of a plane curve the system reduces to the ODE (recall that in two dimensions the curvature has a sign)

$$T'(s) = -k(s)\,N(s). \tag{6.56}$$

If the continuous function $k(s)$ is given, (6.56) can be easily integrated. Given $s_0 \in I$, set

$$\vartheta(s) = \int_{s_0}^{s} k(r)dr, \qquad \forall\, s \in I. \tag{6.57}$$

The curve $\gamma : I \to \mathbb{R}^2$ of equations

$$\gamma(s) = \left(\int_{s_0}^{s} \cos \vartheta(r)dr \,, \int_{s_0}^{s} \sin\, \vartheta(r)dr \right) \qquad s \in I,$$

is regular, of class C^2. Moreover,

$$\gamma'(s) = (\cos \vartheta(s) \,, \sin \vartheta(s)) = T(s), \tag{6.58}$$

so the parameter s is the arclength. From (6.57) and (6.58) we obtain

$$T'(s) = k(s)\,(-\sin\, \vartheta(s) \,, \cos\, \vartheta(s)) = -k(s)N(s), \qquad s \in I,$$

so, by (6.27), $k(s)$ is precisely the curvature of γ at $\gamma(s)$.



Differential One-Forms

7

7.1 Vector Fields. Work. Conservative Fields

Let $A \subseteq \mathbb{R}^3$ be a connected open set. In Mechanics a *force field*, or *vector field*, defined on A is a map $F : x \in A \to (F_1(x), F_2(x), F_3(x)) \in \mathbb{R}^3$ that associates with any point x in A a vector $F(x)$ representing the *force* acting on a point particle at $x \in A$. If $h \in \mathbb{R}^3$ is a vector of sufficiently small norm $|h|$, the *work* needed to displace a particle along the line segment between x and $x + h$ is approximately the inner product $(F(x), h)$ between the force acting on the particle at x and the displacement h.

In general, if γ is a piecewise-regular curve whose support Γ is contained in A and P_0, P_1 are the curve's endpoints, the *work* W spent to move a particle along the curve from position P_0 to position P_1 is

$$W = \int_\gamma (F(x), T(x))\, ds\,, \tag{7.1}$$

where $T(x)$ is the curve's unit tangent vector, and the curve is oriented from P_0 to P_1. Clearly if the particle moves along the curve in the opposite direction, from P_1 to P_0, the unit tangent vector at any point x will be $-T(x)$, and so by (7.1) the work in this case will equal

$$-\int_\gamma (F(x), T(x))\, ds\,.$$

Consider for example the *vector field* F generated by a point particle of mass M at the origin of the given frame system, and suppose it acts on a particle of mass m. It

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is known that the vector field $F : \mathbb{R}^3 - \{0\} \to \mathbb{R}^3$ is

$$F(x, y, z) = -\frac{GmM}{r^2}\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right) \tag{7.2}$$

where G is the gravitational constant and $r = \sqrt{x^2 + y^2 + z^2}$ the distance of the particle of mass m from the origin.

Let $\gamma : [a, b] \to \mathbb{R}^3$ be a piecewise-regular curve and $r_1 = |\gamma(a)|$, $r_2 = |\gamma(b)|$ the distances of the curve's endpoints to the origin (see Fig. 7.1).

If $\gamma(t) = (x(t), y(t), z(t))$ are the curve's parametric equations, the work W to displace the particle along γ from $\gamma(a)$ to $\gamma(b)$ is, by (7.1),

$$W = \int_a^b \left(F(\gamma(t)), \left(\frac{x'(t)}{|\gamma'(t)|}, \frac{y'(t)}{|\gamma'(t)|}, \frac{z'(t)}{|\gamma'(t)|}\right)\right) |\gamma'(t)|\, dt\,,$$

where the vector field F is defined by (7.2). Computing explicitly the integrand's inner product gives

$$\begin{aligned} W &= -GmM \int_a^b \frac{x(t)x'(t) + y(t)y'(t) + z(t)z'(t)}{\left(x^2(t) + y^2(t) + z^2(t)\right)^{3/2}}\, dt = \\ &= -GmM \int_a^b \frac{d}{dt}\left(-\left(x^2(t) + y^2(t) + z^2(t)\right)^{-1/2}\right) dt = \\ &= GmM\left(\frac{1}{|\gamma(b)|} - \frac{1}{|\gamma(a)|}\right) = GmM\left(\frac{1}{r_2} - \frac{1}{r_1}\right). \end{aligned}$$

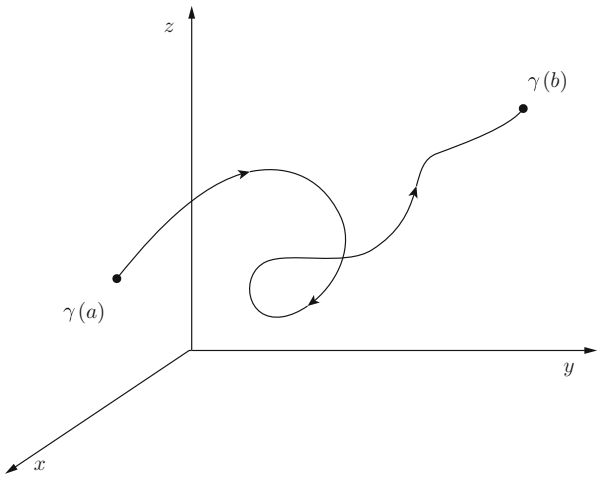


Fig. 7.1

For the vector field (7.2), therefore, the work W needed to displace the particle along γ from $\gamma(a)$ to $\gamma(b)$ does not depend on the particular curve γ along which the particle moves, but only on the initial and final positions.

This property does not hold for any vector field.

Consider for example the vector field

$$F(x, y, z) = (xy, y, 0)$$

and the circles γ_1, γ_2, in Fig. 7.2, of equations

$$\gamma_1(t) = (\cos t, \sin t, 0), \qquad t \in [0, 2\pi],$$
$$\gamma_2(t) = (2 + \cos t, \sin t, 0), \qquad t \in [-\pi, \pi].$$

The work needed for a particle to travel around γ_1 once starting from $P_0 = (1, 0, 0)$ is

$$W_1 = \int_0^{2\pi} ((\cos t \sin t, \sin t, 0), (-\sin t, \cos t, 0))\ dt = 0,$$

while the work to go around γ_2 once, from the same initial point P_0 is

$$W_2 = \int_{-\pi}^{\pi} ((2 \sin t + \cos t \sin t, \sin t, 0), (-\sin t, \cos t, 0))\ dt = -2\pi .$$

Hence $W_1 \neq W_2$.

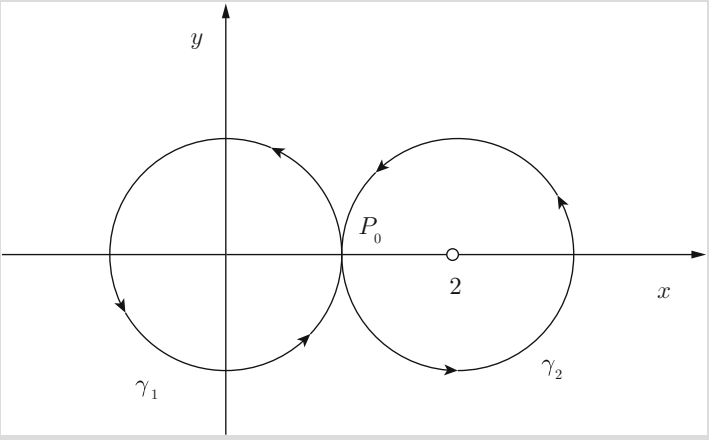


Fig. 7.2

One says a vector field $F : A \to \mathbb{R}^3$ is *conservative* if the work required to displace a particle from a point $P_0 \in A$ to a point $P_1 \in A$ does not depend on the curve γ of motion, but only depends on the curve's endpoints P_0, P_1 and on the orientation.

7.2 Differential 1-Forms. Line Integrals

We know from Linear Algebra (also see Sect. 2.5, Chap. 2) that a generic element L in the dual space $(\mathbb{R}^n)^*$ of $\mathbb{R}^n$, i.e. a generic linear functional on $\mathbb{R}^n$, can always be represented in the form

$$L = \sum_{i=1}^{n} a_i e^i = \sum_{i=1}^{n} a_i dx_i \,, \tag{7.3}$$

where $a_1, a_2, \ldots, a_n$ are real numbers and $e^i = dx_i$, $i = 1, 2, \ldots, n$, are the functionals mapping $h \in \mathbb{R}^n$ to its ith component h_i , i.e. $e^i(h) = dx_i(h) = h_i$, for any $h \in \mathbb{R}^n$. Hence the functional $L : \mathbb{R}^n \to \mathbb{R}$, for any $h \in \mathbb{R}^n$, equals

$$L(h) = \sum_{i=1}^{n} a_i \, e^i(h) = \sum_{i=1}^{n} a_i \, dx_i(h) = \sum_{i=1}^{n} a_i \, h_i \,.$$

Recall the functionals $\{e^1, e^2, \ldots, e^n\} = \{dx_1, dx_2, \ldots, dx_n\}$ form a basis of $(\mathbb{R}^n)^*$, so (7.3) is a unique representation.

Consider now an open subset A in $\mathbb{R}^n$. We call *differential* 1*-form* (or *differential one-form*, often shortened to 1*-form*) a map $\omega : A \to (\mathbb{R}^n)^*$ sending an element $x \in A$ to the linear functional on $\mathbb{R}^n$

$$\omega(x) = \sum_{i=1}^{n} a_i(x) \; dx_i \,.$$

If h is an arbitrary vector in $\mathbb{R}^n$, for any x in A the value that the functional $\omega(x)$ assumes at h is

$$\omega(x)(h) = \sum_{i=1}^{n} a_i(x) \; dx_i(h) = \sum_{i=1}^{n} a_i(x) h_i \,.$$

The functions $a_1(x), a_2(x), \ldots, a_n(x)$ are the *coefficients* of the differential form ω. A differential form is of class C^k, $k \geq 0$, when its coefficients are C^k functions.

A generic 1-form $\omega : A \to (\mathbb{R}^2)^*$, where A is an open set in the plane, is therefore representable as

$$\omega(x, y) = a(x, y)\, dx + b(x, y)\, dy \,.$$

Similarly, if A is an open set in $\mathbb{R}^3$, one typically writes

$$\omega(x, y, z) = a(x, y, z)\, dx + b(x, y, z)\, dy + c(x, y, z)\, dz$$

for a generic 1-form defined on $A \subseteq \mathbb{R}^3$. We may think of associating with any such 1-form the vector field $F : A \subseteq \mathbb{R}^3 \to \mathbb{R}^3$

$$F(x, y, z) = (a(x, y, z), b(x, y, z), c(x, y, z)) \,,$$

whose components are the coefficients of ω, in the same order.

Observe that if $f : A \to \mathbb{R}$, with A open in $\mathbb{R}^n$, is a *differentiable function* on A (see Chap. 3), the differential

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x)\, dx_i = \sum_{i=1}^{n} f_{x_i}(x)\, dx_i$$

is a 1-form whose coefficients are the partial derivatives $f_{x_1}(x),\ f_{x_2}(x), \ldots,\ f_{x_n}(x)$ of f. We must stress that *not every differential form arises as the differential of some function* f.

For instance, the differential form

$$\omega(x, y) = 3x^2\, dx - xy\, dy \tag{7.4}$$

does not coincide with the differential of any function $f : \mathbb{R}^2 \to \mathbb{R}^2$. If, in fact, there existed a differentiable function f such that $df = \omega = 3x^2\, dx - xy\, dy$, then necessarily

$$\frac{\partial f}{\partial x} = 3x^2, \qquad \frac{\partial f}{\partial y} = -xy\,. \tag{7.5}$$

But integrating the first relation in $x \in \mathbb{R}$ gives $f(x, y) = x^3 + \varphi(y)$, for some differentiable function φ (as f is differentiable by assumption). Then the partial derivative of f in y would be $f_y(x, y) = \varphi'(y)$, i.e. constant in x, which clashes with the fact that we should have $f_y(x, y) = -xy$, from (7.5).

Let A be an open set in $\mathbb{R}^n$. Fix a continuous 1-form $\omega : A \to (\mathbb{R}^n)^*$ and a piecewise-regular curve $\gamma : [a, b] \to A \subset \mathbb{R}^n$ of equations $x_i = x_i(t)$, $i = 1, 2, \ldots, n$. We call *line integral of the differential form along* γ, written

$$\int_\gamma \omega\,,$$

the integral

$$\int_\gamma \omega(x)\,(T(x))\ ds = \sum_{i=1}^{n} \int_a^b a_i(x_1(t), x_2(t), \ldots, x_n(t)) x_i'(t)\, dt\,, \tag{7.6}$$

where $T(x)$ is the unit tangent vector to γ at x.

If $-\gamma$ denotes the curve equivalent to γ but with the opposite orientation, the unit tangent vector at any point on $-\gamma$ is opposite the unit tangent vector to γ at the same point. Hence from (7.6)

$$\int_{-\gamma} \omega = -\int_\gamma \omega\,. \tag{7.7}$$

Suppose now that we break γ into piecewise-regular curves $\gamma_r : [t_{r-1}, t_r] \to A$, with $a = t_0 < t_1 < \ldots < t_N = b$, $\gamma_r(t) = \gamma(t)$ in $[t_{r-1}, t_r]$, for any $r = 1, 2, \ldots, N$. Applying (6.23) we obtain

$$\int_\gamma \omega = \int_{\gamma_1} \omega + \int_{\gamma_2} \omega + \ldots + \int_{\gamma_N} \omega \,. \tag{7.8}$$

Analogously, from (6.21) we deduce that if ω and ζ are continuous 1-forms on an open set $A \subseteq \mathbb{R}^n$ and α, β are real numbers,

$$\int_\gamma (\alpha\omega + \beta\zeta) = \alpha \int_\gamma \omega + \beta \int_\gamma \zeta \,, \tag{7.9}$$

where $(\alpha\omega + \beta\zeta)(x) = \alpha\omega(x) + \beta\zeta(x)$ for any x in A. At last, (6.22) also implies

$$\left| \int_\gamma \omega \right| \leq L(\gamma) \max_\gamma \sqrt{a_1^2(x) + a_2^2(x) + \ldots + a_n^2(x)} \,,$$

where $L(\gamma)$ is the length of γ.

Consider the three-dimensional case and let

$$\omega(x, y, z) = a(x, y, z)\, dx + b(x, y, z)\, dy + c(x, y, z)\, dz$$

be a continuous differential form defined on an open set $A \subseteq \mathbb{R}^3$. Take a piecewise-regular curve $\gamma(t) = (x(t), y(t), z(t))$, $t \in [a, b]$, with support in A. From (7.6), $n = 3$, we have

$$\begin{aligned} \int_\gamma \omega &= \int_\gamma \omega(x)(T(x))\, ds = \\ &= \int_a^b \left\{ a(x(t), y(t), z(t))x'(t) + b(x(t), y(t), z(t))y'(t) + c(x(t), y(t), z(t))z'(t) \right\} dt \,. \end{aligned} \tag{7.10}$$

This line integral coincides with the *work*

$$W = \int_\gamma (F,\ T)\ ds$$

of the *vector field*

$$F(x, y, z) = (a(x, y, z), b(x, y, z), c(x, y, z))$$

required to displace a particle from $\gamma(a)$ to $\gamma(b)$ along the curve.

For instance, for $n = 2$, let us compute

$$\int_\gamma y\,dx - xy\,dy\,,$$

where $\omega = y\,dx - xy\,dy$ and γ is the unit semi-circle centred at the origin contained in the upper half-plane and oriented clockwise.

Let ψ be the curve of equations $x = \cos t$, $y = \sin t$, $t \in [0, \pi]$. It is equivalent to γ but oppositely oriented. By (7.7), (7.9) we obtain

$$\int_\gamma y\,dx - xy\,dy = -\int_\psi y\,dx - xy\,dy = \int_\psi -y\,dx + xy\,dy\,.$$

The last line integral, from (7.10) when $z = 0$, equals

$$\int_0^\pi \left\{-y(t)x'(t) + x(t)y(t)y'(t)\right\} dt =$$
$$= \int_0^\pi \left\{\sin^2 t + \sin t \cos^2 t\right\} dt = \frac{3\pi + 4}{6}\,.$$

7.3 Exact 1-Forms

A 1-form ω on an open subset A in $\mathbb{R}^n$ is said to be *exact on* A when there is a differentiable function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$, called a *primitive* of ω, such that

$$\omega = df\,. \tag{7.11}$$

Equivalently, a differentiable function f on an open set $A \subseteq \mathbb{R}^n$ is a *primitive* of a differential form ω of coefficients $a_1(x), a_2(x), \ldots, a_n(x)$, i.e.

$$\omega(x) = \sum_{i=1}^n a_i(x)\,dx_i\,, \qquad \forall\, x \in A,$$

whenever

$$\frac{\partial f}{\partial x_i}(x) = a_i(x)\,, \qquad \forall i = 1, 2, \ldots, n.$$

This amounts to

$$\omega = df = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x)\,dx_i\,, \qquad \forall\, x \in A.$$

If f is a primitive of a differential form ω, i.e. f verifies (7.11), then also $f + c$, where c is any real constant, is a primitive of ω. Conversely, supposing A is a *connected open set*, if two primitives f, g of ω satisfy

$$df = dg\,,$$

then $d(f-g) = 0$, and so the gradient $D(f-g)$ is null on A (since $f-g$ has zero partial derivatives). Therefore (see Sect. 3.8)

$$f(x) - g(x) = \text{constant}, \qquad \forall\, x \in A\,.$$

Eventually, *if A is a connected open set in $\mathbb{R}^n$ and f is a primitive on A of the differential form ω, then any primitive of ω arises by adding to f a constant.*

Consider $n = 1$. Let A be a real open interval and $a : A \subseteq \mathbb{R} \to \mathbb{R}$ a given function. Let us look at the differential form $\omega = a\,dx$ defined by

$$\omega(x)(h) = a(x)h$$

for any $x \in A$ and $h \in \mathbb{R}$. It is evident that finding a primitive for ω is the same as finding a primitive of the function $a(x)$. *The notion of primitive of a 1-form thus generalises to $\mathbb{R}^n$ the notion of primitive of a function of one real variable.*

As we saw for the form (7.4), it is not always the case that a 1-form is exact. So it would be useful to have necessary and sufficient conditions for that to happen. Let us start proving an interesting property of exact differential forms.

Theorem 4 *Let ω be an exact differential form that is continuous on the open set $A \subset \mathbb{R}^n$, and let x_0, x be points in A. If γ is a piecewise-regular curve in A, joining the two points and oriented from x_0 to x, then*

$$\int_\gamma \omega = f(x) - f(x_0)\,, \tag{7.12}$$

where f is an arbitrary primitive of ω.

Proof Suppose $f : A \to \mathbb{R}$ is a primitive of ω and $x_i = x_i(t)$, $t \in [a,b]$, $i = 1, 2, \ldots, n$, are the parametric equations of γ. By (7.6) we have

$$\begin{aligned}
\int_\gamma \omega = \int_\gamma df = \int_a^b \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x_1(t), x_2(t), \ldots, x_n(t)) x_i'(t)\, dt = \\
= \int_a^b \frac{d}{dt}\big[f(x_1(t), x_2(t), \ldots, x_n(t))\big]\, dt = \\
= f(x_1(b), x_2(b), \ldots, x_n(b)) - f(x_1(a), x_2(a), \ldots, x_n(a)) = \\
= f(x) - f(x_0)\,.
\end{aligned}$$

This theorem shows that if ω is exact, its line integral along any curve γ does not depend upon the curve, but only on the endpoints and obviously on its orientation. We want to prove that this property characterises exact forms.

Theorem 5 (Characterisation of Exact Forms) *Let $A \subseteq \mathbb{R}^n$ be a connected open set, ω a continuous 1-form on A and γ, γ_1, γ_2 piecewise-regular curves contained in A. The following properties are equivalent:*

(i) *ω is exact on A;*
(ii) *$\int_\gamma \omega = 0$ for any closed curve γ contained in A;*
(iii) *if γ_1, γ_2 have the same endpoints and same orientation, $\int_{\gamma_1} \omega = \int_{\gamma_2} \omega$.*

Proof

(i) $\Rightarrow$ (ii): This is immediate from (7.12), since if γ is closed, the integration endpoints x_0 and x_1 coincide.

(ii) $\Rightarrow$ (iii): If γ_1, γ_2 have the same endpoints x_0, x_1 and the same orientation, say from x_0 to x_1, then the curve $\gamma_1 - \gamma_2$, obtained by moving along γ_1 from x_0 to x_1 and then along γ_2 backwards from x_1 to x_0, is piecewise regular and closed. By assumption, and using (7.7), (7.8), we obtain

$$0 = \int_{\gamma_1 - \gamma_2} \omega = \int_{\gamma_1} \omega + \int_{-\gamma_2} \omega = \int_{\gamma_1} \omega - \int_{\gamma_2} \omega \,,$$

and so $\int_{\gamma_1} \omega = \int_{\gamma_2} \omega$.

(iii) $\Rightarrow$ (i): fix x_0 in A and define

$$f(x) = \int_\gamma \omega \,, \tag{7.13}$$

for any x in A, where γ is an arbitrary piecewise-regular curve contained in A joining x_0 and x and oriented from x_0 to x. As A is connected, and therefore path connected (see Sect. 2.11), there is always one such γ. The function f is then well defined, since the line integral in (7.13) does not depend on γ by assumption (the endpoints and orientation being given). We claim f is differentiable on A and

$$df = \omega \,,$$

i.e. f is a *primitive* of the given differential form ω.

Let $x \in A$, and take $h \in \mathbb{R}$, $h \neq 0$, such that $x + he_1 \in A$. Consider

$$f(x + he_1) - f(x) = \int_{\gamma_1} \omega - \int_\gamma \omega \,,$$

where γ is a curve from x_0 to x and γ_1 is the curve obtained attaching to γ the line segment φ from x to $x + he_1$ (see Fig. 7.3).

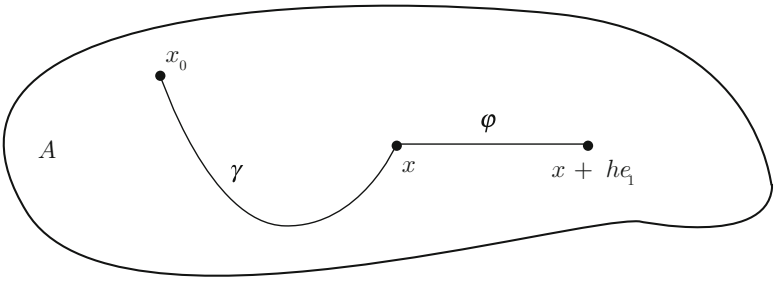


Fig. 7.3

By (7.7), (7.8) we have

$$f(x + he_1) - f(x) = \int_{\varphi} \omega \,. \tag{7.14}$$

In the following we shall consider $h > 0$. If $h < 0$ it suffices to replace $[0, h]$ by $[h, 0]$. Using the parametrisation

$$x_1(t) = x_1 + t, \quad x_2(t) = x_2, \ldots, \quad x_n(t) = x_n, \qquad t \in [0, h],$$

of the segment φ from x to $x + he_1$, and dividing (7.14) by h, we find

$$\frac{f(x + he_1) - f(x)}{h} = \frac{1}{h} \int_0^h a_1(x_1 + t,\, x_2, \ldots, x_n)\, dt \,.$$

Now take the limit as h goes to zero. By the fundamental theorem of calculus (which holds since a_1 is continuous) we have

$$\frac{\partial f}{\partial x_1}(x) = a_1(x) \,.$$

Similarly, one can prove the other partial derivatives exist too, and $\partial f / \partial x_i = a_i$ for any $i = 2, \ldots, n$. The function f is then C^1, hence differentiable, and is a primitive of ω.

It follows from this result that if ω is exact and of class C^k, $k = 0, 1, 2, \ldots$, on the connected open set $A \subseteq \mathbb{R}^n$, its primitive (7.13) (and so any other primitive) is of class $C^{k+1}(A)$.

Consider for example the exact form

$$\omega = d\left(x^2 y + \frac{y^3}{3}\right) = 2xy\,dx + (x^2 + y^2)\,dy\,,$$

defined on the whole plane. We want to show (7.13) does indeed give us a primitive of ω. Set

$$f(x, y) = \int_\gamma \omega\,,$$

where γ is the line segment from $(0, 0)$ to (x, y). As γ may be parametrised as (tx, ty), $t \in [0, 1]$, we find that

$$f(x, y) = \int_0^1 \left\{2t^2x^2y + t^2(x^2 + y^2)y\right\}\,dt =$$

$$= \left[\frac{2}{3}t^3x^2y + \frac{1}{3}t^3(x^2 + y^2)y\right]_{t=0}^{t=1} = x^2y + \frac{y^3}{3}$$

is a primitive of ω.

Let $F(x) = (F_1(x), F_2(x), F_3(x))$ be a vector field defined on the connected open set $A \subseteq \mathbb{R}^3$. We remarked that the work needed to displace a point particle along a curve γ under the action of F is the integral along γ of the differential form $\omega = F_1\,dx + F_2\,dy + F_3\,dz$. The above theorem then implies that *the vector field F is conservative if and only if ω is exact*. In this situation a primitive f of ω is called a *potential* of the vector field F.

7.4 Exact 1-Forms on the Plane. Simply Connected Open Sets in $\mathbb{R}^2$

Let $\omega = a(x, y)\,dx + b(x, y)\,dy$ be an *exact* differential form of class C^1 defined on an open subset A of the plane, let and $f : A \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a primitive function. By definition of primitive,

$$\frac{\partial f}{\partial x} = a(x, y)\,, \quad \frac{\partial f}{\partial y} = b(x, y)\,, \qquad \forall\,(x, y) \in A. \tag{7.15}$$

As the coefficients a, b of ω are of class $C^1(A)$, therefore, differentiating the first relation in (7.15) with respect to y and the second one with respect to x, the Schwarz theorem forces

$$\frac{\partial a}{\partial y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial b}{\partial x}\,, \qquad \forall\,(x, y) \in A.$$

Hence if the differential form $a\,dx + b\,dy$ is exact and its coefficients are C^1, the following happens:

$$\frac{\partial a}{\partial y} = \frac{\partial b}{\partial x}\,, \qquad \forall\,(x, y) \in A. \tag{7.16}$$

A differential form $\omega = a\,dx + b\,dy$ of class C^1 on some open set $A \subset \mathbb{R}^2$ is said to be *closed on* A if its coefficients a, b satisfy (7.16). If so, the previous discussion proves that *an exact form is closed*.

It is rather natural to ask whether (7.16) is sufficient to ensure the form is exact. Consider for example

$$\omega = \frac{-y}{x^2 + y^2}\,dx + \frac{x}{x^2 + y^2}\,dy, \tag{7.17}$$

defined on the open set $\mathbb{R}^2 - \{(0, 0)\}$, and whose coefficients meet (7.16) (the reader should check this, and also the upcoming formula (7.18)). Hence the form ω *is closed* on $\mathbb{R}^2 - \{(0, 0)\}$. If γ is the counter-clockwise unit circle centred at the origin, we obtain

$$\int_\gamma \omega = 2\pi\,. \tag{7.18}$$

Then ω *is not exact* on $\mathbb{R}^2 - \{(0, 0)\}$, because its line integral along the closed curve γ is different from zero.

On the other hand, if we take the open subset

$$A = \left\{(x, y) \in \mathbb{R}^2 : x > 0\right\}$$

and restrict ω to A (abusing the notation, we shall still call this by ω), the function

$$f(x, y) = \operatorname{arctg} \frac{y}{x}$$

is clearly a primitive of ω on A (the reader should compute the partial derivatives). Therefore ω *is exact* on A.

In order to conclude that a given form is exact, it is not enough to have (7.16). With other words, it is not enough for the form to be closed on the open set, but we need to take into account the *geometry* of the open set where the differential form is defined.

Let us start by examining a simple situation. Call R an open *rectangle* in the plane, i.e. $R = (a, b) \times (c, d)$, where the intervals (a, b), (c, d) might be unbounded. In particular R may be the whole plane $\mathbb{R}^2$.

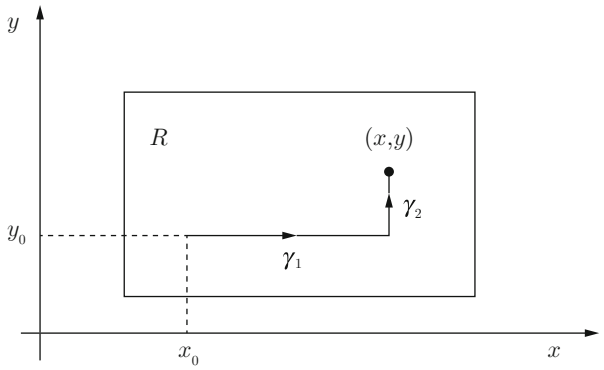


Fig. 7.4

Theorem 1 (Differential Forms on a Rectangle in $\mathbb{R}^2$) *If a 1-form $\omega = a\,dx + b\,dy$ of class C^1 on an open rectangle $R \subseteq \mathbb{R}^2$ is closed (it satisfies (7.16)), then it is exact on R.*

Proof Let us construct a primitive f of the given form. Fix (x_0, y_0) in R and define

$$f(x, y) = \int_{\gamma} \omega$$

for any $(x, y) \in R$, where γ is the polygonal path formed by the segment γ_1 from (x_0, y_0) to (x, y_0) and the segment γ_2 from (x, y_0) to (x, y), as in Fig. 7.4. Then

$$f(x, y) = \int_{\gamma_1} \omega + \int_{\gamma_2} \omega = \int_{x_0}^{x} a(t, y_0)\,dt + \int_{y_0}^{y} b(x, t)\,dt\,. \tag{7.19}$$

We can differentiate inside the integral (see Sect. 3.10) and obtain

$$\frac{\partial f}{\partial x} = a(x, y_0) + \int_{y_0}^{y} \frac{\partial b}{\partial x}(x, t)\,dt\,; \tag{7.20}$$

$$\frac{\partial f}{\partial y} = b(x, y)\,. \tag{7.21}$$

As the form is closed ((7.16) holds), from (7.20) we find

$$\frac{\partial f}{\partial x} = a(x, y_0) + \int_{y_0}^{y} \frac{\partial a}{\partial y}(x, t)dt = a(x, y_0) + \big[a(x, t)\big]_{t=y_0}^{t=y} = a(x, y)\,.$$

This, together with (7.21), implies

$$df = \omega\,.$$

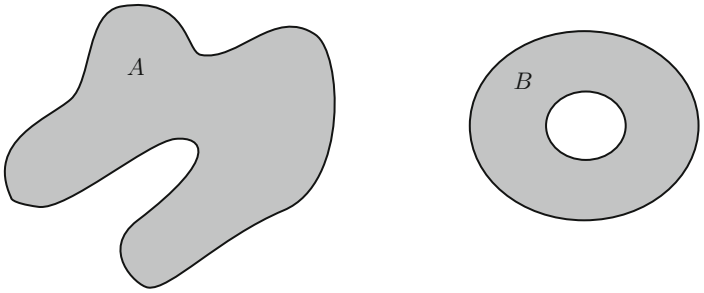


Fig. 7.5 A is simply connected, whereas B is not

Under the hypotheses of Theorem 1, formula (7.19) is a useful tool to compute primitives explicitly.

In view of the next crucial definition we recall (also see Sect. 3.1) that a *domain* $D \subseteq \mathbb{R}^n$ is the closure of an open set. Hence a domain is a closed set, and the union of an open set and its boundary.

A connected open set A in the plane is said to be *simply connected* when every piecewise-regular, simple closed curve γ in A is the boundary of a bounded domain D entirely contained in A (see Fig. 7.5).

The open set $A = \mathbb{R}^2 - \{(0,0)\}$, on which we considered the form ω in (7.17), is not simply connected. An open rectangle $R \subseteq \mathbb{R}^2$, as of Theorem 1, is a simply connected set.

Theorem 2 (Differential Forms on a Simply Connected Open Set in $\mathbb{R}^2$) *A 1-form $\omega = a\,dx + b\,dy$ defined on a simply connected open set A, of class C^1 and closed (i.e. (7.16)), is exact on A.*

For the proof see Sect. 8.3, Chap. 8.

In the previous example we looked at the differential form ω defined by (7.17) on the open set $A = \mathbb{R}^2 - \{(0,0)\}$. The example shows how the above theorem does not hold if we drop the simply connectedness of A. Nevertheless, in certain special cases this condition turns out to be superfluous. The next result describes one such case.

Let us recall (see also Sect. 3.9) a set $A \subseteq \mathbb{R}^2$ is a *cone* if

$$x \in A \quad \Rightarrow \quad tx \in A\,, \qquad \forall\, t > 0\,.$$

So let A be a cone in $\mathbb{R}^2$ and α a real number. A function $f : A \to \mathbb{R}$ is *homogeneous of degree* α on A if

$$f(t\,x) = t^\alpha f(x), \qquad \forall\, x \in A, \quad \forall\, t > 0.$$

Theorem 3 (Differential Forms in $\mathbb{R}^2$ with Homogeneous Coefficients) *Let A be an open cone in the plane and $\omega = a\,dx + b\,dy$ a 1-form whose coefficients are homogeneous of degree $\alpha \neq -1$ and C^1. If the form is closed then it is exact on A. Moreover,*

$$f(x, y) = \frac{xa(x, y) + yb(x, y)}{\alpha + 1} \tag{7.22}$$

is a primitive of ω.

Proof With f given by (7.22) we have

$$\frac{\partial f}{\partial x} = \frac{1}{\alpha + 1}\left[a(x, y) + x\frac{\partial a}{\partial x}(x, y) + y\frac{\partial b}{\partial x}(x, y)\right].$$

As the form is closed, i.e. $\partial b/\partial x = \partial a/\partial y$,

$$\frac{\partial f}{\partial x} = \frac{1}{\alpha + 1}\left[a(x, y) + x\frac{\partial a}{\partial x}(x, y) + y\frac{\partial a}{\partial y}(x, y)\right]. \tag{7.23}$$

By Euler's theorem on homogeneous functions (Sect. 3.9)

$$x\frac{\partial a}{\partial x}(x, y) + y\frac{\partial a}{\partial y}(x, y) = \alpha a(x, y)\,, \qquad \forall\,(x, y) \in A, \tag{7.24}$$

and then (7.23), (7.24) imply

$$\frac{\partial f}{\partial x} = a(x, y)\,, \qquad \forall\,(x, y) \in A.$$

One can prove similarly that $\partial f/\partial y = b(x, y)$ for any $(x, y) \in A$, resulting in $df = \omega$.

The above theorem does not hold for 1-forms whose coefficients are (C^1 and) homogeneous of degree $\alpha = -1$. For example, the form ω in (7.17) has homogeneous coefficients of degree $\alpha = -1$, it is closed on the open cone $A = \mathbb{R}^2 - \{(0, 0)\}$, but it is not exact on A.

On the contrary, on the open cone $A = \mathbb{R}^2 - \{(0, 0)\}$ the differential form

$$\omega = \frac{x}{\left(x^2 + y^2\right)^2}\,dx + \frac{y}{\left(x^2 + y^2\right)^2}\,dy$$

is closed (the reader should check (7.16)) and the coefficients are homogeneous of degree -3. By (7.22) a primitive of this ω is

$$f(x, y) = \frac{-1}{2\left(x^2 + y^2\right)}\,.$$

7.5 One-Forms in Space. Irrotational Vector Fields

Let $\omega = a\,dx + b\,dy + c\,dz$ be a C^1 differential form on the open set A in $\mathbb{R}^3$ and f a primitive. We proceed as in the previous section: by definition of primitive

$$\frac{\partial f}{\partial x} = a\,, \quad \frac{\partial f}{\partial y} = b\,, \quad \frac{\partial f}{\partial z} = c\,, \qquad \forall\,(x, y, z) \in A.$$

Schwarz's theorem gives

$$\frac{\partial a}{\partial y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial b}{\partial x}\,, \qquad \forall\,(x, y, z) \in A$$

and similarly for the pairs x, z and y, z. In this way we obtain

$$\frac{\partial c}{\partial y} = \frac{\partial b}{\partial z}\,, \quad \frac{\partial a}{\partial z} = \frac{\partial c}{\partial x}\,, \quad \frac{\partial b}{\partial x} = \frac{\partial a}{\partial y}\,, \tag{7.25}$$

generalising the condition $\partial b/\partial x = \partial a/\partial y$ for the plane.

A differential form $\omega = a\,dx + b\,dy + c\,dz$, of class C^1 on an open set A in $\mathbb{R}^3$, is said to be *closed* if its coefficients a, b, c verify (7.25). We have then shown that under the above assumptions *any exact form is closed*.

The form $2xyz\,dx + x^2z\,dy + x^2yz\,dz$ is not exact on $\mathbb{R}^3$ since it is not closed. Letting $a = 2xyz$, $c = x^2yz$, we have

$$\frac{\partial a}{\partial z} \neq \frac{\partial c}{\partial x}\,.$$

Let now $F = (F_1, F_2, F_3)$ be a vector field. It is *conservative* if and only if $F_1\,dx + F_2\,dy + F_3\,dz$ is an exact form. Now Eqs. (7.25) read

$$\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} = 0\,, \quad \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} = 0\,, \quad \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0\,. \tag{7.26}$$

If F is a vector field, the vector field whose components are the left-hand sides in (7.26) is called the *curl* of F, indicated by curl F:

$$\operatorname{curl} F = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\,, \quad \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\,, \quad \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right).$$

The curl of F may be thought of as the vector field obtained expanding along the first column the *formal determinant*

$$\begin{vmatrix} e_1 & \dfrac{\partial}{\partial x} & F_1 \\ e_2 & \dfrac{\partial}{\partial y} & F_2 \\ e_3 & \dfrac{\partial}{\partial z} & F_3 \end{vmatrix},$$

where e_1, e_2, e_3 is the canonical basis of $\mathbb{R}^3$. Then (7.26) becomes

$$\operatorname{curl} F = 0\,.$$

In Physics a vector field whose curl vanishes is said to be *curl-free* or *irrotational*.

There is a suitable notion of *simple connectedness* also for three-dimensional open sets, with which one proves a result similar to the plane one; we will do this in the chapter's Appendix. Here we shall limit ourselves to considering special situations, that generalise the previous sections' results for the plane.

Theorems 1 and 3 of the previous section immediately extend to forms in $\mathbb{R}^3$, with the same proof. We remind that a rectangle in $\mathbb{R}^3$ is the Cartesian product of three real intervals.

Theorem 1 (Differential Forms on a Rectangle in $\mathbb{R}^3$) *A C^1 form $\omega = a\,dx + b\,dy + c\,dz$ that is closed (i.e. fulfilling (7.25)) on an open rectangle R, is exact on R.*

Theorem 2 (Differential Forms on $\mathbb{R}^3$ with Homogeneous Coefficients) *Let A be an open cone in space, and $\omega = a\,dx + b\,dy + c\,dz$ a 1-form with coefficients of class C^1 and homogeneous of degree $\alpha \neq -1$. If the form is closed then it is exact on A. Moreover,*

$$f(x, y, z) = \frac{x\,a(x, y, z) + y\,b(x, y, z) + z\,c(x, y, z)}{\alpha + 1}$$

is a primitive of ω.

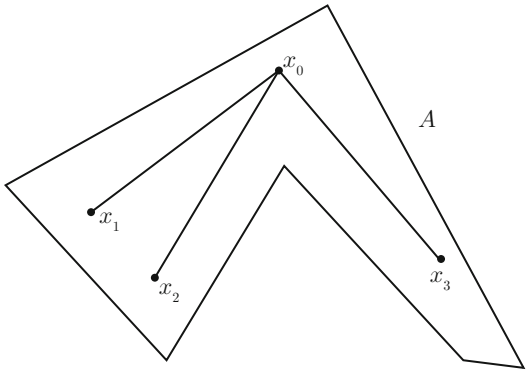
Let us now define a further class of open sets in $\mathbb{R}^3$, that allows to use differential forms without involving the concept of simple connectedness (as we said, the latter will be examined in the chapter's Appendix, i.e. in the next section).

An open subset $A \subseteq \mathbb{R}^3$ is *star-shaped* if there is a point $x_0 \in A$ such that, for any $x \in A$, the line segment between x_0 and x is contained in A.

An open convex set is patently star-shaped, because we may choose as x_0 any point of A. As Fig. 7.6 shows, though, a star-shaped open set might not be convex.

The next result extends Theorem 1 on forms over rectangles in $\mathbb{R}^3$, which are clearly star-shaped.

Fig. 7.6



Theorem 3 (Differential Forms on Star-Shaped Open Sets in $\mathbb{R}^3$) *If A is star-shaped and open, and ω a closed differential form on A, then ω is exact on A.*

Proof Assume for simplicity that the origin belongs to A, and that for any given point $(x, y, z) \in A$ the segment γ of equations

$$x(t) = tx, \quad y(t) = ty, \quad z(t) = tz, \qquad t \in [0, 1],$$

is contained in A (this is always possible up to translating suitably). Put

$$f(x, y, z) = \int_\gamma \omega = \int_\gamma a\,dx + b\,dy + c\,dz \tag{7.27}$$

and let us show this function is a primitive. Computing the line integral we have

$$f(x, y, z) = \int_0^1 \left\{a(tx, ty, tz)x + b(tx, ty, tz)y + c(tx, ty, tz)z\right\} dt.$$

Now differentiate inside the integral (see Sect. 3.10),

$$\frac{\partial f}{\partial x}(x, y, z) = \int_0^1 \left\{a(tx, ty, tz) + \frac{\partial a}{\partial x}(tx, ty, tz)tx + \right.$$
$$\left. + \frac{\partial b}{\partial x}(tx, ty, tz)ty + \frac{\partial c}{\partial x}(tx, ty, tz)tz\right\} dt\,.$$

At this point we use the fact ω is closed on A. By (7.25)

$$\frac{\partial f}{\partial x}(x,y,z) = \int_0^1 \Big\{ a(tx,ty,tz) + \frac{\partial a}{\partial x}(tx,ty,tz)tx + \\ + \frac{\partial a}{\partial y}(tx,ty,tz)ty + \frac{\partial a}{\partial z}(tx,ty,tz)tz \Big\} dt = \\ = \int_0^1 \frac{d}{dt}\big(ta(tx,ty,tz)\big)dt = \big[ta(tx,ty,tz)\big]_{t=0}^{t=1} = a(x,y,z)\,.$$

Similarly one can prove $\dfrac{\partial f}{\partial y} = b$ and $\dfrac{\partial f}{\partial z} = c$, whence the claim follows.

Note that (7.27) gives us an explicit formula for the primitive f for a star-shaped open set. It is also evident that the definition of star-shaped set, and the above theorem as well, extend in the obvious manner to open sets in $\mathbb{R}^n$.

Appendix to Chap. 7

7.6 Simply Connected Open Sets in $\mathbb{R}^n$ and Exact 1-Forms

Let $\omega(x) = \sum_{i=1}^{n} a_i(x)\,dx_i$ be an exact differential form of class $C^1(A)$ on some open set A in $\mathbb{R}^n$. Applying to the primitive of ω the Schwarz theorem, as we did earlier for $n = 2, 3$, it is easy to see that

$$\frac{\partial a_i}{\partial x_j}(x) = \frac{\partial a_j}{\partial x_i}(x)\,, \qquad \forall\, i,\, j = 1, 2, \ldots, n,$$

for any $x \in A$. This property of ω, which generalises (7.25) to arbitrary dimensions, is phrased by saying the 1-form ω is *closed*.

Now we introduce the concept of *homotopy* of curves of class C^2, with the purpose of extending to $\mathbb{R}^n$ the notion of a simply connected open set given in two dimensions.

Let A be an open set in $\mathbb{R}^n$ and $\varphi_0, \varphi_1 : [a,b] \to A$ two closed curves of class C^2. We say they are *homotopic in* A if there exists a C^2 map (the *homotopy*) $\Phi : [0,1] \times [a,b] \to A$ such that

$$\Phi(0,t) = \varphi_0(t)\,, \qquad \forall\, t \in [a,b],$$
$$\Phi(1,t) = \varphi_1(t)\,, \qquad \forall\, t \in [a,b],$$
$$\Phi(s,a) = \Phi(s,b)\,, \qquad \forall\, s \in [0,1].$$

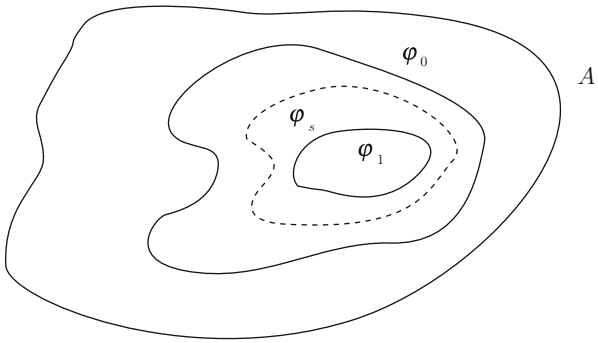


Fig. 7.7

Let us emphasise that the map $\varphi_s(t) = \Phi(s,t)$ is for any $s \in (0,1)$ a closed C^2 curve contained in A (see Fig. 7.7).

If A is convex, the closed curves $\varphi_0, \varphi_1 : [a,b] \to A$ are always homotopic in A, for it suffices to take as homotopy the convex combination

$$\Phi(s,t) = s\varphi_1(t) + (1-s)\varphi_0(t)\,, \qquad \forall\, s \in [0,1],\ t \in [a,b].$$

Furthermore, even if φ_0, φ_1 are regular, the above definition does not require that $\varphi_s(t) = \Phi(s,t)$ are regular curves for $0 < s < 1$.

An open set A in $\mathbb{R}^n$ is called *simply connected* if any closed curve $\varphi : [a,b] \to A$ of class C^2 is homotopic to a point.

It can be proved that in the plane this definition coincides with the one of Sect. 7.4. Observe, though, that while an annulus in the plane is not simply connected, a spherical shell in dimension $n \geq 3$ is.

Theorem (Differential Forms on Simply Connected Sets in $\mathbb{R}^n$) *If $A \subseteq \mathbb{R}^n$ is a simply connected open set and ω is a closed form of class $C^1(A)$, then ω is exact on A.*

The proof is postponed to Sect. 12.9.



Multiple Integrals

8

In this chapter we present the theory of integration for functions of n real variables. We start in Sects. 8.1–8.4, with the simpler case $n = 2$, namely *double integrals* of continuous functions over *normal domains*. Section 8.5 addresses *triple integrals* ($n = 3$). Finally, from Sect. 8.6 we study the general case ($n \geq 2$) of *multiple integrals*, and discuss the general theory of Riemann integration.

If the subject is being approached for the first time, it may be enough to concentrate on the cases $n = 2, 3$, dealt with in Sects. 8.1–8.5. Sections from 8.6 onwards are directed at readers wishing to deepen the study of Riemann integration in more generality.

8.1 Double Integrals on Normal Domains

Let $\alpha = \alpha(x)$ and $\beta = \beta(x)$ be two continuous functions on a closed bounded interval $[a, b] \subset \mathbb{R}$, $a < b$, such that

$$\alpha(x) \leq \beta(x)\,, \qquad \forall\, x \in [a, b].$$

The subset in $\mathbb{R}^2$ (Fig. 8.1)

$$D = \big\{(x, y) \in [a, b] \times \mathbb{R} : \ \ \alpha(x) \leq y \leq \beta(x)\big\}$$

is called a *normal domain with respect to the variable* x (or simply *with respect to* x).

The formula expressing the area of D is known from the theory of integration of one real variable. The *area*, or *measure*, $m(D)$ of the set D equals

$$m(D) = \int_a^b (\beta(x) - \alpha(x))\, dx\,. \tag{8.1}$$

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N. Fusco et al., *Mathematical Analysis*, La Matematica per il 3+2 137,
https://doi.org/10.1007/978-3-031-04151-8_8

Fig. 8.1

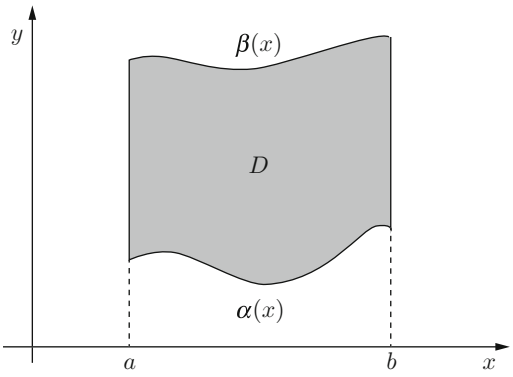
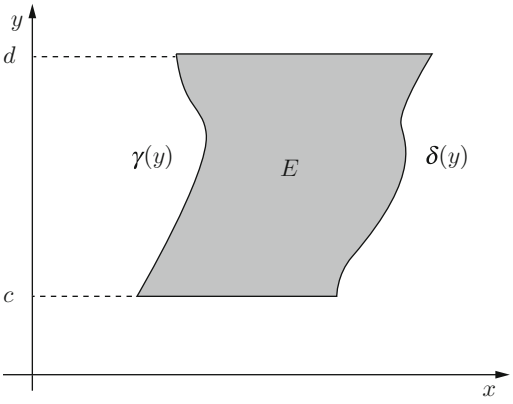


Fig. 8.2



Analogously, if $\gamma = \gamma(y)$ and $\delta = \delta(y)$ are continuous functions on the closed bounded interval $[c, d]$ such that

$$\gamma(y) \le \delta(y)\,, \qquad \forall\, y \in [c, d],$$

the subset in $\mathbb{R}^2$ (Fig. 8.2)

$$E = \big\{(x, y) \in \mathbb{R} \times [c, d] : \ \ \gamma(y) \le x \le \delta(y)\big\}$$

is a *normal domain with respect to* y, and its *area*, or *measure*, $m(E)$ is

$$m(E) = \int_c^d (\delta(y) - \gamma(y))\, dy\,. \tag{8.2}$$

Note that the word *domain* (closure of an open set) for the sets $D, E \subset \mathbb{R}^2$ is justified only when $\alpha(x), \beta(x)$ (or $\gamma(y), \delta(y)$) do not coincide on some subset of $[a, b]$ (respectively $[c, d]$) with non-empty interior. We shall nonetheless use the

term *normal domain* without distinguishing whether the inequality between $\alpha(x)$ and $\beta(x)$ (or $\gamma(y), \delta(y)$) is strict or not, since this fact will not make any difference in the theory.

Observe moreover that if D is a normal domain with respect to both axes x and y, formulas (8.1) and (8.2) for the domain's measure yield the same result (see Sect. 8.8).

Let D be a normal domain, say with respect to x. A *partition* of D in normal domains is a finite collection $\{D_1, D_2, \dots, D_h\}$ of normal domains with respect to x, all contained in D, whose pairwise intersections have empty interiors and whose union is D.

Lemma 1 *Let D be a normal domain with respect to, say, x. For any $\delta > 0$ there exists a partition of D in normal domains with respect to x, such that the diameter of every subdomain is less than δ.*

Proof Observe first that the construction of the partition, as stated, is not unique. We may for example consider a partition obtained by slicing D with lines parallel to the axes. The construction described below has the advantage of being more symmetric in terms of the subdomains.

So let us divide D in subsets D_{ij}. We consider a subdivision of the interval $[a, b]$ given by $k + 1$ points

$$a = x_0 < x_1 < x_2 < \cdots < x_k = b\,.$$

For our purposes it suffices to choose the x_i, $i = 0, 1, \dots, k$, to be equidistant:

$$x_i = a + \frac{i}{k}(b - a), \qquad \forall i = 0, 1 \dots, k\,.$$

Moreover, consider $k + 1$ continuous functions on $[a, b]$, say $\varphi_j = \varphi_j(x)$, $j = 0, 1, \dots, k$, such that

$$\alpha(x) = \varphi_0(x) \le \varphi_1(x) \le \varphi_2(x), \dots, \varphi_k(x) = \beta(x) \qquad \forall x \in [a, b]\,.$$

Here, as well, we may pick equidistant functions $\varphi_j(x)$:

$$\varphi_j(x) = \alpha(x) + \frac{j}{k}(\beta(x) - \alpha(x)), \qquad \forall j = 0, 1, \dots, k\,, \tag{8.3}$$

as in Fig. 8.3.

Consider the domain

$$D_{ij} = \left\{(x, y) \in \mathbb{R}^2 : x_{i-1} \le x \le x_i,\ \varphi_{j-1}(x) \le y \le \varphi_j(x)\right\} \tag{8.4}$$

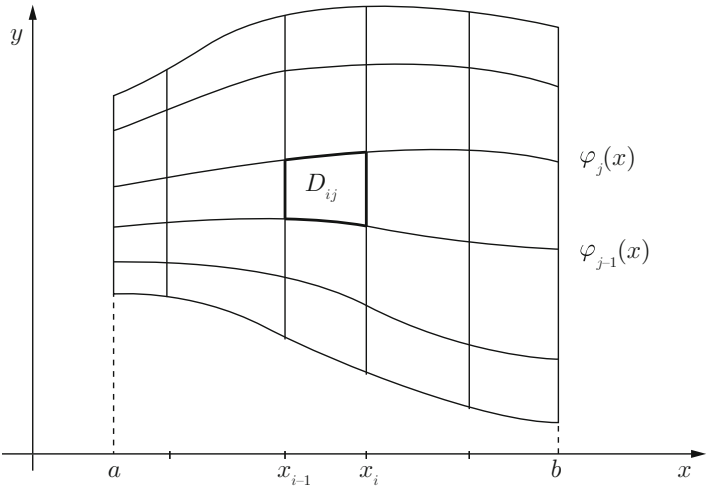


Fig. 8.3

for any $i, j \in \{1, 2, \ldots, k\}$. This is normal with respect to x. We want to show that for any $\delta > 0$ it is possible to choose $k \in \mathbb{N}$ so that the diameter of D_{ij}, for any $i, j \in \{1, 2, \ldots, k\}$, is less than δ. For that we remind that α and β are continuous on $[a, b]$ and hence uniformly continuous. Given $\delta > 0$, therefore, there is $k_0 \in \mathbb{N}$ such that for $k > k_0$

$$|\alpha(x') - \alpha(x'')| \leq \frac{\delta}{2}, \quad |\beta(x') - \beta(x'')| \leq \frac{\delta}{2} \qquad \forall x', x'' \in [a, b] \text{ such that } |x' - x''| \leq \frac{b-a}{k}. \tag{8.5}$$

But for any $i = 1, 2, \ldots, k$

$$x_i - x_{i-1} = \frac{1}{k}(b - a),$$

so from this, (8.3) and (8.5) we deduce that when $k > k_0$,

$$|\varphi_j(x') - \varphi_j(x'')| = \left|\left(1 - \frac{j}{k}\right)(\alpha(x') - \alpha(x'')) + \frac{j}{k}(\beta(x') - \beta(x''))\right| \leq$$
$$\leq \left(1 - \frac{j}{k}\right)|\alpha(x') - \alpha(x'')| + \frac{j}{k}|\beta(x') - \beta(x'')| \leq \frac{\delta}{2}$$

for any $j = 0, 1, \dots, k$ and any $x', x'' \in [x_{i-1}, x_i]$. Call m and M respectively the minimum of $\alpha(x)$ and the maximum of $\beta(x)$ on $[a, b]$, for $j = 1, 2, \dots, k$ and $x', x'' \in [x_{i-1}, x_i]$, $i = 1, 2, \dots, k$. The previous inequality and (8.3) say that

$$|\varphi_j(x') - \varphi_{j-1}(x'')| \le |\varphi_j(x') - \varphi_j(x'')| + |\varphi_j(x'') - \varphi_{j-1}(x'')| \le \frac{\delta}{2} + \frac{1}{k}(M - m) . \tag{8.6}$$

Now choose $(x', y'), (x'', y'') \in D_{ij}$. Without loss of generality we may assume $y' \ge y''$. Then (8.6) implies

$$|(x', y') - (x'', y'')| = \sqrt{(x' - x'')^2 + (y' - y'')^2} \le \sqrt{\left(\frac{b-a}{k}\right)^2 + (\varphi_j(x') - \varphi_{j-1}(x''))^2} \le$$

$$\le \sqrt{\left(\frac{b-a}{k}\right)^2 + \left(\frac{\delta}{2} + \frac{1}{k}(M - m)\right)^2} \le \delta ,$$

as long as we choose $k > k_0$ large enough. Hence

$$\operatorname{diam} D_{ij} \le \delta .$$

Additivity of the Measure *Let D be a normal domain (for example) with respect to x and $\{D_1, D_2, \dots, D_h\}$ a partition of D into normal domains with respect to x. Then*

$$m(D) = \sum_{i=1}^{h} m(D_i) . \tag{8.7}$$

Recall that, by definition, the pairwise intersections of the partitioning sets $D_1, D_2, \dots, D_h$ do not have interior points. As $D = \bigcup_{i=1}^{h} D_i$, we may recast (8.7) as

$$m\left(\bigcup_{i=1}^{h} D_i\right) = \sum_{i=1}^{h} m(D_i) .$$

Then we say that the measure m on D is *finitely additive*.

Proof Readers should convince themselves of the validity of formula (8.7) when the partition consists of two sets D_1, D_2 obtained from D by dividing the base interval $[a, b]$ using a point c, or dividing $[\alpha(x), \beta(x)]$ with an intermediate function $\gamma(x)$. More generally, we can say the following.

We retain the previous lemma's notations and limit ourselves to the case in which the partition is $P = \{D_{ij} : i, j \in \{1, 2, \dots, k\}\}$, where the normal domains in x are defined, as in (8.4), by

$$D_{ij} = \left\{(x, y) \in \mathbb{R}^2 : x_{i-1} \le x \le x_i,\ \varphi_{j-1}(x) \le y \le \varphi_j(x)\right\}.$$

As a matter of fact one can prove that it is always possible to reduce to this situation by successive subdivisions of the given partition. Now, we need to prove that

$$m(D) = m\left(\bigcup_{i,j=1}^{k} D_{ij}\right) = \sum_{i,j=1}^{k} m\left(D_{ij}\right). \tag{8.8}$$

The measure $m(D_{ij})$ is

$$m(D_{ij}) = \int_{x_{i-1}}^{x_i} \left(\varphi_j(x) - \varphi_{j-1}(x)\right) dx\,,$$

so the integral's additivity implies

$$\begin{aligned}\sum_{i,j=1}^{k} m(D_{ij}) &= \sum_{i,j=1}^{k} \int_{x_{i-1}}^{x_i} \left(\varphi_j(x) - \varphi_{j-1}(x)\right) dx = \\ &= \sum_{j=1}^{k} \left(\sum_{i=1}^{k} \int_{x_{i-1}}^{x_i} \left(\varphi_j(x) - \varphi_{j-1}(x)\right) dx \right) = \sum_{j=1}^{k} \int_a^b \left(\varphi_j(x) - \varphi_{j-1}(x)\right) dx = \\ &= \int_a^b \sum_{j=1}^{k} \left(\varphi_j(x) - \varphi_{j-1}(x)\right) dx = \int_a^b (\beta(x) - \alpha(x))\, dx = m(D)\,,\end{aligned}$$

i.e. (8.8).

Let D be a normal domain in $\mathbb{R}^2$, for example with respect to x. Suppose $f : D \subset \mathbb{R}^2 \to \mathbb{R}$ is a *bounded* function on D. For any partition $P = \{D_1, D_2, \dots, D_h\}$ of D in normal domains with respect to x define the *integral sums* $s(P)$, $S(P)$:

$$s(P) = \sum_{i=1}^{h} m(D_i) \inf\{f(x) : \quad x \in D_i\};$$

$$S(P) = \sum_{i=1}^{h} m(D_i) \sup\{f(x) : \quad x \in D_i\}.$$

They are respectively called *lower (integral) sum* and *upper (integral) sum* relative to the partition P.

Using

$$\inf\{f(x): \quad x \in D_i\} = \inf f(D_i)\,,$$
$$\sup\{f(x): \quad x \in D_i\} = \sup f(D_i)\,,$$

the lower and upper sums read

$$s(P) = \sum_{i=1}^{h} m(D_i) \inf f(D_i)\,; \qquad S(P) = \sum_{i=1}^{h} m(D_i) \sup f(D_i)\,.$$

Let D_1, D_2 be domains in $\mathbb{R}^2$ that are normal with respect to the same variable, say x:

$$D_1 = \{(x, y) \in [a_1, b_1] \times \mathbb{R}: \quad \alpha_1(x) \le y \le \beta_1(x)\}\,,$$
$$D_2 = \{(x, y) \in [a_2, b_2] \times \mathbb{R}: \quad \alpha_2(x) \le y \le \beta_2(x)\}\,,$$

as in Fig. 8.4. Is it straightforward to see that the *intersection* $D_1 \cap D_2$, when non-empty, is still a normal domain in x of the form

$$D_1 \cap D_2 = \{(x, y) \in [a_3, b_3] \times \mathbb{R}: \quad \alpha_3(x) \le y \le \beta_3(x)\}$$

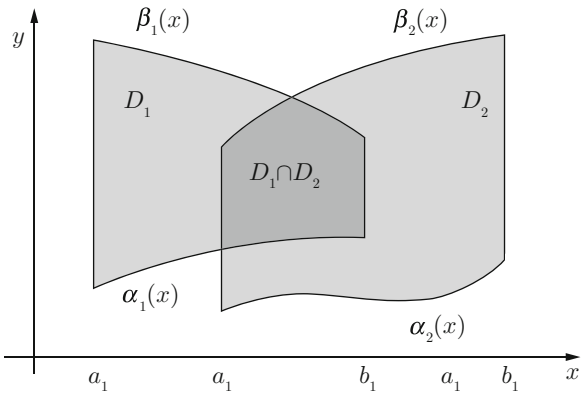
(possibly with empty interior), where

$$a_3 = \max\{a_1; a_2\}\,, \qquad b_3 = \min\{b_1; b_2\}\,,$$
$$\alpha_3(x) = \max\{\alpha_1(x); \alpha_2(x)\}\,, \qquad \beta_3(x) = \min\{\beta_1(x); \beta_2(x)\}\,.$$

If $P_1 = \{D_1, D_2, \ldots, D_h\}$ and $P_2 = \{E_1, E_2, \ldots, E_k\}$ are partitions of D, the partition P_{12} *generated* by P_1, P_2 consists of the intersections of the elements of P_1 with the elements of P_2:

$$P_{12} = \{D_1 \cap E_1, \ldots, D_1 \cap E_k\,, D_2 \cap E_1, \ldots, D_h \cap E_1, \ldots, D_h \cap E_k\},$$

Fig. 8.4



or

$$P_{12} = \{D_i \cap E_j : \quad i \in \{1, 2, \dots, h\},\ j \in \{1, 2, \dots, k\}\}$$

for short. Note that some of the intersections might be empty, or have empty interior, and hence have measure zero.

Lemma 2 *Let $P_1 = \{D_1, D_2, \dots, D_h\}$ and $P_2 = \{E_1, E_2, \dots, E_k\}$ be partitions of the normal domain D with respect to x made by normal domains with respect to x. The lower and upper sums relative to the partition P_{12} generated by P_1, P_2 satisfy*

$$s(P_1) \le s(P_{12}) \le S(P_{12}) \le S(P_2)\,. \tag{8.9}$$

Proof First of all, the partition P_{12} generated by P_1, P_2 consists of normal domains with respect to x. For $i = 1, 2, \dots, h$ and $j = 1, 2, \dots, k$ set

$$D_{ij} = D_i \cap E_j\,,$$

so

$$\bigcup_{i=1}^{h} D_{ij} = E_j\,, \qquad \bigcup_{j=1}^{k} D_{ij} = D_i\,.$$

By the additivity of the measure m,

$$\begin{aligned} s(P_1) &= \sum_{i=1}^{h} m(D_i) \inf f(D_i) = \sum_{i=1}^{h} m\left(\bigcup_{j=1}^{k} D_{ij}\right) \inf f(D_i) = \\ &= \sum_{i=1}^{h} \left(\sum_{j=1}^{k} m(D_{ij})\right) \inf f(D_i) \le \sum_{i=1}^{h}\sum_{j=1}^{k} m(D_{ij}) \inf f(D_{ij}) = s(P_{12})\,, \end{aligned} \tag{8.10}$$

where we used the inequalities $\inf f(D_i) \le \inf f(D_{ij})$, valid for any $i = 1, 2, \dots, h$ and any $j = 1, 2, \dots, k$ (a consequence of $f(D_{ij}) \subseteq f(D_i)$). Similarly,

$$\begin{aligned} S(P_2) &= \sum_{j=1}^{k} m(E_j) \sup f(E_j) = \sum_{j=1}^{k} m\left(\bigcup_{i=1}^{h} D_{ij}\right) \sup f(E_j) = \\ &= \sum_{j=1}^{k} \left(\sum_{i=1}^{h} m(D_{ij})\right) \sup f(E_j) \ge \sum_{i=1}^{h}\sum_{j=1}^{k} m(D_{ij}) \sup f(D_{ij}) = s(P_{12})\,. \end{aligned} \tag{8.11}$$

Since $s(P_{12}) \leq S(P_{12})$, now (8.9) follows from (8.10) and (8.11).

By Lemma 2, as we let the partitions P_1 , P_2 of the normal domain (with respect to x) D vary, the collections of all lower and upper sums

$$\{s(P_1)\}, \qquad \{S(P_2)\}$$

are *separated* sets. In other words $s(P_1) \leq S(P_2)$ for any pair of partitions P_1, P_2 made of normal domains with respect to x.

If the above sets are additionally *contiguous*, meaning

$$\sup_{P_1} \{s(P_1)\} = \inf_{P_2} \{S(P_2)\},$$

we say f is *integrable* on D. The (unique) separating element is called *(double) integral* of $f = f(x, y)$ over the set $D \subset \mathbb{R}^2$, and is denoted by

$$\iint_D f(x, y)\, dx\, dy\,.$$

If $f = f(x, y)$ is integrable in the above sense, f is then Riemann integrable (see Sect. 8.7). Moreover, one can prove that the converse holds, too, i.e. a Riemann integrable function is integrable in the aforementioned sense. Hence if D is a normal domain with respect to both variables, the previous definition of integral does not depend on the choice of axis used for partitioning.

If the function $f = f(x, y)$ is non-negative as (x, y) varies in D, the double integral of f over D represents the *volume* of the solid body in $\mathbb{R}^3$ bounded by D on the xy-plane, by the graph of f (which is the support of a surface in $\mathbb{R}^3$ inside $\{(x, y, z) \in \mathbb{R}^3 : z \geq 0\}$) and by the vertical segments rising from the boundary of D.

With the next result we prove the integrability of continuous functions over normal domains.

Integrability of Continuous Functions *Let D be a normal domain in $\mathbb{R}^2$ and $f : D \subset \mathbb{R}^2 \to \mathbb{R}$ a continuous function. Then f is integrable on D.*

Proof A normal domain D is compact in $\mathbb{R}^2$. By Cantor's theorem f is then uniformly continuous on D, so given $\varepsilon > 0$, there is $\delta > 0$ such that

$$(x, y),\ (x', y') \in D, \qquad \sqrt{(x - x')^2 + (y - y')^2} < \delta \tag{8.12}$$

forces

$$\left| f(x, y) - f(x', y') \right| < \varepsilon\,. \tag{8.13}$$

Let D be normal with respect to x. By Lemma 1 there is a partition $P = \{D_1, D_2, \ldots, D_h\}$ of D into normal domains with respect to x of diameter less than δ. For any $i = 1, 2, \ldots, h$, call (x_i, y_i), (x_i', y_i') the maximum and minimum points (respectively) of $f(x, y)$ on the compact set D_i (which exist by the Weierstrass theorem). Due to (8.12) and (8.13),

$$\max f(D_i) - \min f(D_i) = f(x_i, y_i) - f(x_i', y_i') < \varepsilon\,.$$

Hence the measure's additivity forces

$$S(P) - s(P) = \sum_{i=1}^{h} m(D_i)[\max f(D_i) - \min f(D_i)] <$$

$$< \varepsilon \sum_{i=1}^{h} m(D_i) = \varepsilon m(D),$$

making $\{s(P)\}$, $\{S(P)\}$ contiguous.

An analogous argument proves the following theorem for the integral of continuous functions.

Theorem (Representing Integrals as Limits) *Let f be a continuous function on a normal domain $D \subset \mathbb{R}^2$. For any partition $P = \{D_1, D_2, \ldots, D_h\}$ of D denote by (x_1, y_1), $(x_2, y_2), \ldots, (x_h, y_h)$, h points in D such that $(x_i, y_i) \in D_i$ for any $i = 1, 2, \ldots, h$. The integral of f on D is the limit*

$$\lim_{\operatorname{diam} P \to 0} \sum_{i=1}^{h} m(D_i) f(x_i, y_i) = \iint_D f(x, y)\, dx\, dy\,,$$

in the sense that for any $\varepsilon > 0$ there is $\delta > 0$ such that

$$\left| \iint_D f(x, y)\, dx\, dy - \sum_{i=1}^{h} m(D_i) f(x_i, y_i) \right| < \varepsilon$$

for every partition $P = \{D_1, D_2, \ldots, D_h\}$ of D into normal domains D_i, $i = 1, 2, \ldots, h$ whose largest diameter (called diameter diam P *of the partition P) is less than δ.*

Proof Let D be a normal domain, say with respect to x. As D is compact in $\mathbb{R}^2$, by Cantor's theorem the function f is uniformly continuous on D. So, given $\varepsilon > 0$, there exists $\delta > 0$ such that if

$$(x, y), (x', y') \in D, \quad \sqrt{(x - x')^2 + (y - y')^2} < \delta\,, \tag{8.14}$$

then

$$\left| f(x, y) - f(x', y') \right| < \varepsilon \,. \tag{8.15}$$

Let $P = \{D_1, D_2, \ldots, D_h\}$ be a partition of D into normal domains in x with diameter less than δ. Pick $(x_i, y_i) \in D_i$ for any $i = 1, 2, \ldots, h$. Indicate by (x_i', y_i') the minimum point of $f(x, y)$ on the compact set D_i, which exists by the Weierstrass theorem (we could have chosen maximum points as well). By (8.14) and (8.15)

$$f(x_i, y_i) - \min f(D_i) = f(x_i, y_i) - f(x_i', y_i') < \varepsilon \,.$$

Since the measure is additive,

$$\begin{aligned} \sum_{i=1}^{h} m(D_i) f(x_i, y_i) - s(P) = \sum_{i=1}^{h} m(D_i)[f(x_i, y_i) - \min f(D_i)] < \\ < \varepsilon \sum_{i=1}^{h} m(D_i) = \varepsilon m(D) \,. \end{aligned} \tag{8.16}$$

Finally, from the previous theorem we know f is integrable on D, and since the diameter of the partition P is less than δ, we have

$$\iint_D f(x, y)\, dx\, dy - s(P) \le S(P) - s(P) < \varepsilon m(D) \,. \tag{8.17}$$

Now from (8.16) and (8.17) we obtain the desired conclusion

$$\begin{aligned} &\left| \iint_D f(x, y)\, dx\, dy - \sum_{i=1}^{h} m(D_i)\, f(x_i\,, y_i) \right| \le \\ \le &\left| \iint_D f(x, y)\, dx\, dy - s(P) \right| + \left| s(P) - \sum_{i=1}^{h} m(D_i) f(x_i, y_i) \right| < 2\varepsilon m(D) \,. \end{aligned}$$

8.2 Reduction Formulas for Double Integrals

We shall discuss in this section a few very useful formulas for calculating double integrals of continuous functions over normal domains.

Reduction Formulas *Let $[a, b]$ be a closed bounded interval and D the normal domain with respect to x defined by*

$$D = \{(x, y) \in [a, b] \times \mathbb{R} : \ \alpha(x) \le y \le \beta(x)\},$$

where $\alpha(x) \le \beta(x)$ are continuous on $[a, b]$. Then

$$\iint_D f(x, y)dx\, dy = \int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x, y)\, dy \tag{8.18}$$

for any continuous function $f : D \subset \mathbb{R}^2 \to \mathbb{R}$. If E is a normal domain with respect to y defined by

$$E = \{(x, y) \in \mathbb{R} \times [c, d] : \ \gamma(y) \le x \le \delta(y)\}$$

with $\gamma(y) \le \delta(y)$ continuous on $[c, d]$, and $f : E \subset \mathbb{R}^2 \to \mathbb{R}$ is a continuous function on E, then

$$\iint_E f(x, y)\, dx\, dy = \int_c^d dy \int_{\gamma(y)}^{\delta(y)} f(x, y)\, dx\,. \tag{8.19}$$

Proof Let us prove (8.18). First, the function

$$x \in [a, b] \to \int_{\alpha(x)}^{\beta(x)} f(x, y)\, dy$$

is continuous, due to the continuity of integral functions (Sect. 3.10). Therefore it is integrable on $[a, b]$. As in the proof of Lemma 1 in the previous section (see Fig. 8.3 as well), we divide D into subsets D_{ij} by taking a subdivision of $[a, b]$ made by $k + 1$ points x_i such that

$$a = x_0 < x_1 < x_2 < \ldots < x_k = b\,,$$

and $k + 1$ continuous functions on $[a, b]$, $\varphi_j = \varphi_j(x)$, $j = 0, 1, \ldots, k$, such that

$$\alpha(x) = \varphi_0(x) \le \varphi_1(x) \le \varphi_2(x) \le \ldots \le \varphi_k(x) = \beta(x), \qquad \forall\, x \in [a, b].$$

For any $i, j = 1, 2, \ldots, k$ consider the normal domain with respect to x

$$D_{ij} = \left\{(x, y) \in \mathbb{R}^2 : \ x_{i-1} \le x \le x_i,\ \varphi_{j-1}(x) \le y \le \varphi_j(x)\right\}.$$

As in the aforementioned Lemma 1, we can choose the x_i and the functions $\varphi_j(x)$ so that the maximum diameter of the D_{ij} tends to zero as $k \to +\infty$.

The additivity of the integral in dx tells that

$$\int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x, y)\, dy = \sum_{i=1}^{k} \int_{x_{i-1}}^{x_i} dx \int_{\alpha(x)}^{\beta(x)} f(x, y)\, dy\,. \tag{8.20}$$

The additivity of the integral in dy, moreover, gives

$$\int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy = \sum_{j=1}^{k} \int_{\varphi_{j-1}(x)}^{\varphi_j(x)} f(x,y)\,dy\,. \tag{8.21}$$

From (8.20) and (8.21) we then have

$$\int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy = \sum_{i,j=1}^{k} \int_{x_{i-1}}^{x_i} dx \int_{\varphi_{j-1}(x)}^{\varphi_j(x)} f(x,y)\,dy\,. \tag{8.22}$$

Now let m_{ij}, M_{ij} be the minimum and maximum of f on the compact set $D_{ij} \subset \mathbb{R}^2$. For $x \in [x_{i-1}, x_i]$ we have

$$m_{ij}\,\big(\varphi_j(x) - \varphi_{j-1}(x)\big) \le \int_{\varphi_{j-1}(x)}^{\varphi_j(x)} f(x,y)\,dy \le M_{ij}\,\big(\varphi_j(x) - \varphi_{j-1}(x)\big),$$

whence

$$m_{ij} \int_{x_{i-1}}^{x_i} \big(\varphi_j(x) - \varphi_{j-1}(x)\big)\,dx \le \int_{x_{i-1}}^{x_i} dx \int_{\varphi_{j-1}(x)}^{\varphi_j(x)} f(x,y)\,dy \le$$

$$\le M_{ij} \int_{x_{i-1}}^{x_i} \big(\varphi_j(x) - \varphi_{j-1}(x)\big)\,dx\,.$$

Consider the measure of D_{ij},

$$m(D_{ij}) = \int_{x_{i-1}}^{x_i} \big(\varphi_j(x) - \varphi_{j-1}(x)\big)\,dx\,.$$

Summing over $i, j = 1, 2, \ldots, k$,

$$\sum_{i,j=1}^{k} m_{ij} m(D_{ij}) \le \sum_{i,j=1}^{k} \int_{x_{i-1}}^{x_i} dx \int_{\varphi_{j-1}(x)}^{\varphi_j(x)} f(x,y)\,dy \le \sum_{i,j=1}^{k} M_{ij} m(D_{ij})$$

and keeping (8.22) in account,

$$\sum_{i,j=1}^{k} m_{ij} m(D_{ij}) \le \int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy \le \sum_{i,j=1}^{k} M_{ij} m(D_{ij})\,. \tag{8.23}$$

The two sums in (8.23) are integral sums of f (lower and upper, respectively), so we obtain

$$\sum_{i,j=1}^{k} m_{ij} m(D_{ij}) \le \iint_D f(x,y)\,dx\,dy \le \sum_{i,j=1}^{k} M_{ij} m(D_{ij})\,. \tag{8.24}$$

Then by (8.23) and (8.24) we have

$$\left| \iint_D f(x,y)dx\,dy - \int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy \right| \le \sum_{i,j=1}^{k} \big(M_{ij} - m_{ij}\big) m(D_{ij})\,. \tag{8.25}$$

Now, f is uniformly continuous on the compact set D, so for any $\varepsilon > 0$ there exists $\delta > 0$ such that the oscillation of f is less than ε on any subset of D of diameter less than δ. Choosing the D_{ij}, $i, j = 1, 2, \ldots, k$, to have diameter smaller than δ, we obtain

$$M_{ij} - m_{ij} < \varepsilon\,, \qquad \forall\, i, j \in \{1, 2, \ldots, k\}. \tag{8.26}$$

Then (8.25) and (8.26) force

$$\left| \iint_D f(x,y)dx\,dy - \int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy \right| < \varepsilon \sum_{i,j=1}^{k} m(D_{ij}) = \varepsilon\, m(D)\,,$$

and hence the claim, since ε is arbitrary.

Sometimes the integration domain is normal with respect to both coordinates, so (8.18) and (8.19) hold simultaneously. In this case *the order of integration can be interchanged*:

$$\int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy = \int_c^d dy \int_{\gamma(y)}^{\delta(y)} f(x,y)\,dx\,. \tag{8.27}$$

Let us emphasise two special cases of (8.27). In the first instance D is an interval $[a,b] \times [c,d]$ in $\mathbb{R}^2$, where $\alpha(x) \equiv c$, $\beta(x) \equiv d$, $\gamma(y) \equiv a$, $\delta(y) \equiv b$ and

$$\iint_{[a,b]\times[c,d]} f(x,y)\,dx\,dy = \int_a^b dx \int_c^d f(x,y)\,dy = \int_c^d dy \int_a^b f(x,y)\,dx\,.$$

In the second case

$$D = \Big\{(x,y) \in [a,b] \times [a,b] \subset \mathbb{R}^2 : y \le x\Big\}$$

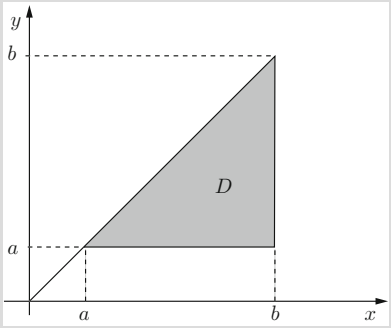
is a triangle, represented in Fig. 8.5. This time we choose $\alpha(x) \equiv a$, $\beta(x) = x$, $\gamma(y) = y$, $\delta(y) \equiv b$, and

$$\iint_D f(x,y)\,dx\,dy = \int_a^b dx \int_a^x f(x,y)\,dy = \int_a^b dy \int_y^b f(x,y)\,dx\,. \tag{8.28}$$

We shall refer to formula (8.28) as *Dirichlet's formula* for interchanging integrals.

(continued)

Fig. 8.5



As application of the reduction formulas let us calculate the volume of a solid body. Using coordinates x, y, z, consider the set $S \subset \mathbb{R}^3$ bounded by the paraboloid $z = x^2 + y^2$, the cylinder $x^2 + y^2 = 9$ and contained in the *first octant* $x \geq 0$, $y \geq 0$, $z \geq 0$, see Fig. 8.6.

Let us determine the volume $m_3(S)$ of S. If D denotes the quarter of the disc $x^2+y^2 \leq 9$ in the first quadrant of the xy-plane, we may regard D as a normal domain with respect to y (the reader should note the symmetry in the first two coordinates), and represent it as (see Fig. 8.7)

$$D = \left\{(x, y) \in \mathbb{R}^2 : \ 0 \leq y \leq 3, \ 0 \leq x \leq \sqrt{9 - y^2}\right\}.$$

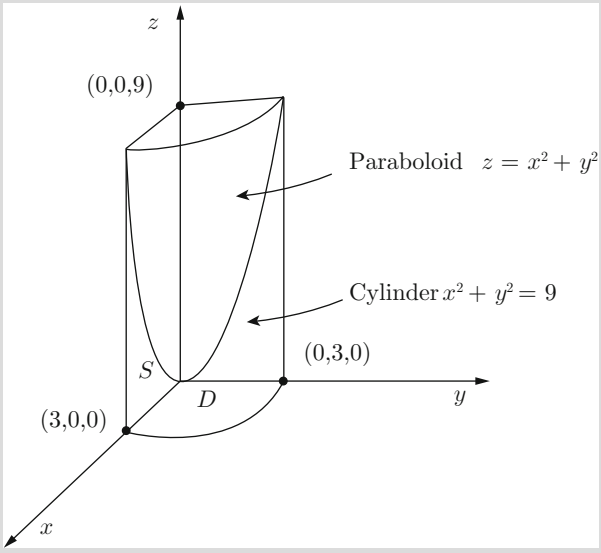


Fig. 8.6

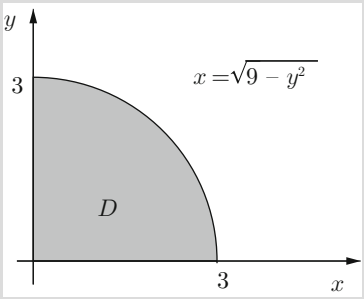
(continued)

The reduction formulas tell

$$m_3(S) = \iint_D \left(x^2 + y^2\right) dx\, dy =$$

$$= \int_0^3 dy \int_0^{\sqrt{9-y^2}} \left(x^2 + y^2\right) dx = \int_0^3 \left\{ \frac{1}{3}\left(9 - y^2\right)^{3/2} + y^2\sqrt{9 - y^2} \right\} dy\,.$$

Fig. 8.7



The substitution $y = 3\sin\vartheta$ produces

$$m_3(S) = \int_0^{\pi/2} \left(9\cos^3\vartheta + 27\sin^2\vartheta\cos\vartheta\right) \cdot 3\cos\vartheta\, d\vartheta =$$

$$= 27\int_0^{\pi/2} \left(\cos^4\vartheta + 3\sin^2\vartheta \cdot \cos^2\vartheta\right) d\vartheta = 27\int_0^{\pi/2} \cos^2\vartheta\left(1 + 2\sin^2\vartheta\right) d\vartheta =$$

$$= 27\int_0^{\pi/2} \left(\frac{1 + \cos 2\vartheta}{2} + \frac{1 - \cos 4\vartheta}{4}\right) d\vartheta = \frac{81}{8}\pi\,.$$

If D is a normal domain in the plane, the point (x_0, y_0) defined by

$$x_0 = \frac{1}{m(D)} \iint_D x\, dx\, dy\,, \qquad y_0 = \frac{1}{m(D)} \iint_D y\, dx\, dy \tag{8.29}$$

is called *centroid* of D. The centroid might not belong to D (think of an annulus, for instance), but the reader can easily show that $(x_0\,, y_0) \in D$ when D is convex.

The next result will be proved in a more general form in Sect. 9.8, Chap. 9.

First Guldinus Theorem *Let S be the solid generated by the rotation by angle α of a normal domain D about an axis r not intersecting D. The volume of S is the product of the area of D times the length of the circular arc traced out by the rotation of the centroid.*

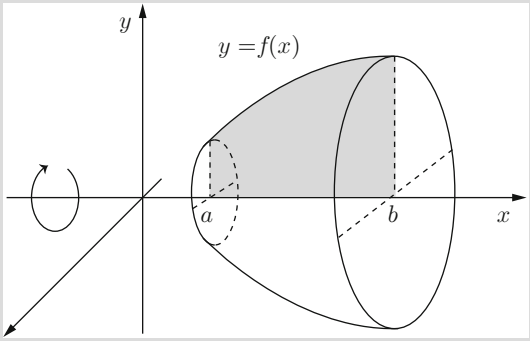
For example, suppose D makes one complete turn around the x-axis. The centroid $(x_0\,,\,y_0)$ of D travels along a circle of radius y_0, so by (8.29) and the Guldinus theorem the volume of S is

$$2\pi \iint_D y\,dx\,dy\,. \tag{8.30}$$

Let D be the trapezoid determined by a non-negative continuous function f over $[a, b]$. If S is the solid of revolution about the x-axis generated by D (Fig. 8.8), from (8.30) we deduce that the volume of S equals

$$2\pi \int_a^b dx \int_0^{f(x)} y\,dy = \pi \int_a^b f^2(x)\,dx\,. \tag{8.31}$$

Fig. 8.8



Using that, we can in particular find the volume of the ball S in $\mathbb{R}^3$ with centre 0 and radius $r > 0$. We may view S as generated by the revolution around the x-axis of the trapezoid over $[-r, r]$ of the function

$$f(x) = \sqrt{r^2 - x^2}\,, \qquad x \in [-r, r].$$

Applying formula (8.31) we obtain *the volume of the ball*

$$m_3(S) = \pi \int_{-r}^{r} \left(r^2 - x^2\right) dx = \frac{4}{3}\pi\, r^3\,.$$

In closing, let us remark that the concepts of measure of a normal domain D and integral of a function $f(x, y)$ over D can be extended in a natural way to domains

that are decomposable in a finite number of normal domains $D_1,\ D_2,\ \ldots,\ D_N$ with pairwise-disjoint interiors. In this case one calls

$$\iint_D f(x,y)dx\,dy = \sum_{i=1}^{N} \iint_{D_i} f(x,y)\,dx\,dy$$

the *double integral of* f *over* D, and one proves that it does not depend on the chosen decomposition.

8.3 Gauss-Green Formulas. The Divergence Theorem. Stokes's Formula

Let D be the normal domain with respect to x

$$D = \{(x,y) \in [a,b] \times \mathbb{R} :\ \alpha(x) \le y \le \beta(x)\}.$$

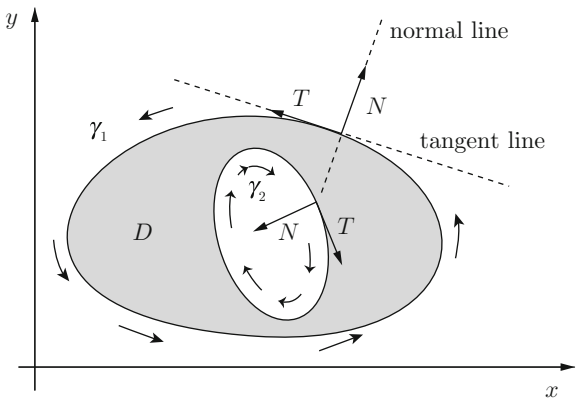
We say that D is a *regular normal domain* when $\alpha(x)$, $\beta(x)$ are C^1 functions on $[a,b]$ and $\alpha(x) < \beta(x)$ for any $x \in (a,b)$.

A *regular domain* D is by definition the union of finitely many regular normal domains (with respect to x or y) $D_1, D_2, \ldots, D_N$ with pairwise-disjoint interiors.

In the following, to ease the presentation, when dealing with a regular domain D we shall always assume that *its boundary* ∂D *is the union of a finite number of piece wise regular curves*. In this way ∂D admits a unit tangent vector T at all points except for a finite number at most. In correspondence with the latter the unit normal N to ∂D is well defined. About this fact, recall that (see Sect. 6.5) if γ is a plane curve with unit tangent vector T at some point, the unit normal N is defined so that the basis (N, T) is oriented as the canonical basis (see Fig. 8.9).

That said, our convention is to endow the boundary of D with the orientation for which the above unit normal N *points to the outer region of* ∂D at each point. Such

Fig. 8.9



orientation is called *positive orientation* of the boundary of D, and we shall use the symbol $+\partial D$.

Figure 8.9 shows a domain D whose boundary is the union of two curves γ_1, γ_2. The positive orientation of ∂D makes the outer curve γ_1 counter-clockwise and the inner curve γ_2 clockwise.

Let D be a regular domain and consider a *parametrisation*

$$\begin{cases} x = x(t) \\ y = y(t), \end{cases} \qquad t \in [a, b], \tag{8.32}$$

of a portion γ of the boundary ∂D. This induces an orientation on γ. Changing parameter with $s = -t$ gives parametric equations

$$\begin{cases} x = x(-s) \\ y = y(-s), \end{cases} \qquad s \in [-b, -a],$$

for the same curve γ, but now oriented in the opposite direction. We shall assume that the parametric equations (8.32) induce on γ the orientation of $+\partial D$.

If so, as we saw in Chap. 6, the *unit tangent vector* T is

$$T(t) = \frac{\gamma'(t)}{|\gamma'(t)|},$$

where $\gamma : [a, b] \to \mathbb{R}^2$ has components $\gamma(t) = (x(t), y(t))$. Equivalently, in coordinates, the unit tangent vector T is

$$T(t) = \left(\frac{x'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2}}, \frac{y'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2}} \right)$$

for any $t \in [a, b]$ such that $x(t)$, $y(t)$ are differentiable with continuous derivatives, and $\big(x'(t)\big)^2 + \big(y'(t)\big)^2 \neq 0$.

The *unit normal* corresponding to T (the *outward* unit normal) is

$$N(t) = \left(\frac{y'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2}}, \frac{-x'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2}} \right). \tag{8.33}$$

For example, let D be the closed unit disc centred at the origin of $\mathbb{R}^2$. Its *counter-clockwise* boundary can be represented by

$$\begin{cases} x = \cos t \\ y = \sin t, \end{cases} \qquad t \in [0, 2\pi],$$

(continued)

so by (8.33) the unit normal

$$N(t) = (\cos t, \sin t)$$

points outside the disc D.

Having recalled the above concepts and notations, we are ready for the *Gauss-Green theorem*, whereby a double integral over a regular domain is transformed into a line integral along the domain's boundary.

Gauss-Green Formulas *Let $D \subset \mathbb{R}^2$ be a regular domain and $f = f(x, y)$ a function of class $C^1(D)$. Then*

$$\iint_D \frac{\partial f}{\partial x}\, dx\, dy = \int_{+\partial D} f\, dy\,; \tag{8.34}$$

$$\iint_D \frac{\partial f}{\partial y}\, dx\, dy = -\int_{+\partial D} f\, dx\,. \tag{8.35}$$

Proof We shall only prove (8.34), since (8.35) is completely similar. For simplicity we also suppose f is C^1 on an open set A containing D, which enables us to differentiate integral functions also along the boundary of D.

Let us begin with the case where D is normal with respect to y:

$$D = \{(x, y) \in \mathbb{R} \times [c, d] :\ \gamma(y) \le x \le \delta(y)\}\,.$$

By the reduction formulas (previous section)

$$\iint_D \frac{\partial f}{\partial x}\, dx\, dy = \int_c^d dy \int_{\gamma(y)}^{\delta(y)} \frac{\partial f}{\partial x}\, dx = \int_c^d \{f(\delta(y), y) - f(\gamma(y), y)\}\, dy\,. \tag{8.36}$$

Let γ_1, γ_2, γ_3, γ_4 be the curves bounding D (Fig. 8.10). The claim follows from formula (8.36) by observing

$$\int_{+\partial D} f\, dy = \int_{\gamma_3} f\, dy + \int_{\gamma_1} f\, dy = \int_c^d \{f(\delta(y), y) - f(\gamma(y), y)\}\, dy\,.$$

If D is normal with respect to x instead,

$$D = \{(x, y) \in [a, b] \times \mathbb{R} :\ \alpha(x) \le y \le \beta(x)\}\,,$$

with α, β of class $C^1([a, b])$, we introduce the integral function

$$F(x, y) = \int_{\gamma_{x,y}} f\, dy = \int_a^x f(t, \alpha(t))\alpha'(t)\, dt + \int_{\alpha(x)}^y f(x, t)\, dt\,,$$

where, for any x, $y \in D$, $\gamma_{x,y}$ is the curve of Fig. 8.11.

Fig. 8.10

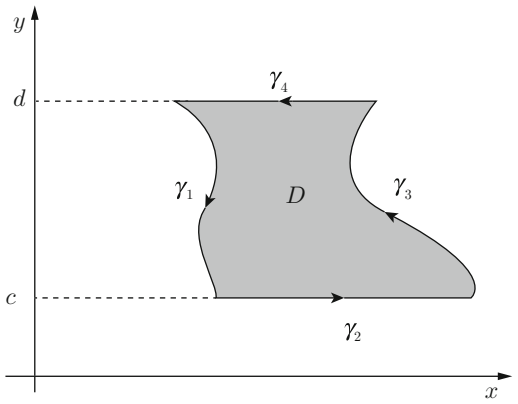
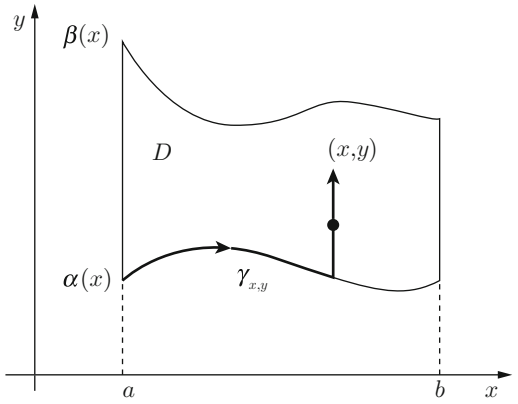


Fig. 8.11



By the formulas of Sect. 3.10, the partial derivatives of F are

$$\frac{\partial F}{\partial x} = f(x, \alpha(x))\alpha'(x) + \int_{\alpha(x)}^{y} \frac{\partial f}{\partial x}(x,t)dt - f(x, \alpha(x))\alpha'(x) = \int_{\alpha(x)}^{y} \frac{\partial f}{\partial x}(x,t)\, dt; \quad (8.37)$$

$$\frac{\partial F}{\partial y} = f(x, y)\,. \quad (8.38)$$

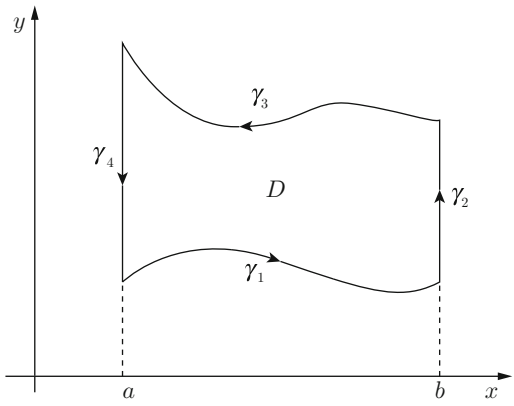
But the differential form $F_x\, dx + F_y\, dy$ is exact, so its line integral along the boundary of D is zero (see Theorem 5, Sect. 7.3). Hence

$$\int_{+\partial D} \frac{\partial F}{\partial x}\, dx + \frac{\partial F}{\partial y}\, dy = 0\,. \quad (8.39)$$

If γ_1, γ_2, γ_3, γ_4 are the curves in Fig. 8.12, from (8.37) we deduce

$$\int_{+\partial D} \frac{\partial F}{\partial x}\, dx = \int_{\gamma_3} \frac{\partial F}{\partial x}\, dx = -\int_a^b dx \int_{\alpha(x)}^{\beta(x)} \frac{\partial f}{\partial x}(x,t)\, dt\,.$$

Fig. 8.12



Then, using (8.38) and (8.39) we obtain

$$\int_a^b dx \int_{\alpha(x)}^{\beta(x)} \frac{\partial f}{\partial x}(x,t)\,dt = \int_{+\partial D} f(x,y)\,dy\,.$$

Therefore (8.34) follows from the reduction formulas, since

$$\iint_D \frac{\partial f}{\partial x}\,dx\,dy = \int_a^b dx \int_{\alpha(x)}^{\beta(x)} \frac{\partial f}{\partial x}(x,t)\,dt\,.$$

Finally, let D be a *regular* domain, i.e. the union of finitely many regular normal domains (with respect to x or y) $D_1, D_2, \ldots, D_N$, whose pairwise intersections have empty interiors (see Fig. 8.13; note that each D_i, besides normal, is also regular).

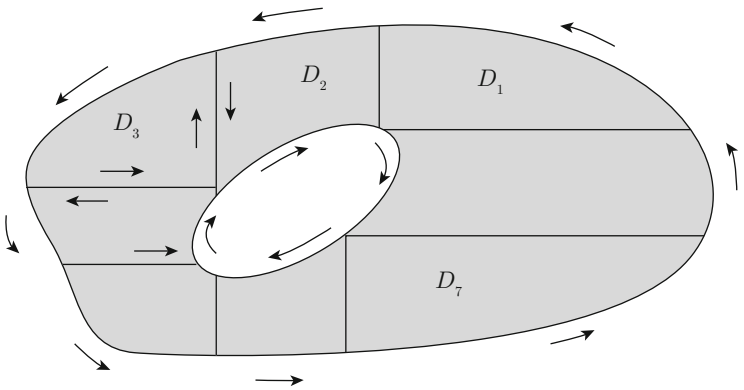


Fig. 8.13

Evidently the double integral in the Gauss-Green formula (8.34) is the sum of the corresponding double integrals over the D_i, and therefore

$$\iint_D \frac{\partial f}{\partial x}\, dx\, dy = \sum_{i=1}^{N} \iint_{D_i} \frac{\partial f}{\partial x}\, dx\, dy = \sum_{i=1}^{N} \int_{+\partial D_i} f\, dy\,. \tag{8.40}$$

Now, the boundary arcs of adjacent domains appear twice in the right-hand sum of (8.40) but with opposite orientation. Hence they cancel each other out, and

$$\sum_{i=1}^{N} \int_{+\partial D_i} f\, dy = \int_{+\partial D} f\, dy\,,$$

returns (8.34), because of (8.40).

In Chap. 12 we will prove a suitable generalisation of the Gauss-Green theorem for domains in $\mathbb{R}^n$ (while the three-dimensional case is addressed in Chap. 10).

The Gauss-Green formulas imply the divergence theorem.

Divergence Theorem *Let D be a regular domain in the plane and $F = (F_1, F_2)$ a map from D to $\mathbb{R}^2$ of class $C^1(D)$. Then*

$$\iint_D \operatorname{div} F\; dx\, dy = \int_{\partial D} (F, N)\, ds\,,$$

where

$$\operatorname{div} F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}$$

is the divergence of the vector $F(x, y) = (F_1(x, y), F_2(x, y))$, (F, N) is the inner product of F and the outward unit normal N to ∂D, and s is the arclength parameter of the boundary of D.

Proof We use the Gauss-Green formulas, to be precise: (8.34) with F_1 replacing f, and (8.35) with F_2 in place of f. Adding the two relations produces

$$\iint_D \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx\, dy = \int_{+\partial D} -F_2\, dx + F_1\, dy\,. \tag{8.41}$$

If the boundary of D is a piecewise-regular curve of parametric equations $x = x(t)$, $y = y(t)$, $t \in [a, b]$, and the induced orientation is that of $+\partial D$, the outward unit normal N is (8.33). Then by definition of line integral

$$\int_{\partial D} (F, N)\, ds = \int_a^b \left(\frac{F_1 y'}{\sqrt{(x')^2 + (y')^2}} - \frac{F_2 x'}{\sqrt{(x')^2 + (y')^2}} \right) \sqrt{(x')^2 + (y')^2}\, dt = \tag{8.42}$$
$$= \int_a^b \left(F_1 y' - F_2 x' \right) dt = \int_{+\partial D} F_1\, dy - F_2\, dx\,.$$

Comparing (8.41) with (8.42) proves the claim when ∂D is a piecewise-regular curve. If ∂D is a finite union of piecewise-regular curves (e.g., in an annulus) we argue as in the final part of the proof of the Gauss-Green formulas: we divide D in regular normal domains without common interior points.

The divergence theorem can be formulated as follows.

Stokes's Formula *Let $F : D \to \mathbb{R}^2$ be a C^1 map on the regular domain $D \subset \mathbb{R}^2$. Then*

$$\int_{+\partial D} F_1\, dx + F_2\, dy = \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx\, dy\,. \tag{8.43}$$

Proof It suffices to apply (8.41) to the vector-valued function $D \to \mathbb{R}^2$ of components $(F_2, -F_1)$.

As a corollary to the Stokes formula we shall prove a result on differential forms, which was stated in Chap. 7 (Theorem 2 in Sect. 7.4).

Theorem (Differential Forms on Simply Connected Open Sets in $\mathbb{R}^2$) *Let $\omega = a\, dx + b\, dy$ be a 1-form of class C^1 defined on a simply connected open set A. If the form is closed, i.e.*

$$\frac{\partial b}{\partial x} = \frac{\partial a}{\partial y}\,,$$

then it is exact on A.

Proof We must show that for any curve γ, piecewise regular, closed and contained in A,

$$\int_\gamma \omega = 0\,, \tag{8.44}$$

and then apply Theorem 2 (characterising exact forms) from Sect. 7.3. For simplicity we shall prove (8.44) when γ is additionally a simple curve.

Given γ, let D be the bounded domain with boundary γ (the existence of such a D, albeit intuitively evident, can be proved rigorously; the result is known as *Jordan's curve theorem*). Moreover, it is not restrictive to assume γ oriented counter-clockwise. Then

$$\int_\gamma \omega = \int_\gamma ((a(x,y), b(x,y)), T(x,y))ds = \int_{\partial D} ((b(x,y), -a(x,y)), N(x,y))ds\,,$$

where $T(x,y)$ and $N(x,y)$ are the unit tangent and normal vectors of γ, hence of the boundary ∂D. The unit normal N points outside of D. The divergence theorem tells

$$\int_\gamma \omega = \iint_D \left(\frac{\partial b}{\partial x} - \frac{\partial a}{\partial y}\right) dx\,dy\,,$$

and the double integral is zero since the form is closed. Hence (8.44) is proved, and the assertion with it.

Let us now present a few applications of the Gauss-Green formulas.

Theorem (Integration by Parts) *If f and g are C^1 functions on the regular domain $D \subset \mathbb{R}^2$, then*

$$\iint_D f\frac{\partial g}{\partial x}\,dx\,dy = \int_{+\partial D} fg\,dy - \iint_D \frac{\partial f}{\partial x} g\,dx\,dy\,;$$

$$\iint_D f\frac{\partial g}{\partial y}\,dx\,dy = -\int_{+\partial D} fg\,dy - \iint_D \frac{\partial f}{\partial y} g\,dx\,dy\,.$$

Proof It is enough to apply the Gauss-Green formulas (8.34) and (8.35) with fg in place of f.

If D is a regular domain in $\mathbb{R}^2$, its area $m(D)$ is the double integral over D of the constant function 1:

$$m(D) = \iint_D dx\,dy\,.$$

The next theorem gives a formula for the area $m(D)$ of a regular domain $D \subset \mathbb{R}^2$. The expression is particularly useful when D is defined by some parametrisation of the boundary.

Area Formulas *If D is a regular domain in $\mathbb{R}^2$ then*

$$m(D) = \int_{+\partial D} x\,dy = -\int_{+\partial D} y\,dx\,. \tag{8.45}$$

For any pair $\alpha, \beta \in \mathbb{R}$ such that $\alpha + \beta \neq 0$, moreover, we have

$$m(D) = \frac{1}{\alpha + \beta} \int_{+\partial D} \alpha x \, dy - \beta y \, dx \, ; \tag{8.46}$$

in particular

$$m(D) = \frac{1}{2} \int_{+\partial D} x \, dy - y \, dx \, . \tag{8.47}$$

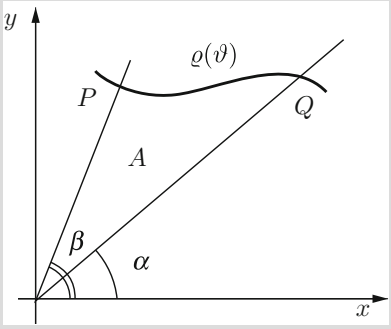
Proof To obtain (8.45) we just apply (8.34) to $f(x, y) = x$ and (8.35) to $f(x, y) = y$. Formula (8.46) is a linear combination of (8.45). Formula (8.47) is a consequence of (8.46) when $\alpha = \beta = 1$.

Let us apply the previous formulas to calculate the area of a *sector* in the plane in polar coordinates. Precisely, let $[\alpha, \beta]$ be a subinterval of $[0, 2\pi]$ and $\varrho = \varrho(\vartheta)$ a positive continuous function on $[\alpha, \beta]$. The set D of points (ϱ, ϑ) in polar coordinates satisfying

$$\alpha \leq \vartheta \leq \beta, \qquad 0 \leq \varrho \leq \varrho(\vartheta) \, ,$$

is called a *sector* (Fig. 8.14).

Fig. 8.14



Call Q and P the points of polar coordinates $(\varrho(\alpha), \alpha)$ and $(\varrho(\beta), \beta)$ respectively. The arc PQ of the boundary of D has parametric equations

$$\begin{cases} x = \varrho(\vartheta) \cos \vartheta \\ y = \varrho(\vartheta) \sin \vartheta \, , \end{cases} \qquad \vartheta \in [\alpha, \beta].$$

As ϑ goes from α to β, the arc is oriented counter-clockwise, corresponding to the positive orientation. If ϱ is piecewise C^1, we obtain

$$x \, dy - y \, dx = \varrho \cos \vartheta \big(\varrho' \sin \vartheta + \varrho \cos \vartheta\big) - \varrho \sin \vartheta \big(\varrho' \cos \vartheta - \varrho \sin \vartheta\big) =$$
$$= \varrho^2 \Big(\cos^2 \vartheta + \sin^2 \vartheta\Big) = \varrho^2$$

(continued)

along the curve $\varrho = \varrho(\vartheta)$, while along the line segments OQ and OP we have $x\,dy - y\,dx = 0$. Then from (8.47)

$$m(D) = \frac{1}{2}\int_{+\partial D} x\,dy - y\,dx = \frac{1}{2}\int_{\alpha}^{\beta} \varrho^2(\vartheta)\,d\vartheta\,.$$

8.4 Variable Change in Double Integrals

To calculate integrals in one variable the *substitution method* is often useful for simplifying the integrand's expression. There exist practical changes of the variables also for double integrals, which turn out to be helpful especially to "*simplify*" the integration domain.

Let T be a *regular domain* in the plane with coordinates u, v. Given two functions

$$\begin{cases} x = x(u, v) \\ y = y(u, v)\,, \end{cases} \qquad (u, v) \in T,$$

of class $C^1(T)$, we denote by Φ the map

$$\Phi : (u, v) \in T \to (x(u, v), y(u, v))$$

and let $D = \Phi(T)$ be the image of Φ. Indicate by

$$\det \frac{\partial(x, y)}{\partial(u, v)}$$

the *Jacobian determinant* of Φ (see Sect. 3.14), i.e. the determinant of the Jacobian matrix of Φ:

$$\det \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\,.$$

Variable Change in Double Integrals *Let T, D be regular domains in $\mathbb{R}^2$. Suppose $\Phi : T \to D$ is an invertible C^1 map with non-zero Jacobian determinant on T. Then for any continuous function $f : D = \Phi(T) \to \mathbb{R}$*

$$\iint_{\Phi(T)} f(x, y)\,dx\,dy = \iint_T f(x(u, v), y(u, v)) \left|\det \frac{\partial(x, y)}{\partial(u, v)}\right| du\,dv\,. \qquad (8.48)$$

The above hypotheses are equivalent to the following set of conditions:

(i) $\Phi : T \to \mathbb{R}^2$ is a C^1 map on the regular domain T;

(ii) the Jacobian determinant of Φ is non-zero on T;
(iii) the map Φ induces a 1-1 correspondence between the boundary of T and the boundary of $D = \Phi(T)$.

Under the above hypotheses, in fact, not only T, but also D is a domain (see Proposition 2, Sect. 11.3). Moreover, Φ is invertible (compare with the global inverse function theorem of Sect. 11.3). Finally, it can be proved that D is regular too.

Put $f(x, y) \equiv 1$ in (8.48), so to obtain the formula for the *area of the image* $\Phi(T)$

$$m(\Phi(T)) = \iint_T \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv\,. \tag{8.49}$$

The Jacobian determinant has then the geometric meaning of the *area element* of the image under Φ. If $(u_0, v_0) \in T$, in fact, (8.49) implies

$$\left| \det \frac{\partial(x, y)}{\partial(u, v)}(u_0, v_0) \right| = \lim_{r \to 0} \frac{m(\Phi(I_r))}{m(I_r)}\,,$$

where I_r is the circle with centre (u_0, v_0) and radius r.

Proof of the Variable Change in Double Integrals To simplify the proof we shall assume some extra regularity on f and Φ. More precisely, we suppose $f \in C^1(\mathbb{R}^2)$ and that Φ is C^2. Letting $F = F(x, y)$ be a primitive of $f(x, y)$ in x, i.e.

$$\frac{\partial F}{\partial x} = f\,,$$

it is straightforward to see that F is $C^1(\mathbb{R}^2)$.

By Proposition 1 in Sect. 11.3 the boundary ∂D is in 1-1 correspondence with ∂T. Suppose finally that ∂T is a piecewise-regular curve parametrised by

$$\begin{cases} u = u(t) \\ v = v(t)\,, \end{cases} \qquad t \in [a, b].$$

Then the composite parametric equations of ∂D are

$$\begin{cases} x = x(u(t), v(t)) \\ y = y(u(t), v(t))\,, \end{cases} \qquad t \in [a, b]. \tag{8.50}$$

By the Gauss-Green theorem

$$\begin{aligned} \iint_D f(x, y)\, dx\, dy = \iint_D \frac{\partial F}{\partial x}\, dx\, dy = \int_{+\partial D} F\, dy = \\ = \pm \int_a^b F \frac{dy}{dt} dt = \pm \int_a^b F \left(\frac{\partial y}{\partial u}\frac{du}{dt} + \frac{\partial y}{\partial v}\frac{dv}{dt} \right) dt\,, \end{aligned} \tag{8.51}$$

where we choose the $+$ or $-$ sign depending on whether the orientation of ∂D given by (8.50) coincides with $+\partial D$ or not. We postpone the discussion of the choice of sign to the end of the proof.

By definition of line integral, from (8.51) we obtain

$$\iint_D f(x,y)\,dx\,dy = \pm \int_{+\partial T} F\frac{\partial y}{\partial u}du + F\frac{\partial y}{\partial v}\,dv\,. \tag{8.52}$$

Furthermore,

$$\begin{aligned} f(x(u,v),y(u,v))\det\frac{\partial(x,y)}{\partial(u,v)} &= \frac{\partial F}{\partial x}\left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v}-\frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right) = \\ &= \frac{\partial F}{\partial x}\left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v}-\frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right)+\frac{\partial F}{\partial y}\left(\frac{\partial y}{\partial u}\frac{\partial y}{\partial v}-\frac{\partial y}{\partial v}\frac{\partial y}{\partial u}\right) = \\ &= \left(\frac{\partial F}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial F}{\partial y}\frac{\partial y}{\partial u}\right)\frac{\partial y}{\partial v}-\left(\frac{\partial F}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial F}{\partial y}\frac{\partial y}{\partial v}\right)\frac{\partial y}{\partial u} = \\ &= \frac{\partial F}{\partial u}\frac{\partial y}{\partial v}-\frac{\partial F}{\partial v}\frac{\partial y}{\partial u} = \frac{\partial}{\partial u}\left(F\frac{\partial y}{\partial v}\right)-\frac{\partial}{\partial v}\left(F\frac{\partial y}{\partial u}\right). \end{aligned}$$

In the last equality above we have used the identity

$$\frac{\partial^2 y}{\partial v\,\partial u} = \frac{\partial^2 y}{\partial u\,\partial v}\,,$$

warranted by Φ being C^2. Now, applying again the Gauss-Green theorem,

$$\begin{aligned} &\iint_T f(x(u,v),y(u,v))\det\frac{\partial(x,y)}{\partial(u,v)}\,du\,dv = \\ &= \iint_T \left\{\frac{\partial}{\partial u}\left(F\frac{\partial y}{\partial v}\right)-\frac{\partial}{\partial v}\left(F\frac{\partial y}{\partial u}\right)\right\}du\,dv = \int_{+\partial T} F\frac{\partial y}{\partial u}\,du + F\frac{\partial y}{\partial v}\,dv\,. \end{aligned} \tag{8.53}$$

At last, comparing (8.52) and (8.53) we find, for a suitable choice of sign,

$$\iint_{\Phi(T)} f(x,y)\,dx\,dy = \pm\iint_T f(x(u,v),y(u,v))\det\frac{\partial(x,y)}{\partial(u,v)}\,du\,dv\,. \tag{8.54}$$

In order to understand the right-hand side of (8.48) recall that the Jacobian determinant is by assumption continuous on T and never zero, so it will be everywhere either positive or negative.

But *the choice of sign is independent of* f (it only depends on the orientation of the curve in (8.50)), so we may take $f(x,y)\equiv 1$. Then (8.54) reads

$$m(D) = \pm\iint_T \det\frac{\partial(x,y)}{\partial(u,v)}\,du\,dv\,. \tag{8.55}$$

On the left we have a positive quantity, so if the determinant is positive in (8.55), hence in (8.54), we must choose the $+$ sign. Conversely, if the determinant is negative we must choose the $-$ sign. In any case (8.54) becomes

$$\iint_{\Phi(T)} f(x,y)\,dx\,dy = \iint_T f(x(u,v),y(u,v))\left|\det\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv\,.$$

The variable-change formula (8.48) holds under more general assumptions than the given ones. These assumptions cover, as a special case, the important variable change from Cartesian to polar coordinates. To this end let us introduce the following hypotheses:

(j) the functions $x = x(u,v)$ and $y = y(u,v)$ are of class $C^1(T)$, where T is a finite union of normal domains in the uv-plane, and the image $D = \Phi(T)$ of $\Phi : (u,v) \in T \to (x(u,v), y(u,v))$ is a union of normal domains in the xy-plane;
(jj) there exists a sequence $T_k \subseteq T$ of regular domains such that the restriction of Φ to T_k satisfies the formula of variable change;
(jjj) putting $D_k = \Phi(T_k)$, we have

$$\lim_{k\to+\infty} m(D_k) = m(D)\,, \qquad \lim_{k\to+\infty} m(T_k) = m(T)\,.$$

Theorem *Under assumptions* (j), (jj), (jjj), *the variable-change formula* (8.48) *holds for any continuous function* $f : D \to \mathbb{R}$.

Proof This is a direct application of the previous result governing variable changes in double integrals. Putting

$$M = \max\{|f(x,y)| \,:\, (x,y) \in D\}$$

we have

$$\left|\iint_D f(x,y)\,dx\,dy - \iint_{D_k} f(x,y)\,dx\,dy\right| \le M \cdot m(D - D_k)\,,$$

so (jjj) implies

$$\lim_{k\to+\infty} \iint_{D_k} f(x,y)\,dx\,dy = \iint_D f(x,y)\,dx\,dy\,. \tag{8.56}$$

Similarly one proves

$$\begin{aligned}\lim_{k\to+\infty}\iint_{T_k} f(x(u,v),y(u,v))\left|\det\frac{\partial(x,y)}{\partial(u,v)}\right|du\,dv = \\ = \iint_{T} f(x(u,v),y(u,v))\left|\det\frac{\partial(x,y)}{\partial(u,v)}\right|du\,dv\,.\end{aligned} \tag{8.57}$$

As the double integrals in (8.56), (8.57), over D_k and T_k respectively, are equal, the claim follows, i.e. the corresponding double integrals on D and T coincide.

Notice that (j), (jj), (jjj) do not exclude the possibility that the Jacobian of Φ might vanish on a set "of measure zero", for example a regular curve contained in T, or that Φ might not be bijective. Let us check what we have claimed in the next important example.

Consider the circle D with centre the origin O and radius $r > 0$ in the xy-plane. Define the *polar coordinates* ϱ, ϑ, where O is called *pole* (or *centre*) and the x-axis is the *polar axis*. The formulas for changing coordinates are

$$\begin{cases} x = \varrho\cos\vartheta \\ y = \varrho\sin\vartheta\,, \end{cases} \qquad \varrho\in[0,r],\ \ \vartheta\in[0,2\pi].$$

The map $\Phi : (\varrho,\vartheta)\to(x,y)$ is not a bijection between

$$T = \{(\varrho,\vartheta)\in\mathbb{R}^2 :\ 0\le\varrho\le r,\ 0\le\vartheta\le 2\pi\}$$

and the image $D = \Phi(T)$, since $O = (0,0)\in D$ corresponds to every point $(\varrho,\vartheta) = (0,\vartheta)$ of T when $\vartheta\in[0,2\pi]$. Furthermore, the Jacobian determinant

$$\det\frac{\partial(x,y)}{\partial(\varrho,\vartheta)} = \begin{vmatrix}\cos\vartheta & -\varrho\sin\vartheta \\ \sin\vartheta & \varrho\cos\vartheta\end{vmatrix} = \varrho$$

vanishes along the edge $\varrho = 0$ of T. However, as shown in Fig. 8.15, setting

$$T_k = \left[\frac{1}{k}, r\right]\times\left[0, 2\pi - \frac{1}{k}\right],$$
$$D_k = \Phi(T_k)$$

for any $k\in\mathbb{N}$, we have

$$m(T_k) = \left(r-\frac{1}{k}\right)\left(2\pi-\frac{1}{k}\right),$$
$$m(D_k) = \pi r^2 - \frac{\pi}{k^2} - \frac{\pi}{2k}\left(r^2-\frac{1}{k^2}\right).$$

Hence

$$\lim_{k\to+\infty} m(T_k) = 2\pi r = m(T), \qquad \lim_{k\to+\infty} m(D_k) = \pi r^2 = m(D)\,,$$

(continued)

so (jjj) holds. It is not hard to show that also (j), (jj) are valid. By the previous theorem we obtain the *formula for passing from Cartesian to polar coordinates*

$$\iint_D f(x, y)\, dx\, dy = \iint_T f(\varrho \cos \vartheta, \varrho \sin \vartheta)\, \varrho\, d\varrho\, d\vartheta\,.$$

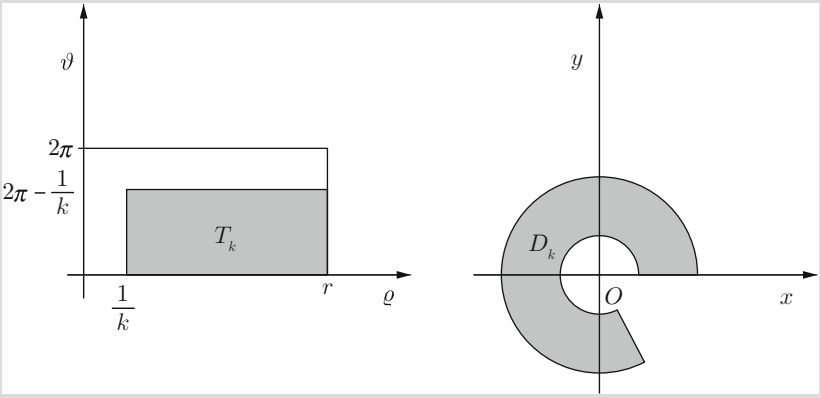


Fig. 8.15

Using the reduction formulas,

$$\iint_D f(x, y)\, dx\, dy = \int_0^{2\pi} d\vartheta \int_0^r f(\varrho \cos \vartheta, \varrho \sin \vartheta)\, \varrho\, d\varrho = \\ = \int_0^r \varrho\, d\varrho \int_0^{2\pi} f(\varrho \cos \vartheta, \varrho \sin \vartheta)\, d\vartheta\,.$$

More generally, one can prove that if T is the set in the plane

$$T = \{(\varrho, \vartheta) \in \mathbb{R}^2 :\ \alpha \le \vartheta \le \beta, \quad \varphi(\vartheta) \le \varrho \le \psi(\vartheta)\}\,,$$

and $D = \Phi(T)$ is as before, then

$$\iint_D f(x, y)\, dx\, dy = \int_\alpha^\beta d\vartheta \int_{\varphi(\vartheta)}^{\psi(\vartheta)} f(\varrho \cos \vartheta, \varrho \sin \vartheta)\, \varrho\, d\varrho\,.$$

8.5 Triple Integrals

We want to explain now an easy way to define the (triple) integral of functions $f = f(x, y, z)$ of three real variables over a *normal domain in* $\mathbb{R}^3$. In the rest of the section we will present the main properties (reduction formulas and variable change) without proof, due to the similarities with double integrals.

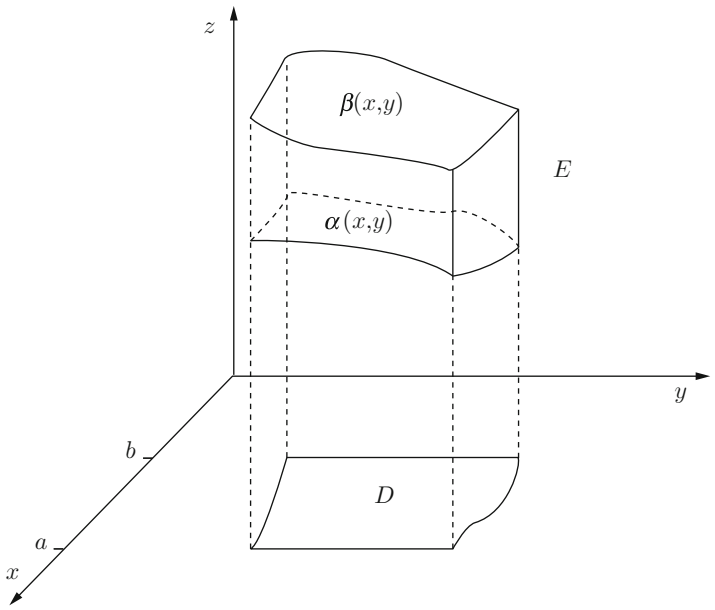


Fig. 8.16

Let D be a normal domain with respect to the xy-plane and $\alpha = \alpha(x, y), \beta = \beta(x, y)$ two continuous functions on D, such that

$$\alpha(x, y) \leq \beta(x, y)\,, \qquad \forall\,(x, y) \in D.$$

The set in $\mathbb{R}^3$ (see Fig. 8.16)

$$E = \{(x, y, z) \in D \times \mathbb{R} :\ \alpha(x, y) \leq z \leq \beta(x, y)\} \tag{8.58}$$

is called a *normal domain with respect to the xy-plane*.

The *volume* of $E \subset \mathbb{R}^3$, or *measure* of E, written $m(E)$, is given by the double integral

$$m(E) = \iint_D (\beta(x, y) - \alpha(x, y))\, dx\, dy\,.$$

Normal domains with respect to the yz-plane or the xz-plane are defined similarly. We shall use the generic term *normal domain in* $\mathbb{R}^3$ when there is no need to specify the variables of reference.

We shall also say E is a regular normal domain if D is a regular normal domain in the plane and the functions α, β are C^1 on D.

Let E be a normal domain in $\mathbb{R}^3$, for example with respect to the xy-plane. Let $f : E \subset \mathbb{R}^3 \to \mathbb{R}$ be a *bounded* function on E. For any partition $P =$

$\{E_1, E_2, \dots, E_h\}$ of E in normal domains with respect to the xy-plane define the *integral sums* $s(P)$, $S(P)$:

$$s(P) = \sum_{i=1}^{h} m(E_i) \inf\{f(x) : \ x \in E_i\};$$

$$S(P) = \sum_{i=1}^{h} m(E_i) \sup\{f(x) : \ x \in E_i\}.$$

They are respectively called *lower (integral) sum* and *upper (integral) sum* relative to the partition P.

Exactly as in two dimensions, the numerical sets

$$\{s(P_1)\}, \qquad \{S(P_2)\},$$

as the partitions P_1, P_2 of the normal domain E (with respect to the xy-plane) vary, are *separated*. That is, $s(P_1) \le S(P_2)$ for any partitions P_1, P_2 made of normal domains with respect to the xy-plane. If these sets are *contiguous* as well, i.e.

$$\sup_{P_1}\{s(P_1)\} = \inf_{P_2}\{S(P_2)\},$$

the function f is said to be *integrable* on E, and the separating element is the *triple integral* of $f(x, y, z)$ over $E \subset \mathbb{R}^3$, written

$$\iiint_E f(x, y, z)\, dx\, dy\, dz.$$

As for functions of two variables, also continuous functions in three variables are integrable on normal domains of $\mathbb{R}^3$. Namely,

Integrability of Continuous Functions *Let E be a normal domain in $\mathbb{R}^3$ and $f : E \subset \mathbb{R}^3 \to \mathbb{R}$ a continuous function. Then f is integrable on E.*

Continuous functions satisfy the *reduction formulas*. For example, if $f(x, y, z)$ is continuous on a domain E of $\mathbb{R}^3$, normal with respect to the xy-plane, as in (8.58), then we have a reduction formula (of the triple integral to a double integral)

$$\iiint_E f(x, y, z)\, dx\, dy\, dz = \iint_D dx\, dy \int_{\alpha(x,y)}^{\beta(x,y)} f(x, y, z)\, dz.$$

Suppose $D \subset \mathbb{R}^2$ itself is normal with respect to x:

$$D = \left\{(x, y) \in \mathbb{R}^2 : \ a \le x \le b, \ g(x) \le y \le h(x)\right\},$$

with g, h continuous on $[a, b]$. Then the double integral over D reduces to two integrals in one variable, resulting in

$$\iiint_E f(x, y, z)\, dx\, dy\, dz = \int_a^b dx \int_{g(x)}^{h(x)} dy \int_{\alpha(x,y)}^{\beta(x,y)} f(x, y, z)\, dz\,.$$

We shall just mention the *variable change* for triple integrals, and consider only two particular significant cases. As for two dimensions, a domain E in space is called *regular* if it can be split into a finite union of regular normal domains whose pairwise intersections have empty interiors.

Let T, E be regular domains in $\mathbb{R}^3$. Suppose

$$x = x(u, v, w)\,, \quad y = y(u, v, w)\,, \quad z = z(u, v, w) \tag{8.59}$$

are of class $C^1(T)$, and that the map $\Phi : (u, v, w) \in T \to (x, y, z) \in \Phi(T) = E$ defined by (8.59) is invertible with non-zero Jacobian determinant

$$\det \frac{\partial(x, y, z)}{\partial(u, v, w)}.$$

Then

$$\begin{aligned} &\iiint_{\Phi(T)} f(x, y, z)\, dx\, dy\, dz = \\ &= \iiint_T f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \det \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du\, dv\, dw\,. \end{aligned} \tag{8.60}$$

Let us discuss two special cases. First, for the *spherical coordinates* defined by the transformation

$$\begin{cases} x = \varrho \sin\varphi \cos\vartheta \\ y = \varrho \sin\varphi \sin\vartheta \\ z = \varrho \cos\varphi \end{cases} \tag{8.61}$$

the Jacobian determinant is easily

$$\det \frac{\partial(x, y, z)}{\partial(\varrho, \varphi, \vartheta)} = \varrho^2 \sin\varphi\,.$$

Consider the solid E in Fig. 8.17, image under (8.61) of the domain

$$T = \{(\varrho, \vartheta, \varphi) : \ \varrho_1 \le \varrho \le \varrho_2 \,, \ \vartheta_1 \le \vartheta \le \vartheta_2 \,, \ \varphi_1 \le \varphi \le \varphi_2\} \,.$$

If $f = f(x, y, z)$ is the continuous function

$$f(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$$

for $(x, y, z) \neq (0, 0, 0)$, by formula (8.60)

$$\iiint_E f(x, y, z)\, dx\, dy\, dz =$$
$$= \iiint_T f(\varrho \sin\varphi \cos\vartheta, \varrho \sin\varphi \sin\vartheta, \varrho \cos\varphi)\, \varrho^2 \sin\varphi \, d\varrho \, d\vartheta \, d\varphi \,.$$

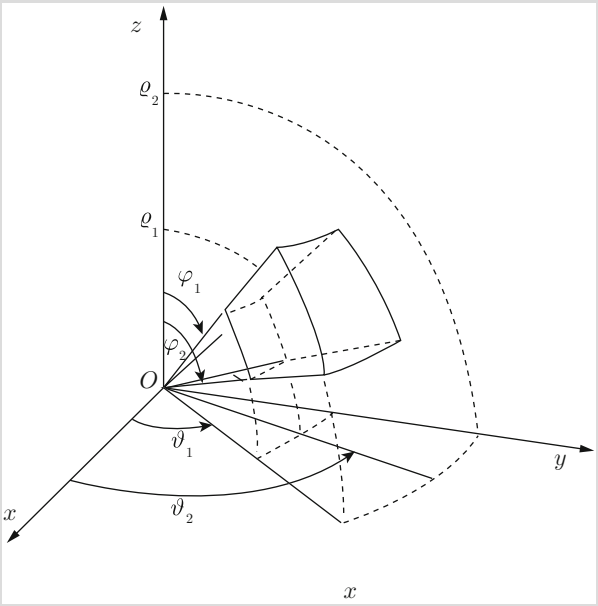


Fig. 8.17

After the substitution, the reduction formulas yield

$$\iiint_E f(x, y, z)\, dx\, dy\, dz = \iiint_T \sin\varphi \, d\varrho \, d\vartheta \, d\varphi =$$
$$= \int_{\varrho_1}^{\varrho_2} d\varrho \int_{\vartheta_1}^{\vartheta_2} d\vartheta \int_{\varphi_1}^{\varphi_2} \sin\varphi \, d\varphi = (\varrho_2 - \varrho_1)(\vartheta_2 - \vartheta_1)(\cos\varphi_1 - \cos\varphi_2) \,.$$

Let us finally introduce *cylindrical coordinates*, defined by the transformation

$$\begin{cases} x = \varrho \cos \vartheta \\ y = \varrho \sin \vartheta \\ z = z \,, \end{cases} \tag{8.62}$$

whose Jacobian determinant equals

$$\det \frac{\partial(x, y, z)}{\partial(\varrho, \vartheta, z)} = \varrho \,.$$

Let E be the image under (8.62) of the solid domain

$$T = \{(\varrho, \vartheta, z) : \ \varrho_1 \le \varrho \le \varrho_2, \ \vartheta_1 \le \vartheta \le \vartheta_2, \ z_1 \le z \le z_2\}$$

in $\varrho\vartheta z$-space. For any continuous function $f : E \to \mathbb{R}$,

$$\iiint_E f(x, y, z)\, dx\, dy\, dz = \iiint_T f(\varrho \cos \vartheta, \varrho \sin \vartheta, z)\ \varrho\, d\varrho\, d\vartheta\, dz \,. \tag{8.63}$$

If E is bounded by the cylinder $x^2 + y^2 = x$ and the cone $x^2 + y^2 = z^2$, to compute

$$\iiint_D \left(x^2 + y^2\right)^{1/2} dx\, dy\, dz$$

we pass to cylindrical coordinates. In this system the cylinder has equation $\varrho = \cos \vartheta$ and the cone $\varrho = |z|$. The corresponding solid E is the image under (8.62) of the region

$$T = \left\{(\varrho, \vartheta, z) : \ 0 \le \varrho \le \cos \vartheta, \ -\frac{\pi}{2} \le \vartheta \le \frac{\pi}{2}, \ -\varrho \le z \le \varrho\right\}.$$

From (8.63) we then obtain

$$\begin{aligned} &\iiint_E \left(x^2 + y^2\right)^{1/2} dx\, dy\, dz = \iiint_T \varrho \cdot \varrho\, d\varrho\, d\vartheta\, dz = \\ &= \int_{-\pi/2}^{\pi/2} d\vartheta \int_0^{\cos \vartheta} d\varrho \int_{-\varrho}^{\varrho} \varrho^2\, dz = 2 \int_{-\pi/2}^{\pi/2} d\vartheta \int_0^{\cos \vartheta} \varrho^3\, d\varrho = \\ &= \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos^4 \vartheta\, d\vartheta = \frac{3}{16}\pi \,. \end{aligned}$$

8.6 Peano-Jordan Measurable Subsets of $\mathbb{R}^n$

Starting from this section we begin the systematic study of *multiple integrals* in general, and present the theory of *Riemann integration*. We set off by introducing the *Peano-Jordan measure*.

Let $I = [a_1, b_1) \times [a_2, b_2) \times \ldots \times [a_n, b_n)$ (with $a_i \leq b_i$ for any $i \in \{1, 2, \ldots, n\}$) be a *(right) semi-open interval* in $\mathbb{R}^n$. We call (*elementary*) *measure* of I the positive number

$$m(I) = (b_1 - a_1) \cdot (b_2 - a_2) \cdot \ldots \cdot (b_n - a_n)$$

when I is non-empty. By convention, $m(\emptyset) = 0$. In the sequel, when it will be necessary to emphasise the dependence of the measure m on the dimension n, we shall also write $m_n(I)$.

We call *(right) semi-open pluri-interval* a finite union P of semi-open intervals as above. The (*elementary*) *measure* of such P is the non-negative number

$$m(P) = \sum_{r=1}^{h} m(I_r)\,, \tag{8.64}$$

where $\{I_1, I_2, \ldots, I_h\}$ is a partition of P made of semi-open intervals.

Although for brevity we shall not do it, it can be proved that if $\{J_1, J_2, \ldots, J_k\}$ is another partition of P into semi-open intervals, then

$$\sum_{r=1}^{h} m(I_r) = \sum_{r=1}^{k} m(J_r)\,;$$

the measure of P therefore is well defined, as it does not depend on the partition chosen for representing P.

If $\mathcal{P}$ is the set of *semi-open pluri-intervals* in $\mathbb{R}^n$, formula (8.64) defines a function

$$m : \mathcal{P} \to [0, +\infty)$$

that is *finitely additive* on $\mathcal{P}$, i.e.

$$P \cap P' = \emptyset \quad \Rightarrow \quad m(P \cup P') = m(P) + m(P')\,, \tag{8.65}$$

for any pluri-intervals P, P' of $\mathbb{R}^n$.

Let, in fact, $\{I_1, I_2, \ldots, I_h\}$ be a partition of P and $\{J_1, J_2, \ldots, J_k\}$ a partition of P', both made of semi-open intervals. Then by definition

$$m(P) = \sum_{r=1}^{h} m(I_r)\,; \qquad m(P') = \sum_{r=1}^{k} m(J_r)\,. \tag{8.66}$$

Since $\{I_1, I_2, \dots, I_h, J_1, J_2, \dots, J_k\}$ is a partition of the pluri-interval $P \cup P'$, still by definition we have

$$m(P \cup P') = \sum_{r=1}^{h} m(I_r) + \sum_{r=1}^{k} m(J_r)\,. \tag{8.67}$$

Now (8.65) follows from comparing (8.66) and (8.67).

Let us state two consequences of the finite additivity (8.65) of the measure m: for any $P, P' \in \mathcal{P}$,

$$m(P - P') = m(P) - m(P \cap P')\,; \tag{8.68}$$

$$m(P \cup P') = m(P) + m(P') - m(P \cap P')\,. \tag{8.69}$$

Formula (8.68) is proved observing that $P = (P - P') \cup (P \cap P')$ and $P - P'$ and $P \cap P'$ are disjoint elements in $\mathcal{P}$, so

$$m(P) = m(P - P') + m(P \cap P')\,.$$

For (8.69) we start from $P \cup P' = P' \cup (P - P')$, then use finite additivity and (8.68) to obtain

$$m(P \cup P') = m(P') + m(P - P') = m(P') + m(P) - m(P \cap P')\,.$$

In the special case where P' is contained in P, (8.68) becomes

$$m(P - P') = m(P) - m(P')\,.$$

It follows that the measure m is *increasing*, in the sense that

$$P' \subseteq P \quad \Rightarrow \quad m(P') \le m(P)\,.$$

Also notice that (8.69) implies that m is *subadditive*, i.e.

$$m(P \cup P') \le m(P) + m(P')$$

for any $P, P' \in \mathcal{P}$. Let now X be a *bounded* subset of $\mathbb{R}^n$. The *inner Peano-Jordan measure* $\underline{m}(X)$ and the *outer Peano-Jordan measure* $\overline{m}(X)$ of X are the quantities

$$\underline{m}(X) = \sup\{m(P) :\ P \in \mathcal{P}\,,\ P \subseteq X\}\,, \tag{8.70}$$

$$\overline{m}(X) = \inf\{m(P) :\ P \in \mathcal{P}\,,\ P \supseteq X\}\,, \tag{8.71}$$

respectively. In general,

$$\underline{m}(X) \leq \overline{m}(X)\,.$$

The set $X \subset \mathbb{R}^n$ is said to be *Peano-Jordan measurable* whenever

$$\underline{m}(X) = \overline{m}(X)\,;$$

then, the common value is denoted by $m(X)$ and called *Peano-Jordan measure of* X. Let $\mathcal{M}$ denote the collection of *bounded and Peano-Jordan measurable* subsets of $\mathbb{R}^n$.

If P is a semi-open pluri-interval, its Peano-Jordan measure coincides with (8.64). It is easy to prove that the closure $\overline{P}$ and the interior $\overset{\circ}{P}$ of P are measurable, and

$$m(\overset{\circ}{P}) = m(P) = m(\overline{P})\,.$$

If X is a bounded and measurable set, i.e. $X \in \mathcal{M}$, its measure $m(X)$ is clearly the separating element of two collections of elementary measures: the measures of pluri-intervals contained in X and the measures of pluri-intervals containing X. In other words, *X is measurable if and only if, for any $\varepsilon > 0$, there exist two semi-open pluri-intervals P', P'' such that*

$$P' \subseteq X \subseteq P'' \quad \text{and} \quad m(P'') - m(P') < \varepsilon\,. \tag{8.72}$$

Example 1 A Set that is not Peano-Jordan Measurable. Consider the subset $X = [0, 1] \cap \mathbb{Q}$ in $\mathbb{R}$. The only pluri-interval P' contained in X is the empty set, while any pluri-interval P'' containing X must contain $[0, 1)$ and hence it must have measure $m(P'') \geq 1$. Therefore X is *not measurable*.

In Chap. 9 the reader will see that X is Lebesgue measurable (with measure zero). According to Lebesgue's theory all open and closed subsets of $\mathbb{R}^n$ are measurable, in contrast to the Peano-Jordan theory. For example, the *generalised Cantor set* K_s, described in Example 4, Sect. 9.4, is a compact set that, for $s > 3$, is not Peano-Jordan measurable. The reader can check that the Cantor set (K_s when $s = 3$) is Peano-Jordan measurable and its measure is zero.

The following Propositions 1 and 2 are rather useful.

Proposition 1 *A bounded subset $X \subset \mathbb{R}^n$ is measurable if and only if, for any $\varepsilon > 0$, there exist two measurable sets X', X'' such that*

$$X' \subseteq X \subseteq X'' \quad \textit{and} \quad m(X'') - m(X') < \varepsilon\,. \tag{8.73}$$

Proof If X is bounded and measurable, we have already remarked that an immediate consequence of the definition of measurable set is the existence of two semi-open pluri-intervals P', P'' satisfying (8.72). That is, (8.73) holds for $X' = P'$ and $X'' = P''$.

Conversely, if (8.73) is true, given $\varepsilon > 0$ there are P' , $P'' \in \mathcal{P}$ such that

$$P' \subseteq X', \;\; X'' \subseteq P'',$$
$$m(X') - m(P') < \varepsilon, \;\; m(P'') - m(X'') < \varepsilon\,.$$

Then

$$P' \subseteq X \subseteq P'' \quad \text{and} \quad m(P'') - m(P') < 3\varepsilon,$$

and X is measurable.

Proposition 2 *The Peano-Jordan measure m is finitely additive on $\mathcal{M}$:*

$$X, Y \in \mathcal{M}, \;\; X \cap Y = \emptyset \quad \Rightarrow \quad \begin{cases} X \cup Y \quad \textit{is measurable} \\ m(X \cup Y) = m(X) + m(Y)\,. \end{cases} \tag{8.74}$$

Proof Given $\varepsilon > 0$, let P'_X , P''_X , P'_Y , P''_Y be semi-open pluri-intervals such that

$$\begin{aligned} &P'_X \subseteq X \subseteq P''_X\,, \;\; m(P''_X) - m(P'_X) < \varepsilon\,; \\ &P'_Y \subseteq Y \subseteq P''_Y\,, \;\; m(P''_Y) - m(P'_Y) < \varepsilon\,. \end{aligned} \tag{8.75}$$

Now, $X \cap Y = \emptyset$ implies $P'_X \cap P'_Y = \emptyset$. Finite additivity on $\mathcal{P}$ (see (8.65)) forces

$$m(P'_X) + m(P'_Y) = m(P'_X \cup P'_Y)\,. \tag{8.76}$$

Since $P'_X \cup P'_Y \subseteq X \cup Y \subseteq P''_X \cup P''_Y$, by subadditivity and (8.75), (8.76), we then have

$$m(P''_X \cup P''_Y) - m(P'_X \cup P'_Y) \leq m(P''_X) + m(P''_Y) - \big(m(P'_X) + m(P'_Y)\big) < 2\varepsilon\,,$$

making $X \cup Y$ measurable. By definitions (8.70) and (8.71) it follows that

$$m(P'_X \cup P'_Y) \leq m(X \cup Y) \leq m(P''_X \cup P''_Y)\,.$$

The above relationships imply

$$m(P'_X) + m(P'_Y) \;\leq\; m(X \cup Y) \leq m(P''_X) + m(P''_Y)\,. \tag{8.77}$$

Furthermore, as $P'_X \subseteq X \subseteq P''_X$, we have $m(P'_X) \leq m(X) \leq m(P''_X)$, and similarly $m(P'_Y) \leq m(Y) \leq m(P''_Y)$. Adding these, we deduce

$$m(P'_X) + m(P'_Y) \leq m(X) + m(Y) \leq m(P''_X) + m(P''_Y)\,. \tag{8.78}$$

From (8.77), (8.78), and using (8.75) again,

$$\big|m(X \cup Y) - [m(X) + m(Y)]\big| \leq m(P''_X) + m(P''_Y) - \big[m(P'_X) + m(P'_Y)\big] < 2\varepsilon.$$

But ε is arbitrary, so (8.74) follows.

Proposition 3 *For any bounded measurable sets X, Y, the sets $X \cup Y$, $X \cap Y$ and $X - Y$ are measurable and*

$$m(X - Y) = m(X) - m(X \cap Y)\,; \tag{8.79}$$

$$m(X \cup Y) = m(X) + m(Y) - m(X \cap Y)\,. \tag{8.80}$$

Moreover, the measure m is increasing:

$$Y \subseteq X \quad \Rightarrow \quad m(Y) \leq m(X)\,. \tag{8.81}$$

Proof Given $\varepsilon > 0$, take semi-open pluri-intervals P'_X , P''_X , P'_Y , P''_Y such that

$$P'_X \subseteq X \subseteq P''_X\,, \quad m(P''_X) - m(P'_X) < \varepsilon\,;$$
$$P'_Y \subseteq Y \subseteq P''_Y\,, \quad m(P''_Y) - m(P'_Y) < \varepsilon\,.$$

Then $P'_X - P''_Y \subseteq X - Y \subseteq P''_X - P'_Y$. But

$$m(P''_X - P'_Y) - m(P'_X - P''_Y) = m\big((P''_X - P'_Y) - (P'_X - P''_Y)\big) \leq$$

$$\leq m\big((P''_X - P'_X) \cup (P''_Y - P'_Y)\big) \leq m(P''_X - P'_X) + m(P''_Y - P'_Y) < 2\varepsilon\,,$$

so $X - Y$ is measurable. The measurability of $X \cap Y$ and $X \cup Y$ follows from the above together with Proposition 2, since

$$X \cap Y = X - (X - Y)\,, \qquad X \cup Y = X \cup (Y - X)\,.$$

Formulas (8.79), (8.80) and (8.81) descend from Proposition 2 by arguing as in the pluri-interval case.

Theorem *A bounded subset X of $\mathbb{R}^n$ is measurable if and only if its boundary ∂X is measurable and has zero measure.*

Proof If X is measurable, given $\varepsilon > 0$ there are $P, Q \in \mathcal{P}$ such that

$$P \subseteq X \subseteq Q \quad \text{and} \quad m(Q) - m(P) < \varepsilon \,.$$

Then

$$\partial X = \overline{X} - \overset{\circ}{X} \subseteq \overline{Q} - \overset{\circ}{P} \,, \tag{8.82}$$

because the boundary ∂X is contained in $\overline{Q}$ (which contains $\overline{X}$) and is disjoint from $\overset{\circ}{P}$ (which is contained in $\overset{\circ}{X}$).

Recall that the sets $\overline{Q}$ and $\overset{\circ}{P}$ are measurable, and $m(\overline{Q}) = m(Q), m(\overset{\circ}{P}) = m(P)$. As $\overset{\circ}{P} \subset \overline{Q}$, by finite additivity

$$m(\overline{Q}) = m(\overset{\circ}{P}) + m(\overline{Q} - \overset{\circ}{P}) \,,$$

and so

$$m(\overline{Q} - \overset{\circ}{P}) = m(\overline{Q}) - m(\overset{\circ}{P}) = m(Q) - m(P) < \varepsilon \,.$$

Then (8.82) gives $m(\partial X) = 0$.

Conversely, suppose $m(\partial X) = 0$. Given $\varepsilon > 0$, there exists then a semi-open pluri-interval P containing ∂X and such that $m(P) < \varepsilon$.

Let I be a semi-open interval containing X, and $\{X_1, X_2, \ldots, X_h\}$ a partition of $I - P$ made of semi-open intervals. Let P' be the union (possibly empty) of the intervals X_i that are disjoint from $I - X$, i.e. such that $X_i \cap (I - X) = \emptyset$. Then $P' \subseteq X$. The set $P'' = P \cup P'$ is a semi-open pluri-interval containing X. In other words $P' \subseteq X \subseteq P''$, and so

$$m(P'') - m(P') < \varepsilon \,,$$

making X measurable.

Up to this moment it was not necessary to indicate the explicit dependence of m on the dimension n of $\mathbb{R}^n$. We need to do so when, as in the next proposition, the measure m_{n+h} on $\mathbb{R}^{n+h}$ of a Cartesian product $X \times Y$ ($X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^h$) is put in relationship with the measures $m_n(X)$ on $\mathbb{R}^n$ and $m_h(Y)$ on $\mathbb{R}^h$.

Proposition 4 *Let X be a measurable set in $\mathbb{R}^n$ and Y a measurable set in $\mathbb{R}^h$. The Cartesian product $X \times Y$ is measurable on $\mathbb{R}^{n+h} = \mathbb{R}^n \times \mathbb{R}^h$, and*

$$m_{n+h}(X \times Y) = m_n(X) \cdot m_h(Y) \,. \tag{8.83}$$

Proof By assumption X and Y are bounded in $\mathbb{R}^n$ and $\mathbb{R}^h$ respectively, so there exist two intervals (of finite measure) $I \supseteq X$ in $\mathbb{R}^n$ and $J \supseteq Y$ in $\mathbb{R}^h$.

Given $\varepsilon > 0$, let P', $P'' \subseteq I$ be semi-open pluri-intervals such that

$$P' \subseteq X \subseteq P'' \quad \text{and} \quad m_n(P'') - m_n(P') < \frac{\varepsilon}{m_h(J)} .$$

Similarly, let Q', $Q'' \subseteq J$ be semi-open pluri-intervals such that

$$Q' \subseteq Y \subseteq Q'' \quad \text{and} \quad m_h(Q'') - m_h(Q') < \frac{\varepsilon}{m_n(I)} .$$

Clearly $P' \times Q' \subseteq X \times Y \subseteq P'' \times Q''$, and

$$\begin{aligned} & m_{n+h}(P'' \times Q'') - m_{n+h}(P' \times Q') = \\ &= m_{n+h}(P'' \times Q'') - m_n(P'') \cdot m_h(Q') + m_n(P'') \cdot m_h(Q') - m_{n+h}(P' \times Q') = \\ &= m_n(P'')\big[m_h(Q'') - m_h(Q')\big] + m_h(Q')\big[m_n(P'') - m_n(P')\big] < 2\varepsilon . \end{aligned}$$

Hence $X \times Y \subset \mathbb{R}^n \times \mathbb{R}^h$ is measurable. The proof of (8.83) follows from the previous inequalities, observing that it holds for pluri-intervals.

Measurability can be defined for *unbounded sets* as well. An unbounded subset X in $\mathbb{R}^n$ is *measurable* if $X \cap Y \in \mathcal{M}$ for any $Y \in \mathcal{M}$. In such a case the measure of X (possibly infinite) is defined by

$$m(X) = \sup\{m(Y) : \ Y \in \mathcal{M}, \ Y \subseteq X\} .$$

Proposition 5 *If X is a measurable set and I_k is an increasing sequence of open intervals in $\mathbb{R}^n$ whose union is $\mathbb{R}^n$, then*

$$m(X) = \lim_{k \to +\infty} m(X \cap I_k) .$$

Proof Immediately $m(X) \geq \lim_{k \to +\infty} m(X \cap I_k)$. For the opposite inequality, consider $\lambda < m(X)$. There exists a subset Y of X, bounded and measurable ($Y \in \mathcal{M}$), such that

$$m(Y) > \lambda . \tag{8.84}$$

Since the union of the I_k is $\mathbb{R}^n$, there exists ν such that

$$I_k \supseteq Y , \qquad \forall\, k \geq \nu,$$

whence $X \cap I_k \supseteq X \cap Y = Y$, for any $k \geq \nu$, and therefore

$$m(X \cap I_k) \geq m(Y)\,, \qquad \forall\, k \geq \nu.$$

From (8.84) we deduce

$$\lim_{k \to +\infty} m(X \cap I_k) > \lambda\,,$$

and the conclusion follows when $\lambda \to m(X)$.

8.7 The Riemann Integral in $\mathbb{R}^n$

Let X be a Peano-Jordan measurable and bounded subset of $\mathbb{R}^n$, and $f : X \to \mathbb{R}$ a bounded function. For any partition $P = \{X_1, X_2, \ldots, X_h\}$ of X made of measurable sets, we put

$$s(P) = \sum_{i=1}^{h} m(X_i) \inf\{f(x) :\ x \in X_i\}$$

$$S(P) = \sum_{i=1}^{h} m(X_i) \sup\{f(x) :\ x \in X_i\}\,,$$

called *lower Riemann sum* and *upper Riemann sum* relative to the partition P.

As we said for two dimensions, we may write more compactly

$$\inf\{f(x) :\ x \in X_i\} = \inf f(X_i)\,; \qquad \sup\{f(x) :\ x \in X_i\} = \sup f(X_i)\,.$$

Thus the lower and upper Riemann sums read

$$s(P) = \sum_{i=1}^{h} m(X_i) \inf f(X_i)\,; \qquad S(P) = \sum_{i=1}^{h} m(X_i) \sup f(X_i)\,.$$

If $P_1 = \{X_1, X_2, \ldots, X_h\}$ and $P_2 = \{Y_1, Y_2, \ldots, Y_k\}$ are partitions of X, the partition P_{12} generated by P_1, P_2 consists of the intersections of the elements of P_1 and P_2:

$$P_{12} = \{X_1 \cap Y_1, \ldots, X_1 \cap Y_k\,, X_2 \cap Y_1, \ldots, X_h \cap Y_1, \ldots, X_h \cap Y_k\},$$

or

$$P_{12} = \{X_i \cap Y_j :\ i \in \{1, 2, \ldots, h\},\ \ j \in \{1, 2, \ldots, k\}\}\,.$$

Lemma *Let $P_1 = \{X_1, X_2, \dots, X_h\}$ and $P_2 = \{Y_1, Y_2, \dots, Y_k\}$ be partitions of the set X made of measurable subsets. Then the lower and upper sums of the partition P_{12} generated by P_1, P_2 satisfy*

$$s(P_1) \le s(P_{12}) \le S(P_{12}) \le S(P_2)\,. \tag{8.85}$$

Proof For $i = 1, 2, \dots, h$ and $j = 1, 2, \dots, k$ we put

$$X_{ij} = X_i \cap Y_j\,,$$

so

$$\bigcup_{i=1}^{h} X_{ij} = Y_j\,, \qquad \bigcup_{j=1}^{k} X_{ij} = X_i\,.$$

By the additivity of m

$$\begin{aligned} s(P_1) &= \sum_{i=1}^{h} m(X_i) \inf f(X_i) = \sum_{i=1}^{h} m\left(\bigcup_{j=1}^{k} X_{ij}\right) \inf f(X_i) = \\ &= \sum_{i=1}^{h} \left(\sum_{j=1}^{k} m(X_{ij})\right) \inf f(X_i) \le \sum_{i=1}^{h}\sum_{j=1}^{k} m(X_{ij}) \inf f(X_{ij}) = s(P_{12})\,. \end{aligned} \tag{8.86}$$

In (8.86) we exploited the inequalities $\inf f(X_i) \le \inf f(X_{ij})$, valid for any $i = 1, 2, \dots, h$ and $j = 1, 2, \dots, k$, which follow from the inclusions $f(X_{ij}) \subseteq f(X_i)$. Similarly,

$$\begin{aligned} S(P_2) &= \sum_{j=1}^{k} m(Y_j) \sup f(Y_j) = \sum_{j=1}^{k} m\left(\bigcup_{i=1}^{h} X_{ij}\right) \sup f(Y_i) = \\ &= \sum_{j=1}^{k} \left(\sum_{i=1}^{h} m(X_{ij})\right) \sup f(Y_j) \ge \sum_{i=1}^{h}\sum_{j=1}^{k} m(X_{ij}) \sup f(X_{ij}) = S(P_{12})\,. \end{aligned} \tag{8.87}$$

As, clearly, $s(P_{12}) \le S(P_{12})$, our claim (8.85) now follows from (8.86) and (8.87).

The above lemma implies that the sets

$$\{s(P_1)\}, \qquad \{S(P_2)\},$$

respectively containing the lower and upper sums as P_1, P_2 vary among all partitions, are *separated*, i.e. $s(P_1) \leq S(P_2)$ for any P_1, P_2. If these sets are also *contiguous*:

$$\sup_{P_1} \{s(P_1)\} = \inf_{P_2} \{S(P_2)\}\,, \tag{8.88}$$

we say f is *Riemann integrable* on X, and the (only) separating element is the *definite integral* of f over X. The symbols for it are

$$\int_X f(x)\, dm\,, \tag{8.89}$$

or

$$\int_X f(x)\, dx\,, \qquad \int_X f(x_1, x_2, \ldots, x_n)\, dx_1 dx_2 \ldots dx_n\,. \tag{8.90}$$

If $f = f(x, y)$ is a function of two real variables, as in the first part of the chapter, we also use the *double integral*

$$\iint_X f(x, y)\, dx\, dy\,,$$

while for $f = f(x, y, z)$ of three variables we may use the *triple integral*

$$\iiint_X f(x, y, z)\, dx\, dy\, dz\,.$$

In general, we shall say that the integrals in (8.89) or (8.90) are *multiple integrals* if the dimension n is larger than 1.

Integrability Criterion *Let f be a bounded function on the bounded and measurable set $X \subset \mathbb{R}^n$. Then f is integrable on X if and only if, for any $\varepsilon > 0$, there exists a partition $P = \{X_1, X_2, \ldots, X_h\}$ of X, made of measurable sets, such that*

$$S(P) - s(P) < \varepsilon\,, \tag{8.91}$$

i.e.

$$\sum_{i=1}^{h} m(X_i)\big[\sup f(X_i) - \inf f(X_i)\big] < \varepsilon\,.$$

Proof Suppose f is integrable on X and denote by $I \in \mathbb{R}$ the definite integral's value. By definition (8.88), for any $\varepsilon > 0$, there exist two partitions P_1, P_2 such that

$$S(P_2) < I + \frac{\varepsilon}{2}\,, \qquad s(P_1) > I - \frac{\varepsilon}{2}\,,$$

and subtracting,

$$S(P_2) - s(P_1) < \varepsilon\,.$$

Now let $P = P_{12}$ be the partition generated by P_1 and P_2, the one consisting of the intersections of sets in P_1 and P_2. By the previous lemma

$$S(P) - s(P) \le S(P_2) - s(P_1) < \varepsilon\,,$$

and so we have (8.91). Conversely, if (8.91) holds it is clear that (8.88) is valid too, i.e. f is integrable on X.

If we consider the trivial partition of X made by X only, by the definition of integral if f is integrable on X then

$$m(X)\inf f(X) \le \int_X f(x)\,dx \le m(X)\sup f(X)\,. \tag{8.92}$$

If f is a *constant function* on X, i.e. $f(x) = c$ for any $x \in X$ for some $c \in \mathbb{R}$, then f is integrable on X, and $\inf f(X) = \sup f(X) = c$. Hence

$$\int_X f(x)\,dx = c \cdot m(X)\,,$$

and in particular

$$\int_X dx = \int_X 1\,dx = m(X)\,.$$

Let f be a bounded integrable function on X. The *integral function* F of f is

$$F(Y) = \int_Y f(x)\,dx\,,$$

as Y varies among measurable subsets of X. By (8.92) the *integral function is absolutely continuous*, i.e. for any $\varepsilon > 0$ there is $\delta > 0$ such that

$$Y \subseteq X\,, \quad m(Y) < \delta \qquad \Rightarrow \qquad \left|\int_Y f(x)\,dx\right| < \varepsilon\,. \tag{8.93}$$

To prove (8.93) observe first of all that the integral function F is well defined, since for any measurable subset $Y \subset X$, f is integrable on Y. Since, by assumption, f is bounded on X, there exists a constant $M > 0$ such that $|f(x)| \le M$ for any $x \in X$. Then (8.92) implies

$$\left|\int_Y f(x)\,dx\right| \le M\,m(Y) \le M\,\varepsilon\,,$$

for any $Y \subseteq X$ such that $m(Y) < \varepsilon$.

Just like for definite integrals in one variable, also the Riemann integral obeys a *linearity property*

$$\int_X \{a\, f(x) + b\, g(x)\}\, dx = a \int_X f(x)\, dx + b \int_X g(x)\, dx\,,$$

and a *monotonicity propriety*

$$f(x) \le g(x), \quad \forall\, x \in X \quad \Rightarrow \quad \int_X f(x)\, dx \le \int_X g(x)\, dx\,,$$

for any Riemann integrable functions f, g on X and any $a, b \in \mathbb{R}$.

As for functions of one real variable, the integral has a remarkable geometric meaning. Consider a *non-negative* function f defined on a bounded and measurable subset $X \subset \mathbb{R}^n$. To simplify the explanations we shall use the term *cylindroid* over X to refer to the set C_f in $\mathbb{R}^{n+1}$ given by

$$C_f = \{(x, y) \in X \times \mathbb{R} : \quad 0 \le y \le f(x)\}\,.$$

Figures 8.18 and 8.19 represent two instances, for $n = 1$ and $n = 2$. In particular, for $n = 1$ (Fig. 8.18) a cylindroid is a *trapezoid*, as is well known.

The next proposition is a special case of formula (8.99), which will be proved in the following section.

Fig. 8.18

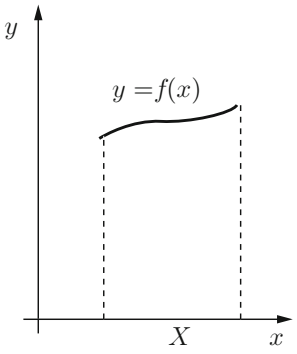
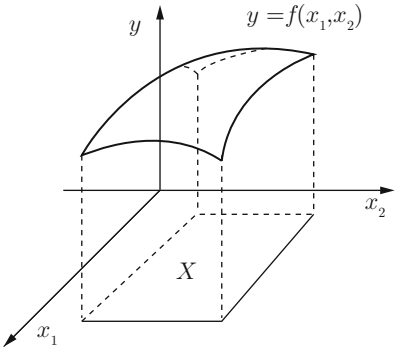


Fig. 8.19



Proposition *If $f : X \to [0, +\infty)$ is integrable on a bounded and measurable set X, then $C_f \subset \mathbb{R}^{n+1}$ is measurable, with measure*

$$m_{n+1}(C_f) = \int_X f(x)\, dx\,. \tag{8.94}$$

Example 1 (A Bounded Function that is not Riemann Integrable) Consider the *Dirichlet function*

$$\begin{cases} 0 & \text{if } x \in [0,1] \cap \mathbb{Q} \\ 1 & \text{if } x \in [0,1] - \mathbb{Q}, \end{cases}$$

and let us show that it is not Riemann integrable on [0, 1].

First, note that if $X \subset [0,1] - \mathbb{Q}$ is measurable, then $m(X) = 0$. By contradiction, if $m(X) > 0$, then X would have to contain one open interval and therefore infinitely many rational numbers. Similarly, if $X \subset [0,1] \cap \mathbb{Q}$ is measurable, for the same reason $m(X) = 0$.

Given a partition $P = \{X_1, X_2, \ldots, X_h\}$ of [0, 1] made of measurable sets, we always have

$$s(P) = \sum_{i=1}^{h} m(X_i) \inf f(X_i) = 0\,.$$

In fact if X_i contains no rationals, by the above argument $m(X_i) = 0$. If $X_i \cap \mathbb{Q} \neq \emptyset$ instead, then $\inf f(X_i) = 0$. In either case the product $m(X_i) \inf f(X_i)$ is zero.

We claim

$$S(P) = \sum_{i=1}^{h} m(X_i) \sup f(X_i) = 1\,. \tag{8.95}$$

(continued)

Example 1 (continued)
Indicate with $X'_1, X'_2, \ldots, X'_k$, $1 \le k \le h$, the partition's elements containing irrational points. By the previous part, the remaining partition elements have measure zero, so

$$\sum_{i=1}^{k} m(X'_i) = 1 .$$

But $\sup f(X'_i) = 1$ for any $i \in \{1, 2, \ldots, k\}$, and (8.95) follows.

In Sect. 9.6 we will prove that, on the contrary, the Dirichlet function is Lebesgue integrable.

The following theorem on the Riemann integrability of continuous functions on compact sets is very relevant in the theory of integration.

Integrability of Continuous Functions *Let X be a compact measurable subset of $\mathbb{R}^n$ and $f : X \to \mathbb{R}$ a continuous function on X. Then f is integrable on X.*

Proof If the measure $m(X)$ of X is null the function f is measurable, since the upper sums are zero and the integral of f on X is zero too. Suppose then $m(X) > 0$.

As f is uniformly continuous on X, Cantor's theorem ensures that given $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x, y \in X,\ |x - y| < \delta \quad \Rightarrow \quad |f(x) - f(y)| < \frac{\varepsilon}{m(X)} . \tag{8.96}$$

Let us show that there is a finite partition made of measurable sets with diameter less than δ. By assumption X is bounded, so there is a semi-open interval I containing X. We decompose I (for example using hyperplanes parallel to the coordinate hyperplanes) into a finite number of semi-open intervals I_i, $i = 1, 2, \ldots, h$, of diameter less than δ.

The sets $X_i = X \cap I_i$ are measurable, pairwise disjoint, of diameter less than δ, and their union is X. Put

$$\ell_i = \inf f(X_i) , \qquad L_i = \sup f(X_i) ,$$

so from (8.96)

$$L_i - \ell_i < \frac{\varepsilon}{m(X)} . \qquad \forall\, i = 1, 2, \ldots, h.$$

Hence

$$\sum_{i=1}^{h} m(X_i)\big[\sup f(X_i) - \inf f(X_i)\big] =$$

$$= \sum_{i=1}^{h} m(X_i)[L_i - \ell_i] < \frac{\varepsilon}{m(X)} \sum_{i=1}^{h} m(X_i) =$$

$$= \frac{\varepsilon}{m(X)} m(X) = \varepsilon\,.$$

The claim follows from the previous section's integrability criterion.

Reasoning as in the above theorem (the two-dimensional case was proved in full detail in Sect. 8.3) we obtain the following result.

Integral of a Continuous Function as Limit *Let X be compact and measurable in $\mathbb{R}^n$ and $f : X \to \mathbb{R}$ continuous on X. For any partition $P = \{X_1, X_2, \ldots, X_h\}$ of X choose h points $x_1, x_2, \ldots, x_h \in X$ such that $x_i \in X_i$ for any $i = 1, 2, \ldots, h$. Then the integral of f on X is given by the limit*

$$\lim_{\text{diam } P \to 0} \sum_{i=1}^{h} m(X_i)\, f(x_i) = \int_X f(x)\, dx\,.$$

This means that for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\left| \int_X f(x)dx - \sum_{i=1}^{h} m(X_i)\, f(x_i) \right| < \varepsilon$$

irrespective of the partition P into sets X_i, $i = 1, 2, \ldots, h$, whose maximum diameter (called diameter of P) is less than δ.

More generally, we have the following theorem on the integrability of functions that are bounded but may not be continuous on the entire set X. Here is the statement, without proof.

A function $f : X \to \mathbb{R}$ is said to be *almost everywhere (a.e.) Peano-Jordan continuous* if it is continuous on $X - Y$ for some set Y of zero Peano-Jordan measure.

Integrability of Almost Everywhere Continuous Functions *Let X be compact and measurable in $\mathbb{R}^n$ and $f : X \to \mathbb{R}$ bounded and a.e. Peano-Jordan continuous on X. Then f is integrable on X.*

In Sect. 9.6 we will prove the *Vitali-Lebesgue criterion*, a complete characterisation of bounded, Riemann integrable functions.

We close the section with a mean value theorem for integrals of continuous functions on connected sets.

Theorem (Integral Mean Value) *Let X be a bounded, connected and measurable domain of $\mathbb{R}^n$, and $f : X \to \mathbb{R}$ a continuous function on X. There exists a point $x_0 \in X$ such that*

$$\int_X f(x)dx = f(x_0) \cdot m(X) .$$

Proof If $m(X) = 0$ the result holds for any x_0 in X. Then consider $m(X) > 0$. By (8.92)

$$\inf f(X) \leq \frac{1}{m(X)} \int_X f(x)\, dx \leq \sup f(X) . \tag{8.97}$$

As X is compact, the Weierstrass theorem guarantees there are two points in X, x_1 and x_2, such that

$$f(x_1) = \inf f(X) , \qquad f(x_2) = \sup f(X) .$$

Hence (8.97) is equivalent to

$$f(x_1) \leq \frac{1}{m(X)} \int_X f(x)\, dx \leq f(x_2) .$$

But X is connected, so the continuous function f assumes every value between $f(x_1)$ and $f(x_2)$ (see the existence of intermediate values, Sect. 2.11). Therefore there is a point $x_0 \in X$ such that

$$\frac{1}{m(X)} \int_X f(x)\, dx = f(x_0) .$$

8.8 Properties of Riemann Integrals

This section aims at establishing *reduction formulas*, using which the integration of a bounded function over a measurable bounded set in $\mathbb{R}^{n+1}$ can be reduced to an (iterated) integral of functions of one variable.

We shall need some definitions, that generalise the analogous definitions in two and three dimensions for normal *domains* seen in the first part of the chapter. Let $\alpha = \alpha(x)$ and $\beta = \beta(x)$ be continuous functions on a closed, bounded and measurable set X of $\mathbb{R}^n$, and such that

$$\alpha(x) \leq \beta(x) , \qquad \forall\, x \in X.$$

The subset of $\mathbb{R}^{n+1}$

$$D = \{(x, y) \in X \times \mathbb{R} : \ \alpha(x) \leq y \leq \beta(x)\} \tag{8.98}$$

will be called *normal set (with respect to the first n coordinates) over X*. Analogous notions exist for any collection of n coordinates.

Measurability of Normal Sets *Let $X \subset \mathbb{R}^n$ be closed, bounded and measurable, and D the normal set over X defined in (8.98). Then D is measurable and its measure is*

$$m_{n+1}(D) = \int_X \{\beta(x) - \alpha(x)\}\, dx\,. \tag{8.99}$$

Proof First we consider the case $\alpha(x) \equiv 0$. Let $P = \{X_1, X_2, \ldots, X_h\}$ be a partition of X into measurable sets. Correspondingly,

$$A_i = X_i \times [0, \inf \beta(X_i))\,, \qquad B_i = X_i \times [0, \sup \beta(X_i)]\,,$$

are measurable in $\mathbb{R}^{n+1}$ with measure

$$m_{n+1}(A_i) = m_n(X_i) \cdot \inf \beta(X_i)\,, \qquad m_{n+1}(B_i) = m_n(X_i) \cdot \sup \beta(X_i)\,.$$

Moreover, the sets

$$\mathcal{A} = \bigcup_{i=1}^{h} A_i\,, \qquad \mathcal{B} = \bigcup_{i=1}^{h} B_i\,,$$

are measurable in $\mathbb{R}^{n+1}$ and

$$m_{n+1}(\mathcal{A}) = \sum_{i=1}^{h} m_n(X_i) \cdot \inf \beta(X_i)\,, \qquad m_{n+1}(\mathcal{B}) = \sum_{i=1}^{h} m_n(X_i) \cdot \sup \beta(X_i)\,,$$

$$\mathcal{A} \subseteq D \subseteq \mathcal{B}\,.$$

Now, as β is integrable on X, given $\varepsilon > 0$ there exists a partition $P = \{X_1, X_2, \ldots, X_h\}$ of X made of measurable sets such that

$$S(P) - s(P) = \sum_{i=1}^{h} m_n(X_i) \cdot \sup \beta(X_i) - \sum_{i=1}^{h} m_n(X_i) \cdot \inf \beta(X_i) < \varepsilon\,.$$

Then

$$m_{n+1}(\mathcal{B}) - m_{n+1}(\mathcal{A}) < \varepsilon\,,$$

and so D is measurable. Its measure is the separating element of the collections $\{m_{n+1}(\mathcal{B})\}$ and $\{m_{n+1}(\mathcal{A})\}$, i.e., its measure is (8.99).

The proof for a generic $\alpha(x)$ descends from the above argument by using the following lemma. The given normal set, in fact, can be represented (modulo translations) as difference of two cylindroids, one of which does not include the defining graph.

Lemma *If X is measurable and bounded in $\mathbb{R}^n$ and f is a bounded integrable function on X, the graph*

$$G_f = \{(x, y) \in X \times \mathbb{R} : \ y = f(x)\}$$

has zero Peano-Jordan measure.

Proof Without loss of generality we may assume f is non-negative. By the first part of the previous proof, the cylindroid C_f of f over X is measurable. Moreover, the graph G_f is contained in the boundary of C_f. By the theorem of Sect. 8.6 the boundary of a measurable set has measure zero, and the claim is proved.

Before we get to the *reduction theorems* we shall prove two results about the continuous dependence of multiple integrals, similar to those for one-variable integrals seen in Sect. 3.10.

Let $X \subseteq \mathbb{R}^n$, $Y \subseteq \mathbb{R}^h$ be compact and measurable, and $f : X \times Y \to \mathbb{R}$ a continuous function. Fix x in X, and consider the function

$$f(x, \cdot) : \ y \in Y \to f(x, y) \tag{8.100}$$

of the variable y. As (8.100) is continuous for any $x \in X$ and hence integrable on the compact set Y,

$$F(x) = \int_Y f(x, y)\, dy \tag{8.101}$$

is well defined on X. If the above function is continuous on X, we say that the integral in (8.101) depends continuously on the parameter x. The next two lemmas, about the continuity of integrals with respect to some parameters, generalise the similar properties in one variable seen in Chap. 3 (Sect. 3.10), as already said.

Lemma 1 (Continuous Dependence) *If $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^h$ are compact and measurable, the integral $F(x)$ in (8.101) depends continuously on the parameter $x \in X$.*

Proof By Cantor's theorem $f = f(x, y)$ is uniformly continuous on $X \times Y$. Fix $\varepsilon > 0$, so there exists $\delta > 0$ such that, for any pair (x_1, y_1), (x_2, y_2) in $X \times Y$ with distance

$$d_{\mathbb{R}^n \times \mathbb{R}^h}((x_1, y_1), (x_2, y_2)) = \sqrt{|x_1 - x_2|^2 + |y_1 - y_2|^2} < \delta$$

in $\mathbb{R}^n \times \mathbb{R}^h$, it follows that

$$|f(x_1, y_1) - f(x_2, y_2)| < \frac{\varepsilon}{m_h(Y)}\,.$$

Now, for any $y \in Y$ the distance between (x_1, y) and (x_2, y) equals $|x_1 - x_2|$. Then for any $x_1, x_2 \in X$ with $|x_1 - x_2| < \delta$ we have

$$\left| \int_Y f(x_1, y)\, dy - \int_Y f(x_2, y)\, dy \right| \leq \int_Y |f(x_1, y) - f(x_2, y)|\, dy \leq \frac{\varepsilon}{m_h(Y)} \cdot m_h(Y) = \varepsilon\,.$$

Lemma 2 (Continuous Dependence) *Let $X \subset \mathbb{R}^n$ and $[c, d] \subset \mathbb{R}$ be compact intervals, and $f : X \times [c, d] \to \mathbb{R}$ a continuous function. Then*

$$G(x, s, t) = \int_s^t f(x, y)\, dy$$

is continuous on $X \times [c, d] \times [c, d]$.

Proof Let $M > 0$ be such that $|f(x, y)| \leq M$ for any $x \in X$ and any $y \in [c, d]$. Given $(x_0, s_0, t_0) \in X \times [c, d] \times [c, d]$, for any (x, s, t) in the same set

$$\begin{aligned} &|G(x, s, t) - G(x_0, s_0, t_0)| \leq \\ &\leq \left| \int_s^{s_0} |f(x, y)|\, dy \right| + \left| \int_{s_0}^{t_0} |f(x, y) - f(x_0, y)|\, dy \right| + \left| \int_{t_0}^{t} |f(x, y)|\, dy \right| \leq \\ &\leq M|s - s_0| + \left| \int_{s_0}^{t_0} |f(x, y) - f(x_0, y)|\, dy \right| + M|t - t_0|\,. \end{aligned} \tag{8.102}$$

As $|f(x, y) - f(x_0, y)|$ is continuous at (x, y), by Lemma 1 also

$$D(x) = \int_c^d |f(x, y) - f(x_0, y)|\, dy$$

is continuous at $x \in X$. But $D(x_0) = 0$, so given $\varepsilon > 0$ there exists an open ball $I \subset \mathbb{R}^n$ centred at x_0 such that

$$|D(x) - D(x_0)| = |D(x)| < \varepsilon\,, \qquad \forall\, x \in I \cap X.$$

Hence, for any $x \in I \cap X$,

$$\left| \int_{s_0}^{t_0} |f(x, y) - f(x_0, y)|\, dy \right| \leq \int_c^d |f(x, y) - f(x_0, y)|\, dy = |D(x)| < \varepsilon\,. \tag{8.103}$$

Call J and K two open intervals in $\mathbb{R}$, with respective centres s_0, t_0 and length 2ε. From (8.102), (8.103), for any $(x, s, t) \in X \times [c, d] \times [c, d]$ such that $(x, s, t) \in I \times J \times K$, we have

$$|G(x, s, t) - G(x_0, s_0, t_0)| \leq M\varepsilon + \varepsilon + M\varepsilon = (2M + 1)\varepsilon\,,$$

i.e. G is continuous at (x_0, s_0, t_0).

The next theorem provides a useful *reduction formula* for the integral over a normal set. The proof was given in Sect. 8.2 for continuous functions of two variables on normal domains, and for brevity we shall not prove the obvious generalisation.

Reduction Theorem for Multiple Integrals *Let X be a compact and measurable subset of $\mathbb{R}^n$ and D the normal set over X defined by*

$$D = \{(x, y) \in X \times \mathbb{R} :\ \alpha(x) \leq y \leq \beta(x)\}\,,$$

where $\alpha(x) \leq \beta(x)$ are continuous on X. Then for any continuous function $f : D \to \mathbb{R}$

$$\int_D f(x, y) dm_{n+1} = \int_X \left(\int_{\alpha(x)}^{\beta(x)} f(x, y)\, dy \right) dm_n\,. \tag{8.104}$$

We warn the reader that formula (8.104) is often presented in the more expressive form

$$\iint_D f(x, y)\, dx\, dy = \int_X dx \int_{\alpha(x)}^{\beta(x)} f(x, y)\, dy\,.$$

A similar formula to (8.104) holds when D is a normal set of the type

$$D = \{(x, y) \in \mathbb{R} \times Y :\ \gamma(y) \leq x \leq \delta(y)\}\,.$$

Then

$$\iint_D f(x, y)\, dm_{n+1} = \int_Y dy \int_{\gamma(y)}^{\delta(y)} f(x, y)\, dx\,.$$

In the same spirit of the reduction theorems we shall provide a formula for computing the measure of subsets in $\mathbb{R}^n$.

(continued)

Let S be a subset of $\mathbb{R}^{n+1}$. Given $t \in \mathbb{R}$ and $i \in \{1, 2, \ldots, n, n+1\}$, the set

$$S_i^t = \left\{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : (x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_n) \in S\right\}$$

is the ith *cross-section of* S *at* t. The ith *projection* $\pi_i : S \to \mathbb{R}$ is the function

$$\pi_i(x) = x_i \ , \qquad \forall\, x \in S.$$

Proposition *Let $S \subset \mathbb{R}^{n+1}$ be bounded and measurable set whose cross-sections $S_i^t \subset \mathbb{R}^n$ are measurable, for any $t \in \mathbb{R}$. Then*

$$m_{n+1}(S) = \int_{I_i} m_n(S_i^t)\, dt \ ,$$

where I_i is an arbitrary interval containing the ith projection $\pi_i(S)$ of S.

We briefly mention how to change variables in multiple integrals. The following formula generalises the analogous two- and three-dimensional ones.

Suppose T is a bounded measurable domain of $\mathbb{R}^n$ and $\Phi : T \to D$ a transformation of class $C^1(T)$ with non-zero Jacobian determinant $\det J_\Phi$ on $\overset{\circ}{T}$. If D is a measurable domain and Φ is injective, then

$$\int_T f(x)\, dx = \int_D f(\Phi(u))|\det J_\Phi(u)|\, du \ .$$

for any continuous function $f : D \to \mathbb{R}$.

8.9 Summable Functions

We would like to extend the notion of integral to *real functions that are not necessarily bounded*, and defined on *measurable sets that may not be bounded*.

Throughout this section X will be a measurable domain in $\mathbb{R}^n$ and f a *generically continuous* function on X, meaning continuous on X except for finitely many points at most.

Start from a non-negative function $f : X \to [0, +\infty)$. Denote by $\mathcal{L}(f)$ the collection of bounded measurable subsets of X on each of which f is bounded and integrable. Then we may define the map

$$F : Y \in \mathcal{L}(f) \ \to \int_Y f(x)\, dx \ . \tag{8.105}$$

If X is bounded and f is bounded on X, the map F in (8.105) is the *integral function* of f. As f is non-negative on X,

$$\int_X f(x)\,dx = \sup\left\{\int_Y f(x)\,dx \;:\; Y \in \mathcal{L}(f)\right\}. \tag{8.106}$$

More generally, suppose now X is a measurable set, but possibly unbounded. Let f be a non-negative function which may or not be bounded on X. We shall say that f is *summable* (or *integrable in generalised sense*) over X if

$$\sup\left\{\int_Y f(x)\,dx \;:\; Y \in \mathcal{L}(f)\right\} < +\infty\,.$$

If so, *the integral of f over X* is, by definition, the above least upper bound. This integral, also referred to as *improper integral*, is denoted by the usual symbol, and hence equals the real number in (8.106).

Let now $f : X \to [0, +\infty)$ be a generically continuous function, and suppose Y_k is an increasing sequence of sets in $\mathcal{L}(f)$ such that

$$m\left(X - \bigcup_{k=1}^{\infty} Y_k\right) = 0\,. \tag{8.107}$$

Characterisation of Non-Negative Summable Functions *A function f is summable over X if and only if*

$$\lim_{k\to+\infty} \int_{Y_k} f(x)\,dx = \ell < +\infty\,, \tag{8.108}$$

and in that case

$$\ell = \int_X f(x)\,dx\,. \tag{8.109}$$

Proof The limit in (8.108) exists since Y_k is increasing and $f \geq 0$. So it suffices to show that the limit of the integrals over Y_k equals the integral in (8.109) if f is summable over X, while it equals $+\infty$ if f is not summable over X.

Suppose f is summable. For any $\varepsilon > 0$ there exists $Y \subseteq X$, measurable and bounded, such that

$$\int_Y f(x)\,dx > \int_X f(x)\,dx - \varepsilon\,.$$

The restriction of f to Y has absolutely continuous integral function, so there exists $\delta > 0$ such that

$$Z \subseteq Y,\ m(Z) < \delta \quad \Rightarrow \quad \int_Z f(x)\,dx < \varepsilon\,.$$

On the other hand (8.107) implies

$$\lim_{k\to+\infty} m(Y - Y_k) = 0\,,$$

so taking ν such that $m(Y - Y_k) < \delta$ for any $k \geq \nu$, we have

$$\int_{Y-Y_k} f(x)\,dx < \varepsilon\,, \qquad \forall\, k \geq \nu. \tag{8.110}$$

The above relations give

$$\int_X f(x)\,dx - \int_{Y_k} f(x)\,dx < \int_Y f(x)\,dx - \int_{Y\cap Y_k} f(x)\,dx + \varepsilon = \int_{Y-Y_k} f(x)\,dx + \varepsilon < 2\varepsilon\,,$$

for any $k \geq \nu$, i.e. the claim.

If f is not summable over X, for any $M > 0$ there exists $Y \subseteq X$, measurable and bounded, such that

$$\int_Y f(x)\,dx > M\,.$$

Given $\varepsilon \leq 1$, pick ν so that (8.110) holds, and then

$$\int_{Y_k} f(x)\,dx \geq \int_Y f(x)\,dx - \int_{Y-Y_k} f(x)\,dx > M - \varepsilon \geq M - 1\,,$$

for any $k \geq \nu$. In the end

$$\lim_{k\to+\infty} \int_{Y_k} f(x)\,dx = +\infty\,.$$

By virtue of the above theorem we can extend the Riemann integral's monotonicity and linearity to non-negative summable functions. In particular, we can prove the following facts.

Proposition 1 *Let f, g be generically continuous non-negative functions such that $g(x) \leq f(x)$ for any $x \in X$. If f is summable over X then so is g, and*

$$\int_X g(x)\,dx \leq \int_X f(x)\,dx\,.$$

Proposition 2 *If f and g are generically continuous, non-negative and summable over X, and $a, b \in \mathbb{R}$,*

$$\int_X \{a\, f(x) + b\, g(x)\}\,dx = a \int_X f(x)\,dx + b \int_X g(x)\,dx\,.$$

Let us pass to the *summability of functions of arbitrary sign*. If X is a measurable domain of $\mathbb{R}^n$ and $f : X \to \mathbb{R}$ is generically continuous, we say f is *summable* over X if $|f|$ is summable over X. Define functions f^+ and f^- by

$$f^+(x) = \max\{0, f(x)\},$$
$$f^-(x) = -\min\{0, f(x)\}.$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-,$$

and the summability of $|f|$ implies the summability of f^+ and f^-, by Proposition 1. Hence if f is summable over X, and we call *integral of f over X* the real number

$$\int_X f(x)\,dx = \int_X f^+\,dx - \int_X f^-\,dx\,.$$

Directly out of the characterisation of non-negative summable functions we have the following theorem.

Integral of a Summable Function *If f is summable over X and Y_k is an increasing sequence of bounded and measurable sets in X, on each of which f is bounded and such that*

$$m\left(X - \bigcup_{k=1}^{\infty} Y_k\right) = 0\,,$$

then

$$\lim_{k\to+\infty} \int_{Y_k} f(x)\,dx = \int_X f(x)\,dx\,.$$

Now we present a number of summability and non-summability criteria. To this end we start with a few remarkable examples of summable functions on bounded and unbounded set.

Using results from Sect. 3.1 we establish the summability properties of a function of two real variables like

$$f(x, y) = \frac{1}{\left(\sqrt{(x - x_0)^2 + (y - y_0)^2}\right)^{\alpha}}\,, \tag{8.111}$$

where (x_0, y_0) is a given point in the plane and α is a positive number.

The following two propositions generalise to $n \in \mathbb{N}$ real variables (see Propositions 5 and 6), so we shall emphasise $n = 2$ in the statements. The general case holds by removing the constraint $n = 2$.

Proposition 3 deals with unbounded functions on bounded sets, while Proposition 4 treats bounded functions on unbounded sets.

Proposition 3 *Let I_r be a ball of centre $(x_0\,, y_0)$ and radius $r > 0$, and let $f(x, y)$ be the function defined in (8.111). Then*

$$\iint_{I_r} f(x, y)\, dx\, dy = \begin{cases} \dfrac{2\pi r^{2-\alpha}}{2-\alpha} & \text{if } \alpha < n = 2 \\ +\infty & \text{if } \alpha \geq n = 2\,. \end{cases} \tag{8.112}$$

Proof Under the change of variables $u = x - x_0$, $v = y - y_0$, whose Jacobian determinant is identically equal to 1, the integral does not change, while the point (x_0, y_0) in the xy-plane is mapped to $(0, 0)$ in the uv-plane. Hence we can assume $(x_0, y_0) = (0, 0)$ with no loss of generality.

For an integer $k > 1/r$ let D_k be the annulus bounded by the circles of centre $(0, 0)$ and radii $1/k$ and r. Let us compute

$$\iint_{D_k} f(x, y)\, dx\, dy\,. \tag{8.113}$$

In polar coordinates ϱ, ϑ the annulus D_k is defined by

$$D_k = \left\{ (\varrho, \vartheta) \in \mathbb{R}^2 : \frac{1}{k} \leq \varrho \leq r, 0 \leq \vartheta \leq 2\pi \right\}.$$

Hence the integral in (8.113) is

$$\iint_{D_k} f(x, y)\, dx\, dy = \int_0^{2\pi} d\vartheta \int_{1/k}^{r} \frac{1}{\varrho^\alpha} \varrho\, d\varrho = \begin{cases} \dfrac{2\pi}{2-\alpha}\left[r^{2-\alpha} - \left(\dfrac{1}{k}\right)^{2-\alpha} \right] & \text{if } \alpha \neq 2 \\ 2\pi \log rk & \text{if } \alpha = 2\,. \end{cases}$$

In the limit when $k \to +\infty$ we obtain (8.112).

Proposition 4 *Let I_r be a ball of centre $(x_0\,, y_0)$ and radius $r > 0$, and $f(x, y)$ the function in (8.111). Then*

$$\iint_{\mathbb{R}^n - I_r} f(x, y)\, dx\, dy = \begin{cases} \dfrac{2\pi r^{2-\alpha}}{\alpha - 2} & \text{if } \alpha > n = 2 \\ +\infty & \text{if } \alpha \leq n = 2\,. \end{cases} \tag{8.114}$$

Proof As before we may just consider $(x_0, y_0) = (0, 0)$. For any $k \in \mathbb{N}$ let D_k be the annulus bounded by the circles centred at $(0, 0)$ with radii r and $r + k$. Let us

calculate the integral of $f(x, y)$ on D_k. Proceeding as in Proposition 3,

$$\iint_{D_k} f(x, y)\, dx\, dy = \int_0^{2\pi} d\vartheta \int_r^{r+k} \frac{1}{\varrho^\alpha} \varrho\, d\varrho = \begin{cases} \frac{2\pi}{2-\alpha}\left[(r+k)^{2-\alpha} - r^{2-\alpha}\right] & \text{if } \alpha \neq 2 \\ 2\pi \log\left(1 + \frac{k}{r}\right) & \text{if } \alpha = 2\,. \end{cases}$$

When $k \to +\infty$ we obtain (8.114).

As was mentioned, we may prove the following Propositions 5 and 6, valid for functions of $n \in \mathbb{N}$ real variables.

Proposition 5 *Let x_0 be a point of $\mathbb{R}^n$ and α a positive number. Then the function defined on $\mathbb{R}^n - \{x_0\}$ by*

$$f(x) = \frac{1}{|x - x_0|^\alpha} \tag{8.115}$$

is summable over any bounded measurable neighbourhood I of x_0 if and only if α is less than n.

Proposition 6 *Let x_0 be a point of $\mathbb{R}^n$ and α a positive number. Then the function $f(x)$ defined on $\mathbb{R}^n - \{x_0\}$ by (8.115), for any bounded measurable neighbourhood I of x_0, is summable over $\mathbb{R}^n - I$ if and only if α is larger than n.*

The comparison criteria imply the following facts.

Proposition 7 *Let x_0 be a point of $\mathbb{R}^n$, I a neighbourhood of x_0 and f a continuous function on $I - \{x_0\}$. If there exist $\alpha \in (0, n)$ and $M > 0$ such that*

$$|f(x)| \leq \frac{M}{|x - x_0|^\alpha}\,, \qquad \forall x \in I - \{x_0\}, \tag{8.116}$$

then f is summable over I. Conversely, if

$$|f(x)| \geq \frac{M}{|x - x_0|^\alpha}\,, \qquad \forall x \in I - \{x_0\}$$

for some $\alpha \geq n$, $M > 0$, then f is not summable over I.

Note that (8.116) holds near x_0 when, for example

$$\lim_{x \to x_0} |f(x)| \cdot |x - x_0|^\alpha = \ell < +\infty \tag{8.117}$$

for some $\alpha < n$, and so f is summable over I if that happens. Furthermore, (8.117) holds on a neighbourhood I of x_0 if, for example,

$$\lim_{x \to x_0} |f(x)| \cdot |x - x_0|^\alpha = +\infty$$

for some $\alpha \geq n$. In this case f is not summable.

Analogous properties hold on the complement of any neighbourhood of x_0, but for brevity we leave their formulation to the reader.

Appendix to Chap. 8

8.10 Jensen's Inequality

This section is devoted to the *integral Jensen inequality*, namely the continuous version of the *discrete Jensen inequality* proved for convex functions of several variables in Sect. 3.15, Appendix to Chap. 3.

We start by defining the integral of a vector-valued function $\varphi : X \subseteq \mathbb{R}^n \to \mathbb{R}^m$, of components

$$(\varphi_1, \varphi_2, \ \dots \ , \varphi_m) \, ,$$

defined on a measurable and bounded subset $X \subset \mathbb{R}^n$. If the components φ_k, $k = 1, 2, \dots, m$, are integrable on X, we say that φ is *integrable* on X. By definition, the *integral* of φ on X is the point in $\mathbb{R}^m$ of components

$$\left(\int_X \varphi_1 \, dx, \int_X \varphi_2 \, dx, \dots, \int_X \varphi_m \, dx \right) ,$$

as usual denoted by

$$\int_X \varphi(x) \, dx \, .$$

(Integral) Jensen Inequality *Let X be a compact and measurable set in $\mathbb{R}^n$ and $\varphi : X \subseteq \mathbb{R}^n \to A \subseteq \mathbb{R}^m$ a continuous function on X. Suppose K is a closed convex set in $\mathbb{R}^m$ such that $\varphi(X) \subseteq K$, and $f : K \subseteq \mathbb{R}^m \to \mathbb{R}$ is a convex function on K. Then*

$$\frac{1}{m(X)} \int_X \varphi(x) \, dx \in K \, ,$$

$$f\left(\frac{1}{m(X)} \int_X \varphi(x) \, dx \right) \leq \frac{1}{m(X)} \int_X f(\varphi(x)) \, dx \, .$$

Proof Consider a partition $P = \{X_1, X_2, \dots, X_h\}$ of X made of measurable sets, and let $x_1, x_2, \dots, x_h$ be h points of X such that $x_i \in X_i$ for any $i = 1, 2, \dots, h$. The

integral of a continuous function can be computed as a suitable limit (see Sect. 8.1), so the integral of φ on X is

$$\lim_{\operatorname{diam}\{X_i\}\to 0} \sum_{i=1}^{h} m(X_i)\,\varphi(x_i) = \int_X \varphi(x)\,dx\,,$$

when the diameter of $P = \{X_i\}$ tends to zero. By assumption $\varphi(x_i) \in K$ for any $i = 1, 2, \ldots, h$, so the discrete Jensen inequality gives

$$\frac{\displaystyle\sum_{i=1}^{h} m(X_i)\varphi(x_i)}{m(X)} = \frac{\displaystyle\sum_{i=1}^{h} m(X_i)\varphi(x_i)}{\displaystyle\sum_{i=1}^{h} m(X_i)} \in K\,,$$

and then

$$f\left(\frac{\displaystyle\sum_{i=1}^{h} m(X_i)\varphi(x_i)}{m(X)}\right) = f\left(\frac{\displaystyle\sum_{i=1}^{h} m(X_i)\varphi(x_i)}{\displaystyle\sum_{i=1}^{h} m(X_i)}\right) \leq$$

$$\leq \frac{\displaystyle\sum_{i=1}^{h} m(X_i) f(\varphi(x_i))}{\displaystyle\sum_{i=1}^{h} m(X_i)} = \frac{\displaystyle\sum_{i=1}^{h} m(X_i)\, f(\varphi(x_i))}{m(X)}\,.$$

Since K is closed, the conclusion follows from the limit as the diameter of $P = \{X_i\}$ tends to zero.

8.11 The Gamma Function. The Measure of the Unit Ball in $\mathbb{R}^n$

For any $t > 0$ define

$$\Gamma(t) = \int_0^{+\infty} x^{t-1} e^{-x}\,dx\,. \tag{8.118}$$

The function $\Gamma : (0, +\infty) \to (0, +\infty)$ thus defined is called *Gamma function*. Note that for any $t > 0$, the improper integral defining $\Gamma(t)$ is finite.

The above function has interesting applications in Mathematical Analysis. Here we shall illustrate a few properties of Γ and show how the use of this function allows to compute the measure of the unit ball $B = \{x \in \mathbb{R}^n : |x| \leq 1\}$ in $\mathbb{R}^n$.

The Γ function generalises the well-known concept of *factorial* of a positive integer. For any $t > 0$, in fact,

$$\Gamma(t+1) = \int_0^{+\infty} x^t\, e^{-x}\, dx = \left[-x^t\, e^{-x}\right]_{x=0}^{x\to+\infty} + t \int_0^{+\infty} x^{t-1} e^{-x}\, dx$$

and so

$$\Gamma(t+1) = t\Gamma(t)\,, \qquad \forall\, t > 0. \tag{8.119}$$

Since

$$\Gamma(1) = \int_0^{+\infty} e^{-x}\, dx = 1\,, \tag{8.120}$$

from (8.119) and (8.120) we deduce that

$$\Gamma(k+1) = k!\,, \qquad \forall\, k = 0, 1, 2, \ldots$$

Using Γ we may obtain a formula to compute integrals of the type

$$I_\alpha = \int_{-1}^{1} (1-x^2)^\alpha\, dx\,, \qquad \alpha > -1. \tag{8.121}$$

For that, let us prove the following.

Proposition 1 *Given $s, t > 0$, set*

$$B(s,t) = 2\int_0^{\pi/2} (\cos\vartheta)^{2s-1} (\sin\vartheta)^{2t-1}\, d\vartheta\,. \tag{8.122}$$

Then

$$B(s,t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)}\,. \tag{8.123}$$

The function in (8.122) is also known as *Beta function*.

Proof Given $t > 0$, the change of variable $x = u^2/2$ in (8.118) produces

$$\Gamma(t) = 2^{1-t} \int_0^{+\infty} u^{2t-1} e^{-u^2/2}\, du\,. \tag{8.124}$$

For $s, t > 0$, then, we find

$$\begin{aligned}\Gamma(s)\Gamma(t) &= 2^{2-s-t}\int_0^{+\infty} u^{2s-1}e^{-u^2/2}\,du\cdot\int_0^{+\infty} v^{2t-1}e^{-v^2/2}\,dv = \\ &= 2^{2-s-t}\iint_X u^{2s-1}v^{2t-1}e^{-(u^2+v^2)/2}\,du\,dv\,,\end{aligned} \tag{8.125}$$

where $X = \{(u, v) \in \mathbb{R}^2 : u > 0, v > 0\}$ is the first quadrant of $\mathbb{R}^2$. Passing to polar coordinates in the last integral,

$$\Gamma(s)\Gamma(t) = 2^{2-s-t}\int_0^{+\infty} \varrho^{2(s+t)-1}e^{-\varrho^2/2}\,d\varrho\cdot\int_0^{\pi/2}(\cos\vartheta)^{2s-1}(\sin\vartheta)^{2t-1}\,d\vartheta\,,$$

so (8.124), (8.125) and definition (8.122) give

$$\Gamma(s)\Gamma(t) = 2\Gamma(s+t)\int_0^{\pi/2}(\cos\vartheta)^{2s-1}(\sin\vartheta)^{2t-1}\,d\vartheta = \Gamma(s+t)B(s,t)\,,$$

which is precisely (8.123).

The above easily implies

Proposition 2 *For any $\alpha > -1$ the integral I_α in (8.121) equals*

$$I_\alpha = \int_{-1}^{1}(1-x^2)^\alpha\,dx = \frac{\Gamma(1/2)\Gamma(\alpha+1)}{\Gamma(\alpha+3/2)}\,.$$

Proof The change of variable $x = \sin\vartheta$ in (8.121) gives

$$I_\alpha = 2\int_0^{\pi/2}(\cos\vartheta)^{2\alpha+1}\,d\vartheta = B(\alpha+1, 1/2) \tag{8.126}$$

by definition (8.122). The claim follows now from this and (8.123).

Also note that if we choose $s = t = 1/2$ in (8.123), and recalling that $\Gamma(1) = 1$, we deduce in particular

$$(\Gamma(1/2))^2 = B(1/2, 1/2) = 2\int_0^{\pi/2} d\vartheta = \pi\,,$$

and hence

$$\Gamma(1/2) = \sqrt{\pi}\,. \tag{8.127}$$

Now let us prove the following

Formula for the Measure of the Unit Ball in $\mathbb{R}^n$ *For any $n \in \mathbb{N}$ denote by ω_n the measure $m_n(B)$ of the unit ball $B = \{x \in \mathbb{R}^n : |x| \le 1\}$ in $\mathbb{R}^n$. Then*

$$\omega_n = m_n(B) = \frac{\pi^{n/2}}{(n/2)\Gamma(n/2)}\,, \qquad n = 1, 2, 3, \dots \tag{8.128}$$

The reader should notice that, due to (8.127), (8.120) and (8.119),

$$\Gamma(1/2) = \sqrt{\pi}\,, \quad \Gamma(1) = 1\,, \quad \Gamma(3/2) = \frac{1}{2}\sqrt{\pi}\,,$$

and for $n = 1, 2, 3$ formula (8.128) returns, as expected, the well-known values

$$\omega_1 = 2\,, \quad \omega_2 = \pi\,, \quad \omega_3 = \frac{4}{3}\pi\,.$$

Observe moreover that (8.128) and (8.119) imply

$$\frac{\omega_{n+2}}{\omega_n} = \frac{n\pi}{n+2}\,\frac{\Gamma(n/2)}{\Gamma(n/2+1)} = \frac{2\pi}{n+2}\,,$$

so for even indices we have

$$\omega_{2k} = \frac{\pi}{k}\omega_{2k-2} = \dots = \frac{\pi^{k-1}}{k!}\omega_2 = \frac{\pi^k}{k!}\,, \tag{8.129}$$

whilst for odd indices

$$\omega_{2k+1} = \frac{2\pi}{2k+1}\omega_{2k-1} = \dots = \frac{2^k\pi^k}{(2k+1)(2k-1)\cdots 3}\omega_1 = \frac{2^{k+1}\pi^k}{(2k+1)!!}\,, \tag{8.130}$$

where $(2k+1)!! = (2k+1)\cdot(2k-1)\cdots 3\cdot 1$.

In particular

$$\lim_{n\to+\infty} \omega_n = 0\,.$$

This fact is surprising, since the measure of the unit hypercube $Q = [-1, 1]^n$, in which B is inscribed, equals 2^n, and so it diverges as the dimension increases!

Proof of (8.128) We argue by induction on the dimension. We have already seen the formula for $n = 1$. When $n \ge 1$, the reduction formulas of Sect. 8.8 imply

$$\omega_{n+1} = m_{n+1}(B) = \int_{-1}^{1} m_n(B_t)\,dt\,, \tag{8.131}$$

where $B_t = \{x \in \mathbb{R}^n : (x,t) \in B\} = \{x \in \mathbb{R}^n : |x|^2 \le 1 - t^2\}$. For any $t \in (-1,1)$, therefore, B_t is a ball of radius $r = \sqrt{1-t^2}$, so

$$m_n(B_t) = \omega_n \, (1-t^2)^{n/2} = \frac{\pi^{n/2}}{(n/2)\Gamma(n/2)} (1-t^2)^{n/2} \, . \tag{8.132}$$

By (8.131), (8.132) and (8.126) we then have

$$\omega_{n+1} = \int_{-1}^{1} m_n(B_t)\, dt = \frac{\pi^{n/2}}{(n/2)\Gamma(n/2)} \int_{-1}^{1} (1-t^2)^{n/2}\, dt =$$
$$= \frac{\pi^{n/2}}{(n/2)\Gamma(n/2)} \frac{\Gamma(1/2)\Gamma(n/2+1)}{\Gamma(n/2+3/2)} \, .$$

But

$$\Gamma\left(\frac{n}{2}+1\right) = \frac{n}{2}\Gamma\left(\frac{n}{2}\right) , \qquad \Gamma\left(\frac{n}{2}+\frac{3}{2}\right) = \Gamma\left(\frac{n+1}{2}+1\right) = \frac{n+1}{2}\Gamma\left(\frac{n+1}{2}\right) ,$$

so we obtain

$$\omega_{n+1} = \frac{\pi^{\frac{n+1}{2}}}{\frac{n+1}{2}\Gamma\left(\frac{n+1}{2}\right)}$$

and then the claim.

Example 10 in Sect. 9.8 will furnish an alternative proof for (8.129) and (8.130).



The Lebesgue Integral

9

9.1 Introduction

In the previous chapter we introduced the theory of the Peano-Jordan measure and a definition of integral that essentially goes back to Riemann (1854). The years following this first definition witnessed many attempts to generalise that notion.

The theory put forward by Lebesgue at the beginning of the last century (1902) is perhaps the most sophisticated one, but without doubt the most flexible and satisfying one as well. First of all, while the Peano-Jordan measure of a subset $E \subset \mathbb{R}^n$ is defined by an approximation process involving pluri-intervals, the Lebesgue measure is built using a double approximation. The first step consists in defining the measure of an arbitrary open set A by approximating it from within using pluri-intervals, and the measure of a compact set K by approximating it with pluri-intervals but from outside. Next, one defines the measure of an arbitrary set E by approximating it with bigger open sets and smaller compact sets. The consequence is that all open sets and all compact sets in $\mathbb{R}^n$ are "*measurable*", which is not true in the Peano-Jordan theory.

Moreover, the Lebesgue measure enjoys the remarkable property of being *countably additive* (Theorem 1, Sect. 9.4). This, too, is false for the Peano-Jordan measure.

Particularly relevant in Lebesgue's theory are the sets of zero measure, precisely because countable additivity guarantees that the union of a sequence of zero-measure sets E_k has zero measure. Furthermore (see Sect. 9.4) any measurable set with finite measure coincides, up to zero-measure sets, with a countable union of compact sets and with a countable intersection of open sets.

Enlarging the family of measurable sets (any Peano-Jordan measurable set is Lebesgue measurable) enables us to integrate many more functions than what we can do in Riemann's theory. The definition of integral suggested by Lebesgue, though, has the disadvantage of not providing a concrete procedure for computing

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integrals explicitly. On the other hand, Fubini's theorem allows to reduce a multiple integral to integrals in one variable.

But the Lebesgue theory's true strength is encapsulated by the theorems ensuring when limits and integrals can be interchanged, which are way more general than what the Riemann integral allows for. These results alone justify the use of Lebesgue's theory to address the most complex and delicate issues of *Mathematical Analysis*.

9.2 Pluri-Intervals. Open Sets. Compact Sets

In the sequel we shall call

$$I = [a_1, b_1] \times [a_2, b_2] \times \ldots \times [a_n, b_n] \tag{9.1}$$

a *closed interval* of $\mathbb{R}^n$, and a *closed pluri-interval* P will be a finite union of closed intervals. The measure $m(I)$ of I is defined as

$$m(I) = \prod_{i=1}^{n} (b_i - a_i)\,, \tag{9.2}$$

so it equals the product of the lengths of the factors $[a_i, b_i]$, $i = 1, 2, \ldots, n$.

If I is the interval in (9.1), setting

$$I_\delta = \prod_{i=1}^{n} [a_i - \delta, b_i + \delta]\,, \qquad \text{with } \delta > 0,$$

we have

$$I \subset \overset{\circ}{I}_\delta, \qquad \lim_{\delta \to 0^+} m(I_\delta) = m(I)\,.$$

That is to say: given an interval I, for any $\varepsilon > 0$ there is an interval J such that

$$I \subset \overset{\circ}{J}, \qquad m(J) < m(I) + \varepsilon\,. \tag{9.3}$$

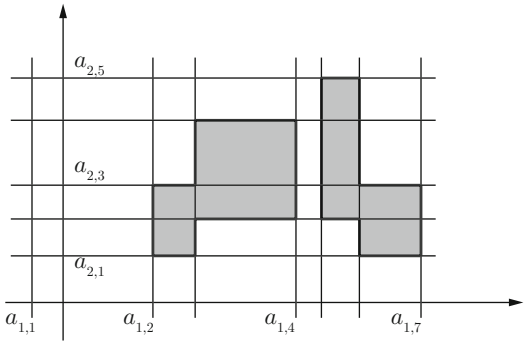
Similarly, if $\overset{\circ}{I}$ is non-empty, for any $\varepsilon > 0$ there is an interval J' such that

$$J' \subset \overset{\circ}{I}, \qquad m(J') > m(I) - \varepsilon\,. \tag{9.4}$$

For any $i = 1, 2, \ldots, n$, fix real numbers

$$a_{i,0} < a_{i,1} < \ldots < a_{i,N_i}\,.$$

Fig. 9.1



Given i and j, consider the hyperplanes

$$H_{ij} = \left\{x \in \mathbb{R}^n : \; x_i = a_{i,j}\right\}.$$

Their union $\mathcal{G}$ is called a *grid*. A grid determines $N_1 \times N_2 \times \ldots \times N_n$ distinct closed intervals, with endpoints lying on the hyperplanes:

$$J = \prod_{i=1}^{n} [a_{i,s_i-1}, a_{i,s_i}], \tag{9.5}$$

where $s_i \in \{1, 2, \ldots, N_i\}$ for any $i = 1, 2, \ldots, n$.

We say that a pluri-interval P is *determined by the grid* $\mathcal{G}$ when it is a union of intervals of type (9.5) (see Fig. 9.1).

So if $P = \bigcup_{k=1}^{N} J_k$, with $J_1, J_2, \ldots, J_N$ distinct intervals determined by $\mathcal{G}$, we let the measure of P be the quantity

$$m(P) = \sum_{k=1}^{N} m(J_k). \tag{9.6}$$

This sum does not depend on the particular grid $\mathcal{G}$ chosen to represent P. It is straightforward to see that by adding to $\mathcal{G}$ any other hyperplane, possibly generating a different representation for P, the sum in (9.6) remains unchanged. By induction then, if $\mathcal{G}'$ is any grid containing $\mathcal{G}$, the quantity in (9.6) stays the same.

If, furthermore, P can be expressed as union of intervals coming from two grids $\mathcal{G}_1, \mathcal{G}_2$, the measures of P using $\mathcal{G}_1$ and $\mathcal{G}_2$ coincide, since both are equal to the measure of P coming from the grid $\mathcal{G}_1 \cup \mathcal{G}_2$.

If P_1 and P_2 are pluri-intervals determined by $\mathcal{G}_1$ and $\mathcal{G}_2$, we have

$$\overset{\circ}{P}_1 \cap \overset{\circ}{P}_2 = \emptyset \quad \Rightarrow \quad m(P_1 \cup P_2) = m(P_1) + m(P_2). \tag{9.7}$$

This follows from the definition, because if P_1, P_2 do not have common interior points they are determined by distinct intervals of $\mathcal{G}_1 \cup \mathcal{G}_2$.

Similarly, writing arbitrary pluri-intervals P_1 and P_2 as unions of intervals of $\mathcal{G}_1 \cup \mathcal{G}_2$, by (9.6) we deduce

$$m(P_1 \cup P_2) \leq m(P_1) + m(P_2)\,. \tag{9.8}$$

Now we define the measure of open and compact subsets in $\mathbb{R}^n$. If $A \subseteq \mathbb{R}^n$ is *open* we define its measure to be

$$m(A) = \sup\{m(P) :\ P \subset A,\ \ P \text{ is a pluri-interval}\}\,. \tag{9.9}$$

Note that if P is a pluri-interval, the measure of its interior $\overset{\circ}{P}$, according to (9.9), coincides with the measure of P from (9.6):

$$m(\overset{\circ}{P}) = m(P)\,. \tag{9.10}$$

The inequality $m(\overset{\circ}{P}) \leq m(P)$ is clear. Vice versa, suppose $P = \bigcup_{k=1}^{N} I_k$, with $I_1, I_2, \ldots, I_N$ distinct intervals determined by $\mathcal{G}$. Given $\varepsilon > 0$, by (9.4) there are intervals $J_1, J_2, \ldots, J_N$ such that

$$J_k \subset \overset{\circ}{I}_k, \quad m(J_k) > m(I_k) - \frac{\varepsilon}{N}\,, \qquad \forall\, k = 1, 2, \ldots, N.$$

Put $Q = \bigcup_{k=1}^{N} J_k$, so $Q \subset \overset{\circ}{P}$ and

$$m(Q) = \sum_{k=1}^{N} m(J_k) > m(P) - \varepsilon\,.$$

From this it follows that $m(\overset{\circ}{P}) > m(P) - \varepsilon$, whence (9.10).

Definition (9.9) says that m is *monotone* on open sets, so

$$A \subseteq B \quad \Rightarrow \quad m(A) \leq m(B)$$

for any open sets A, B. If $K \subset \mathbb{R}^n$ is *compact* we define its measure by

$$m(K) = \inf\left\{m(P) :\ \overset{\circ}{P} \supset K,\ \ P \text{ is a pluri-interval}\right\}. \tag{9.11}$$

The reasoning leading to (9.10) also shows that, using (9.3), if P is a pluri-interval then (9.11) gives as measure of P exactly the number in (9.6).

Evidently m is *monotone* on compact sets as well, i.e.

$$H \subseteq K \quad \Rightarrow \quad m(H) \leq m(K) \tag{9.12}$$

for any compact sets K, H.

The measure is *invariant under translations* on open sets and compact sets alike. For instance, if A is open and $x_0 \in \mathbb{R}^n$,

$$m(x_0 + A) = m(A)\,, \tag{9.13}$$

where $x_0 + A = \{x \in \mathbb{R}^n : x = x_0 + z, z \in A\}$. Property (9.13) holds for intervals, hence for pluri-intervals. A similar observation goes for *homotheties*: for any $\lambda > 0$ and any open (or compact) set $A \subseteq \mathbb{R}^n$ we have

$$m(\lambda A) = \lambda^n\, m(A)\,, \tag{9.14}$$

where $\lambda\, A = \{x \in \mathbb{R}^n : x = \lambda z,\ z \in A\}$.

Before we move on to arbitrary subsets, let us establish a few properties of m, namely: *subadditivity* on open sets and *superadditivity* on compact sets. To do that we start with a lemma.

Lemma 1 *Take open sets A, B in $\mathbb{R}^n$ and $K \subset A \cup B$ compact. There exists $r > 0$ such that, if $x \in K$, the ball $B_r(x)$ is entirely contained in one of the open sets.*

Proof By Proposition 4, Sect. 2.3, the function

$$f(x) = d(x, \mathbb{R}^n - A) + d(x, \mathbb{R}^n - B)\,, \qquad x \in \mathbb{R}^n,$$

is continuous. As K is contained in $A \cup B$, $f(x)$ is never zero on K, so

$$M = \min\{f(x) : x \in K\}$$

must be positive. Given $r \in (0, M/2)$, taking $x \in K$ implies $f(x) > 2r$ and hence either $d(x, \mathbb{R}^n - A) > r$ or $d(x, \mathbb{R}^n - B) > r$. In the former case $B_r(x) \subseteq A$, in the latter $B_r(x) \subseteq B$.

Finite Subadditivity on Open Sets *Let A, A_1, A_2 be open sets in $\mathbb{R}^n$ with $A \subseteq A_1 \cup A_2$. Then*

$$m(A) \leq m(A_1) + m(A_2)\,. \tag{9.15}$$

Proof Let $P \subset A$ be a pluri-interval. By Lemma 1 there is $r > 0$ such that, if $x \in P$, $B_r(x)$ is contained in one of A_1, A_2 at least. Choose a grid $\mathcal{G}$ such that $P = \bigcup_{k=1}^{N} I_k$,

where $I_1, I_2, \ldots, I_N$ are determined by $\mathcal{G}$. Possibly adding to $\mathcal{G}$ a finite number of hyperplanes we may assume that all intervals determined by $\mathcal{G}$ have diameter smaller than r. Now put

$$P_1 = \bigcup \{I_k : I_k \subset A_1\}, \qquad P_2 = \bigcup \{I_k : I_k \subset A_2\},$$

so $P_1 \cup P_2 = P$, and by (9.8),

$$m(P) \le m(P_1) + m(P_2) \le m(A_1) + m(A_2)\,.$$

As P is arbitrary, the claim follows.

Countable Subadditivity on Open Sets *Let $A, A_1, A_2, \ldots, A_k, \ldots$ be open sets in $\mathbb{R}^n$, with $A = \bigcup_{k=1}^{\infty} A_k$. Then*

$$m(A) \le \sum_{k=1}^{+\infty} m(A_k)\,.$$

Proof A pluri-interval $P \subset A$ is compact, so by the characterisation of compact sets of Sect. 2.12 there is an integer N such that $P \subset \bigcup_{k=1}^{N} A_k$. Then (9.15) implies

$$m(P) \le m\left(\bigcup_{k=1}^{N} A_k\right) \le \sum_{k=1}^{N} m(A_k) \le \sum_{k=1}^{+\infty} m(A_k)$$

and hence the assertion.

Finite Superadditivity on Compact Sets *Let H, K be compact in $\mathbb{R}^n$, with $H \cap K = \emptyset$. Then*

$$m(H) + m(K) \le m(H \cup K)\,. \tag{9.16}$$

Proof Set $M = M(H, K) = \min\{d(x, K) : x \in H\}$. As H is compact and disjoint from K, we have $M > 0$.

Let P be a pluri-interval such that $\overset{\circ}{P} \supset K \cup H$. As in the proof of subadditivity on open sets, we may suppose $P = \bigcup_{k=1}^{N} I_k$, with distinct intervals $I_1, I_2, \ldots, I_N$ determined by a suitable grid $\mathcal{G}$ and of diameter less than M. Each interval then intersects one compact set at most. Letting

$$P_1 = \bigcup \{I_k : I_k \cap H \neq \emptyset\}, \qquad P_2 = \bigcup \{I_k : I_k \cap K \neq \emptyset\},$$

we have $H \subseteq P_1$, $K \subseteq P_2$ and $\overset{\circ}{P}_1 \cap \overset{\circ}{P}_2 = \emptyset$. By (9.7) and (9.12) then,

$$m(H) + m(K) \leq m(P_1) + m(P_2) = m(P_1 \cup P_2) \leq m(P)$$

and hence (9.16).

In conclusion observe that

$$K \subset A \quad \Rightarrow \quad m(K) \leq m(A) \tag{9.17}$$

if K is compact and A open. This follows directly from definitions (9.9), (9.11) and the following fact.

Lemma 2 *If K is compact and A open, $K \subset A$, there exists a pluri-interval P such that $K \subset \overset{\circ}{P} \subset P \subset A$.*

Proof For any $x \in K$ there is a closed interval $I_x \subset A$ with $x \in \overset{\circ}{I}_x$. By compactness, we can find a finite number of such intervals, say $I_1, I_2, \ldots, I_N$, so that $K \subset \bigcup_{k=1}^{N} \overset{\circ}{I}_k$. Now choose $P = \bigcup_{k=1}^{N} I_k$.

9.3 Bounded Measurable Sets

From the measure of compact and open sets we can construct the *outer measure* m_e and the *inner measure* m_i of any bounded subset E in $\mathbb{R}^n$:

$$\begin{aligned} m_e(E) &= \inf\{m(A) : \ A \supseteq E, \ A \text{ is open}\}; \\ m_i(E) &= \sup\{m(K) : \ K \subseteq E, \ K \text{ is compact}\}. \end{aligned} \tag{9.18}$$

Note that m_i and m_e are monotone, i.e. if $E \subseteq F$ and F is bounded

$$m_i(E) \leq m_i(F), \qquad m_e(E) \leq m_e(F).$$

By (9.17), moreover,

$$m_i(E) \leq m_e(E)$$

for any bounded E. A bounded subset E in $\mathbb{R}^n$ is called *measurable* if the inner and outer measures coincide. Their common value, written

$$m(E) = m_i(E) = m_e(E)\,,$$

is the (*Lebesgue*) *measure* of E.

In other terms, E is measurable if and only if, for any $\varepsilon > 0$, there exist a compact set $K \subseteq E$ and an open set $A \supseteq E$ such that

$$m(A) - m(K) < \varepsilon\,. \tag{9.19}$$

Immediately, if A is open and bounded, A is measurable. In fact, by (9.9), for any $\varepsilon > 0$ there is a closed pluri-interval such that $m(A) - m(P) < \varepsilon$. Since $m(P) \le m_i(A) \le m_e(A) = m(A)$, the Lebesgue measure of A coincides with (9.9). Similarly, any compact set K is measurable, and its Lebesgue measure coincides with (9.11).

If E is bounded and we write $\overline{m}(E)$, $\underline{m}(E)$ for the outer and inner Peano-Jordan measures, then

$$\underline{m}(E) \le m_i(E) \le m_e(E) \le \overline{m}(E)\,.$$

So, if E is Peano-Jordan measurable then it is Lebesgue measurable as well. The converse (see Example 1, next section) is typically false.

We pass to the additivity of m on bounded measurable sets, starting from a lemma.

Lemma *Let E, F be bounded. Then*

$$m_e(E \cup F) \le m_e(E) + m_e(F)\,. \tag{9.20}$$

If, further, $E \cap F = \emptyset$, then

$$m_i(E) + m_i(F) \le m_i(E \cup F)\,. \tag{9.21}$$

Proof Given $\varepsilon > 0$, consider open sets A, B such that $A \supseteq E$, $B \supseteq F$ and

$$m(A) < m_e(E) + \varepsilon\,, \qquad m(B) < m_e(F) + \varepsilon\,.$$

As $E \cup F \subseteq A \cup B$, by subadditivity on open sets

$$m_e(E \cup F) \le m(A \cup B) \le m(A) + m(B) < m_e(E) + m_e(F) + 2\varepsilon,$$

whence (9.20). Property (9.21) is proved in a similar way, using superadditivity on compact sets.

Properties (9.20) and (9.21) immediately imply finite additivity on measurable sets.

Finite Additivity on Measurable Sets *Let E, F be measurable, bounded and disjoint. Then $E \cup F$ is measurable and*

$$m(E \cup F) = m(E) + m(F)\,. \tag{9.22}$$

Proposition *If E, F are bounded and measurable, also $E \cup F$, $E - F$ and $E \cap F$ are measurable.*

Proof We claim $E - F$ is measurable. Given $\varepsilon > 0$ there exist open sets A, B and compact sets H, K such that $H \subseteq E \subseteq A$, $K \subseteq F \subseteq B$, and

$$m(A) - m(H) < \varepsilon\,, \qquad m(B) - m(K) < \varepsilon\,.$$

We can always assume A and B bounded, possibly intersecting them with open intervals containing E and F. Note that $H - B \subseteq E - F \subseteq A - K$. By the additivity on bounded measurable sets

$$\begin{aligned} m(A - K) - m(H - B) &= m\left((A - K) - (H - B)\right) \leq \\ &\leq m((A - H) \cup (B - K))\,, \end{aligned}$$

so from (9.15) and (9.22) we have

$$\begin{aligned} m(A - K) - m(H - B) &\leq m(A - H) + m(B - K) = \\ &= m(A) - m(H) + m(B) - m(K) < 2\varepsilon\,. \end{aligned}$$

By (9.19), $E - F$ is then measurable.

Now as $E \cap F = E - (E - F)$ it follows that $E \cap F$ is measurable, while $E \cup F = E \cup (F - E)$ is measurable by finite additivity.

Let us now consider countably many measurable sets.

Countable Additivity on Measurable Sets *Let E_k be a collection of pairwise-disjoint bounded measurable sets such that $E = \bigcup_{k=1}^{\infty} E_k$ is bounded. Then E is measurable and*

$$m(E) = \sum_{k=1}^{+\infty} m(E_k)\,.$$

Proof Repeatedly applying (9.22) gives

$$\sum_{k=1}^{N} m(E_k) = m(E_1 \cup E_2 \cup \ldots \cup E_N) =$$

$$= m_i(E_1 \cup E_2 \cup \ldots \cup E_N) \leq m_i(E)\,, \qquad \forall\, N \in \mathbb{N},$$

so as $N \to +\infty$,

$$\sum_{k=1}^{+\infty} m(E_k) \leq m_i(E)\,. \tag{9.23}$$

Conversely, given $\varepsilon > 0$, for any $k \in \mathbb{N}$ let $A_k \supseteq E_k$ be an open set such that $m(A_k) < m(E_k) + \varepsilon/2^k$. By countable subadditivity on open sets,

$$m_e(E) \leq m\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{+\infty} m(A_k) \leq \sum_{k=1}^{+\infty} m(E_k) + \varepsilon\,,$$

implying, as ε is arbitrary,

$$m_e(E) \leq \sum_{k=1}^{+\infty} m(E_k)\,. \tag{9.24}$$

The claim follows from (9.23) and (9.24).

Corollary *If E_k is a collection of measurable sets such that $E = \bigcup_{k=1}^{\infty} E_k$ is bounded, then E is measurable.*

Proof Put

$$F_1 = E_1, \quad F_2 = E_2 - E_1, \ldots, F_k = E_k - \bigcup_{i=1}^{k-1} E_i\,.$$

The F_k are measurable, pairwise disjoint and

$$E = \bigcup_{k=1}^{\infty} F_k\,.$$

9.4 Unbounded Measurable Sets

Consider an *unbounded* subset E in $\mathbb{R}^n$. Call

$$B_r = B_r(0) = \{x \in \mathbb{R}^n : \ |x| < r\}$$

the ball at the origin with radius $r > 0$. We say E is *measurable* if, for any $r > 0$, $E \cap B_r$ is measurable, and in this case we define the measure of E as

$$m(E) = \lim_{r \to +\infty} m(E \cap B_r)\,. \tag{9.25}$$

The limit in (9.25) exists since the function $r \to m(E \cap B_r)$ is increasing. Moreover, it coincides with the measure of E when E is bounded and measurable.

We observed earlier that an open bounded set A is measurable and its Lebesgue measure coincides with (9.9). The same happens if A is unbounded. For any $r > 0$, in fact, $A \cap B_r$ is open, bounded and hence measurable. Therefore A is measurable in the above sense. Setting

$$\lambda = \sup\{m(P) : \ P \subset A, \ P \text{ is a pluri-interval}\}\,,$$

for any $r > 0$

$$m(A \cap B_r) = \sup\{m(P) : \ P \subset A \cap B_r\,, \ P \text{ is a pluri-interval}\}\,,$$

and hence

$$\lim_{r \to +\infty} m(A \cap B_r) \leq \lambda\,. \tag{9.26}$$

Conversely, if $P \subset A$ is a pluri-interval, there is $r_0 > 0$ such that $P \subset A \cap B_r$ for any $r > r_0$. Therefore

$$m(P) \leq \lim_{r \to +\infty} m(A \cap B_r)\,,$$

and then

$$\lambda \leq \lim_{r \to +\infty} m(A \cap B_r)\,. \tag{9.27}$$

By (9.26) and (9.27) we obtain

$$\lambda = \lim_{r \to +\infty} m(A \cap B_r)\,,$$

showing that the Lebesgue measure of A coincides with (9.9) in this case, too.

The next result summarises the main properties of measurable sets.

Theorem 1 *If E is a measurable subset of $\mathbb{R}^n$, its complement $\mathbb{R}^n - E$ is measurable. If E_k is a family of measurable sets, both $\bigcap_{k=1}^{\infty} E_k$ and $\bigcup_{k=1}^{\infty} E_k$ are measurable. Moreover*

$$m\left(\bigcup_{k=1}^{+\infty} E_k\right) \leq \sum_{k=1}^{+\infty} m(E_k)\,, \tag{9.28}$$

and if the E_k are pairwise disjoint,

$$m\left(\bigcup_{k=1}^{+\infty} E_k\right) = \sum_{k=1}^{+\infty} m(E_k)\,. \tag{9.29}$$

Proof If E is measurable, for any $r > 0$

$$(\mathbb{R}^n - E) \cap B_r = B_r - (B_r \cap E)$$

so the previous section's proposition says that $(\mathbb{R}^n - E) \cap B_r$ is measurable. Hence $\mathbb{R}^n - E$ is measurable.

If E_k is a collection of measurable sets, for any $r > 0$

$$\left(\bigcup_{k=1}^{\infty} E_k\right) \cap B_r = \bigcup_{k=1}^{\infty} (E_k \cap B_r),$$

making the above measurable by the previous section's corollary. Hence $\bigcup_{k=1}^{\infty} E_k$ is measurable. Finally,

$$\bigcap_{k=1}^{\infty} E_k = \mathbb{R}^n - \bigcup_{k=1}^{\infty} (\mathbb{R}^n - E_k)$$

is measurable by what was said above.

Let us prove (9.28). Setting

$$E = \bigcup_{k=1}^{\infty} E_k\,, \quad F_1 = E_1\,, \quad F_2 = E_2 - E_1, \ldots, F_k = E_k - \bigcup_{i=1}^{k-1} E_i, \ldots,$$

the F_k are measurable, pairwise disjoint and $E = \bigcup_{k=1}^{\infty} F_k$. For any $r > 0$ then, by the countable additivity on bounded measurable sets (previous section),

$$m(E \cap B_r) = \sum_{k=1}^{+\infty} m(F_k \cap B_r) \leq \sum_{k=1}^{+\infty} m(E_k \cap B_r) \leq \sum_{k=1}^{+\infty} m(E_k)\,,$$

and hence (9.28) when computing the limit as $r \to +\infty$. If the E_k are pairwise disjoint, given N, by finite additivity on bounded measurable sets we have

$$\sum_{k=1}^{N} m(E_k \cap B_r) = m\left(\bigcup_{k=1}^{N}(E_k \cap B_r)\right) \leq m(E)\,,$$

for any $r > 0$. As $r \to +\infty$ we find

$$\sum_{k=1}^{N} m(E_k) \leq m(E)\,,$$

and since N is arbitrary,

$$\sum_{k=1}^{+\infty} m(E_k) \leq m(E)\,.$$

Then (9.29) follows from the above inequality and (9.28).

Remarks (On Sets of Zero Measure) It will become clearer in the sequel, when we introduce the concept of integral, that *zero-measure sets* play a central role in Lebesgue's theory. For the time being observe that if $E \subset \mathbb{R}^n$ is measurable with zero measure, any subset F of E is measurable and has measure zero. We express this property by saying that the Lebesgue measure is *complete*.

If E is measurable and E_0 has measure zero, moreover, also $E \cup E_0$ and $E - E_0$ are measurable and

$$m(E \cup E_0) = m(E - E_0) = m(E)\,.$$

That is, *adding to E or subtracting from E a zero-measure set produces a measurable set with the same measure as E*. For this reason, if we are interested in a certain measure-related property on the points of E, it will be irrelevant that the property holds at every point on E rather than just on $E - E_0$, where E_0 has measure zero. One says that a certain property holds *almost everywhere* on E, or *for almost every $x \in E$*, if it holds on $E - E_0$, where $m(E_0) = 0$. It is customary to abbreviate this by "*a.e.*".

We end these observations on zero-measure sets by noting that

$$m(E_k) = 0, \qquad \forall\, k \in \mathbb{N} \quad \Rightarrow \quad m\left(\bigcup_{k=1}^{+\infty} E_k\right) = 0\,,$$

i.e. the union of a sequence of zero-measure sets has measure zero.

Example 1 It is immediate to see that if $x \in \mathbb{R}^n$ then $m(\{x\}) = 0$, i.e. singletons have measure zero. From what we said above, the measure of $E = \{x_1, x_2, \ldots, x_k, \ldots\}$ is zero. Then also

$$F = \{q \in [0,1]^n : \ q = (q_1, q_2, \ldots, q_n) \in \mathbb{Q}\},$$

has $m(F) = 0$. Note that this set is not Peano-Jordan measurable, since its outer measure is 1 whereas the inner measure is 0 (see the example in Sect. 8.6).

Example 2 Let x_k be the collection of rational numbers in $[0, 1]$. Given $\varepsilon \in (0, 1/2)$, define

$$A = (0,1) \cap \bigcup_{k=1}^{\infty} \left(x_k - \frac{\varepsilon}{2^{k+1}}, x_k + \frac{\varepsilon}{2^{k+1}} \right).$$

Now, A is open and its measure is bounded:

$$m(A) \leq \sum_{k=1}^{+\infty} \frac{\varepsilon}{2^k} = \varepsilon\,.$$

Put $K = [0,1] - A$, i.e. $K = \partial A$ is the boundary of A. As $m(K) = 1 - m(A) \geq 1 - \varepsilon > 1/2$, the boundary's measure is larger than the measure of A.

Based on this example the reader may construct an open set whose Lebesgue measure is less than a given positive number ε and whose boundary has infinite measure.

In Lebesgue's theory is it quite easy to work with monotone sequences of sets. A sequence E_k of subsets of $\mathbb{R}^n$ is called *increasing* if

$$E_k \subseteq E_{k+1}\,, \qquad \forall\, k \in \mathbb{N},$$

and *decreasing* when

$$E_k \supseteq E_{k+1}\,, \qquad \forall\, k \in \mathbb{N}.$$

Theorem 2 (Monotone Sequences of Measurable Sets) *If E_k is an increasing sequence of measurable sets,*

$$m\left(\bigcup_{k=1}^{\infty} E_k \right) = \lim_{k \to +\infty} m(E_k)\,. \tag{9.30}$$

If E_k is a decreasing sequence of measurable sets and $m(E_1) < +\infty$, then

$$m\left(\bigcap_{k=1}^{\infty} E_k\right) = \lim_{k\to+\infty} m(E_k)\,. \tag{9.31}$$

Proof Let us prove (9.30). Define $E = \bigcup_{k=1}^{\infty} E_k$, so

$$\lim_{k\to+\infty} m(E_k) \le m(E) \tag{9.32}$$

(the limit exists by monotonicity). If

$$\lim_{k\to+\infty} m(E_k) = +\infty,$$

inequality (9.32) implies (9.30). Otherwise, in particular, $m(E_k) < +\infty$ for any $k \in \mathbb{N}$. Define

$$F_1 = E_1\,, \quad F_2 = E_2 - E_1, \ldots, F_k = E_k - E_{k-1}, \ldots.$$

These are measurable, pairwise disjoint and $E = \bigcup_{k=1}^{\infty} F_k$, so

$$m(E) = \sum_{k=1}^{+\infty} m(F_k) = \lim_{k\to+\infty} \sum_{i=1}^{k} m(F_i) =$$
$$= \lim_{k\to+\infty} m(F_1 \cup F_2 \cup \ldots \cup F_k) = \lim_{k\to+\infty} m(E_k)\,,$$

proving (9.30). As for (9.31), observe that if E_k is decreasing, $E_1 - E_k$ is an increasing sequence. Hence

$$m(E_1) - m\left(\bigcap_{k=1}^{\infty} E_k\right) = m\left(E_1 - \bigcap_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty}(E_1 - E_k)\right) =$$
$$= \lim_{k\to+\infty} m(E_1 - E_k) = m(E_1) - \lim_{k\to+\infty} m(E_k)\,,$$

and (9.31) follows from $m(E_1) < +\infty$.

Example 3 Formula (9.31) might not hold if $m(E_1) = +\infty$. In fact, taking $E_k = [k, +\infty)$ for any $k = 1, 2, \ldots$, we have $\bigcap_{k=1}^{\infty} E_k = \emptyset$, so $m\left(\bigcap_{k=1}^{\infty} E_k\right) = 0$ while $m(E_k) = +\infty$ for any $k \in \mathbb{N}$.

Example 4 (Generalised Cantor Set) Fix a real number $s \geq 3$ and define

$$C_1 = [0,1] - I\,,$$

where I is the open interval with midpoint $1/2$ and length $1/s$:

$$I = \left(\frac{1}{2} - \frac{1}{2s}, \frac{1}{2} + \frac{1}{2s}\right);$$

C_1 is then the union of two closed intervals (Fig. 9.2). Now let C_2 be the set obtained removing from each interval a concentric open interval of length $1/s^2$ (see Fig. 9.2). Iterating for any $k \in \mathbb{N}$, call C_{k+1} the set remaining after we remove from each of the 2^k closed intervals in C_k a concentric open interval of length $1/s^{k+1}$. In this way we generate a decreasing sequence C_k of measurable sets.

Let us find the measure of C_k. For any k

$$\begin{aligned} m(C_{k+1}) &= m(C_k) - \frac{2^k}{s^{k+1}} = m(C_{k-1}) - \frac{2^{k-1}}{s^k} - \frac{2^k}{s^{k+1}} = \\ &= \ldots = m(C_1) - \frac{2}{s^2} - \frac{2^2}{s^3} - \ldots - \frac{2^k}{s^{k+1}} = 1 - \sum_{i=0}^{k} \frac{2^i}{s^{i+1}}\,. \end{aligned} \tag{9.33}$$

For any $s \geq 3$ define

$$K_s = \bigcap_{k=1}^{\infty} C_k\,.$$

K_3 is called *Cantor set*, while K_s is the *generalised Cantor set* for $s > 3$. For any $s \geq 3$, K_s is compact with empty interior. By (9.31) and (9.33),

$$m(K_s) = 1 - \sum_{k=0}^{\infty} \frac{2^k}{s^{k+1}} = 1 - \frac{1}{s} \sum_{k=0}^{\infty} \frac{2^k}{s^k} = 1 - \frac{1}{s} \cdot \frac{1}{1 - 2/s} = \frac{s-3}{s-2}\,.$$

In particular, the Lebesgue measure of the Cantor set is zero, while $m(K_s) > 0$ for any $s > 3$. Therefore for any $s > 3$ the generalised Cantor set K_s is compact with empty interior, but it has positive measure.

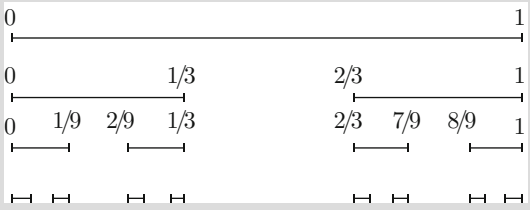


Fig. 9.2 Constructing the Cantor set

The next result will show that unbounded measurable sets may also be approximated, in a measure-theoretical sense, by open and compact sets.

Theorem 3 *If E is measurable,*

$$m(E) = \inf\{m(A) : \ A \supseteq E, \ A \text{ open}\}, \tag{9.34}$$

$$m(E) = \sup\{m(K) : \ K \subseteq E, \ K \ \text{ compact}\}. \tag{9.35}$$

Proof Property (9.34) is obvious if $m(E) = +\infty$. Otherwise, for any k consider $E_k = E \cap (B_k - B_{k-1})$, with $B_0 = \emptyset$. Given $\varepsilon > 0$, by (9.18) we know that for any k there is an open set $A_k \supseteq E_k$ such that $m(A_k) < m(E_k) + \varepsilon/2^k$. Set $A = \bigcup_{k=1}^{\infty} A_k$, so

$$m(A) \leq \sum_{k=1}^{+\infty} m(A_k) < \sum_{k=1}^{+\infty} m(E_k) + \varepsilon = m(E) + \varepsilon,$$

and now (9.34) follows. To prove (9.35) define $F_k = E \cap B_k$ and pick compact sets K_k contained in F_k such that $m(K_k) > m(F_k) - 1/k$. Then

$$\lim_{k \to +\infty} m(K_k) = \lim_{k \to +\infty} m(F_k) = m(E),$$

proving (9.35).

Example 5 The theorem just proved implies that if E is any measurable subset of $\mathbb{R}^n$, there exists a family of open sets A'_k such that $A'_k \supseteq E$ for any k and $m(E) = \lim_{k \to +\infty} m(A'_k)$. For any k, let $A_k = \bigcap_{i=1}^{k} A'_i$. This is open and

$$A_k \supseteq A_{k+1} \supseteq E, \qquad \forall k \in \mathbb{N}, \qquad \lim_{k \to +\infty} m(A_k) = m(E).$$

In the same way we can construct an increasing sequence of compact subsets K_k such that $K_k \subseteq E$ for any k and $m(E) = \lim_{k \to +\infty} m(K_k)$. If, in particular, E has finite measure then

$$m\left(\bigcup_{k=1}^{\infty} K_k\right) = m(E) = m\left(\bigcap_{k=1}^{\infty} A_k\right), \qquad \bigcup_{k=1}^{\infty} K_k \subseteq E \subseteq \bigcap_{k=1}^{\infty} A_k.$$

Hence E coincides, up to a zero-measure set, with an increasing union of compact sets and with a decreasing intersection of open sets.

Example 6 Theorem 3 and (9.13) easily imply that the set $x_0 + E = \{x \in \mathbb{R}^n : x = x_0 + z,\ z \in E\}$, obtained displacing a measurable set E, is measurable, and

$$m(x_0 + E) = m(E)\,.$$

The above relation says that the Lebesgue measure is *translation-invariant*. Similarly, from (9.14) we immediately see that for any $\lambda > 0$, $\lambda E = \{x \in \mathbb{R}^n : x = \lambda z, z \in E\}$ is measurable and

$$m(\lambda E) = \lambda^n \cdot m(E)\,. \tag{9.36}$$

Example 7 (A Non-Measurable Set) Consider on $\mathbb{R}$ the relation

$$x \sim y \quad \text{if and only if} \quad x - y \quad \text{is rational.}$$

This is manifestly an equivalence relation, and for any $x \in \mathbb{R}$ there is a real number $y \in (0, 1)$ such that $x \sim y$. So let $E \subset (0, 1)$ be a set containing exactly one point in each equivalence class of $\sim$. The existence of such a set is warranted by the axiom of choice. We claim that

$$r, s \in \mathbb{Q},\ r \neq s \quad \Rightarrow \quad (r + E) \cap (s + E) = \emptyset\,.$$

If there existed $x \in (r+E)\cap(s+E)$, then $x = r+y$, $x = s+z$ with $y, z \in E$. Hence $y \sim z$ and $y - z = s - r \neq 0$, so E would contain two distinct elements of the same equivalence class, against the definition of E. Furthermore, for any $x \in (0, 1)$ there is $r \in (-1, 1) \cap \mathbb{Q}$ such that

$$x \in r + E\,. \tag{9.37}$$

In fact if $x \in (0, 1)$, there is $y \in E$ such that $x - y = r \in \mathbb{Q}$, and since $E \subset (0, 1)$, then $r \in (-1, 1)$.

Let us show E is not measurable. If it were, calling $\alpha = m(E)$ and $F = \bigcup\{r + E : r \in (-1, 1) \cap \mathbb{Q}\}$, we would have $F \subset (-1, 2)$ and hence $m(F) \leq 3$. But F is a disjoint union of translates of E, and since $m(F)$ cannot be $+\infty$, necessarily $m(F) = 0$, whence $\alpha = m(E) = 0$. On the other hand (9.37) implies $F \supseteq (0, 1)$, so $m(F) \geq 1$. The contradiction forces E to be non-measurable.

9.5 Measurable Functions

In the sequel we shall consider functions with values in the extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$. The usual operations of sum and product between real numbers

extend to $\overline{\mathbb{R}}$ if we adopt the following conventions

$$(+\infty) + (+\infty) = +\infty,$$
$$(+\infty) + x = x + (+\infty) = +\infty, \qquad \forall\, x \in \mathbb{R},$$
$$(+\infty) \cdot x = x \cdot (+\infty) = \begin{cases} +\infty & \text{if } x > 0 \\ -\infty & \text{if } x < 0, \end{cases} \qquad \forall\, x \in \mathbb{R} - \{0\}.$$

Similar conventions hold for $-\infty$. Note however that the symbols $(+\infty) - (+\infty)$, $(+\infty) \cdot 0$, $0 \cdot (+\infty)$ are not defined.

We say that $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ is *measurable* if for any real number t the set

$$L_t = \{x \in \mathbb{R}^n : \ f(x) > t\}$$

is measurable.

Example 1 Let χ_E denote the *characteristic function* of a subset $E \subseteq \mathbb{R}^n$:

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E. \end{cases}$$

Since for any $t \in \mathbb{R}$ the set $\{x \in \mathbb{R}^n : \ \chi_E(x) > t\}$ can only be $\emptyset$, E or $\mathbb{R}^n$, we immediately deduce that χ_E is measurable if and only if E is a measurable set.

If $f : \mathbb{R}^n \to \mathbb{R}$ is *a continuous function*, then f *is measurable as well*. In fact for any $t \in \mathbb{R}$ the set $L_t = f^{-1}((t, +\infty))$ is open (see the characterisation of continuity in Sect. 2.12) and hence measurable.

Proposition 1 *Given $f : \mathbb{R}^n \to \overline{\mathbb{R}}$, the following are equivalent:*

(i) $L_t = \{x \in \mathbb{R}^n : \ f(x) > t\}$ *is measurable for any* $t \in \mathbb{R}$*;*
(ii) $L'_t = \{x \in \mathbb{R}^n : \ f(x) \geq t\}$ *is measurable for any* $t \in \mathbb{R}$*;*
(iii) $S_t = \{x \in \mathbb{R}^n : \ f(x) < t\}$ *is measurable for any* $t \in \mathbb{R}$*;*
(iv) $S'_t = \{x \in \mathbb{R}^n : \ f(x) \leq t\}$ *is measurable for any* $t \in \mathbb{R}$*.*

Proof (i) $\Rightarrow$ (ii). It is enough to notice that

$$L'_t = \bigcap_{k=1}^{\infty} L_{t-\frac{1}{k}} \ .$$

(ii) $\Rightarrow$ (iii). This follows from the identity $S_t = \mathbb{R}^n - L'_t$.

(iii) $\Rightarrow$ (iv). Just observe that

$$S'_t = \bigcap_{k=1}^{\infty} S_{t+\frac{1}{k}} .$$

(iv) $\Rightarrow$ (i) . Follows from $L_t = \mathbb{R}^n - S'_t$.

Example 2 If $f : \mathbb{R}^n \to \mathbb{R}$ is measurable and (a, b) is an open bounded real interval, by Proposition 1 the set

$$f^{-1}((a, b)) = L_a \cap S_b$$

is measurable. If $g : \mathbb{R} \to \mathbb{R}$ is continuous, the composite $g \circ f$ is measurable. In fact for any $t \in \mathbb{R}$

$$\{x \in \mathbb{R}^n : \ g(f(x)) > t\} = f^{-1}(\{y \in \mathbb{R} : \ g(y) > t\}) .$$

Since $\{y \in \mathbb{R} : \ g(y) > t\}$ is open, there is a sequence of open intervals (a_k, b_k) such that

$$\{y \in \mathbb{R} : \ g(y) > t\} = \bigcup_{k=1}^{\infty} (a_k, b_k) .$$

Hence

$$\{x \in \mathbb{R}^n : \ g(f(x)) > t\} = \bigcup_{k=1}^{\infty} f^{-1}(a_k, b_k)$$

and the set on the left is measurable.

Let us briefly recall the notions of *limit inferior* and *limit superior* of a sequence a_k of (extended) real numbers (also known as *lower limit* and *upper limit* respectively). Define

$$b_k = \inf_{i \geq k} a_i \ , \qquad \forall\, k \in \mathbb{N}.$$

The sequence b_k is increasing so it admits a limit. By definition, the *limit inferior of* a_k is the limit as $k \to +\infty$ of b_k, written

$$\liminf_{k \to +\infty} a_k, \qquad \liminf_{k \to +\infty}{}' \, a_k \, .$$

Therefore

$$\liminf_{k \to +\infty} a_k = \lim_{k \to +\infty} \inf_{i \geq k} a_i = \sup_{k \in \mathbb{N}} \inf_{i \geq k} a_i \, .$$

The *limit superior*, denoted by

$$\limsup_{k\to+\infty} a_k\,, \qquad \lim''_{k\to+\infty} a_k\,,$$

is defined in a similar fashion by

$$\limsup_{k\to+\infty} a_k = \lim_{k\to+\infty} \sup_{i\geq k} a_i = \inf_{k\in\mathbb{N}} \sup_{i\geq k} a_i\,.$$

Easily the sequence a_k admits limit as $k \to +\infty$ if and only if the limit inferior and the limit superior of a_k coincide; if so,

$$\lim_{k\to+\infty} a_k = \liminf_{k\to+\infty} a_k = \limsup_{k\to+\infty} a_k\,.$$

Example 3 A function $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ is called *lower semi-continuous* if, for any $t \in \mathbb{R}$, the set $L_t = \{x \in \mathbb{R}^n :\ f(x) > t\}$ is open. The argument used in the proposition of Sect. 2.12 shows that f is lower semi-continuous if and only if, for any x_0 and any sequence x_k tending to x_0 ,

$$\liminf_{k\to+\infty} f(x_k) \geq f(x_0)\,.$$

Similarly, f is *upper semi-continuous* if, for any $t \in \mathbb{R}$, the set $S_t = \{x \in \mathbb{R}^n :\ f(x) < t\}$ is open. Equivalently, for any x_0 and any sequence x_k tending to x_0,

$$\limsup_{k\to+\infty} f(x_k) \leq f(x_0)\,.$$

A lower (or upper) semi-continuous function is therefore measurable.

We leave it to the reader to verify that the characteristic function χ_A of an open set $A \subseteq \mathbb{R}^n$ is lower semi-continuous, while if $C \subseteq \mathbb{R}^n$ is closed then χ_C is upper semi-continuous.

The class of measurable functions is closed under addition, multiplication, supremum and infimum.

Theorem

(i) If f is measurable and $c \in \mathbb{R}$, then $f + c$ and $c \cdot f$ are measurable;
(ii) if f, g are measurable, $f + g$, $f \cdot g$ and f/g (if $g(x) \neq 0$ for any x) are measurable;
(iii) if f_k is a sequence of measurable functions, the functions

$$S(x) = \sup_k f_k(x)\,, \quad s(x) = \inf_k f_k(x)\,, \qquad x \in \mathbb{R}^n,$$

are measurable;

(iv) if f_k is a sequence of measurable functions, so are

$$f(x) = \liminf_{k\to+\infty} f_k(x)\,, \quad g(x) = \limsup_{k\to+\infty} f_k(x)\,, \qquad x \in \mathbb{R}^n.$$

Proof Property (i) follows directly from the definition. For (ii), for any $t \in \mathbb{R}$ we have

$$\{x :\ f(x) + g(x) > t\} = \bigcup_{q_k\in\mathbb{Q}} \{x :\ f(x) > q_k\} \cap \{x :\ g(x) > t - q_k\}\,.$$

Hence if f and g are measurable, the sum $f + g$ is measurable.

We claim that $h^2(x)$ is measurable when $h(x)$ is. For this,

$$\{x :\ h^2(x) > t\} = \{x :\ h(x) > \sqrt{t}\} \cup \{x :\ h(x) < -\sqrt{t}\}\,,$$

for any $t \geq 0$, while if $t < 0$, $\{x :\ h^2(x) > t\} = \mathbb{R}^n$. So if f, g are measurable, as

$$f \cdot g = \frac{1}{4}\Big[(f+g)^2 - (f-g)^2\Big]\,,$$

also $f \cdot g$ is measurable. The proof for the quotient f/g reduces to the measurability of $1/g$, which is left to the reader.

To show (iii) observe that

$$\{x :\ S(x) > t\} = \bigcup_{k=1}^{\infty} \{x :\ f_k(x) > t\}\,, \qquad \forall\, t \in \mathbb{R}.$$

The measurability of $s(x)$ goes like the measurability of $S(x)$, or alternatively using

$$s(x) = -\sup_k\,(-f_k(x))\,, \qquad x \in \mathbb{R}^n.$$

Finally, (iv) follows from the definitions of limsup and liminf

$$\liminf_{k\to+\infty} f_k(x) = \sup_k \inf_{h\geq k} f_h(x)\,, \qquad x \in \mathbb{R}^n;$$

$$\limsup_{k\to+\infty} f_k(x) = \inf_k \sup_{h\geq k} f_h(x)\,, \qquad x \in \mathbb{R}^n.$$

This theorem implies that for f, g measurable, the set

$$\{x \in \mathbb{R}^n :\ f(x) \neq g(x)\} = (f-g)^{-1}(\mathbb{R} - \{0\})$$

is measurable. Furthermore, under the same hypothesis

$$f \vee g = \max\{f, g\}, \quad f \wedge g = \min\{f, g\}, \tag{9.38}$$

are measurable. In particular

$$f^+ = f \vee 0, \quad f^- = -f \wedge 0, \quad |f| = f^+ + f^- \tag{9.39}$$

are measurable functions.

Also note that if f is measurable and $g : \mathbb{R}^n \to \overline{\mathbb{R}}$ is any function such that

$$g(x) = f(x)\,, \qquad \text{for almost every } x \in \mathbb{R}^n,$$

then g is measurable. Therefore, modifying the values of a measurable function f on a zero-measure set still gives a measurable function.

Another consequence of the theorem is this: if f_k is a sequence of measurable functions such that $\lim_{k\to+\infty} f_k(x)$ exist for almost every $x \in \mathbb{R}^n$, then any f such that

$$f(x) = \lim_{k\to+\infty} f_k(x)\,, \qquad \text{for a.e. } x \in \mathbb{R}^n$$

is measurable. In the sequel we shall encounter functions defined on subsets of $\mathbb{R}^n$. If E is a measurable subset of $\mathbb{R}^n$, we say $f : E \to \overline{\mathbb{R}}$ is *measurable on* E if the set

$$\{x \in E : \ f(x) > t\}$$

is measurable for any $t \in \mathbb{R}$. When f is measurable on E, the extension $\overline{f}$ obtained by setting f equal to zero outside of E,

$$\overline{f}(x) = \begin{cases} f(x) & \text{if } x \in E \\ 0 & \text{if } x \notin E, \end{cases} \tag{9.40}$$

is measurable, as is clear using the definition. Conversely, if $f : \mathbb{R}^n \to \overline{\mathbb{R}}$ is measurable, its restriction to a measurable set E is measurable on E. Evidently all the properties of measurable functions on $\mathbb{R}^n$ hold for measurable functions on E. In the sequel we shall often speak of a "measurable function" without necessarily saying whether it is defined on $\mathbb{R}^n$ or on some measurable subset.

Now we introduce a special subclass of measurable functions. We say a measurable function $s : \mathbb{R}^n \to [0, +\infty)$ is *simple* if its range consists of finitely many distinct values $c_1, c_2, \ldots, c_N$. For every $i = 1, 2, \ldots, N$ the set

$$E_i = \{x \in \mathbb{R}^n : \ s(x) = c_i\}$$

is measurable. Furthermore, $E_1, E_2, \ldots, E_N$ are pairwise disjoint and

$$s(x) = \sum_{i=1}^{N} c_i \, \chi_{E_i}(x) \,, \qquad \forall\, x \in \mathbb{R}^n.$$

The next result shows how non-negative measurable functions can be characterised as pointwise limits of sequences of simple functions.

Proposition 2 *If $f : \mathbb{R}^n \to [0, +\infty]$ is a measurable function, there is an increasing sequence s_k of simple functions such that*

$$f(x) = \lim_{k \to +\infty} s_k(x) \,, \qquad \forall\, x \in \mathbb{R}^n. \tag{9.41}$$

Proof Fix $k \in \mathbb{N}$ and put

$$E_k = f^{-1}([k, +\infty]) \,,$$
$$E_{k,i} = f^{-1}\Big([(i-1)/2^k, i/2^k)\Big) \,, \qquad i = 1, 2, \ldots, k2^k.$$

By Proposition 1 these sets are measurable, so

$$s_k(x) = k \, \chi_{E_k}(x) + \sum_{i=1}^{k2^k} \frac{i-1}{2^k} \chi_{E_{k,i}}(x)$$

is simple and $s_k(x) \le f(x)$ for any $x \in \mathbb{R}^n$.

Pick $x \in E_{k,i}$, $i = 1, 2, \ldots, k2^k$. Since

$$\left[\frac{i-1}{2^k}, \frac{i}{2^k}\right) = \left[\frac{2i-2}{2^{k+1}}, \frac{2i-1}{2^{k+1}}\right) \cup \left[\frac{2i-1}{2^{k+1}}, \frac{2i}{2^{k+1}}\right) ,$$

i.e.

$$E_{k,i} = E_{k+1,2i-1} \cup E_{k+1,2i} \,,$$

we have

$$s_{k+1}(x) = \begin{cases} \dfrac{2i-2}{2^{k+1}} & \text{if } x \in E_{k+1,2i-1} \\ \dfrac{2i-1}{2^{k+1}} & \text{if } x \in E_{k+1,2i-1}. \end{cases}$$

Therefore

$$s_k(x) \le s_{k+1}(x) \,, \qquad \forall\, x \in E_{k,i} \,.$$

A similar inequality holds if $x \in E_k$. Then the sequence s_k is increasing. Now, (9.41) is obvious if $f(x) = +\infty$, because $s_k(x) = k$ for any $k \in \mathbb{N}$. Instead if $f(x) < k_0$, for any $k > k_0$

$$0 \le f(x) - s_k(x) < \frac{1}{2^k}\,. \tag{9.42}$$

Note that if f is bounded, (9.42) implies that the sequence s_k converges to f uniformly. Clearly if s_k is an increasing sequence of simple functions, the function

$$f(x) = \lim_{k\to+\infty} s_k(x)\,, \qquad \forall\, x \in \mathbb{R}^n$$

is measurable. Then Proposition 2 completely characterises non-negative measurable functions as pointwise limits of increasing sequences of simple functions.

9.6 The Lebesgue Integral. Interchanging Limits and Integrals

Consider the simple function $s : \mathbb{R}^n \to [0, +\infty)$,

$$s(x) = \sum_{i=1}^{N} c_i\, \chi_{E_i}(x)\,, \qquad \forall\, x \in \mathbb{R}^n,$$

where $c_1, c_2, \ldots, c_N$ are distinct range values of $s(x)$ and

$$E_i = \{x \in \mathbb{R}^n :\ s(x) = c_i\}\,, \qquad \forall\, i = 1, 2, \ldots, N.$$

If E is a measurable subset of $\mathbb{R}^n$, we call *Lebesgue integral of* $s(x)$ *on* E the extended real number

$$\int_E s(x)\,dx = \sum_{i=1}^{N} c_i\, m(E_i \cap E)\,, \tag{9.43}$$

where the product $c_i\, m(E_i \cap E)$ is equal to zero if $c_i = 0$, irrespective of the value $m(E_i \cap E)$.

Suppose $f : \mathbb{R}^n \to [0, +\infty]$ is measurable. The *Lebesgue integral of* f *on* E is the quantity

$$\int_E f(x)\,dx = \sup\left\{\int_E s(x)\,dx :\ s \text{ is simple and } 0 \le s \le f \text{ on } \mathbb{R}^n\right\}. \tag{9.44}$$

If the integral of f on E is finite, then f is *summable on* E.

The integral of f on E is indicated by

$$\int_E f\,dm\,, \qquad \int_E f\,dx_1\,dx_2\,\ldots\,dx_n\,,$$

or, when $n = 2, 3$, by

$$\iint_E f\,dx\,dy \qquad \iiint_E f\,dx\,dy\,dz$$

respectively.

Note that if f is simple, definition (9.44) recovers the integral defined by (9.43).

The next proposition is an immediate consequence of the definition of integral, and its easy proof is left to the reader.

Proposition 1 *If $f, g : \mathbb{R}^n \to [0, +\infty]$ are measurable, E, F are measurable subsets in $\mathbb{R}^n$ and $c > 0$, then*

$$\int_E f\,dx = \int_{\mathbb{R}^n} \chi_E \cdot f\,dx\,; \tag{9.45}$$

$$f(x) \le g(x)\,, \quad \forall\, x \in E \quad \Rightarrow \quad \int_E f\,dx \le \int_E g\,dx\,; \tag{9.46}$$

$$F \subseteq E \quad \Rightarrow \quad \int_F f\,dx \le \int_E f\,dx\,;$$

$$\int_E c \cdot f\,dx = c \int_E f\,dx\,;$$

$$f(x) = 0\,,\ \forall\, x \in E \quad \Rightarrow \quad \int_E f\,dx = 0\,; \tag{9.47}$$

$$m(E) = 0 \quad \Rightarrow \quad \int_E f\,dx = 0\,.$$

If $g, h : \mathbb{R}^n \to [0, +\infty]$ are measurable and $g(x) = h(x)$ for any $x \in E$, with E measurable, (9.45) implies

$$\int_E g\,dx = \int_E h\,dx\,. \tag{9.48}$$

If $f : E \to [0, +\infty]$ is measurable on the measurable set E, we define the integral of f on E by

$$\int_E \overline{f}\,dx\,,$$

where $\overline{f}$ is the trivial extension of f to $\mathbb{R}^n$ (see (9.40)). By (9.48),

$$\int_E \overline{f}\, dx = \int_E g\, dx\,,$$

irrespective of the measurable extension g of f to $\mathbb{R}^n$. The common value will be denoted by

$$\int_E f\, dx\,.$$

Proposition 2 *Let s be a simple function, E a measurable set, E_k a sequence of pairwise-disjoint measurable sets such that $E = \bigcup_{k=1}^{\infty} E_k$. Then*

$$\int_E s\, dx = \sum_{k=1}^{+\infty} \int_{E_k} s\, dx\,. \tag{9.49}$$

Proof Suppose $s = \sum_{i=1}^{N} c_i\, \chi_{F_i}$, where $c_1, c_2, \ldots, c_N$ are the values of s and

$$F_i = \{x \in \mathbb{R}^n :\ s(x) = c_i\}\,, \qquad \forall\, i = 1, 2, \ldots, N.$$

Definition (9.43) and the Lebesgue measure's additivity imply

$$\int_E s(x)\, dx = \sum_{i=1}^{N} c_i\, m(F_i \cap E) =$$

$$= \sum_{i=1}^{N} \sum_{k=1}^{\infty} c_i\, m(F_i \cap E_k) = \sum_{k=1}^{\infty} \sum_{i=1}^{N} c_i\, m(F_i \cap E_k) = \sum_{k=1}^{\infty} \int_{E_k} s\, dx\,.$$

This proves (9.49).

Proposition 3 *Let s, t be simple functions and E a measurable set. Then*

$$\int_E (s + t)\, dx = \int_E s\, dx + \int_E t\, dx\,.$$

Proof Put

$$s = \sum_{i=1}^{N} c_i\, \chi_{F_i}\,, \qquad t = \sum_{j=1}^{M} d_j\, \chi_{G_j}\,,$$

where $c_1, c_2, \dots, c_N$ are the values of s and $d_1, d_2, \dots, d_M$ are the values of t. For any $i = 1, 2, \dots, N$, $j = 1, 2, \dots, M$, set

$$F_i = \{x \in \mathbb{R}^n : \ s(x) = c_i\}\,, \qquad G_j = \{x \in \mathbb{R}^n : \ t(x) = d_j\}\,.$$

As $\bigcup_{i=1}^{N} F_i = \bigcup_{j=1}^{M} G_j = \mathbb{R}^n$, (9.49) implies

$$\begin{aligned}\int_E (s+t)\,dx &= \sum_{i=1}^{N}\sum_{j=1}^{M}\int_{E\cap F_i\cap G_j} (s+t)\,dx = \\ &= \sum_{i=1}^{N}\sum_{j=1}^{M}(c_i + d_j)m(E\cap F_i\cap G_j) = \\ &= \sum_{i=1}^{N} c_i\, m(E\cap F_i) + \sum_{j=1}^{M} d_j m(E\cap G_j) = \int_E s\,dx + \int_E t\,dx\,.\end{aligned}$$

One good feature of the Lebesgue integral is to make it extremely easy to prove theorems on limits of integrals under very general hypotheses.

Beppo Levi's Monotone Convergence Theorem *Let f_k be an increasing sequence of non-negative measurable functions on $\mathbb{R}^n$ and*

$$f(x) = \lim_{k\to+\infty} f_k(x)\,, \qquad \forall\, x \in \mathbb{R}^n.$$

Then

$$\int_{\mathbb{R}^n} f\,dx = \lim_{k\to+\infty}\int_{\mathbb{R}^n} f_k\,dx\,. \tag{9.50}$$

Let us point out that, due to property (9.45) in Proposition 1, both the Beppo Levi theorem and the following Fatou lemma still hold for functions defined on a measurable set E instead of $\mathbb{R}^n$, and if the integrals are taken over E.

Proof By property (9.46), Proposition 1, as $k \to +\infty$ the numerical sequence in the right-hand side of (9.50) admits limit, and

$$\lim_{k\to+\infty}\int_{\mathbb{R}^n} f_k\,dx \le \int_{\mathbb{R}^n} f\,dx\,. \tag{9.51}$$

Let s be a simple function, with $s \le f$. Given $\alpha \in \mathbb{R}$, $0 < \alpha < 1$, we define $E_0 = \emptyset$ and

$$E_k = \{x : \ f_k(x) \ge \alpha\, s(x)\}\,, \qquad \forall\, k \in \mathbb{N}.$$

Then $E_k \subseteq E_{k+1}$ for any $k \in \mathbb{N}$ and $\bigcup_{k=1}^{\infty} E_k = \mathbb{R}^n$, and so

$$\int_{\mathbb{R}^n} f_k\, dx \ge \int_{E_k} f_k\, dx \ge \alpha \int_{E_k} s\, dx = \alpha \sum_{i=1}^{k} \int_{E_i - E_{i-1}} s\, dx\,,$$

for any $k \in \mathbb{N}$. Since the sets $E_i - E_{i-1}$ are pairwise disjoint, by Proposition 2

$$\lim_{k\to+\infty} \int_{\mathbb{R}^n} f_k\, dx \ge \alpha \sum_{i=1}^{\infty} \int_{E_i - E_{i-1}} s\, dx = \alpha \int_{\mathbb{R}^n} s\, dx\,.$$

As $\alpha \to 1^-$ we have

$$\lim_{k\to+\infty} \int_{\mathbb{R}^n} f_k\, dx \ge \int_{\mathbb{R}^n} s\, dx$$

and since s is arbitrary,

$$\lim_{k\to+\infty} \int_{\mathbb{R}^n} f_k\, dx \ge \int_{\mathbb{R}^n} f\, dx\,. \tag{9.52}$$

The claim now follows from (9.51), (9.52).

Fatou Lemma *Let f_k be a sequence of non-negative measurable functions, and set*

$$f(x) = \liminf_{k\to+\infty} f_k(x)\,, \qquad \forall\, x \in \mathbb{R}^n.$$

Then

$$\int_{\mathbb{R}^n} f\, dx \le \liminf_{k\to+\infty} \int_{\mathbb{R}^n} f_k\, dx\,. \tag{9.53}$$

As we said above, the Beppo Levi theorem and the Fatou lemma still hold when the integrals are taken over measurable sets $E \subseteq \mathbb{R}^n$, instead of $\mathbb{R}^n$.

Obviously the Fatou lemma conclusion (9.53) reads (without mentioning f)

$$\int_{\mathbb{R}^n} \liminf_{k\to+\infty} f_k(x)\, dx \le \liminf_{k\to+\infty} \int_{\mathbb{R}^n} f_k(x)\, dx\,.$$

Proof For any $k = 1, 2, \ldots$ consider

$$g_k(x) = \inf_{i \geq k} f_i(x)\,, \qquad \forall\, x \in \mathbb{R}^n.$$

The sequence g_k is increasing and

$$\lim_{k \to +\infty} g_k(x) = \liminf_{k \to +\infty} f_k(x) = f(x)\,, \qquad \forall\, x \in \mathbb{R}^n.$$

By the Beppo Levi theorem

$$\int_{\mathbb{R}^n} f\,dx = \lim_{k \to +\infty} \int_{\mathbb{R}^n} g_k\,dx\,.$$

As $g_k \leq f_k$ for any $k = 1, 2, \ldots$, the above now implies our claim:

$$\int_{\mathbb{R}^n} f\,dx = \lim_{k \to +\infty} \int_{\mathbb{R}^n} g_k\,dx \leq \liminf_{k \to +\infty} \int_{\mathbb{R}^n} f_k\,dx\,.$$

Example 1 In (9.53) there is no equality, in general. The sequence

$$f_k(x) = k \cdot \chi_{(0,1/k)}(x)\,, \qquad \forall\, k \in \mathbb{N},\ \forall\, x \in \mathbb{R}$$

converges to $f(x) = 0$ for any $x \in \mathbb{R}$, while

$$\int_{\mathbb{R}} f\,dx = 0 < \lim_{k \to +\infty} \int_{\mathbb{R}} f_k\,dx = 1\,.$$

Theorem 1 *If $f, g : \mathbb{R}^n \to [0, +\infty]$ are measurable functions and E is a measurable set,*

$$\int_E (f + g)\,dx = \int_E f\,dx + \int_E g\,dx\,. \tag{9.54}$$

Moreover, if f_k is a sequence of non-negative measurable functions on $\mathbb{R}^n$,

$$\int_E \sum_{k=1}^{\infty} f_k\,dx = \sum_{k=1}^{\infty} \int_E f_k\,dx\,. \tag{9.55}$$

Proof By Proposition 2, Sect. 9.5, there exist two increasing sequences s_k, t_k of simple functions satisfying $s_k(x) \to f(x)$ and $t_k(x) \to g(x)$ for any $x \in \mathbb{R}^n$.

Beppo Levi's theorem and Proposition 3 imply

$$\int_E (f+g)\,dx = \lim_{k\to+\infty} \int_E (s_k + t_k)\,dx = \\ = \lim_{k\to+\infty} \int_E s_k\,dx + \lim_{k\to+\infty} \int_E t_k\,dx = \int_E f\,dx + \int_E g\,dx\,.$$

Now (9.55) follows from (9.54) and Beppo Levi's theorem, because

$$\int_E \sum_{k=1}^{\infty} f_k\,dx = \int_E \lim_{k\to+\infty} \sum_{i=1}^{k} f_i\,dx = \lim_{k\to+\infty} \sum_{i=1}^{k} \int_E f_i\,dx = \sum_{k=1}^{\infty} \int_E f_k\,dx\,.$$

Corollary 1 *If f is non-negative and measurable, E_k is a sequence of measurable sets and $E = \bigcup_{k=1}^{\infty} E_k$, then:*

(i) if the E_k are pairwise disjoint

$$\int_E f\,dx = \sum_{k=1}^{\infty} \int_{E_k} f\,dx\,; \tag{9.56}$$

(ii) if E_k is an increasing sequence

$$\int_E f\,dx = \lim_{k\to+\infty} \int_{E_k} f\,dx\,; \tag{9.57}$$

(iii) if E_k is a decreasing sequence and

$$\int_{E_1} f\,dx < +\infty\,, \qquad F = \bigcap_{k=1}^{\infty} E_k\,,$$

then

$$\int_F f\,dx = \lim_{k\to+\infty} \int_{E_k} f\,dx\,. \tag{9.58}$$

Proof Claims (i), (ii) come from (9.55) and Beppo Levi's theorem on $f_k = f \cdot \chi_{E_k}$ respectively.

Property (iii) is a consequence of (ii) applied to the increasing sequence $E_1 - E_k$. For this we use the Lebesgue integral's additivity with respect to the integration

domain, established in (i):

$$\int_{E_1 - E_k} f\,dx = \int_{E_1} f\,dx - \int_{E_k} f\,dx\,, \qquad \forall\, k \in \mathbb{N},$$

and similarly for $E_1 - F$.

Using a similar argument the reader can easily prove the following

Corollary 2 *If f_k is a decreasing sequence of non-negative measurable functions and*

$$\int_E f_1\,dx < +\infty\,,$$

with $E \subset \mathbb{R}^n$ measurable, then

$$\int_E f\,dx = \lim_{k\to+\infty} \int_E f_k\,dx$$

where

$$f(x) = \lim_{k\to+\infty} f_k(x)\,, \qquad \forall\, x \in E.$$

Remark (On the Role of Zero-Measure Sets) We point out that if $f : \mathbb{R}^n \to [0, +\infty]$ is measurable, E is measurable and E_0 has measure zero, by (9.56) we have

$$\int_{E\cup E_0} f\,dx = \int_{E - E_0} f\,dx = \int_E f\,dx\,.$$

In other words, *the integral of f does not change if we add or remove from the integration domain E a zero-measure set*. For this reason all the results proved in this section, and the previous one, hold when we replace the expression "for any x" with "for almost every x" in the statements. This is manifest, for instance, regarding (9.46) and (9.47) in Proposition 1.

Suppose now f_k is a sequence of measurable functions such that for any k

$$f_k(x) \le f_{k+1}(x)\,, \qquad \text{for a.e. } x \in E,$$

where E is a given measurable set. Let

$$Z_k = \{x \in E :\ f_k(x) > f_{k+1}(x)\}$$

for any $k \in \mathbb{N}$. By definition $m(Z_k) = 0$, and

$$Z = \bigcup_{k=1}^{\infty} Z_k$$

has measure zero. As the sequence is monotone on the set E, the limit of $f_k(x)$ exists as $k \to +\infty$ for any $x \in E - Z$. Call this limit $f(x)$, and extend f outside of $E - Z$ in an arbitrary way. Clearly we still have

$$\int_E f\,dx = \lim_{k\to+\infty} \int_E f_k\,dx\,.$$

Similar considerations apply to the Fatou lemma and Corollary 2.

Example 2 [Comparison with the Riemann Integral] Consider a continuous function $f : [0, 1] \to [0, +\infty)$. For any $k \in \mathbb{N}$ we take

$$s_k = \sum_{i=1}^{2^k} m_{i,k}\ \chi_{\left[\frac{i-1}{2^k}, \frac{i}{2^k}\right]}\,,$$

where

$$m_{i,k} = \min\left\{ f(x) :\ \frac{i-1}{2^k} \le x \le \frac{i}{2^k} \right\}.$$

The sequence s_k is increasing and converges (uniformly) to f on $[0, 1]$. By Beppo Levi's theorem

$$\int_0^1 f\,dx = \lim_{k\to+\infty} \sum_{i=1}^{2^k} \frac{m_{i,k}}{2^k}\,. \tag{9.59}$$

By definition, the right-hand side of (9.59) is precisely the Riemann integral of f. Hence for continuous functions on $[0, 1]$ the two notions of integral coincide. In the same way the reader may show that *if $f : K \to [0, +\infty)$ is continuous on a Peano-Jordan measurable compact set K, the Riemann integral and the Lebesgue integral of f on K coincide*.

Example 3 Let $x_0 \in \mathbb{R}^n$ and $\alpha > 0$. We claim that

$$\int_{B_1(x_0)} \frac{dx}{|x - x_0|^\alpha}$$

(continued)

Example 3 (continued)
is finite if and only if $\alpha < n$. For this consider the annulus

$$E_k = B_{2^{-k}}(x_0) - B_{2^{-k-1}}(x_0)$$

for any $k = 0, 1, 2, \ldots$. If ω_n denotes the measure of the unit ball in $\mathbb{R}^n$, from (9.36) we obtain

$$m(E_k) = \omega_n \left(\frac{1}{2^{kn}} - \frac{1}{2^{(k+1)n}} \right). \tag{9.60}$$

If $\alpha \geq n$,

$$\frac{1}{|x - x_0|^\alpha} > 2^{kn}, \qquad \forall\, x \in E_k$$

for any $k = 0, 1, 2, \ldots$, so

$$\frac{1}{|x - x_0|^\alpha} > \sum_{k=0}^{\infty} 2^{kn}\, \chi_{E_k}(x), \qquad \forall\, x \in B_1(x_0) - \{x_0\}.$$

By (9.60) and (9.55), then,

$$\int_{B_1(x_0)} \frac{dx}{|x - x_0|^\alpha} \geq \int_{B_1(x_0)} \sum_{k=0}^{\infty} 2^{kn} \chi_{E_k} dx =$$

$$= \sum_{k=0}^{\infty} 2^{kn}\, m(E_k) = \omega_n \left(1 - \frac{1}{2^n}\right) \sum_{k=0}^{\infty} 1 = +\infty.$$

If $0 < \alpha < n$, instead, arguing as above

$$\frac{1}{|x - x_0|^\alpha} < \sum_{k=0}^{\infty} 2^{(k+1)\,\alpha}\, \chi_{E_k}(x), \qquad \forall\, x \in B_1(x_0) - \{x_0\}.$$

and so

$$\int_{B_1(x_0)} \frac{dx}{|x - x_0|^\alpha} \leq \int_{B_1(x_0)} \sum_{k=0}^{\infty} 2^{(k+1)\,\alpha}\, \chi_{E_k}\, dx =$$

$$= \sum_{k=0}^{\infty} 2^{(k+1)\,\alpha}\, m(E_k) = 2^\alpha\, \omega_n \left(1 - \frac{1}{2^n}\right) \sum_{k=0}^{\infty} \frac{1}{2^{(n-\alpha)k}} < +\infty.$$

(continued)

Example 3 (continued)
Similarly one proves that

$$\int_{\mathbb{R}^n - B_1(x_0)} \frac{dx}{|x - x_0|^\alpha}$$

is finite if and only if $\alpha > n$. To see this it suffices to consider annuli

$$F_k = B_{2^{k+1}}(x_0) - B_{2^k}(x_0)\,, \qquad k = 0, 1, 2, \ldots,$$

and argue as above.

Example 4 Let us compute the limit

$$\lim_{k\to+\infty} \int_0^k \left(1 + \frac{x}{k}\right)^k e^{-2x}\, dx\,.$$

Define, for any $k \in \mathbb{N}$,

$$f_k(x) = \left(1 + \frac{x}{k}\right)^k e^{-2x}\, \chi_{[0,k]}(x)\,.$$

This is an increasing sequence that converges for any $x \in \mathbb{R}$ to

$$f(x) = e^{-x}\, \chi_{[0,+\infty)}(x)\,.$$

Beppo Levi's theorem forces

$$\lim_{k\to+\infty} \int_0^k \left(1 + \frac{x}{k}\right)^k e^{-2x}\, dx = \int_0^{+\infty} e^{-x}\, dx = 1\,.$$

Example 5 Let $f : E \to [0, +\infty]$ be a measurable function on the measurable set E. Suppose

$$\int_E f\, dx < +\infty$$

and let us show

$$f(x) < +\infty\,, \qquad \text{for a.e. } x \in E.$$

(continued)

Example 5 (continued)
Put

$$E_\infty = \{x \in E : \ f(x) = +\infty\}\,,$$
$$E_k = \{x \in E : \ f(x) \geq k\}\,, \qquad \forall\, k \in \mathbb{N},$$

so that

$$k\, m(E_k) \leq \int_{E_k} f\, dx \leq \int_E f\, dx\,. \tag{9.61}$$

Since $E_k \supseteq E_{k+1}$ for any $k \in \mathbb{N}$, and because $E_\infty = \bigcap_{k=1}^{\infty} E_k$, from (9.61) and (9.31) we deduce

$$m(E_\infty) = \lim_{k\to+\infty} m(E_k) \leq \lim_{k\to+\infty} \frac{1}{k} \int_E f\, dx = 0\,.$$

Example 6 Let us compute

$$\int_0^1 \frac{\log x}{x^2-1}\, dx\,.$$

Recall first the sum of the geometric series,

$$\frac{1}{1-x^2} = \sum_{k=0}^{+\infty} x^{2k}\,, \qquad \forall\, x \in [0,1).$$

By (9.55), then,

$$\int_0^1 \frac{\log x}{x^2-1}\, dx = \int_0^1 -\log x \sum_{k=0}^{+\infty} x^{2k}\, dx = \sum_{k=0}^{+\infty} \int_0^1 -x^{2k} \log x\, dx\,. \tag{9.62}$$

Since for any $k = 0, 1, 2, \ldots$

$$\int_0^1 -x^{2k} \log x\, dx = \frac{1}{(2k+1)^2}\,,$$

(9.62) and (1.63) imply

$$\int_0^1 \frac{\log x}{x^2-1}\, dx = \sum_{k=0}^{+\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}\,.$$

Example 7 It is well known that the *Dirichlet function*

$$f(x) = \begin{cases} 0 & \text{if } x \in [0,1] \cap \mathbb{Q} \\ 1 & \text{if } x \in [0,1] - \mathbb{Q} \end{cases}$$

(see the example in Sect. 8.7) is a bounded function that is not Riemann integrable. The reader should think about why, in Lebesgue's theory, computing the integral of f is effortless. As f coincides almost everywhere with the constant function 1, it is measurable, and so

$$\int_0^1 f\,dx = 1\,.$$

The Dirichlet function is discontinuous everywhere on its domain. The next theorem, which we shall not prove, gives a criterion for the *Riemann* integrability of a function in terms of its discontinuity set.

Vitali-Lebesgue Criterion on Riemann Integrability *Let X be a Peano-Jordan measurable bounded subset of $\mathbb{R}^n$, and $f : X \to \mathbb{R}$ a bounded function. Then f is Riemann integrable on X if and only if the discontinuity set has zero Lebesgue measure.*

Let us now define the Lebesgue integral of functions with arbitrary sign. First we recall the notations (compare with (9.38) and (9.39))

$$f^+ = \max\{f, 0\}, \qquad f^- = -\min\{f, 0\},$$

whence

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-\,.$$

As for non-negative functions it is convenient to consider functions with values in the extended reals. So let $f : E \to \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ be a function defined on a measurable set $E \subseteq \mathbb{R}^n$ and measurable on it. We say f is *summable* on E if

$$\int_E |f|\,dx < +\infty\,.$$

In this case the *integral* of f on E is

$$\int_E f\,dx = \int_E f^+\,dx - \int_E f^-\,dx\,.$$

The above right-hand side is well defined and finite, since $0 \le f^-,\, f^+ \le |f|$.

Theorem 2 *Let f, g be summable functions on E and $c \in \mathbb{R}$ a constant. Then $f+g$ and cf are summable on E, and*

$$\int_E cf\,dx = c\int_E f\,dx\,; \tag{9.63}$$

$$\int_E (f+g)\,dx = \int_E f\,dx + \int_E g\,dx\,. \tag{9.64}$$

Proof Claim (9.63) is obvious. To prove (9.64) note that $|f+g| \le |f|+|g|$ implies

$$\int_E |f+g|\,dx \le \int_E |f|\,dx + \int_E |g|\,dx\,,$$

so $f+g$ is summable on E. Furthermore,

$$(f+g)^+ - (f+g)^- = f+g = f^+ - f^- + g^+ - g^-,$$

implying

$$(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+\,.$$

As the integral is additive on non-negative functions,

$$\begin{aligned}\int_E (f+g)^+ dx + \int_E f^- dx + \int_E g^- dx = \\ = \int_E (f+g)^- dx + \int_E f^+ dx + \int_E g^+ dx\,.\end{aligned}$$

All integrals being finite, (9.64) is proved.

Corollary 3 *If f is summable on E*

$$\left|\int_E f\,dx\right| \le \int_E |f|\,dx\,.$$

Proof Just compute

$$\begin{aligned}\left|\int_E f\,dx\right| \le \left|\int_E f^+\,dx - \int_E f^-\,dx\right| \le \\ \le \int_E f^+\,dx + \int_E f^-\,dx = \int_E |f|\,dx\,.\end{aligned}$$

Observe explicitly that while (9.55) is not true in general for summable functions of any sign, Corollary 1 holds if f is summable, as is easy to see by applying Corollary 1 to f^+ and f^- separately.

Example 8 (Absolute Continuity of Integrals) Let E be a measurable set and f a summable function on E. We want to prove that, for any $\varepsilon > 0$, we can find a number $\delta > 0$ such that

$$F \subseteq E,\ m(F) < \delta \quad \Rightarrow \quad \int_F |f|\,dx < \varepsilon\,. \tag{9.65}$$

If not, there would exist $\varepsilon_0 > 0$ and, for any $k \in \mathbb{N}$, a measurable subset $F_k \subseteq E$ such that

$$m(F_k) < \frac{1}{2^k}\,, \qquad \int_{F_k} |f|\,dx \geq \varepsilon_0\,.$$

Take $E_k = \bigcup_{i=k}^{\infty} F_i$, so

$$E_k \supseteq E_{k+1}\,, \qquad \int_{E_k} |f|\,dx \geq \varepsilon_0\,, \qquad \forall\, k \in \mathbb{N}. \tag{9.66}$$

Since

$$m\left(\bigcap_{k=1}^{\infty} E_k\right) = \lim_{k\to+\infty} m(E_k) \leq \lim_{k\to+\infty} \sum_{i=k}^{\infty} \frac{1}{2^i} = \lim_{k\to+\infty} \frac{1}{2^{k-1}} = 0\,,$$

by Corollary 1 we would have

$$\lim_{k\to+\infty} \int_{E_k} |f|\,dx = 0\,,$$

contradicting (9.66).

Also for summable functions the limit of an integral equals the integral of the limit under very general assumptions.

Lebesgue's Dominated Convergence Theorem *Let f_k be a sequence of summable functions over the measurable set E. Suppose that the limit*

$$f(x) = \lim_{k\to+\infty} f_k(x) \quad \textit{exists for a.e. } x \in E.$$

If there is a summable function g on E such that

$$|f_k(x)| \leq g(x)\,, \qquad \forall\, k \in \mathbb{N} \textit{ and a.e. } x \in E,$$

then

$$\lim_{k\to+\infty}\int_E |f_k - f|\,dx = 0\,. \tag{9.67}$$

Since

$$\left|\int_E f_k\,dx - \int_E f\,dx\right| \le \int_E |f_k - f|\,dx\,, \qquad \forall\, k\in\mathbb{N},$$

from (9.67) we deduce in particular

$$\lim_{k\to+\infty}\int_E f_k\,dx = \int_E f\,dx\,.$$

Proof As $|f_k(x)-f(x)|\le 2g(x)$ a.e. on E, and $\lim_{k\to+\infty}|f_k(x)-f(x)|=0$ a.e. on E, the Fatou lemma implies

$$\int_E 2g\,dx \le \liminf_{k\to+\infty}\int_E \{2g - |f_k - f|\}\,dx = \int_E 2g\,dx - \limsup_{k\to+\infty}\int_E |f_k - f|\,dx$$

and therefore

$$\limsup_{k\to+\infty}\int_E |f_k - f|\,dx \le 0,$$

proving the claim.

Example 9 Consider on $(0,1)$ the sequence

$$f_k(x) = \sqrt{k}(kx)^{-k}\,e^{k-\frac{1}{x}}\,.$$

Immediately, $f_k(x)\to 0$ for any $x\in(0,1)$, but the sequence does not converge uniformly because

$$f_k\left(\frac{1}{k}\right) = \sqrt{k}\to+\infty\,.$$

We claim that

$$\lim_{k\to+\infty}\int_0^1 f_k\,dx = 0\,. \tag{9.68}$$

(continued)

Example 9 (continued)
For this we seek a summable function bounding the sequence from above. Consider for any k the function $g_k(x) = \sqrt{x}\, f_k(x)$, and note $x_k = 2/(2k-1)$ is a maximum point. For any $x \in (0,1)$, then,

$$\sqrt{x}\, f_k(x) \le g_k(x_k) = \sqrt{\frac{2k}{2k-1}} \left(\frac{2k}{2k-1}\right)^{-k} e^{1/2} \le \sqrt{e}$$

hence for any $k \in \mathbb{N}$ and any $x \in (0,1)$

$$0 \le f_k(x) \le \sqrt{\frac{e}{x}}\,.$$

Now (9.68) follows from the dominated convergence theorem.

Example 10 The sequences

$$f_k(x) = \frac{k}{1+k+k^2x^2}\,, \qquad g_k(x) = \frac{k}{k^2+x^2}\,,$$

both converge to zero a.e. on $\mathbb{R}$. More precisely, $f_k(x)$ converges to zero for any $x \in \mathbb{R} - \{0\}$ and does not converge uniformly, whereas $g_k(x)$ converges to zero uniformly on $\mathbb{R}$. Yet,

$$\lim_{k\to+\infty} \int_{\mathbb{R}} f_k\, dx = 0\,, \qquad \lim_{k\to+\infty} \int_{\mathbb{R}} g_k\, dx = \pi\,.$$

This should not surprise us. The first sequence, although it tends to zero only pointwise, is dominated by the function $1/(1+x^2)$, which is summable on $\mathbb{R}$. The second sequence, albeit uniformly convergent to zero, is not uniformly bounded by any summable function on $\mathbb{R}$.

Example 11 Let f_k be a sequence of summable functions, on a measurable set E, such that

$$\sum_{k=1}^{+\infty} \int_E |f_k|\, dx < +\infty\,.$$

We want to show that the series of general term $f_k(x)$ converges absolutely to some function $f(x)$ for almost every $x \in E$, that f is summable on E and that

$$\int_E f\, dx = \sum_{k=1}^{+\infty} \int_E f_k\, dx\,. \tag{9.69}$$

(continued)

Example 11 (continued)
Define

$$g(x) = \sum_{k=1}^{+\infty} |f_k(x)|\,, \qquad \forall\, x \in E,$$

so Beppo Levi's theorem implies

$$\int_E g\,dx = \sum_{k=1}^{+\infty} \int_E |f_k|\,dx < +\infty\,.$$

Hence g is summable on E, and from Example 5, $g(x) < +\infty$ for a.e. $x \in E$. This proves that for a.e. $x \in E$ the series of general term $f_k(x)$ converges absolutely. Moreover, for any $k \in \mathbb{N}$,

$$\left|\sum_{i=1}^{k} f_i(x)\right| \le \sum_{i=1}^{k} |f_i(x)| \le g(x)\,, \qquad \forall\, x \in E,$$

and now (9.69) follows from the dominated convergence theorem.

Sometimes it is convenient to use the notion of integral also for non-summable functions of arbitrary sign. Suppose then $f : E \to \overline{\mathbb{R}}$ is measurable, and defined on the measurable set E. We shall say f is *integrable* if at least one of the integrals

$$\int_E f^+ dx\,, \qquad \int_E f^-\,dx$$

is finite. The integral of f on E will then be

$$\int_E f\,dx = \int_E f^+\,dx - \int_E f^-\,dx\,,$$

which is a well-defined element of the extended real line.

Proposition 4 *Let f be a measurable function on E. If*

$$\int_F f\,dx = 0$$

for any measurable subset $F \subset E$, then $f(x) = 0$ for a.e. $x \in E$.

Proof For any $k \in \mathbb{N}$ set $E_k = \{x \in E :\ f(x) > 1/k\}$. Then

$$\frac{1}{k} m(E_k) \le \int_{E_k} f\,dx = 0,$$

and since

$$\{x \in E : \ f(x) > 0\} = \bigcup_{k=1}^{\infty} E_k \,,$$

we have $m(\{x \in E : \ f(x) > 0\}) = 0$, i.e. $f(x) \leq 0$ for almost every $x \in E$. Similarly one proves that $f(x) \geq 0$ for almost every $x \in E$, whence the claim.

9.7 Measure and Integration on Product Spaces

The main result of this section is the Fubini theorem, which will enable us to calculate integrals on $\mathbb{R}^n$ by suitable iterations of n one-dimensional integrals. Among the consequences we obtain a cascade of useful formulas for areas, volumes and other applications of physical flavour.

We shall work in Euclidean spaces of different dimensions, so we shall often write m_h, instead of just m, for the Lebesgue measure on $\mathbb{R}^h$. The subscript will be dropped when no ambiguity arises.

Lemma *For any open set $A \subseteq \mathbb{R}^n$ it is possible to find an increasing sequence of closed pluri-intervals P_k such that*

$$\bigcup_{k=1}^{\infty} P_k = A \,. \tag{9.70}$$

For any compact set $K \subset \mathbb{R}^n$ there exists a decreasing sequence of closed pluri-intervals P'_k such that

$$\bigcap_{k=1}^{\infty} \overset{\circ}{P}{}'_k = K \,. \tag{9.71}$$

Proof For any $k \in \mathbb{N}$ we denote by $\mathcal{Q}_k$ the family of closed n-dimensional cubes of the form

$$\prod_{i=1}^{n} \left[a_i \,, a_i + \frac{1}{2^k} \right] ,$$

where, for any $i = 1, 2, \ldots, n$,

$$a_i = \frac{z_i}{2^k} \,, \qquad z_i \in \mathbb{Z}.$$

The following properties are straightforward:

(i) any $Q \in \mathcal{Q}_k$ decomposes as a union of 2^n cubes in the family $\mathcal{Q}_{k+1}$;
(ii) given $x_0 \in \mathbb{R}^n$, for any $r > 0$ there exist an integer k and a cube $Q \in \mathcal{Q}_k$ such that $x_0 \in Q \subset B_r(x_0)$.

Given an open set $A \subseteq \mathbb{R}^n$, for any $k \in \mathbb{N}$ we call Q_k the union of all cubes of $\mathcal{Q}_k$ contained in A. From (i) and (ii) we have

$$Q_k \subseteq Q_{k+1}\,, \qquad \forall\, k \in \mathbb{N}, \qquad \bigcup_{k=1}^{\infty} Q_k = A\,.$$

Then (9.70) follows from taking, for every k,

$$P_k = Q_k \cap [-k, k]^n\,.$$

Given a compact set K, choose $N > 0$ so that $K \subset (-N, N)^n$, and consider an increasing sequence P_k of closed pluri-intervals such that

$$\bigcup_{k=1}^{\infty} P_k = (-N, N)^n - K\,.$$

We obtain (9.71) by putting

$$P'_k = [-N, N]^n - \mathring{P}_k$$

for any k.

Measure of Cartesian Products *Let $E \subseteq \mathbb{R}^n$, $F \subseteq \mathbb{R}^h$ be measurable sets. Then $E \times F \subseteq \mathbb{R}^n \times \mathbb{R}^h$ is measurable and*

$$m_{n+h}(E \times F) = m_n(E) \cdot m_h(F)\,, \tag{9.72}$$

as long as the left-hand-side product is well defined.

Proof Formula (9.72) is a direct consequence of definition (9.2) when E and F are closed intervals. By the additivity of the Lebesgue measure, it also holds when E, F are closed pluri-intervals.

If E and F are both open, by the previous lemma we can find increasing sequences P_k and Q_k of closed pluri-intervals such that

$$E = \bigcup_{k=1}^{\infty} P_k\,, \qquad F = \bigcup_{k=1}^{\infty} Q_k\,.$$

From (9.30) we have

$$m_{n+h}(E \times F) = \lim_{k \to +\infty} m_{n+h}(P_k \times Q_k) = \lim_{k \to +\infty} m_n(P_k) \cdot m_h(Q_k) = m_n(E) \cdot m_h(F)\,.$$

Similarly, using the previous lemma and (9.31), one proves (9.72) if E and F are both compact.

Suppose now E, F are measurable and bounded. If they have positive measure, given $0 < \varepsilon < \min\{m_n(E), m_h(F)\}$ there exist, by (9.19), open sets A, B and compact sets K, H such that $K \subseteq E \subseteq A$, $H \subseteq F \subseteq B$, with

$$m_n(A) - m_n(K) < \varepsilon\,, \qquad m_h(B) - m_h(H) < \varepsilon.$$

Using the outer and inner measures m_e, m_i of $\mathbb{R}^n \times \mathbb{R}^h$, we have

$$m_e(E \times F) \le m_{n+h}(A \times B) = m_n(A) \cdot m_h(B) \le (m_n(E) + \varepsilon) \cdot (m_h(F) + \varepsilon)\,, \tag{9.73}$$

and similarly

$$m_i(E \times F) \ge (m_n(E) - \varepsilon) \cdot (m_h(F) - \varepsilon)\,. \tag{9.74}$$

As ε is arbitrary, (9.73) and (9.74) imply that $E \times F$ is measurable and that (9.72) holds. The claim follows in the same way if either $m_n(E) = 0$ or $m_h(F) = 0$.

Finally if E, or F, is unbounded then

$$E \times F = \bigcup_{k=1}^{\infty} \left(E \cap [-k,k]^n\right) \times \left(F \cap [-k,k]^h\right)$$

is measurable. From (9.30) we have

$$\begin{aligned} m_{n+h}(E \times F) &= \lim_{k \to +\infty} m_{n+h}\left(E \times F \cap [-k,k]^{n+h}\right) = \\ &= \lim_{k \to +\infty} m_n\left(E \cap [-k,k]^n\right) \cdot m_h\left(F \cap [-k,k]^h\right) = m_n(E) \cdot m_h(F)\,. \end{aligned}$$

Note that by (9.72), $m_n(E) = 0$ clearly implies $m_{n+h}(E \times \mathbb{R}^h) = 0$. Since the Lebesgue measure is complete (see Sect. 9.4), $E \times F$ is measurable and has measure zero on $\mathbb{R}^n \times \mathbb{R}^h$ irrespective of what $F \subseteq \mathbb{R}^h$ is (even if it is not measurable).

As consequence of the above theorem we recover the well-known geometric interpretation of the Riemann integral of a non-negative function of one real variable.

Theorem 1 *Let $f : E \to \overline{\mathbb{R}}$ be a measurable function on the measurable set $E \subset \mathbb{R}^n$. Then the set*

$$\mathcal{G} = \{(x,t) \in E \times \mathbb{R} : \ t < f(x)\}$$

is measurable on $\mathbb{R}^{n+1}$. Moreover, if f is non-negative, the sets

$$\mathcal{R} = \{(x,t) \in E \times \mathbb{R} : \ 0 < t < f(x)\},$$
$$\widetilde{\mathcal{R}} = \{(x,t) \in E \times \mathbb{R} : \ 0 < t \leq f(x)\},$$

are measurable and

$$m_{n+1}(\mathcal{R}) = m_{n+1}(\widetilde{\mathcal{R}}) = \int_E f\,dx\,. \tag{9.75}$$

Proof For any $t \in \mathbb{R}$, define $L_t = \{x \in E : \ f(x) > t\}$, so

$$\mathcal{G} = \bigcup_{r \in \mathbb{Q}} [L_r \times (-\infty, r)]\,.$$

Evidently, $L_r \times (-\infty, r) \subset \mathcal{G}$ for any r. Conversely, if $(x,t) \in \mathcal{G}$, there is $r \in \mathbb{Q}$ such that $t < r < f(x)$ and so $(x,t) \in L_r \times (-\infty, r)$. Since, for any r, L_r is measurable on $\mathbb{R}^n$, the measurability of $\mathcal{G}$ on $\mathbb{R}^{n+1}$ descends from the previous theorem.

Suppose now E is measurable on $\mathbb{R}^n$ and let

$$s = \sum_{i=1}^{N} c_i\, \chi_{E_i}$$

be a simple function with values $c_1, c_2, \ldots, c_N$. Take

$$E_i = \{x \in \mathbb{R}^n : \ s(x) = c_i\}\,, \qquad \forall\, i = 1, 2, \ldots, N.$$

Assuming $c_1 = 0$ and $c_i > 0$ for $i = 2, 3, \ldots, N$, we have

$$\{(x,t) \in E \times \mathbb{R} : \ 0 < t < s(x)\} = \bigcup_{i=2}^{N} (E \cap E_i) \times (0, c_i)$$

and so

$$m_{n+1}(\{(x,t) \in E \times \mathbb{R} : \ 0 < t < s(x)\}) = \sum_{i=2}^{N} c_i\, m_n(E \cap E_i) = \int_E s\,dx\,. \tag{9.76}$$

Pick a non-negative measurable function f defined on E. Indicating by f the extension of f that is zero outside of E, by Proposition 2, Sect. 9.5, there is an increasing sequence of simple functions s_k that converges to f on $\mathbb{R}^n$ pointwise. Hence

$$\mathcal{R} = \bigcup_{k=1}^{\infty} \mathcal{R}_k \,,$$

where $\mathcal{R}_k = \{(x,t) \in E \times \mathbb{R} : \ 0 < t < s_k(x)\}$. But $\mathcal{R}_k$ is an increasing sequence, so (9.30) and the Beppo Levi theorem imply, using (9.76),

$$m_{n+1}(\mathcal{R}) = \lim_{k \to +\infty} m_{n+1}(\mathcal{R}_k) = \lim_{k \to +\infty} \int_E s_k \, dx = \int_E f \, dx \,.$$

Now, for any $\alpha > 1$

$$\mathcal{R} \subset \widetilde{\mathcal{R}} \subset \{(x,t) \in E \times \mathbb{R} : \ 0 < t < \alpha f(x)\} \,,$$

so (9.75) gives

$$\int_E f \, dx \le m_{n+1}(\widetilde{\mathcal{R}}) \le \alpha \int_E f \, dx \,.$$

Letting α tend to 1 produces (9.75).

We now pass to the proof of Fubini's theorem. To simplify the presentation we have split the classical statement in two: the first part is about cross-sections of measurable sets in $\mathbb{R}^n \times \mathbb{R}^h$, while the second is the proper Fubini theorem.

In the rest of the section we shall indicate by (x, y), $x \in \mathbb{R}^n$, $y \in \mathbb{R}^h$ a generic point in $\mathbb{R}^n \times \mathbb{R}^h$. If E is any subset of $\mathbb{R}^n \times \mathbb{R}^h$ and $x \in \mathbb{R}^n$, the *cross-section of E at x* is the set E_x defined by

$$E_x = \left\{ y \in \mathbb{R}^h : \ (x, y) \in E \right\}$$

see Fig. 9.3. Similarly, the cross-section of E at $y \in \mathbb{R}^h$ is

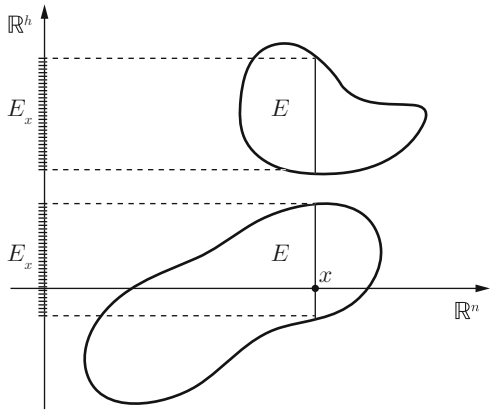
$$E_y = \left\{ x \in \mathbb{R}^n : \ (x, y) \in E \right\} .$$

This definition of *cross-section* generalises that of Sect. 8.8.

Cross-Sections of Measurable Sets *Let E be a measurable subset in $\mathbb{R}^n \times \mathbb{R}^h$. Then E_x is a measurable subset of $\mathbb{R}^h$ for almost every $x \in \mathbb{R}^n$. Moreover the function $x \in \mathbb{R}^n \to m_h(E_x)$ is measurable on $\mathbb{R}^n$, and*

$$\int_{\mathbb{R}^n} m_h(E_x) \, dx = m_{n+h}(E) \,. \tag{9.77}$$

Fig. 9.3



In essence, the theorem says that to compute the measure of E we may go about as follows: first we find the "h-dimensional" measure of all cross-sections at x and then we "add", i.e. integrate with respect to x, the measures thus obtained. The aforementioned theorem clearly holds if we interchange the roles of x and y. So we can say that, almost everywhere on $\mathbb{R}^h$, the cross-section E_y is measurable, the function $y \in \mathbb{R}^h \to m_n(E_y)$ is measurable, and that

$$\int_{\mathbb{R}^h} m_n(E_y)\, dy = m_{n+h}(E)\,.$$

Finally, notice that the theorem is obvious if E is a closed interval or, by additivity, a pluri-interval. More generally, the theorem is self-evident if E is the Cartesian product of a measurable set in $\mathbb{R}^n$ and a measurable set in $\mathbb{R}^h$.

Proof We have split the argument in four steps.

Step One: E is open. Take an increasing sequence P_k of closed pluri-intervals with union E. For any $x \in \mathbb{R}^n$ the set E_x is open in $\mathbb{R}^h$ and hence measurable. Moreover $P_{k,x}$ is an increasing sequence of closed pluri-intervals such that

$$E_x = \bigcup_{k=1}^{\infty} P_{k,x}\,.$$

Then

$$m_h(E_x) = \lim_{k\to+\infty} m_h(P_{k,x})\,, \qquad \forall\, x \in \mathbb{R}^n.$$

By the previous remarks the map $x \to m_h(E_x)$, being the pointwise limit of measurable functions, is measurable. By the Beppo Levi theorem, furthermore,

$$\int_{\mathbb{R}^n} m_h(E_x)\,dx = \lim_{k\to+\infty} \int_{\mathbb{R}^n} m_h(P_{k,x})\,dx = \lim_{k\to+\infty} m_{n+h}(P_k) = m_{n+h}(E)\,,$$

since, as was said before, (9.77) is true if E is a pluri-interval.

Step Two: E has measure zero. By Example 5 in Sect. 9.4 there is a decreasing sequence of open sets A_k containing E and such that $m_{n+h}(A_k) \to 0$ as $k \to +\infty$. By Step One the function

$$f(x) = \lim_{k\to+\infty} m_h(A_{k,x})\,, \qquad \forall\, x \in \mathbb{R}^n, \tag{9.78}$$

is measurable as pointwise limit of a decreasing sequence of measurable functions. The Fatou lemma implies

$$\int_{\mathbb{R}^n} f\,dx \le \liminf_{k\to+\infty} \int_{\mathbb{R}^n} m_h(A_{k,x})\,dx = \lim_{k\to+\infty} m_{n+h}(A_k) = 0\,.$$

Proposition 4 in the previous section allows to conclude that $f(x) = 0$ for almost every $x \in \mathbb{R}^n$. Since $E_x \subseteq A_{k,x}$ for any $x \in \mathbb{R}^n$ and any $k \in \mathbb{N}$, from (9.78) we deduce that E_x is measurable for almost every $x \in \mathbb{R}^n$, and its measure is zero. This proves the claim in this case.

Step Three: E has finite measure. By Example 5, Sect. 9.4, there is a decreasing sequence of open sets A_k such that

$$m_{n+h}(A_1) < +\infty\,, \qquad \lim_{k\to+\infty} m_{n+h}(A_k) = m_{n+h}(E)\,,$$

and

$$E = \bigcap_{k=1}^{\infty} A_k - E_0\,, \qquad \text{with} \quad m_{n+h}(E_0) = 0\,.$$

As

$$E_x = \bigcap_{k=1}^{\infty} A_{k,x} - E_{0,x}\,, \qquad \forall\, x \in \mathbb{R}^n,$$

Steps One and Two say that E_x is measurable for almost every $x \in \mathbb{R}^n$. Moreover, since

$$\int_{\mathbb{R}^n} m_h(A_{1,x})\,dx = m_{n+h}(A_1) < +\infty\,, \tag{9.79}$$

we have $m_h(A_{1,x}) < +\infty$ and $m_h(E_{0,x}) = 0$ for almost every $x \in \mathbb{R}^n$. Now (9.31) implies

$$m_h(E_x) = \lim_{k \to +\infty} m_h(A_{k,x})$$

for almost every $x \in \mathbb{R}^n$. The function $x \to m_h(E_x)$ is, for almost every $x \in \mathbb{R}^n$, the limit of a decreasing sequence of measurable functions, hence measurable. Then (9.79) and the previous section's Corollary 2 force

$$\int_{\mathbb{R}^n} m_h(E_x)\, dx = \lim_{k \to +\infty} \int_{\mathbb{R}^n} m_h(A_{k,x})\, dx = \lim_{k \to +\infty} m_{n+h}(A_k) = m_{n+h}(E)\,.$$

Step Four: E is measurable with infinite measure. Set

$$E_k = E \cap \left\{ z \in \mathbb{R}^{n+h} : \ |z| < k \right\}\,, \qquad \forall\, k \in \mathbb{N},$$

so that

$$E_x = \bigcup_{k=1}^{\infty} E_{k,x}\,, \qquad \forall\, x \in \mathbb{R}^n$$

and, by Step Three, E_x is the union of an increasing sequence of measurable sets for almost every $x \in \mathbb{R}^n$. Therefore E_x is measurable for almost every x, and

$$m_h(E_x) = \lim_{k \to +\infty} m_h(E_{k,x})\,.$$

This shows that the function $x \in \mathbb{R}^n \to m_h(E_x)$ is measurable. Now (9.77) follows from the above relation and Step Three, by applying the Beppo Levi theorem.

Next we use measurable cross-sections together with Theorem 1 to show the following characterisation of measurable functions.

Theorem 2 *Let $E \subseteq \mathbb{R}^n$ be measurable. A function $f : E \to \overline{\mathbb{R}}$ is measurable if and only if the set*

$$\mathcal{G} = \{(x,t) \in E \times \mathbb{R} : \ t < f(x)\}$$

is measurable on $\mathbb{R}^{n+1}$.

Proof By Theorem 1 it suffices to show that f is measurable on E if $\mathcal{G}$ is measurable on $\mathbb{R}^{n+1}$. So given $t \in \mathbb{R}$, since the cross-section

$$\mathcal{G}_s = \{x \in E : \ (x,s) \in \mathcal{G}\} = \{x \in E : \ s < f(x)\}$$

is measurable for almost every $s \in \mathbb{R}$, there is an increasing sequence t_k, tending to t, such that $\mathcal{G}_{t_k}$ is measurable for any k. This makes

$$\{x \in E : \ f(x) > t\} = \bigcup_{k=1}^{\infty} \mathcal{G}_{t_k}$$

measurable, and in turn f is measurable on E.

Corollary *Let $E \subseteq \mathbb{R}^n$ be a Peano-Jordan measurable and bounded set, and $f : E \to \mathbb{R}$ a Riemann integrable, bounded function. Then f is (Lebesgue) integrable, and the Riemann integral of f coincides with the Lebesgue integral.*

Proof First of all note that, in the light of the remark made in Sect. 9.3, E is (Lebesgue) measurable. To prove the claim it is enough to assume $f \geq 0$, and in the general case to consider f^+ and f^-.

By the proposition of Sect. 8.7 and the lemma in Sect. 8.8 the set

$$\mathcal{R} = \{(x, t) \in E \times \mathbb{R} : \ 0 < t < f(x)\}$$

is Peano-Jordan measurable, hence also Lebesgue measurable. Theorem 2 then implies that f is measurable. The last part is a consequence of (9.75) and (8.94).

Let us now present a few concrete applications of the measurability of cross-sections. In these examples, and the following ones as well, we will use for subsets in the plane the term "area" to mean the two-dimensional Lebesgue measure (and for subsets in space, "volume" will refer to the three-dimensional Lebesgue measure).

Example 1 A subset E of the plane is called *normal with respect to x* when

$$E = \left\{(x, y) \in \mathbb{R}^2 : \ a \leq x \leq b, \ \alpha(x) \leq y \leq \beta(x)\right\},$$

for some measurable functions $\alpha, \beta : [a, b] \to \overline{\mathbb{R}}$ (see Fig. 9.4).

By Theorem 1 E is measurable. Since, for any $x \in \mathbb{R}^n$,

$$E_x = \begin{cases} [\alpha(x), \beta(x)] & \text{if } a \leq x \leq b \\ \emptyset & \text{otherwise} \end{cases},$$

the result on the measurability of cross-sections says that the area of E equals

$$\int_a^b m_1(E_x)\, dx = \int_a^b [\beta(x) - \alpha(x)]\, dx\,.$$

(continued)

Example 1 (continued)
Similarly, $F \subset \mathbb{R}^2$ is normal with respect to y if

$$F = \left\{(x, y) \in \mathbb{R}^2 : \ c \le y \le d, \ \gamma(y) \le x \le \delta(y)\right\},$$

with $\gamma, \delta : [c, d] \to \overline{\mathbb{R}}$ measurable (see Fig. 9.5). Also F is measurable and its area is the integral

$$\int_c^d [\delta(y) - \gamma(y)]\, dy\,.$$

Fig. 9.4

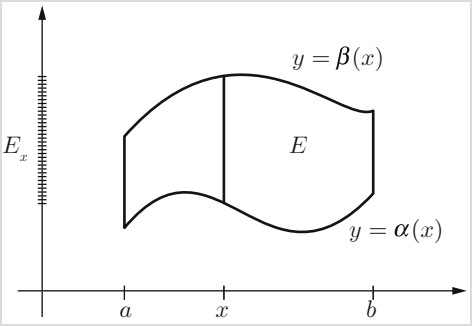
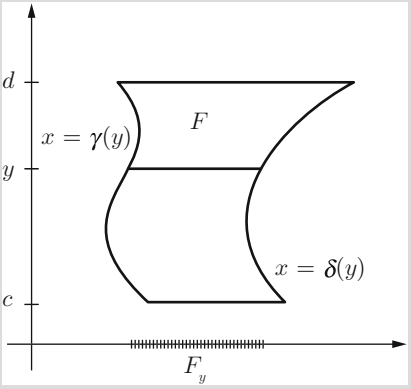


Fig. 9.5



Example 2 A subset E in space is *normal with respect to the* xy-plane if

$$E = \left\{(x, y, z) \in \mathbb{R}^3 : \ (x, y) \in D, \ \alpha(x, y) \le z \le \beta(x, y)\right\},$$

(continued)

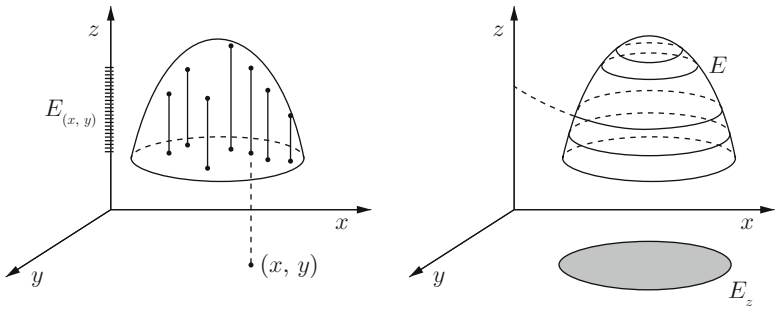


Fig. 9.6

Example 2 (continued)
where D is a measurable subset of the plane and $\alpha, \beta : D \to \overline{\mathbb{R}}$ are measurable functions. The set E is measurable, and since

$$m_1(E_{(x,y)}) = \begin{cases} \beta(x,y) - \alpha(x,y) & \text{if } (x,y) \in D \\ 0 & \text{otherwise}, \end{cases}$$

the volume of E is the integral

$$\iint_D [\beta(x,y) - \alpha(x,y)]\, dx\, dy\,.$$

Normal sets with respect to the xz- or yz-planes are defined in analogy to the above.

There are two strategies to calculate the measure of a subset in $\mathbb{R}^3$. Either we decompose the subset into "wires" (one-dimensional cross-sections) and then integrate the various lengths—as was done in the previous example, or we decompose it into "slices" (two-dimensional cross-sections) and add the resulting areas (see Fig. 9.6). The second method is useful when computing the volumes of certain solids of revolutions.

Example 3 Let f be a non-negative measurable function on $[a, b]$, and consider the set (see Fig. 9.7)

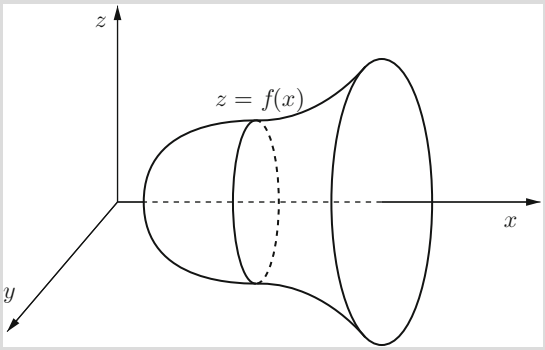
$$E = \left\{(x,y,z) \in \mathbb{R}^3 :\ a \le x \le b,\ y^2 + z^2 \le [f(x)]^2\right\}$$

obtained by a complete rotation about the x-axis of

$$\mathcal{R} = \left\{(x,z) \in \mathbb{R}^2 :\ a \le x \le b,\ 0 \le z \le f(x)\right\}.$$

(continued)

Example 3 (continued)
Fig. 9.7



By (9.77) the volume of E equals

$$\int_a^b m_2(E_x)\,dx\,.$$

Since

$$E_x = \left\{(y,z) \in \mathbb{R}^2 :\ y^2 + z^2 \le [f(x)]^2\right\}, \qquad \forall\, x \in [a,b],$$

the area of E_x is $\pi[f(x)]^2$. Then the volume of E is given by the formula

$$\pi \int_a^b [f(x)]^2\,dx\,. \tag{9.80}$$

The above formula is a special case of a theorem by Guldinus (next section). The reader should try to show, as exercise, that if E is obtained by rotating $\mathcal{R}$ by an angle α, $0 < \alpha < 2\pi$, the volume equals

$$\frac{\alpha}{2} \int_a^b [f(x)]^2\,dx\,. \tag{9.81}$$

For example, let us calculate the volume of the ellipsoid

$$E = \left\{(x,y,z) :\ x^2 + 2(y^2+z^2) \le 4\right\}.$$

As

$$E = \left\{(x,y,z) :\ -2 \le x \le 2,\ y^2 + z^2 \le \frac{4-x^2}{2}\right\},$$

(continued)

Example 3 (continued)

by (9.80) the volume of E is

$$\pi \int_{-2}^{2} \frac{4 - x^2}{2}\, dx = \frac{16\pi}{3}\,.$$

Now we are ready for Fubini's theorem.

Fubini's Theorem *Let $f : \mathbb{R}^n \times \mathbb{R}^h \to \overline{\mathbb{R}}$ be a summable function. Then the function $y \in \mathbb{R}^h \to f(x, y)$ is summable for almost every $x \in \mathbb{R}^n$. Moreover, the integral function*

$$x \in \mathbb{R}^n \to \int_{\mathbb{R}^h} f(x, y)\, dy \tag{9.82}$$

is summable and

$$\int_{\mathbb{R}^{n+h}} f(x, y)\, dx\, dy = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^h} f(x, y)\, dy \right) dx\,. \tag{9.83}$$

There is an analogous statement where the variables x, y are interchanged, so besides (9.83) we also have

$$\int_{\mathbb{R}^{n+h}} f(x, y)\, dx\, dy = \int_{\mathbb{R}^h} \left(\int_{\mathbb{R}^n} f(x, y)\, dx \right) dy\,.$$

Proof Let

$$s = \sum_{i=1}^{N} c_i\, \chi_{E_i}$$

be a simple function, with values $c_1, c_2, \ldots, c_N$, and where

$$E_i = \left\{ (x, y) \in \mathbb{R}^n \times \mathbb{R}^h : \; s(x, y) = c_i \right\}, \qquad \forall\, i = 1, 2, \ldots, N.$$

Since, for any $x \in \mathbb{R}^n$,

$$y \in \mathbb{R}^h \to s(x, y) \tag{9.84}$$

coincides with

$$y \in \mathbb{R}^h \to \sum_{i=1}^{N} c_i \chi_{E_{i,x}}(y)\,,$$

by the measurability of cross-sections we deduce that the function in (9.84) is measurable for almost every $x \in \mathbb{R}^n$. Moreover,

$$x \in \mathbb{R}^n \to \int_{\mathbb{R}^h} s(x,y)\,dy$$

coincides almost everywhere on $\mathbb{R}^n$ with the function

$$x \in \mathbb{R}^n \to \sum_{i=1}^{N} c_i m_h(E_{i,x})\,,$$

and as such it is measurable. From (9.77) we then have

$$\int_{\mathbb{R}^n} dx \int_{\mathbb{R}^h} s(x,y)\,dy = \sum_{i=1}^{N} c_i \int_{\mathbb{R}^n} m_h(E_{i,x})\,dx =$$

$$= \sum_{i=1}^{N} c_i m_{n+h}(E_i) = \int_{\mathbb{R}^{n+h}} s(x,y)\,dx\,dy\,.$$

If f is a non-negative measurable function, we take an increasing sequence s_k of simple functions tending to $f(x,y)$ for almost every $(x,y) \in \mathbb{R}^n \times \mathbb{R}^h$. Since for almost every $x \in \mathbb{R}^n$ the functions $y \to s_k(x,y)$ are measurable for any $k \in \mathbb{N}$, also $y \to f(x,y)$ is measurable for almost every $x \in \mathbb{R}^n$. Beppo Levi's theorem then tells that

$$\int_{\mathbb{R}^h} f(x,y)\,dy = \lim_{k\to+\infty} \int_{\mathbb{R}^h} s_k(x,y)\,dy\,, \qquad \text{for a.e. } x \in \mathbb{R}^n.$$

This shows that

$$x \in \mathbb{R}^n \to \int_{\mathbb{R}^h} f(x,y)\,dy$$

is measurable on $\mathbb{R}^n$. Finally, since the sequence of functions

$$x \in \mathbb{R}^n \to \int_{\mathbb{R}^h} s_k(x,y)\,dy$$

is increasing, applying the Beppo Levi theorem gives

$$\int_{\mathbb{R}^n} dx \int_{\mathbb{R}^h} f(x,y)\,dy = \lim_{k\to+\infty} \int_{\mathbb{R}^n} dx \int_{\mathbb{R}^h} s_k(x,y)\,dy = \\ = \lim_{k\to+\infty} \int_{\mathbb{R}^{n+h}} s_k(x,y)\,dx\,dy = \int_{\mathbb{R}^{n+h}} f(x,y)\,dx\,dy\,.$$

If f is also summable, the above implies that the function in (9.82) is summable on $\mathbb{R}^n$. Therefore (see Example 5, previous section)

$$\int_{\mathbb{R}^h} f(x,y)\,dy < +\infty\,, \qquad \text{for a.e. } x \in \mathbb{R}^n;$$

i.e. $y \in \mathbb{R}^h \to f(x,y)$ is summable for almost every $x \in \mathbb{R}^n$.

The case in which f is an arbitrary summable function follows by applying to f^+ and f^- what was proved for non-negative functions.

Let us point out that Fubini's theorem is valid if we take f only integrable, and we replace "summable" with "integrable" in the conclusion.

Example 4 Let $E = \{(x,y) \in \mathbb{R}^2 :\ a \le x \le b,\ \alpha(x) \le y \le \beta(x)\}$ be a normal set with respect to x. If f is integrable on E, let $\overline{f}$ be the extension of f equal to zero outside of E. Then $\overline{f}$ is integrable on the whole plane, and (9.83) implies

$$\iint_E f(x,y)\,dx\,dy = \iint_{\mathbb{R}^2} \overline{f}(x,y)\,dx\,dy = \\ = \int_{\mathbb{R}} dx \int_{\mathbb{R}} \overline{f}(x,y)\,dy = \int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x,y)\,dy\,. \tag{9.85}$$

Similarly, if $F = \{(x,y) \in \mathbb{R}^2 :\ c \le y \le d,\ \gamma(y) \le x \le \delta(y)\}$ is normal with respect to y and f is integrable on F, we have

$$\iint_F f(x,y)\,dx\,dy = \int_c^d dy \int_{\gamma(y)}^{\delta(y)} f(x,y)\,dx\,. \tag{9.86}$$

Example 5 Let C be the unit semi-circle with centre $(0,1)$ contained in the first quadrant, as in Fig. 9.8. Let us compute

$$\iint_C xy\,dx\,dy\,.$$

(continued)

Example 5 (continued)

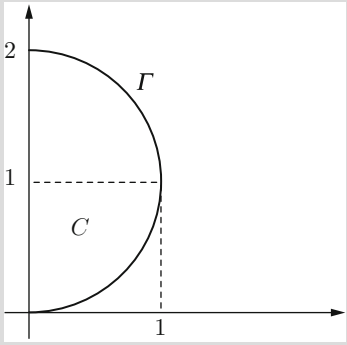


Fig. 9.8 Semi-circle Γ of equation $x^2 + y^2 - 2y = 0$

As $C = \{(x, y) \in \mathbb{R}^2 : \ 0 \le y \le 2, \ 0 \le x \le \sqrt{2y - y^2}\ \}$, this set is normal with respect to y. Hence (9.86) implies

$$\iint_C xy\,dx\,dy = \int_0^2 dy \int_0^{\sqrt{2y-y^2}} xy\,dx =$$

$$= \int_0^2 y\left[\frac{x^2}{2}\right]_0^{\sqrt{2y-y^2}} dy = \frac{1}{2}\int_0^2 y(2y - y^2)\,dy = \frac{2}{3}\,.$$

Note that C is normal with respect to x as well, because

$$C = \left\{(x, y) \in \mathbb{R}^2 : \ 0 \le x \le 1, \ 1 - \sqrt{1 - x^2} \le y \le 1 + \sqrt{1 - x^2}\right\}.$$

Using (9.85) we obtain

$$\iint_C xy\,dx\,dy = \int_0^1 dx \int_{1-\sqrt{1-x^2}}^{1+\sqrt{1-x^2}} xy\,dy = \int_0^1 x\left[\frac{y^2}{2}\right]_{1-\sqrt{1-x^2}}^{1+\sqrt{1-x^2}} dx =$$

$$= 2\int_0^1 x\sqrt{1 - x^2}\,dx = \left[-\frac{2}{3}\sqrt{(1 - x^2)^3}\right]_0^1 = \frac{2}{3}\,.$$

Example 6 Let us compute the double integral

$$\iint_E \frac{dx\,dy}{x(x^2 + y^2)}\,,$$

(continued)

Example 6 (continued)

where $E = \{(x, y) : \ x \geq 1, \ 0 \leq y \leq x^2\}$. If E_k is an increasing sequence of measurable sets whose union is E, by (9.57)

$$\iint_E \frac{dx\,dy}{x(x^2+y^2)} = \lim_{k\to+\infty} \iint_{E_k} \frac{dx\,dy}{x(x^2+y^2)}\,. \tag{9.87}$$

Define $E_k = \{(x, y) : \ 1 \leq x \leq k, \ 0 \leq y \leq x^2\}$, so E_k is normal with respect to x and

$$\iint_{E_k} \frac{dx\,dy}{x(x^2+y^2)} = \int_1^k dx \int_0^{x^2} \frac{dy}{x(x^2+y^2)} = \int_1^k \frac{1}{x^2}\Big[\operatorname{arctg}\frac{y}{x}\Big]_0^{x^2} dx =$$

$$= \int_1^k \frac{\operatorname{arctg} x}{x^2}\,dx\,.$$

From (9.87) the given integral equals

$$\iint_E \frac{dx\,dy}{x(x^2+y^2)} = \lim_{k\to+\infty} \int_1^k \frac{\operatorname{arctg} x}{x^2}\,dx = \int_1^{+\infty} \frac{\operatorname{arctg} x}{x^2}\,dx =$$

$$= \Big[-\frac{\operatorname{arctg} x}{x}\Big]_1^{+\infty} + \int_1^{+\infty} \frac{dx}{x(1+x^2)} = \frac{\pi}{4} + \int_1^{+\infty} \left(\frac{1}{x} - \frac{x}{1+x^2}\right) dx =$$

$$= \frac{\pi}{4} + \left[\log \frac{x}{\sqrt{1+x^2}}\right]_1^{+\infty} = \frac{\pi}{4} + \frac{1}{2}\log 2\,.$$

Example 7 If $E = \{(x, y, z) \in \mathbb{R}^3 : \ (x, y) \in D, \ \alpha(x, y) \leq z \leq \beta(x, y)\}$ is normal with respect to the xy-plane and f is integrable on E,

$$\iiint_E f\,dx\,dy\,dz = \iint_D dx\,dy \int_{\alpha(x,y)}^{\beta(x,y)} f\,dz\,. \tag{9.88}$$

If D is furthermore normal, say with respect to x,

$$D = \Big\{(x, y) \in \mathbb{R}^2 : \ a \leq x \leq b, \ r(x) \leq y \leq s(x)\Big\}\,,$$

then (9.88) becomes

$$\iiint_E f\,dx\,dy\,dz = \int_a^b dx \int_{r(x)}^{s(x)} dy \int_{\alpha(x,y)}^{\beta(x,y)} f(x, y, z)\,dz\,. \tag{9.89}$$

Example 8 Let us compute the triple integral

$$\iiint_E xz\,dx\,dy\,dz\,, \tag{9.90}$$

where $E = \{(x, y, z) \in \mathbb{R}^3 :\ x \geq 0,\ z \geq 0,\ 0 \leq y \leq 2 - x^2 - z^2\}$. The set E, drawn in Fig. 9.9, is normal with respect to the xy-plane.

More precisely, $E = \{(x, y, z) \in \mathbb{R}^3 :\ (x, y) \in D,\ 0 \leq z \leq \sqrt{2 - x^2 - y}\}$, where $D = \{(x, y) \in \mathbb{R}^2 :\ 0 \leq x \leq \sqrt{2},\ 0 \leq y \leq 2 - x^2\}$ is normal with respect to x. From (9.89),

$$\begin{aligned}\iiint_E xz\,dx\,dy\,dz &= \int_0^{\sqrt{2}} dx \int_0^{2-x^2} dy \int_0^{\sqrt{2-x^2-y}} xz\,dz = \\ &= \frac{1}{2}\int_0^{\sqrt{2}} dx \int_0^{2-x^2} x(2 - x^2 - y)dy = \\ &= \frac{1}{2}\int_0^{\sqrt{2}} x\left[(2-x^2)y - \frac{y^2}{2}\right]_0^{2-x^2} dx = \frac{1}{4}\int_0^{\sqrt{2}} x(2-x^2)^2\,dx = \frac{1}{3}\,.\end{aligned}$$

Clearly E can be viewed as a normal set with respect to the xz- or yz-planes, and the reader may calculate (9.90) using this point of view.

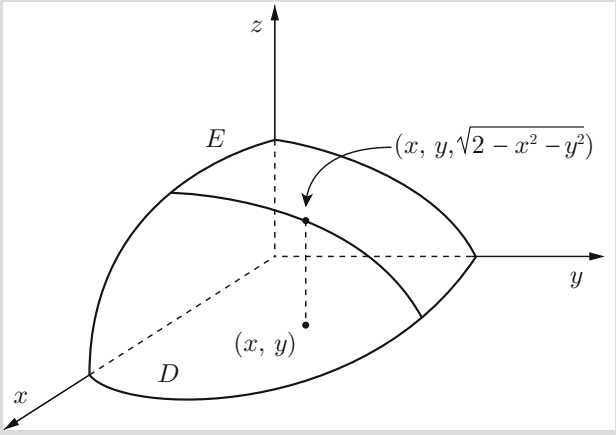


Fig. 9.9

Example 9 Let us compute the multiple integral

$$\int_E xy\,dx\,dy\,dz\,dw$$

on the domain $E = \{(x, y, z, w) \in \mathbb{R}^4 :\ x \geq 0,\ y \geq 0,\ x^2 + y^2 + z^2 + w^2 \leq 1\}$ in $\mathbb{R}^4$. Define

$$D = \left\{(x, y) \in \mathbb{R}^2 :\ x \geq 0,\ y \geq 0,\ x^2 + y^2 \leq 1\right\},$$

so that the cross-section of E at (x, y) is

$$E_{(x,y)} = \begin{cases} \{(z, w) \in \mathbb{R}^2 :\ z^2 + w^2 \leq 1 - x^2 - y^2\} & \text{if } (x, y) \in D \\ \emptyset & \text{otherwise.} \end{cases}$$

Then, if $(x, y) \in D$, the set $E_{(x,y)}$ is the disc centred at the origin with radius $\sqrt{1 - x^2 - y^2}$. By Fubini's theorem our integral equals

$$\int_E xy\,dx\,dy\,dz\,dw = \iint_D xy\,dx\,dy \iint_{E_{(x,y)}} dz\,dw =$$

$$= \iint_D xy\, m_2\big(E_{(x,y)}\big)\,dx\,dy = \pi \iint_D xy\Big(1 - x^2 - y^2\Big)\,dx\,dy\ .$$

As $D = \{(x, y) \in \mathbb{R}^2 :\ 0 \leq x \leq 1,\ 0 \leq y \leq \sqrt{1 - x^2}\}$, the last integral is

$$\pi \int_0^1 x\,dx \int_0^{\sqrt{1-x^2}} y(1 - x^2 - y^2)\,dy = \frac{\pi}{4} \int_0^1 x(1 - x^2)^2\,dx = \frac{\pi}{24}\,.$$

Example 10 Fix $\ell > 0$. An *n-simplex* with edge ℓ is a set of type

$$E_n = \left\{x \in \mathbb{R}^n :\ x_1 + x_2 + \ldots + x_n \leq \ell,\ x_i \geq 0 \quad \forall\, i = 1, 2, \ldots, n\right\}.$$

When $n = 1$, E_1 is the interval $[0, \ell]$, while E_2 is a right triangle whose legs have length ℓ, E_3 is a tetrahedron etc. We claim that

$$m_n(E_n) = \frac{\ell^n}{n!}\,. \tag{9.91}$$

(continued)

Example 10 (continued)
We shall use induction on n and the Fubini theorem. For $n = 1$ formula (9.91) is obvious. Assuming (9.91) true for $n \in \mathbb{N}$, consider

$$E_{n+1} = \left\{ (x,t) \in \mathbb{R}^{n+1} : x_1 + x_2 + \ldots + x_n + t \le \ell,\ t \ge 0,\ x_i \ge 0\ \forall\, i = 1, 2, \ldots, n \right\},$$

and observe that for $t \in \mathbb{R}$, $E_{n+1,t} = \emptyset$ if $t \notin [0, \ell]$, while

$$E_{n+1,t} = \left\{ x \in \mathbb{R}^n :\ x_1 + x_2 + \ldots + x_n \le \ell - t,\ x_i \ge 0\ \forall\, i = 1, 2, \ldots, n \right\},$$

if $t \in [0, \ell]$, i.e. $E_{n+1,t}$ is an n-simplex with edge $\ell - t$. By induction, the measurability of cross-sections implies

$$m_{n+1}(E_{n+1}) = \int_0^\ell m_n(E_{n+1,t})\, dt = \int_0^\ell \frac{(\ell - t)^n}{n!}\, dt = \frac{\ell^{n+1}}{(n+1)!},$$

proving (9.91).

If E is a measurable set in $\mathbb{R}^n$, with positive and bounded measure, the *centroid* of E is the point $x_0 \equiv (x_{0,i}) \in \mathbb{R}^n$ where

$$x_{0,i} = \frac{1}{m(E)} \int_E x_i\, dx\,, \qquad \forall\, i = 1, 2, \ldots, n. \tag{9.92}$$

When $n = 2, 3$ this is precisely the physical *centre of mass*, or *barycentre*, of a *uniform distribution of mass* on E.

Example 11 Let us compute the centroid of the semi-disc

$$C = \{(x,y) \in \mathbb{R}^2 :\ x^2 + y^2 \le 1,\ y > 0\}\,.$$

Letting (x_0, y_0) be the centroid, by symmetry reasons $x_0 = 0$. The second coordinate is

$$y_0 = \frac{1}{m_2(C)} \iint_C y\, dx\, dy\,.$$

But $m_2(C) = \pi/2$, so

$$y_0 = \frac{2}{\pi} \int_{-1}^1 dx \int_0^{\sqrt{1-x^2}} y\, dy = \frac{1}{\pi} \int_{-1}^1 (1 - x^2)\, dx = \frac{4}{3\pi}\,.$$

In Physics the *moment of inertia* of a solid body $E \subset \mathbb{R}^3$, about a given axis r, is the scalar quantity

$$\int_E [d(x,r)]^2 \, dm_3 \,,$$

where $d(x,r)$ is the distance of the generic point $x \in \mathbb{R}^3$ to the axis r.

Example 12 Let us compute the moment of inertia about the z-axis of the cone

$$C = \left\{ (x,y,z) \in \mathbb{R}^3 : \; x^2 + y^2 \le 1, \; \sqrt{x^2+y^2} \le z \le 1 \right\}.$$

Since the distance of a point $(x,y,z) \in \mathbb{R}^3$ from the reference axis is $\sqrt{x^2+y^2}$, we must calculate the triple integral

$$\begin{aligned} \iiint_C (x^2+y^2)\, dx\, dy\, dz &= \iint_D dx\, dy \int_{\sqrt{x^2+y^2}}^1 (x^2+y^2)\, dz = \\ &= \iint_D (x^2+y^2)\left(1 - \sqrt{x^2+y^2}\right) dx\, dy \,, \end{aligned} \tag{9.93}$$

where $D = \{(x,y) : \; x^2 + y^2 \le 1\}$. As D is normal, for the last integral in (9.93) we may apply Fubini's theorem. This computation is rather lengthy, so in the next section we will show how polar coordinates can simplify the calculation.

9.8 Changing Variables in Multiple Integrals

Let A and B be open sets in $\mathbb{R}^n$. A map $\Phi : A \to B$ is a *diffeomorphism* if

(i) Φ is invertible;
(ii) Φ and the inverse $\Phi^{-1} : B \to A$ are of class C^1.

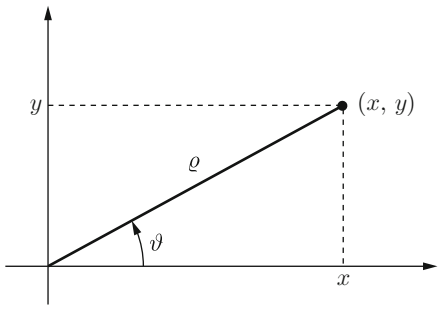
In Chap. 3 we have seen that if $x_0 \in A$ and $y_0 = \Phi(x_0)$, the Jacobian matrix $D\Phi^{-1}(y_0)$ is the the inverse of the Jacobian matrix of Φ at x_0 (see (3.70)):

$$D\Phi^{-1}(y_0) = [D\Phi(x_0)]^{-1} \,.$$

Moreover, if J_Φ, $J_{\Phi^{-1}}$ denote the Jacobian determinants of Φ and Φ^{-1}, then (see (3.71))

$$J_{\Phi^{-1}}(y_0) = \frac{1}{J_\Phi(x_0)} \,.$$

Fig. 9.10



In this section we will prove the two following remarkable results, and show the main applications to integral calculus.

Change of Measure Under Diffeomorphisms *Let $\Phi : A \to B$ be a diffeomorphism between open sets A, B in $\mathbb{R}^n$. If $E \subseteq A$ is measurable then $\Phi(E)$ is measurable, and*

$$m(\Phi(E)) = \int_E |J_\Phi| \, dx \, . \tag{9.94}$$

In more generality,

Change of Variables in Multiple Integrals *Let $\Phi : A \to B$ be a diffeomorphism. If $f : B \to \overline{\mathbb{R}}$ is an integrable function, then $f \circ \Phi$ is integrable on A and*

$$\int_B f(y) \, dy = \int_A f(\Phi(x)) |J_\Phi(x)| \, dx \, . \tag{9.95}$$

Notice how (9.95) reduces to (9.94) if f is chosen to be the characteristic function of $\Phi(E)$. As a matter of fact the above results are equivalent, and they can be proved simultaneously. Before doing that—the proof is long and technical—we will describe a number of examples and consequences of formulas (9.94) and (9.95).

We start from a very important case of change of variables, and introduce *polar coordinates*. Consider a point $(x, y) \in \mathbb{R}^2$ other than the origin. Its polar coordinates (ϱ, ϑ) are defined by $\varrho = \sqrt{x^2 + y^2}$ being the distance from the origin, and $\vartheta \in [0, 2\pi)$ the angle between the positive x-axis and the line from $(0, 0)$ to (x, y) (see Fig. 9.10).

Hence

$$\begin{cases} x = \varrho \cos \vartheta \\ y = \varrho \sin \vartheta \, . \end{cases}$$

The above relations define a map $\Phi : S \to \mathbb{R}^2$, where S is the strip $S = \{(\varrho, \vartheta) : \varrho \geq 0, \ 0 \leq \vartheta < 2\pi\}$. It is of class C^1, surjective, but not injective. To obtain a

diffeomorphism we must restrict Φ to the open set

$$A = \left\{(\varrho, \vartheta) \in \mathbb{R}^2 : \ \varrho > 0, \ 0 < \vartheta < 2\pi\right\}.$$

In this way we obtain a diffeomorphism from A to the open set $B = \mathbb{R}^2 - E_0$, where $E_0 = \{(x, y) : \ x \geq 0, \ y = 0\}$. The Jacobian matrix of Φ is

$$\begin{pmatrix} \cos\vartheta & -\varrho\sin\vartheta \\ \sin\vartheta & \varrho\cos\vartheta \end{pmatrix}$$

and so $J_\Phi(\varrho, \vartheta) = \varrho$ for any (ϱ, ϑ).

If E is a measurable subset in $\mathbb{R}^2$ and f is integrable on E, we can apply (9.95) to $f \cdot \chi_{E-E_0}$, obtaining

$$\iint_{E-E_0} f(x, y)\, dx\, dy = \iint_{\Phi^{-1}(E-E_0)} f(\varrho\cos\vartheta, \varrho\sin\vartheta)\varrho\, d\varrho\, d\vartheta\,. \quad (9.96)$$

But $m_2(E_0) = 0$, $\Phi^{-1}(E_0) = \{(\varrho, \vartheta) : \ \varrho \geq 0, \ \vartheta = 0\}$, and then $m_2(\Phi^{-1}(E_0)) = 0$. Hence (9.96) gives

$$\iint_E f(x, y)\, dx\, dy = \iint_{\Phi^{-1}(E)} f(\varrho\cos\vartheta, \varrho\sin\vartheta)\varrho\, d\varrho\, d\vartheta\,, \quad (9.97)$$

for any $E \subseteq \mathbb{R}^2$ measurable and any f integrable on E.

Example 1 Using polar coordinates we compute the integral in (9.93)

$$\iint_D (x^2 + y^2)\left(1 - \sqrt{x^2 + y^2}\right) dx\, dy\,,$$

where $D = \{(x, y) : \ x^2 + y^2 \leq 1\}$. As $\Phi^{-1}(D) = \{(\varrho, \vartheta) : \ 0 \leq \varrho \leq 1, \ 0 \leq \vartheta \leq 2\pi\}$, by (9.97) the integral equals

$$\int_0^{2\pi} d\vartheta \int_0^1 \varrho^3(1 - \varrho)\, d\varrho = 2\pi \int_0^1 (\varrho^3 - \varrho^4)\, d\varrho = \frac{\pi}{10}\,.$$

Example 2 Let us compute

$$\iint_C \sqrt{x^2 + y^2}\, dx\, dy\,, \quad (9.98)$$

(continued)

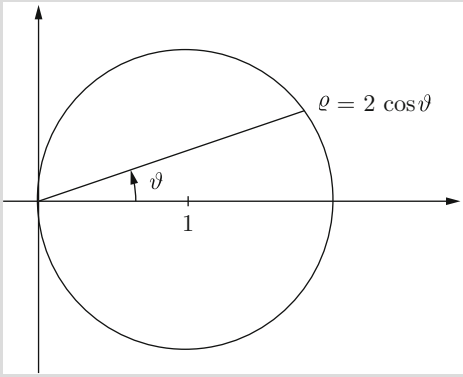
Example 2 (continued)
where C is the disc of centre $(1, 0)$ and radius 1:

$$C = \left\{(x, y) \in \mathbb{R}^2 : \; x^2 + y^2 - 2x \le 0\right\}.$$

In polar coordinates the inequality $x^2 + y^2 - 2x \le 0$ reads $\varrho^2 - 2\varrho\cos\vartheta \le 0$. Hence $\Phi^{-1}(C)$ is normal with respect to ϱ (see Fig. 9.11),

$$\Phi^{-1}(C) = \left\{(\varrho, \vartheta) : \; 0 \le \varrho \le 2\cos\vartheta, \; -\frac{\pi}{2} \le \vartheta \le \frac{\pi}{2}\right\}.$$

Fig. 9.11



Then from (9.97) the integral in (9.98) equals

$$\int_{-\pi/2}^{\pi/2} d\vartheta \int_0^{2\cos\vartheta} \varrho^2 \, d\varrho = \frac{8}{3}\int_{-\pi/2}^{\pi/2} \cos^3\vartheta \, d\vartheta = \frac{32}{9}.$$

Example 3 (Area of a Sector) Fix $\vartheta_1, \vartheta_2 \in [0, 2\pi]$. Consider two measurable functions $\varrho_1, \varrho_2 : [\vartheta_1, \vartheta_2] \to [0, +\infty)$ such that $\varrho_1(\vartheta) \le \varrho_2(\vartheta)$ for any $\vartheta \in [\vartheta_1, \vartheta_2]$. The set S of points in the plane whose polar coordinates (ϱ, ϑ) satisfy

$$\vartheta_1 \le \vartheta \le \vartheta_2, \qquad \varrho_1(\vartheta) \le \varrho \le \varrho_2(\vartheta),$$

is called a *sector* (see Fig. 9.12).

From (9.94) the area of S is

$$m_2(S) = \int_{\vartheta_1}^{\vartheta_2} d\vartheta \int_{\varrho_1(\vartheta)}^{\varrho_2(\vartheta)} \varrho \, d\varrho = \frac{1}{2}\int_{\vartheta_1}^{\vartheta_2} \left\{\varrho_2^2(\vartheta) - \varrho_1^2(\vartheta)\right\} d\vartheta. \tag{9.99}$$

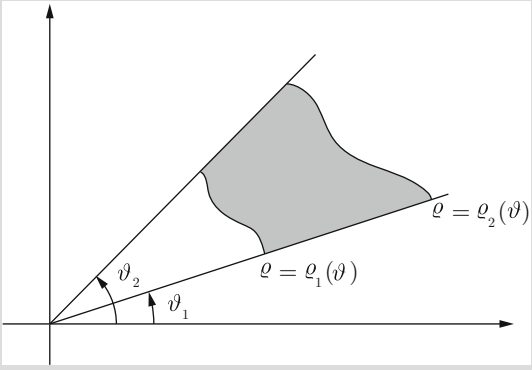
(continued)

Example 3 (continued)
For example, the area of the region enclosed by the *cardioid*, of polar equation $\varrho = a(1 + \cos\vartheta)$ with a positive and $\vartheta \in [0, 2\pi]$ (Fig. 6.6), is

$$\frac{1}{2}\int_0^{2\pi} a^2(1+\cos\vartheta)^2\,d\vartheta = \frac{3\pi a^2}{2}\,,$$

by (9.99).

Fig. 9.12



The next two examples show how the appropriate change of variables is often suggested by the shape of the sets we are working with.

Example 4 Let E be the set in the plane bounded by the parabolas $y = 2x^2$, $y = x^2$ and the hyperbolas $xy = 1$ and $xy = 2$, as in Fig. 9.13. Let us compute the area of E using the following change of variables

$$\begin{cases} s = xy \\ t = y/x^2. \end{cases} \tag{9.100}$$

In this way E becomes the square $Q = \{(s,t) :\ 1 \le s \le 2,\ 1 \le t \le 2\}$. Hence $E = \Phi(Q)$, where Φ is the inverse transformation of (9.100)

$$\begin{cases} x = \sqrt[3]{s/t} \\ y = \sqrt[3]{s^2 \cdot t}\,. \end{cases}$$

The Jacobian matrix of Φ is

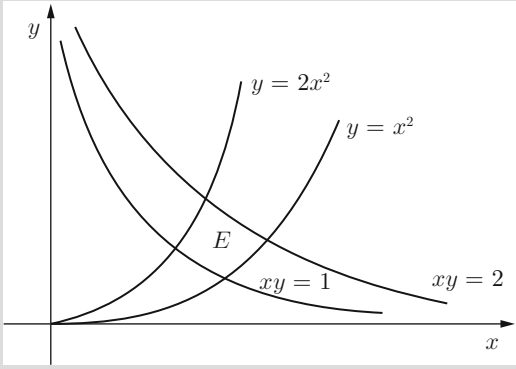
$$\begin{pmatrix} \frac{1}{3}s^{-2/3}\,t^{-1/3} & -\frac{1}{3}s^{1/3}\,t^{-4/3} \\ \frac{2}{3}s^{-1/3}\,t^{1/3} & \frac{1}{3}s^{2/3}\,t^{-2/3} \end{pmatrix}$$

(continued)

Example 4 (continued)
and so the Jacobian determinant of Φ equals $1/(3t)$. Now (9.94) gives the area of E:

$$\int_1^2 ds \int_1^2 \frac{1}{3t}\, dt = \frac{1}{3}\log 2\,.$$

Fig. 9.13



Example 5 Let us compute

$$\iint_E \frac{x}{y^2}\, dx\, dy\,,$$

where E is the region in the plane between the straight lines $y = x$, $y = 2x$, $x + y = 1$ and $x + 2y = 1$, as in Fig. 9.14.

The change of variables

$$\begin{cases} s = y/x \\ t = y/(1-x) \end{cases} \tag{9.101}$$

transforms E in the rectangle $R = \{(s,t) :\ 1 \le s \le 2,\ 1/2 \le t \le 1\}$.

Calling Φ the inverse transformation of (9.101),

$$\begin{cases} x = t/(s+t) \\ y = st/(s+t) \end{cases}$$

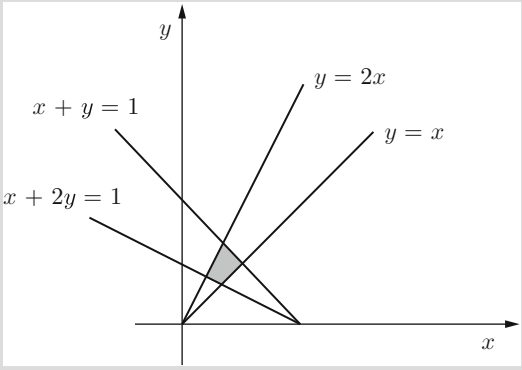
we have $E = \Phi(R)$ and the Jacobian matrix of Φ is

$$\begin{pmatrix} \dfrac{-t}{(s+t)^2} & \dfrac{s}{(s+t)^2} \\ \dfrac{t^2}{(s+t)^2} & \dfrac{s^2}{(s+t)^2} \end{pmatrix}.$$

(continued)

Example 5 (continued)

Fig. 9.14



The Jacobian determinant is then

$$J_\Phi = \frac{-st}{(s+t)^3}\,,$$

and from (9.95) the given integral equals

$$\begin{aligned}\iint_E \frac{x}{y^2}\,dx\,dy &= \int_1^2 ds \int_{1/2}^1 \frac{dt}{s(s+t)^2} = \\ &= \int_1^2 \frac{1}{s}\left(\frac{2}{1+2s} - \frac{1}{1+s}\right) ds = \\ &= \int_1^2 \left(\frac{1}{s} - \frac{4}{1+2s} + \frac{1}{1+s}\right) ds = \log\frac{27}{25}\,.\end{aligned}$$

Now we consider two systems of space coordinates of frequent use: cylindrical coordinates and spherical coordinates. The cylindrical coordinates (r, ϑ, z) are related to the Cartesian ones by the relationships

$$\begin{cases} x = r\cos\vartheta \\ y = r\sin\vartheta \\ z = z\,, \end{cases} \tag{9.102}$$

where (r, ϑ) are the polar coordinates of the Cartesian point (x, y) on the plane. The Jacobian matrix of Φ, defined in (9.102), is

$$\begin{pmatrix} \cos\vartheta & -r\sin\vartheta & 0 \\ \sin\vartheta & r\cos\vartheta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and so the Jacobian determinant of Φ equals r.

Example 6 Let us compute the volume of

$$E = \left\{(x, y, z) \in \mathbb{R}^3 : \ x^2 + y^2 \le z \le \sqrt{3 - 2(x^2 + y^2)}\right\}.$$

In cylindrical coordinates $E = \Phi(S)$, where

$$S = \left\{(r, \vartheta, z) \in \mathbb{R}^3 : \ 0 \le \vartheta \le 2\pi, \ 0 \le r \le 1, \ r^2 \le z \le \sqrt{3 - 2r^2}\right\},$$

so (9.94) gives for the volume of E

$$\int_0^{2\pi} d\vartheta \int_0^1 dr \int_{r^2}^{\sqrt{3-2r^2}} r \, dz = 2\pi \int_0^1 r\left(\sqrt{3 - 2r^2} - r^2\right) dr = \pi\left(\sqrt{3} - \frac{5}{6}\right).$$

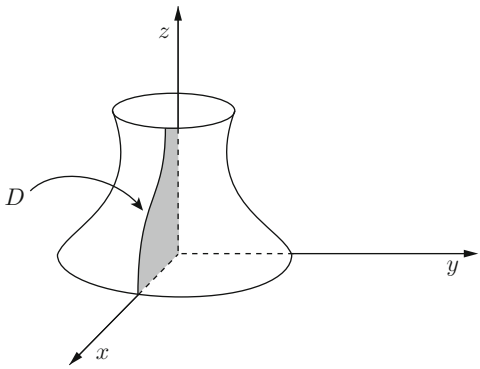
Using cylindrical coordinates we may prove

Guldinus Theorem on the Volume of a Solid of Revolution *The volume of the solid obtained rotating around some axis a region D in the plane equals the product of the area of D times the length of the circular arc described by its centroid.*

Proof Suitably choosing the frame system we can always suppose that D lies on the xz-plane and the axis of revolution is the z-axis (see Fig. 9.15).

Call $\alpha \in (0, 2\pi]$ the angle of rotation, so that the solid E arising from the α-rotation of D is described, in cylindrical coordinates, by $0 \le \vartheta \le \alpha$, $(r, z) \in D$. Its volume is then

Fig. 9.15



$$m_3(E) = \int_0^\alpha d\vartheta \iint_D r\,dr\,dz = \alpha \iint_D r\,dr\,dz = \\ = m_2(D)\cdot\alpha\cdot\frac{1}{m_2(D)}\iint_D x\,dx\,dz = m_2(D)\cdot\alpha\cdot x_0\,, \tag{9.103}$$

where x_0 is the first coordinate of the centroid of D (see (9.92)). The claim follows from (9.103), since $\alpha \cdot x_0$ is the length of the arc described by the centroid during the rotation.

If D is the trapezoid defined by $a \le x \le b$, $0 \le z \le f(x)$, the volume obtained rotating D about the x-axis by α equals, by the Guldinus theorem,

$$\alpha \iint_D z\,dx\,dz = \alpha \int_a^b dx \int_0^{f(x)} z\,dz = \frac{\alpha}{2}\int_a^b [f(x)]^2\,dx\,.$$

We thus recover in a different manner formula (9.81), seen in Example 3 of the previous section.

Example 7 Let us compute the volume of the solid E obtained by a complete revolution about the y-axis of the set $D = \{(x, y) \in \mathbb{R}^2 :\ (x^2 + y^2)^2 \le x^2 y,\ x \ge 0\}$. The Guldinus theorem implies that the volume of E is

$$m_3(E) = 2\pi \iint_D x\,dx\,dy\,.$$

In polar coordinates D is given by

$$0 \le \vartheta \le \frac{\pi}{2}\,, \qquad 0 \le \varrho \le \cos^2\vartheta \sin\vartheta\,.$$

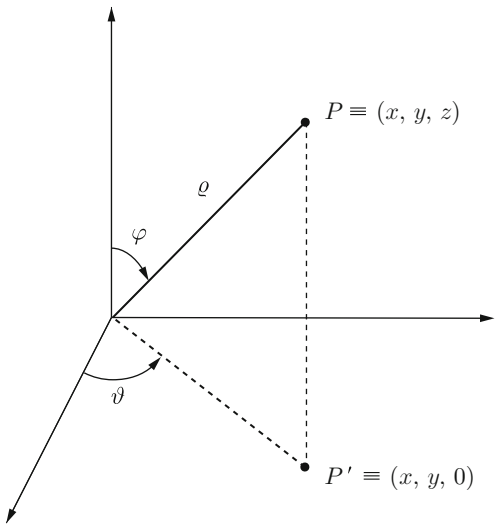
Hence

$$m_3(E) = 2\pi \int_0^{\pi/2} d\vartheta \int_0^{\cos^2\vartheta\sin\vartheta} \varrho^2 \cos\vartheta\,d\varrho = \frac{2\pi}{3}\int_0^{\pi/2} \cos^7\vartheta\,\sin^3\vartheta\,d\vartheta = \frac{\pi}{60}\,.$$

We introduce the *spherical coordinates* in space by setting

$$\begin{cases} x = \varrho\sin\varphi\cos\vartheta \\ y = \varrho\sin\varphi\sin\vartheta \\ z = \varrho\cos\varphi\,, \end{cases} \tag{9.104}$$

Fig. 9.16



for any $P \equiv (x, y, z) \in \mathbb{R}^3$. Referring to Fig. 9.16, $\varrho = \sqrt{x^2 + y^2 + z^2}$ is the distance of P to the origin, φ is the angle between the z-axis and the segment OP, and ϑ is the angle between the x-axis and the segment OP', where P' is the projection of P on the xy-plane.

Now, (9.104) defines a diffeomorphism $\Phi : A \to B$, where

$$A = \left\{(\varrho, \varphi, \vartheta) \in \mathbb{R}^3 : \ \varrho > 0, \ 0 < \varphi < \pi, \ 0 < \vartheta < 2\pi\right\},$$
$$B = \mathbb{R}^3 - E_0\,, \qquad E_0 = \left\{(x, y, z) \in \mathbb{R}^3 : \ x \geq 0, \ y = 0\right\}.$$

The Jacobian matrix of Φ is

$$\begin{pmatrix} \sin\varphi\cos\vartheta & \varrho\cos\varphi\cos\vartheta & -\varrho\sin\varphi\sin\vartheta \\ \sin\varphi\sin\vartheta & \varrho\cos\varphi\sin\vartheta & \varrho\sin\varphi\cos\vartheta \\ \cos\varphi & -\varrho\sin\varphi & 0 \end{pmatrix}$$

so the Jacobian determinant equals $J_\Phi = \varrho^2 \sin\varphi$. Since $m_3(E_0) = 0$, similar considerations to those made for polar coordinates lead to show that if $E \subseteq \mathbb{R}^3$ is a measurable set and f an integrable function on E,

$$\begin{aligned} &\iiint_E f\,dx\,dy\,dz = \\ &= \iiint_{\Phi^{-1}(E)} f(\varrho\sin\varphi\cos\vartheta, \varrho\sin\varphi\sin\vartheta, \varrho\cos\varphi)\cdot\varrho^2\sin\varphi\,d\varrho\,d\varphi\,d\vartheta\,. \end{aligned} \tag{9.105}$$

Example 8 We calculate the volume of the region E contained in the sphere

$$\left\{(x, y, z) \in \mathbb{R}^3 : \ x^2 + y^2 + z^2 \le 2z\right\} \tag{9.106}$$

and lying outside of the paraboloid

$$\left\{(x, y, z) \in \mathbb{R}^3 : \ z \ge 2(x^2 + y^2)\right\}. \tag{9.107}$$

In spherical coordinates, the sphere (9.106) becomes

$$\left\{(\varrho, \varphi, \vartheta) : \ 0 \le \vartheta < 2\pi, \ 0 \le \varphi \le \frac{\pi}{2}, \ \varrho \le 2\cos\varphi\right\}$$

and the paraboloid (9.107) is

$$\left\{(\varrho, \varphi, \vartheta) : \ 0 \le \vartheta \le 2\pi, \ 0 \le \varphi \le \frac{\pi}{2}, \ \varrho \le \frac{\cos\varphi}{2\sin^2\varphi}\right\}.$$

Hence the set E outside of the paraboloid turns into

$$\left\{(\varrho, \varphi, \vartheta) : \ 0 \le \vartheta < 2\pi, \ \frac{\pi}{6} \le \varphi \le \frac{\pi}{2}, \ \frac{\cos\varphi}{2\sin^2\varphi} \le \varrho \le 2\cos\varphi\right\}.$$

By (9.105) we have

$$\begin{aligned}
m_3(E) &= \int_0^{2\pi} d\vartheta \int_{\pi/6}^{\pi/2} d\varphi \int_{\frac{\cos\varphi}{2\sin^2\varphi}}^{2\cos\varphi} \varrho^2 \sin\varphi \, d\varrho = \\
&= \frac{2\pi}{3} \int_{\pi/6}^{\pi/2} \left\{8\cos^3\varphi \sin\varphi - \frac{\cos^3\varphi}{8\sin^6\varphi} \sin\varphi\right\} d\varphi = \\
&= \frac{2\pi}{3} \int_{\pi/6}^{\pi/2} \left\{8\cos^3\varphi \sin\varphi - \frac{\cos\varphi}{8\sin^5\varphi} + \frac{\cos\varphi}{8\sin^3\varphi}\right\} d\varphi = \\
&= \frac{2\pi}{3} \left[-2\cos^4\varphi + \frac{1}{32\sin^4\varphi} - \frac{1}{16\sin^2\varphi}\right]_{\varphi=\pi/6}^{\varphi=\pi/2} = \frac{9\pi}{16}.
\end{aligned}$$

Example 9 Let $x_1, x_2, \ldots, x_n \in \mathbb{R}^n$ be n linearly independent vectors, and P the parallelepiped generated by them:

$$P = \left\{x \in \mathbb{R}^n : \ x = \sum_{i=1}^{n} t_i x_i \, , \ 0 \le t_i \le 1, \ \forall\, i = 1, 2, \ldots, n\right\}.$$

(continued)

Example 9 (continued)
Calling $L : \mathbb{R}^n \to \mathbb{R}^n$ the linear transformation mapping the canonical basis $\{e_1, e_2, \ldots, e_n\}$ to the vectors $\{x_1, x_2, \ldots, x_n\}$:

$$L(e_i) = x_i\,, \qquad \forall\, i = 1, 2, \ldots, n,$$

we have $P = L([0,1]^n)$. Then from (9.94)

$$m(P) = |\det L|\,,$$

where L also denotes the matrix of the above linear map, i.e. the matrix having $x_1, x_2, \ldots, x_n$ as column vectors.

Example 10 Let us compute the measure of the n-dimensional unit ball using Fubini's theorem. The reader should compare this with the method adopted in the Appendix to Chap. 8 (Sect. 8.11).

Let $B = \{x \in \mathbb{R}^n : |x| \le 1\}$ be the unit ball in $\mathbb{R}^n$, with $n \ge 3$. For any $x \in \mathbb{R}^n$ define $x' = (x_1, x_2)$, $x'' = (x_3, x_4, \ldots, x_n)$. For any $x' \in \mathbb{R}^2$ the cross-section of B at x' is

$$B_{x'} = \left\{x'' \in \mathbb{R}^{n-2} : \; x_3^2 + x_4^2 + \ldots + x_n^2 \le 1 - x_1^2 + x_2^2\right\}$$

when $x_1^2 + x_2^2 \le 1$, otherwise $B_{x'} = \emptyset$. In the former case ($x_1^2 + x_2^2 \le 1$), $B_{x'}$ is then the ball in $\mathbb{R}^{n-2}$ with centre at the origin and radius $\sqrt{1 - x_1^2 - x_2^2}$. If ω_n denotes the measure of the unit ball in $\mathbb{R}^n$, (9.77) implies

$$\omega_n = \iint_{\{x_1^2+x_2^2\le 1\}} m_{n-2}(B_{x'})\, dx' =$$

$$= \omega_{n-2} \iint_{\{x_1^2+x_2^2\le 1\}} \left(1 - x_1^2 - x_2^2\right)^{\frac{n-2}{2}} dx_1\, dx_2\,.$$

Using polar coordinates the last integral is

$$\omega_n = 2\pi\,\omega_{n-2} \int_0^1 \varrho\left(1 - \varrho^2\right)^{\frac{n-2}{2}} d\varrho = \frac{2\pi}{n}\omega_{n-2}\,.$$

When $n = 2k$, the above formula tells

$$\omega_{2k} = \frac{\pi}{k}\omega_{2k-2} = \ldots = \frac{\pi^{k-1}}{k!}\omega_2 = \frac{\pi^k}{k!}\,.$$

(continued)

Example 10 (continued)
When $n = 2k + 1$,

$$\omega_{2k+1} = \frac{2\pi}{2k+1}\omega_{2k-1} = \ldots = \frac{2^k \pi^k}{(2k+1)(2k-1)\cdots 3}\,\omega_1 = \frac{2^{k+1}\pi^k}{(2k+1)!!}\,,$$

where $(2k+1)!! = (2k+1)\cdot(2k-1)\cdots 3\cdot 1$.

Let us now pass to proving the theorems stated at the beginning. We have split the proof in a series of lemmas. The first two address the case where Φ is a linear map, and for this we introduce certain special linear transformations.

If $L : \mathbb{R}^n \to \mathbb{R}^n$ is a linear map, we shall also call L the square matrix $(a_{ij})_{i,j=1,2,\ldots,n}$ representing it. We say L is an *elementary transformation* if it satisfies one of the following requirements:

(i) there exist $h, k \in \{1, 2, \ldots, n\}$, $h \neq k$, such that, setting $y = Lx$, $x \in \mathbb{R}^n$, we have $y_h = x_k$, $y_k = x_h$ and $y_i = x_i$ for any $i \neq h, k$. Hence the map L interchanges the hth and kth coordinates of x, while leaving the other ones fixed. In this case the matrix L has determinant -1, and its entries are

$$\begin{aligned} a_{ij} &= \delta_{ij} && \text{if } i \neq h, k, \\ a_{hj} = \delta_{kj}\,, \qquad a_{kj} &= \delta_{hj}\,, && \forall\, j = 1, 2, \ldots, n; \end{aligned}$$

(ii) there exist $h, k \in \{1, 2, \ldots, n\}$, $h \neq k$, and a real number $c \neq 0$, such that if $y = Lx$, $x \in \mathbb{R}^n$, then $y_i = x_i$ for any $i \neq h$, while $y_h = x_h + cx_k$. Here the matrix L has determinant 1 and elements

$$\begin{aligned} a_{ij} &= \delta_{ij}\,, && \text{if } i \neq h,\ \forall\, j = 1, 2, \ldots, n, \\ a_{hk} = c\,, \qquad a_{hj} &= \delta_{hj}\,, && \text{if } j \neq k; \end{aligned}$$

(iii) L is a non-singular diagonal matrix, so

$$a_{ij} = c_i\,\delta_{ij}\,, \qquad \forall\, i, j = 1, 2, \ldots, n,$$

where $c_i \neq 0$ for any $i = 1, 2, \ldots, n$, and hence $\det L = \prod_{i=1}^{n} c_i$.

Lemma 1 *If L is an invertible linear map of $\mathbb{R}^n$, there exists a finite number of elementary transformations $M_1, M_2, \ldots, M_m$ such that*

$$L = M_1 \circ M_2 \circ \ldots \circ M_m\,.$$

Proof Let $v_1, v_2, \dots, v_n$ be the column vectors of the matrix representing L, and $a_1, a_2, \dots, a_n$ the row vectors. If M is an elementary transformation of type (i), it is easy to see that the matrix $M \circ L$ arises from L by interchanging the hth row with the kth one and leaving the others as they are, while $L \circ M$ is obtained interchanging the hth column with the kth one, and leaving the rest untouched.

Similarly, it is easy to see that if M is of type (ii), then $M \circ L$ has the same rows as L except for the hth one, which equals $a_h + ca_k$, while $L \circ M$ has the same columns as L except for the kth one, which is $v_k + cv_h$.

Clearly, then, we can always find finitely many elementary transformations N_1, $N_2, \dots, N_s, N_{s+1}, \dots, N_m$ of type (i) or (ii) so that

$$N_1 \circ N_2 \circ \dots \circ N_s \circ L \circ N_{s+1} \circ \dots \circ N_m = D\,,$$

where D is diagonal of type (iii). This immediately implies the claim, since the inverse of map of type (i) is of type (i), and the same holds for type (ii).

Lemma 2 *If L is a linear map and $K \subset \mathbb{R}^n$ a compact set,*

$$m(L(K)) = |\det L|\, m(K)\,. \tag{9.108}$$

If $F : \mathbb{R}^n \to \mathbb{R}^n$ is an affine transformation, i.e.

$$F(x) = y_0 + L(x)\,, \qquad \forall\, x \in \mathbb{R}^n,$$

with L linear, since the Lebesgue measure is translation-invariant, by (9.108) we have

$$m(F(K)) = |\det L|\, m(K) \tag{9.109}$$

for any compact subset $K \subset \mathbb{R}^n$.

Proof The claim is obvious when $\det L = 0$, because in that case $L(\mathbb{R}^n)$ is a subspace of dimension $n-1$ at most, so its n-dimensional measure is easily zero.

Suppose then $\det L \neq 0$. We start by showing that for an elementary transformation L and a closed interval I,

$$m(L(I)) = |\det L|\, m(I)\,. \tag{9.110}$$

This is obvious if L is of type (i). In that case we immediately have $m(L(I)) = m(I)$ and $\det L = -1$. If L is of type (iii), setting

$$I = \prod_{i=1}^{n} [a_i, b_i]\,, \tag{9.111}$$

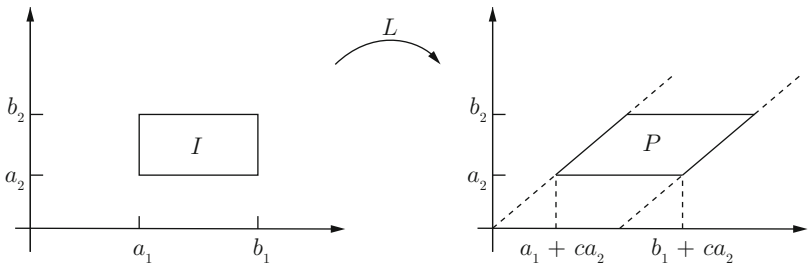


Fig. 9.17

$L(I)$ is the Cartesian product of real intervals of length

$$|c_1|(b_1 - a_1), |c_2|(b_2 - a_2), \ldots, |c_n|(b_n - a_n)\,.$$

Hence

$$m(L(I)) = m(I) \cdot \prod_{i=1}^{N} |c_i| = m(I)|\det L|\,,$$

and (9.108) is proved. Finally, if L is of type (ii), to fix ideas suppose $h = 1, k = 2$. If I is the interval given by (9.111), one can see that

$$L(I) = P \times \prod_{i=3}^{n} [a_i, b_i]\,,$$

where P is the parallelogram in Fig. 9.17, defined by

$$P = \left\{(y_1, y_2) \in \mathbb{R}^2 : \; a_1 + cy_2 \le y_1 \le b_1 + cy_2\,,\; a_2 \le y_2 \le b_2\right\}.$$

From Fubini's theorem we obtain

$$m(L(I)) = m_2(P) \cdot \prod_{i=3}^{n} (b_i - a_i)\,,$$

as since the area of P equals $m_2(P) = (b_1 - a_1)(b_2 - a_2)$, we have $m(L(I)) = m(I)$. As $\det L = 1$, (9.110) is again proved. The same reasoning gives (9.110) also when I is open.

By additivity we can also conclude that (9.108) holds when L is an elementary transformation and K a closed pluri-interval. If $\overset{\circ}{P}$ is an open pluri-interval, since

$\overset{\circ}{P} = I - Q$ for some open interval I and some closed pluri-interval Q, we still have

$$m(L(\overset{\circ}{P})) = |\det L|\, m(\overset{\circ}{P})\,. \tag{9.112}$$

If K is compact, by the lemma of Sect. 9.7 there is a decreasing sequence of open pluri-intervals $\overset{\circ}{P}_k$, whose intersection is K, satisfying (9.112). Now (9.108) follows from (9.112).

At last, suppose L, M are elementary transformations and K is compact. As $L(K)$ is compact, from what was proved we obtain

$$\begin{aligned} m((M \circ L)(K)) &= |\det M| m(L(K)) = \\ &= |\det M| \cdot |\det L| m(K) = |\det(M \circ L)| m(K)\,, \end{aligned}$$

since $\det(M \circ L) = \det M \cdot \det L$. Claim (9.108) for an invertible L now follows from what we proved above and Lemma 1.

In the sequel, if $x_0 \in \mathbb{R}^n$ and $r > 0$, we shall denote by $Q_r(x_0)$ the closed n-dimensional cube with edge $2r$ and centre x_0:

$$Q_r(x_0) = \prod_{i=1}^{n} [x_{0,i} - r, x_{0,i} + r]\,.$$

Lemma 3 *Let A be open in $\mathbb{R}^n$, $\Phi : A \to B$ a diffeomorphism and U an open bounded set such that $\overline{U} \subset A$. For any $\sigma > 0$ there is $r_0 > 0$ such that, if $x_0 \in U$ and $0 < r < r_0$,*

$$\Phi(Q_r(x_0)) \subseteq F\big(Q_{r(1+\sigma)}(x_0)\big)\,, \tag{9.113}$$

where F is the affine transformation

$$F(x) = \Phi(x_0) + D\Phi(x_0)(x - x_0)\,, \qquad \forall\, x \in \mathbb{R}^n. \tag{9.114}$$

Proof Let d be a positive number smaller than the distance of the closure of U from the boundary of A, i.e. such that $0 < d < \operatorname{dist}(\overline{U}, \partial A)$ (and $d > 0$ arbitrary if $A = \mathbb{R}^n$). Putting

$$K = \big\{x \in \mathbb{R}^n :\ \operatorname{dist}(x, \overline{U}) \le d\big\}\,,$$

by the continuity of the distance function K is compact, and furthermore $K \subset A$. Hence the generalised Weierstrass theorem (Sect. 2.12) ensures that $\Phi(K) \subset B$ is compact, and there is a constant c_0 such that

$$\left|D(\Phi^{-1})_i(y)\right| \le c_0\,, \qquad \forall\, y \in \Phi(K),\ \forall\, i = 1, 2, \ldots, n, \tag{9.115}$$

where $(\Phi^{-1})_i$ is the ith component of Φ^{-1}. Moreover, let $r_0 \in (0, d/\sqrt{n})$ be such that

$$x', x'' \in K,\ \left|x' - x''\right| \le r_0\sqrt{n} \quad \Rightarrow \quad \left|D\Phi_i(x') - D\Phi_i(x'')\right| \le \frac{\sigma}{c_0\, n^{3/2}}\,, \tag{9.116}$$

for any $i = 1, 2, \ldots, n$. Given $x_0 \in U$, call F the affine transformation defined by (9.114). Since $L = D\Phi(x_0)$ is invertible and $L^{-1} = D\Phi^{-1}(y_0)$ where $y_0 = \Phi(x_0)$, by (9.115) it follows

$$\left|x_i' - x_i''\right| = \left|D(\Phi^{-1})_i(y_0)\big(L(x' - x'')\big)\right| \le c_0\left|L(x' - x'')\right| \tag{9.117}$$

for any $x', x'' \in \mathbb{R}^n$ and any $i = 1, 2, \ldots, n$. To prove (9.113) consider $y \in \Phi(Q_r(x_0))$, with $0 < r < r_0$. Set $x = \Phi^{-1}(y)$, $\overline{x} = F^{-1}(y)$, so clearly $\Phi(x) = F(\overline{x})$. Then (9.117) implies

$$\begin{aligned} |x_i - \overline{x}_i| &\le c_0\,|L(x - \overline{x})| = c_0|F(x) - F(\overline{x})| = \\ &= c_0|F(x) - \Phi(x)| \le c_0 \sum_{i=1}^{n} |F_i(x) - \Phi_i(x)|\,, \end{aligned} \tag{9.118}$$

where F_i, Φ_i, are the ith components of F and Φ. By the mean value theorem, for any $i \in \{1, 2, \ldots, n\}$, there is a point ξ_i on the line segment between x and x_0 such that

$$F_i(x) - \Phi_i(x) = \Phi_i(x_0) - \Phi_i(x) + (D\Phi_i(x_0), x - x_0) = (D\Phi_i(x_0) - D\Phi_i(\xi_i), x - x_0)\,.$$

Since $|\xi_i - x_0| \le |x - x_0| \le r\sqrt{n} < d$, $\xi_i \in K$, by (9.116) we have

$$|F_i(x) - \Phi_i(x)| \le \frac{\sigma}{c_0 n^{3/2}}|x - x_0| \le \frac{\sigma r}{c_0 n}\,.$$

Hence we obtain from (9.118)

$$|x_i - \overline{x}_i| \le \sigma r, \qquad \forall\, i = 1, 2, \ldots, n,$$

implying $\overline{x} \in Q_{r(1+\sigma)}(x_0)$ and finally $y \in F(Q_{r(1+\sigma)}(x_0))$.

Lemma 4 *Let $\Phi : A \to B$ be a diffeomorphism and $E \subseteq A$ a measurable set. Then $\Phi(E)$ is measurable and*

$$m(\Phi(E)) \le \int_E |J_\Phi|\, dx\,. \tag{9.119}$$

Proof Let us first prove (9.119) when $I \subset A$ is a closed interval

$$I = \prod_{i=1}^{n} [a_i\,,\, b_i]\,,$$

with rational vertices $(a_1, a_2, \ldots, a_n)$, $(b_1, b_2, \ldots, b_n)$. Clearly I can be written as union of a finite number N of closed cubes, all with edges of the same length, and with pairwise-disjoint interiors. Finally, let U be an open bounded set containing I and such that $\overline{U} \subset A$.

Given $\sigma > 0$, choose r_0 as in Lemma 3. Then there is $r \in [0, r_0)$ such that

$$I = \bigcup_{i=1}^{N} Q_r(x_i)\,, \tag{9.120}$$

where $\overset{\circ}{Q}_r(x_i) \cap \overset{\circ}{Q}_r(x_j) = \emptyset$ if $i \neq j$. Moreover, it is possible to choose r so that, if $x', x'' \in I$ and $\left|x' - x''\right| \leq r\sqrt{n}$, then

$$\left|J_\Phi(x') - J_\Phi(x'')\right| < \sigma\,. \tag{9.121}$$

For any $x \in \mathbb{R}^n$ and any $i = 1, 2, \ldots, N$ we set $F_i(x) = \Phi(x_i) + (D\Phi(x_i), x - x_i)$. By (9.120), (9.113) and (9.109) we obtain

$$\begin{aligned} m(\Phi(I)) &\leq \sum_{i=1}^{N} m(\Phi(Q_r(x_i))) \leq \\ &\leq \sum_{i=1}^{N} m\big(F_i(Q_{r(1+\sigma)}(x_i))\big) = 2^n(1+\sigma)^n r^n \sum_{i=1}^{N} |J_\Phi(x_i)| \end{aligned}$$

and so, from (9.121),

$$\begin{aligned} m(\Phi(I)) &\leq 2^n(1+\sigma)^n r^n \sum_{i=1}^{N} |J_\Phi(x_i)| = \\ &= (1+\sigma)^n \sum_{i=1}^{N} \int_{Q_r(x_i)} |J_\Phi(x_i)|\,dx \leq \\ &\leq (1+\sigma)^n \sum_{i=1}^{N} \int_{Q_r(x_i)} |J_\Phi(x_i) - J_\Phi(x)|\,dx + (1+\sigma)^n \int_I |J_\Phi(x)|\,dx \leq \\ &\leq \sigma(1+\sigma)^n m(I) + (1+\sigma)^n \int_I |J_\Phi(x)|\,dx\,. \end{aligned} \tag{9.122}$$

Now (9.119) is a consequence of (9.122) as σ tends to zero.

Next, if $I \subset A$ is any closed interval, letting I_k be a decreasing sequence of closed intervals contained in A, with rational vertices, and whose intersection is I, we have

$$m(\Phi(I)) = \lim_{k\to+\infty} m(\Phi(I_k)) \le \lim_{k\to+\infty} \int_{I_k} |J_\Phi|\, dx = \int_I |J_\Phi|\, dx\,.$$

We conclude that (9.119) holds for any closed interval I, and hence any closed pluri-interval P.

If $U \subset A$ is open, letting P_k be an increasing sequence of closed pluri-intervals with union U, from what we have just proved

$$m\,(\Phi(U)) = \lim_{k\to+\infty} m(\Phi(P_k)) \le \lim_{k\to+\infty} \int_{P_k} |J_\Phi|\, dx = \int_U |J_\Phi|\, dx\,.$$

Hence (9.119) holds for E open.

Now fix an open bounded set U, with $\overline{U} \subset A$, and let us show (9.119) holds for any measurable set $E \subseteq U$. If E has zero measure, taking a decreasing sequence of open sets V_k such that $E \subseteq V_k \subseteq U$ for any $k \in \mathbb{N}$ and $m(V_k) \to 0$ as $k \to +\infty$, by (9.58) we have

$$m_e(\Phi(E)) \le \lim_{k\to+\infty} m(\Phi(V_k)) \le \lim_{k\to+\infty} \int_{V_k} |J_\Phi|\, dx = 0\,.$$

This proves $m(\Phi(E)) = 0$, and so $\Phi(E)$ is measurable. If $E \subseteq U$ is measurable, from Example 5, Sect. 9.4, there is a decreasing sequence V_k of open sets contained in U such that

$$m(E) = \lim_{k\to+\infty} m(V_k)\,, \qquad E = \bigcap_{k=1}^{\infty} V_k - E_0\,,$$

with $E_0 \subset U$ of measure zero. From this,

$$\Phi(E) = \bigcap_{k=1}^{\infty} \Phi(V_k) - \Phi(E_0)$$

is measurable and (9.119) holds.

At last, if $E \subseteq A$ is any measurable subset, the claim follows by the observation that, setting $E_k = E \cap U_k$ with

$$U_k = \{x \in A :\ |x| < k,\ \operatorname{dist}(x, \partial A) > 1/k\}\,,$$

we have $\overline{U_k} \subset A$ for any k, $U_k \subseteq U_{k+1}$ and $A = \bigcup_{k=1}^{\infty} U_k$. Since, for any $k \in \mathbb{N}$, $\Phi(E_k)$ is measurable, also $\Phi(E) = \bigcup_{k=1}^{\infty} \Phi(E_k)$ is measurable. Furthermore, from what was proved above

$$m(\Phi(E_k)) \leq \int_{E_k} |J_\Phi| \, dx \leq \int_E |J_\Phi| \, dx \,,$$

and then, since $\Phi(E_k) \leq \Phi(E_{k+1})$ for any $k \in \mathbb{N}$, (9.119) follows by taking the limit as $k \to +\infty$.

Now we pass to the proof of (9.95). As already said, this relationship implies (9.94) and therefore the theorem on the change of measure under diffeomorphisms: by Lemma 4, we know $\Phi(E)$ is measurable for any measurable set $E \subseteq A$.

We begin by showing that if $f : B \to [0, +\infty]$ is measurable, then $f(\Phi(x))$ is measurable on A and

$$\int_B f(y) \, dy \leq \int_A f(\Phi(x)) |J_\Phi(x)| \, dx \,. \tag{9.123}$$

This is true if

$$f(y) = \sum_{i=1}^{N} c_i \chi_{F_i}(y)$$

is a simple function. In fact by Lemma 4 (applied to Φ^{-1}), $E_i = \Phi^{-1}(F_i)$ is measurable for any $i = 1, 2, \ldots, N$. Hence

$$f(\Phi(x)) = \sum_{i=1}^{N} c_i \chi_{E_i}(x)$$

is measurable as well. By (9.119) we obtain

$$\int_B f(y) \, dy = \sum_{i=1}^{N} c_i \, m(F_i) = \sum_{i=1}^{N} c_i m(\Phi(E_i)) \leq$$

$$\leq \sum_{i=1}^{N} c_i \int_{E_i} |J_\Phi(x)| \, dx = \int_A f(\Phi(x)) |J_\Phi(x)| \, dx \,.$$

The general case follows by an approximation process, considering an increasing sequence of simple functions s_k that tends to f pointwise on B. Thus (9.123) is

proved for any diffeomorphism $\Phi : A \to B$ and any measurable and non-negative function f on B.

To prove the equality, given a measurable function $f : B \to [0, +\infty]$, we apply (9.123) to the diffeomorphism Φ^{-1} and the function $g : A \to [0, +\infty]$, $g(x) = f(\Phi(x))|J_\Phi|$, which is measurable. By (9.123), therefore,

$$\int_A f(\Phi(x))|J_\Phi(x)|\,dx \le$$
$$\le \int_B f\Big(\Phi(\Phi^{-1}(y))\Big)\Big|J_\Phi(\Phi^{-1}(y))\Big| \cdot |J_{\Phi^{-1}}(y)|\,dy = \int_B f(y)\,dy\,.$$

So we have (9.95) for a non-negative measurable function. The general case follows considering the positive and negative parts of f.

Appendix to Chap. 9

9.9 L^p Spaces

Let E be a measurable subset of $\mathbb{R}^n$, of positive measure. If $p \ge 1$ is a real number, one indicates by $L^p(E)$ the family of measurable functions f on E such that $|f|^p$ is summable:

$$L^p(E) = \left\{ f : E \to \overline{\mathbb{R}} \text{ measurable}: \int_E |f|^p\,dx < +\infty \right\}. \tag{9.124}$$

Notice that $L^1(E)$ is exactly the set of summable functions on E, so with the usual operations of sum of functions and product of a function by a scalar, it is a vector space.

Also $L^p(E)$, for $p > 1$, is a vector space. In fact if $f \in L^p(E)$ and $c \in \mathbb{R}$, obviously $cf \in L^p(E)$. If $f, g \in L^p(E)$, the inequality

$$|f(x) + g(x)|^p \le 2^{p-1}\left(|f(x)|^p + |g(x)|^p\right), \qquad \text{for a.e. } x \in E, \tag{9.125}$$

forces $f + g \in L^p(E)$.

We show (9.125) by proving that if $a, b \ge 0$, $p \ge 1$, then

$$(a+b)^p \le 2^{p-1}\left(a^p + b^p\right). \tag{9.126}$$

The function $\varphi(t) = 2^{p-1}(1 + t^p) - (1 + t)^p$ has absolute minimum for $t \ge 0$ at $t_0 = 1$, with minimum value $\varphi(t_0) = 0$. Hence $\varphi(t) \ge 0$ for any $t \ge 0$, and so

$$(1+t)^p \le 2^{p-1}\left(1 + t^p\right), \qquad \forall\, t \ge 0. \tag{9.127}$$

Now (9.126) follows from (9.127) with $t = b/a$.

If f is a measurable function on E, we say that f is *essentially bounded* on E when

$$m(\{x \in E : |f(x)| > M\}) = 0$$

for some constant $M > 0$. That is, $|f(x)| \leq M$ almost everywhere on E.

Consider, for any $t \geq 0$, the set $E_t = \{x \in E : |f(x)| > t\}$. If f is essentially bounded on E, the quantity

$$\operatorname{ess\,sup}_E |f| = \inf\{t > 0 : m(E_t) = 0\} \tag{9.128}$$

is called *essential supremum* of $|f|$ on E.

Writing λ for the essential supremum, let t_k be a strictly decreasing numerical sequence converging to λ. Since

$$m(E_{t_k}) = 0\,, \qquad \forall k \in \mathbb{N}\,, \qquad \text{and} \qquad E_\lambda = \bigcup_{k=1}^{\infty} E_{t_k}\,,$$

we also have $m(E_\lambda) = 0$. Therefore the essential supremum of $|f|$ is the smallest real number $\lambda \geq 0$ such that $|f(x)| \leq \lambda$ almost everywhere on E.

If f is not essentially bounded one sets $\operatorname{ess\,sup}_E |f| = +\infty$.

Now define

$$L^\infty(E) = \left\{ f : E \to \overline{\mathbb{R}},\ f \text{ measurable}, \operatorname{ess\,sup}_E |f| < +\infty \right\}. \tag{9.129}$$

Also $L^\infty(E)$ is a vector space, and we leave to the reader the easy proof.

If f is a measurable function on E, we indicate by $\|f\|_{L^p(E)}$, or simply $\|f\|_{L^p}$ or $\|f\|_p$, the quantity

$$\|f\|_{L^p(E)} = \left\{ \int_E |f|^p \, dx \right\}^{1/p}, \tag{9.130}$$

in case $p \in [1, +\infty)$, while for $p = +\infty$

$$\|f\|_{L^\infty(E)} = \operatorname{ess\,sup}_E |f|\,. \tag{9.131}$$

These quantities do not define a norm on $L^p(E)$, because if f is zero almost everywhere on E, then $\|f\|_{L^p(E)} = 0$ even when f is not the zero function. To overcome this problem one introduces, on the set of measurable functions on E, the equivalence relation

$$f \sim g \quad \Leftrightarrow \quad f(x) = g(x)\,, \qquad \text{for a.e. } x \in E.$$

In the sequel we shall identify a measurable function f with *the equivalence class* it determines. In this way the space $L^p(E)$ is now more precisely *the quotient set* (with respect to the relation $\sim$) of the space in (9.124) (or (9.129) if $p = +\infty$). Two elements f, g in $L^p(E)$ will then be equal whenever the functions f and g coincide almost everywhere on E.

Under that identification, if $f \in L^p(E)$ then

$$\|f\|_{L^p(E)} = 0 \qquad \Leftrightarrow \qquad f = 0\,. \tag{9.132}$$

If $c \in \mathbb{R}$ and $f \in L^p(E)$, it is immediate to see that

$$\|cf\|_{L^p(E)} = |c| \cdot \|f\|_{L^p(E)}\,. \tag{9.133}$$

Let us prove the *triangle inequality*: if $f, g \in L^p(E)$

$$\|f + g\|_{L^p(E)} \le \|f\|_{L^p(E)} + \|g\|_{L^p(E)}\,. \tag{9.134}$$

Formula (9.134) is straightforward when $p + \infty$ or $p = 1$. For $1 < p < +\infty$, (9.134) is called *Minkowski inequality*. To prove it we need another remarkable inequality, for which we introduce some definitions.

When $p \in [1, +\infty]$, we say $p' \in [1, +\infty]$ is *the conjugate exponent* of p if

$$\frac{1}{p} + \frac{1}{p'} = 1\,,$$

where, if $p = +\infty$, we set $1/p = 0$ i.e. $p' = 1$, and similarly for p'.

Hölder's Inequality *If f, g are measurable on E and $p, p' \in [1, +\infty]$ are conjugate exponents, then*

$$\int_E |f \cdot g|\, dx \le \|f\|_{L^p(E)} \cdot \|g\|_{L^{p'}(E)}\,. \tag{9.135}$$

Proof Formula (9.135) is immediate in case $p = 1$, $p' = +\infty$, or $p = +\infty$, $p' = 1$. Suppose then $1 < p, p' < +\infty$.

Again, (9.135) is obvious if the right-hand side equals $+\infty$. If not, by Young's inequality (Sect. 2.7)

$$ab \le \frac{a^p}{p} + \frac{b^{p'}}{p'} \qquad \forall\, a, b \ge 0,$$

we have

$$\frac{|f(x)\cdot g(x)|}{\|f\|_p\cdot\|g\|_{p'}}\le\frac{1}{p}\frac{|f(x)|^p}{\|f\|_p^p}+\frac{1}{p'}\frac{|g(x)|^{p'}}{\|g\|_{p'}^{p'}}\qquad\text{for a.e. } x\in E.$$

Integrating the above inequality over E gives

$$\frac{1}{\|f\|_p\cdot\|g\|_{p'}}\int_E|f(x)\cdot g(x)|\,dx\le$$
$$\le\frac{1}{p\|f\|_p^p}\int_E|f(x)|^p\,dx+\frac{1}{p'\|g\|_{p'}^{p'}}\int_E|g(x)|^{p'}\,dx=\frac{1}{p}+\frac{1}{p'}=1\,,$$

and hence (9.135).

Now let us prove Minkowski's inequality (9.134) for $p\in(1,+\infty)$. As $p'=p/(p-1)$, from (9.135) we obtain

$$\int_E|f+g|^p\,dx\le\int_E|f+g|^{p-1}|f|\,dx+\int_E|f+g|^{p-1}|g|\,dx\le \tag{9.136}$$
$$\le\|f\|_p\left\{\int_E|f+g|^p\,dx\right\}^{\frac{p-1}{p}}+\|g\|_p\left\{\int_E|f+g|^p\,dx\right\}^{\frac{p-1}{p}},$$

and then (9.134) follows by dividing both sides of (9.136) by

$$\left\{\int_E|f+g|^p\,dx\right\}^{\frac{p-1}{p}}.$$

Example 1 If the set E has finite measure and $p\in[1,+\infty]$, putting $g=1$ in Hölder's inequality (9.135) gives

$$\int_E|f(x)|\,dx\le\|f\|_{L^p(E)}\cdot\{m(E)\}^{\frac{p-1}{p}}\,, \tag{9.137}$$

where $(p-1)/p=1$ if $p=+\infty$. In particular, $m(E)<+\infty$ implies $L^p(E)\subseteq L^1(E)$. More generally, if $m(E)<+\infty$ and $1\le r<s\le+\infty$

$$\|f\|_{L^r(E)}\le\|f\|_{L^s(E)}\cdot\{m(E)\}^{\frac{1}{r}-\frac{1}{s}}\,. \tag{9.138}$$

This follows from (9.137) if we write $|f|^r$ in place of f, and $p=s/r$.

(continued)

Example 1 (continued)
In conclusion, if $m(E) < +\infty$,

$$1 \le r < s \le +\infty \quad \Rightarrow \quad L^s(E) \subseteq L^r(E)\,. \tag{9.139}$$

Example 2 If $E = (0,1) \subset \mathbb{R}$ and $f(x) = |x|^{-\alpha}$ with $0 < \alpha < 1$, then $f \in L^p(E)$ if and only if $1 \le p < 1/\alpha$. This example shows that, in general, if $m(E) < +\infty$ and $r < s$, then $L^r(E) \neq L^s(E)$ (but as we showed above, $L^s(E) \subset L^r(E)$). As a matter of fact one can prove that this is always the case, for every measurable set $E \subseteq \mathbb{R}^n$ with $0 < m(E) < +\infty$.
In general, if $m(E) < +\infty$, we also have

$$\bigcup_{s>r} L^s(E) \subsetneqq L^r(E)\,.$$

For instance

$$f(x) = \frac{1}{x|\log\, x|^2}\,,$$

is summable on $(0,1)$, i.e. $f \in L^1((0,1))$, but $f \notin L^p((0,1))$ for any $p > 1$.

Example 3 If $m(E) = +\infty$, (9.139) is not true. For example the function $f(x) = 1/x$ is in $L^p((1,+\infty))$ for any $p > 1$, but not in $L^1((1,+\infty))$.

We have nonetheless the following *interpolation* result:

Proposition 1 *If $1 \le s < r < t \le +\infty$, then*

$$\|f\|_{L^r(E)} \le \|f\|^{\vartheta}_{L^s(E)} \cdot \|f\|^{1-\vartheta}_{L^t(E)}\,, \tag{9.140}$$

for any measurable function f on E, where $\vartheta \in (0,1)$ is such that

$$\frac{1}{r} = \frac{\vartheta}{s} + \frac{1-\vartheta}{t}\,. \tag{9.141}$$

By (9.140) it follows in particular

$$L^s(E) \cap L^t(E) \subseteq L^r(E)\,.$$

As a matter of fact also this inclusion is strict.

Proof From (9.141)

$$\frac{1}{s/(r\vartheta)} + \frac{1}{t/(r(1-\vartheta))} = 1\,;$$

so $s/(r\vartheta)$ and $t/(r(1-\vartheta))$ are conjugate exponents. Hölder's inequality (9.135) implies

$$\|f\|_r^r = \int_E |f|^r \, dx = \int_E |f|^{r\vartheta} \, |f|^{r(1-\vartheta)} \, dx \leq$$

$$\leq \left\{ \int_E |f|^s \, dx \right\}^{\frac{r\vartheta}{s}} \cdot \left\{ \int_E |f|^t \, dx \right\}^{\frac{r(1-\vartheta)}{t}} = \|f\|_s^{r\vartheta} \cdot \|f\|_t^{r(1-\vartheta)} \, .$$

Then (9.140) follows by taking the rth root of the first and last side in the above inequality.

Example 4 If E is has finite measure and f is a measurable function on E, then $\|f\|_{L^p(E)}$ admits limit as $p \to +\infty$, and

$$\lim_{p\to+\infty} \|f\|_{L^p(E)} = \|f\|_{L^\infty(E)} \, . \tag{9.142}$$

From (9.138), in fact,

$$\limsup_{p\to+\infty} \|f\|_{L^p(E)} \leq \|f\|_{L^\infty(E)} \cdot \lim_{p\to+\infty} (m(E))^{1/p} = \|f\|_{L^\infty(E)} \, . \tag{9.143}$$

If f does not vanish almost everywhere (otherwise (9.142) is obvious), given a positive number t smaller than $\|f\|_{L^\infty(E)}$, by definition (9.128) of essential supremum we have $m(E_t) > 0$. As $|f(x)| > t$ for almost every $x \in E_t$, moreover,

$$\|f\|_{L^p(E)} \geq \left\{ \int_{E_t} |f|^p \, dx \right\}^{1/p} \geq t \, (m(E_t))^{1/p} \, .$$

Taking the limit inferior as $p \to +\infty$ gives

$$\liminf_{p\to+\infty} \|f\|_{L^p(E)} \geq t \, , \qquad \forall t < \|f\|_{L^\infty(E)} .$$

Then letting t tend to $\|f\|_{L^\infty(E)}$ we obtain

$$\liminf_{p\to+\infty} \|f\|_{L^p(E)} \geq \|f\|_{L^\infty(E)} \, .$$

This, together with (9.143), gives the conclusion (9.142).

Let E be a measurable set. A sequence of functions f_k *converges almost everywhere on* E to a function f if, as $k \to +\infty$, $f_k(x) \to f(x)$ for almost every $x \in E$. If $1 \le p < +\infty$, a sequence f_k in $L^p(E)$ *converges in* L^p *norm* to f if and only if

$$\lim_{k\to+\infty} \|f_k - f\|^p_{L^p(E)} = \lim_{k\to+\infty} \int_E |f_k - f|^p \, dx = 0 \,.$$

The study of conditions equivalent to the convergence in L^p norm goes beyond the purposes of this textbook. Let us just mention that Lebesgue's dominated convergence theorem immediately implies

Proposition 2 *Let f_k be a sequence in $L^p(E)$ that converges almost everywhere on E to f. If, for some $p \in [1, +\infty)$, there is a function $g \in L^p(E)$ such that*

$$|f_k(x)| \le g(x) \,, \qquad \forall\, k \in \mathbb{N} \ \text{ and for a.e. } x \in E,$$

then f_k converges to f in $L^p(E)$.

If f_k is a sequence in $L^\infty(E)$, to say that f_k converges to f in $L^\infty(E)$ is equivalent to saying, by definition of norm, that for any $k \in \mathbb{N}$ there exist zero-measure sets $E_k \subset E$ such that

$$\lim_{k\to+\infty} \sup\{|f_k(x) - f(x)| : x \in E - E_k\} = 0 \,.$$

Since $m\left(\bigcup_{k=1}^{\infty} E_k\right) = 0$, we have proved the following result.

Proposition 3 *A sequence f_k in $L^\infty(E)$ converges to $f \in L^\infty(E)$ if and only if there is a zero-measure set $E_0 \subset E$ such that f_k converges to f uniformly on $E - E_0$.*

The next result establishes one of the most important properties of the spaces $L^p(E)$.

Theorem *For any $p \in [1, +\infty]$, $L^p(E)$ is a Banach space for the norm defined in* (9.130), (9.131).

We have already seen, in Example 5, Sect. 2.8, that a sequence f_k of continuous functions, that is Cauchy for the L^p norm, may not converge to a continuous function. In other words C^0 is not complete for the L^p norm. The previous theorem, however, guarantees that the sequence f_k admits limit in $L^p(E)$.

Proof We have already proved (see (9.132), (9.133) and (9.134)) the properties

$$\begin{aligned} \|f\|_{L^p(E)} = 0 \quad &\Leftrightarrow \quad f = 0\,; \\ \|cf\|_{L^p(E)} = |c| \cdot \|f\|_{L^p(E)}\,, \qquad &\forall\, c \in \mathbb{R},\ \forall\, f \in L^p(E); \\ \|f+g\|_{L^p(E)} \le \|f\|_{L^p(E)} + \|g\|_{L^p(E)}\,, \qquad &\forall\, f, g \in L^p(E) \end{aligned}$$

for the norm defined in (9.130) and (9.131). These properties make $L^p(E)$, for any $p \in [1, +\infty]$, a normed space. There remains to prove completeness.

If $p = +\infty$, by Proposition 3 completeness reduces to the uniform Cauchy criterion seen in Sect. 1.2. So let $1 \le p < +\infty$. Consider a Cauchy sequence f_k in $L^p(E)$. By Proposition 1, Sect. 2.8, to show that f_k converges in $L^p(E)$ it suffices to find a convergent subsequence.

Let then k_r be a strictly increasing sequence of positive integers such that

$$\|f_h - f_k\|_{L^p(E)} < \frac{1}{2^{r+r/p}}\,, \qquad \forall\, h, k \ge k_r\,. \tag{9.144}$$

Define

$$g(x) = \sum_{r=1}^{\infty} 2^{rp} \left| f_{k_r}(x) - f_{k_{r+1}}(x) \right|^p\,, \qquad \forall\, x \in E. \tag{9.145}$$

Since by (9.144)

$$\|g\|_{L^1(E)} = \sum_{r=1}^{\infty} 2^{rp} \int_E \left| f_{k_r}(x) - f_{k_{r+1}}(x) \right|^p dx < \sum_{r=1}^{\infty} \frac{1}{2^r} = 1,$$

the function g is summable on E, so $g(x) < +\infty$ for almost every $x \in E$. From (9.145) we obtain

$$\left| f_{k_r}(x) - f_{k_{r+1}}(x) \right| \le \frac{1}{2^r} g(x)^{1/p} < +\infty \qquad \forall\, r \in \mathbb{N}, \quad \text{for a.e. } x \in E.$$

Then, for any $r \in \mathbb{N}$ and any $s > r$,

$$\begin{aligned} \left| f_{k_s}(x) - f_{k_r}(x) \right| &\le \sum_{i=0}^{s-r-1} \left| f_{k_{r+i}}(x) - f_{k_{r+i+1}}(x) \right| \le \\ &\le \sum_{i=0}^{s-r-1} \frac{1}{2^{r+i}} g(x)^{1/p} \le \frac{1}{2^{r-1}} g(x)^{1/p} \qquad \text{for a.e. } x \in E. \end{aligned} \tag{9.146}$$

By the (pointwise) Cauchy criterion the sequence f_{k_r} converges pointwise, for almost every $x \in E$, to some function $f(x)$. From (9.146), letting s tend to $+\infty$,

$$\left|f(x) - f_{k_r}(x)\right|^p \le \frac{1}{2^{(r-1)p}} g(x)\,, \qquad \forall\, r \in \mathbb{N},\ \text{for a.e. } x \in E.$$

Integrating over E gives $f \in L^p(E)$. Moreover, as $r \to +\infty$, we see that f_{k_r} converges to f in $L^p(E)$. The claim is then proved.

We finish by observing that the above proof allows to deduce in particular the following

Corollary *If f_k converges to f in $L^p(E)$ for some $p \in [1, +\infty]$, there exists a subsequence f_{k_r} that converges to f almost everywhere on E.*

9.10 Differentiability of Monotone Functions

When studying Lebesgue's indefinite integral

$$F : x \in [a, b] \to \int_a^x f(t)\, dt \tag{9.147}$$

as a function of the endpoint x of integration, one observes that if f is summable and non-negative then F is increasing. If f is summable but has arbitrary sign, it is the difference of two non-negative summable functions (see (9.39))

$$f(t) = f^+(t) - f^-(t)$$

and (9.147) is the difference of two increasing functions $F_1(x)$ and $F_2(x)$

$$F_1(x) = \int_a^x f^+(t)\, dt \qquad F_2(x) = \int_a^x f^-(t)\, dt\,.$$

Therefore the study of Lebesgue's indefinite integral (9.147) reduces to the study of the difference of two increasing functions, which is not monotone in general.

We recall notions and notations regarding *monotone real functions* defined on a closed interval $[a, b]$ in $\mathbb{R}$. If F is increasing on $[a, b]$ and $x_0 \in [a, b)$, by $F(x_0^+)$ we denote the right limit

$$\lim_{x \to x_0^+} F(x),$$

and similarly for $x_0 \in (a, b]$ we set

$$F(x_0^-) = \lim_{x \to x_0^-} F(x) .$$

When $x_0 \in (a, b)$ we call the non-negative number $F(x_0) - F(x_0^-)$ the *left jump* of F at x, and the non-negative number $F(x_0^+) - F(x_0)$ the *right jump*. The *jump of* F at x_0 is

$$F(x_0^+) - F(x_0^-) .$$

The next lemma contains an upper bound for the *sum of all jumps* of an increasing function.

Lemma 1 *Let F be increasing on $[a, b]$ and $x_1, x_2, \ldots, x_k$ points in (a, b). Then*

$$[F(a^+) - F(a)] + \sum_{j=1}^{k} [F(x_j^+) - F(x_j^-)] + [F(b) - F(b^-)] \leq F(b) - F(a) . \tag{9.148}$$

Proof Supposing

$$a = x_0 < x_1 < x_2 < \ldots < x_k < x_{k+1} = b$$

we pick points $y_0, y_1, \ldots, y_k$ such that

$$x_j < y_j < x_{j+1} \qquad j = 0, 1, 2, \ldots, k .$$

Then

$$\begin{aligned} F(x_j^+) - F(x_j^-) &\leq F(y_j) - F(y_{j-1}) \\ F(a^+) - F(a) &\leq F(y_0) - F(a) \\ F(b) - F(b^-) &\leq F(b) - F(y_k) \end{aligned}$$

and adding the inequalities gives (9.148).

Corollary 1 *If $F(x)$ is increasing on $[a, b]$, for any $\varepsilon > 0$ there exist at most finitely many discontinuity points at which the jump of F is bigger than ε.*

Proof If the points $x_1, x_2, \ldots, x_k$ internal to $[a, b]$ are discontinuities for F with jump larger than ε, then

$$k\varepsilon < F(b) - F(a)\,,$$

and so k cannot be arbitrarily large.

Proposition 1 *The discontinuity set of a monotone function F is at most countable.*

Proof Let D be the discontinuity set of F, and D_k, $k \in \mathbb{N}$, the subset of points where the jump is bigger than or equal to $1/k$. Evidently

$$D = \underset{k\in\mathbb{N}}{\cup} D_k$$

and so D is at most countable, by Corollary 1.

If (x_j) is the sequence of elements of D, by (9.148) it follows (as $k \to \infty$) that

$$F(a^+) - F(a) + \sum_{j=1}^{\infty} [F(x_j^+) - F(x_j^-)] + F(b) - F(b^-) \le F(b) - F(a)\,. \tag{9.149}$$

Let now $F : [a, b] \to \mathbb{R}$ be an increasing function. Define the jump function $S(x)$ of $F(x)$ by:

$$S(x) = \begin{cases} 0 & \text{for} \quad x = a \\ F(a^+) - F(a) + \sum\limits_{x_k < x} [F(x_k^+) - F(x_k^-)] + & \\ \qquad + [F(x) - F(x^-)] & \text{for} \quad a < x \le b \end{cases} \tag{9.150}$$

This is an increasing function.

Proposition 2 *The difference*

$$\Delta(x) = F(x) - S(x)$$

is increasing and continuous.

Proof Take $a \le x < y \le b$ and apply (9.149) to the interval $[x, y]$. Then

$$S(y) - S(x) \le F(y) - F(x) \tag{9.151}$$

so $\Delta(x) \le \Delta(y)$, and Δ is increasing. Taking the limit in (9.151) as $y \to x^+$ gives

$$S(x^+) - S(x) \le F(x^+) - F(x)\,. \tag{9.152}$$

By definition of $S(x)$, on the other hand, for $y > x$

$$F(x^+) - F(x) \le S(y) - S(x)\,.$$

When $y \to x^+$

$$F(x^+) - F(x) \le S(x^+) - S(x)\,. \tag{9.153}$$

From (9.152), (9.153) we deduce

$$F(x^+) - F(x) = S(x^+) - S(x)\,,$$

i.e.

$$\Delta(x^+) = \Delta(x)\,.$$

With a similar argument one proves

$$\Delta(x^-) = \Delta(x),$$

so Δ is continuous.

Covering Theorems
Let X be a subset of $\mathbb{R}$ and $\mathcal{F}$ a family of closed non-degenerate intervals.

Definition 1 The family $\mathcal{F}$ is a Vitali covering of X if, for any $x \in X$ and any $\varepsilon > 0$, there exists $I \in \mathcal{F}$ such that

$$x \in I \quad \text{and} \quad m(I) < \varepsilon\,. \tag{9.154}$$

In other words every point in X is contained in some closed interval of $\mathcal{F}$ whose measure is arbitrarily small.

Now we state a classical covering theorem, whose proof we omit.

Theorem 1 (Vitali Coverings) *Let $X \subset \mathbb{R}$ be a measurable set and $\mathcal{F}$ a Vitali covering of X. There exist a finite or countable subfamily $\mathcal{F}' \subset \mathcal{F}$ made of pairwise-disjoint intervals such that*

$$m\left(X \setminus \bigcup_{I \in \mathcal{F}'} I\right) = 0\,.$$

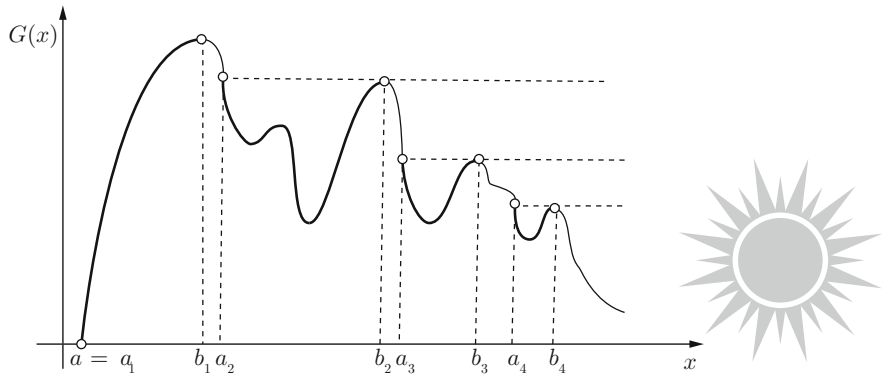


Fig. 9.18 The points on the thicker parts of the graph of G are those whose x-coordinate lies in A and are not hit by the rising sun's light

Remark 1 If in Theorem 1 we assume that X has finite measure, and we call $(I_j)_{j\in\mathbb{N}}$ the (possibly empty) intervals of $\mathcal{F}'$, by (9.154) we have

$$\lim_{k\to+\infty} m\left(X \setminus \bigcup_{j=1}^{k} I_j\right) = 0.$$

Hence we conclude that for any $\varepsilon > 0$ there is a finite set $\{I_1, \ldots, I_k\} \subset \mathcal{F}$ of pairwise-disjoint closed intervals such that

$$m\left(X \setminus \bigcup_{j=1}^{k} I_j\right) < \varepsilon\,.$$

The following result is a version of the 'Rising Sun Lemma', originally due to F. Riesz. The statement is depicted in Fig. 9.18.

Lemma 2 (Riesz) *Let $G : [a, b] \to \mathbb{R}$ be a continuous function and define*

$$A = \{x \in (a, b) : \ \exists\, y > x : \ G(y) > G(x)\}\,.$$

Then A is the union of a finite, or at most countable, number of pairwise-disjoint open intervals (a_j, b_j) such that

$$G(a_j) \le G(b_j)\,. \tag{9.155}$$

Similarly, setting

$$B = \{x \in (a, b) : \ \exists\, y < x : \ G(y) > G(x)\}\,,$$

B is a finite or at most countable union of pairwise-disjoint, open intervals (α_j, β_j) such that

$$G(\alpha_j) \geq G(\beta_j)\,. \tag{9.156}$$

Proof We will just prove the first part, as the second assertion is completely similar.

First of all, the continuity of G implies that A is open. Hence A is a union, at most countable, of pairwise-disjoint intervals (a_j, b_j). In particular, for any j the endpoints a_j, b_j cannot belong to A.

To prove (9.155) we argue by contradiction, supposing that for some j the opposite inequality

$$G(a_j) > G(b_j) \tag{9.157}$$

holds. By (9.157) there is $x_0 \in (a_j, b_j)$ such that $G(x_0) > G(b_j)$. Let y be the largest $x \in (a_j, b_j)$ for which $G(x) = G(x_0)$. This maximum exists due to (9.157), the continuity of G and the fact that $y \geq x_0$. Since y belongs to A, there exists $z \in (y, b)$ such that $G(z) > G(y)$. Moreover z cannot lie in (a_j, b_j), because otherwise, due to $G(z) > G(y) > G(b_j)$, the intermediate value theorem would give us $x \in (z, b_j)$ with $G(x) = G(y) = G(x_0)$, against the definition of y.

So we have found a point $z \in (b_j, b)$ such that $G(z) > G(y) > G(b_j)$. Then b_j, too, belongs in A. But this is absurd since, as observed above, $b_j \notin A$. Therefore (9.155) holds for any j.

Among the most important results in the theory of monotone functions is

Theorem 2 (Lebesgue) *An increasing function $F : [a, b] \to \mathbb{R}$ is differentiable almost everywhere on (a, b), and its derivative F' is measurable on (a, b).*

We shall prove the theorem under the additional assumption that F is continuous. The proof of this particular case is pretty much as the general one, but without the technical complications of the latter.

Let us begin with a few definitions and a preliminary lemma. If $x_0 \in (a, b)$, we denote by $D^+F(x_0)$ the upper right derivative of F at x_0, defined by

$$D^+F(x_0) = \limsup_{x \to x_0^+} \frac{F(x) - F(x_0)}{x - x_0}\,. \tag{9.158}$$

The lower right derivative $D_+F(x_0)$ is defined substituting in (9.158) the limit superior $\limsup_{x \to x_0^+}$ with $\liminf_{x \to x_0^+}$. In the obvious way one also defines the upper left derivative $D^-F(x_0)$ and the lower left derivative $D_-F(x_0)$.

Remark 2 By the definitions just given it follows that if F is continuous on $[a, b]$ (and not necessarily monotone) all of its upper and lower derivatives are measurable functions. For example,

$$D^+F(x) = \lim_{k\to+\infty} \sup\left\{\frac{F(q)-F(x)}{q-x} : \ x < q < x + \frac{1}{k}, \ q \in \mathbb{Q} \cap (a, b)\right\}. \tag{9.159}$$

Observe that if $q \in \mathbb{Q} \cap (a, b)$ the function

$$G_{k,q}(x) = \frac{F(q)-F(x)}{q-x}\chi_{\left(x,x+\frac{1}{k}\right)}(q) = \frac{F(q)-F(x)}{q-x}\chi_{\left(q-\frac{1}{k},q\right)}(x)\,,$$

for any $x \in (a, b)$, is measurable as product of two measurable functions. By (9.159) we have

$$D^+F(x) = \lim_{k\to+\infty} \sup_{q\in\mathbb{Q}\cap(a,b)} G_{k,q}(x)$$

so D^+F is measurable, being the limit of sequences of suprema of measurable functions.

Also note that if F is increasing, all right and left derivatives, both upper and lower, are non-negative.

Lemma 3 *Let $F : [a, b] \to \mathbb{R}$ be an increasing continuous function. Given real numbers p and q with $0 < p < q$, set*

$$E_{p,q} = \{x \in (a, b) : \ D_-F(x) < p, \ D^+F(x) > q\}. \tag{9.160}$$

For any interval $(\alpha, \beta) \subset (a, b)$,

$$m\big(E_{p,q} \cap (\alpha, \beta)\big) \le \frac{p}{q}(\beta - \alpha)\,. \tag{9.161}$$

Proof Pick $x_0 \in (\alpha, \beta)$ such that $D_-F(x_0) < p$. Then there exists $y \in (a, x_0)$ such that

$$\frac{F(y)-F(x_0)}{y-x_0} < p$$

and so

$$F(y) - py > F(x_0) - px_0\,.$$

The above says that $G(y) > G(x_0)$ for some $y < x_0$, where we have set $G(y) = F(y) - py$. Applying Lemma 2 to G, there are finitely many, or countably many at most, pairwise-disjoint intervals $(a_j, b_j) \subset (\alpha, \beta)$, with $j \in J \subset \mathbb{N}$, such that

$$\{x \in (\alpha, \beta) : \ D_- F(x) < p\} \subset \bigcup_{j \in J} (a_j, b_j)\,. \tag{9.162}$$

Moreover (9.156) implies that for any $j \in J$

$$F(a_j) - p\, a_j \geq F(b_j) - p\, b_j\,. \tag{9.163}$$

Similarly, using (9.155), one proves that for any $j \in J$ there exists a finite or at most countable number of pairwise-disjoint intervals $(a_{j,h}, b_{j,h}) \subset (a_j, b_j)$, with $h \in J_j \subset \mathbb{N}$, such that

$$\{x \in (a_j, b_j) : \ D^+ F(x) > q\} \subset \bigcup_{h \in J_j} (a_{j,h}, b_{j,h}), \tag{9.164}$$

$$F(a_{j,h}) - q\, a_{j,h} \leq F(b_{j,h}) - q\, b_{j,h} \qquad \forall h \in J_j \tag{9.165}$$

for any j. Hence, from inclusions (9.162), (9.164) and inequalities (9.163) and (9.165), we conclude

$$\begin{aligned} m\big(E_{p,q} \cap (\alpha, \beta)\big) &\leq \sum_{j \in J} \sum_{h \in J_j} (b_{j,h} - a_{j,h}) \leq \frac{1}{q} \sum_{j \in J} \sum_{h \in J_j} (F(b_{j,h}) - F(a_{j,h})) \leq \\ &\leq \frac{1}{q} \sum_{j \in J} (F(b_j) - F(a_j)) \leq \frac{p}{q} \sum_{j \in J} (b_j - a_j) \leq \frac{p}{q} (\beta - \alpha) \end{aligned}$$

because F is increasing. This gives (9.161).

Proof of Theorem 2 As already mentioned we shall assume F continuous. We claim that

$$D^+ F(x) < \infty \qquad \text{for a.e. } x \in (a, b)\,. \tag{9.166}$$

Exactly as in Lemma 3 (see (9.164) and (9.165)) one proves that for any $q > 0$ there exists a finite or at most countable number of pairwise-disjoint intervals $(a_j, b_j) \subset (a, b)$, $j \in J \subset \mathbb{N}$, such that

$$\{x \in (a, b) : \ D^+ F(x) > q\} \subset \bigcup_{j \in J} (a_j, b_j),$$

$$F(a_j) - q\, a_j \leq F(b_j) - q\, b_j \qquad \forall j \in J\,.$$

Hence

$$m\big(\{x \in (a,b) : \ D^+F(x) > q\}\big) \le \sum_{j\in J}(b_j - a_j) \le$$
$$\le \frac{1}{q}\sum_{j\in J}(F(b_j) - F(a_j)) \le$$
$$\le \frac{1}{q}(F(b) - F(a))\,,$$

and letting q go to $+\infty$ we find

$$m\big(\{x \in (a,b) : \ D^+F(x) = +\infty\}\big) = 0$$

whence (9.166). Now we prove

$$D_-F(x) \ge D^+F(x) \qquad \text{for a.e. } x \in (a,b)\,. \tag{9.167}$$

As $D_-F \ge 0$, we have

$$\{x \in (a,b) : \ D_-F(x) < D^+F(x)\} = \bigcup_{q,p\in\mathbb{Q},\, 0<p<q} E_{p,q}\,,$$

where $E_{p,q}$ is defined as in (9.160). To achieve (9.167) it suffices to prove that for any $0 < p < q$

$$m(E_{p,q}) = 0\,. \tag{9.168}$$

Given $\varepsilon > 0$, let A be an open set such that $E_{p,q} \subset A \subset (a,b)$ and

$$m(A) < m(E_{p,q}) + \varepsilon\,. \tag{9.169}$$

Such an open set always exists by the definition of a bounded set's Lebesgue measure. Let now (α_j, β_j) be a finite or countable collection of pairwise-disjoint intervals with union A. Applying Lemma 3 to these intervals, from (9.169) we have

$$m(E_{p,q}) \le \sum_j m\big(E_{p,q} \cap (\alpha_j, \beta_j)\big) \le \sum_j \frac{p}{q}(\beta_j - \alpha_j) =$$
$$= \frac{p}{q}m(A) \le \frac{p}{q}(m(E_{p,q}) + \varepsilon)\,.$$

Letting ε tend to zero in this inequality, and recalling $0 < p < q$, produces (9.168) immediately, and therefore (9.167).

To finish the proof set $F^*(x) = -F(-x)$ for any $x \in [-b, -a]$. Then F^* is continuous and increasing, so applying (9.167) to F^* gives

$$D_- F^*(x) \geq D^+ F^*(x) \qquad \text{for a.e. } x \in (-b, -a)\,.$$

But this relation is the same as saying

$$D_+ F(x) \geq D^- F(x) \qquad \text{for a.e. } x \in (a, b)\,.$$

From the above inequality, together with (9.166), (9.167) and the fact F is increasing, we deduce that

$$0 \leq D^+ F(x) \leq D_- F(x) \leq D^- F(x) \leq D_+ F(x) \leq D^+ F(x) < +\infty$$

for almost every $x \in (a, b)$. Therefore F is differentiable almost everywhere on (a, b). The measurability of F' follows from the fact that $F' = D^+ F$ almost everywhere on (a, b), and this is, as noticed above, measurable.

We wrap up this survey on monotone functions with the following remarkable result.

Theorem 3 *Let $F : [a, b] \to \mathbb{R}$ be increasing. Then*

$$\int_a^b F'(x)\,dx \leq F(b) - F(a), \tag{9.170}$$

so F' is summable.

Proof Let us extend F to $[a, b+1]$ by mapping the interval $[b, b+1]$ to the value $F(b)$. The extension, which we still call F, is measurable since increasing. Let F_k be the function on $[a, b]$:

$$F_k(x) = \frac{F\left(x + \dfrac{1}{k}\right) - F(x)}{\dfrac{1}{k}}\,.$$

Then

$$\lim_{k \to \infty} F_k(x) = F'(x) \tag{9.171}$$

almost everywhere on $[a, b]$. As F is increasing and hence $F'(x) \geq 0$ a.e. on $[a, b]$, we consider the integral of F' on $[a, b]$. By (9.171), the Fatou lemma (Sect. 9.6) implies

$$\int_a^b F'(x)\,dx \leq \sup_k \int_a^b F_k(x)\,dx = \sup_k k \int_a^b \left[F\left(x + \frac{1}{k}\right) - F(x) \right] dx\,.$$

Since the function $x \to F\left(x + \frac{1}{k}\right)$ is summable,

$$\int_a^b \left[F\left(x + \frac{1}{k}\right) - F(x)\right] dx = \int_b^{b+1/k} F(x)\, dx - \int_a^{a+1/k} F(x)\, dx\,.$$

On the other hand $F(x) = F(b)$ when $b \le x \le b + \frac{1}{k}$, and $F(a) \le F(x)$ for $a \le x \le \frac{1}{k}$, so

$$\int_a^b \left[F\left(x + \frac{1}{k}\right) - F(x)\right] dx \le \frac{1}{k}[F(b) - F(a)]\,.$$

Now (9.170) follows from the above inequalities.

Remark 3 The fundamental formula of integral calculus

$$\int_a^b F'(t)\, dt = F(b) - F(a),$$

where F is continuous with Riemann integrable derivative F', is not valid in the more general setting of monotone functions F: these typically only satisfy inequality (9.170).

Example 2, Sect. 9.12, defines an increasing continuous function $F : [0, 1] \to [0, 1]$, with almost everywhere null derivative and such that $F(0) = 0$ and $F(1) = 1$, for which inequality (9.170) is strict.

9.11 Functions with Bounded Variation

In the previous section we stressed the importance of the class of functions F that can be written as *difference of increasing functions*. The aim of this section is to describe this class precisely, by means of functions with *bounded variation*.

Let $F : [a, b] \to \mathbb{R}$ be any function. Consider a subdivision Π of the interval $[a, b]$ by points

$$x_0 = a < x_1 < \ldots < x_k = b\,. \tag{9.172}$$

We define

$$v_F(\Pi) = \sum_{j=1}^{k} |F(x_j) - F(x_{j-1})|$$

and call *total variation* of F on $[a, b]$ the supremum

$$V_F[a, b] = \sup_{\Pi} v_F(\Pi)$$

over all possible subdivisions Π.

Definition 1 (Functions with Bounded Variation) A function $F : [a, b] \to \mathbb{R}$ is said to have bounded variation if

$$V_F[a, b] < \infty .$$

Proposition 1 *Every monotone function F has bounded variation, and*

$$V_F[a, b] = |F(b) - F(a)| . \tag{9.173}$$

Proof If F is increasing and Π is given by the points (9.172),

$$v_F(\Pi) = \sum_{j=1}^{k} [F(x_j) - F(x_{j-1})] = F(b) - F(a)$$

so

$$V_F[a, b] = F(b) - F(a) .$$

If F is decreasing then

$$V_F[a, b] = -[F(b) - F(a)]$$

whence (9.173) again follows.

Proposition 2 *If $F : [a, b] \to \mathbb{R}$ is Lipschitz, i.e. there exists $L > 0$ such that*

$$|F(x_1) - F(x_2)| \le L|x_1 - x_2|$$

for all $x_1, x_2 \in [a, b]$, then F has bounded variation and

$$V_F[a, b] \le L(b - a) . \tag{9.174}$$

Proof It is enough to note that given x_j as in (9.172),

$$|F(x_j) - F(x_{j-1})| \le L(x_j - x_{j-1})$$

for any $j = 1, 2, \ldots, k$. Summing over j gives (9.174).

Proposition 3 *If F has bounded variation then it is bounded.*

Proof Given $x \in (a, b)$, using the subdivision Π of $[a, b]$ determined by a, x, b shows that

$$|F(x) - F(a)| + |F(b) - F(x)| \leq V_F[a, b],$$

and so

$$|F(x)| \leq |F(a)| + V_F[a, b]\,.$$

Example 1 Here is a function defined and continuous on $[0, 2/\pi]$, hence bounded, whose variation is not bounded:

$$F(x) = x \sin \frac{1}{x} \qquad x \neq 0$$

and $F(0) = 0$.

For $k \in \mathbb{N}$

$$\left| F\left(\frac{1}{(k + 1/2)\pi}\right) - F\left(\frac{1}{(k + 1 + 1/2)\pi}\right)\right| =$$

$$= \frac{1}{(k + 1/2)\pi} + \frac{1}{(k + 3/2)\pi} > \frac{1}{(k + 1)\pi}$$

so, for any $n \in \mathbb{N}$,

$$V_F[0, 2/\pi] > \sum_{k=0}^{n-1} \frac{1}{(k + 1)\pi} = \frac{1}{\pi} \sum_{k=1}^{n} \frac{1}{k}\,.$$

Therefore F does not have bounded variation.

The following is easy to prove:

Proposition 4 *Sums, differences and products of two functions with bounded variation on $[a, b]$ have bounded variation on $[a, b]$.*

Proof We will only prove that the product

$$P(x) = F(x)G(x)$$

has bounded variation if $F(x)$ and $G(x)$ both have bounded variation. Let

$$A = \sup_{x\in[a,b]} |F(x)|\,, \qquad B = \sup_{x\in[a,b]} |G(x)|\,,$$

so

$$\begin{aligned} P(x_{j+1}) - P(x_j) &\le |F(x_{j+1})G(x_{j+1}) - F(x_j)G(x_{j+1})| + \\ &\quad + |F(x_j)G(x_{j+1}) - F(x_j)G(x_j)| \le \\ &\le B|F(x_{j+1}) - F(x_j)| + A|G(x_{j+1}) - G(x_j)|. \end{aligned}$$

Evidently then

$$V_P[a,b] \le BV_F[a,b] + AV_G[a,b]$$

and P has bounded variation.

Proposition 5 *If $F : [a,b] \to \mathbb{R}$ has bounded variation, for any $c \in (a,b)$*

$$V_F[a,b] = V_F[a,c] + V_F[c,b]\,.$$

Proof Let Π be a subdivision of $[a,b]$ as in (9.172), and let $h \in \{1,\dots,k\}$ be such that $x_{h-1} < c \le x_h$. Then

$$\begin{aligned} &\sum_{j=1}^{k} |F(x_j) - F(x_{j-1})| = \sum_{j\neq h} |F(x_j) - F(x_{j-1})| + |F(x_h) - F(x_{h-1})| \le \\ &\le \sum_{j\neq h} |F(x_j) - F(x_{j-1})| + |F(c) - F(x_{h-1})| + |F(x_h) - F(c)| = \\ &= \left(\sum_{j<h-1} |F(x_j) - F(x_{j-1})| + |F(c) - F(x_{h-1})| \right) + \\ &\quad + \left(|F(x_h) - F(c)| + \sum_{j>h} |F(x_j) - F(x_{j-1})| \right) \le \\ &\le V_F[a,c] + V_F[c,b]\,, \end{aligned}$$

and since Π is arbitrary,

$$V_F[a,b] \le V_F[a,c] + V_F[c,b]\,. \tag{9.175}$$

For the opposite inequality, given $\varepsilon > 0$ choose $x_0 = a < x_1 < \cdots < x_k = c$ to divide $[a, c]$ and $y_0 = c < y_1 < \cdots < y_h = b$ to divide $[c, b]$, in such a way that

$$V_F[a, c] < \sum_{i=1}^{k} |F(x_i) - F(x_{i-1})| + \varepsilon, \quad V_F[c, b] < \sum_{j=1}^{h} |F(y_j) - F(y_{j-1})| + \varepsilon .$$

Adding these inequalities gives

$$V_F[a, c] + V_F[c, b] < \sum_{i=1}^{k} |F(x_i) - F(x_{i-1})| + \\ + \sum_{j=1}^{h} |F(y_j) - F(y_{j-1})| + 2\varepsilon \leq V_F[a, b] + 2\varepsilon ,$$

and as ε goes a zero, we find the opposite inequality to (9.175), whence the assertion.

Proposition 6 *If $F : [a, b] \to \mathbb{R}$ has bounded variation, the function*

$$V : x \in [a, b] \to V_F[a, x]$$

is increasing.

Proof Take $x_1 < x_2$, so by Proposition 5

$$V_F[a, x_2] = V_F[a, x_1] + V_F[x_1, x_2] \geq V_F[a, x_1]$$

since $V_F[x_1, x_2] \geq 0$.

Remark 1 Proposition 5 easily implies that if the points c_j with

$$a = c_0 < c_1 < \ldots < c_r < c_{r+1} = b$$

determine a subdivision of $[a, b]$ and on each interval $[c_j, c_{j+1}]$ a function $F : [a, b] \to \mathbb{R}$ has bounded variation, the same F has bounded variation on $[a, b]$.

Let $BV[a, b]$ be the set of functions with bounded variation on $[a, b]$. One shows that the sum and product of functions F, G in $BV[a, b]$ belongs in $BV[a, b]$. For $\alpha \in \mathbb{R}$, moreover,

$$V_{\alpha F}[a, b] = |\alpha| V_F[a, b]$$

and

$$V_{F+G}[a,b] \le V_F[a,b] + V_G[a,b]\,.$$

One of the main results on functions with bounded variation is

Theorem 1 (Jordan) *$F \in BV[a,b]$ if and only if F can be written as difference of two increasing functions.*

Proof For $x \in [a,b]$ define

$$V(x) = V_F[a,x]$$

and $V(a) = 0$. Proposition 5 tells that V is increasing. Then

$$W(x) = V(x) - F(x) \tag{9.176}$$

is increasing too, because by Proposition 5 if $x < y$ then

$$W(y) = V(y) - F(y) = V(x) + V_F[x,y] - F(y)$$

and so

$$W(y) - W(x) = V_F[x,y] - [F(y) - F(x)]\,.$$

On the other hand, by definition we have

$$F(y) - F(x) \le V_F[x,y]$$

and so $W(y) - W(x) \ge 0$. Hence $F(x)$, by (9.176), is the difference of the increasing functions V and W:

$$F(x) = V(x) - W(x)\,.$$

Due to Theorem 1, functions with bounded variation inherit several of the properties enjoyed by increasing functions (Propositions 1 and 2, Theorem 2 in Sect. 9.10).

Corollary 1 *The discontinuity set of a function with bounded variation is at most countable.*

Corollary 2 *A function with bounded variation on $[a,b]$ is almost everywhere differentiable, and its derivative F' is summable over $[a,b]$.*

What is more, let us define the jump function of a function with bounded variation F according to (9.150). Then Proposition 2, Sect. 9.10, implies

Proposition 7 *A function F with bounded variation is the sum of its jump function and a **continuous** function with bounded variation.*

Theorem 2 *A function $F \in BV[a,b]$ is continuous at x_0 if and only if the function*

$$V : x \in [a,b] \to V_F[a,x] \tag{9.177}$$

is continuous at x_0.

Proof Take $x_0 < b$ and let us prove V is right continuous at x_0. Given $\varepsilon > 0$ pick $\delta > 0$ such that

$$|F(x) - F(x_0)| < \varepsilon \tag{9.178}$$

for every $x \in (a,b]$ satisfying $0 < x - x_0 < \delta$. Let Π be a subdivision of $[x_0,b]$ by points

$$x_0 < x_1 < \ldots < x_k = b,$$

so that

$$v_F(\Pi) = \sum_{i=1}^{k} |F(x_i) - F(x_{i-1})| > V_F[x_0,b] - \varepsilon\,.$$

We may assume $x_1 - x_0 < \delta$ because the sum in $v_F(\Pi)$ increases when we add new subdivision points. Therefore

$$V_F[x_0,b] < \varepsilon + |F(x_1) - F(x_0)| + \sum_{i=2}^{k} |F(x_i) - F(x_{i-1})| \le 2\varepsilon + V_F[x_1,b]$$

and so

$$V_F[x_0,x_1] < 2\varepsilon,$$

i.e.

$$V(x_0^+) - V(x_0) \le V(x_1) - V(x_0) < 2\varepsilon\,.$$

This implies

$$V(x_0^+) = V(x_0),$$

in other words the claim.

As for the left continuity at $x_0 > a$, given $\varepsilon > 0$ choose $\delta > 0$ such that (9.178) holds for $x \in [a, b)$ with $0 < x_0 - x < \delta$. Let Π be the subdivision of $[a, x_0]$ given by points

$$a = x_n < x_{n-1} < \cdots < x_1 < x_0,$$

so that

$$V_F[a, x_0] < \sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| + \varepsilon\,.$$

As before, we may suppose $x_0 - x_1 < \delta$ without loss of generality. Then

$$V_F[a, x_0] < \sum_{i=2}^{n} |F(x_i) - F(x_{i-1})| + |F(x_1) - F(x_0)| + \varepsilon < V_F[a, x_1] + 2\varepsilon\,.$$

Letting x_1 tend to x_0 we find

$$V(x_0) \le V(x_0^-) + 2\varepsilon\,,$$

so now for $\varepsilon \to 0$ we obtain

$$V(x_0) \le V(x_0^-) \le V(x_0),$$

whence the assertion follows.

Conversely, if V is continuous at x_0, the continuity of F at x_0 follows by observing that

$$|F(x) - F(x_0)| \le V_F[x_0, x] = V(x) - V(x_0) \tag{9.179}$$

if $x > x_0$, and similarly, if $x < x_0$,

$$|F(x_0) - F(x)| \le V(x_0) - V(x)\,.$$

Corollary 3 *A continuous function with bounded variation can be written as difference of two continuous, increasing functions.*

We conclude our review of the elementary properties of functions with bounded variation with the following

Proposition 8 *If $F \in BV[a,b]$ then*

$$|F'(x)| \le V'(x) \qquad \text{for a.e. } x \in (a,b)\,, \tag{9.180}$$

where V is total variation function defined in (9.177)*.*

Proof Pick $x_0 \in (a,b)$ where both F and V are differentiable. If $x > x_0$, immediately

$$\left|\frac{F(x)-F(x_0)}{x-x_0}\right| \le \frac{V(x)-V(x_0)}{x-x_0}\,,$$

see (9.179). Then (9.180) follows from the above inequality in the limit as $x \to x_0^+$.

One of the reasons why continuous functions with bounded variation $f : [a,b] \to \mathbb{R}$ are relevant is that they are precisely the functions whose graph

$$G_f = \{(x,y) \in [a,b] \times \mathbb{R} : \quad y = f(x)\}$$

is a *rectifiable* simple curve in the plane (see Chap 6). This term denotes a curve whose finite length $l(G_f)$ satisfies

$$l(G_f) = \sup_{\mathcal{P}} l(\mathcal{P})\,,$$

where $\mathcal{P}$ ranges over all possible polygonal paths inscribed in the parametric curve G_f of equation

$$\varphi(t) = (\varphi_1(t), \varphi_2(t)) = (t, f(t)) \quad \text{for} \quad t \in [a,b]\,.$$

Namely, we have the remarkable result:

Theorem 3 (Jordan) *A simple (open or closed) curve $\Gamma \subset \mathbb{R}^2$ is rectifiable if and only if the coordinates (φ_1, φ_2) of any parametrisation $\varphi : I = [a,b] \to \Gamma$ have bounded variation.*

Precisely, for $j = 1, 2$

$$V_{\varphi_j}(I) \le L(\varphi) \le V_{\varphi_1}(I) + V_{\varphi_2}(I)\,, \tag{9.181}$$

where $L(\varphi)$ is the curve's length, defined as in (6.9)*.*

Proof Consider a subdivision of $[a, b]$

$$a = t_0 < t_1 < t_2 < \ldots < t_n = b\,,$$

and call $\mathcal{P}$ the polygonal path determined by the points $\varphi(a), \varphi(t_1), \ldots, \varphi(b)$.

Evidently

$$\begin{aligned} l(\mathcal{P}) &= \sum_{i=1}^{n-1} \left[|\varphi_1(t_i) - \varphi_1(t_{i+1})|^2 + |\varphi_2(t_i) - \varphi_2(t_{i+1})|^2 \right]^{1/2} \leq \\ &\leq \sum_{i=1}^{n-1} \left[|\varphi_1(t_i) - \varphi_1(t_{i+1})| + |\varphi_2(t_i) - \varphi_2(t_{i+1})| \right] = \\ &= \sum_{i=1}^{n-1} |\varphi_1(t_i) - \varphi_1(t_{i+1})| + \sum_{i=1}^{n-1} |\varphi_2(t_i) - \varphi_2(t_{i+1})| \leq \\ &\leq V_{\varphi_1}[a, b] + V_{\varphi_2}[a, b]\,. \end{aligned} \tag{9.182}$$

The second inequality in (9.181) now follows from (9.182) due to $\mathcal{P}$ being arbitrary.

To prove the first inequality in (9.181) it suffices to recall the obvious relationship

$$|\varphi_j(t_i) - \varphi_j(t_{i+1})| \leq \left[|\varphi_1(t_i) - \varphi_1(t_{i+1})|^2 + |\varphi_2(t_i) - \varphi_2(t_{i+1})|^2 \right]^{1/2},$$

which implies for $j = 1, 2$

$$\sum_{i=1}^{n-1} |\varphi_j(t_i) - \varphi_j(t_{i+1})| \leq l(\mathcal{P}) \leq L(\varphi)\,.$$

9.12 Absolutely Continuous Functions

A function $F : [a, b] \to \mathbb{R}$ is absolutely continuous if $\forall\, \varepsilon > 0, \exists\, \delta > 0$ such that

$$\sum_{j=1}^{k} |F(b_j) - F(a_j)| < \varepsilon$$

for any family of pairwise-disjoint open intervals $(a_j, b_j) \subset [a, b]$, $j = 1, \ldots, k$ such that

$$\sum_{j=1}^{k} (b_j - a_j) < \delta\,. \tag{9.183}$$

Evidently a function that is absolutely continuous on $[a, b]$ is necessarily (uniformly) continuous on $[a, b]$, since we can take $k = 1$. We claim that in the definition we can replace the variation $|F(b_j) - F(a_j)|$ with the oscillation

$$\operatorname{osc}_F[a_j, b_j] = M_j - m_j.$$

Above,

$$M_j = \max_{[a_j, b_j]} F = F(\beta_j)$$

and

$$m_j = \min_{[a_j, b_j]} F = F(\alpha_j), \tag{9.184}$$

where $\alpha_j, \beta_j \in [a_j, b_j]$ are minimum and maximum points of F, respectively, on $[a_j, b_j]$.

It is enough to note that the sum of the lengths of the intervals (α_j, β_j) does not exceed the sum of the lengths of the $[a_j, b_j]$, so

$$\sum_{j=1} |F(\beta_j) - F(\alpha_j)| < \varepsilon\,. \tag{9.185}$$

Therefore if F is absolutely continuous on $[a, b]$, for any $\varepsilon > 0$, $\exists\, \delta > 0$ such that, for any (finite or countable) subdivision of $[a, b]$ by pairwise-disjoint intervals (a_j, b_j) satisfying

$$\sum_j (b_j - a_j) < \delta,$$

we have

$$\sum_j (M_j - m_j) < \varepsilon\,.$$

Evidently a Lipschitz function F is also absolutely continuous on $[a, b]$:

$$\operatorname{Lip}[a, b] \subset AC[a, b]$$

where $AC[a, b]$ indicates the set of absolutely continuous functions on $[a, b]$. It is easy to see that sums and products of absolutely continuous functions are absolutely continuous.

The following proposition proves the inclusion

$$AC[a,b] \subset BV[a,b],$$

where $BV[a,b]$ is the set of functions with bounded variation.

Proposition 1 *An absolutely continuous function* $F : [a,b] \to \mathbb{R}$ *has bounded variation.*

Proof Take $F \in AC[a,b]$ and let $\delta > 0$ be such that

$$\sum_{j=1}^{k} |F(b_j) - F(a_j)| < 1\,.$$

for any finite set of pairwise-disjoint intervals $(a_j, b_j) \subset [a,b]$ satisfying (9.183).

Choose points

$$\alpha = c_0 < c_1 < \ldots < c_h = b$$

in $[a,b]$ such that

$$c_{j+1} - c_j < \delta \qquad j = 0, 1, \ldots, h-1\,.$$

Then for any subdivision of $[c_j, c_{j+1}]$ the sum of the variations of F will be less than 1. Hence

$$V_F[c_j, c_{j+1}] \leq 1$$

and so

$$V_F[a,b] \leq h,$$

implying $F \in BV[a,b]$.

Using Proposition 1 we find, as consequence of Corollary 2, Sect. 9.11, the following result.

Corollary 1 *If* F *is absolutely continuous on* $[a,b]$*, its derivative exists almost everywhere and is summable on* $[a,b]$.

Now we discuss an example of function F that is continuous on $[a,b]$, differentiable on (a,b), but whose derivative is not summable. Naturally F is not absolutely continuous on $[a,b]$, nor with bounded variation.

Example 1 For $x \in [0, 1]$ define

$$F(x) = x^2 \cos(\pi/x^2) \quad \text{for} \quad x \neq 0$$

and $F(0) = 0$. Then $F'(x)$ is finite at any point in $[a, b]$ but not summable. If $0 < \alpha < \beta < 1$, $F'(x)$ is bounded on $[\alpha, \beta]$ and

$$\int_\alpha^\beta F'(t)\,dt = \beta^2 \cos\frac{\pi}{\beta^2} - \alpha^2 \cos\frac{\pi}{\alpha^2}$$

In particular, if $\alpha_k = \sqrt{2/(4k+1)}$, $\beta_k = 1/\sqrt{2k}$,

$$\int_{\alpha_k}^{\beta_k} F'(t)\,dt = \frac{1}{2k}$$

and so

$$\int_{\bigcup_{k\in\mathbb{N}} [\alpha_k, \beta_k]} |F'(t)|\,dt \geq \sum_{k=1}^{\infty} \frac{1}{2k} = +\infty\,.$$

Proceeding in the study of the *properties of the total variation function*

$$V : x \in [a, b] \to V_F[a, x], \tag{9.186}$$

we will see in the next section that for $F \in BV[a, b]$

$$\int_a^b |F'(t)|\,dt \leq V_F[a, b],$$

while for $F \in AC[a, b]$, and only these,

$$\int_a^b |F'(t)|\,dt = V_F[a, b]\,.$$

We have proved that if $F \in BV[a, b]$ the function V in (9.186) is increasing (Proposition 6, Sect. 9.11) and continuous at $x_0 \in [a, b]$ if and only if F is continuous at x_0 (Theorem 2, Sect. 9.11). Furthermore, still for $F \in BV[a, b]$ we have V differentiable almost everywhere and (see the previous section "Functions with Bounded Variation", Proposition 8)

$$|F'(x)| \leq V'(x).$$

The next result is noteworthy.

Proposition 2 *Let $F : [a, b] \to \mathbb{R}$ be a function. Then the function V in (9.186) is absolutely continuous on $[a, b]$ if and only if $F \in AC[a, b]$.*

Proof We will only prove that $F \in AC[a, b]$ implies $V \in AC[a, b]$. The converse is immediate once we note that for any interval $[\alpha, \beta] \subset [a, b]$

$$|F(\beta) - F(\alpha)| \leq V(\beta) - V(\alpha)\,.$$

Given $\varepsilon > 0$, choose $\delta > 0$ so that

$$\sum_{j=1}^{n} (b_j - a_j) < \delta \quad \Rightarrow \quad \sum_{j=1}^{n} |F(b_j) - F(a_j)| < \varepsilon,$$

in agreement with the definition of absolute continuity of F.

Consider the sum

$$\sum_{j=1}^{n} |V(b_j) - V(a_j)|\,. \tag{9.187}$$

It represents the supremum of the sums

$$\sum_{k=1}^{n} \sum_{j=1}^{m_k} |F(x_{k,j}) - F(x_{k,j-1})| \tag{9.188}$$

over all finite subdivisions

$$\begin{aligned} a_1 = x_{1,0} &< x_{1,1} \ldots < x_{1,m_1} = b_1 \\ a_2 = x_{2,0} &< x_{2,1} \ldots < x_{2,m_2} = b_2 \\ &\cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \\ a_n = x_{n,0} &< x_{n,1} \ldots < x_{n,m_n} = b_n \end{aligned}$$

of the intervals $(a_1, b_1), \ldots, (a_n, b_n)$. Since the sum of the lengths of the $(x_{k,j-1}, x_{k,j})$, over which we sum in (9.188), is less than δ, all these sums are less than or equal to ε. Therefore (9.187), being the least upper bound, does not exceed ε.

Proposition 3 *If F is absolutely continuous, there exist two absolutely continuous and increasing functions F_1, F_2 such that*

$$F = F_1 - F_2\,.$$

Proof Recalling the Jordan theorem (Theorem 1, Sect. 9.11) whereby

$$F_1 = V \qquad F_2 = V - F,$$

the claim now follows from Proposition 2.

Very important for AC functions is the following result.

Theorem 1 *If F is absolutely continuous on $[a, b]$ and its derivative satisfies*

$$F'(x) = 0 \qquad a.e.\ on\ (a, b),$$

then F is constant.

Proof To prove the theorem it is enough to show that $F(a) = F(b)$, because we can then apply that to any subinterval $[a', b'] \subset [a, b]$.

Defining

$$X = \{x \in (a, b) : F'(x) = 0\},$$

we are supposing $m(X) = b - a$. Given $\varepsilon > 0$, since

$$\lim_{h \to 0} \frac{F(x + h) - F(x)}{h} = 0$$

for $x \in X$, we know that for any $\eta > 0$ there exists $I_x = (a_x, b_x) \subset (a, b)$, with $x \in I_x$, such that

$$|F(b_x) - F(a_x)| \leq \varepsilon (b_x - a_x)$$

and

$$(b_x - a_x) < \eta\,.$$

As $x \in X$ varies, the family of intervals (a_x, b_x) is a Vitali covering of X (Definition 1, Sect. 9.10). Hence by Remark 1, Sect. 9.10, for any $\delta > 0$ there is a finite disjoint subfamily of intervals $I_j = (a_j, b_j)$, $1 < j \leq n$, such that

$$\sum_{j=1}^{n} m(I_j) \geq m(X) - \delta = b - a - \delta\,. \tag{9.189}$$

For $j = 1, \dots, n$

$$|F(b_j) - F(a_j)| \le \varepsilon(b_j - a_j)$$

and adding we obtain

$$\sum_{j=1}^{n} |F(b_j) - F(a_j)| \le \varepsilon(b - a)\,.$$

Consider the set

$$[a, b] \setminus \bigcup_{j=1}^{n} I_j$$

consisting in the union

$$\bigcup_{k=1}^{r} [\alpha_k, \beta_k]$$

of total length $\le \delta$, due to (9.189). By the absolute continuity of F, if we choose δ suitably depending on ε, we have

$$\sum_{k=1}^{r} |F(\beta_k) - F(\alpha_k)| \le \varepsilon\,.$$

Therefore

$$|F(b) - F(a)| \le \sum_{i=1}^{n} |F(b_i) - F(a_i)| + \sum_{k=1}^{r} |F(\beta_k) - F(\alpha_k)| \le \varepsilon(b - a) + \varepsilon\,.$$

But ε is arbitrary, so $F(b) - F(a) = 0$.

The following definition, due to N. N. Lusin, is of great interest.

Definition (Lusin's (N) Property) *A continuous function $F : [a, b] \subset \mathbb{R} \to \mathbb{R}$ satisfies Lusin's (N) property if every zero-measure set N_0 has image $F(N_0)$ of measure zero.*

In this respect, the following result holds.

Theorem 2 (Banach) *A continuous function F with bounded variation is absolutely continuous if and only if it satisfies the (N) property:*

$$m(N_0) = 0 \quad \Rightarrow \quad m(F(N_0)) = 0\,. \tag{9.190}$$

Proof For simplicity we shall only prove that any $F \in AC[a,b]$ satisfies the (N) property. Take $N_0 \subset (a,b)$ with $m(N_0) = 0$. Given $\varepsilon > 0$, pick $\delta > 0$ so that for any subdivision by pairwise-disjoint intervals (a_j, b_j) such that

$$\sum_j (b_j - a_j) < \delta$$

we have

$$\sum_j (M_j - m_j) < \varepsilon$$

with m_j and M_j as in (9.184) and (9.185).

As $m(N_0) = 0$ there is an open set A such that

$$N_0 \subset A \subset (a,b)\,, \qquad m(A) < \delta\,.$$

Evidently A is a union of intervals (a_j, b_j) whose lengths add up to less than δ. This implies

$$F(N_0) \subset F(A) \subseteq \bigcup_j F\left([a_j, b_j]\right)$$

and the outer measure of $F(N_0)$ can be bounded as follows

$$m_e(F(N_0)) \leq \sum_j m_e\left(F\left([a_j, b_j]\right)\right).$$

Recalling that $F\left([a_j, b_j]\right) = [m_j, M_j]$, we then have

$$m_e(F(N_0)) \leq \sum_j (M_j - m_j) \leq \varepsilon\,.$$

But ε is arbitrary, so (9.190) follows.

Another sufficient condition warranting Lusin's property is given in the following proposition.

Proposition 4 *Let $F : [a,b] \to \mathbb{R}$ be continuous and differentiable at **every** point in (a,b). Then F has the (N) property.*

Proof Suppose $m(N_0) = 0$. For $j, k \in \mathbb{N}$ let $N_{j,k}$ be the set of points $x \in N_0$ such that, for any interval I containing x and of length less than $1/k$, we have

$$m(F(I)) \leq j\, m(I)\,.$$

Since F is differentiable everywhere on (a, b),

$$N_0 = \bigcup_{j,k=1}^{\infty} N_{j,k}$$

so it will suffice to show

$$m(F(N_{j,k})) = 0 \tag{9.191}$$

to prove the claim. Given $\varepsilon > 0$, $N_{j,k}$ is contained in the union of at most countably many intervals of length less than $1/k$ such that the lengths' sum does not exceed ε/j. Therefore

$$m(F(N_{j,k})) \leq \varepsilon$$

and, as ε is arbitrary, (9.191) follows. Therefore

$$m(F(N_0)) = 0\,.$$

Example 2 (The Cantor Function) Recall that the *Cantor set*, here indicated by K (but called K_3 in Sect. 9.4), is a subset of the interval [0, 1] with the following properties:

(i) K is closed, it has empty interior and is perfect, i.e. any point is a limit point;
(ii) K has the cardinality of the continuum;
(iii) $m(K) = 0$.

The Cantor set, and the Cantor function

$$C : [0, 1] \to [0, 1]$$

defined below, are important because they provide counter-intuitive examples of subsets of real numbers and functions of one real variable.

The Cantor function has the following properties:

(j) C is continuous, increasing on [0, 1], $C(0) = 0$, $C(1) = 1$;
(jj) C is constant on each interval "contiguous" to the Cantor set K;

(continued)

Example 2 (continued)

(jjj) C is not constant on any interval containing points of K;
(jv) its derivative is zero on $[0, 1] \setminus K$, hence $C'(x) = 0$ almost everywhere;
(v) C is not differentiable on an uncountable set of measure zero.

Furthermore C has bounded variation and it is Riemann integrable (as continuous) but not absolutely continuous.

We recall that the Cantor set K is constructed by removing from $[0, 1]$ the (open) middle third $(1/3, 2/3)$, then removing from each intervals in the union $[0, 1/3] \cup [2/3, 1]$ the (open) middle thirds $(1/9, 2/9)$ and $(7/9, 8/9)$, and so on.

What remains in $[0, 1]$ after removing all these open intervals is the Cantor set K. In other terms, if C_k is the closed union of the 2^k closed intervals (each of length 3^{-k}) remaining after the kth deletion, we have a decreasing sequence of sets $(C_k)_{k\in\mathbb{N}}$, and we can express the Cantor set as

$$K = \bigcap_{k\in\mathbb{N}} C_k = \bigcap_{k\in\mathbb{N}} \bigcup_{a_i\in\{0,1\}} \left[2\sum_{i=1}^{k} \frac{a_i}{3^i}\,,\ 2\sum_{i=1}^{k} \frac{a_i}{3^i} + \frac{1}{3^k} \right].$$

The set $A_k = [0, 1] \setminus C_k$ is the union of $2^k - 1$ intervals I_j^k (ordered from left to right) removed after k deletions in the construction of K. Let

$$c_k : [0, 1] \to [0, 1]$$

be the continuous function satisfying the following requirements: $c_k(0) = 0$, $c_k(1) = 1$,

$$c_k(x) = \frac{j}{2^k} \quad \text{on} \quad I_j^k$$

for $j = 1, 2, \ldots, 2^k - 1$, so that c is constant on the deleted intervals $(1/3, 2/3)$, $(1/9, 2/9)$ and $(7/9, 8/9)$, …, and c_k is linear on each interval of C_k. In Fig. 9.19 we have plotted the graph of $c_2(x)$.

Each c_k is increasing, and $c_{k+1} = c_k$ on I_k^j $(j = 1, 2, \ldots, 2^k - 1)$. Moreover

$$|c_k(x) - c_{k+1}(x)| < \frac{1}{2^k}\,.$$

Consequently the series $\sum_k (c_k - c_{k+1})$ is uniformly convergent on $[0, 1]$, so (c_k) converges uniformly on $[0, 1]$ to some function

$$C : [0, 1] \to [0, 1]$$

that is increasing, continuous, with $C(0) = 0$ and $C(1) = 1$, and constant on every open interval deleted from K.

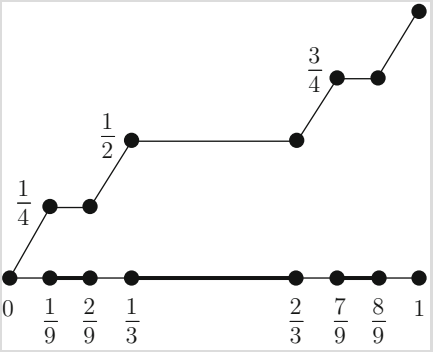
Note that $C(x)$ is constant on each interval in $[0, 1] \setminus K$, so

$$C'(x) = 0 \qquad \text{a.e. on} \quad [0, 1]\,.$$

(continued)

Example 2 (continued)

Fig. 9.19



9.13 The Indefinite Integral in Lebesgue's Theory

Let $f : [a, b] \to \mathbb{R}$ be a summable function. The function

$$G(x) = c + \int_a^x f(t)\, dt \qquad (c \in \mathbb{R}) \tag{9.192}$$

is the Lebesgue indefinite integral of f.

Using the absolute continuity of the Lebesgue (definite) integral (see (9.65)), we prove

Theorem 1 (Absolute Continuity of the Indefinite Integral) *If f is summable on $[a, b]$, the indefinite integral $G(x)$ is an absolutely continuous function.*

Proof Given $\varepsilon > 0$ (Example 8, Sect. 9.6), there is $\delta > 0$ such that

$$m(E) < \delta \quad \Rightarrow \quad \int_E |f(t)|\, dt < \varepsilon$$

for any measurable set $E \subset [a, b]$. If (a_j, b_j) is a collection of pairwise-disjoint open intervals contained in $[a, b]$ such that

$$\sum_j (b_j - a_j) < \delta$$

we will have

$$\sum_{j=1}^{k} \int_{a_j}^{b_j} |f(t)|\, dt < \varepsilon\,.$$

As

$$G(b_j) - G(a_j) = \int_{a_j}^{b_j} f(t)\,dt$$

it follows that

$$\sum_{j=1}^{k} |G(b_j) - G(a_j)| < \varepsilon,$$

making $G(x)$ absolutely continuous.

Therefore an indefinite integral is an absolutely continuous function. One of the main results proved here is the converse, namely: *every absolutely continuous function is an indefinite integral* (Theorem 3).

The previous theorem and Corollary 1 in Sect. 9.12 imply that the derivative G' of (9.192) is summable. This derivative is actually described by the following result.

Theorem 2 (Fundamental Theorem of Calculus) *If f is summable on $[a, b]$ and we define*

$$G(x) = \int_a^x f(t)\,dt,$$

then G is differentiable and

$$G'(x) = f(x) \qquad \text{a.e. on} \quad [a, b]\,. \tag{9.193}$$

Proof We claim that

$$G'(x) \le f(x) \qquad \text{a.e.}$$

Given two rational numbers $r < s$, let $X_{r,s}$ be the set of points $x \in (a, b)$ at which G is differentiable and such that

$$f(x) < r < s < G'(x)\,.$$

Clearly $X_{r,s}$ is measurable, since f and G' are. Let us show

$$m(X_{r,s}) = 0\,.$$

The set of rational pairs (r, s) is countable, so

$$m(X) = 0$$

where

$$X = \{x \in [a, b] : \ f(x) < G'(x)\}$$

because clearly

$$X = \bigcup_{(r,s)} X_{r,s}$$

with $r, s \in \mathbb{Q}$ and $r < s$.

Fix $\varepsilon > 0$ and take $\delta > 0$ so that

$$m(Y) < \delta \quad \Rightarrow \quad \left| \int_Y f(t)\, dt \right| < \varepsilon \,. \tag{9.194}$$

Let $A \subset [a, b]$ be an open subset such that

$$X_{r,s} \subset A \,, \qquad m(A) < m(X_{r,s}) + \delta \,. \tag{9.195}$$

If $x \in X_{r,s}$, when ξ is close enough to x, $\xi > x$, we have

$$\frac{G(\xi) - G(x)}{\xi - x} > s$$

i.e.

$$G(\xi) - s\xi > G(x) - sx \,.$$

From this, applying the Rising Sun Lemma (Lemma 2, Sect. 9.10) to $y \to G(y) - sy$, there exists a disjoint, at most countable collection of intervals (a_j, b_j), $j \in J \subset \mathbb{N}$, such that

$$X_{r,s} \subset B \subset A \qquad G(b_j) - sb_j \geq G(a_j) - sa_j$$

where $B = \bigcup_{j \in J} (a_j, b_j)$. In other words,

$$\int_{a_j}^{b_j} f(t)\, dt = G(b_j) - G(a_j) \geq s(b_j - a_j) \,.$$

Adding over j,

$$\int_B f(t)\,dt \geq s\; m(B)\,. \tag{9.196}$$

On the other hand, by (9.194) and (9.195) we obtain

$$\int_B f(t)\,dt = \int_{X_{r,s}} f(t)\,dt + \int_{B\setminus X_{r,s}} f(t)\,dt \leq \tag{9.197}$$
$$r\; m(X_{r,s}) + \varepsilon \leq r\; m(B) + \varepsilon\,.$$

From (9.196) and (9.197) we deduce

$$r\; m(B) + \varepsilon \geq s\; m(B)$$

and so

$$m(B) \leq \frac{\varepsilon}{s-r}\,.$$

Hence $X_{r,s}$ is contained in an open set of arbitrarily small measure, i.e. $m(X_{r,s}) = 0$. Eventually

$$G'(x) \leq f(x) \qquad \text{a.e.} \tag{9.198}$$

Replacing f with $-f$ one can similarly prove

$$-f(x) \geq -G'(x) \tag{9.199}$$

and so

$$f(x) \leq G'(x)\,.$$

Now (9.198) and (9.199) yield (9.193).

Theorem 3 *An absolutely continuous function F is an indefinite integral of its derivative, i.e.*

$$F(x) = F(a) + \int_a^x F'(t)\,dt\,.$$

Proof We know from Corollary 1, Sect. 9.12, that F is differentiable a.e. and F' is summable. Define

$$G(x) = F(a) + \int_a^x F'(t)\,dt.$$

Note G, too, is absolutely continuous by Theorem 2, and

$$G'(x) = F'(x) \qquad \text{a.e.}$$

By Theorem 1, Sect. 9.12, we deduce that

$$G(x) - F(x) = \text{ constant},$$

and since $G(a) - F(a) = 0$,

$$G(x) = F(x) \qquad \text{a.e.}$$

The following theorem is useful. It asserts that *the total variation of an absolutely continuous function is the integral of the derivative's absolute value.*

Theorem 4 *Let f be summable on $[a, b]$. If*

$$F(x) = \int_a^x f(t)\, dt$$

then

$$V_F[a, b] = \int_a^b |f(t)|\, dt\,. \tag{9.200}$$

Proof Let us begin by showing that the left-hand side is less than or equal to the right-hand side. Let $x_0 = a < x_1 < x_2 < \ldots < x_k = b$ be a subdivision of $[a, b]$, so

$$\sum_{j=1}^{k} |F(x_j) - F(x_{j-1})| = \sum_{j=1}^{k} \left| \int_{x_{j-1}}^{x_j} f(t)\, dt \right| \leq$$

$$\leq \sum_{j=1}^{k} \int_{x_{j-1}}^{x_j} |f(t)|\, dt = \int_a^b |f(t)|\, dt$$

and then

$$V_F[a, b] \leq \int_a^b |f(t)|\, dt\,. \tag{9.201}$$

For the opposite inequality fix $\varepsilon > 0$. By the integral's absolute continuity properties, see Example 8 in Sect. 9.6, there exists $\delta > 0$ such that

$$m(E) < \delta \quad \Longrightarrow \quad \int_E |f(t)|\,dt < \varepsilon \tag{9.202}$$

for any measurable set $E \subset [a,b]$. Define, for any integer h,

$$E_h^+ = \left\{x \in (a,b) :\ 0 < f(x) \le \frac{1}{h}\right\}, \quad E_h^- = \left\{x \in (a,b) :\ -\frac{1}{h} \le f(x) < 0\right\} \tag{9.203}$$

and observe

$$\bigcap_{h=1}^{\infty} E_h^+ = \emptyset, \quad \bigcap_{h=1}^{\infty} E_h^- = \emptyset\,.$$

Hence there is h such that

$$m(E_h^+) < \delta, \quad m(E_h^-) < \delta\,. \tag{9.204}$$

Finally let

$$A_h^+ = \left\{x \in (a,b) :\ f(x) > \frac{1}{h}\right\}, \quad A_h^- = \left\{x \in (a,b) :\ f(x) < -\frac{1}{h}\right\}.$$

The sets A_h^+, A_h^- are open, so we can find for each one a finite number of pairwise-disjoint intervals $(a_i,b_i) \subset A_h^+$, $i = 1,\dots,n$, and $(\alpha_j,\beta_j) \subset A_h^-$, $j = 1,\dots,m$, such that

$$m\left(A_h^+ \setminus \bigcup_{i=1}^{n}(a_i,b_i)\right) < \delta, \quad m\left(A_h^- \setminus \bigcup_{j=1}^{m}(\alpha_j,\beta_j)\right) < \delta \tag{9.205}$$

Now, (9.202), (9.203) and (9.204)–(9.205) easily imply

$$\begin{aligned}\int_a^b |f(t)|\,dt &= \int_{E_h^+} |f(t)|\,dt + \int_{E_h^-} |f(t)|\,dt + \int_{A_h^+} |f(t)|\,dt + \int_{A_h^-} |f(t)|\,dt \le \\ &\le 2\varepsilon + \sum_{i=1}^{n}\int_{a_i}^{b_i} f(t)\,dt + \varepsilon + \sum_{j=1}^{m}\int_{\alpha_j}^{\beta_j}(-f(t))\,dt + \varepsilon = \\ &= 4\varepsilon + \sum_{i=1}^{n}(F(b_i)-F(a_i)) + \sum_{j=1}^{m}(F(\alpha_j)-F(\beta_j)) \le V_F[a,b] + 4\varepsilon.\end{aligned}$$

As ε goes a zero, we recover the opposite inequality to (9.201), whence the claim.

In the sequel we shall assume that, beside having bounded variation, F is also *continuous* on $[a, b]$:

$$F \in BV[a, b] \cap C^0[a, b]\,. \tag{9.206}$$

We have the following result, saying that such an F decomposes as a sum of an absolutely continuous function and a *singular* function (i.e. a function with zero derivative almost everywhere).

Theorem 5 *Let F satisfy assumption (9.206). Then there exist $G \in AC[a, b]$ and R singular such that*

$$F(x) = G(x) + R(x), \tag{9.207}$$

and the decomposition is unique up to constants.

Proof As F' is well defined and summable, we set

$$\begin{aligned} G(x) &= F(a) + \int_a^x F'(t)\,dt \\ R(x) &= F(x) - G(x), \end{aligned} \tag{9.208}$$

thus obtaining decomposition (9.207) where: G is absolutely continuous (Theorem 1), while R has bounded variation, is continuous and satisfies

$$R'(x) = 0 \qquad \text{a.e.}$$

Let us prove the *uniqueness* of the representation. If there were another sum

$$F(x) = G_1(x) + R_1(x)$$

we would have

$$G(x) - G_1(x) = R_1(x) - R(x)$$

and so

$$D(G - G_1)(x) = 0 \qquad \text{a.e.}$$

But $G - G_1$ is absolutely continuous, so constant. Hence there is $c \in \mathbb{R}$ such that

$$G(x) = G_1(x) + c$$

implying

$$R(x) = R_1(x) - c\,.$$

If F is increasing, both G and R must be increasing. That is because $F' \geq 0$, so $G(x)$ defined in (9.208) is increasing, and by Theorem 4, Sect. 9.10,

$$\int_x^y F'(t)\,dt \leq F(y) - F(x)$$

for $x < y$. Consequently

$$G(y) - G(x) \leq F(y) - F(x)$$

and so

$$R(x) \leq R(y)\,.$$

Theorem 5 easily gives us the characterisation of absolutely continuous functions (Theorem 3) that are further assumed *increasing*.

Proposition 1 *A continuous, increasing function F is absolutely continuous if and only if*

$$\int_a^b F'(x)\,dx = F(b) - F(a)\,. \tag{9.209}$$

Proof By Theorem 3 is suffices to show that (9.209) implies $F \in AC[a, b]$.

By contradiction, suppose F is not absolutely continuous, and take G, R as in Theorem 5. Then

$$F(b) - F(a) = G(b) - G(a) + R(b) - R(a)$$

i.e.

$$F(b) - F(a) = \int_a^b f'(t)\,dt + R(b) - R(a)\,.$$

But R is increasing and non-constant, so $R(a) < R(b)$, which makes (9.209) impossible.

In Sect. 9.11 we saw (Proposition 7) that every function F with bounded variation can be expressed as a sum of its jump function S and a *continuous* function with

bounded variation. In the light of Theorem 5 we can now say that *any function F with bounded variation may be written as*

$$F(x) = F_a(x) + F_c(x) + F_s(x) \qquad (\textit{Lebesgue decomposition}) \quad (9.210)$$

with $F_a = G$, $F_c = R$, $F_s = S$ such that:

(i) F_a is *absolutely continuous*;
(ii) F_c is *continuous and singular*: $F_c'(x) = 0$ a.e. (sometimes F_c is called the "Cantor part" of F);
(iii) F_s is the *jump function*.

Furthermore, the summands are uniquely determined up to an additive constant.

Differentiating (9.210) gives

$$F'(x) = (F_a)'(x)\,, \qquad \text{a.e.}$$

since $(F_c)'$ and $(F_s)'$ vanish almost everywhere. Consequently, integrating the derivative of a function $F \in BV[a,b]$ does not give the function itself, but only its absolutely continuous part.

We close this section with another theorem characterising absolutely continuous functions.

Theorem 6 *Let F be a function on $[a,b]$ with bounded variation. Then*

$$\int_a^b |F'(x)|\,dx \le V_F[a,b]\,. \qquad (9.211)$$

Moreover F is absolutely continuous on $[a,b]$ if and only if there is equality in (9.211).

Proof If F has bounded variation, putting $V(x) = V_F[a,x]$ for $x \in [a,b]$, from Proposition 8, Sect. 9.11, we have

$$\int_a^b |F'(x)|\,dx \le \int_a^b V'(x)\,dx \le V(b) - V(a) = V_F[a,b]$$

and so (9.211) follows.

If F is absolutely continuous we take

$$G(x) = \int_a^x F'(t)\,dt$$

for any $x \in [a, b]$, so $V_F[a, x] = V_G[a, x]$ and by (9.200)

$$V_F[a, b] = V_G[a, b] = \int_a^b |F'(x)|\, dx\,.$$

Conversely, if in (9.211) we have equality, applying (9.211) to the intervals $[a, x]$ and $[x, b]$, for any $x \in [a, b]$, we find

$$\int_a^x |F'(t)|\, dt \le V_F[a, x] = V_F[a, b] - V_F[x, b] \le$$

$$\le \int_a^b |F'(t)|\, dt - \int_x^b |F'(t)|\, dt = \int_a^x |F'(t)|\, dt\,.$$

Therefore

$$V(x) = \int_a^x |F'(t)|\, dt \qquad \forall x \in [a, b]\,.$$

Then V is absolutely continuous, and by Proposition 2 in Sect. 9.12 also F is absolutely continuous.

Check for updates

Surfaces and Surface Integrals

10

10.1 Regular Surfaces

We recall that a *domain* in $\mathbb{R}^n$ is the closure of an open set and, by definition, a *connected domain* in $\mathbb{R}^n$ is the closure of a connected open set.
Let D be a connected domain in the plane. We call *parametrised regular surface* (see Fig. 10.1) a map $\varphi : D \to \mathbb{R}^3$ of class C^1 on D that satisfies the two conditions:

(i) the restriction of φ to $\overset{\circ}{D}$ is invertible;
(ii) for any (u, v) in $\overset{\circ}{D}$ the Jacobian matrix

$$D\varphi(u,v) = \frac{\partial(\varphi_1, \varphi_2, \varphi_3)}{\partial(u,v)} = \begin{pmatrix} \frac{\partial \varphi_1}{\partial u}(u,v) & \frac{\partial \varphi_1}{\partial v}(u,v) \\ \frac{\partial \varphi_2}{\partial u}(u,v) & \frac{\partial \varphi_2}{\partial v}(u,v) \\ \frac{\partial \varphi_3}{\partial u}(u,v) & \frac{\partial \varphi_3}{\partial v}(u,v) \end{pmatrix}$$

has *rank* 2.
Once we introduce the vectors

$$\varphi_u(u,v) = \frac{\partial \varphi}{\partial u}(u,v) = \left(\frac{\partial \varphi_1}{\partial u}(u,v), \frac{\partial \varphi_2}{\partial u}(u,v), \frac{\partial \varphi_3}{\partial u}(u,v) \right),$$

$$\varphi_v(u,v) = \frac{\partial \varphi}{\partial v}(u,v) = \left(\frac{\partial \varphi_1}{\partial v}(u,v), \frac{\partial \varphi_2}{\partial v}(u,v), \frac{\partial \varphi_3}{\partial v}(u,v) \right),$$

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N. Fusco et al., *Mathematical Analysis*, La Matematica per il 3+2 137,
https://doi.org/10.1007/978-3-031-04151-8_10

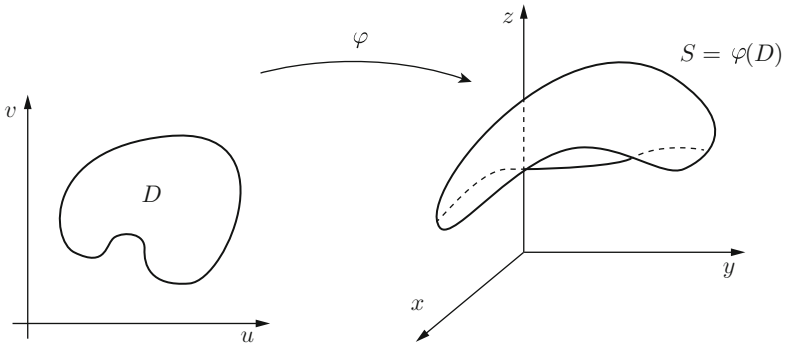


Fig. 10.1

condition (ii) is equivalent to demanding that $\varphi_u(u, v)$, $\varphi_v(u, v)$ are linearly independent for any $(u, v) \in \overset{\circ}{D}$ or, using the *cross product* of $\varphi_u(u, v)$ and $\varphi_v(u, v)$ (see Sect. 6.6), that

$$\frac{\partial \varphi}{\partial u}(u, v) \wedge \frac{\partial \varphi}{\partial v}(u, v) \neq 0 \qquad \forall\, (u, v) \in \overset{\circ}{D}. \tag{10.1}$$

As we will see, this condition ensures the existence, at each point $\varphi(u, v) \in \mathbb{R}^3$ with $(u, v) \in \overset{\circ}{D}$, of a unique tangent plane to the surface's *support* S, i.e. the image $\varphi(D)$ of φ.

A parametrised surface, just as a regular curve, is determined by the map φ and not by its support S, even though the word "surface" is customarily used to denote the set S, in which case the convention is that the map φ is given, albeit not explicitly.

The equations

$$\begin{cases} x = \varphi_1(u, v) \\ y = \varphi_2(u, v) \\ z = \varphi_3(u, v)\,, \end{cases} \qquad (u, v) \in D,$$

are called *parametric equations* of the surface.

Example 1 Let D be a connected domain in the plane and $f : D \to \mathbb{R}$ a C^1 function on D. The map $\varphi : D \to \mathbb{R}^3$, of parametric equations

$$\begin{cases} x = u \\ y = v \\ z = f(u, v)\,, \end{cases} \qquad (u, v) \in D,$$

(continued)

Example 1 (continued)
is a regular surface. Since $\varphi(u, v) = (u, v, f(u, v))$, in fact, φ is invertible on D and the vectors

$$\varphi_u = \left(1\,, 0\,, \frac{\partial f}{\partial u}\right), \qquad \varphi_v = \left(0\,, 1\,, \frac{\partial f}{\partial v}\right),$$

are linearly independent, i.e. the Jacobian matrix

$$D\varphi(u, v) = \frac{\partial(x, y, z)}{\partial(u, v)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{pmatrix}$$

has rank 2 (irrespective of f). We call a surface whose support is the graph of a function f as above a *Cartesian surface*.

Example 2 Consider the disc $D = \{(u, v) \in \mathbb{R}^2 : \ u^2 + v^2 \leq r^2\}$. The image of the map $\varphi(u, v) = \left(u, v, \sqrt{r^2 - (u^2 + v^2)}\right)$, $(u, v) \in D$, is the closed *hemisphere* of centre the origin and radius r contained in $H = \{(x, y, z) \in \mathbb{R}^3 : \ z \geq 0\}$. It is not a regular surface since φ is not C^1 on D.

Example 3 Let $r > 0$, $D = \{(u, v) \in \mathbb{R}^2 : \ u^2 + v^2 \leq r^2\}$ and $k \neq 0$. The map $\varphi : D \to \mathbb{R}^3$, $\varphi(u, v) = \left(u, v, k\sqrt{u^2 + v^2}\right)$, with $(u, v) \in D$, is not a regular surface because φ is not differentiable at $(0, 0)$. The image S coincides with the lateral surface of a *half-cone*, with vertex at the origin. If $k > 0$ then S is contained in $H = \{(x, y, z) \in \mathbb{R}^3 : \ z \geq 0\}$, and if $k < 0$ it is contained in the opposite half-space.

In either case the map $\psi : [0, r] \times [0, 2\pi] \to \mathbb{R}^3$ defined by

$$\begin{cases} x = u \cos v \\ y = u \sin v \\ z = k\, u\,, \end{cases} \qquad (u, v) \in D, \tag{10.2}$$

is a parametrised regular surface, whose support coincides with that of φ. Putting $T = [0, r] \times [0, 2\pi]$, if (u_1, v_1) and (u_2, v_2) are distinct points in $\overset{\circ}{T}$ and $u_1 \neq u_2$, immediately $\psi(u_1, v_1) \neq \psi(u_2, v_2)$; while if $u_1 = u_2$ and $v_1 \neq v_2$, since $v_1\,, v_2 \in (0, 2\pi)$ we have $(\cos v_1, \sin v_1) \neq (\cos v_2, \sin v_2)$ and so $\psi(u_1, v_1) \neq \psi(u_2, v_2)$. The map ψ is therefore invertible on $\overset{\circ}{T}$. Moreover

$$\psi_u = (\cos v, \sin v, k), \qquad \psi_v = (-u \sin v, u \cos v, 0),$$
$$\psi_u \wedge \psi_v = (-k\, u \cos v, -k\, u \sin v, u),$$

(continued)

Example 3 (continued)

so $|\psi_u \wedge \psi_v| = u\sqrt{1+k^2} > 0$ on $\overset{\circ}{T}$ and (10.1) holds too.

We emphasise that ψ is not injective on all of T, since it maps the points on the edge $\{0\} \times [0, 2\pi] \subset \mathbb{R}^2$ to the origin in $\mathbb{R}^3$, and the edges $[0, r] \times \{0\}$ and $[0, r] \times \{2\pi\}$ to the same segment on the cone (Fig. 10.2).

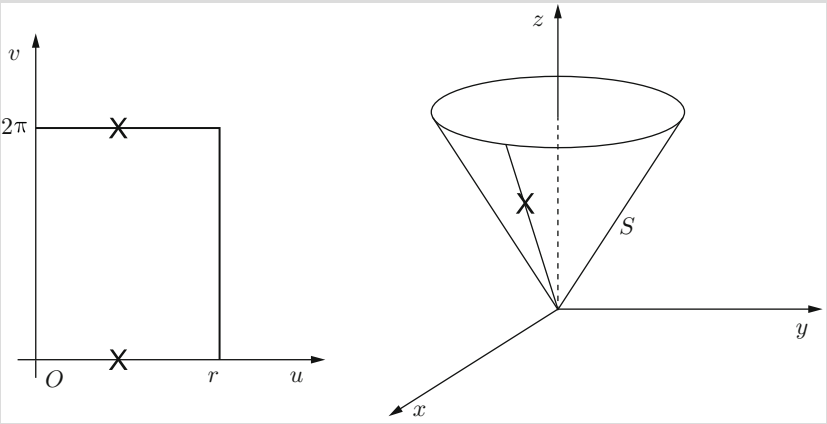


Fig. 10.2

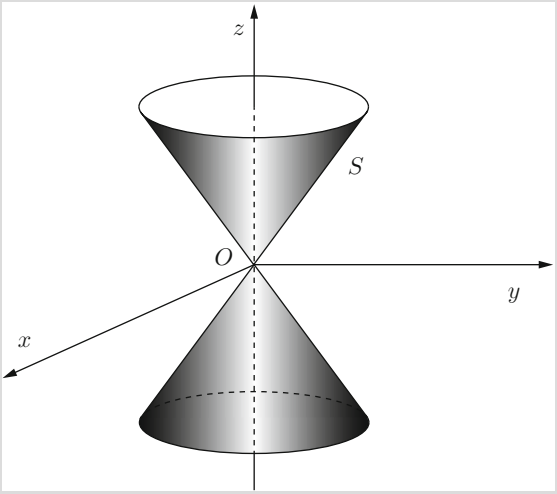
The previous example highlights that, in analogy to what happened for curves, a surface's regularity depends in an essential way on the parametrisation, rather than on the support S (cf. Example 5, Sect. 6.1).

Example 4 Let $r > 0$ and $D = [-r, r] \times [0, 2\pi]$. The surface $\varphi : D \to \mathbb{R}^3$ parametrised by (10.2) is not regular. The support S, for $k \neq 0$, is the lateral surface of the *double cone* in Fig. 10.3, while for $k = 0$ it is a *disc* in the Oxy plane counted "twice".

(continued)

Example 4 (continued)

Fig. 10.3



In either case φ is not 1-1 on $\overset{\circ}{D}$: the segment $\{0\} \times [0, 2\pi]$ is mapped by φ to the origin. Furthermore, the vector $\varphi_u \wedge \varphi_v$ is null at every point of that segment.

Example 5 Let $D = [0, \pi] \times [-\pi, \pi]$, $r > 0$ and $\varphi : D \to \mathbb{R}^3$ the surface parametrised by

$$\begin{cases} x = r \sin\psi \cos\vartheta \\ y = r \sin\psi \sin\vartheta \\ z = r \cos\psi\,, \end{cases} \qquad (\psi, \vartheta) \in D.$$

The image of φ is the sphere $S \subseteq \mathbb{R}^3$ with centre at the origin and radius r. We claim that φ is a regular surface.

Condition (i) is proved using an argument similar to the one employed for Example 3. Since

$$\varphi_\psi(\psi, \vartheta) = r(\cos\psi \cos\vartheta, \cos\psi \sin\vartheta, -\sin\psi),$$

$$\varphi_\vartheta(\psi, \vartheta) = r(-\sin\psi \sin\vartheta, \sin\psi \cos\vartheta, 0),$$

$$\varphi_\psi \wedge \varphi_\vartheta(\psi, \vartheta) = r^2\Big(\sin^2\psi \cos\vartheta, \sin^2\psi \sin\vartheta, \sin\psi \cos\psi\Big),$$

the quantity

$$\big|\varphi_\psi \wedge \varphi_\vartheta(\psi, \vartheta)\big| = r^2 \sin\psi \tag{10.3}$$

(continued)

Example 5 (continued)

is never zero on $\overset{\circ}{D}$.

The parameters ψ and ϑ have an interesting geometric interpretation. If $P_0 = (x_0, y_0, z_0)$ is a point on S different from the sphere's *poles* $(0, 0, \pm r)$, there exist a unique $\psi_0 \in (0, \pi)$ and a unique $\vartheta_0 \in (-\pi, \pi]$ such that $P_0 = \varphi(\psi_0, \vartheta_0)$. Precisely, $\psi_0 = \arccos(z/r)$, $\vartheta_0 = \operatorname{Arg}((x_0, y_0))$; in fact if O is the origin, ψ_0 measures the angle formed by the segment OP_0 and the positive z-axis, while ϑ_0 measures the angle between the segment OQ_0 and the positive x-axis, where Q_0 is the projection of P_0 onto the Oxy plane (Fig. 10.4).
The angles ϑ_0 and ψ_0 are called *longitude* and *colatitude* of P_0 respectively. The angle $\pi/2 - \psi_0$ is the *latitude* of P_0. At $(0, 0, r)$ and $(0, 0, -r)$ the latitude equals $\pi/2$ and $-\pi/2$ respectively, while the longitude is undefined. Finally, note that under φ, the segment $\{0\} \times [-\pi, \pi]$ is mapped to the "north pole" $(0, 0, r)$ and the segment $\{\pi\} \times [-\pi, \pi]$ to the "south pole" $(0, 0, -r)$, while the segments $[0, \pi] \times \{\pi\}$ and $[0, \pi] \times \{-\pi\}$ are mapped to the curve Γ on S consisting of points of the form $(x, 0, z)$ with $x \leq 0$.

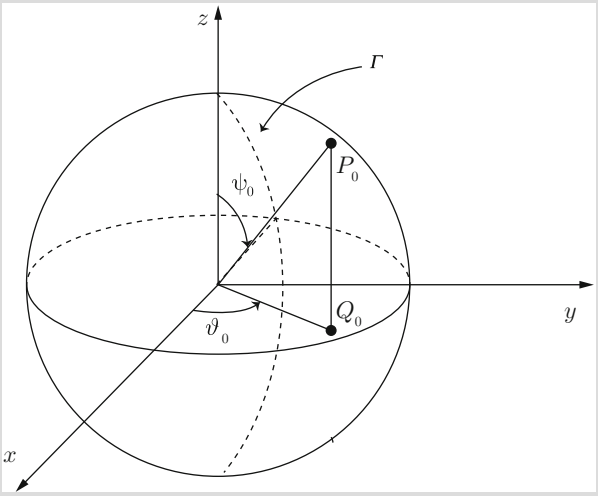


Fig. 10.4

Example 6 Consider the *cylindrical helix* of equations

$$\gamma(u) = (r\cos u, r\sin u, k\,u), \qquad u \in [a, b],$$

where $r > 0$ and $k \neq 0$. From every point on the helix let us draw a horizontal segment to the z-axis. This creates a regular surface called *helicoid*, see Fig. 10.5.
The helicoid has parametric equations

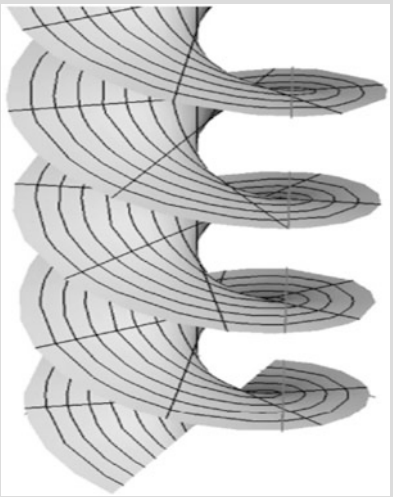
$$\varphi(u, v) = (v\cos u, v\sin u, k\,u), \qquad (u, v) \in [a, b] \times [0, r].$$

(continued)

Example 6 (continued)

Condition (i) is straightforward, and $|\varphi_u \wedge \varphi_v\,(u, v)| = \sqrt{v^2 + k^2} > 0$ for any (u, v).

Fig. 10.5



The next example provides another useful parametrisation of the sphere.

Example 7 (*Stereographic Projection*) Let S be the sphere of radius $r > 0$, of equation $x^2 + y^2 + (z - r)^2 = r^2$. For any point $P \equiv (x, y, z) \in S$ other than the "*north pole*" $N \equiv (0, 0, 2r)$ consider the intersection $Q(P)$ between the line through P and N and the Oxy plane, as in Fig. 10.6.

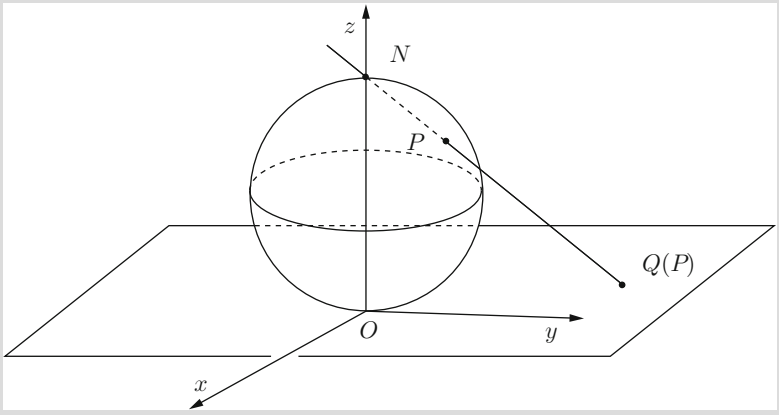


Fig. 10.6

(continued)

Example 7 (continued)
This line has parametric equation $r(t) = (tx, ty, 2r + t(z - 2r))$. Choosing t so that $2r + t(z - 2r) = 0$ gives the coordinates of $Q(P)$ on Oxy:

$$Q(P) = \left(\frac{2rx}{2r - z}, \frac{2ry}{2r - z} \right).$$

Setting

$$u = \frac{2rx}{2r - z}, \qquad v = \frac{2ry}{2r - z},$$

and using the sphere's equation $x^2 + y^2 + (z - r)^2 = r^2$, we recover the coordinates of P in terms of the coordinates (u, v) of $Q(P)$:

$$\begin{cases} x = \dfrac{4r^2 u}{u^2 + v^2 + 4r^2} \\ y = \dfrac{4r^2 v}{u^2 + v^2 + 4r^2} \\ z = \dfrac{2r(u^2 + v^2)}{u^2 + v^2 + 4r^2} \end{cases} \tag{10.4}$$

If we let (u, v) vary in a connected domain D of the plane, (10.4) are the parametric equations of a regular surface whose support is a closed portion S' of $S - \{N\}$.

Given a parametrised regular surface $\varphi : D \to \mathbb{R}^3$ and an internal point $(u_0, v_0) \in D$, one can prove that the restriction of φ to some neighbourhood I of the point is a homeomorphism of I onto $\varphi(I)$, where the latter inherits the standard topology of $\mathbb{R}^3$. For this reason in *Differential Geometry* one prefers the following definition for regular surfaces: *a subset S of $\mathbb{R}^3$, equipped with the induced topology of $\mathbb{R}^3$, is a regular surface if for any point $P_0 \in S$ there exist a neighbourhood U of P_0, an open set A in the plane and a C^1 map $g : A \to U$ such that*

(j) *g is a homeomorphism between A and $U \cap S$;*
(jj) *the Jacobian matrix Dg has rank 2 on A.*

It is easy to convince ourselves that the above definition *is not equivalent* to ours. The definition just given has the advantage of being more natural, since it starts from the geometric object S rather than from a map from $\mathbb{R}^2$ to $\mathbb{R}^3$; but it has the evident limitation of being technically less tractable. These considerations, together with a clear idea of the present book's goals, led us to prefer the notion of regular surface given at the beginning of the section. Note, however, that if φ is a parametrised regular surface, every point (u_0, v_0) internal to D does have a neighbourhood I such that $\varphi(I)$ is a regular surface in the second sense (see Proposition 2, Sect. 12.1).

10.2 Local Coordinates and Change of Parameters

Let $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a regular surface. A transformation $\Phi : T \to D$, where T is a connected domain in the plane, is called a *reparametrisation* when it is invertible, of class C^1, with C^1 inverse, and its Jacobian determinant J_Φ is non-zero on $\overset{\circ}{T}$.

Easily, the map $\psi : T \to \mathbb{R}^3$ given by

$$\psi(s,t) = \varphi(\Phi(s,t)), \qquad (s,t) \in T,$$

is a parametrised regular surface. In fact ψ is clearly invertible on $\overset{\circ}{T}$ and

$$\frac{\partial \psi}{\partial s} = \frac{\partial \varphi}{\partial u}\frac{\partial \Phi_1}{\partial s} + \frac{\partial \varphi}{\partial v}\frac{\partial \Phi_2}{\partial s},$$
$$\frac{\partial \psi}{\partial t} = \frac{\partial \varphi}{\partial u}\frac{\partial \Phi_1}{\partial t} + \frac{\partial \varphi}{\partial v}\frac{\partial \Phi_2}{\partial t},$$

so

$$\frac{\partial \psi}{\partial s}(s,t) \wedge \frac{\partial \psi}{\partial t}(s,t) = \left(\frac{\partial \varphi}{\partial u}(\Phi(s,t)) \wedge \frac{\partial \varphi}{\partial v}(\Phi(s,t))\right)\left(\frac{\partial \Phi_1}{\partial s}\frac{\partial \Phi_2}{\partial t} - \frac{\partial \Phi_1}{\partial t}\frac{\partial \Phi_2}{\partial s}\right),$$

i.e.

$$\frac{\partial \psi}{\partial s} \wedge \frac{\partial \psi}{\partial t} = J_\Phi \cdot \left(\frac{\partial \varphi}{\partial u} \wedge \frac{\partial \varphi}{\partial v}\right). \tag{10.5}$$

Since $J_\Phi \neq 0$ on $\overset{\circ}{T}$, it follows that $\dfrac{\partial \psi}{\partial s} \wedge \dfrac{\partial \psi}{\partial t} \neq 0$ on $\overset{\circ}{T}$, so the regularity condition (10.1) holds.

It is quite natural, as we did for curves, to give the following definition. We say that two regular surfaces $\varphi : D \to \mathbb{R}^3$ and $\psi : T \to \mathbb{R}^3$ are *equivalent* if there is a reparametrisation $\Phi : T \to D$ such that $\psi = \varphi \circ \Phi$, or equivalently $\varphi = \psi \circ \Phi^{-1}$.

The symbol $\sim$ will denote the above relation, clearly an equivalence relation. In the sequel we shall say *regular surface* to mean both a given parametrisation φ or the equivalence relation it determines.

Consider now a regular surface $\varphi : D \to \mathbb{R}^3$, with support S, and a point $P_0 = \varphi(u_0, v_0)$ of S where $(u_0, v_0) \in \overset{\circ}{D}$. The straight lines $v = v_0$, $u = u_0$ are mapped by φ to the curves

$$u \to \varphi(u, v_0), \qquad v \to \varphi(u_0, v), \tag{10.6}$$

which lie on S and pass through P_0 (Fig. 10.7).

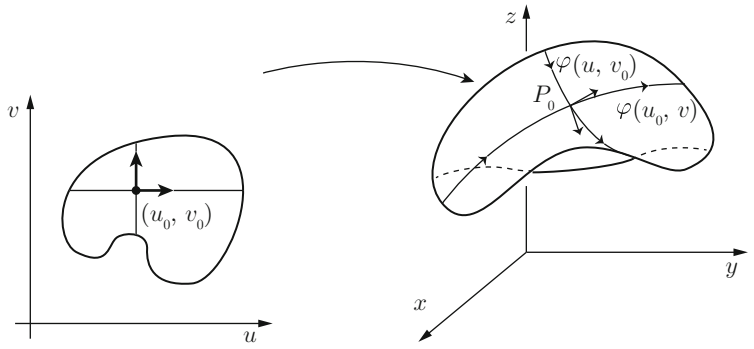


Fig. 10.7

These curves are regular, because formula (10.1) implies in particular that the vectors $\varphi_u(u, v_0)$ and $\varphi_v(u_0, v)$ are never zero when (u, v_0) and (u_0, v) belong to $\overset{\circ}{D}$.

The curves defined in (10.6) are called *coordinate lines of the surface*, while the variables u and v are usually called *local coordinates* of the surface. In general the coordinates lines on S are not orthogonal and depend upon the particular parametrisation chosen for the surface.

Example 1 Let $\gamma : [a, b] \to \mathbb{R}^2$ be a regular simple curve of equations $\gamma(u) = (x(u), y(u))$. Given $D = [a, b] \times [c, d]$, the map $\varphi(u, v) = (x(u), y(u), v)$, $(u, v) \in D$, is a regular surface. In fact φ is invertible on $(a, b) \times [c, d]$ and

$$\varphi_u \wedge \varphi_v = \big(y'(u), -x'(u), 0\big) \tag{10.7}$$

is not zero on $\overset{\circ}{D}$, by virtue of the regularity of γ. The support S of φ is called *cylindrical surface* (Fig. 10.8).

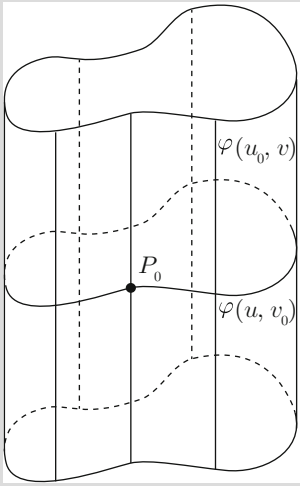
We call a coordinate line $u \to \varphi(u, v_0)$ on the cylindrical surface S a *generating curve*, while $v \to \varphi(u_0, v)$ is a *directrix* (line). Note that any two such are orthogonal, since

$$(\varphi_u\,, \varphi_v) = \big((x', y', 0), (0, 0, 1)\big) = 0\,.$$

(continued)

Example 1 (continued)

Fig. 10.8



Example 2 Let γ be a plane curve and r a straight line lying on the same plane. The *surface of revolution generated by the curve* γ is the geometric locus of the trajectories described by the points of γ during an α-rotation about the axis r.

Let us fix a Cartesian frame $Oxyz$ on $\mathbb{R}^3$ so that the Oxz plane coincides with the plane containing the curve and the z-axis becomes the axis of rotation. Suppose γ is a regular simple curve parametrised by

$$\begin{cases} x = x(t) \\ z = z(t)\,, \end{cases} \qquad t \in [a, b]\,,$$

with $x(t) \neq 0$ for any $t \in (a, b)$. The surface of revolution φ, of parametric equations

$$\begin{cases} x = x(t)\cos\vartheta \\ y = x(t)\sin\vartheta \\ z = z(t)\,, \end{cases} \qquad (t, \vartheta) \in D,$$

with $D = [a, b] \times [\alpha_1, \alpha_2]$, $0 < \alpha_2 - \alpha_1 \leq 2\pi$, is a regular surface. The regularity condition (i) is immediate, while to see (ii) we note that

$$\varphi_t \wedge \varphi_\vartheta = \big(-x(t)z'(t)\cos\vartheta, -x(t)z'(t)\sin\vartheta, x(t)x'(t)\big) \tag{10.8}$$

and so

$$|\varphi_t \wedge \varphi_\vartheta| = |x(t)|\sqrt{[x'(t)]^2 + [z'(t)]^2} > 0\,, \qquad \forall\,(t, \vartheta) \in \overset{\circ}{D}, \tag{10.9}$$

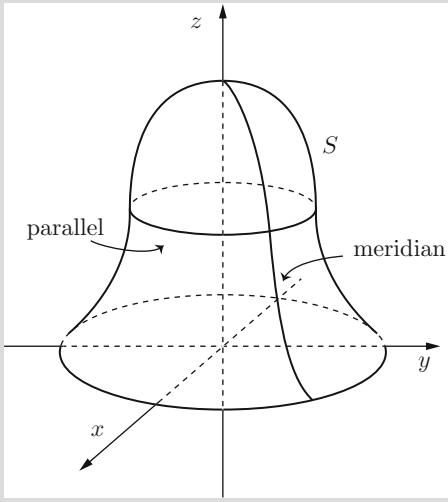
(continued)

Example 2 (continued)
due to the regularity of γ and the fact that $x(t) \neq 0$ for any $t \in (a, b)$.

The coordinate lines $t \to \varphi(t, \vartheta_0)$, $\vartheta \to \varphi(t_0, \vartheta)$ are called *meridians* and *parallels*, respectively, of the given surface of revolution. Meridians and parallels arise by intersecting S with planes through the z-axis and planes perpendicular to the z-axis respectively. Hence they are always orthogonal (see Fig. 10.9), as is easily confirmed by the inner product

$$(\varphi_t \, , \varphi_\vartheta) = \big((x'(t)\cos\vartheta, x'(t)\sin\vartheta, z'(t)), (-x(t)\sin\vartheta, x(t)\cos\vartheta, 0)\big) = 0\,.$$

Fig. 10.9



The sphere is clearly a particular surface of revolution obtained rotating a semi-circle around its diameter. If we rotate from $-\pi$ to π the semi-circle γ of equations $x(t) = \sin t$, $z(t) = \cos t$, with $t \in [0, \pi]$, we obtain precisely the parametrisation of Example 5 in the previous section.

Example 3 If $z = f(x)$, $x \in [a, b]$ with $a \geq 0$, is a C^1 function, the surface generated by a complete turn of the graph of f around the z-axis has parametric equations

$$\varphi(u, v) = (u\cos v, u\sin v, f(u)) \qquad (u, v) \in [a, b] \times [0, 2\pi].$$

For example, for $f(x) = hx$, $h \neq 0$, $x \in [a, b]$ with $a = 0$, we obtain a *right circular cone* with vertex at the origin. Taking $x \in [a, b]$ with $a > 0$ generates a *truncated cone* (Fig. 10.10).
Choosing $f(x) = hx^2$, $h \neq 0$, $x \in [0, b]$, produces a *paraboloid of revolution* (a special type of *elliptic paraboloid*), as in Fig. 10.11.

(continued)

Example 3 (continued)

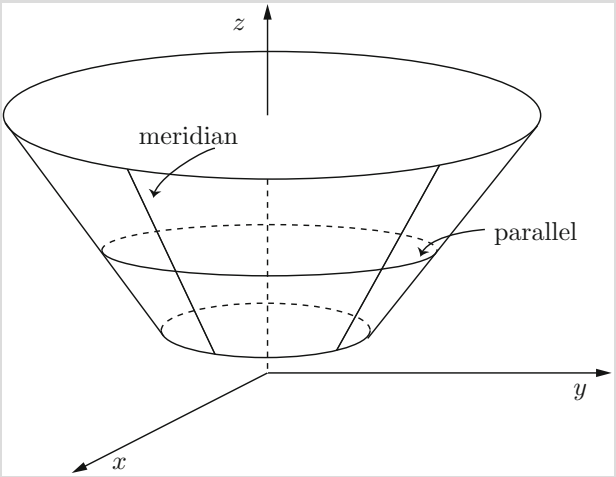


Fig. 10.10

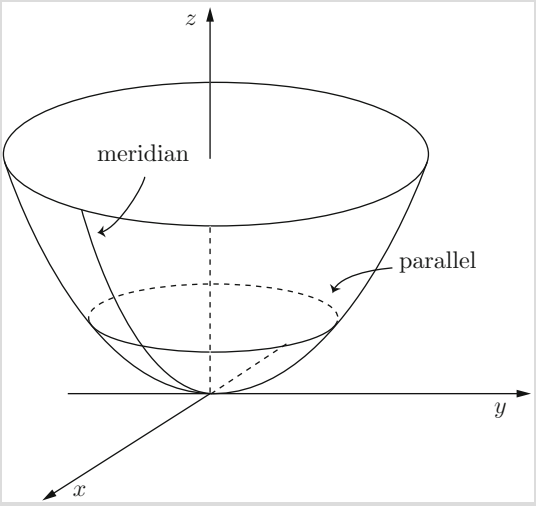


Fig. 10.11

(continued)

Example 4 (continued)

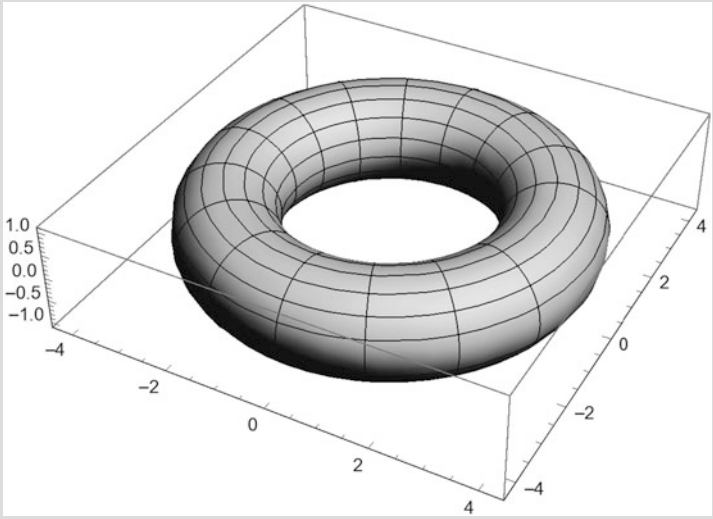


Fig. 10.12

Example 4 Another example of surface of revolution is the *torus*, which can be defined by the complete rotation of a circle γ around an external coplanar axis. If γ is the circle on the Oxz plane of radius r and centre $(R, 0, 0)$, with $R > r$, the parametric equations of the torus, plotted in Fig. 10.12, are

$$\varphi(t, \vartheta) = ((R + r\cos t)\cos\vartheta, (R + r\cos t)\sin\vartheta, r\sin t)\,,$$

as (t, ϑ) varies in the square $[0, 2\pi] \times [0, 2\pi]$.
Let us remark that the condition $r < R$ is saying that γ never intersects the axis of rotation. This constraint guarantees the regularity condition (ii) for the Jacobian matrix, because (the reader should check this)

$$|\varphi_t \wedge \varphi_\vartheta| = r\,|R + r\cos t|\,, \tag{10.10}$$

and since $r < R$ and $|\cos t| \leq 1$ it follows that $R + r\cos t \geq R - r > 0$. Hence the right-hand side of (10.10) is never zero, making the torus a regular surface.

Examples 1 and 2 showed that cylindrical surfaces and surfaces of revolution admit explicit parametrisations in which the coordinate lines are orthogonal. This property is actually true (at least locally) for any regular surface. Indeed the following result holds, which we just state.

Proposition *If $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$ is a regular surface and $(u_0, v_0) \in \overset{\circ}{D}$, there exists a neighbourhood of (u_0, v_0) on which the surface can be reparametrised so that the new coordinate lines become orthogonal.*

10.3 The Tangent Plane and the Unit Normal

Consider a regular surface of equations

$$\varphi(u, v) = (x(u, v), y(u, v), z(u, v)), \qquad (u, v) \in D.$$

Set $S = \varphi(D)$ and consider a point P_0 on S of local coordinates $(u_0, v_0) \in \overset{\circ}{D}$. Let $\gamma : [a, b] \to \overset{\circ}{D}$ be a regular curve contained in $\overset{\circ}{D}$ and passing through (u_0, v_0), and $t_0 \in [a, b]$ a number such that $(u_0, v_0) = \gamma(t_0)$. It is easy to see that φ transforms γ into a regular curve $\widetilde{\gamma}$ contained in S and passing through P_0 (see Fig. 10.13). In fact if γ has equations $\gamma(t) = (u(t), v(t))$, the equations of $\widetilde{\gamma}$ read

$$\widetilde{\gamma}(t) = \begin{cases} x = x(u(t), v(t)) \\ y = y(u(t), v(t)) \\ z = z(u(t), v(t)), \end{cases} \qquad t \in [a, b]. \tag{10.11}$$

The regularity of γ forces $\widetilde{\gamma}$ to be C^1 on $[a, b]$. Moreover,

$$\widetilde{\gamma}'(t) = \left(x_u\, u' + x_v\, v',\ y_u\, u' + y_v\, v',\ z_u\, u' + z_v\, v'\right) = \varphi_u\, u' + \varphi_v v'. \tag{10.12}$$

For any $t \in (a, b)$, $\widetilde{\gamma}'$ is then a linear combination of the linearly independent vectors $\varphi_u = \varphi_u(u(t), v(t))$ and $\varphi_v = \varphi_v(u(t), v(t))$, with non zero coefficients $u' = u'(t)$

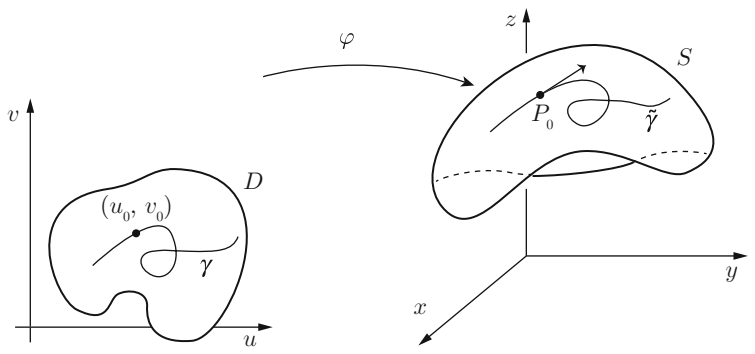


Fig. 10.13

and $v' = v'(t)$. Therefore the vector $\widetilde{\gamma}'$ is non-zero for any $t \in (a, b)$, and $\widetilde{\gamma}$ is regular.

One can actually prove that any regular curve contained in S and passing through P_0 is (locally) the image, under φ, of a regular curve γ contained in $\overset{\circ}{D}$ and passing through (u_0, v_0). Then Eq. (10.11) is, locally, the equation of a generic regular curve on S through the point P_0.

Since

$$\widetilde{\gamma}(t_0) = (x(u_0, v_0), y(u_0, v_0), z(u_0, v_0)) = P_0 \,,$$

from (10.11) and (10.12) we obtain in particular

$$\widetilde{\gamma}'(t_0) = \varphi_u(u_0, v_0)\, u'(t_0) + \varphi_v(u_0, v_0)\, v'(t_0) \,,$$

i.e. $\widetilde{\gamma}'(t_0)$ is a linear combination of $\varphi_u(u_0, v_0)$ and $\varphi_v(u_0, v_0)$. It then becomes evident that the tangent lines at P_0 to a generic regular curve on S through P_0 all lie on the plane determined by $\varphi_u(u_0, v_0)$ and $\varphi_v(u_0, v_0)$.

This plane is called the *tangent plane* to the surface at P_0. Since the cross product of two vectors is orthogonal to the plane they span, the surface's tangent plane at P_0 is orthogonal to the vector $\varphi_u(u_0, v_0) \wedge \varphi_v(u_0, v_0)$, and hence orthogonal to the unit vector

$$\nu(P_0) = \frac{\varphi_u(u_0, v_0) \wedge \varphi_v(u_0, v_0)}{|\varphi_u(u_0, v_0) \wedge \varphi_v(u_0, v_0)|} \,.$$

The latter is called the *surface unit normal* at P_0 (Fig. 10.14).

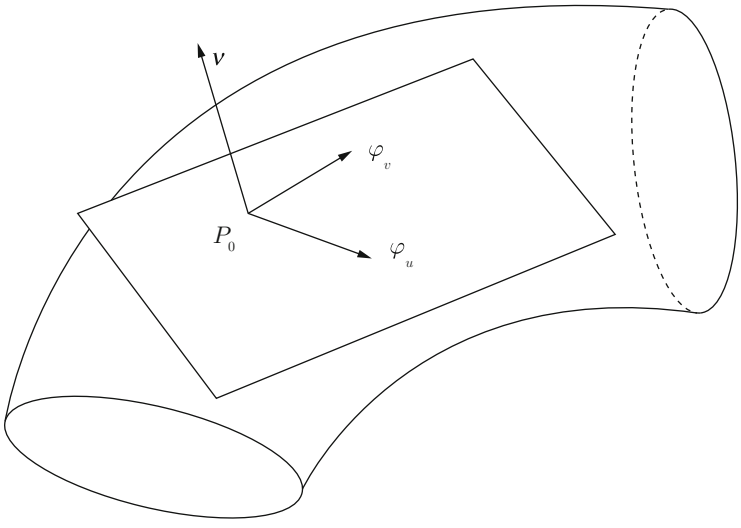


Fig. 10.14

Write $A = A(u, v)$, $B = B(u, v)$, $C = C(u, v)$ for the components of $\varphi_u \wedge \varphi_v$:

$$A(u, v) = \det \frac{\partial(y, z)}{\partial(u, v)}, \quad B(u, v) = \det \frac{\partial(z, x)}{\partial(u, v)}, \quad C(u, v) = \det \frac{\partial(x, y)}{\partial(u, v)}. \tag{10.13}$$

Then the equation of the tangent plane to S at P_0 is

$$A_0(x - x_0) + B_0\,(y - y_0) + C_0(z - z_0) = 0\,, \tag{10.14}$$

where $A_0 = A(u_0, v_0)$, $B_0 = B(u_0, v_0)$, $C_0 = C(u_0, v_0)$, and (x_0, y_0, z_0) are the coordinates of P_0.

If $\psi : T \to \mathbb{R}^3$ is another parametrisation of the regular surface, obtained by the map $\Phi : T \to D$, from (10.5) we deduce

$$\frac{\psi_s \wedge \psi_t}{|\psi_s \wedge \psi_t|} = \frac{J_\Phi}{|J_\Phi|} \cdot \frac{\varphi_u \wedge \varphi_v}{|\varphi_u \wedge \varphi_v|}.$$

When changing parametrisations, therefore, the normal vector's orientation may change, but its direction will not. The tangent plane does not change either, obviously.

Example 1 Consider the Cartesian surface S given by the graph of $f : D \to \mathbb{R}$, and a point $(x_0\,, y_0\,, f(x_0\,, y_0)) \in S$ with $(x_0\,, y_0) \in \overset{\circ}{D}$. We have seen that S is regular if we represent it by

$$\begin{cases} x = u \\ y = v \\ z = f(u, v)\,, \end{cases} \qquad (u, v) \in D.$$

The unit normal to S at P_0 is

$$\nu(P_0) = \frac{1}{\sqrt{1 + |Df(x_0\,, y_0)|^2}} \big(-f_x(x_0\,, y_0)\,, -f_y(x_0\,, y_0)\,, 1\big)\,.$$

Then by (10.13), (10.14) the tangent plane to S at P_0 has equation

$$-f_x(x_0\,, y_0)\,(x - x_0) - f_y(x_0\,, y_0)\,(y - y_0) + z - f(x_0\,, y_0) = 0\,,$$

exactly the expression seen in Sect. 3.5 on functions of several variables.

Example 2 Let S be the cylindrical surface of equations

$$\varphi(u, v) = (x(u), y(u), v)$$

and γ the generating curve (see Example 1, Sect. 10.2). By (10.7) the normal to S at the point P_0 of coordinates $(x(u_0), y(u_0), v_0)$ is

$$\nu(P_0) = \frac{1}{\sqrt{[x'(u_0)]^2 + [y'(u_0)]^2}} \big(y'(u_0), -x'(u_0), 0\big),$$

which coincides with the normal to the curve parallel to γ and passing through P_0.

Example 3 Let S be the surface generated by rotating $\gamma(t) = (x(t), z(t))$, with $x(t) > 0$, around the z-axis. By (10.8) the normal line to S at some point P_0 on the generating curve γ coincides with the normal to γ at P_0. The normal to S at any other point on the parallel traced by P_0 is obtained simply by rotating the normal at P_0 around the axis.

Example 4 Let $\gamma : I \subset \mathbb{R} \to \mathbb{R}^3$ be a biregular simple curve of class C^3, parametrised by arclength s. Recall (Sect. 6.7) that γ is said *biregular* when the curvature $k(s)$ is everywhere strictly positive. If $[a, b]$ is a closed interval internal to I, there is $r > 0$ such that $r < 1/k(s)$ for any $s \in [a, b]$. With this choice of r it is easy to show that the equations

$$\varphi(s, t) = \gamma(s) + r\,(N(s)\cos t + B(s)\sin t)\,, \qquad s \in [a, b],\ t \in [0, 2\pi],$$

where $N(s)$ and $B(s)$ are the normal and binormal to γ at $\gamma(s)$ respectively, define a regular surface. It is called *tubular surface* of radius r (Fig. 10.15).

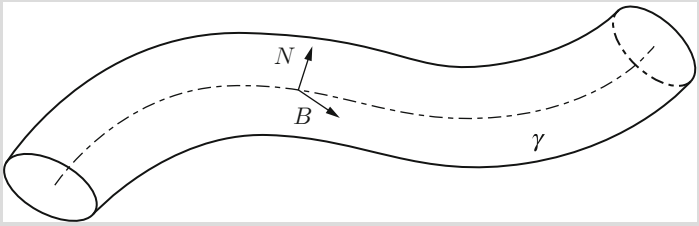


Fig. 10.15

(continued)

Example 4 (continued)
Using the Frenet formulas (6.55) we compute the normal $\nu(s,t)$ at an arbitrary point of the surface. For any $s \in [a,b]$ and $t \in [0, 2\pi]$

$$\varphi_s(s,t) = \gamma'(s) + r\big(N'(s)\cos t + B'(s)\sin t\big) =$$
$$= (1 - rk\cos t)\,T(s) + r\tau N(s)\sin t - r\tau B(s)\cos t\,,$$

where $k = k(s)$ and $\tau = \tau(s)$ are the curvature and torsion of γ at $\gamma(s)$. Moreover

$$\varphi_t(s,t) = r\,(-N(s)\sin t + B(s)\cos t)\,,$$

so

$$\varphi_s \wedge \varphi_t = r(1 - rk\cos t)T \wedge (-N\sin t + B\cos t) =$$
$$= -r(1 - rk\cos t)(N\cos t + B\sin t)\,.$$

The surface normal at $\varphi(s,t)$ is then

$$\nu(s,t) = -\,(N(s)\cos t + B(s)\sin t)\ .$$

10.4 The Area of a Surface

Starting from this section until the end of the chapter we shall assume D is a regular domain (see the definition in Sect. 8.3).
Let $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a regular surface with support S. For any point (u,v) in $\overset{\circ}{D}$ and any vector (λ,μ) of $\mathbb{R}^2$ consider the *quadratic form* in (λ,μ)

$$\begin{aligned}\mathcal{E} = \mathcal{E}(u,v)(\lambda,\mu) &= |\varphi_u(u,v)\lambda + \varphi_v(u,v)\mu|^2 = \\ &= |\varphi_u(u,v)|^2\,\lambda^2 + 2\,(\varphi_u(u,v),\varphi_v(u,v))\,\lambda\mu + |\varphi_v(u,v)|^2\,\mu^2 = \\ &= E(u,v)\,\lambda^2 + 2\,F(u,v)\,\lambda\mu + G(u,v)\,\mu^2\,,\end{aligned} \tag{10.15}$$

where

$$E(u,v) = |\varphi_u(u,v)|^2\,,$$
$$F(u,v) = (\varphi_u(u,v),\varphi_v(u,v))\,,$$
$$G(u,v) = |\varphi_v(u,v)|^2\,.$$

By definition (10.15), $\mathcal{E}$ is a non-negative quadratic form. Since φ is regular, the vectors $\varphi_u(u,v)$, $\varphi_v(u,v)$ are linearly independent and hence $\mathcal{E}(u,v)(\lambda,\mu) > 0$ for any $(\lambda,\mu) \neq 0$, meaning $\mathcal{E}$ is positive definite.

The quadratic form $\mathcal{E}$ defined by (10.15) is called the *first fundamental form* of the surface. The 2×2 matrix associated with $\mathcal{E}$ is

$$\mathcal{A} = \begin{pmatrix} E(u,v) & F(u,v) \\ F(u,v) & G(u,v) \end{pmatrix} = \begin{pmatrix} |\varphi_u|^2 & (\varphi_u, \varphi_v) \\ (\varphi_u, \varphi_v) & |\varphi_v|^2 \end{pmatrix}.$$

The determinant of $\mathcal{A}$

$$\det \mathcal{A} = EG - F^2 = |\varphi_u|^2 \cdot |\varphi_v|^2 - (\varphi_u, \varphi_v)^2$$

is non-negative by the Cauchy-Schwarz inequality. More precisely, since the surface is regular the determinant of $\mathcal{A}$ is positive on $\overset{\circ}{D}$, because $\varphi_u(u,v), \varphi_v(u,v)$ are linearly independent. This is another reason (cf. Sect. 3.12) to say that the quadratic form $\mathcal{E}$ is positive definite on $\overset{\circ}{D}$. The identity

$$EG - F^2 = |\varphi_u|^2 \cdot |\varphi_v|^2 - (\varphi_u, \varphi_v)^2 = |\varphi_u \wedge \varphi_v|^2$$

holds, because any two vectors $a = (a_1, a_2, a_3)$, $b = (b_1, b_2, b_3)$ in $\mathbb{R}^3$ satisfy

$$|a \wedge b|^2 = |a|^2 \cdot |b|^2 - (a,b)^2 . \tag{10.16}$$

This relation is a consequence of Proposition 4, Sect. 6.6. Here is a direct poof of (10.16). Since

$$a \wedge b = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1) ,$$

we have

$$\begin{aligned} |a \wedge b|^2 &= (a_2 b_3 - a_3 b_2)^2 + (a_3 b_1 - a_1 b_3)^2 + (a_1 b_2 - a_2 b_1)^2 = \\ &= a_1^2 (b_2^2 + b_3^2) + a_2^2 (b_1^2 + b_3^2) + a_3^2 (b_1^2 + b_2^2) - 2(a_1 a_2 b_1 b_2 + a_2 a_3 b_2 b_3 + a_1 a_3 b_1 b_3). \end{aligned} \tag{10.17}$$

But

$$(a,b)^2 = a_1^2 b_1^2 + a_2^2 b_2^2 + a_3^2 b_3^2 + 2\,(a_1 a_2 b_1 b_2 + a_2 a_3 b_2 b_3 + a_1 a_3 b_1 b_3) ,$$

so (10.17) implies

$$\begin{aligned} |a \wedge b|^2 &= a_1^2\,(b_2^2 + b_3^2) + a_2^2\,(b_1^2 + b_3^2) + a_3^2\,(b_1^2 + b_2^2) + \\ &\quad + a_1^2 b_1^2 + a_2^2 b_2^2 + a_3^2 b_3^2 - (a,b)^2 = \\ &= (a_1^2 + a_2^2 + a_3^2) \cdot (b_1^2 + b_2^2 + b_3^2) - (a,b)^2 = |a|^2 \cdot |b|^2 - (a,b)^2 . \end{aligned}$$

If $\gamma(t) = (u(t), v(t))$, $t \in [a, b]$, is a regular curve contained in $\overset{\circ}{D}$ and $\widetilde{\gamma} = \varphi \circ \gamma$ is the image on the surface S, by (10.12) the length of $\widetilde{\gamma}$ is

$$\int_a^b \left|\widetilde{\gamma}'(t)\right| dt = \int_a^b \sqrt{E(u')^2 + 2Fu'v' + G(v')^2}\, dt\,.$$

Consider, in particular, a point moving along γ from $\gamma(t)$ to $\gamma(t + \Delta t)$, $\Delta t > 0$. It describes an arc of approximate length

$$\left|\gamma'(t)\right|\Delta t = \Delta t\sqrt{(u'(t))^2 + (v'(t))^2}\,.$$

The image of the point describes on $\widetilde{\gamma}$ an arc of length approximatively equal to

$$\Delta t\sqrt{E(u')^2 + 2Fu'v' + G(v')^2}\,.$$

The ratio of these quantities is precisely

$$\sqrt{\mathcal{E}(u(t), v(t))(T(t))}\,,$$

where $T(t)$ is the unit tangent vector to γ at $(u(t), v(t))$. Given a unit vector (h, k) on the plane, we can then interpret

$$\sqrt{\mathcal{E}(u, v)\,(h, k)} = \sqrt{E\,h^2 + 2F\,hk + G\,k^2}$$

as the measure of the variation of the *distance* (or *metric*) on the surface when a point on S, image of a point in $\overset{\circ}{D}$, moves from (u, v) along the direction (h, k).

Similarly, consider the rectangle R determined by points (u, v), $(u+\Delta u, v+\Delta v)$ of $\overset{\circ}{D}$ and its image $\widetilde{R}$ on S. We may think that a good approximation of the "*area*" of $\widetilde{R}$ is the area of the parallelogram Π with a vertex at $\varphi(u, v)$ and edges parallel and congruent to the vectors $\xi = \varphi_u(u, v)\Delta u$ and $\eta = \varphi_v(u, v)\,\Delta v$ (Fig. 10.16). The area of R is $|\Delta u| \cdot |\Delta v|$, while the area of Π equals

$$|\xi \wedge \eta| = |\Delta u| \cdot |\Delta v|\sqrt{EG - F^2}\,.$$

Hence we may interpret $\sqrt{E(u, v)G(u, v) - F^2(u, v)}$ as a measure of how much a region on S around the point P, image of $(u, v) \in \overset{\circ}{D}$, get distorted.

Based on these considerations it is natural to define the *area* $A(\varphi)$ *of the regular surface* φ as the number

$$A(\varphi) = \iint_D |\varphi_u \wedge \varphi_v|\, du\, dv = \iint_D \sqrt{EG - F^2}\, du\, dv\,. \tag{10.18}$$

Note that the above quantity is independent of the parametrisation. If $\Phi : T \to D$ is a reparametrisation and we define $\psi : T \to \mathbb{R}^2$, $\psi(s, t) = \varphi(\Phi(s, t))$, from (10.5)

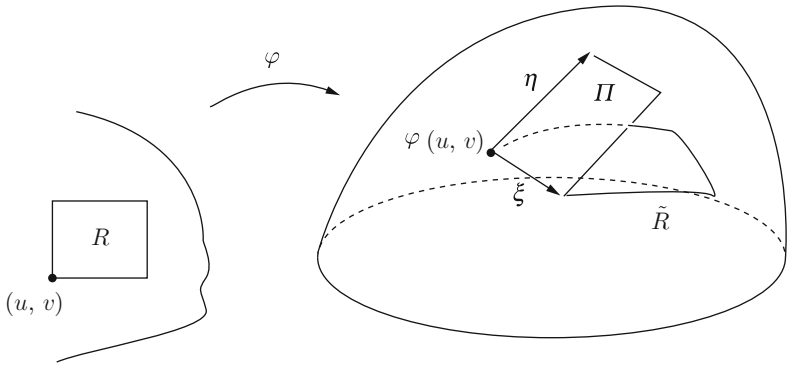


Fig. 10.16

and the formula for changing variables in double integrals we obtain

$$\begin{aligned}\iint_T |\psi_s \wedge \psi_t|\, ds\, dt &= \iint_T |(\varphi_u \wedge \varphi_v)\,(\Phi(s,t))| \cdot |J_\Phi|\; ds\, dt = \\ &= \iint_D |\varphi_u \wedge \varphi_v| \cdot \left|J_\Phi\left(\Phi^{-1}(u,v)\right)\right| \cdot |J_{\Phi^{-1}}(u,v)|\; du\, dv = \\ &= \iint_D |\varphi_u \wedge \varphi_v|\; du\, dv\,.\end{aligned}$$

In the particular case where the surface has equations $\varphi(x, y) = (x, y, f(x, y))$, with $f : D \to \mathbb{R}$ of class C^1, since $\varphi_x = (1, 0, f_x)$, $\varphi_y = (0, 1, f_y)$ we obtain $\varphi_x \wedge \varphi_y = (-f_x, -f_y, 1)$. Hence (10.18) reads

$$A(\varphi) = \iint_D \sqrt{1 + (f_x)^2 + (f_y)^2}\, dx\, dy = \iint_D \sqrt{1 + |Df|^2}\, dx\, dy. \qquad (10.19)$$

Here is another explanation for (10.18). Suppose φ is parametrised by $\varphi(x, y) = (x, y, f(x, y))$ with $f : D \to \mathbb{R}$ of class C^1. Consider a decomposition of D in normal domains $\{D_1, D_2, \ldots, D_N\}$ without common interior points, and let

$$\varphi_i = \varphi_{|_{D_i}}\,, \qquad i = 1, 2, \ldots, N,$$

be the corresponding Cartesian surfaces. Fix for every $i = 1, 2, \ldots, N$ a point $(x_i\,, y_i) \in \overset{\circ}{D}_i$, with image $P_i = (x_i\,, y_i\,, f(x_i\,, y_i))$ on the surface φ_i, and let Π_i be the part of the tangent plane to S at P_i that projects onto D_i. It is easy to convince ourselves that the area of Π_i is related to the area of D_i by

$$A(D_i) = A(\Pi_i) \cos \omega_i\,, \qquad (10.20)$$

where ω_i is the angle between the normal to the plane containing Π_i and the z-axis. As the unit normal to S at P_i is

$$\frac{1}{\sqrt{1+|Df(x_i\,,\,y_i)|^2}}(-f_x(x_i\,,\,y_i),-f_y(x_i\,,\,y_i),1)\,,$$

we find

$$\cos\omega_i = \frac{1}{\sqrt{1+|Df(x_i\,,\,y_i)|^2}}\,. \tag{10.21}$$

Consider the sum σ of all the areas of these parts of tangent planes Π_i. By (10.20) and (10.21)

$$\sigma = \sum_{i=1}^{N}\sqrt{1+|Df(x_i\,,\,y_i)|^2}\,A(D_i)\,.$$

The natural thing is to define the area of φ as the limit of these sums as the maximum diameter δ of the D_i goes to zero. In this way

$$A(\varphi) = \lim_{\delta\to 0^+}\sigma = \iint_D \sqrt{1+|Df|^2}\,dx\,dy\,.$$

The above coincides with (10.19).

One could also explain (10.18) by showing (see Sect. 12.1) that if $\varphi : D \to \mathbb{R}^3$ is a regular surface and $P_0 = \varphi(u_0\,,\,v_0)$, with $(u_0\,,\,v_0) \in \overset{\circ}{D}$, we can reparametrise φ on a neighbourhood of P_0 to become Cartesian, also recalling that $\iint_D |\varphi_u \wedge \varphi_v|\,du\,dv$ is independent of the parametrisation, as we have already remarked.

Example 1 Using the parametrisation of the sphere introduced in Example 5, Sect. 10.1, we compute the area of the *sphere* of radius r. By (10.3)

$$A(\varphi) = \int_0^{\pi} d\psi \int_{-\pi}^{\pi} r^2\sin\psi\,d\vartheta = 2\pi r^2\int_0^{\pi}\sin\psi\,d\psi = 4\pi r^2\,.$$

Example 2 Let us find the area of the *torus* generated by the rotation of a circle of radius r around an axis at distance $a > r$ from the circle's centre. With the parametrisation of Example 4, Sect. 10.2

$$\varphi(t\,,\,\vartheta) = ((a+r\cos t)\cos\vartheta\,,\,(a+r\cos t)\sin\vartheta\,,\,r\sin t)$$

(continued)

Example 2 (continued)
where $(t, \vartheta) \in [0, 2\pi] \times [0, 2\pi]$, we obtain

$$|(\varphi_t \wedge \varphi_\vartheta)(t, \vartheta)| = r(a + r\cos t)$$

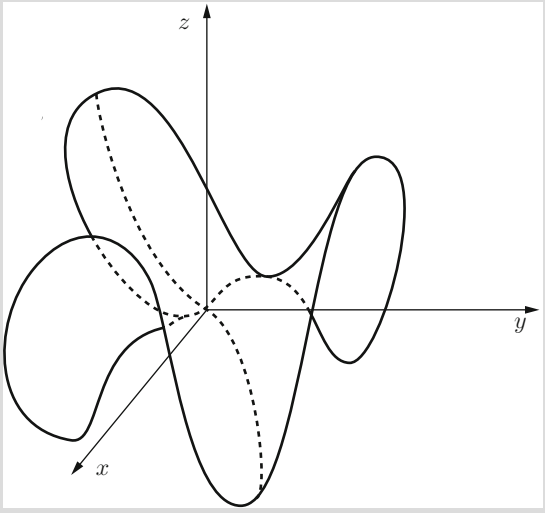
by (10.10). Hence

$$A(\varphi) = \int_0^{2\pi} dt \int_0^{2\pi} r(a + r\cos t)\, d\vartheta = 4\pi^2 ar\,.$$

Example 3 Let φ be the graph of the function $f(x, y) = x^3 - 3xy^2$, where $(x, y) \in D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$ (see Fig. 10.17, or the computer-generated Fig. 3.23). Then

$$\begin{aligned} A(\varphi) &= \iint_D \sqrt{1 + (3x^2 - 3y^2)^2 + 36x^2y^2}\, dx\, dy = \\ &= \iint_D \sqrt{1 + 9(x^2 + y^2)^2}\, dx\, dy = \\ &= 2\pi \int_0^1 \varrho\sqrt{1 + 9\varrho^2}\, d\varrho = \frac{2\pi}{27}(10\sqrt{10} - 1)\,. \end{aligned}$$

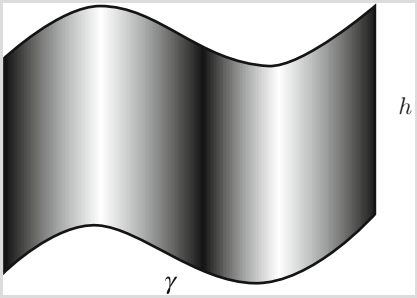
Fig. 10.17



(continued)

Example 4 (continued)

Fig. 10.18 $A(\varphi) = h\, L(\gamma)$



Example 4 Let φ be the cylindrical surface of equations $\varphi(u, v) = (x(u), y(u), v)$, with $u \in [a, b]$, $v \in [0, h]$, generated by the plane curve γ. By (10.7)

$$A(\varphi) = \int_0^h dv \int_a^b \sqrt{(x'(u))^2 + (y'(u))^2}\, du = h\, L(\gamma)\,.$$

That is to say, *the area of φ equals h times the length $L(\gamma)$ of the generating curve γ* (Fig. 10.18).

Consider now the surface of revolution generated by rotating a regular curve γ, of equations $\gamma(u) = (x(u), z(u))$, $u \in [a, b]$, $x(u) > 0$ for any $u \in (a, b)$, by $\alpha \in (0, 2\pi]$. From (10.9) the area $A(\varphi)$ of the regular surface φ thus obtained is

$$\begin{aligned} A(\varphi) &= \int_0^\alpha dv \int_a^b x(u)\sqrt{(x'(u))^2 + (z'(u))^2}\, du = \alpha \int_\gamma x\, ds = \\ &= \alpha \cdot \ell \cdot x_B\,, \end{aligned} \tag{10.22}$$

where ℓ is the length of the rotating curve γ, while

$$x_B = \frac{1}{\ell} \int_\gamma x\, ds$$

is the first coordinate of the centroid B of γ (see Sect. 6.4), which in this case coincides with the distance of B to the axis. Then (10.22) can be formulated as follows.

Guldinus Theorem for Surfaces of Revolution *The area of the surface generated by rotating a regular curve γ by an angle α equals the length the curve times the length of circular arc travelled by the centroid.*

Example 5 The Guldinus theorem immediately gives us the area of the torus φ, already computed in Example 2. The length of the rotating curve γ is $2\pi r$, while its centroid, i.e. the centre of the circle γ, has distance to the axis equal to a, so it describes a circle of length $2\pi a$. Therefore

$$A(\varphi) = 2\pi r \cdot 2\pi a = 4\pi^2 ar\,.$$

The area formula generalises to piecewise-regular surface. We say that φ is a *piecewise-regular surface* when $\varphi : D \to \mathbb{R}^3$ is continuous and D decomposes into regular domains $\{D_1, D_2, \ldots, D_N\}$, with pairwise-disjoint interiors, so that

$$\varphi_i = \varphi_{|_{D_i}}$$

(φ_i is the restriction of φ to D_i) is a regular surface for any $i = 1, 2, \ldots, N$. If φ is a piecewise-regular surface one defines

$$A(\varphi) = \sum_{i=1}^{N} \iint_{D_i} |\varphi_u \wedge \varphi_v|\, du\, dv\,.$$

This definition can be shown to be independent of the specific parametrisation and also of the chosen decomposition of D.

10.5 Orientable Surfaces: Surfaces with Boundary

Let $\varphi : D \to \mathbb{R}^3$ be a regular surface with support S. In Sect. 10.3 we defined the normal to S at any point of S_0, the image under φ of the interior points of the domain D. We say that φ is an *orientable surface* if the unit normal vector field can be extended from S_0 to S so that the resulting map

$$\nu : P \in S \to \nu(P) \in \mathbb{R}^3$$

is continuous on S.

If $\psi : T \to \mathbb{R}^3$ is a parametrisation of the orientable surface φ obtained by the reparametrisation $\Phi : T \to D$, we have seen that

$$\frac{\psi_s \wedge \psi_t}{|\psi_s \wedge \psi_t|} = \frac{J_\Phi}{|J_\Phi|} \cdot \frac{\varphi_u \wedge \varphi_v}{|\varphi_u \wedge \varphi_v|}\,.$$

That is, when changing parametrisation the normal vector field either stays the same or changes sign. Then we can define the following: the surfaces φ, ψ are called *equivalent under the relation* $\overset{\circ}{\sim}$ if the Jacobian determinant of Φ is positive on $\overset{\circ}{T}$.

Evidently $\overset{\circ}{\sim}$ is an equivalence relation on the set of orientable surfaces, and is stronger than the relation $\sim$ introduced in Sect. 10.2. Given an orientable surface φ, the $\sim$-equivalence class $[\varphi]$ consists of two equivalence classes under $\overset{\circ}{\sim}$, each of which is called an *orientation of the surface*.

Example 1 The Cartesian surface φ of equations $\varphi(x, y) = (x, y, f(x, y))$, with $f : D \to \mathbb{R}$, is clearly orientable. The two possible orientations correspond to the unit normal pointing upwards, i.e. $(\nu, e_3) > 0$ at every point on S, or downwards, i.e. $(\nu, e_3) < 0$ (see Fig. 10.19).

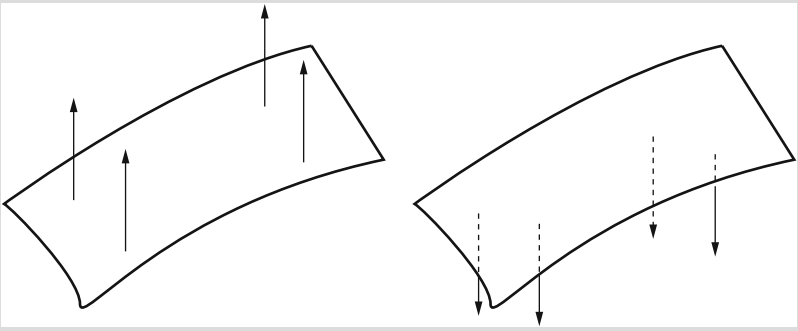


Fig. 10.19

Example 2 The sphere is an orientable surface. For example, if $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = r^2\}$, the two orientations correspond to the unit normal field being

$$P \equiv (x, y, z) \in S \to \nu(P) = \left(\frac{x}{\sqrt{x^2 + y^2 + z^2}}, \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

or the opposite, as in Fig. 10.20.

(continued)

Example 2 (continued)

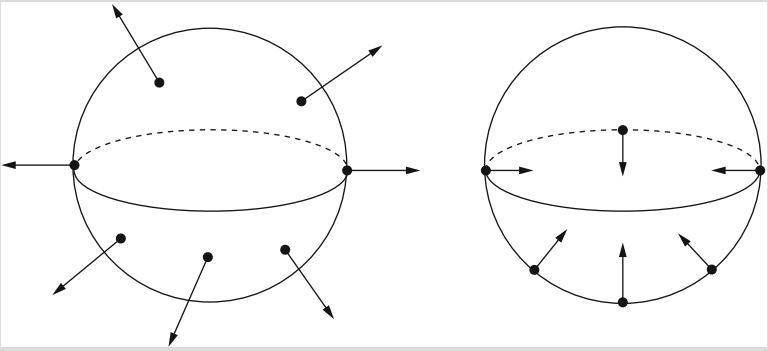


Fig. 10.20

In general, consider a surface φ generated by the rotation, around a coplanar axis, of a simple and regular plane curve γ. If γ does not meet the axis, the surface is orientable, as is immediate from (10.8).

Example 3 A remarkable non-orientable surface is the *Möbius strip* S. We can parametrise it by $\varphi : [0, 2\pi] \times [-1, 1] \to \mathbb{R}^3$,

$$\varphi(u, v) = \left(\left(2 - v \sin \frac{u}{2}\right) \sin u, \left(2 - v \sin \frac{u}{2}\right) \cos u, v \cos \frac{u}{2}\right).$$

The surface S can be constructed taking the segment AB in the yz-plane defined by $y = 2$ and $|z| \leq 1$ and the circle γ in the xy-plane with centre the origin and radius 2. Let the midpoint C of AB move along γ so that when it has travelled an angle u around the z-axis, the segment AB gets tilted by an angle $u/2$. In this way, when C has gone around the z-axis once, the segment AB returns to its initial vertical position but upside down (the endpoints are interchanged, see Fig. 10.21 and the computer-generated Fig. 10.22).

(continued)

Example 3 (continued)

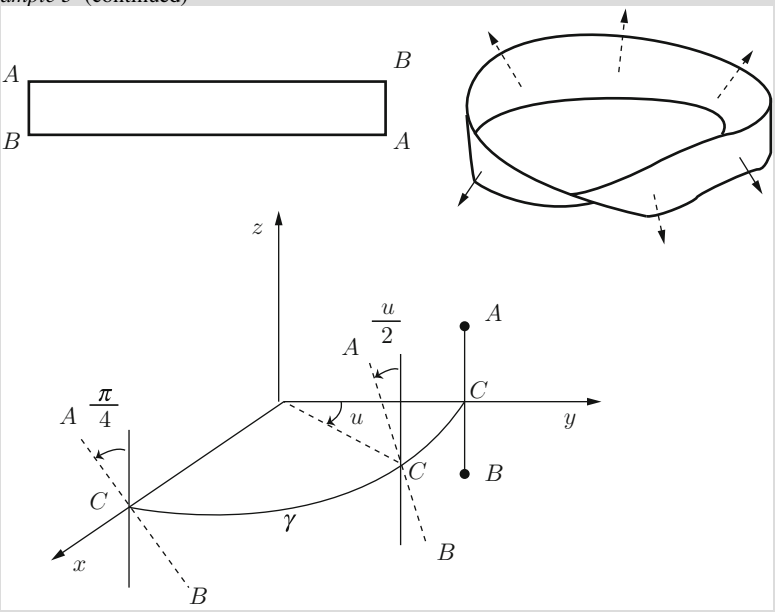


Fig. 10.21

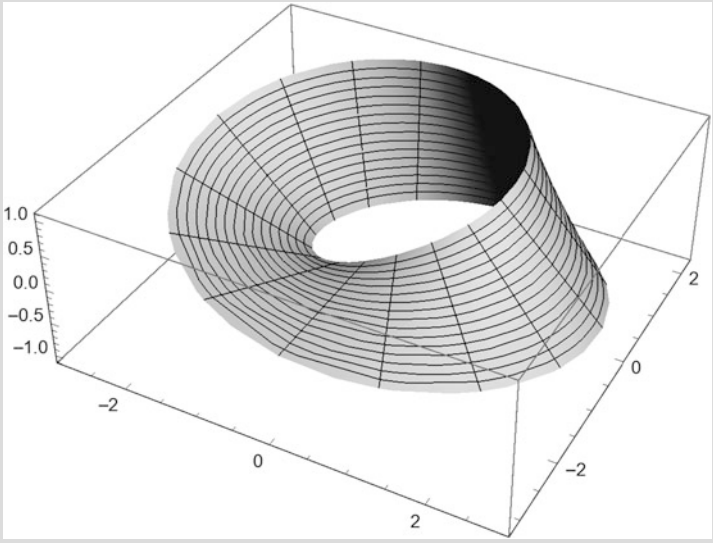


Fig. 10.22

(continued)

Example 3 (continued)
Let us find the unit normal to S at $\varphi(u, 0)$, with $u \in (0, 2\pi)$, corresponding to the points on the circle γ reached by C during the rotation. Since

$$\varphi_u(u, 0) = 2(\cos u, -\sin u, 0)\,,$$

$$\varphi_v(u, 0) = \left(-\sin u \sin\frac{u}{2}, -\cos u \sin\frac{u}{2}, \cos\frac{u}{2}\right)\,,$$

$$\varphi_u(u, 0) \wedge \varphi_v(u, 0) = -2\left(\sin u \cos\frac{u}{2}, \cos u \cos\frac{u}{2}, \sin\frac{u}{2}\right)\,,$$

the unit normal to S at $\varphi(u, 0)$ is

$$N(u, 0) = -\left(\sin u \cos\frac{u}{2}, \cos u \cos\frac{u}{2}, \sin\frac{u}{2}\right)\,.$$

Note however

$$\lim_{u\to 0^+} N(u, 0) = -e_2\,, \qquad \lim_{u\to 2\pi^-} N(u, 0) = e_2\,,$$

so after a complete revolution around the z-axis the unit normal at C has changed orientation. The same happens to every other point on AB. It is then evident that the unit normal vector field cannot be extended continuously to the entire surface, which makes S non-orientable.

Given a connected and regular domain D on the plane, a *regular surface with boundary* is a map $\varphi : D \to \mathbb{R}^3$, restriction to D of a C^1 map on an open set $A \supset D$, satisfying the following conditions:

(i) φ is one-to-one on D;
(ii) the Jacobian matrix $D\varphi(u, v)$ has rank 2 for any $(u, v) \in D$.

A regular surface with boundary is clearly regular. It is also always orientable. The image of the boundary ∂D of D is a regular curve on the surface's support S, called *boundary ∂S of the surface*.

We saw that the parametrisation φ determines an orientation of the surface; now we will show that it also induces an orientation on its boundary ∂S.

Let $\gamma : [a, b] \to \mathbb{R}^2$ be a piecewise-regular curve with support ∂D. If its positive orientation coincides with the positive orientation of the boundary of D, we say that the curve $\varphi \circ \gamma : [a, b] \to \mathbb{R}^3$ *induces the (positive) orientation on the boundary* ∂S of the surface. In the sequel we shall denote this orientation with $\partial^+ S$.

In a similar way one defines the boundary's positive orientation in case the boundary of D is a finite union of piecewise-regular curves (Fig. 10.23).

Example 4 Consider a C^1 function $f : D \to \mathbb{R}$ over a regular domain D in the plane. The Cartesian surface $\varphi(x, y) = (x, y, f(x, y))$, for $(x, y) \in D$, is a regular surface with boundary. Figure 10.23 shows the orientation of the surface and the boundary's positive orientation given by the parametrisation φ.

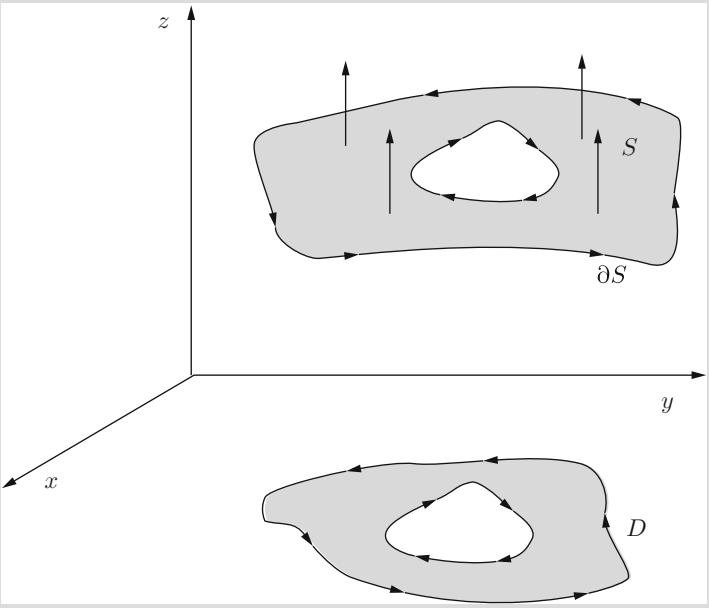


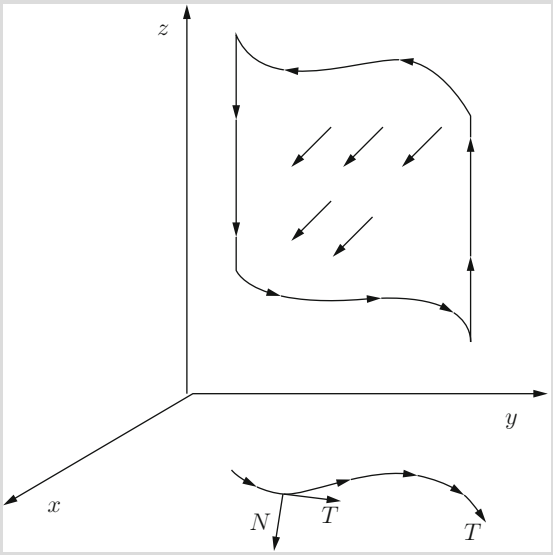
Fig. 10.23

Example 5 Consider the cylindrical surface of equations $\varphi(u, v) = (x(u), y(u), v)$, with $(u, v) \in [a, b] \times [c, d]$, generated by the regular, open, simple curve $\gamma(u) = (x(u), y(u))$, $u \in [a, b]$. Figure 10.24 shows the orientation of γ, and the orientations of the cylindrical surface and its boundary induced by the parametrisation φ (cf. the analytical expression of the surface's normal, computed in Example 2, Sect. 10.3).

(continued)

Example 5 (continued)

Fig. 10.24



10.6 Surface Integrals

Let $\varphi : D \to \mathbb{R}^3$ be a regular surface defined on the connected regular domain D, and call $S = \varphi(D)$ the support. Given a continuous function $f : S \to \mathbb{R}$, the quantity

$$\int_S f \, d\sigma = \iint_D f(\varphi(u, v)) \cdot |\varphi_u(u, v) \wedge \varphi_v(u, v)| \, du \, dv \tag{10.23}$$

is the *(surface) integral* of f *over* S. Let us point out that this notion can be introduced in a similar way to how we defined integrals along curves in Sect. 6.4, which the readers can pursue by themselves in this situation. We also emphasise that surface integrals satisfy, apart from the properties of linearity, monotonicity and additivity that are common to all types of integrals seen so far, the following relationships:

$$\int_S d\sigma = A(\varphi) \, ;$$

$$\left| \int_S f \, d\sigma \right| \leq \int_S |f| \, d\sigma \leq \max_S |f| \cdot A(\varphi) \, .$$

Definition (10.23) does not depend on the particular parametrisation φ of the surface. If $\Phi : T \to D$ is a change of variables and $\psi = \varphi \circ \Phi : T \to \mathbb{R}^3$ the resulting parametrisation of S, by (10.5) and the formula of change of variables in double integrals we have

$$\begin{aligned}
&\iint_T f(\psi(s,t)) \cdot |(\psi_s \wedge \psi_t)(s,t)| \, ds\, dt = \\
&= \iint_T f(\psi(s,t)) \cdot |(\varphi_u \wedge \varphi_v)(\Phi(s,t))| \cdot |J_\Phi(s,t)| \, ds\, dt = \\
&= \iint_D f(\varphi(u,v)) \cdot |(\varphi_u \wedge \varphi_v)(u,v)| \cdot \left| J_\Phi\left(\Phi^{-1}(u,v)\right) \right| \cdot |J_{\Phi^{-1}}(u,v)| \, du\, dv = \\
&= \iint_D f(\varphi(u,v)) \cdot |(\varphi_u \wedge \varphi_v)(u,v)| \, du\, dv.
\end{aligned}$$

Before we pass to some applications of surface integrals, with interest especially in Physics, observe that (10.23) generalises in the obvious manner to piecewise-regular surfaces.

Let $\varrho : S \to [0, +\infty)$ be the *density* of a distribution of mass on the surface S. The total mass m is then

$$m = \int_S \varrho \, d\sigma \,,$$

while the distribution's *centre of mass* C has coordinates

$$x_C = \frac{1}{m} \int_S x\, \varrho \, d\sigma \,, \qquad y_C = \frac{1}{m} \int_S y \varrho \, d\sigma \,, \qquad z_C = \frac{1}{m} \int_S z\, \varrho \, d\sigma \tag{10.24}$$

In case of a uniform mass distribution, i.e. when ϱ is constant, formulas (10.24) give the coordinates of the surface's *centroid* B

$$x_B = \frac{1}{\text{area}\,(S)} \int_S x \, d\sigma \,, \quad y_B = \frac{1}{\text{area}\,(S)} \int_S y \, d\sigma \,, \quad z_B = \frac{1}{\text{area}\,(S)} \int_S z \, d\sigma \,.$$

The *moment of inertia* of the given distribution about a certain axis r is the quantity

$$\mathcal{M}_r = \int_S \varrho \, d^2 \, d\sigma \,,$$

where $d = d(x, y, z)$ is the distance of the point (x, y, z) from r.

Example 1 Consider a uniform mass distribution, with density $\varrho = 1$, on the lateral surface S of the cone with vertex at the origin and based on the circle $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 1, z = 1\}$. Let us calculate the moment of inertia about the x-axis. Since $\varrho = 1$ and $d^2 = y^2 + z^2$, we must compute

$$\mathcal{M}_x = \int_S (y^2 + z^2)\, d\sigma\,. \tag{10.25}$$

Now, S can be parametrised by (see Example 3, Sect. 10.1) the map $\varphi : [0, 1] \times [0, 2\pi] \to \mathbb{R}^3$,

$$\varphi(u, v) = (u \cos v, u \sin v, u)\,.$$

As $|(\varphi_u \wedge \varphi_v)\,(u, v)| = \sqrt{2}\, u$, from (10.23) and (10.25) it follows that

$$\mathcal{M}_x = \int_0^1 du \int_0^{2\pi} \left(u^2 \sin^2 v + u^2\right) \sqrt{2} u\, dv = \\ = \sqrt{2} \int_0^1 u^3\, du \cdot \int_0^{2\pi} \left(\sin^2 v + 1\right) dv = \frac{3\sqrt{2}\pi}{4}\,.$$

Let $\varphi : D \to \mathbb{R}^3$ be an orientable regular surface of equations $\varphi(u, v) = (x(u, v), y(u, v), z(u, v))$, and let $\nu(u, v)$ be the unit normal at the point $\varphi(u, v)$. If $F : S \to \mathbb{R}^3$ is a continuous vector field defined on the support S of φ, we call *flux of the vector field F across S* the integral

$$\int_S (F, \nu)\, d\sigma\,. \tag{10.26}$$

As

$$\nu = \frac{(\varphi_u \wedge \varphi_v)\,(u, v)}{|(\varphi_u \wedge \varphi_v)(u, v)|}\,,$$

and recalling that the components of $(\varphi_u \wedge \varphi_v)(u, v)$ are

$$\det \frac{\partial(y, z)}{\partial(u, v)}\,, \qquad \det \frac{\partial(z, x)}{\partial(u, v)}\,, \qquad \det \frac{\partial(x, y)}{\partial(u, v)}\,,$$

we find that the surface integral (10.26) equals the double integral

$$\iint_D \left\{ F_1(\varphi(u, v)) \det \frac{\partial(y, z)}{\partial(u, v)} + F_2(\varphi(u, v)) \det \frac{\partial(z, x)}{\partial(u, v)} + \right. \\ \left. + F_3(\varphi(u, v)) \det \frac{\partial(x, y)}{\partial(u, v)} \right\} du\, dv\,. \tag{10.27}$$

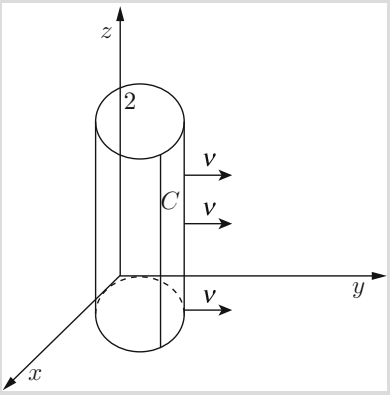
Notice that changing the orientation of S will reverse the unit normal ν and hence (10.26) changes sign. If S is the boundary of a domain T in space, one speaks of *ingoing flux* or *outgoing flux* depending on whether the normal vector field to S points outside (outer normal) or inside (inner normal) of the domain T.

Example 2 Consider the curve γ in the Oxy plane of polar equation

$$\varrho = \sin\vartheta\cos\vartheta\,, \qquad 0 \le \vartheta \le \frac{\pi}{2}\,.$$

Let C be the right cylinder of height $h = 2$ over the plane domain enclosed by γ, contained in the half-space $H = \{(x, y, z) \in \mathbb{R}^3 : z \ge 0\}$, as in Fig. 10.25.

Fig. 10.25



Example 3 Let us find the ingoing flux of the vector field $F = (\sqrt{x^2+y^2}, 0, z^2)$ across the cylinder's lateral surface S. The latter can be parametrised by

$$\varphi(u,v) = \begin{cases} x = \sin u \cos^2 u \\ y = \sin^2 u \cos u \\ z = v\,, \end{cases} \qquad (u,v) \in \left[0, \frac{\pi}{2}\right] \times [0,2]\,.$$

The components of $\varphi_u \wedge \varphi_v$ are then

$$\begin{aligned} \det \frac{\partial(y,z)}{\partial(u,v)} &= 2\sin u\cos^2 u - \sin^3 u\,, \\ \det \frac{\partial(z,x)}{\partial(u,v)} &= -\cos^3 u + 2\sin^2 u\cos u\,, \\ \det \frac{\partial(x,y)}{\partial(u,v)} &= 0\,. \end{aligned} \tag{10.28}$$

(continued)

Example 3 (continued)
At the points of S of local coordinates $(\pi/4, v)$ we have

$$(\varphi_u \wedge \varphi_v)\Big(\frac{\pi}{4}, v\Big) = \Big(\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, 0\Big),$$

so there the normal

$$\nu\Big(\frac{\pi}{4}, v\Big) = \Big(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0\Big)$$

points outwards. The ingoing flux across S is then

$$-\int_S (F, \nu)\, d\sigma$$

which, by (10.27) and (10.28), equals

$$-\int_0^{\pi/2} du \int_0^2 \sin u \cos u \det \frac{\partial(y, z)}{\partial(u, v)}\, dv =$$
$$= -2\int_0^{\pi/2} \sin u \cos u (2 \sin u \cos^2 u - \sin^3 u)\, du = -\frac{2}{15}\,.$$

10.7 Stokes's Formula and the Divergence Theorem

Recall (see Sect. 7.5) that the *curl* of a *vector field* $F = (F_1, F_2, F_3)$ is the vector field of components

$$\operatorname{curl} F = \Big(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\Big)\,.$$

Stokes's Formula *Let D be a regular domain and $\varphi : D \to \mathbb{R}^3$ a regular surface with C^2 boundary. Suppose that $F : A \to \mathbb{R}^3$ is a C^1 vector field on an open set $A \subseteq \mathbb{R}^3$ containing the surface's support S. Then*

$$\int_S (\operatorname{curl} F, \nu)\, d\sigma = \int_{\partial^+ S} (F, T)\, ds\,, \tag{10.29}$$

where ν is the normal vector field to S and T is the tangent vector field to the boundary $\partial^+ S$ equipped with the induced orientation of S.

Relation (10.29), which as we have already said is called *Stokes's formula*, formalises the fact that the line integral of F along the boundary of S (the right-hand-side integral) equals the *flux* of the field's curl across the surface (the left-hand-side integral).

Example 1 Let φ be the surface of equations

$$\begin{cases} x = u - v \\ y = u \\ z = u^2 + v^2 \,, \end{cases}$$

where $(u, v) \in D = \{u^2 + v^2 \le 1\}$ (the reader should check that this surface is a portion of a paraboloid). Using the Stokes formula we compute

$$\int_{\partial^+ S} x^2 z \, dx + y \, dy + yz \, dz \,. \tag{10.30}$$

Calling F the vector field of components $(x^2 z, y, yz)$, the integral in (10.30) is equal to

$$\int_{\partial^+ S} (F, T) \, ds \,.$$

Since curl $F = (z, x^2, 0)$, by (10.29) the required integral equals

$$\iint_D (u^2 + v^2) \det \frac{\partial(y, z)}{\partial(u, v)} + (u - v)^2 \det \frac{\partial(z, x)}{\partial(u, v)} \, du \, dv \,, \tag{10.31}$$

and because

$$\det \frac{\partial(y, z)}{\partial(u, v)} = 2v \,, \qquad \det \frac{\partial(z, x)}{\partial(u, v)} = -2(u + v) \,,$$

(10.31) reduces to

$$2 \iint_D (u^2 + v^2)v - (u - v)^2(u + v) \, du \, dv = 2 \iint_D u(v^2 - u^2 + 2uv) \, du \, dv = 0 \,.$$

Proof of Stokes's Formula. For simplicity we shall prove (10.29) when the boundary ∂D is the support of a regular curve $\gamma : [a, b] \to \mathbb{R}^2$ of equations $\gamma(t) = (u(t), v(t))$. We leave it to the reader to verify that the formula still holds if ∂D is a finite union of piecewise-regular curves.

Suppose the orientation induced on ∂D by the parametrisation γ coincides with the positive orientation of ∂D. If $X(x, y, z)$, $Y(x, y, z)$, $Z(x, y, z)$ are the components of the vector field F, formula (10.29) becomes

$$\begin{aligned} &\int_{\partial^+ S} \omega = \\ &= \iint_D \left\{ (Z_y - Y_z) \det \frac{\partial(y, z)}{\partial(u, v)} + (X_z - Z_x) \det \frac{\partial(z, x)}{\partial(u, v)} + (Y_x - X_y) \det \frac{\partial(x, y)}{\partial(u, v)} \right\} du dv \,, \end{aligned} \tag{10.32}$$

where $\omega = X\,dx + Y\,dy + Z\,dz$ and $(x(u,v), y(u,v), z(u,v))$ is a parametrisation of the surface. Since the oriented curve $\partial^+ S$ has equations $x = x(u(t), v(t))$, $y = y(u(t), v(t))$, $z = z(u(t), v(t))$, $t \in [a, b]$, we have

$$\int_{\partial^+ S} \omega = \int_a^b \left\{ (X\,x_u + Y\,y_u + Z\,z_u)\,u' + (X\,x_v + Y\,y_v + Z\,z_v)\,v' \right\} dt =$$
$$= \int_{+\partial D} (X\,x_u + Y\,y_u + Z\,z_u)\,du + (X\,x_v + Y\,y_v + Z\,z_v)\,dv\,.$$

Applying to the last integral the Stokes formula for the plane (see Sect. 8.3) we deduce

$$\int_{\partial^+ S} \omega = \iint_D \left\{ \frac{\partial}{\partial u}(X\,x_v + Y\,y_v + Z\,z_v) - \frac{\partial}{\partial v}(X\,x_u + Y\,y_u + Z\,z_u) \right\} du dv =$$
$$= \iint_D \left\{ \left(\frac{\partial X}{\partial u} x_v + \frac{\partial Y}{\partial u} y_v + \frac{\partial Z}{\partial u} z_v \right) - \left(\frac{\partial X}{\partial v} x_u + \frac{\partial Y}{\partial v} y_u + \frac{\partial Z}{\partial v} z_u \right) \right\} du dv\,. \tag{10.33}$$

Notice that

$$\frac{\partial X}{\partial u} x_v - \frac{\partial X}{\partial v} x_u =$$
$$= \big(X_x\,x_u + X_y\,y_u + X_z\,z_u\big)\,x_v - \big(X_x\,x_v + X_y\,y_v + X_z\,z_v\big)\,x_u =$$
$$= -X_y\,(x_u\,y_v - x_v\,y_u) + X_z\,(z_u\,x_v - z_v\,x_u)\ ,$$

and similarly

$$\frac{\partial Y}{\partial u} y_v - \frac{\partial Y}{\partial v} y_u = Y_x (x_u\,y_v - x_v\,y_u) - Y_z (y_u\,z_v - y_v\,z_u)$$
$$\frac{\partial Z}{\partial u} z_v - \frac{\partial Z}{\partial v} z_u = -Z_x (z_u\,x_v - z_v\,x_u) + Z_y (y_u\,z_v - y_v\,z_u)\,.$$

By (10.33), the above relations give

$$\int_{\partial^+ S} \omega = \iint_D \Big\{ \big(Z_y - Y_z\big)(y_u\,z_v - y_v\,z_u) +$$
$$+ (X_z - Z_x)(z_u\,x_v - z_v\,x_u) + \big(Y_x - X_y\big)(x_u\,y_v - x_v\,y_u) \Big\}\, du dv\,,$$

and (10.32) is proved.

Let us remark that the C^2 hypothesis is not needed to state Stokes's formula. It is only used in the proof, to be precise for writing (10.33). As a matter of fact formula (10.29) is valid when φ is only of class C^1.

Now let us discuss another key result. Recall that *regular domains* were defined in Sect. 8.5.

Divergence Theorem *Let T be a regular domain in $\mathbb{R}^3$. If $F : T \to \mathbb{R}^3$ is a C^1 vector field, then*

$$\iiint_T \operatorname{div} F \, dx\, dy\, dz = \int_S (F, \nu_e)\, d\sigma \,, \tag{10.34}$$

where ν_e is the outer normal of S and the function

$$\operatorname{div} F = \frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} + \frac{\partial Z}{\partial z}$$

is called the divergence of $F = (X, Y, Z)$.

Formula (10.34) expresses the fact that the integral over T of the *divergence* of F equals the outgoing *flux*.

Example 2 Let $T = \{(x, y, z) \in \mathbb{R}^3 : \ x^2 + y^2 \le z^2, \ x^2 + y^2 + z^2 \le 2y, \ z \ge 0\}$ (see Figs. 10.26 and 10.27). We shall find the outgoing flux of the vector field $F = (0, yz, x)$. The set T is the intersection between the cone with vertex the origin with generating lines forming an angle $\pi/2$, and the unit hemisphere with centre $(0, 1, 0)$ in the upper half-space $H = \{(x, y, z) \in \mathbb{R}^3 : \ z \ge 0\}$.

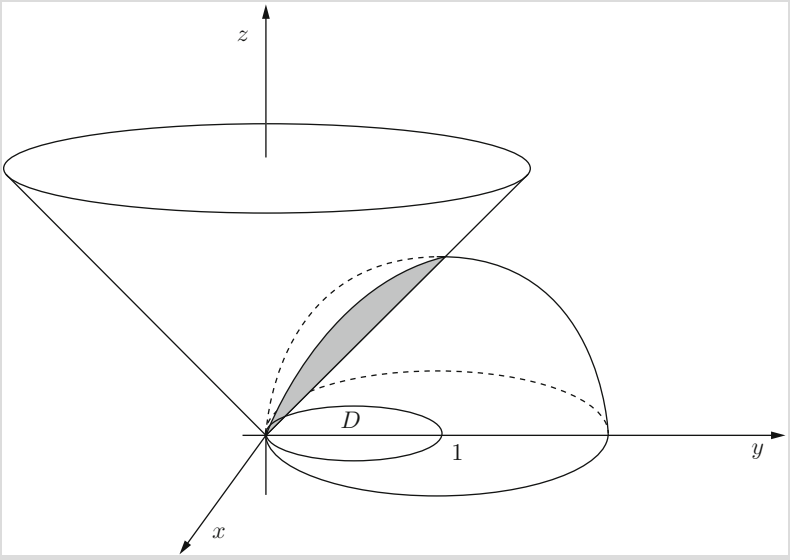


Fig. 10.26

(continued)

Example 2 (continued)

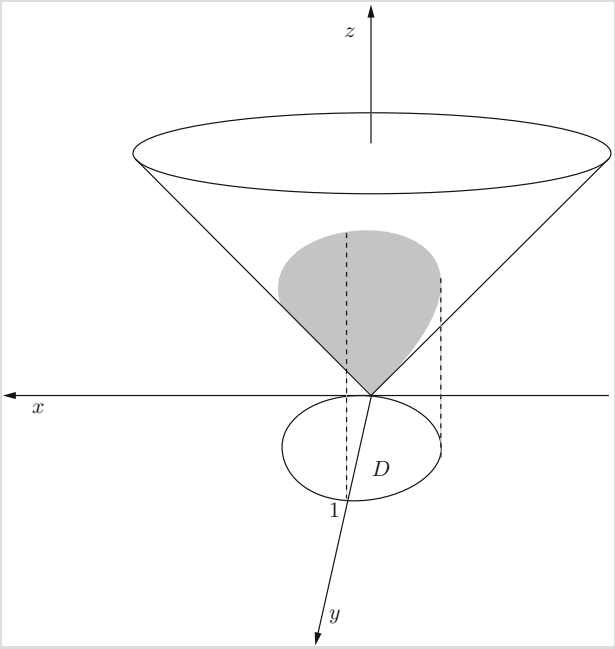


Fig. 10.27

The projection of T on Oxy is the disc $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 - y \le 0\}$, so T may be viewed as the normal domain defined by

$$\sqrt{x^2 + y^2} \le z \le \sqrt{2y - x^2 - y^2}\,, \qquad (x, y) \in D$$

(see Fig. 10.26). As $\operatorname{div} F = z$, by (10.34) we have

$$\int_{\partial T} (F, \nu_e)\, d\sigma = \iiint_T z\, dx\, dy\, dz =$$

$$= \iint_D dx\, dy \int_{\sqrt{x^2+y^2}}^{\sqrt{2y-x^2-y^2}} z\, dz = \iint_D \left(y - x^2 - y^2\right) dx\, dy\,.$$

The last integral can be computed in polar coordinates, so that

$$\int_{\partial T} (F, \nu_e)\, d\sigma = \int_0^{\pi} d\vartheta \int_0^{\sin\vartheta} \left(\varrho^2 \sin\vartheta - \varrho^3\right) d\varrho = \frac{1}{12} \int_0^{\pi} \sin^4\vartheta\, d\vartheta = \frac{\pi}{32}\,.$$

Proof of the Divergence Theorem We shall prove (10.34) in the simplest case, where the domain T is regular and normal with respect to all three coordinate planes. Call $X(x, y, z)$, $Y(x, y, z)$, $Z(x, y, z)$ the components of the vector field F. Then

$$\iiint_T \operatorname{div} F \, dx\, dy\, dz = \iiint_T \left(\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} + \frac{\partial Z}{\partial z} \right) dx\, dy\, dz \,. \tag{10.35}$$

As T is normal with respect to the xy-plane, there exist a regular domain D in the plane and two functions $f_1, f_2 : D \to \mathbb{R}$, of class $C^1(D)$, such that (see Fig. 10.28)

$$T = \{(x, y, z) : \ (x, y) \in D, \ \ f_1(x, y) \le z \le f_2(x, y)\} \,.$$

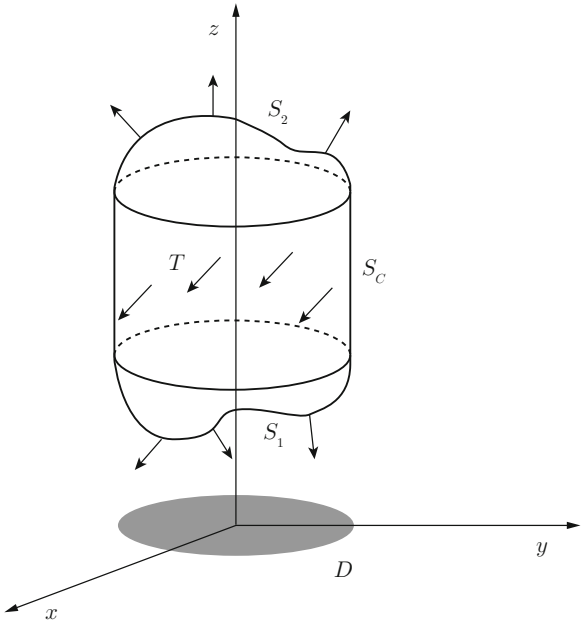
By the reduction formulas for triple integrals (seen in Sect. 8.5)

$$\begin{aligned} \iiint_T \frac{\partial Z}{\partial z} \, dx\, dy\, dz &= \iint_D dx\, dy \int_{f_1(x,y)}^{f_2(x,y)} \frac{\partial Z}{\partial z} \, dz = \\ &= \iint_D \{Z(x, y, f_2(x, y)) - Z(x, y, f_1(x, y))\} \, dx\, dy \,. \end{aligned} \tag{10.36}$$

Now let us compute

$$\int_S (F, \nu_e) \, d\sigma = \int_{\partial T} (F, \nu_e) \, d\sigma = \int_{\partial T} \{X(e_1, \nu_e) + Y(e_2, \nu_e) + Z(e_3, \nu_e)\} \, d\sigma \,. \tag{10.37}$$

Fig. 10.28



Let us focus on the last summand

$$\int_{\partial T} Z(e_3, \nu_e)\, d\sigma\,,$$

where ν_e is the outer normal of ∂T, and ∂T is the union of the graphs S_1, S_2 of f_1, f_2 and the cylindrical surface

$$S_C = \Big\{(x, y, z) \in \mathbb{R}^3 :\ (x, y) \in \partial D,\ \ f_1(x, y) \le z \le f_2(x, y)\Big\}\,.$$

As the normal to S_C is orthogonal to e_3, we have

$$\int_{\partial T} Z(e_3, \nu_e)\, d\sigma = \int_{S_1} Z(e_3, \nu_e)\, d\sigma + \int_{S_2} Z(e_3, \nu_e)\, d\sigma\,.$$

At every point of S_1 the outer unit normal is

$$\nu_{e|_{S_1}} = \frac{1}{\sqrt{1 + |Df_1|^2}}\left(\frac{\partial f_1}{\partial x}\,,\ \frac{\partial f_1}{\partial y}\,, -1\right),$$

while on S_2 the outer unit normal is given by

$$\nu_{e|_{S_2}} = \frac{1}{\sqrt{1 + |Df_2|^2}}\left(-\frac{\partial f_2}{\partial x}\,, -\frac{\partial f_2}{\partial y}\,,\ 1\right).$$

Recalling the definition of surface integral, we then have

$$\int_{\partial T} Z(e_3, \nu_e)\, d\sigma = -\iint_D Z(x, y, f_1(x, y))\, dx\, dy + \iint_D Z(x, y, f_2(x, y))\, dx\, dy\,.$$

Therefore (10.36) implies

$$\iiint_T \frac{\partial Z}{\partial z}\, dx\, dy\, dz = \int_{\partial T} Z(e_3, \nu_e)\, d\sigma\,.$$

In a similar manner one proves that

$$\iiint_T \frac{\partial Y}{\partial y}\, dx\, dy\, dz = \int_{\partial T} Y(e_2, \nu_e)\, d\sigma\,,$$
$$\iiint_T \frac{\partial X}{\partial x}\, dx\, dy\, dz = \int_{\partial T} X(e_1, \nu_e)\, d\sigma\,.$$

Comparing the last equalities with (10.35) and (10.37) proves the claim.



Implicit Functions

11

11.1 The Implicit Function Theorem for Equations

Let $F(x, y)$ be a function of two real variables x, y. When we consider the equation

$$F(x, y) = 0 \tag{11.1}$$

and we fix a value of x, there could be either *no* value of y solving the equation, or *exactly one* y, or even *several* values of y satisfying (11.1). It is then of interest to establish under which conditions Eq. (11.1) allows to associate with x a *unique* value of y, i.e. when y can be expressed *as a function of* x, at least for some values of the variables. Said more precisely, we shall determine conditions for the existence of an interval $I \subseteq \mathbb{R}$ and of a unique function $f : I \to \mathbb{R}$ such that

$$F(x, f(x)) = 0\,, \qquad \forall\, x \in I. \tag{11.2}$$

After presenting a few examples of the possible scenarios we will show that under certain assumptions, if the point (x_0, y_0) is such that

$$F(x_0, y_0) = 0\,,$$

then there exist an open interval I containing x_0 and a unique function $f : I \to \mathbb{R}$ that satisfies (11.2) and $f(x_0) = y_0$.

In this way the zero set

$$Z = \{(x, y) :\ F(x, y) = 0\}$$

of the function F, restricted to a neighbourhood of (x_0, y_0), will be a *Cartesian graph* over the interval I, as in Fig. 11.1. The function $f : I \to \mathbb{R}$ is defined implicitly on I by Eq. (11.1), and therefore it is called an *implicit function*.

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https://doi.org/10.1007/978-3-031-04151-8_11

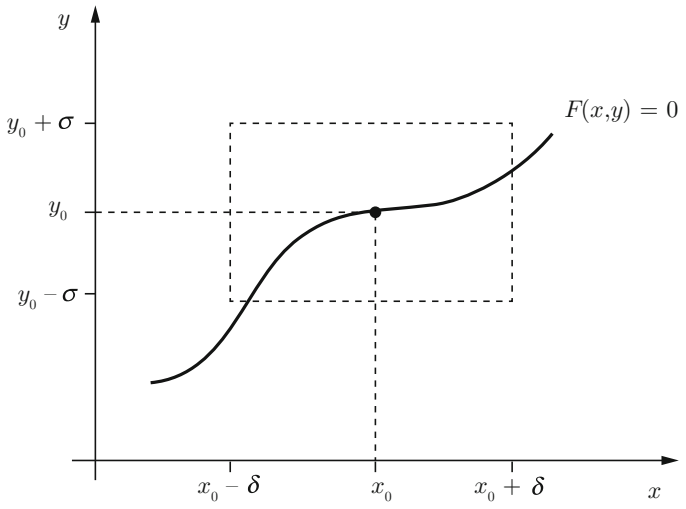


Fig. 11.1

Example 1 Consider $F(x, y) = y^5 - x$, with $(x, y) \in \mathbb{R}^2$. If $I = \mathbb{R}$, clearly the only function implicitly defined by (11.1), i.e. such that

$$F(x, f(x)) = 0\,, \qquad \forall\, x \in \mathbb{R},$$

is $f(x) = \sqrt[5]{x}$.

Example 2 Take $F(x, y) = x^2 + y^2 - 1$ with $x, y \in \mathbb{R}$. Consider the interval $I = [-1, 1]$ and let f_1, f_2 be two functions defined on I by

$$f_1(x) = \sqrt{1 - x^2}\,, \qquad f_2(x) = -\sqrt{1 - x^2}\,.$$

Evidently $f_1 \neq f_2$ and

$$F(x, f_1(x)) = 0\,, \qquad F(x, f_2(x)) = 0\,, \qquad \forall\, x \in I.$$

So here we have two distinct implicit functions defined on I by the same equation $F(x, y) = 0$.

Example 3 Now consider $F(x, y) = x^4 + y^4 + 1$ for $x, y \in \mathbb{R}$. As $F(x, y) \neq 0$ for any $(x, y) \in \mathbb{R}^2$, for any real interval I there is no function $f : I \to \mathbb{R}$ such that $F(x, f(x)) = 0$ when $x \in I$.

Example 4 Suppose $F(x, y) = x^3 + y^3 + x^2 y - 3y^2$. Equation (11.1) implicitly defines a unique function f defined on $I = (-\infty, -\sqrt{3}]$. In fact, if $x \leq -\sqrt{3}$ the function $y \in \mathbb{R} \to F(x, y)$ is strictly increasing (the reader should show this) and moreover

$$\lim_{y \to \pm\infty} F(x, y) = \pm\infty\,.$$

Hence, for any $x \leq -\sqrt{3}$ there is a unique value y depending on x, i.e. $y = f(x)$, such that $F(x, f(x)) = 0$.

Similarly one proves that, in this situation, (11.1) implicitly defines a unique function f on $[\sqrt{3}, +\infty)$. Notice that the equation $F(0, y) = 0$ has, instead, two solutions $y = 0$ and $y = 3$, while $F(1, y) = 0$ has three solutions: $y = 1$, $y = 1 \pm \sqrt{2}$.

The following theorem prescribes conditions under which $F(x, y) = 0$ expresses y in terms of x for certain values of the variables x, y.

Implicit Function Theorem *Let $F(x, y)$ be a continuous function with continuous partial derivative F_y on an open set $A \subseteq \mathbb{R}^2$. Suppose there is a point (x_0, y_0) in A where*

$$F(x_0, y_0) = 0 \quad \textit{and} \quad F_y(x_0, y_0) \neq 0\,, \tag{11.3}$$

Then there exist positive numbers δ, σ such that the equation

$$F(x, y) = 0$$

implicitly defines a unique function

$$f : (x_0 - \delta, x_0 + \delta) \;\to\; (y_0 - \sigma, y_0 + \sigma)\,,$$

i.e. a function such that

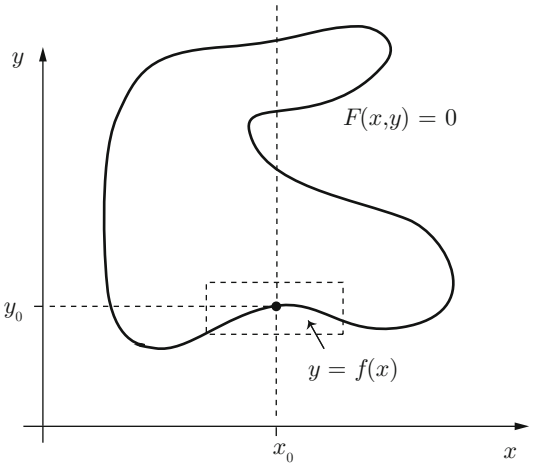
$$F(x, f(x)) = 0\,, \qquad \forall\, x \in (x_0 - \delta, x_0 + \delta).$$

Furthermore, f is continuous and $f(x_0) = y_0$.

Figure 11.2 shows the geometric meaning of the implicit function theorem. In practice, the theorem says that there exists an open rectangle

$$R = (x_0 - \delta, x_0 + \delta) \times (y_0 - \sigma, y_0 + \sigma)$$

Fig. 11.2



whose intersection with the zero set $Z = \{(x, y) \in A : \ F(x, y) = 0\}$ is the graph of a certain continuous function $y = f(x)$. Naturally the "vertical" line $x = x_0$ may cross the zero set at more than one point.

Proof To fix ideas, suppose $F_y(x_0, y_0) > 0$. Referring to Fig. 11.3, by the continuity of $F_y(x, y)$ there is a positive number σ such that

$$F_y(x, y) > 0\,,$$

for any point $(x, y) \in A$ satisfying

$$|x - x_0| \leq \sigma\,, \qquad |y - y_0| \leq \sigma\,.$$

Then the function $F(x_0, y)$, of the sole variable y, is strictly increasing on $[y_0 - \sigma, y_0 + \sigma]$. As $F(x_0, y_0) = 0$, necessarily

$$F(x_0, y_0 - \sigma) < 0\,, \qquad F(x_0, y_0 + \sigma) > 0\,.$$

Since the sign of continuous maps is locally constant, for the two functions $F(x, y_0 - \sigma)$ and $F(x, y_0 + \sigma)$, of the sole variable x, there exists $\delta > 0$ (smaller than or equal to σ) such that

$$F(x, y_0 - \sigma) < 0\,, \qquad F(x, y_0 + \sigma) > 0\,, \tag{11.4}$$

for any $x \in (x_0 - \delta, x_0 + \delta)$. Therefore for any x in such interval, the function

$$y \ \to \ F(x, y)$$

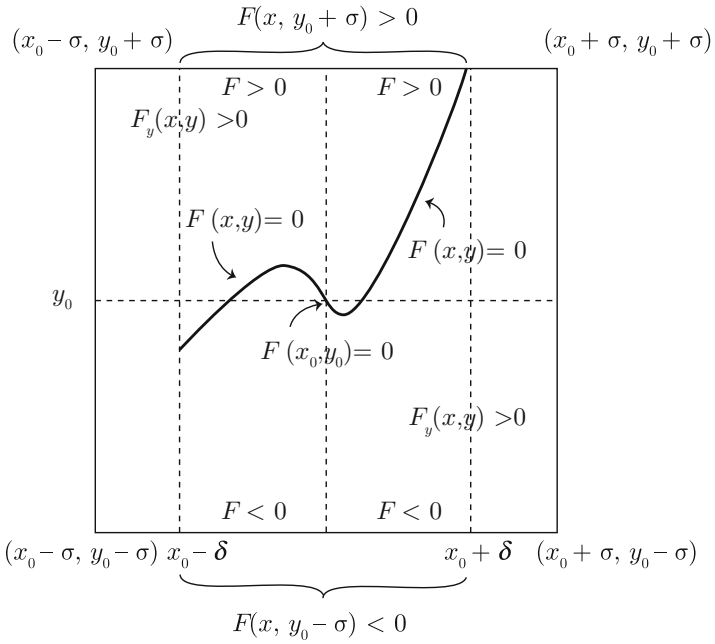


Fig. 11.3

is strictly increasing if $y \in [y_0 - \sigma, y_0 + \sigma]$. By (11.4) it assumes opposite signs at the endpoints of $[y_0 - \sigma, y_0 + \sigma]$, so the intermediate value theorem tells that there is a unique $y \in (y_0 - \sigma, y_0 + \sigma)$ such that $F(x, y) = 0$, for any $x \in (x_0 - \delta, x_0 + \delta)$. This proves the existence of a unique implicit function $y = f(x)$, defined on $(x_0 - \delta, x_0 + \delta)$ and with range contained in $(y_0 - \sigma, y_0 + \sigma)$. By construction $f(x_0) = y_0$.

Now let us show that f is continuous on $(x_0 - \delta, x_0 + \delta)$. To simplify notations call I, J the open intervals

$$I = (x_0 - \delta, x_0 + \delta)\,, \qquad J = (y_0 - \sigma, y_0 + \sigma)\,,$$

and fix $\overline{x} \in I$. We shall prove that for any $\varepsilon > 0$,

$$|f(x) - f(\overline{x})| < \varepsilon \tag{11.5}$$

whenever $x \in I$ is close enough to $\overline{x}$. As $f(\overline{x})$ is internal to J, we may choose ε so that $[f(\overline{x}) - \varepsilon, f(\overline{x}) + \varepsilon] \subset J$.

Recalling that $F(\overline{x}, f(\overline{x})) = 0$ and that the function $y \to F(\overline{x}, y)$ is strictly increasing when $y \in [f(\overline{x}) - \varepsilon, f(\overline{x}) + \varepsilon]$, we have

$$F(\overline{x}, f(\overline{x}) - \varepsilon) < 0 < F(\overline{x}, f(\overline{x}) + \varepsilon)\,.$$

Arguing as above we prove the existence of a positive number δ_1 for which

$$F(x, f(\overline{x}) - \varepsilon) < 0 < F(x, f(\overline{x}) + \varepsilon),$$

for any $x \in I$ such that $|x - \overline{x}| < \delta_1$. But $F(x, f(x)) = 0$, and for any $x \in I$ the function $y \to F(x, y)$ is strictly increasing. Hence

$$f(\overline{x}) - \varepsilon < f(x) < f(\overline{x}) + \varepsilon,$$

and therefore (11.5) follows.

Suppose we replace, in the implicit function theorem, the hypotheses F_y continuous and $F_y(x_0, y_0) \neq 0$ with F_x continuous and $F_x(x_0, y_0) \neq 0$. Then the implicit equation $F(x, y) = 0$ defines a function $x = g(y)$ on a neighbourhood of y_0. Now the zero set $Z = \{(x, y) \in A : \quad F(x, y) = 0\}$ is, on a neighbourhood of (x_0, y_0), a graph over some interval of the y-axis.

Example 5 For any $(x, y) \in \mathbb{R}^2$ let $F(x, y) = x^2 + y^2 - 1$. Pick a point (x_0, y_0) on the circle Γ of equation $x^2 + y^2 = 1$. If $x_0 \in (-1, 1)$ then $y_0 \neq 0$ and so $F_y(x_0, y_0) = 2y_0 \neq 0$. Hence the implicit function theorem says that there is a rectangular neighbourhood R of (x_0, y_0) such that $\Gamma \cap R$ is the graph of a continuous function $f(x)$, defined on $x \in (x_0 - \delta, x_0 + \delta)$ for a suitable $\delta > 0$.

Instead, if $x_0 = 1$, so $y_0 = 0$ and $F_y(1, 0) = 0$, the implicit function theorem cannot be applied. Nonetheless we have $F_x(1, 0) = 2 \neq 0$, so by the theorem Γ coincides, on some rectangular neighbourhood R' of $(1, 0)$, with the graph of a continuous function $g(y)$ defined on a neighbourhood of $y_0 = 0$ (Fig. 11.4).

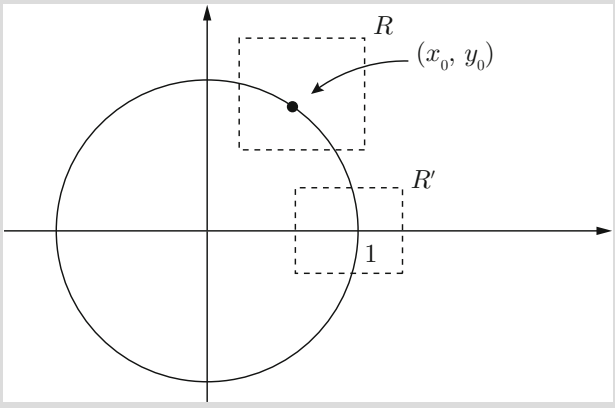


Fig. 11.4

Example 6 Consider $F(x, y) = x^2 - y^2$. At the origin

$$F(0,0) = F_x(0,0) = F_y(0,0) = 0\,.$$

Taking $(0, 0)$, the implicit function theorem's assumptions are not satisfied. Nor is the conclusion, in this case, because there is no neighbourhood of the origin where the equation implicitly defines a unique function in either x or y.

Let us go back to the statement of the implicit function theorem and suppose f is the continuous function from $(x_0 - \delta, x_0 + \delta)$ to $(y_0 - \sigma, y_0 + \sigma)$ defined implicitly by (11.1). Assume F is of class C^1 on A. If f is differentiable, differentiating with respect to x the relation

$$F(x, f(x)) = 0\,, \qquad \forall\, x \in (x_0 - \delta, x_0 + \delta),$$

we obtain

$$F_x(x, f(x)) + F_y(x, f(x))\, f'(x) = 0\,,$$

and in case $F_y \neq 0$,

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))}\,, \qquad \forall\, x \in (x_0 - \delta, x_0 + \delta)\,. \tag{11.6}$$

The next result shows that the above formula is actually true if F is of class C^1 on A.

Derivatives of Implicit Functions *Assuming, in the implicit function theorem, that F is additionally C^1 on the open set A, then the implicit function f is of class C^1 on $(x_0 - \delta, x_0 + \delta)$, and the derivative f' is given by (11.6). Moreover, if F is C^k on A for some $k \in \mathbb{N}$, the implicit function f is C^k on $(x_0 - \delta, x_0 + \delta)$.*

Proof Let us go back to the proof of the implicit function theorem. In particular

$$F_y(x, y) \neq 0\,, \qquad \forall\, (x, y) \in R, \tag{11.7}$$

where R is the rectangle

$$R = (x_0 - \delta, x_0 + \delta) \times (y_0 - \sigma, y_0 + \sigma)\,.$$

Fix x and $h \neq 0$ so that x and $x + h$ belong to $(x_0 - \delta, x_0 + \delta)$. By definition of f and the mean value theorem,

$$\begin{aligned} 0 &= F(x + h, f(x + h)) - F(x, f(x)) = \\ &= F_x(\xi, \eta)h + F_y(\xi, \eta)[f(x + h) - f(x)]\,, \end{aligned}$$

where (ξ, η) is an internal point of the segment between $(x, f(x))$ and $(x+h, f(x+h))$, and hence belongs to R. From this relation and (11.7) we obtain

$$\frac{f(x+h)-f(x)}{h} = -\frac{F_x(\xi,\eta)}{F_y(\xi,\eta)}\,. \tag{11.8}$$

But f is continuous, so

$$\lim_{h\to 0}(\xi,\eta) = (x, f(x))\,,$$

and taking the limit in (11.8) as h tends to zero gives (11.6), because F is C^1.

If F is C^k for some $k > 1$, one argues by induction. Suppose the statement is true for $k-1$. If F is C^k, it is also C^{k-1}, and by induction f is C^{k-1}. Hence the right-hand side of (11.6) is a C^{k-1} function, and therefore $f' \in C^{k-1}$, i.e. f is of class C^k.

Remark 1 Let F be C^1 on the open set $A \subseteq \mathbb{R}^2$ and consider

$$Z = \{(x, y) \in A :\ F(x, y) = 0\}\,.$$

If (x_0, y_0) is a point in Z at which the gradient of F does not vanish, on a neighbourhood of (x_0, y_0) the zero set Z coincides with the support of a regular simple curve Γ, whose tangent line at the point has equation

$$F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)\,(y - y_0) = 0\,. \tag{11.9}$$

In fact, suppose for instance $F_y(x_0, y_0) \neq 0$. By the implicit function theorem Z is, around (x_0, y_0), the graph of a C^1 function $f(x)$ such that $f(x_0) = y_0$, and whose derivative at x_0 is

$$f'(x_0) = -\frac{F_x(x_0, y_0)}{F_y(x_0, y_0)}$$

by (11.6). But the tangent line to the graph of f has equation

$$y = y_0 + f'(x_0)(x - x_0) = y_0 - \frac{F_x(x_0, y_0)}{F_y(x_0, y_0)}(x - x_0)$$

and (11.9) follows. The latter can also be proved assuming $F_x(x_0, y_0) \neq 0$. In that case Z is, around (x_0, y_0), the graph of a C^1 function $g(y)$ for which

$$g'(y_0) = -\frac{F_y(x_0, y_0)}{F_x(x_0, y_0)}\,,$$

so the tangent to the graph of g at y_0 is

$$x = x_0 + g'(y_0)(y - y_0) = x_0 - \frac{F_y(x_0, y_0)}{F_x(x_0, y_0)}(y - y_0)\,,$$

whence we find (11.9) again.

A zero point $(x_0, y_0) \in Z$ is called *regular* if $DF(x_0, y_0) \neq 0$, while we say it is *singular* in case $DF(x_0, y_0) = 0$.

On a neighbourhood of a regular point Z is the graph of a function. From Example 6 is its evident that on the neighbourhood of a singular point, Z might not be a simple curve. The chapter's Appendix (see Sect. 11.5) contains a more thorough discussion on singular points.

Suppose now that $F(x, y)$ satisfies at (x_0, y_0) the implicit function theorem's assumptions (11.3): $F(x_0, y_0) = 0$ and $F_y(x_0, y_0) \neq 0$, and that $f(x)$ is defined implicitly by $F(x, y) = 0$ on a neighbourhood I of x_0 , i.e.

$$F(x, f(x)) = 0\,, \qquad \forall\, x \in I.$$

Assume $F \in C^2(A)$. Differentiating

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))}\,, \qquad \forall\, x \in I,$$

gives

$$f''(x) = -\frac{(F_{xx} + F_{xy} \cdot f')\, F_y - (F_{yx} + F_{yy} \cdot f')\, F_x}{(F_y)^2}\,, \qquad \forall\, x \in I,$$

and replacing f' with the right-hand side of (11.6),

$$f''(x) = -\frac{F_{xx}(F_y)^2 - 2\, F_{xy} F_x F_y + F_{yy}(F_x)^2}{(F_y)^3}\,, \qquad \forall\, x \in I, \tag{11.10}$$

where the partial derivatives are evaluated at $(x, f(x))$.

Recall that x_0 is a local maximum (or minimum) point of f if $f'(x_0) = 0$, i.e. $F_x(x_0, y_0) = 0$, and $f''(x_0) < 0$ (respectively, $f''(x_0) > 0$). Then if (x_0, y_0) solves the system

$$\begin{cases} F(x, y) = 0 \\ F_x(x, y) = 0, \end{cases} \tag{11.11}$$

and if

$$\frac{F_{xx}(x_0, y_0)}{F_y(x_0, y_0)} > 0\,, \qquad (\text{respectively } < 0),$$

then x_0 is a *local maximum* (respectively, *local minimum*) point of f.

In fact, if $F_x(x_0, y_0) = 0$, from (11.10) we obtain

$$f''(x_0) = -\frac{F_{xx}(x_0, y_0)}{F_y(x_0, y_0)}\,. \tag{11.12}$$

Example 7 In Example 4 we showed that

$$F(x, y) = x^3 + y^3 + x^2 y - 3y^2 = 0$$

implicitly defines a unique function $f : (-\infty, -\sqrt{3}] \to \mathbb{R}$. Moreover, when $x \le -\sqrt{3}$, we have $F_y(x, y) > 0$ for any $y \in \mathbb{R}$. Let us find the local extrema of f by solving system (11.11), which at present reads

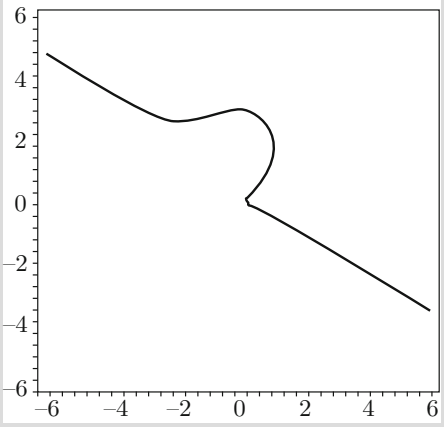
$$\begin{cases} x^3 + y^3 + x^2 y - 3y^2 = 0 \\ 3x^2 + 2xy = 0\,. \end{cases} \tag{11.13}$$

The second equation is solved by $x = 0$ and $x = -2y/3$. The first value must be discarded because f is defined for $x \le -\sqrt{3}$, while substituting the other solution in the first equation of (11.13) gives $x_0 = -54/31$, $y_0 = f(x_0) = 81/31$, and so $f'(x_0) = 0$. But $F_{xx} = 6x + 2y$, so (11.12) implies

$$f''(x_0) = -\frac{F_{xx}(x_0, y_0)}{F_y(x_0, y_0)} = \frac{162/31}{F_y(x_0, y_0)} > 0\,.$$

Hence x_0 is a local minimum point of $f(x)$. Figure 11.5 highlights the local minimum point (x_0, y_0), where $x_0 \simeq -2$ and $2 < y_0 < 3$. It is possible to prove that the curve has a cusp at the origin, which the computer picture only roughly represents.

Fig. 11.5



Example 8 Let a be a given positive number. We seek the local extrema of the function $y = f(x)$ defined implicitly by

$$F(x, y) = x^4 + y^4 + a\,(x^2 - y^2)\,,$$

(continued)

Example 8 (continued)
ignoring the singular points. For that, consider the system

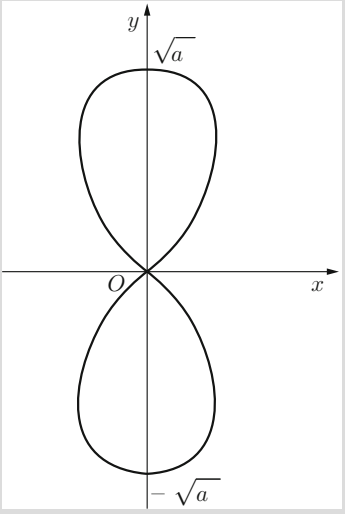
$$\begin{cases} F(x, y) = x^4 + y^4 + a\,(x^2 - y^2) = 0 \\ F_x(x, y) \;= 2x\,(2x^2 + a) = 0\,. \end{cases} \tag{11.14}$$

It admits only one solution $(0, 0)$ at which $F_y(x, y) = 2y\,(2y^2 - a)$ vanishes, so this is a singular point. The other solutions of (11.14) are $(0, \pm\sqrt{a}\,)$. The expression

$$\frac{F_{xx}(x, y)}{F_y(x, y)} = \frac{6x^2 + a}{y\,(2y^2 - a)}$$

is positive at $(x, y) = (0, \sqrt{a}\,)$ and negative at $(x, y) = (0, -\sqrt{a}\,)$. Hence the implicit function $y = f_1(x)$ that has value $y = \sqrt{a}$ at $x = 0$ has a maximum at $x = 0$. The implicit function $y = f_2(x)$, which evaluated at $x = 0$ gives $y = -\sqrt{a}$, has a minimum at $x = 0$. Figure 11.6 shows the zero set Z of F.

Fig. 11.6



Implicit Function Theorem in Several Variables *Let* $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, $y \in \mathbb{R}$ *and suppose* $F(x, y) = F(x_1, x_2, \ldots, x_n, y)$ *is a continuous function with continuous partial derivative* F_y *on the open set* $A \subseteq \mathbb{R}^{n+1}$. *If at the point* $(x_0, y_0) = (x_{01}, x_{02}, \ldots, x_{0n}, y_0)$ *of* A

$$F(x_0, y_0) = 0\,, \qquad F_y(x_0, y_0) \neq 0\,,$$

then there exist $\delta, \sigma > 0$ *and a unique function* $f : I_\delta(x_0) \to (y_0 - \sigma, y_0 + \sigma)$, *where* $I_\delta(x_0)$ *is the ball of centre* x_0 *and radius* δ, *such that*

$$F(x, f(x)) = 0\,, \qquad \forall\, x \in I_\delta(x_0). \tag{11.15}$$

Moreover, f *is continuous and* $f(x_0) = y_0$. *If* F *is of class* C^1 *so is* f, *and*

$$\frac{\partial f}{\partial x_i}(x) = -\frac{F_{x_i}(x, f(x))}{F_y(x, f(x))} \qquad \forall\, i = 1, 2, \ldots, n \tag{11.16}$$

for any $x \in I_\delta(x_0)$. *Finally, if* F *is of class* $C^k(A)$ *for some* $k \in \mathbb{N}$, *then* $f \in C^k(I_\delta(x_0))$.

The proof of the implicit function theorem in several variables goes as the two-variable case seen earlier: the argument relies on the monotonicity of $F(x, y)$ in $y \in \mathbb{R}$ and the continuity of $F(x, y)$ in $x \in \mathbb{R}^n$.

Below we will give another proof of the existence of the implicit function f satisfying (11.15), based on the *method of successive approximation*. Notice the proof's similarity with the one for systems in the following section.

For $(x, y) \in A$ put

$$G(x, y) = y - y_0 - \frac{F(x, y)}{F_y(x_0, y_0)}\,,$$

and observe how (11.15) is equivalent to

$$f(x) = y_0 + G(x, f(x))\,, \qquad \forall\, x \in I_\delta(x_0), \tag{11.17}$$

which is what we set out to prove. Since A is open, $G_y(x, y)$ is continuous on A, and

$$G_y(x_0, y_0) = 0\,,$$

we can find $\delta > 0$ and $\sigma > 0$ such that $(x, y) \in A$ when $|x - x_0| \le \delta$ and $|y - y_0| \le \sigma$. For such points (x, y), moreover,

$$\big|G_y(x, y)\big| \le \frac{1}{2}\,. \tag{11.18}$$

But $G(x_0, y_0) = 0$, so we can choose δ so that

$$|G(x, y_0)| \le \frac{\sigma}{2}\,, \qquad \forall\, x :\ |x - x_0| \le \delta \tag{11.19}$$

without loss of generality.

To define a sequence f_k of *successive approximations* of the solution $f(x)$ to (11.17), set

$$f_1(x) = y_0 + G(x, y_0)\,, \qquad \forall\, x:\ |x - x_0| \le \delta.$$

We have $f_1(x_0) = y_0$. Since, by (11.19),

$$|f_1(x) - y_0| \ \le\ \frac{\sigma}{2} < \sigma\,, \qquad |x - x_0| \le \delta, \tag{11.20}$$

the point $(x,\, f_1(x))$ belongs to A if $|x - x_0| \le \delta$. Hence

$$f_2(x) = y_0 + G(x,\, f_1(x))\,, \qquad \forall\, x:\ |x - x_0| \le \delta$$

is well defined. Now $f_1(x_0) = y_0$, so $f_2(x_0) = y_0$. By (11.19)

$$\begin{aligned} |f_2(x) - y_0| = |G(x,\, f_1(x))| &\le |G(x,\, y_0)| + |G(x,\, f_1(x)) - G(x,\, y_0)| \le \\ &\le \frac{\sigma}{2} + |G(x,\, f_1(x)) - G(x,\, y_0)|\ . \end{aligned}$$

The mean value theorem implies that there exists ξ between y_0 and $f_1(x)$ so that

$$G(x,\, f_1(x)) - G(x,\, y_0) = G_y(x, \xi) \cdot (f_1(x) - y_0)$$

for any x such that $|x - x_0| \le \delta$. Then (11.18), (11.20) force

$$\begin{aligned} |f_2(x) - y_0| &\le \frac{\sigma}{2} + |G(x,\, f_1(x)) - G(x,\, y_0)| \le \\ &\le \frac{\sigma}{2} + \big|G_y(x, \xi)\big| \cdot |f_1(x) - y_0| < \frac{\sigma}{2} + \frac{1}{2} \cdot \sigma = \sigma\,. \end{aligned}$$

So if $|x - x_0| \le \delta$, the point $(x,\, f_2(x))$ belongs to A and

$$f_3(x) = y_0 + G(x,\, f_2(x))\,, \qquad \forall\, x : |x - x_0| \le \delta$$

is well defined. Similarly,

$$|f_3(x) - y_0| < \sigma\,, \qquad |x - x_0| \le \delta\,,$$

and $f_3(x_0) = y_0$.

In this way we construct a recursive sequence of continuous functions $f_k(x)$, such that $f_0 = y_0$ and

$$f_{k+1}(x) = y_0 + G(x,\, f_k(x))\,, \qquad \forall\, x : |x - x_0| \le \delta, \tag{11.21}$$

satisfying

$$|f_k(x) - y_0| < \sigma\,, \qquad |x - x_0| \le \delta; \tag{11.22}$$

$$f_k(x_0) = y_0 \tag{11.23}$$

for $k = 0, 1, 2, \ldots$. Once we prove that the sequence $f_k(x)$ converges uniformly on $|x - x_0| \le \delta$ to, say, a continuous function $f(x)$, by taking the limit in (11.21) and (11.23) as $k \to +\infty$ we will obtain the claim

$$f(x) = y_0 + G(x, f(x))\,, \qquad \forall\, x : |x - x_0| \le \delta,$$
$$f(x_0) = y_0$$

due to the continuity of G.

So there remains to show that the sequence $f_k(x)$ converges uniformly on $|x - x_0| \le \delta$. For this, note

$$|f_{k+1}(x) - f_k(x)| = |G(x, f_k(x)) - G(x, f_{k-1}(x))|\,,$$

for any $k \in \mathbb{N}$ and for any x such that $|x - x_0| \le \delta$. By the mean value theorem, for those k and x there exists ξ between $f_{k-1}(x)$ and $f_k(x)$ such that

$$G(x, f_k(x)) - G(x, f_{k-1}(x)) = G_y(x, \xi) \cdot (f_k(x) - f_{k-1}(x))\,,$$

so (11.18) implies

$$\begin{aligned} |f_{k+1}(x) - f_k(x)| &= \left|G_y(x, \xi)\right| \cdot |f_k(x) - f_{k-1}(x)| \le \\ &\le \frac{1}{2}\, |f_k(x) - f_{k-1}(x)|\,, \qquad \forall\, k \in \mathbb{N},\ \forall\, x :\ |x - x_0| \le \delta. \end{aligned} \tag{11.24}$$

From (11.24) induction gives

$$|f_{k+1}(x) - f_k(x)| < \frac{\sigma}{2^k}\,, \tag{11.25}$$

for any $k = 0, 1, 2, \ldots$ and any x such that $|x - x_0| \le \delta$. In fact, when $k = 0$ inequality (11.25) is nothing but (11.22). Moreover, assuming (11.25) for a given k and using (11.24) we obtain

$$|f_{k+2}(x) - f_{k+1}(x)| \le \frac{1}{2}\ |f_{k+1}(x) - f_k(x)| < \frac{1}{2} \cdot \frac{\sigma}{2^k} = \frac{\sigma}{2^{k+1}}\,,$$

i.e. (11.25) with $k + 1$ replacing k.

Because of this, the series of functions

$$y_0 + \sum_{k=0}^{\infty} (f_{k+1}(x) - f_k(x))$$

(whose kth partial sum is $f_{k+1}(x)$) converges totally when $|x - x_0| \le \delta$. Therefore the sequence of functions $f_k(x)$ converges uniformly on that set.

What is more, the limit $f(x)$ of $f_k(x)$ as $k \to +\infty$ is continuous. By (11.22), (11.23) this limit satisfies $f(x_0) = y_0$ and the inequality

$$|f(x) - y_0| \le \sigma\,, \qquad \forall\, x:\ |x - x_0| \le \delta.$$

This implies the existence of the implicit function $f : I_{\delta_1}(x_0) \to (y_0 - \sigma, y_0 + \sigma)$, with δ_1 such that $I_{\delta_1}(x_0) \subseteq f^{-1}(y_0 - \sigma, y_0 + \sigma)$.

Example 9 Consider $F(x, y, z) = e^z - x^2 - y^2 - z^2$. As

$$F(1, 0, 0) = 0\,, \qquad F_z(1, 0, 0) = 1 \neq 0\,,$$

by the previous theorem the equation $F(x, y, z) = 0$ implicitly defines a unique function $f(x, y)$ for which $f(1, 0) = 0$. By (11.16) we also have

$$f_x(1, 0) = 2\,, \qquad f_y(1, 0) = 0\,.$$

Since $F_x(1, 0, 0) = -2 \neq 0$, equation $F(x, y, z) = 0$ defines on a neighbourhood of $(0, 0)$ a unique function $x = g(y, z)$ for which $g(0, 0) = 1$. By the implicit function theorem the partial derivatives of g are

$$g_y(0, 0) = -\frac{F_y(1, 0, 0)}{F_x(1, 0, 0)} = 0\,, \qquad g_z(0, 0) = -\frac{F_z(1, 0, 0)}{F_x(1, 0, 0)} = \frac{1}{2}\,.$$

Note that g can be made explicit:

$$g(y, z) = \sqrt{e^z - y^2 - z^2}$$

on a neighbourhood of $(0, 0)$. At last, observe that $F_y(0, 0, 0) = 0$ and that $F(x, y, z) = 0$ does not determine, on any neighbourhood of $(1, 0)$, a unique function $y = h(x, z)$ for which $h(1, 0) = 0$.

Remark 2 Let $F(x, y, z)$ be C^1 on an open set A in $\mathbb{R}^3$ and call Z the zero set:

$$Z = \{(x, y, z) \in A :\ \ F(x, y, z) = 0\}\,.$$

If (x_0, y_0, z_0) belongs to Z and $DF(x_0, y_0, z_0) \neq 0$, the argument of Remark allows to show easily that on a neighbourhood of (x_0, y_0, z_0) the set Z is a regular surface whose tangent plane at the point has equation

$$F_x(x_0, y_0, z_0)\,(x - x_0) + F_y(x_0, y_0, z_0)\,(y - y_0) + F_z(x_0, y_0, z_0)\,(z - z_0) = 0\,.$$

11.2 The Implicit Function Theorem for Systems

Consider a system of two equations in four unknowns x, y, u, v

$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases} \tag{11.26}$$

and suppose that the first equation defines u implicitly, as a function of x, y, v. Let us write this function as

$$u = h(x, y, v) \tag{11.27}$$

and substitute it in the second equation. This gives an equation in x, y, v

$$H(x, y, v) = G(x, y, h(x, y, v), v) = 0\,.$$

If the above defines v implicitly in terms of x and y :

$$v = g(x, y)\,, \tag{11.28}$$

we can substitute this v in (11.27) and obtain a function of x, y of the form

$$u = f(x, y) = h(x, y, g(x, y))\,. \tag{11.29}$$

All in all we have, at least formally, solved system (11.26) for u, v in terms of x, y, using (11.28), (11.29). In view of these relations, system (11.26) becomes

$$\begin{cases} F(x, y, f(x, y), g(x, y)) = 0 \\ G(x, y, f(x, y), g(x, y)) = 0\,. \end{cases} \tag{11.30}$$

In analogy to the case of equations, we will say that the functions $u = f(x, y)$ and $v = g(x, y)$ are *defined implicitly by system* (11.26).

Suppose now that we are allowed to differentiate in x the equations of (11.30), to obtain

$$\begin{cases} F_x + F_u f_x + F_v g_x = 0 \\ G_x + G_u f_x + G_v g_x = 0\,, \end{cases}$$

a linear system in the unknowns f_x and g_x. If the coefficients' determinant $F_u G_v - F_v G_u$ is non-zero, Cramer's rule gives

$$f_x = -\frac{\begin{vmatrix} F_x & F_v \\ G_x & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}\,, \qquad g_x = -\frac{\begin{vmatrix} F_u & F_x \\ G_u & G_x \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}\,. \tag{11.31}$$

The expressions of f_y, g_y can be found in a similar way.

Above we did not really care about the passages that led to (11.30) and (11.31); we only worked formally, in analogy to the implicit function theorem for equations. Now we shall discuss a result that clarifies the hypotheses under which the previous manipulations are allowed. It will be a important extension of the implicit function theorem to systems of equations of the type

$$\begin{cases} F_1(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_h) = 0 \\ F_2(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_h) = 0 \\ \dots\ \dots\ \dots\dots\ \dots\ \dots\dots\ \dots\ \dots \\ F_h(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_h) = 0\,. \end{cases} \tag{11.32}$$

To simplify the notation let us put

$$x = (x_1, x_2, \dots, x_n)\,, \qquad y = (y_1, y_2, \dots, y_h)\,,$$

and call (x, y) the point in $\mathbb{R}^{n+h}$

$$(x, y) = (x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_h)\,.$$

If A is open in $\mathbb{R}^{n+h}$, we let $F = F(x, y)$ be the map, from A to $\mathbb{R}^h$, whose components $F_1(x, y)$, $F_2(x, y), \dots, F_h(x, y)$ are defined for $(x, y) \in A$ and real-valued. Thus system (11.32) becomes

$$F(x, y) = 0\,,$$

where 0 is as usual the null vector in $\mathbb{R}^h$.

Implicit Function Theorem for Systems *Let A be an open set in $\mathbb{R}^{n+h}$, $F = F(x, y)$ a C^1 map from A to $\mathbb{R}^h$ and (x_0, y_0) a point in A such that*

$$F(x_0, y_0) = 0\,, \qquad \text{and} \qquad \det\left(\frac{\partial(F_1, F_2, \dots, F_h)}{\partial(y_1, y_2, \dots, y_h)}\,(x_0, y_0)\right) \neq 0\,. \tag{11.33}$$

Then there exist a ball I around x_0, a ball J around y_0 and a unique function $f : I \subset \mathbb{R}^n \to J \subset \mathbb{R}^h$ such that $f(x_0) = y_0$ and

$$F(x, f(x)) = 0\,, \qquad \forall\, x \in I.$$

Moreover, f is of class C^1 and the Jacobian matrix of $f = (f_1, f_2, \dots, f_h)$ at x is

$$\begin{aligned} &\frac{\partial(f_1, f_2, \dots, f_h)}{\partial(x_1, x_2, \dots, x_n)}(x) = \\ &= -\left[\frac{\partial(F_1, F_2, \dots, F_h)}{\partial(y_1, y_2, \dots, y_h)}(x, f(x))\right]^{-1} \circ \frac{\partial(F_1, F_2, \dots, F_h)}{\partial(x_1, x_2, \dots, x_n)}(x, f(x)) \end{aligned} \tag{11.34}$$

for any $x \in I$.

Remark 3 Let $F = (F_1(x, y, z), F_2(x, y, z))$ be a C^1 function on an open set $A \subseteq \mathbb{R}^3$. Let $P_0 \equiv (x_0, y_0, z_0)$ be a point of A where $F(P_0) = 0$ and the Jacobian determinants

$$\begin{aligned} J_1 &= \det \frac{\partial(F_1, F_2)}{\partial(y, z)}(P_0), \\ J_2 &= \det \frac{\partial(F_1, F_2)}{\partial(z, x)}(P_0), \\ J_3 &= \det \frac{\partial(F_1, F_2)}{\partial(x, y)}(P_0), \end{aligned} \tag{11.35}$$

are not simultaneously all zero. Then the zero set

$$Z = \{(x, y, z) \in A : \quad F(x, y, z) = 0\}$$

coincides, on a neighbourhood of P_0, with a *regular simple curve* Γ, whose tangent line at P_0 is parallel to the vector of components J_1, J_2, J_3.

To fix ideas, suppose $J_1 \neq 0$. By the implicit function theorem for systems, stated above, around P_0 the set Z is the graph of a C^1 function $f(x)$ defined on a neighbourhood $(x_0 - \delta, x_0 + \delta)$ of x_0 and with values in $\mathbb{R}^2$. Hence Z coincides with the support of the regular curve Γ of parametric equations $(x, f_1(x), f_2(x))$, with $x \in (x_0 - \delta, x_0 + \delta) = I$. Since, for any $x \in I$,

$$\begin{cases} F_1(x, f_1(x), f_2(x)) = 0 \\ F_2(x, f_1(x), f_2(x)) = 0, \end{cases} \qquad \forall x \in I,$$

differentiating in x gives the linear system

$$\begin{cases} \dfrac{\partial F_1}{\partial x} + \dfrac{\partial F_1}{\partial y} f_1' + \dfrac{\partial F_1}{\partial z} f_2' = 0 \\ \dfrac{\partial F_2}{\partial x} + \dfrac{\partial F_2}{\partial y} f_1' + \dfrac{\partial F_2}{\partial z} f_2' = 0 \end{cases}$$

in f_1', f_2'. Once solved, the system will give, for $x = x_0$,

$$f_1'(x_0) = \frac{J_2}{J_1}\,, \qquad f_2'(x_0) = \frac{J_3}{J_1}$$

also recalling that $f_1(x_0) = y_0$, $f_2(x_0) = z_0$. On a neighbourhood of P_0, therefore, the curve Γ, defined implicitly by $F(x, y, z) = 0$, has unit tangent vector parallel to (J_1, J_2, J_3). The reader should show that the condition by which the vector (J_1, J_2, J_3) is non-zero is saying that Γ is locally the intersection of the two regular surfaces of equations $F_1(x, y, z) = 0$ and $F_2(x, y, z) = 0$, and that the corresponding tangent planes along Γ are distinct.

Example 1 Let us show that on a neighbourhood of (0, 1, 1) the equations

$$\begin{cases} F_1(x, y, z) = x^2 + y^2 + z^2 - 2 = 0 \\ F_2(x, y, z) = x^2 + y^2 - z^2 - 2x = 0, \end{cases} \tag{11.36}$$

implicitly define a regular simple curve Γ. Let us compute the determinants J_1, J_2, J_3 defined in (11.35):

$$J_1 = \begin{vmatrix} 2 & 2 \\ 2 & -2 \end{vmatrix}\,, \qquad J_2 = \begin{vmatrix} 2 & 0 \\ -2 & -2 \end{vmatrix}\,, \qquad J_3 = \begin{vmatrix} 0 & 2 \\ -2 & 2 \end{vmatrix}\,,$$

so $J_1 = -8$, $J_2 = -4$, $J_3 = 4$. The above observations tell that on a neighbourhood of (0, 1, 1) Eqs. (11.36) define a regular curve Γ, whose tangent at the point is parallel to $(-2, -1, 1)$. Note that this curve is the intersection of the sphere of equation $F_1 = 0$ with the cone of equation $F_2 = 0$.

Proof of the Implicit Function Theorem for Systems Let M and N denote the Jacobian matrices

$$M = \frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(x_1, x_2, \ldots, x_n)}\,(x_0, y_0)\,, \qquad N = \frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(y_1, y_2, \ldots, y_h)}\,(x_0, y_0)\,.$$

In the rest of the proof we will use the fact that N is an invertible square matrix. As F is differentiable at (x_0, y_0) and $F(x_0, y_0) = 0$, by the definition of differential we have

$$F(x, y) = M\,(x - x_0) + N\,(y - y_0) + \omega(x, y)\,, \tag{11.37}$$

for any $(x, y) \in A$, where $\omega(x, y)$ is a C^1 function from A to $\mathbb{R}^h$ satisfying

$$\lim_{(x,y)\to(x_0,y_0)} \frac{\omega(x, y)}{\sqrt{|x - x_0|^2 + |y - y_0|^2}} = 0\,.$$

Recall that we are making use of the notation

$$|x| = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}, \qquad |y| = \left(\sum_{i=1}^{h} y_i^2\right)^{1/2},$$

to indicate the usual norms on the respective Euclidean spaces. Due to assumption (11.33) the matrix N is invertible. Applying to (11.37) the inverse N^{-1} gives

$$\begin{aligned} y - y_0 &= N^{-1}(N\,(y - y_0)) = \\ &= N^{-1}(F(x, y)) - N^{-1}(M\,(x - x_0)) - N^{-1}(\omega(x, y))\,. \end{aligned} \tag{11.38}$$

The map $L : \mathbb{R}^n \to \mathbb{R}^h$, defined by

$$L\,x = -N^{-1}(M\,x)\,, \qquad \forall\, x \in \mathbb{R}^n,$$

is linear. Moreover the map $G : A \subseteq \mathbb{R}^n \times \mathbb{R}^h \to \mathbb{R}^h$,

$$G(x, y) = -N^{-1}(\omega(x, y))\,, \qquad \forall\,(x, y) \in A \subseteq \mathbb{R}^n \times \mathbb{R}^h,$$

is of class $C^1(A)$, and satisfies

$$\lim_{(x,y)\to(x_0,y_0)} \frac{G(x, y)}{\sqrt{|x - x_0|^2 + |y - y_0|^2}} = 0\,. \tag{11.39}$$

Since $N^{-1} : \mathbb{R}^h \to \mathbb{R}^h$ is invertible, we have $N^{-1}z = 0$ if and only if $z = 0$. With the above notations, then, from (11.38) we deduce that

$$F(x, y) = 0 \quad \Leftrightarrow \quad y = y_0 + L\,(x - x_0) + G(x, y)\,.$$

Hence the existence of a unique implicitly defined function $f : I \subseteq \mathbb{R}^n \to J \subseteq \mathbb{R}^h$ is granted once we show that there exist a neighbourhood I of x_0 in $\mathbb{R}^n$ and a neighbourhood J of y_0 in $\mathbb{R}^h$ such that the map

$$y \to y_0 + L\,(x - x_0) + G(x, y) \qquad y \in J,$$

admits a *unique fixed point* in J, for any $x \in I$.

For that we use Banach's fixed-point theorem, proved in Sect. 2.9. For any $\varrho > 0$ define

$$\overline{I}_\varrho = \left\{x \in \mathbb{R}^n : |x - x_0| \le \varrho\right\}, \qquad \overline{J}_\varrho = \left\{y \in \mathbb{R}^h : |y - y_0| \le \varrho\right\}.$$

As $G(x_0, y_0) = 0$, (11.39) implies $DG(x_0, y_0) = 0$. But G is of class C^1, so there is a positive number ϱ_0 such that

$$|DG(x, y)| \leq \frac{1}{2\sqrt{h}}, \qquad \forall\, (x, y) \in \overline{I}_{\varrho_0} \times \overline{J}_{\varrho_0}.$$

As G has h components,

$$|G(x_1, y_1) - G(x_2, y_2)| \leq \frac{1}{2}\,[|x_1 - x_2| + |y_1 - y_2|]\,, \tag{11.40}$$

for any $x_1\,, x_2 \in \overline{I}_{\varrho_0}$ and for any $y_1\,, y_2 \in \overline{J}_{\varrho_0}$. Choose ϱ_1 so that

$$0 < \varrho_1 < \frac{\varrho_0}{1 + 2\,|L|}\,,$$

where $|L|$ is the norm of L. If $x \in \overline{I}_{\varrho_1}$, since $G(x_0, y_0) = 0$,

$$\begin{aligned} |(y_0 + L(x - x_0) + G(x, y)) - y_0| &\leq |L(x - x_0)| + |G(x, y)| \leq \\ &\leq |L| \cdot |x - x_0| + \frac{1}{2}\,[|x - x_0| + |y - y_0|] \leq \\ &\leq \left(|L| + \frac{1}{2}\right) \varrho_1 + \frac{1}{2}\,\varrho_0 < \varrho_0 \end{aligned}$$

for any $y \in \overline{J}_{\varrho_0}$. Therefore $y \;\to\; y_0 + L\,(x - x_0) + G(x, y)$ maps $\overline{J}_{\varrho_0}$ into itself (actually, into its interior J_{ϱ_0}). Moreover, for $x \in \overline{I}_{\varrho_1}$ and $y_1\,, y_2 \in \overline{J}_{\varrho_0}$, (11.40) implies

$$|(y_0 + L(x - x_0) + G(x, y_1)) - (y_0 + L(x - x_0) + G(x, y_2))| \leq \frac{1}{2}\,|y_1 - y_2|\,.$$

Hence, for any $x \in \overline{I}_{\varrho_1}$, the map $y \to y_0 + L\,(x - x_0) + G(x, y)$ is a contraction, with constant $1/2$, on the metric space $\overline{J}_{\varrho_0} \subseteq \mathbb{R}^h$ with the usual Euclidean metric. Then this map from $\overline{J}_{\varrho_0}$ to J_{ϱ_0} has, for any $x \in \overline{I}_{\varrho_1}$, a unique fixed point in J_{ϱ_0}, say $f(x)$. By construction, $f(x_0) = y_0$.

We claim that $f : x \in \overline{I}_{\varrho_1} \;\to\; f(x) \in J_{\varrho_0}$ is Lipschitz. For $x_1\,, x_2 \in \overline{I}_{\varrho_1}$, from (11.40) we deduce

$$\begin{aligned} &|f(x_1) - f(x_2)| = \\ &= |(y_0 + L(x_1 - x_0) + G(x_1, f(x_1))) - (y_0 + L(x_2 - x_0) + G(x_2, f(x_2)))| \leq \\ &\leq |L| \cdot |x_1 - x_2| + |G(x_1, f(x_1)) - G(x_2, f(x_2))| \leq \\ &\leq |L| \cdot |x_1 - x_2| + \frac{1}{2}\,[|x_1 - x_2| + |f(x_1) - f(x_2)|]\,, \end{aligned}$$

whence

$$|f(x_1) - f(x_2)| \leq (2|L| + 1) \cdot |x_1 - x_2| \,. \tag{11.41}$$

Then the function

$$x \in \overline{I}_{\varrho_1} \to \det\left(\frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(y_1, y_2, \ldots, y_h)}(x, f(x))\right)$$

is continuous. But as

$$\det\left(\frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(y_1, y_2, \ldots, y_h)}\,(x_0, y_0)\right) \neq 0\,,$$

there is $\varrho_2 \in (0, \varrho_1]$ such that

$$\det\left(\frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(y_1, y_2, \ldots, y_h)}(x, f(x))\right) \neq 0\,, \qquad \forall\, x \in I_{\varrho_2}. \tag{11.42}$$

Let us now show that $f(x)$ is differentiable for any $x \in I_{\varrho_2}$. Take x and x_1 in I_{ϱ_2}. For any $i = 1, 2, \ldots, h$ there is a point P_i internal to the segment between $(x, f(x))$ and $(x_1\,, f(x_1))$ for which

$$F_i(x_1\,, f(x_1)) - F_i(x, f(x)) =$$

$$= (D_x F_i(P_i), x_1 - x) + (D_y F_i(P_i), f(x_1) - f(x))\,,$$

where $D_x F_i$ is the gradient of F_i in the variable $x \in \mathbb{R}^n$, while $D_y F_i$ is the gradient of F_i in $y \in \mathbb{R}^h$. Hence

$$\begin{aligned} 0 = &\left[\frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(x_1, x_2, \ldots, x_n)}\,(x, f(x))\right](x_1 - x) + \\ &+ \left[\frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(y_1, y_2, \ldots, y_h)}\,(x, f(x))\right](f(x_1) - f(x)) + \\ &+ R\,(x_1 - x) + S\,(f(x_1) - f(x)), \end{aligned} \tag{11.43}$$

where the matrices R, S have coefficients

$$R_{ij} = \frac{\partial F_i}{\partial x_j}(P_i) - \frac{\partial F_i}{\partial x_j}(x, f(x))\,, \qquad i = 1, 2, \ldots, h,\ \ j = 1, 2, \ldots, n,$$

$$S_{ij} = \frac{\partial F_i}{\partial y_j}(P_i) - \frac{\partial F_i}{\partial y_j}(x, f(x))\,, \qquad i = 1, 2, \ldots, h,\ \ j = 1, 2, \ldots, h.$$

Now, from (11.43) we have

$$f(x_1)-f(x)=-\left(N_x^{-1}\circ M_x\right)(x_1-x)-N_x^{-1}\left[R(x_1-x)+S(f(x_1)-f(x))\right],$$

where

$$M_x=\frac{\partial(F_1,F_2,\ldots,F_h)}{\partial(x_1,x_2,\ldots,x_n)}\,(x,f(x))$$

and N_x is the matrix

$$N_x=\frac{\partial(F_1,F_2,\ldots,F_h)}{\partial(y_1,y_2,\ldots,y_h)}\,(x,f(x))$$

(invertible, by (11.42)). To prove (11.34) there remains to observe that, due to the continuity of f,

$$\lim_{x_1\to x}|R|=0\,,\qquad \lim_{x_1\to x}|S|=0\,,$$

so (11.41) implies

$$\lim_{x_1\to x}\frac{\left|N_x^{-1}\left[R\,(x_1-x)+S\,(f(x_1)-f(x))\right]\right|}{|x_1-x|}\le$$

$$\le\left|N_x^{-1}\right|\lim_{x_1\to x}\left[|R|+|S|\,(2\,|L|+1)\right]=0\,.$$

We end the section with an observation. In the implicit function theorem for systems, if F is chosen of class C^k on A, then also f is of class C^k. The proof follows from the above theorem using an easy induction argument on k.

11.3 Local and Global Invertibility

One consequence of the implicit function theorem is the *inverse function theorem in one real variable*: if A is a open subset of $\mathbb{R}$ and $f:A\to\mathbb{R}$ is a C^1 function such that

$$f(x_0)=y_0\,,\qquad f'(x_0)\neq 0\,,$$

there is a neighbourhood I of y_0 on which the inverse function f^{-1} is well defined. Moreover, f^{-1} has on I continuous derivative

$$\left(f^{-1}\right)'(y)=\frac{1}{f'(x)},\quad\text{with}\quad y=f(x).$$

The proof is straightforward if we apply the implicit function theorem to

$$F(x, y) = f(x) - y\,,$$

since $F(x_0, y_0) = 0$, $F_x(x_0, y_0) \neq 0$.

The goal of this section is to prove a similar statement for functions $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$, as a consequence of the implicit function theorem for systems.

Let $f = (f_1, f_2, \dots, f_n) : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a function defined on the open set A in $\mathbb{R}^n$, of class $C^1(A)$. We say that f is *locally invertible around* $x_0 \in A$ if there exists a neighbourhood $I \subseteq A$ of x_0 such that the restriction of f to I is invertible as a function from I to $f(I)$.

If the above restriction is of class C^1 together with its inverse, then f is called a *local diffeomorphism around* x_0.

Local Invertibility Theorem *Let A be an open set in $\mathbb{R}^n$, $x_0 \in A$ a point and $f = (f_1, f_2, \dots, f_n) : A \to \mathbb{R}^n$ a function of class $C^1(A)$. Suppose the Jacobian determinant is non-zero at x_0 :*

$$\det \frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}\,(x_0) \neq 0\,. \tag{11.44}$$

Then there exist open neighbourhoods I of x_0 and J of $f(x_0)$ such that $f : I \to J$ is invertible, and the inverse $f^{-1} : J \to I$ is of class C^1. For any $y \in J$, furthermore,

$$\frac{\partial((f^{-1})_1, (f^{-1})_2, \dots, (f^{-1})_n)}{\partial(y_1, y_2, \dots, y_n)}\,(y) = \left[\frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}\,(f^{-1}(y))\right]^{-1}. \tag{11.45}$$

Proof Let us show that there is a ball I_1 around x_0 on which f is one-to-one (hence invertible). If not, in fact, for any $k \in \mathbb{N}$ there exist x'_k , $x''_k \in B_{1/k}(x_0)$, with $x'_k \neq x''_k$ and $f(x'_k) = f(x''_k)$. By the mean value theorem applied to the components of f, for any $i = 1, 2, \dots, n$ there is a point $x^i_k \in B_{1/k}(x_0)$ such that $\big(Df_i(x^i_k), x''_k - x'_k\big) = 0$. Setting $v_k = \big(x''_k - x'_k\big)/\big|x''_k - x'_k\big|$, there is a subsequence v_{k_r} converging to some vector v_0 with $|v_0| = 1$. As $x^i_{k_r} \to x_0$, in the limit $(Df_i(x_0), v_0) = 0$ for any $i = 1, 2, \dots, n$. But this is absurd, since $Df(x_0)$ is an $n \times n$ matrix with non-zero determinant.

Now let us prove that, for a suitable open neighbourhood I of x_0 , the image $f(I)$ is open and on that set f^{-1} is of class C^1 (here we shall use the implicit function theorem). For $x \in A$ and $y \in \mathbb{R}^n$ set

$$F(x, y) = f(x) - y\,,$$

and also $y_0 = f(x_0)$. Since, by (11.44),

$$\det\left(\frac{\partial(F_1, F_2, \dots, F_n)}{\partial(x_1, x_2, \dots, x_n)}(x_0, y_0)\right) = \det \frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}\,(x_0) \neq 0\,,$$

the assumptions (11.33) of the implicit function theorem for systems hold, provided we interchange x and y. Therefore there exist two open neighbourhoods J of y_0 and I_2 of x_0, and a unique function $g(y)$, with $g : J \to I_2$ of class C^1, such that

$$F(g(y), y) = f(g(y)) - y = 0\,, \qquad \forall\, y \in J.$$

We may assume $I_2 \subseteq I_1$, by possibly replacing I_2 with $I_1 \cap I_2$ and J with $g^{-1}(I_1 \cap I_2)$. Since f is injective (so invertible) on I_1, g is the inverse of f restricted to $I = g(J) = f^{-1}(J)$. The set I is open due to the continuity of f. There remains to prove formula (11.45). From (11.34) and the definition of F we have

$$\begin{aligned} &\frac{\partial(g_1, g_2, \dots, g_n)}{\partial(y_1, y_2, \dots, y_n)}(y) = \\ &= -\left[\frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}\,(g(y))\right]^{-1} \circ \frac{\partial(F_1, F_2, \dots, F_n)}{\partial(y_1, y_2, \dots, y_n)}\,(g(y), y)\,. \end{aligned} \tag{11.46}$$

Since

$$\frac{\partial F_i}{\partial y_j} = \begin{cases} 0 & \text{if} \quad i \neq j \\ -1 & \text{if} \quad i = j\,, \end{cases}$$

it follows that

$$\frac{\partial(F_1, F_2, \dots, F_n)}{\partial(y_1, y_2, \dots, y_n)} = -\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{pmatrix}. \tag{11.47}$$

Finally, claim (11.45) follows from (11.46) and (11.47).

Observe that under the above hypotheses we can only say that there exists a neighbourhood J of $y_0 = f(x_0)$, every point of which is the image of a unique point of A that belongs to a suitable neighbourhood I of x_0. This does not exclude the possibility that y_0 and other points of J are themselves images of points of A not in I.

Example 1 Let $A \subset \mathbb{R}^2$ be the open annulus bounded by the circles centred at the origin with radii 1 and 3:

$$A = \{(x, y) \in \mathbb{R}^2 :\ 1 < x^2 + y^2 < 9\}\,,$$

(continued)

Example 1 (continued)
and consider $f : (x, y) \in A \to (u, v) \in \mathbb{R}^2$ defined by

$$f(x, y) = (u(x, y), v(x, y)) = \left(x^2 - y^2, 2xy\right) .$$

The Jacobian of f

$$J_f = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4\,(x^2 + y^2)$$

is non-zero for every $(x, y) \in A$. By the local invertibility theorem f is locally invertible on A. Yet it is not invertible globally on A, because

$$f(2, 0) = f(-2, 0) = (4, 0)$$

for instance.

A natural problem emerges, namely that of assigning conditions on the function f for it to be globally invertible on a domain $D \subseteq \mathbb{R}^n$. One first result in this direction is contained in the following proposition.

Proposition 1 *Let D be a domain in $\mathbb{R}^n$ and $f : D \to \mathbb{R}^n$ a C^1 function on the interior $\overset{\circ}{D}$ of D. Suppose*

$$J_f(x) \neq 0\,, \qquad \forall\, x \in \overset{\circ}{D}.$$

Then f maps the interior of D into the interior of $T = f(D)$.

Proof Let x_0 be an interior point of D and $y_0 = f(x_0)$. By the local invertibility theorem there exist an open neighbourhood I of x_0 in D and an open neighbourhood J of y_0 such that f induces a 1–1 correspondence between I and J. But J is an open subset of $T = f(D)$, so it must be contained in the interior of T. This proves the claim. Note that whilst an interior point of D cannot be mapped to a boundary point of $T = f(D)$, it may happen that a boundary point of D is transformed into an interior point of T.

For example, consider the function

$$f(x, y) = (u(x, y), v(x, y)) = \left(x^2 - y^2, 2xy\right) .$$

(continued)

defined on the semi-disc

$$D = \{(x, y) \in \mathbb{R}^2 : \ x^2 + y^2 \leq 1, \ \ y \geq 0\},$$

introduced in the previous example. Here

$$T = f(D) = \left\{(u, v) \in \mathbb{R}^2 : \ u^2 + v^2 \leq 1\right\}.$$

In particular, the points of D with first coordinate $x \in (-1, 1)$ and second coordinate $y = 0$, which belong to the boundary ∂D, are mapped to points with first coordinate $u \in [0, 1)$ and second coordinate $v = 0$, which are internal to the domain T.

Vice versa, in agreement with Proposition 1, the points (u, v) on the boundary of T, those satisfying $u^2 + v^2 = 1$, are images of points $(x, y) \in D$ such that $x^2 + y^2 = 1$, which lie on ∂D.

Suppose that f is of class $C^1(D)$ and satisfies

$$J_f(x) \neq 0, \qquad \forall\, x \in \overset{\circ}{D}. \tag{11.48}$$

If f is invertible on D with C^1 inverse on $T = f(D)$, the inverse $f^{-1} : T \to D$, too, has non-zero Jacobian determinant on the interior of T, and it maps the interior of T into the interior of D.

For $f : D \to T$ to be globally invertible, therefore, *it is necessary that f maps boundary points of D to boundary points of* $T = f(D)$. This condition is, in some sense, also sufficient, which will become clear after the following remarkable theorem, where we shall suppose the domain D is bounded and connected.

Global Invertibility Theorem *Let* $f : A \to \mathbb{R}^n$ *be a* C^1 *function on the open set* $A \subseteq \mathbb{R}^n$*, and assume* $D \subset A$ *is a connected bounded domain such that*

$$J_f(x) \neq 0, \qquad \forall\, x \in D. \tag{11.49}$$

If f induces a 1–1 correspondence between the boundary of D and the boundary of the image $T = f(D)$*, then T is a connected bounded domain, and f is (globally) invertible over D.*

To prove this fact we shall first discuss a proposition about C^1 maps

$$f = (f_1, f_2, \ldots, f_n) : D \subseteq \mathbb{R}^n \to \mathbb{R}^n$$

defined on bounded domains D and satisfying (11.48).

Proposition 2 *If D is a bounded domain of $\mathbb{R}^n$ and f is a C^1 function satisfying (11.48), then also $T = f(D)$ is a bounded domain.*

Proof The generalised Weierstrass theorem (see Sect. 2.12) implies that $T = f(D)$ is a closed and bounded subset in $\mathbb{R}^n$. To show it is a domain we start by considering $\overline{y} \in \partial T$ and $\overline{x} \in D$ such that $\overline{y} = f(\overline{x})$. Since every neighbourhood of $\overline{x}$ contains interior points of D, by Proposition 1 in any neighbourhood of $\overline{y}$ there will be interior points of T.

Proof of the Global Invertibility Theorem Setting

$$Y = \{y \in T = f(D) : \ \exists\, x_1, x_2,\ x_1 \neq x_2 :\ f(x_1) = f(x_2) = y\}\,,$$

we claim that

$$Y \subseteq \partial T\,, \tag{11.50}$$

i.e. the set of images $y \in T$ of distinct points is contained in the boundary of T.

From that, since f induces a 1–1 correspondence between ∂D and ∂T, it follows that $Y = \emptyset$, so f is invertible on D.

We begin by showing that Y is closed. Let y_0 be a limit point of Y, so y_0 is also a limit point for $T \supseteq Y$. As T is a domain, hence a closed set, $y_0 \in T$. Then there exists $x_0 \in D$ such that $y_0 = f(x_0)$. By assumption (11.49) f is invertible on a neighbourhood of x_0 (even if this point is on the boundary ∂D), so there are open neighbourhoods I_0 of x_0 and J_0 of y_0 such that

$$\forall\, y \in J_0\,, \qquad \exists!\ x \in I_0\ : y = f(x)\,.$$

Since y_0 is a limit point of Y, J_0 contains a sequence y_k with limit y_0 such that every y_k is image of at least two points of D, so image of at least one point of $D - I_0$.

For any $k \in \mathbb{N}$, take $x_k \in D - I_0$ such that $f(x_k) = y_k$. As $D - I_0$ is compact, there is a subsequence x_{k_r} that converges to some $\overline{x} \in D - I_0$. By the continuity of f, therefore, $f(x_{k_r}) \to f(\overline{x}) = y_0$ as $r \to +\infty$, and so $y_0 \in Y$. This shows that Y is closed.

Let us prove that there are no boundary points of Y internal to T, i.e.

$$Y \cap \overset{\circ}{T} \subseteq \overset{\circ}{Y}\,. \tag{11.51}$$

Let $y_1 \in Y \cap \overset{\circ}{T}$, so there exist $x_1\,, x_2 \in D$, $x_1 \neq x_2$ such that $y_1 = f(x_1) = f(x_2)$. By hypothesis f defines a bijection between ∂D and ∂T, so $y_1 \in \overset{\circ}{T}$ implies $x_1\,, x_2 \in \overset{\circ}{D}$.

As f is locally invertible, there are three balls I_1, I_2, J, around x_1, x_2, y_1 respectively, such that

$$\forall\, y \in J\,, \qquad \exists!\, x' \in I_1\,, \qquad \exists!\, x'' \in I_2\,: \qquad y = f(x') = f(x'')\,.$$

We may assume that the radii of I_1 and I_2 are so small that I_1 and I_2 are contained in D and disjoint. Then every point in the neighbourhood J of y_1 is the image of at least two points of D, i.e. $y_1 \in \overset{\circ}{Y}$.

Let us prove $\overset{\circ}{T}$ is connected, making T a connected domain. Suppose by contradiction that $\overset{\circ}{T}$ is not connected. Then there exist non-empty disjoint open sets A_1, B_1 such that $A_1 \cup B_1 = \overset{\circ}{T}$. Define

$$A_2 = f^{-1}(A_1) \cap \overset{\circ}{D}\,, \qquad B_2 = f^{-1}(B_1) \cap \overset{\circ}{D}\,,$$

which are both open because f is continuous (see the Characterization of continuity in Sect. 2.12). If one of them were empty, say A_2, the points of $A_1 \subseteq \overset{\circ}{T}$ would not be image of anything in $\overset{\circ}{D}$, so they would have to arise from boundary points of D. This would violate the 1-1-correspondence between ∂D and ∂T. Moreover A_2, B_2 are disjoint sets whose union is $\overset{\circ}{D}$, against the hypothesis that D is a connected domain, i.e. $\overset{\circ}{D}$ is connected and open.

To prove (11.50), we suppose that, by contradiction, there exists $\overline{y} \in Y$ internal to T. In this case

$$\overset{\circ}{T} \subset Y\,. \tag{11.52}$$

In fact, if an interior point y_1 of T did not belong to Y, since $\overset{\circ}{T}$ is connected the points $\overline{y}$ and y_1 could be joined by a polygonal path P contained in $\overset{\circ}{T}$ (see the Theorem on connected open subsets of $\mathbb{R}^n$ in Sect. 2.11). In addition,

$$\partial Y \cap \overset{\circ}{T} \supseteq \partial Y \cap P \neq \emptyset\,,$$

in contrast with (11.51), $Y \cap \overset{\circ}{T} \subseteq \overset{\circ}{Y}$. Hence (11.52) is proved.

Recall that we are arguing by contradiction and supposing the existence of $\overline{y} \in Y$ interior to T. As Y is closed, from (11.52) all boundary points of T (which are limits of interior points since, by Proposition 2, T is a domain) would have to belong to Y, i.e.

$$\partial T \subset Y\,.$$

But this is absurd, because every boundary point of T is image of only one point of D. Therefore there is no $\overline{y} \in Y$ that is interior to T, i.e. (11.50) holds, $Y \subseteq \partial T$.

11.4 Constrained Maxima and Minima. Lagrange Multipliers

Let $f(x, y)$ and $F(x, y)$ be functions admitting continuous partial derivatives on an open set A of $\mathbb{R}^2$, with

$$\left(\frac{\partial F}{\partial x}(x, y)\right)^2 + \left(\frac{\partial F}{\partial y}(x, y)\right)^2 > 0\,, \qquad \forall\, (x, y) \in A \subseteq \mathbb{R}^2.$$

Supposing that the zero set of F

$$Z = \{(x, y) \in A : \quad F(x, y) = 0\} \tag{11.53}$$

is non-empty, we wish to describe a method for finding the possible maxima and minima of f belonging in the set Z.

This is an instance of a *constrained problem*, because the variables of f are not independent, but rather related by the condition $F(x, y) = 0$, which is also known as a *constraint*.

One first example is when Z is the support of a regular simple curve $\gamma : [a, b] \to A$, of parametric equations

$$\begin{cases} x = x(t) \\ y = y(t)\,, \end{cases} \qquad t \in [a, b]. \tag{11.54}$$

The problem of finding the possible extremum points of f in Z is reduced to the determination of the extrema of the one-variable function

$$H(t) = f(x(t), y(t))\,, \qquad t \in [a, b].$$

Suppose $t_0 \in (a, b)$ is a local maximum (or minimum) point for $H(t)$. Then $H'(t_0) = 0$, and setting $(x_0, y_0) = (x(t_0), y(t_0))$ we will have

$$f_x(x_0, y_0)\, x'(t_0) + f_y(x_0, y_0)\, y'(t_0) = 0\,.$$

In the notations of Chaps. 3 and 6 the above relation can be written in terms of the inner product of the gradient vector $Df = (f_x, f_y)$ and $\gamma' = (x', y')$,

$$\big(Df(x_0, y_0), \gamma'(t_0)\big) = 0\,.$$

Geometrically this relation implies that at a local maximum (or minimum) point $(x_0, y_0) = (x(t_0), y(t_0))$ of $f(x, y)$, with $t_0 \in (a, b)$, the gradient of f must be orthogonal to the curve γ.

Example 1 Let us find the maximum for $x, y > 0$ of the function

$$f(x, y) = xy\,,$$

constrained by $x^2 + y^2 = 1$.

Define $F(x, y) = x^2 + y^2 - 1$, so the zero set of F is the unit circle, parametrised as

$$\begin{cases} x = \cos t \\ y = \sin t\,, \end{cases} \qquad t \in [0, 2\pi].$$

The problem then reduces to finding the maximum of

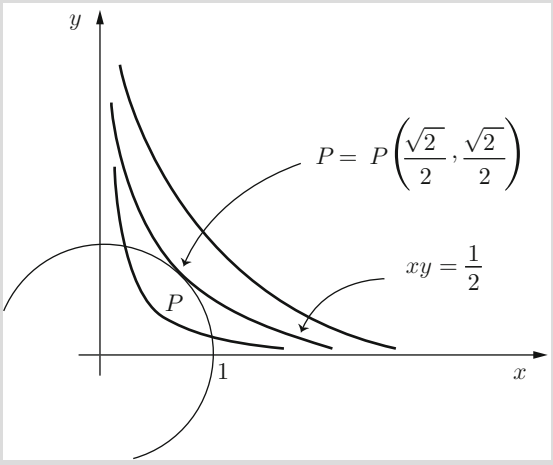
$$H(t) = \sin t\ \cos t = \frac{1}{2}\ \sin 2t\,, \qquad t \in [0, 2\pi].$$

It is immediate to see that $t = \pi/4$ and $t = (5\pi)/4$ are absolute maximum points, and $t = (3\pi)/4, (7\pi)/4$ are the absolute minimum points. Hence the maximum of $f(x, y)$, constrained by the condition $F(x, y) = 0$, is attained for $(x, y) = \pm(\sqrt{2}/2, \sqrt{2}/2)$.

Figure 11.7 shows the unit circle and a few level curves of the function, i.e. the hyperbolas of equation $xy = k$. Observe that the level curve passing through $P \equiv (\sqrt{2}/2, \sqrt{2}/2)$, i.e. the hyperbola of equation $xy = 1/2$, is tangent to the unit circle at P.

In agreement with what was observed above, the gradient of f and the gradient of F, both normal to the unit circle and to the hyperbola at P, are parallel vectors.

Fig. 11.7



In general we should not expect that the set Z, defined by (11.53), will be the support of a simple curve like (11.54): in other words the constraint equation $F(x, y) = 0$ will not be solvable, globally, with respect to one variable in terms of the other. This will happen, though, near a regular point of Z (see Remark , Sect. 11.1).

If (x_0, y_0) is a regular point of Z then

$$F_x(x_0, y_0) \neq 0\,, \quad \text{or} \quad F_y(x_0, y_0) \neq 0\,,$$

so equation $F(x, y) = 0$ defines locally an implicit function $x = g(y)$, or an implicit function $y = h(x)$, with the condition $x_0 = g(y_0)$ or, respectively, $y_0 = h(x_0)$. In this way the points of Z contained in a suitable neighbourhood of (x_0, y_0) form the graph of g (on a neighbourhood of y_0) or of h (on a neighbourhood of x_0), respectively.

Suppose, to fix ideas, that $F_y(x_0, y_0) \neq 0$, that the implicit function is $y = h(x)$, and that x_0 is a local extremum point of

$$x \;\to\; f(x, h(x))\,.$$

Then at x_0 the first derivative vanishes

$$f_x + f_y\, h' = 0\,. \tag{11.55}$$

From (11.6) we obtain the expression of the derivative h' in terms of F

$$h' = -\frac{F_x}{F_y}\,,$$

and substituting in (11.55),

$$f_x - f_y \cdot \frac{F_x}{F_y} = 0\,.$$

If at x_0 we also have $f_y \neq 0$, then at x_0 we can define

$$\lambda_0 = \frac{F_x}{F_y} = \frac{f_x}{f_y}\,.$$

Overall, we can write the following system in $x_0\,,\ y_0\,,\ \lambda_0\,,$

$$\begin{cases} f_x - \lambda_0\, F_x = 0 \\ f_y - \lambda_0\, F_y = 0 \\ F = 0\,. \end{cases} \tag{11.56}$$

In other terms, the extremum points of $f(x, y)$, under the constraint $F(x, y) = 0$, are *critical points* (i.e. points at which the gradient vanishes; see Sect. 3.13), of the function of three variables x, y, λ

$$H(x, y, \lambda) = f(x, y) - \lambda\, F(x, y)\,.$$

The variable λ is called *Lagrange multiplier*, and the *Lagrange method* is the procedure just described, consisting in solving system (11.56) to find the possible extrema of $f(x, y)$ constrained by $F(x, y) = 0$.

Example 2 Consider the quadratic form in two real variables

$$f(x, y) = ax^2 + 2bxy + cy^2 ,$$

and let us determine the minimum and maximum of f constrained by

$$F(x, y) = x^2 + y^2 - 1 = 0 . \tag{11.57}$$

We have already discussed this problem in Sect. 3.13, Chap. 3. Now we address it (with slightly different notations) in order to apply Lagrange's method.

Write the system of equations

$$\begin{cases} \dfrac{\partial}{\partial x} f(x, y) - \lambda \dfrac{\partial}{\partial x} \left(x^2 + y^2 - 1\right) = 0 \\ \dfrac{\partial}{\partial y} f(x, y) - \lambda \dfrac{\partial}{\partial y} \left(x^2 + y^2 - 1\right) = 0 \end{cases} ,$$

which reduces to

$$\begin{cases} (a - \lambda)\, x + b\, y = 0 \\ b\, x + (c - \lambda)\, y = 0 \end{cases} . \tag{11.58}$$

For x, y to satisfy (11.58) and (11.57) – the latter implying that x, y cannot vanish simultaneously – the homogeneous system (11.58) must have determinant zero, i.e. λ must solve

$$\begin{vmatrix} a - \lambda & b \\ b & c - \lambda \end{vmatrix} = \lambda^2 - (a + c)\, \lambda + ac - b^2 = 0 . \tag{11.59}$$

If λ is a root of the above polynomial, multiplying the first equation in (11.58) by x, the second equation by y and adding, we obtain

$$ax^2 + 2bxy + cy^2 = \lambda\, (x^2 + y^2) = \lambda .$$

We conclude that the minimum and maximum of f will be the smallest and respectively largest solution to (11.59). As the equation is quadratic in λ, it can be solved explicitly because $\Delta = (a - c)^2 + 4b^2 \geq 0$. Letting $\lambda_1 \leq \lambda_2$ denote the roots of (11.59), for any $(x, y) \in \mathbb{R}^2$ such that $x^2 + y^2 = 1$ we have

$$\lambda_1 \leq ax^2 + 2bxy + cy^2 \leq \lambda_2 . \tag{11.60}$$

There will be equality above for particular choices of (x, y) on the unit circle.

(continued)

Example 2 (continued)

Finally, if (x, y) is an arbitrary point of $\mathbb{R}^2$ other than the origin, we can apply inequalities (11.60) to $(x/\sqrt{x^2+y^2}\,,\, y/\sqrt{x^2+y^2}\,)$ to obtain

$$\lambda_1\,(x^2+y^2) \le ax^2+2bxy+cy^2 \le \lambda_2\,(x^2+y^2)\,, \qquad \forall\,(x,y)\in\mathbb{R}^2.$$

Lagrange's method can be applied to more general situations. Given $n, h \in \mathbb{N}$, take f together with functions $F_1\,,\, F_2\,,\,\ldots\,,\, F_h$ in $n+h$ variables

$$(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_h) = (x, y) \in \mathbb{R}^n \times \mathbb{R}^h$$

defined and C^1 on the open set A of $\mathbb{R}^{n+h}$. Let Z_0 be the set of points $(x, y) \in A$ satisfying

$$\begin{cases} F_1(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_h) = 0 \\ F_2(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_h) = 0 \\ \ldots\ \ldots\ \ldots\ \ldots\ \ldots\ \ldots\ \ldots\ \ldots \\ F_h(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_h) = 0\,, \end{cases} \tag{11.61}$$

and such that the Jacobian matrix of the $F_i\,,\ i = 1, 2\ldots, h,$

$$\frac{\partial(F_1, F_2, \ldots, F_h)}{\partial(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_h)}\,(x, y) \tag{11.62}$$

has rank h.

We say that f has a *constrained local maximum* (with constraint given by (11.61)) at (x_0, y_0) in Z_0 if there exists a neighbourhood I_0 of the point such that

$$f(x, y) \le f(x_0, y_0) \tag{11.63}$$

for any $(x, y) \in Z_0 \cap I_0$. *Constrained local minima* are defined similarly.

Under the above hypotheses the following result holds.

Theorem (Lagrange Multipliers) *Suppose $(x_0, y_0) \in Z_0$ is a constrained local maximum (or minimum) point of f, with constraint given by system (11.61), and assume that the Jacobian matrix of the F_i, $i = 1, 2\ldots, h$, in (11.62) has rank h. Then there exist h constants $\lambda_1\,,\,\lambda_2\,,\,\ldots\,,\,\lambda_h$ such that the function*

$$f(x, y) - \lambda_1\,F_1(x, y) - \lambda_2\,F_2(x, y) - \ldots - \lambda_h\,F_h(x, y)$$

has zero derivatives at (x_0, y_0).

The proof, seen at the beginning of the chapter in the two-dimensional case, is based on concepts and results that will be discussed in the following chapter.

Proof Suppose (x_0, y_0) is a local maximum point constrained by (11.61), and $I_0 \subseteq A$ is a neighbourhood of the point on which (11.63) holds. By the assumption made on Z_0, in particular since the Jacobian matrix of the F_i, $i = 1, 2 \ldots, h$, in (11.62) has rank h, $Z_0 \cap I_0$ is an n-dimensional manifold.

Let T be a tangent vector to this manifold at (x_0, y_0). Then there exists a regular curve $\gamma(t) : [-\delta, \delta] \to Z_0 \cap I_0$ such that

$$\gamma(0) = (x_0, y_0)\,, \qquad \gamma'(0) = T \in \mathbb{R}^{n+h}\,.$$

Since the composite function $f(\gamma(t))$ has a local maximum at $t = 0$, by the chain rule

$$\frac{d}{dt}\, f(\gamma(t))|_{\,t=0} = (Df(x_0, y_0), T) = 0\,.$$

Hence the gradient of f at (x_0, y_0) is normal to the manifold $Z_0 \cap I_0$ at the point. But the vectors $DF_1(x_0, y_0)$, $DF_2(x_0, y_0)$, $\ldots$, $DF_h(x_0, y_0)$ form a basis of the normal vector space of $Z_0 \cap I_0$ at (x_0, y_0) (see the corollary in Sect. 12.2), so

$$Df(x_0, y_0) = \lambda_1\, DF_1(x_0, y_0) + \lambda_2\, DF_2(x_0, y_0) + \ldots + \lambda_h\, DF_h(x_0, y_0)$$

for suitable constants $\lambda_1, \lambda_2, \ldots, \lambda_h$. The claim then follows.

Example 3 Fix $n \geq 2$ and $p > 0$. We shall determine the minimum of the function f defined, for $x_i > 0$, $i = 1, 2, \ldots, n$, by

$$f(x_1, x_2, \ldots, x_n) = x_1 + x_2 + \ldots + x_n\,,$$

under the constraint that the variables' product is constant:

$$x_1 \cdot x_2 \cdot \ldots \cdot x_n = p\,. \tag{11.64}$$

Define $F(x_1, x_2, \ldots, x_n) = x_1 \cdot x_2 \cdot \ldots \cdot x_n - p$, so we have a minimum problem for f on the set $A = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : x_i > 0,\ i = 1, 2, \ldots, n\}$, with the only constraint $F(x_1, x_2, \ldots, x_n) = 0$. Using Lagrange's method, we first write the n equations

$$\frac{\partial}{\partial x_i}\,(x_1 + x_2 + \ldots + x_n) - \lambda\, \frac{\partial}{\partial x_i}\,(x_1 \cdot x_2 \cdot \ldots \cdot x_n - p) = 0\,,$$

(continued)

Example 3 (continued)

for $i = 1, 2, \ldots, n$. These form a system of n equations in $n + 1$ unknowns (one unknown is the multiplier λ)

$$\begin{cases} x_2 \cdot x_3 \cdot \ldots \cdot x_n = 1/\lambda \\ x_1 \cdot x_3 \cdot \ldots \cdot x_n = 1/\lambda \\ \ldots\ldots\ldots\ldots\ldots \\ x_1 \cdot x_2 \cdot \ldots \cdot x_{n-1} = 1/\lambda\,, \end{cases}$$

giving ∞^1 solutions

$$x_1 = x_2 = \ldots = x_n\,. \tag{11.65}$$

From (11.64), (11.65) then,

$$x_1 = x_2 = \ldots = x_n = \sqrt[n]{p}\,, \tag{11.66}$$

so the minimum of f, of which we shall prove the existence, will be $n \cdot \sqrt[n]{p}$.

We claim that the minimum point of f belongs to the set

$$A_0 = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n :\ x_1 \cdot x_2 \cdot \ldots \cdot x_n = p,\ x_i > 0,\ \forall\, i = 1, 2, \ldots, n\}\,.$$

In fact, (11.66) implies

$$\inf f(A_0) \le f(\sqrt[n]{p}, \sqrt[n]{p}, \ldots, \sqrt[n]{p}) = n \cdot \sqrt[n]{p}\,. \tag{11.67}$$

As $f = x_1 + x_2 + \ldots + x_n$ is a sum of positive terms, to compute the greatest lower bound it suffices, by (11.67), to assume that every summand x_i is less than or equal to $n \cdot \sqrt[n]{p}$. Set $K_0 = \{(x_1, x_2, \ldots, x_n) \in A_0 :\ 0 < x_i \le n \cdot \sqrt[n]{p}\}$, so that $\inf f(A_0) = \inf f(K_0)$. As K_0 is compact (due to the constraint defining A_0; the reader should prove this), there must be a minimum.

Having shown in (11.66) that f has a unique constrained critical point, the latter is necessarily the minimum point. Hence

$$f(x_1, x_2, \ldots, x_n) \ge \min\ f = f(\sqrt[n]{p}, \sqrt[n]{p}, \ldots, \sqrt[n]{p}) = \\ = n \cdot \sqrt[n]{p} = n \cdot \sqrt[n]{x_1 \cdot x_2 \cdot \ldots \cdot x_n}\,,$$

whenever $x_1 \cdot x_2 \cdot \ldots \cdot x_n = p$. More explicitly,

$$n \cdot \sqrt[n]{x_1 \cdot x_2 \cdot \ldots \cdot x_n} \le x_1 + x_2 + \ldots + x_n\,. \tag{11.68}$$

Since either side in the above relationship is homogeneous (of degree 1), inequality (11.68) holds for any $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, where $x_i > 0$, $i = 1, 2, \ldots, n$. To see this just multiply $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ by a factor $k > 0$ so that kx respects the constraint (11.64), then insert kx in (11.68) and, using the homogeneity, divide both sides by k. Observe, further, that if some components x_i, $i \in \{1, 2, \ldots, n\}$, are zero, then (11.68) is trivial.

(continued)

Example 3 (continued)

Finally, note that there will be equality in (11.68) if and only if the coordinates $x_1, x_2, \ldots, x_n$ are all equal. We may recast (11.68) in the more telling form

$$\sqrt[n]{x_1 \cdot x_2 \cdot \ldots \cdot x_n} \leq \frac{x_1 + x_2 + \ldots + x_n}{n}, \qquad x_i \geq 0 \quad \forall\, i = 1, 2, \ldots, n, \tag{11.69}$$

where the left-hand side is the *geometric mean* of the given n positive numbers (or, in general, non-negative numbers), while the right-hand side is their *arithmetic mean*. Hence, *the geometric mean of n positive numbers never exceeds the arithmetic mean*.

Example 4 As in the previous example, one can more generally prove that given n positive numbers $a_1, a_2, \ldots, a_n$, the function

$$f(x_1, x_2, \ldots, x_n) = \sum_{i=1}^{n} a_i\, x_i\,,$$

defined for $x_i > 0, i = 1, 2, \ldots, n$, reaches its minimum under the constraint

$$x_1^{a_1} \cdot x_2^{a_2} \cdot \ldots \cdot x_n^{a_n} = p\,,$$

(where $p > 0$ is a given real number) when

$$x_1 = x_2 = \ldots = x_n = p^{\frac{1}{a_1 + a_2 + \ldots + a_n}}\,.$$

When, in particular, $\sum_{i=1}^{n} a_i = 1$, then

$$\begin{aligned} f(x_1, x_2, \ldots, x_n) &\geq \min\ f = f(p, p, \ldots, p) \\ &= \sum_{i=1}^{n} a_i\, p = p = x_1^{a_1} \cdot x_2^{a_2} \cdot \ldots \cdot x_n^{a_n}\,, \end{aligned}$$

whenever $x_1^{a_1} \cdot x_2^{a_2} \cdot \ldots \cdot x_n^{a_n} = p$. Said more explicitly,

$$x_1^{a_1} \cdot x_2^{a_2} \cdot \ldots \cdot x_n^{a_n} \leq a_1\, x_1 + a_2\, x_2 + \ldots + a_n\, x_n\,. \tag{11.70}$$

Here both sides are homogeneous (of degree 1), so under the hypothesis that $\sum_{i=1}^{n} a_i = 1$, inequality (11.70) holds for any $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, when $x_i > 0,\ i = 1, 2, \ldots, n$. Again, there is equality in (11.70) if (and only if) $x_1, x_2, \ldots, x_n$ are all equal.

(continued)

Example 4 (continued)
If we take

$$a_1 = a_2 = \ldots = a_n = \frac{1}{n}\,,$$

then (11.70) recovers the geometric-arithmetic mean inequality (11.69).

Example 5 As before, suppose $n \geq 2$ and consider the $n \times n$ matrix

$$A(x) = \begin{pmatrix} x_{11} & x_{12} & \ldots & x_{1n} \\ x_{21} & x_{22} & \ldots & x_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ x_{n1} & x_{n2} & \ldots & x_{nn} \end{pmatrix}.$$

It is a function of the n^2 variables x_{ij}, $i, j = 1, 2, \ldots, n$. Let us find the maximum and minimum of the determinant of $A(x)$ (as usual denoted by the symbol $\det A(x)$), over the closed and bounded subset of $\mathbb{R}^{n^2}$ given by the system of equations

$$\begin{cases} x_{11}^2 + x_{12}^2 + \ldots + x_{1n}^2 = b_1^2 \\ x_{21}^2 + x_{22}^2 + \ldots + x_{2n}^2 = b_2^2 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ x_{n1}^2 + x_{n2}^2 + \ldots + x_{nn}^2 = b_n^2\,, \end{cases} \tag{11.71}$$

where b_i, $i = 1, 2, \ldots, n$, are n given positive numbers. Applying the Lagrange method we write the n^2 equations

$$\begin{aligned} \frac{\partial(\det A(x))}{\partial x_{ij}} &= \lambda_1 \frac{\partial}{\partial x_{ij}} (x_{11}^2 + x_{12}^2 + \ldots + x_{1n}^2 - b_1^2) + \\ &+ \lambda_2 \frac{\partial}{\partial x_{ij}} (x_{21}^2 + x_{22}^2 + \ldots + x_{2n}^2 - b_2^2) + \ldots + \\ &+ \lambda_n \frac{\partial}{\partial x_{ij}} (x_{n1}^2 + x_{n2}^2 + \ldots + x_{nn}^2 - b_n^2)\,, \end{aligned}$$

for $i, j = 1, 2, \ldots, n$. The derivative on the left is the cofactor A_{ij} of x_{ij} in A, whereas the derivatives on the right are all zero except for the ijth one, which is $2x_{ij}$. Hence we reduce to

$$A_{ij} = 2\lambda_i x_{ij}\,, \qquad \forall\, i, j = 1, 2, \ldots, n. \tag{11.72}$$

Recalling that the sum of the products of the elements on one row by the cofactors of a different row is always zero, from (11.72) we infer

(continued)

Example 5 (continued)

$$0 = \sum_{j=1}^{n} A_{ij}\, x_{hj} = 2\lambda_i \sum_{j=1}^{n} x_{ij}\, x_{hj}\,, \qquad \forall\, i \neq h\,. \tag{11.73}$$

We claim that at an extremum point the Lagrange multipliers λ_i are all non-zero. In fact if for a maximum point we had $\lambda_i = 0$, then (11.72) would force $A_{ij} = 0$ for any $j = 1, 2, \ldots, n$, hence $\det A(x) = 0$. This cannot be because the maximum of $x \to \det A(x)$, constrained by (11.71), is positive (for instance, if $x_{ij} = b_i\, \delta_{ij}$, then $\det A(x) = b_1 \cdot b_2 \cdot \ldots \cdot b_n > 0$). Eventually, from (11.73) we have

$$\sum_{j=1}^{n} x_{ij}\, x_{hj} = 0\,, \qquad \forall\, i \neq h \tag{11.74}$$

at an extremum point. Now consider the matrix $C = A \cdot A^T$, with elements (c_{ij}). From (11.74) we deduce that at an extremum

$$c_{ih} = \sum_{j=1}^{n} x_{ij}\, x_{hj} = \begin{cases} 0 & \text{if } i \neq h \\ \sum_{j=1}^{n} x_{ij}^2 = b_i^2 & \text{if } i = h \end{cases}$$

and so

$$(\det A(x))^2 = \det C = b_1^2 \cdot b_2^2 \cdot \ldots \cdot b_n^2\,.$$

Therefore the maximum and minimum of $\det A(x)$ are $\pm b_1 \cdot b_2 \cdot \ldots \cdot b_n$.

So now

$$|\det A(x)| \leq b_1 \cdot b_2 \cdot \ldots \cdot b_n\,,$$

and using the geometric-arithmetic mean inequality (11.69), we obtain

$$|\det A(x)|^2 \leq b_1^2 \cdot b_2^2 \cdot \ldots \cdot b_n^2 \leq \frac{1}{n^n} \left(b_1^2 + b_2^2 + \ldots + b_n^2\right)^n\,. \tag{11.75}$$

By definition of matrix norm, then,

$$b_1^2 + b_2^2 + \ldots + b_n^2 = \sum_{i=1}^{n} \left(x_{i1}^2 + x_{i2}^2 + \ldots + x_{in}^2\right) = \sum_{i,j=1}^{n} x_{ij}^2 = |A(x)|^2$$

and finally (11.75) gives

$$|\det A(x)| \leq n^{-n/2}\, |A(x)|^n\,, \tag{11.76}$$

known as *Hadamard's inequality*. The reader should prove (11.76) in two dimensions $(i, j = 1, 2)$ using the elementary inequality $ab \leq (a^2 + b^2)/2$.

Appendix to Chap. 11

11.5 Singular Points of a Plane Curve

Let $F : A \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a C^1 function on the open set $A \subset \mathbb{R}^2$. Call Γ the set of points $(x, y) \in \mathbb{R}^2$ defined by

$$F(x, y) = 0\,. \tag{11.77}$$

Assuming that the partial derivatives F_x , F_y do not vanish simultaneously at any point of Γ, by the implicit function theorem the intersection between Γ and a suitable neighbourhood of any of its points (x_0, y_0) is the graph of a C^1 function in one of the variables x , y. Therefore Γ is *locally a regular curve*, of which Eq. (11.77) is an implicit representation.

We have already shown that the straight line of equation

$$F_x(x_0, y_0)\,(x - x_0) + F_y(x_0, y_0)\,(y - y_0) = 0 \tag{11.78}$$

is the tangent to Γ at (x_0, y_0). Using the gradient vector $DF = (F_x\,, F_y)$, if $P \equiv (x, y)$ is a generic point on the tangent line distinct from $P_0 \equiv (x_0, y_0)$, we may write (11.78) in the vectorial form

$$(DF(P_0),\, P - P_0) = 0\,,$$

which shows that the gradient vector $DF(P_0)$ at P_0 is orthogonal to the tangent line to Γ at P_0. The line through P_0 and parallel to $DF(P_0) = (F_x(P_0)\,, F_y(P_0))$ is the *normal line* to Γ through P_0 , and has equation

$$F_y(x_0, y_0)\,(x - x_0) - F_x(x_0, y_0)\,(y - y_0) = 0\,.$$

We denote by Γ_s the subset of Γ where both partial derivatives of F vanish. Let us suppose Γ_s is non-empty and that its interior is empty (for instance, Γ_s could be a finite number of points). Then Γ is still called a *curve with implicit equation* (11.77), the points in $\Gamma - \Gamma_s$ are said *regular*, while those in Γ_s are *singular points*.

We further assume that on a neighbourhood of $P_0 \equiv (x_0, y_0)$, the function F has continuous first and second derivatives. One calls $P_0 \equiv (x_0, y_0)$ a *double point* of Γ if at P_0 both F_x, F_y vanish but there is at least one non-zero second derivative.

One says that $P_0 \equiv (x_0, y_0)$ is an *isolated singular point* if $H(x_0, y_0) > 0$, where H is the *Hessian (determinant)*

$$H(x, y) = \begin{vmatrix} F_{xx} & F_{xy} \\ F_{xy} & F_{yy} \end{vmatrix} = F_{xx}\,F_{yy} - F_{xy}^2\,,$$

In such a case P_0 is a proper local extremum (maximum or minimum), and there is a neighbourhood of P_0 where F is always positive or always negative. On this neighbourhood, therefore, there are no other points of Γ.

Example 1 Let a be a positive parameter and consider the function

$$F(x, y) = y^2 + ax^2 - x^3 .$$

The origin belongs to the zero set Γ of F, because $F(0, 0) = 0$. We have

$$\frac{\partial F}{\partial x} = 2ax - 3x^2 , \qquad \frac{\partial F}{\partial y} = 2y ,$$

so the origin is a singular point. The Hessian is

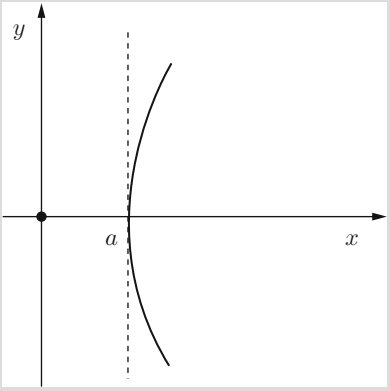
$$H(0, 0) = 4a > 0 ,$$

showing that $(0, 0)$ is an isolated singular point of F. As

$$F(x, y) = 0 \quad \Leftrightarrow \quad y^2 = x^2 (x - a) , \tag{11.79}$$

we have $y^2 = x^2 (x - a) \geq 0$ when $x \geq a$ or $x = 0$. Hence the points of Γ have first coordinate zero or not less than a. Figure 11.8 represents the set Γ.

Fig. 11.8



If $H(x_0, y_0) < 0$, the point $P_0 \equiv (x_0, y_0)$ is neither a maximum nor a minimum for F. As $F(x_0, y_0) = 0$, on any neighbourhood of P_0 the function F is both positive and negative, and the associated quadratic form

$$\mathcal{F}(x, y) = F_{xx}(P_0) (x - x_0)^2 + 2 F_{xy}(P_0) (x - x_0) (y - y_0) + \\ + F_{yy}(P_0) (y - y_0)^2$$

is indefinite. If $F_{yy}(P_0) \neq 0$, the equation

$$\mathcal{F}(x, y) = 0 \tag{11.80}$$

represents the two lines through P_0 of equation

$$y - y_0 = a\,(x - x_0)\,, \qquad y - y_0 = b\,(x - x_0)\,,$$

where a, b are the solutions of (11.80) regarded as a quadratic equation in the unknown $(y - y_0)/(x - x_0)$.

If $F_{yy}(P_0) = 0$, Eq. (11.80) represents the two lines

$$x = x_0\,, \qquad y - y_0 = -\frac{F_{xx}(P_0)}{2\,F_{xy}(P_0)}\,(x - x_0)\,.$$

In the case under exam, where $H(x_0, y_0) < 0$, P_0 is called a *node*, and the distinct tangent lines through it are called *nodal tangents*.

Example 2 Consider

$$F(x, y) = y^2 + ax^2 - x^3$$

from Example 1, but now with $a < 0$. By (11.79) we have

$$F(x, y) = 0 \quad \Leftrightarrow \quad y = \pm f(x)\,,$$

where

$$f(x) = \begin{cases} -x\sqrt{x - a} & \text{if} \quad a \le x \le 0 \\ x\sqrt{x - a} & \text{if} \quad x \ge 0\,. \end{cases}$$

Hence $H(0, 0) = 4a < 0$. The origin is singular, actually a node. To find the nodal tangents we consider Eq. (11.80) and set $m = y/x$, giving

$$F_{yy}\,m^2 + 2\,F_{xy}\,m + F_{xx} = 0\,. \tag{11.81}$$

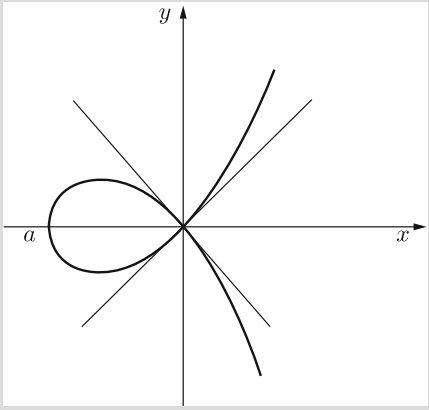
This reduces to $m^2 + a = 0$, and since $a < 0$, there are two branches of Γ passing through the singular point, with distinct tangent lines:

$$y = \sqrt{-a}\,x\,, \qquad y = -\sqrt{-a}\,x\,.$$

(continued)

Example 2 (continued)
Γ is the support of the curve in Fig. 11.9.

Fig. 11.9



At last, if $H(x_0, y_0) = 0$, then $F(x, y)$ may have a proper local extremum at P_0, in which case P_0 would be an isolated singular point. Otherwise Γ has one tangent at P_0, into which the two lines in (11.80) merge. One possibility is that Γ is a regular curve with a *cusp* at P_0.

Example 3 Let us study the singular points of the curve Γ of equation

$$F(x, y) = y^2 - x^3 + x^4 = 0\,.$$

The partial derivatives of F are

$$F_x = x^2\,(4x - 3)\,, \qquad F_y = 2y\,.$$

They both vanish at $(0, 0)$ and $(3/4, 0)$, but $F(3/4, 0) \neq 0$ so the only singular point is the origin. The second partial derivatives are

$$F_{xx} = 6x\,(2x - 1)\,, \qquad F_{xy} = 0\,, \qquad F_{yy} = 2\,,$$

and

$$H(x, y) = 12\,x\,(2x - 1)\,.$$

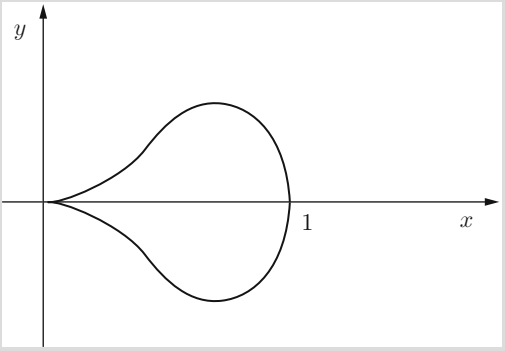
At the origin $H(0, 0) = 0$, so we have a cusp with tangent given by (11.81), i.e. $2\,m^2 = 0$. Hence the x-axis is tangent to the curve at the origin.

(continued)

Example 3 (continued)

The curve's intersections with a generic line $x = a$ occur for $y = \pm a\sqrt{a\,(1-a)}$ and are real when $0 \leq a \leq 1$. Hence the curve with support Γ, represented in Fig. 11.10, has first coordinates in $[0, 1]$.

Fig. 11.10





Manifolds in $\mathbb{R}^n$ and k-Forms

12

In this chapter we shall study *k-dimensional submanifolds of* $\mathbb{R}^n$ and *differential k-forms*. These topics are not usually covered by a standard *Mathematical Analysis II* course, but are discussed in advanced *Geometry* lectures, with different language and symbols.

Nevertheless many results, regarding the integration over curves and surfaces in $\mathbb{R}^n$ especially, are used in higher-level *Analysis* lectures. Examples include the *divergence theorem* and the formula for *integrating by parts* functions in n variables, proved in Sect. 12.4, *Stokes's formula* of Sect. 12.8, or the properties of *closed* and *exact differential forms* on $\mathbb{R}^n$ of Sect. 12.9.

Broadly speaking, students should use this chapter essentially as reference, to look up certain concepts and results, or for future study.

12.1 k-Dimensional Manifolds in $\mathbb{R}^n$

Let M be a subset of $\mathbb{R}^n$, with $n \geq 2$ and $k \in \mathbb{N}$ so that $1 \leq k < n$. We call M a *k-dimensional submanifold of* $\mathbb{R}^n$ when, for any $x_0 \in M$, we can find an open neighbourhood A of x_0 and a C^1 map $F : A \subseteq \mathbb{R}^n \to \mathbb{R}^{n-k}$ such that

(i) the Jacobian matrix $DF(x)$ ha maximal rank for any $x \in A$;
(ii) $M \cap A = \{x \in A : \ F(x) = 0\}$.

We shall often shorten the name to *k-manifold*.

Since the Jacobian matrix $DF(x)$ has $n-k$ rows and n columns, condition (i) is equivalent to requiring that the rank is $n-k$. Moreover, if $DF(x_0)$ has rank $n-k$ at $x_0 \in M$, by the continuity of the first derivatives $DF(x)$ has rank $n-k$ also near x_0. In the end (i) is the same as saying that the Jacobian matrix $DF(x)$ has rank $n-k$ at x_0. With (ii) we demand that on a suitable open neighbourhood A of x_0 the function $F : A \to \mathbb{R}^{n-k}$ vanishes only on the manifold M.

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https://doi.org/10.1007/978-3-031-04151-8_12

If, for any $x_0 \in M$, we can find F of class C^r, with $r = 2, 3, \ldots, \infty$, one says that *the manifold M is of class C^r*.

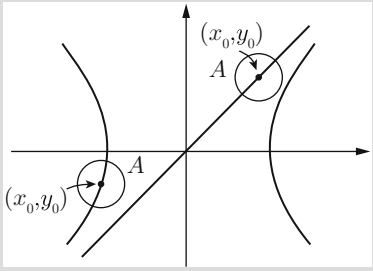
The reader should be aware that the notions of *regular surface* in $\mathbb{R}^3$, according to the definition of Chap. 10, and *two-dimensional submanifold* of $\mathbb{R}^3$ are different (see also the remark at the end of the section).

Example 1 Consider an example with $n = 2$ and $k = 1$. Referring to Fig. 12.1, set

$$M = \{(x, y) \in \mathbb{R}^2 : \ x^2 - y^2 = 1\} \cup \{(x, y) \in \mathbb{R}^2 : \ y = x\}.$$

M is a one-dimensional manifold, because it consists of the straight line r of equation $y = x$, and the hyperbola γ of equation $x^2 - y^2 = 1$. Consider a point $(x_0\,, y_0) \in M$ on the line and a ball A around it that does not meet γ. Setting $F(x, y) = x - y$, conditions (i) and (ii) hold. But if we take $(x_0\,, y_0) \in M$ on the hyperbola γ, and A is a ball around it that does not intersect r, then (i) and (ii) are true for the function $F(x, y) = x^2 - y^2 - 1$.

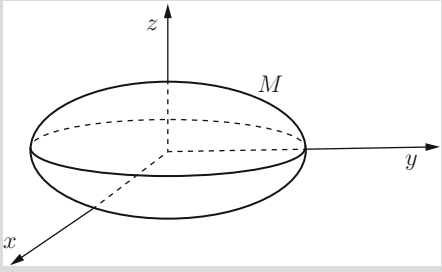
Fig. 12.1



Example 2 Consider a case with $n = 3$ and $k = 2$. Referring to Fig. 12.2, define

$$M = \{(x, y, z) \in \mathbb{R}^3 : \ 2x^2 + y^2 + 8z^2 - 8xz = 4\}\,.$$

Fig. 12.2



(continued)

Example 2 (continued)
M is a two-dimensional manifold, because for every point $(x_0, y_0, z_0) \in M$ the function $F(x, y, z) = 2x^2 + y^2 + 8z^2 - 8xz - 4$ satisfies the requirements. In particular, for $(x_0, y_0, z_0) \in M$,

$$DF(x_0, y_0, z_0) = (4x_0 - 8z_0\,, 2y_0\,, 16z_0 - 8x_0)$$

is a non-zero vector (3×1 matrix), so $DF(x, y, z)$ does not vanish on a suitable open neighbourhood A of the point.

Example 3 Now take $n \geq 2$ generic in $\mathbb{N}$ and $k = n - 1$. Let

$$S^{n-1} = \{x \in \mathbb{R}^n : \ |x| = 1\}$$

be the unit sphere in $\mathbb{R}^n$. For $x_0 \in S^{n-1}$, the gradient of $F(x) = |x|^2 - 1$ is $DF(x) = 2x$, which is non-zero on an open neighbourhood A of $x_0 \in S^{n-1}$. Therefore S^{n-1} is an $(n-1)$-manifold.

Example 4 Let A be an open subset in $\mathbb{R}^n$ and $F : A \to \mathbb{R}^{n-k}$ a map defined on A. If $F \in C^1(A)$, the set

$$M = \{x \in A : \ F(x) = 0 \quad \text{and} \quad \operatorname{rank} DF(x) = n - k\}\,,$$

where $1 \leq k < n$, is a k-manifold. Given $x_0 \in M$ in fact, $DF(x_0)$ has rank $n - k$ by definition of M. The derivatives' continuity implies that $DF(x)$ has rank $n - k$ on a neighbourhood of x_0. There, conditions (i), (ii) hold.

In the special case where $F : A \subseteq \mathbb{R}^n \to \mathbb{R}$, so $k = n - 1$ and

$$M = \{x \in A : \ F(x) = 0\,, \quad DF(x) \neq 0\}\,,$$

M is an $(n - 1)$-submanifold of $\mathbb{R}^n$, as in Examples 2 and 3.

Example 5 Choose $n = 3,\ k = 1$ and define

$$M = \{(x, y, z) \in \mathbb{R}^3 : \ y^2 - 2z = 0\} \cap \{(x, y, z) \in \mathbb{R}^3 : \ x^2 + y^2 - 6z = 1\}\,.$$

We claim that M is a one-dimensional manifold. Based on the previous example it is enough to show that M consists of the zeroes of $F : \mathbb{R}^3 \to \mathbb{R}^2$,

$$F(x, y, z) = (y^2 - 2z, x^2 + y^2 - 6z - 1)\,,$$

(continued)

Example 5 (continued)
and that the Jacobian matrix DF of F has rank 2 everywhere on M. First,

$$DF(x, y, z) = \begin{pmatrix} 0 & 2y & -2 \\ 2x & 2y & -6 \end{pmatrix},$$

and the minor

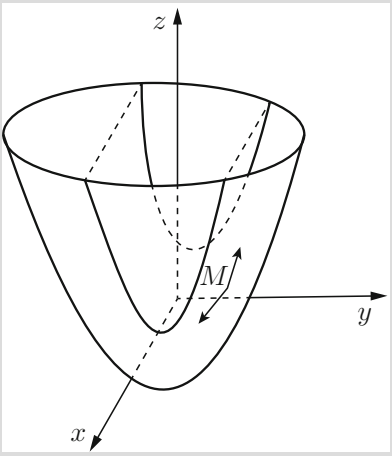
$$\begin{vmatrix} 0 & -2 \\ 2x & -6 \end{vmatrix}$$

is zero if and only if $x = 0$, while the minor

$$\begin{vmatrix} 2y & -2 \\ 2y & -6 \end{vmatrix}$$

vanishes if and only if $y = 0$. As there is no point $(x_0, y_0, z_0) \in M$ for which $x_0 = y_0 = 0$, at least one of the previous minors is non-zero and so rank $DF(x_0, y_0, z_0) = 2$. The reader should also prove that the two sets intersecting in M are 2-dimensional manifolds (Fig. 12.3).

Fig. 12.3



Example 6 Take $n = 4,\ k = 2$ and define

$$M = \{(x, y, z, w) \in \mathbb{R}^4 :\ x^2 + z^2 = 1\,,\ y^2 + w^2 = 1\}\,.$$

(continued)

Example 6 (continued)
M is a two-dimensional manifold in $\mathbb{R}^4$. The map $F : \mathbb{R}^4 \to \mathbb{R}^2$

$$F(x, y, z, w) = (x^2 + z^2 - 1, y^2 + w^2 - 1)$$

has Jacobian matrix

$$DF(x, y, z, w) = \begin{pmatrix} 2x & 0 & 2z & 0 \\ 0 & 2y & 0 & 2w \end{pmatrix},$$

whose rank is 2 at every point of M.

Example 7 Let $f : U \to \mathbb{R}^{n-k}$ be a C^1 map defined on an open set U in $\mathbb{R}^k$. The *graph of* f is

$$M = \left\{x = (x', x'') \in \mathbb{R}^n : \ x' \in U, \ \ x'' = f(x')\right\},$$

where $x' = (x_1, x_2, \ldots, x_k)$, $x'' = (x_{k+1}, x_{k+2}, \ldots, x_n)$. M is a k-manifold, since $M = \{x \in U \times \mathbb{R}^{n-k} : \ F(x) = 0\}$, where $F : U \times \mathbb{R}^{n-k} \to \mathbb{R}^{n-k}$ has components

$$F_i(x) = x_{k+i} - f_i(x_1, x_2, \ldots, x_k), \qquad i = 1, 2, \ldots, n-k.$$

The Jacobian matrix $DF(x)$ has constant rank $n - k$, since

$$\det \frac{\partial(F_1, F_2, \ldots, F_{n-k})}{\partial(x_{k+1}, x_{k+2}, \ldots, x_n)} = 1, \qquad \text{on } U \times \mathbb{R}^{n-k}.$$

An alternative definition of manifold is provided by the following result, itself a consequence of the implicit function theorem for systems (Sect. 11.2) and of what was remarked in the previous example.

Theorem *A subset $M \subseteq \mathbb{R}^n$ is a k-dimensional submanifold if and only if, for any $x_0 \in M$, there is an open neighbourhood A of x_0 such that $M \cap A$ is the graph of a C^1 map $f : U \to \mathbb{R}^{n-k}$ defined on an open subset $U \subseteq \mathbb{R}^k$.*

Proof Let M be a k-manifold and x_0 a point on it. By definition there exist an open neighbourhood A' of x_0 and a function $F : A' \to \mathbb{R}^{n-k}$ of class C^1 such that

$$M \cap A' = \left\{x \in A' : \ F(x) = 0\right\},$$

where $\operatorname{rank} DF(x) = n - k$ on A'. Without loss of generality we may assume

$$\det \frac{\partial(F_1, F_2, \ldots, F_{n-k})}{\partial(x_{k+1}, x_{k+2}, \ldots, x_n)}(x_0) \neq 0,$$

for example. The implicit function theorem (Sect. 11.2) ensures that there exist an open neighbourhood A of x_0, with $A \subseteq A'$, an open neighbourhood U of the point $(x_{01}, x_{02}, \ldots, x_{0k}) \in \mathbb{R}^k$ and a C^1 map $f : U \to \mathbb{R}^{n-k}$ such that

$$M \cap A = \{x \in A : \ F(x) = 0\} = \left\{x \in \mathbb{R}^n : \ x' \in U, \ x'' = f(x')\right\},$$

where $x = (x', x'') \in \mathbb{R}^n$, $x' = (x_1, x_2, \ldots, x_k) \in \mathbb{R}^k$, $x'' = (x_{k+1}, x_{k+2}, \ldots, x_n) \in \mathbb{R}^{n-k}$.

Given $x_0 \in M$, conversely, let us suppose that for some open neighbourhood A' of x_0 the intersection $M \cap A'$ is the graph of a C^1 function $f : U \to \mathbb{R}^{n-k}$, with $U \subseteq \mathbb{R}^k$ open. We may assume that, without any loss of generality, f is a function of the variables $(x_1, x_2, \ldots, x_k)$. Then

$$M \cap A' = \left\{x \in \mathbb{R}^n : \ x' \in U, \ \ x'' = f(x')\right\}.$$

Set $A = U \times \mathbb{R}^{n-k} \cap A'$ and consider the map $F : A \to \mathbb{R}^{n-k}$ of components $F_i(x) = x_{k+1} - f_i(x_1, x_2, \ldots, x_k)$, $i = 1, 2, \ldots, n-k$. Then

$$M \cap A = \{x \in A : \ F(x) = 0\},$$

proving the claim.

Example 8 $M = \{(x, y) \in \mathbb{R}^2 : \ y^2 - x^2 = 0\}$ is the union of the straight lines of equation $y = x$ and $y = -x$, so the origin is a point of M. However, on any neighbourhood of the origin M is neither the graph of a function in x nor in y. By the above theorem M is not a one-dimensional manifold.

The following proposition is a straightforward consequence of the previous theorem.

Proposition 1 *Let M be a k-manifold in $\mathbb{R}^n$. For any $x_0 \in M$ there exist an open subset $U \subseteq \mathbb{R}^k$, an open neighbourhood A of x_0 and a map $\varphi : U \to \mathbb{R}^n$, of class C^1 and injective, such that*

(j) $\varphi(U) = M \cap A$;
(jj) $\operatorname{rank} D\varphi = k$ *on* U.

The map φ, whose existence is granted by Proposition 1, is called a *local parametrisation of the manifold M*.

Proof Fix $x_0 \in M$. By the previous theorem there exist an open neighbourhood A of x_0 and a map $f : U \to \mathbb{R}^{n-k}$, of class C^1 on the open set $U \subset \mathbb{R}^k$, such that

$M \cap A$ is the graph of f. Suppose, for simplicity, that f depends on the variables $(x_1, x_2, \dots, x_k)$. Then

$$M \cap A = \left\{x = (x', x'') \in \mathbb{R}^n : \ x' \in U, \ x'' = f(x')\right\},$$

where $x' = (x_1, x_2, \dots, x_k) \in \mathbb{R}^k$, $x'' = (x_{k+1}, x_{k+2}, \dots, x_n) \in \mathbb{R}^{n-k}$. For any $x' \in U$ we set

$$\varphi(x') = \left(x_1, x_2, \dots, x_k, f_1(x'), f_2(x'), \dots, f_{n-k}(x')\right),$$

so φ is clearly 1-1, the image equals $\varphi(U) = M \cap A$, and condition (jj) holds, because

$$\det \frac{\partial(\varphi_1, \varphi_2, \dots, \varphi_k)}{\partial(x_1, x_2, \dots, x_k)}(x') = 1$$

for any $x' \in U$.

All in all, we may regard a manifold M as being, locally, either the zero set of a suitable function, or the image of an open subset of $\mathbb{R}^k$ under some parametrisation, or the graph of a map defined on an open subset of $\mathbb{R}^k$. In the sequel, when studying the properties of manifolds, we shall use the point of view that is most convenient to treat the situation at hand.

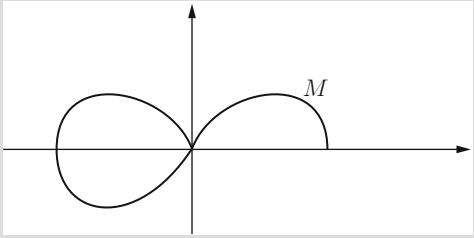
Example 9 Suppose that $\varphi : U \to \mathbb{R}^n$ is a C^1 map on the open set $U \subseteq \mathbb{R}^k$, of class C^1, injective and satisfying (jj). Even so, $\varphi(U)$ will *not be a k-manifold* in general.

For example, consider the curve $\varphi : \left(0, \dfrac{3\pi}{2}\right) \to \mathbb{R}^2$ of equations

$$\varphi(t) = (\cos t, \cos t \sin t), \qquad t \in \left(0, \frac{3\pi}{2}\right),$$

whose support $M = \varphi((0, (3\pi)/2))$ is drawn in Fig. 12.4. As $\varphi'(t) \neq 0$ for any t, φ verifies (jj). But M is not a 1-manifold, because around the origin M is not the graph of a function in x or in y.

Fig. 12.4



To assess the concepts explained in this section the reader should compare the above example with the following result.

Proposition 2 *Let $\varphi : U \to \mathbb{R}^n$ be a C^1 injective map on the open set $U \subseteq \mathbb{R}^k$ that satisfies* (jj)*. Any point $y_0 \in U$ admits an open neighbourhood $V \subseteq U$ such that $\varphi(V)$ is a k-manifold in $\mathbb{R}^n$.*

Proof Fix $y_0 \in U$. Without loss of generality we may assume

$$\det \frac{\partial(\varphi_1, \varphi_2, \ldots, \varphi_k)}{\partial(y_1, y_2, \ldots, y_k)}(y_0) \neq 0\,. \tag{12.1}$$

Define $\Phi : U \to \mathbb{R}^k$ by $\Phi = \Pi \circ \varphi$, where $\Pi : \mathbb{R}^n \to \mathbb{R}^k$ is the projection on the first k coordinates:

$$\Pi(x) = x' = (x_1, x_2, \ldots, x_k)\,, \qquad \forall\, x \in \mathbb{R}^n.$$

By (12.1) the Jacobian of Φ is non-zero at y_0. The local invertibility theorem (Sect. 11.3) guarantees that there exist an open neighbourhood V of y_0 and an open set W in $\mathbb{R}^k$ such that $\Phi(V) = W$, the restriction of Φ to V is invertible and the inverse $\Psi : W \to V$ is also C^1 (Fig. 12.5).

Define $f : W \to \mathbb{R}^{n-k}$ by

$$f_i(x') = \varphi_{k+i}\left(\Psi(x')\right)$$

for any $i = 1, 2, \ldots, n-k$ and $x' \in W$. Clearly f is of class C^1 on W, and $\varphi(V)$ is by definition the graph

$$\left\{x = (x', x'') \in \mathbb{R}^n :\ x' \in W,\ x'' = f(x')\right\},$$

where $x'' = (x_{k+1}, x_{k+2}, \ldots, x_n)$.

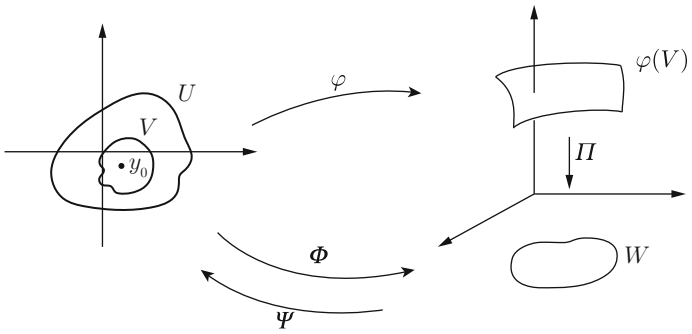


Fig. 12.5

Remark 1 It is clear from Example 9 that if $\varphi : I \to \mathbb{R}^n$ is a regular curve, $\varphi(I)$ need not be a one-dimensional manifold, not even if φ is simple (for example, the simple, closed, regular curve in (6.7) is not a manifold). Proposition 2 nonetheless implies that if φ is regular and simple, for any $t_0 \in \overset{\circ}{I}$ there is an open neighbourhood $J \subseteq \overset{\circ}{I}$ of t_0 such that $\varphi(J)$ is a one-dimensional manifold.

Similar considerations apply to surfaces in space. If $\varphi : D \to \mathbb{R}^3$ is a regular surface, $\varphi(D)$ might not be a two-dimensional manifold (for example, the surface in (10.2) is not). Yet, by Proposition 2, any $(u_0, v_0) \in \overset{\circ}{D}$ has an open neighbourhood $U \subseteq \overset{\circ}{D}$ such that $\varphi(U)$ is a manifold.

12.2 The Tangent Space and the Normal Space of a Manifold

Let M be a k-dimensional manifold in $\mathbb{R}^n$ and x_0 a point of M. A vector $h \in \mathbb{R}^n$ is *tangent* to the manifold at x_0 if, for some $\delta > 0$, there is a regular curve $\gamma : [-\delta, \delta] \to M$ such that

$$\gamma(0) = x_0 \quad \text{and} \quad \gamma'(0) = h\,.$$

The collection of all tangent vectors to M at x_0 is the *tangent space* to the manifold at x_0. We shall use the symbol $T_M(x_0)$.

This definition generalises in a natural way that of tangent plane to a regular surface S seen in Sect. 10.3. In both cases we consider the set of tangent vectors at x_0 to all possible regular curves through x_0 contained in M, or S.

Now suppose $F : A \to \mathbb{R}^{n-k}$ is a C^1 map, defined on the open neighbourhood A of x_0 and satisfying conditions (i) and (ii) of Sect. 12.1. Using F we can describe the tangent space $T_M(x_0)$ explicitly.

Theorem *$T_M(x_0)$ is the kernel of the linear map $DF(x_0) : \mathbb{R}^n \to \mathbb{R}^{n-k}$, i.e.*

$$T_M(x_0) = \left\{h \in \mathbb{R}^n : \ DF(x_0) \cdot h = 0\right\}.$$

Hence $T_M(x_0)$ is a vector subspace of $\mathbb{R}^n$ of dimension k.

Proof Take $h \in T_M(x_0)$ and a regular curve $\gamma : [-\delta, \delta] \to M$ such that $\gamma(0) = x_0$, $\gamma'(0) = h$. Then for any $t \in [-\delta, \delta]$,

$$F(\gamma(t)) = 0\,,$$

so

$$DF(x_0) \cdot h = \frac{d}{dt} F(\gamma(t))\Big|_{t=0} = 0\,.$$

Hence h belongs in the kernel of the linear map $DF(x_0)$.

Now suppose $DF(x_0)\cdot h = 0$. As the matrix $DF(x_0)$ has rank $n-k$, it will have $n-k$ linearly independent columns, and without loss of generality we may assume these to be

$$\frac{\partial F}{\partial x_{k+1}}(x_0)\,, \quad \frac{\partial F}{\partial x_{k+2}}(x_0)\,, \ \ldots\,, \ \frac{\partial F}{\partial x_n}(x_0)\,.$$

Consider the map $\Phi : A \subseteq \mathbb{R}^n \to \mathbb{R}^n$ of components

$$\begin{cases} y_1 = x_1 \\ \cdots\cdots\cdots \\ y_k = x_k \\ y_{k+1} = F_1(x) \\ \cdots\cdots\cdots \\ y_n = F_{n-k}(x)\,. \end{cases} \tag{12.2}$$

The Jacobian matrix

$$D\Phi(x_0) = \begin{pmatrix} 1 & 0 & \ldots & 0 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 & 0 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \ldots & 1 & 0 & \ldots & 0 \\ \frac{\partial F_1}{\partial x_1}(x_0) & \ldots & \ldots & \frac{\partial F_1}{\partial x_k}(x_0) & \ldots & \ldots & \frac{\partial F_1}{\partial x_n}(x_0) \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ \frac{\partial F_{n-k}}{\partial x_1}(x_0) & \ldots & \ldots & \frac{\partial F_{n-k}}{\partial x_k}(x_0) & \ldots & \ldots & \frac{\partial F_{n-k}}{\partial x_n},(x_0) \end{pmatrix}$$

has determinant

$$J_\Phi(x_0) = \det \frac{\partial(F_1, F_2, \ldots, F_{n-k})}{\partial(x_{k+1}, x_{k+2}, \ldots, x_n)}(x_0)$$

different from zero. Then there exist an open set $A' \subseteq A$ containing x_0 and an open neighbourhood U of $y_0 = \Phi(x_0) \in \mathbb{R}^n$, such that $\Phi(A') = U$, Φ restricted to A' is invertible and the inverse $\Psi : U \to A'$ is of class C^1. Moreover, the Jacobian matrix $D\Psi(y_0)$ equals the inverse of $D\Phi(x_0)$.

Let $\overline{h} \in \mathbb{R}^n$ be the vector whose first k components are equal to those of h and the remaining ones all zero:

$$\overline{h} = (h_1\,, h_2\,, \ldots, h_k\,, 0, 0, \ldots, 0)\,.$$

There exists $\delta > 0$ such that $y_0 + t\,\overline{h} \in U$ for any $t \in [-\delta, \delta]$. The curve $\gamma(t) = \Psi(y_0 + t\,\overline{h})$, with $t \in [-\delta, \delta]$, is regular with support contained in M. In fact $\gamma(t) \in A'$, so $\gamma(t) \in A$ for any $t \in [-\delta, \delta]$. Moreover $F(\gamma(t)) = 0$ by definition (12.2) and

because of the choice of $\overline{h}$ (the vector $y_0 + t\,\overline{h}$ has the last $n-k$ components zero). Furthermore

$$\gamma(0) = x_0 \quad \text{and} \quad \gamma'(0) = D\Psi(y_0)\cdot\overline{h}\,.$$

As

$$D\Phi(x_0)\cdot h = (h_1\,,\ldots,h_k\,,DF_1\cdot h,\ldots,DF_{n-k}\cdot h) = \overline{h}\,,$$

and $D\Psi(y_0) = [D\Phi(x_0)]^{-1}$, we obtain $\gamma'(0) = h$. In conclusion, $h \in T_M(x_0)$.

Example 1 Consider $M = \{(x,y,z)\in\mathbb{R}^3 :\ F(x,y,z) = 0\}$ where

$$F(x,y,z) = 2x^2 + y^2 + 8z^2 - 8xz - 4\,.$$

This manifold was considered in Example 2, Sect. 12.1 and is represented in Fig. 12.2. The point $(2,2,1)$ lies on M. As $DF(2,2,1) = (0,4,0)$, a vector $h\in\mathbb{R}^3$ is tangent to M at the given point if and only if the inner product $((0,4,0),h)$ vanishes. Hence the tangent space to M at $(2,2,1)$ is the subspace consisting of the vectors of the Oxz-plane.

Example 2 Let $M = \{(x,y,z)\in\mathbb{R}^3 :\ F(x,y,z) = 0\}$ be defined by $F:\mathbb{R}^3\to\mathbb{R}^2$,

$$F(x,y,z) = (y^2 - 2z, x^2 + y^2 - 6z - 1)\,.$$

This is the one-dimensional manifold considered in Example 5, Sect. 12.1, see Fig. 12.3. We have

$$DF(x,y,z) = \begin{pmatrix} 0 & 2y & -2 \\ 2x & 2y & -6 \end{pmatrix}.$$

The point $(3,2,2)$ belongs to M. A vector $h\in\mathbb{R}^3$ is tangent to M at $(3,2,2)$ if and only if

$$DF(3,2,2)\cdot h = \begin{pmatrix} 0 & 4 & -2 \\ 6 & 4 & -6 \end{pmatrix}\cdot\begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = 0\,,$$

i.e. for

$$\begin{cases} 2h_2 - h_3 = 0 \\ 3h_1 + 2h_2 - 3h_3 = 0\,. \end{cases} \tag{12.3}$$

Putting $h_1 = 4t$, from (12.3) we obtain $h_2 = 3t$ and $h_3 = 6t$. Hence $T_M(3,2,2)$ is the linear space spanned by the vector $(4,3,6)$.

Example 3 Consider the manifold of Example 6, Sect. 12.1, i.e.

$$M = \{(x, y, z, w) \in \mathbb{R}^4 : \ F(x, y, z, w) \ = 0\},$$

where $F : \mathbb{R}^4 \to \mathbb{R}^2$ is defined by

$$F(x, y, z, w) = (x^2 + z^2 - 1, y^2 + w^2 - 1)$$

and has Jacobian matrix

$$DF(x, y, z, w) = \begin{pmatrix} 2x & 0 & 2z & 0 \\ 0 & 2y & 0 & 2w \end{pmatrix}.$$

Let us find the tangent space to M at $(0, 1, -1, 0)$. A vector $h \in \mathbb{R}^4$ belongs to the tangent space at $(0, 1, -1, 0)$ if and only if

$$DF(0, 1, -1, 0) \cdot h = \begin{pmatrix} 0 & 0 & -2 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{pmatrix} = \begin{pmatrix} -2h_3 \\ 2h_2 \end{pmatrix} = 0\,.$$

We conclude that $T_M(0, 1, -1, 0) = \{h \in \mathbb{R}^4 : \ h_2 = h_3 = 0\}$.

In the previous section (Proposition 1) we have shown how a k-manifold can always be expressed, around any of its points, by means of a local parametrisation. Then it becomes useful to be able to describe the tangent space in terms of this parametrisation.

Proposition *Let M be a k-manifold, $x_0 \in M$ a point and $\varphi : U \to \mathbb{R}^n$ a local parametrisation of M around x_0, where $U \subset \mathbb{R}^k$ is open. Call y_0 the point of U such that $\varphi(y_0) = x_0$. Then the vector space $T_M(x_0)$ is generated by the k column vectors*

$$\frac{\partial \varphi}{\partial y_1}(y_0)\,, \quad \frac{\partial \varphi}{\partial y_2}(y_0)\,, \quad \ldots\,, \quad \frac{\partial \varphi}{\partial y_k}(y_0)\,. \tag{12.4}$$

Proof By condition (jj) in the previous section, the vectors (12.4) are linearly independent and so they span a subspace of dimension k. Then it will suffice to prove that

$$\frac{\partial \varphi}{\partial y_i}(y_0) \in T_M(x_0)\,, \qquad \forall\, i = 1, 2, \ldots, k.$$

Fix $i \in \{1, 2, \ldots, k\}$, and let $\delta > 0$ be such that $y_0 + te_i \in U$ for any $t \in [-\delta, \delta]$. The regular curve $\gamma(t) = \varphi(y_0 + te_i)$, $t \in [-\delta, \delta]$, is contained in M, and

$$\gamma(0) = x_0\,, \qquad \gamma'(0) = \frac{\partial \varphi}{\partial y_i}(y_0) \in T_M(x_0)\,.$$

Remark 1 Let $\gamma : [a, b] \to \mathbb{R}^n$ be a regular simple curve. In the previous section we observed that if $t_0 \in (a, b)$ there is an open interval $I \subseteq (a, b)$, with $t_0 \in I$, such that $\gamma(I)$ is a one-dimensional manifold. The above proposition then says that the tangent space to γ at $\gamma(t_0)$ is the vector space spanned by the tangent vector $\gamma'(t_0)$ at $\gamma(t_0)$.

Similarly, if $\varphi : D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ is a regular surface and $(u_0, v_0) \in \overset{\circ}{D}$, there is an open neighbourhood $V \subseteq \overset{\circ}{D}$ of (u_0, v_0) such that $\varphi(V)$ is a two-dimensional manifold M. Again, the previous proposition guarantees that the tangent plane to φ at $P_0 = \varphi(u_0, v_0)$ is the affine subspace $P_0 + T_M(P_0)$.

Example 4 Consider a C^1 function $f : U \to \mathbb{R}$ on the open set $U \subseteq \mathbb{R}^{n-1}$ and its graph M. Define $\varphi(x') = (x', f(x'))$, with $x' = (x_1, x_2, \ldots, x_{n-1}) \in U$, so φ is a parametrisation of the entire manifold M. By the previous proposition, if $x_0 = (x_0', f(x_0'))$ is a point on M, the space $T_M(x_0)$ is generated by the vectors

$$e_i + \frac{\partial f}{\partial x_i}(x_0')\, e_n\,, \qquad i = 1, 2, \ldots, n-1.$$

More generally, if $f : U \to \mathbb{R}^{n-k}$ is of class C^1 on the open set $U \subseteq \mathbb{R}^k$ and M is its graph, define $\varphi(x') = (x', f(x'))$ for any $x' = (x_1, x_2, \ldots, x_k) \in U$. The tangent space to M at $(x_0', f(x_0'))$ is spanned by the vectors

$$e_i + \sum_{j=1}^{n-k} \frac{\partial f_j}{\partial x_i}(x_0') e_{k+j}\,, \qquad i = 1, 2, \ldots, k.$$

Given a k-manifold M and $x_0 \in M$, we call *normal space to M at x_0* the vector space orthogonal to the tangent space $T_M(x_0)$, i.e.

$$T_M^{\perp}(x_0) = \left\{ k \in \mathbb{R}^n : \ (h, k) = 0 \qquad \forall\, h \in T_M(x_0) \right\}.$$

As $\dim T_M(x_0) = k$, the normal space $T_M^{\perp}(x_0)$ has dimension $n - k$.

Remark 2 Returning to Remark 1, it is clear that if $\varphi : D \to \mathbb{R}^3$ is a regular surface and $P_0 = \varphi(u_0, v_0)$, with $(u_0, v_0) \in \overset{\circ}{D}$, the subspace generated by the normal $\nu(P_0)$ to the surface at P_0 is the normal space at P_0 to $M = \varphi(V)$, where V is a suitable neighbourhood of (u_0, v_0).

Let $x_0 \in M$ be a point, A an open neighbourhood of x_0 and $F : A \to \mathbb{R}^{n-k}$ a map of class C^1 satisfying (i) and (ii) from the previous section.

Corollary *The normal space $T_M^{\perp}(x_0)$ is generated by the vectors $DF_i(x_0)$, $i = 1, 2, \ldots, n-k$.*

Proof Because of (ii), the vectors $DF_i(x_0)$, $i = 1, 2, \ldots, n-k$, are linearly independent, so they span an $(n-k)$-dimensional subspace. Hence it suffices to verify that $DF_i(x_0) \in T_M^{\perp}(x_0)$. But this is obvious, since by the previous theorem $DF_i(x_0) \cdot h = 0$ when $h \in T_M(x_0)$.

Example 5 Let $F : A \subseteq \mathbb{R}^n \to \mathbb{R}$ be a C^1 function on the open set A and M the $(n-1)$-dimensional manifold

$$M = \{x \in A : \ F(x) = 0\,, \ \ DF(x) \neq 0\}\,.$$

If $x_0 \in M$, $T_M^{\perp}(x_0)$ is the 1-dimensional space generated by the vector $DF(x_0)$.

Example 6 Consider $f : U \subseteq \mathbb{R}^{n-1} \to \mathbb{R}$ of class C^1 on an open set U. As we saw in Example 7, Sect. 12.1, the graph of f is the manifold

$$M = \{x \in U \times \mathbb{R} : \ F(x) = 0\}\,,$$

where $F(x) = x_n - f(x')$, $x' = (x_1, x_2, \ldots, x_{n-1})$. By the previous example, if $x_0 = (x_0', f(x_0'))$ is a point of M, the normal space to M at x_0 is generated by the vector

$$DF(x_0) = \left(-\frac{\partial f}{\partial x_1}(x_0')\,, -\frac{\partial f}{\partial x_2}(x_0')\,, \ldots, -\frac{\partial f}{\partial x_{n-1}}(x_0'),\ 1\right).$$

Example 7 By what we said in Example 5, the normal space at $P_0 = (2, 2, 1)$ to the manifold M considered in Example 1 is generated by $DF(P_0) = (0, 4, 0)$.

12.3 Measure and Integration on k-Submanifolds in $\mathbb{R}^n$

In this section we shall briefly present some aspects regarding the theory of measure and integration on k-dimensional manifolds. Let us begin with an easy situation.

Let $L : \mathbb{R}^k \to \mathbb{R}^n$, $1 \leq k \leq n$, be a linear map. The symbol L will also denote the $n \times k$ matrix

$$(\ell_{ij})_{\substack{i=1,2,\ldots,n \\ j=1,2,\ldots,k}}\,, \qquad \text{where} \qquad \ell_{ij} = \left(Le_j\,, e_i\right),$$

that represents the map with respect to the canonical bases of $\mathbb{R}^k$ and $\mathbb{R}^n$. If L is injective, $L(\mathbb{R}^k)$ is a k-dimensional subspace of $\mathbb{R}^n$. We intend to give a reasonable definition of k-dimensional measure for certain subsets of $L(\mathbb{R}^k)$.

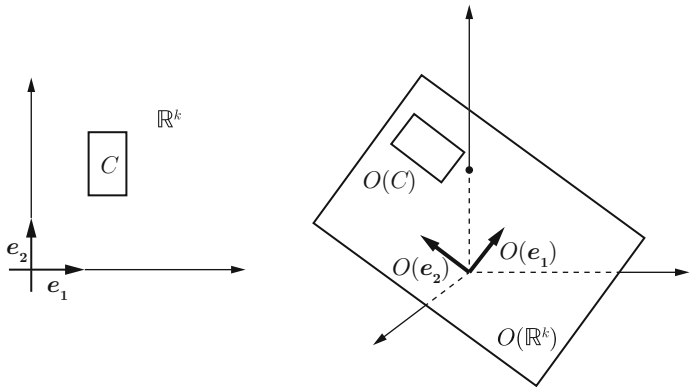


Fig. 12.6

We recall that in the particular case $k = n$, we saw in Sect. 9.8 that if $C \subset \mathbb{R}^n$ is Lebesgue measurable then so is $L(C)$, and

$$m_n(L(C)) = |\det L| \cdot m_n(C)\,, \tag{12.5}$$

where m_n denotes the n-dimensional measure on $\mathbb{R}^n$.

Now we examine the case $1 \leq k < n$ and suppose L is an *orthogonal transformation* $O : \mathbb{R}^k \to \mathbb{R}^n$, i.e. such that

$$(Ox, Oy) = (x, y)\,, \qquad \forall\, x, y \in \mathbb{R}^k. \tag{12.6}$$

From (12.6) we have, for $x, y \in \mathbb{R}^k$,

$$|Ox - Oy| = (Ox - Oy, Ox - Oy)^{1/2} = (x - y, x - y)^{1/2} = |x - y|\,,$$

i.e. O preserves distances, which we express by saying that O is an *isometry* of $\mathbb{R}^k$ (see Fig. 12.6).

If $C \subset \mathbb{R}^k$ is a measurable subset of $\mathbb{R}^k$, the symbol $H^k(O(C))$ will indicate the *k-dimensional measure* of the image $O(C) \subset O(\mathbb{R}^k) \subset \mathbb{R}^n$ of C, by definition equal to

$$H^k(O(C)) = m_k(C)\,. \tag{12.7}$$

All in all, we are assuming that O does not change the measure of the set C.

If $L : \mathbb{R}^k \to \mathbb{R}^n$, $1 \leq k < n$ is a linear map, the result in Linear Algebra known as the *polar decomposition theorem* asserts that there exist an orthogonal map $O : \mathbb{R}^k \to \mathbb{R}^n$ and a symmetric endomorphism $S : \mathbb{R}^k \to \mathbb{R}^k$ such that

$$L = O \circ S\,, \tag{12.8}$$

called *polar decomposition*. Keeping (12.5) and (12.7) in account, it is natural to define the k-dimensional measure of $L(C)$, for a measurable set $C \subset \mathbb{R}^k$, as the quantity

$$H^k(L(C)) = |\det S| \cdot m_k(C)\,. \tag{12.9}$$

As the polar decomposition of a matrix is not unique, we should check that the above definition does not depend on the choice of decomposition.

Proposition 1 *Given $L : \mathbb{R}^k \to \mathbb{R}^n$, $1 \le k < n$, if a symmetric matrix $S : \mathbb{R}^k \to \mathbb{R}^k$ satisfies* (12.8)*, then*

$$|\det S|^2 = \det(L^T \circ L)\,, \tag{12.10}$$

where $L^T : \mathbb{R}^n \to \mathbb{R}^k$ is the transpose of L. Moreover, for any linear map $N : \mathbb{R}^k \to \mathbb{R}^k$,

$$|\det N|^2 \cdot \det(L^T \circ L) = \det\Big((L \circ N)^T \circ (L \circ N)\Big) \tag{12.11}$$

Proof Let O^T be the transpose of the orthogonal map O appearing in (12.8). By definition of transpose and (12.6) it is immediate to see that

$$O^T(Ox) = x \tag{12.12}$$

for any $x \in \mathbb{R}^k$,. For any $i = 1, 2, \dots, n$, $j = 1, 2, \dots, k$, define $O_{ij} = (Oe_j\,, e_i)$. By definition of transpose, $(O^T e_i\,, e_j) = O_{ij}$, so

$$\Big(O^T(Oe_j), e_h\Big) = \left(O^T\Big(\sum_{i=1}^n O_{ij} e_i\Big), e_h\right) = \sum_{i=1}^n O_{ij}\Big(O^T e_i, e_h\Big) =$$

$$= \sum_{i=1}^n O_{ij}\, O_{ih} = \big(Oe_j\,, Oe_h\big) = \delta_{jh}\,, \qquad \forall\, j, h = 1, 2, \dots, k.$$

Hence $O^T(Oe_j) = e_j$ for any $j = 1, 2, \dots, k$, and (12.12) follows.

By (12.12),

$$L^T \circ L = S^T \circ O^T \circ O \circ S = S^T \circ S\,,$$

and since $\det S = \det S^T$, we obtain (12.10). Now, (12.11) is consequence of the determinant of a product being the product of the determinants and that

$$(L \circ N)^T \circ (L \circ N) = N^T \circ L^T \circ L \circ N\,.$$

In the sequel, if $L : \mathbb{R}^k \to \mathbb{R}^n$, $1 \le k \le n$, is a linear map we shall define

$$\widehat{J}_L = \sqrt{\det(L^T \circ L)}\,.$$

When $k = n$, $\widehat{J}_L$ is the absolute value $|\det L|$ of the Jacobian determinant of L.

The next proposition, whose proof we omit for the sake of brevity, is sometimes known as *Cauchy-Binet formula*, and is useful to calculate $\widehat{J}_L$.

Proposition 2 *The quantity $\widehat{J}_L^2$ is equal to the sum of the squares of all $k \times k$ minors of L.*

Now we pass to k-dimensional manifolds M, and start by supposing that we can represent M using only one parametrisation φ. That is to say, there is one map $\varphi : U \to \mathbb{R}^n$, of class C^1 on the open set U in $\mathbb{R}^k$ and injective, such that

(j′) $\varphi(U) = M$;
(jj) $D\varphi$ has rank k on U.

A subset $B \subseteq M$ is said to be *k-measurable* when $\varphi^{-1}(B)$ is measurable. The *k-dimensional measure* of B is

$$H^k(B) = \int_{\varphi^{-1}(B)} \widehat{J}_\varphi \, dy\,, \tag{12.13}$$

where, for any $y \in U$,

$$\widehat{J}_\varphi(y) = \sqrt{\det\left(D\varphi(y)^T \circ D\varphi(y)\right)}\,.$$

Definition (12.13) is justified by the observation that if $y_0 \in U$ and $R \subseteq U$ is a measurable set containing y_0 and of small diameter, then $H^k(\varphi(R))$ is roughly equal to $\widehat{J}_\varphi(y_0)\, m_k(R)$. The latter, due to (12.9) and (12.10), is the k-dimensional measure of the image of R in the tangent space at $\varphi(y_0)$ (compare with the considerations made in Sect. 10.4 regarding regular surfaces).

Note that (12.13) does not depend on the parametrisation φ. In fact if $\psi : V \subseteq \mathbb{R}^k \to \mathbb{R}^n$ is another parametrisation of M, one can prove that the transformation $\Phi : U \to V$, $\Phi = \psi^{-1} \circ \varphi$, is invertible, of class C^1 on U and with C^1 inverse on V. By (12.11), and the formula for changing variables in multiple integrals,

$$\begin{aligned}
\int_{\varphi^{-1}(B)} \widehat{J}_\varphi(y)\, dy &= \int_{\varphi^{-1}(B)} \sqrt{\det\left((D\psi \circ D\Phi)^T \circ (D\psi \circ D\Phi)\right)}\, dy = \\
&= \int_{\varphi^{-1}(B)} |J_\Phi(y)| \sqrt{\det\left(D\psi(\Phi(y))^T \circ D\psi(\Phi(y))\right)}\, dy = \\
&= \int_{\psi^{-1}(B)} \widehat{J}_\psi(z)\, dz\,.
\end{aligned}$$

Example 1 Let $\varphi : [a, b] \to \mathbb{R}^n$ be a C^1 curve such that $\varphi((a, b))$ is a one-dimensional manifold M. In this case

$$\widehat{J_\varphi}(t) = \left|\varphi'(t)\right|,$$
$$H^1(M) = \int_a^b \left|\varphi'(t)\right| dt\,.$$

Hence the manifold's one-dimensional measure coincides with the length of φ.

Example 2 Let $\varphi : D \to \mathbb{R}^3$ be a regular surface of equations

$$\varphi(u, v) = (x(u, v), y(u, v), z(u, v))\,, \qquad (u, v) \in D,$$

such that $M = \varphi(\overset{\circ}{D})$ is a two-dimensional manifold. By Proposition 2

$$\widehat{J}^2_\varphi(u, v) = \left|\det \frac{\partial(x, y)}{\partial(u, v)}\right|^2 + \left|\det \frac{\partial(x, z)}{\partial(u, v)}\right|^2 + \left|\det \frac{\partial(y, z)}{\partial(u, v)}\right|^2,$$

so (12.13) implies that $H^2(M)$ is the area of the regular surface φ.

Example 3 Let $f : U \subseteq \mathbb{R}^k \to \mathbb{R}^{n-k}$ be a C^1 function on the open set U and M the graph of f. A parametrisation of M is $\varphi(x') = (x', f(x'))$, with $x' = (x_1, x_2, \ldots, x_k) \in U$. Hence

$$D\varphi = \begin{pmatrix} 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ldots & 0 & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \ldots & 0 & 1 \\ \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \ldots & \frac{\partial f_1}{\partial x_{k-1}} & \frac{\partial f_1}{\partial x_k} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ \frac{\partial f_{n-k}}{\partial x_1} & \frac{\partial f_{n-k}}{\partial x_2} & \ldots & \frac{\partial f_{n-k}}{\partial x_{k-1}} & \frac{\partial f_{n-k}}{\partial x_k} \end{pmatrix}.$$

Let us compute $\widehat{J}^2_\varphi$ using Proposition 2. We must add the squares of all minors of order k in $D\varphi$. It can be proved that

$$\widehat{J}^2_\varphi(x') = 1 + \sum_{r=1}^{\min\{k, n-k\}} \left|X_r(Df(x'))\right|^2, \tag{12.14}$$

(continued)

Example 3 (continued)
where $X_r(Df(x'))$ is the vector made by the minors of order r in $Df(x')$, for any $r = 1, 2, \ldots, \min\{k, n-k\}$.

If, for example, $f : U \subset \mathbb{R}^{n-1} \to \mathbb{R}$, from (12.13) and (12.14) we obtain

$$H^{n-1}(M) = \int_U \sqrt{1 + |Df(x')|^2}\, dx' . \tag{12.15}$$

For $f : U \subset \mathbb{R}^2 \to \mathbb{R}^2$, $f(x, y) = (u(x, y), v(x, y))$, we have

$$H^2(M) = \iint_U \sqrt{1 + |Du|^2 + |Dv|^2 + \left|\det \frac{\partial(u, v)}{\partial(x, y)}\right|^2}\, dx\, dy .$$

Always assuming that the manifold M is described by one parametrisation φ, if $f : \overline{M} \to \mathbb{R}$ is a continuous function and $B \subseteq M$ is a k-measurable bounded set, we define the *surface integral* of f over B as

$$\int_B f\, dH^k = \int_{\varphi^{-1}(B)} f(\varphi(y))\, \widehat{J}_\varphi(y)\, dy . \tag{12.16}$$

It can be shown that also this definition is independent of the parametrisation.

For the manifold M in Example 1, (12.16) reduces to the definition of integral along a curve, already seen in Sect. 6.4. If M is the manifold of Example 2, we recover the surface integral seen in Sect. 10.6.

Example 4 Consider $M = \{(x, y, z, w) \in \mathbb{R}^4 : x^2 + y^2 = 1,\ z^2 + w^2 = 1,\ x, z > 0\}$. M is a two-dimensional manifold and a parametrisation φ is

$$\varphi(u, v) = (\cos u, \sin u, \cos v, \sin v), \qquad (u, v) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Let us compute

$$\int_M (x^2 + z^2)\, dH^2 . \tag{12.17}$$

As

$$D\varphi = \begin{pmatrix} -\sin u & 0 \\ \cos u & 0 \\ 0 & -\sin v \\ 0 & \cos v \end{pmatrix}, \qquad (D\varphi)^T \circ D\varphi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

we have $\widehat{J}_\varphi = 1$, and (12.17) becomes the integral

$$\int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} (\cos^2 u + \cos^2 v)\, du\, dv = \pi^2 .$$

Lemma (Partition of Unity) *Let M be a compact k-manifold. It is possible to find finitely many local parametrisations $\varphi_i : U_i \to M$ and continuous functions $\eta_i : M \to [0, 1]$, $i = 1, 2, \ldots, N$, such that*

$$\bigcup_{i=1}^{N} \varphi_i(U_i) = M\,,$$

$$\eta_i(x) = 0\,, \qquad \text{if } x \in M - \varphi_i(U_i),$$

$$\sum_{i=1}^{N} \eta_i(x) = 1\,, \qquad \forall\, x \in M.$$

The functions η_i, for $i = 1, 2, \ldots, N$, form a *partition of unity* subordinated to the open sets $\varphi_i(U_i)$ of M.

Proof Proposition 1, Sect. 12.1, implies that for any $x \in M$ there exist an open ball $B_{r_x}(x)$ of radius $r_x > 0$, and a local parametrisation $\varphi_x : U_x \to M$ such that $\varphi_x(U_x) = M \cap B_{r_x}(x)$. As M is compact, we can extract from $\{B_{r_x/2}(x)\}_{x\in M}$ a finite covering.

Hence, there is a finite number of open balls $B_{r_i}(x_i)$, $x_i \in M$, $i = 1, 2, \ldots, N$, and corresponding parametrisations $\varphi_i : U_i \to M$ such that

$$\varphi_i(U_i) = M \cap B_{r_i}(x_i)\,, \qquad M \subset \bigcup_{i=1}^{N} B_{r_i/2}(x_i)\,,$$

where, for any $i = 1, 2, \ldots, N$, $B_{r_i/2}(x_i)$ is the open ball with centre x_i and radius $r_i/2$. Applying the separation theorem of Sect. 2.3 to the metric space $M \subseteq \mathbb{R}^n$ with the induced metric, for any $i = 1, 2, \ldots, N$ we can find a continuous function $\sigma_i : M \to [0, 1]$ such that $\sigma_i(x) = 1$ if $x \in M \cap \overline{B_{r_i/2}(x_i)}$ and $\sigma_i(x) = 0$ if $x \in M - B_{(2r_i)/3}(x_i)$. Define, for any $x \in M$,

$$\sigma(x) = \sum_{i=1}^{N} \sigma_i(x)\,,$$

so $\sigma(x) \geq 1$ on M, and the functions

$$\eta_i(x) = \frac{\sigma_i(x)}{\sigma(x)}\,, \qquad i = 1, 2, \ldots, N,$$

give the required partition of unity.

This lemma allows to extend the concepts of k-measurable set and surface integral to *compact manifolds* M. If we choose the φ_i and the functions η_i as in

the lemma, we will say that $B \subseteq M$ is measurable if $\varphi_i^{-1}(B)$ is measurable in $\mathbb{R}^k$ for any $i = 1, 2, \ldots, N$. We will call k-dimensional measure of B the quantity

$$H^k(B) = \sum_{i=1}^{N} \int_{B \cap \varphi_i(U_i)} \eta_i(x)\, dH^k\,,$$

where the integrals on the right are defined by (12.16).

Finally, if $f : M \to \mathbb{R}$ is continuous, we define

$$\int_B f\, dH^k = \sum_{i=1}^{N} \int_{B \cap \varphi_i(U_i)} f(x)\, \eta_i(x)\, dH^k\,.$$

One can prove that these definitions do not depend on the parametrisations φ_i, nor on the specific partition of unity η_i, $i = 1, 2, \ldots, N$. At any rate it is evident, in case B in contained in one of the open sets $\varphi_i(U_i)$, that we recover the quantities defined in (12.13) and (12.16).

Let $F : A \to \mathbb{R}$ be a C^1 function on the open set A in $\mathbb{R}^n$. For $t \in \mathbb{R}$ the set

$$M_t = \{x \in A :\ F(x) = t\}$$

is typically not a manifold, unless $DF(x) \neq 0$ for any $x \in M_t$. But it can be proved (this fact is known as *Sard's theorem*) that for almost every t (in the sense of the Lebesgue measure)

$$M_t \cap \{x \in A :\ DF(x) = 0\} = \emptyset\,.$$

Therefore M_t is an $(n-1)$-dimensional manifold for almost every t.

The following result holds.

Co-area Formula *Let $F : A \to \mathbb{R}$ be a function of class $C^1(A)$ with $DF \neq 0$ and $f : A \to \mathbb{R}$ a continuous and summable function on A. Then*

$$\int_A f\, dx = \int_{-\infty}^{+\infty} dt \int_{\{x \in A:\ F(x) = t\}} \frac{f}{|DF|}\, dH^{n-1}\,. \tag{12.18}$$

If $F : A \to \mathbb{R}$ is the projection onto one of the coordinate axes, (12.18) is an application of Fubini's theorem. Suppose for example $F(x) = x_n$ for any $x \in A$. Then $M_t = \{x \in A :\ x_n = t\}$ is the cross-section A_t of A over $x_n = t$, for any $t \in \mathbb{R}$. Since $|DF| = 1$, formula (12.18) reads

$$\int_A f\, dx = \int_{-\infty}^{+\infty} dt \int_{A_t} f(x_1, x_2, \ldots, t)\, dx_1\, dx_2\, \ldots\, dx_{n-1}\,.$$

Example 5 Fix $r > 0$ and $B_r = \{x \in \mathbb{R}^n : \ |x| < r\}$. Let us apply (12.18) with $F(x) = |x|$. In this case, for $\varepsilon < t < r$,

$$M_t = \partial B_t = \{x \in \mathbb{R}^n : \ |x| = t\}$$

and if $f : B_r \to \mathbb{R}$ is continuous and summable then

$$\int_{B_r - B_\varepsilon} f \, dx = \int_\varepsilon^r dt \int_{\partial B_t} f \, dH^{n-1} .$$

By letting ε go to zero we obtain

$$\int_{B_r} f \, dx = \int_0^r dt \int_{\partial B_t} f \, dH^{n-1} . \tag{12.19}$$

Let us use this formula with $f \equiv 1$ to find the measure of the sphere $S^{n-1} = \partial B_1$. Denoting by ω_n the unit ball's Lebesgue measure, from (12.19) we obtain

$$\omega_n = \int_{B_1} dx = \int_0^1 dt \int_{\partial B_t} dH^{n-1} = \int_0^1 H^{n-1}(\partial B_t) \, dt .$$

On the other hand it is geometrically evident (consider the homothety mapping the ball of radius t to the unit ball) that

$$H^{n-1}(\partial B_t) = t^{n-1} H^{n-1}(\partial B_1) .$$

The last two equalities imply

$$H^{n-1}(S^{n-1}) = n \, \omega_n .$$

12.4 The Divergence Theorem

In this section we wish to generalise the divergence theorem, which we have already discussed in the plane and in space (Sects. 8.3 and 10.7 respectively), to the case of regular domains in $\mathbb{R}^n$, with n arbitrary.

First, let us provide the notion of a regular domain in $\mathbb{R}^n$. A closed and bounded subset $D \subset \mathbb{R}^n$ is a *regular domain* if for any $x_0 \in \partial D$ there exist an open neighbourhood A of x_0 and a C^1 function $F : A \to \mathbb{R}$ such that $DF(x) \neq 0$ on A and

$$\partial D \cap A = \{x \in A : \ F(x) = 0\} , \tag{12.20}$$

$$\overset{\circ}{D} \cap A = \{x \in A : \ F(x) < 0\} . \tag{12.21}$$

The boundary ∂D of a regular domain is then an $(n-1)$-dimensional manifold, and by (12.21), D locally lies on one side of its boundary.

The definitions of *regular domain* in the plane and in space, given in Chap. 8, do not coincide with the one given above. For instance a square is a regular domain according to the definition of Sect. 8.3, but not according to the one considered in this section, because its boundary is not a C^1 manifold.

From what we saw in Sect. 12.2, the normal space to ∂D at a point $x_0 \in \partial D$ is generated by the gradient vector $DF(x_0)$. We call *outer unit normal of* D *at* x_0 the unit vector

$$\nu(x_0) = \frac{DF(x_0)}{|DF(x_0)|}\,.$$

The word "outer" is explained as follows. If $t > 0$ is sufficiently small, the point $x_0 + t\,\nu(x_0)$ does not belongs to D. In fact if $t > 0$ is small enough then $x_0 + t\,\nu(x_0) \in A$, and since

$$\frac{d}{dt} F(x_0 + t\,\nu(x_0))\Big|_{t=0} = (DF(x_0), \nu(x_0)) = |DF(x_0)| > 0\,,$$

we have $F(x_0 + t\nu(x_0)) > 0$ because $F(x_0) = 0$. Hence by (12.20), (12.21) we have $x_0 + t\nu(x_0) \notin D$ (see Fig. 12.7).

Recall that if $w : D \to \mathbb{R}^n$ is a C^1 map of components $w(x) = (w_1(x), w_2(x), \ldots, w_n(x))$, the *divergence* of w is the function $\operatorname{div} w : D \to \mathbb{R}$,

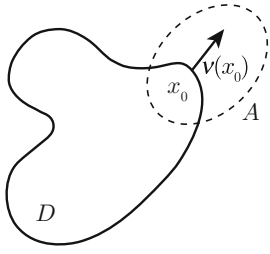
$$\operatorname{div} w(x) = \sum_{i=1}^{n} \frac{\partial w_i}{\partial x_i}\,, \qquad x \in D.$$

Divergence Theorem *Let $D \subset \mathbb{R}^n$ be a regular domain. If $w : D \to \mathbb{R}^n$ is a C^1 map, the following formula holds*

$$\int_D \operatorname{div} w(x)\,dx = \int_{\partial D} (w(x), \nu(x))\,dH^{n-1}\,. \tag{12.22}$$

Before we prove the theorem let us pause for a moment and consider some of the consequences of (12.22).

Fig. 12.7



Let $f, g : D \to \mathbb{R}$ be C^1 functions. Fix an index $i \in \{1, 2, \dots, n\}$ and let $w : D \to \mathbb{R}^n$ be the map of components $(w_1, w_2, \dots, w_n)$, where $w_i(x) = f(x) \cdot g(x)$ and $w_j = 0$ for $j \neq i$. From (12.22) we obtain

$$\int_D \frac{\partial f}{\partial x_i} g \, dx = -\int_D f \frac{\partial g}{\partial x_i} \, dx + \int_{\partial D} f g \nu_i \, dH^{n-1}, \tag{12.23}$$

where ν_i is the ith component of the outer normal. Formula (12.23) is the *integration by parts* for functions of n variables.

Moreover, if $f : D \to \mathbb{R}$ is of class C^2, applying (12.22) to the vector-valued function $w(x) = Df(x)$ gives

$$\int_D \Delta f \, dx = \int_{\partial D} \frac{\partial f}{\partial \nu} \, dH^{n-1},$$

where

$$\Delta f = \operatorname{div} Df = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2},$$

is the *Laplacian* of f (cf. Sect. 3.17) and

$$\frac{\partial f}{\partial \nu}(x) = (Df(x), \nu(x))$$

is the directional derivative of f along the outer normal.

Remark 1 When $w : D \to \mathbb{R}^n$ is null outside a compact set $K \subset \overset{\circ}{D}$, formula (12.22) is easy to prove and holds without any regularity assumption on the domain D. In fact the right-hand side of (12.22) vanishes, while calling $\widetilde{w}$ the extension of w to $\mathbb{R}^n$ that is zero outside D, for the left-hand side we have

$$\int_D \operatorname{div} w(x) \, dx = \int_D \sum_{i=1}^{n} \frac{\partial w_i}{\partial x_i} \, dx = \int_{\mathbb{R}^n} \sum_{i=1}^{n} \frac{\partial \widetilde{w}_i}{\partial x_i} \, dx =$$

$$= \sum_{i=1}^{n} \int_{\mathbb{R}^{n-1}} dx_1 \dots dx_{i-1} \, dx_{i+1} \dots dx_n \int_{-\infty}^{+\infty} \frac{\partial \widetilde{w}_i}{\partial x_i} \, dx_i = 0.$$

The proof of the divergence theorem will go through a series of preliminary lemmas.

Lemma 1 *Let $f : U \to (0, +\infty)$ be of class C^1 on a bounded open set $U \subset \mathbb{R}^{n-1}$ and call C the open cylinder $C = \{x = (x', x_n) : x' \in U, \ 0 < x_n < f(x')\}$. Suppose $u : \overline{C} \to \mathbb{R}$ is a C^1 function on $\overline{C}$, with $u(x', 0) = 0$ for any $x' \in U$. Then*

$$\int_C \frac{\partial u}{\partial x_n} \, dx = \int_M u \nu_n \, dH^{n-1},$$

where ν_n is the nth component of the normal ν to the graph M of f pointing outside of C.

Proof By Fubini's theorem (Sect. 9.7)

$$\int_C \frac{\partial u}{\partial x_n}\,dx = \int_U dx' \int_0^{f(x')} \frac{\partial u}{\partial x_n}\,dx_n = \int_U u(x',\, f(x'))\,dx'\,. \tag{12.24}$$

From Example 6, Sect. 12.2, the normal to M at $(x',\, f(x'))$, pointing outside of C, is

$$\nu(x',\, f(x')) = \frac{1}{\sqrt{1+|Df(x')|^2}}\left(-\frac{\partial f}{\partial x_1}(x')\,, -\frac{\partial f}{\partial x_2}(x')\,, \ldots, -\frac{\partial f}{\partial x_n}(x'),\, 1\right).$$

Recalling that $M = \varphi(U)$, where φ is the parametrisation $\varphi(x') = (x',\, f(x'))$, and that

$$\widehat{J_\varphi}(x',\, f(x')) = \sqrt{1+|Df(x')|^2}$$

on M (see Example 3, Sect. 12.3), from (12.16) we obtain

$$\int_M u\nu_n\,dH^{n-1} = \int_U u(x',\, f(x'))\,dx'\,. \tag{12.25}$$

Then (12.24) and (12.25) imply the claim.

Lemma 2 *Given $x_0 \in \partial D$, there is an open neighbourhood A' of x_0 such that (12.22) holds for any C^1 map $w : \mathbb{R}^n \to \mathbb{R}^n$ that vanishes on the complement of a compact subset of A'.*

Proof Fix $x_0 = (x_{01}, x_{02}, \ldots, x_{0n}) \in \partial D$. We initially suppose

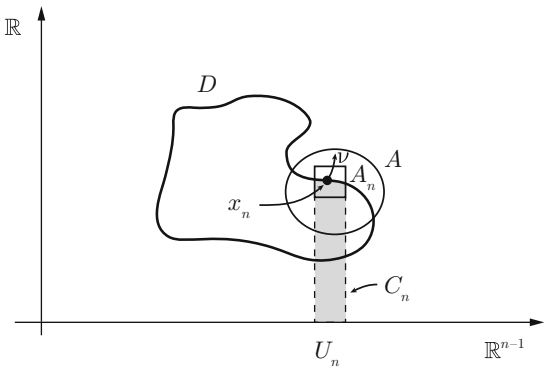
$$x_{0i} > 0\,, \qquad \nu_i(x_0) > 0\,, \qquad \forall\, i = 1, 2, \ldots, n. \tag{12.26}$$

In particular $\nu_n(x_0) > 0$, i.e. $D_n F(x_0) > 0$ where $F : A \to \mathbb{R}$ is the function for which (12.20), (12.21) hold. Referring to Fig. 12.8, the implicit function theorem in several variables (Sect. 11.1) implies that there exist an open set $A_n \subseteq A$ with $x_0 \in A_n$, an open set $U_n \subseteq \mathbb{R}^{n-1}$ and a function $f : U_n \to \mathbb{R}$ such that

$$\partial D \cap A_n = \{x \in A_n :\ x' \in U_n\,,\ \ x_n = f(x')\}\,.$$

Moreover A_n can be fixed so that $x_n > 0$ and $D_n F(x) > 0$ for any $x \in A_n$.

Fig. 12.8



Let C_n be the cylinder

$$C_n = \left\{x = (x', x_n) : \ x' \in U_n \,, \ \ 0 < x_n < f(x')\right\}.$$

If $x \in A_n \cap C_n$, since $F(x', f(x')) = 0$ and $\partial F_n/\partial x_n(x', t) > 0$ for any $x_n < t < f(x')$, we have $F(x) < 0$ and so $x \in A_n \cap \overset{\circ}{D}$. Similarly, if $x \in A_n \cap \overset{\circ}{D}$, due to $F(x) < 0$ and $F(x', f(x')) = 0$, as $D_n F > 0$ on A_n it follows that $x_n < f(x')$, i.e. $x \in C_n$. In conclusion (see Fig. 12.8)

$$A_n \cap C_n = A_n \cap \overset{\circ}{D}\,. \tag{12.27}$$

If w is zero outside a compact set contained in A_n, from (12.27) and Lemma 1 we deduce

$$\begin{aligned}\int_D \frac{\partial w_n}{\partial x_n}\,dx &= \int_{A_n\cap D} \frac{\partial w_n}{\partial x_n}\,dx = \int_{A_n\cap C_n} \frac{\partial w_n}{\partial x_n}\,dx = \\ &= \int_{C_n} \frac{\partial w_n}{\partial x_n}\,dx = \int_{\partial C_n\cap\partial D} w_n\,\nu_n H^{n-1} = \int_{\partial D} w_n\nu_n\ H^{n-1}\,.\end{aligned} \tag{12.28}$$

Similarly one proves that, for any $i = 1, 2, \ldots, n-1$, there is an open set $A_i \subseteq A$, with $x_0 \in A_i$, such that

$$\int_D \frac{\partial w_i}{\partial x_i}\,dx = \int_{\partial D} w_i \nu_i\, H^{n-1}$$

if w is zero outside a compact set contained in A_i. By this and (12.28) the claim follows, if we assume (12.26) and take $A' = A_1 \cap A_2 \cap \ldots \cap A_n$.

The general case is proved by considering a suitable affine map $\Phi(x) = Ox + a$, $x \in \mathbb{R}^n$, where $a \in \mathbb{R}^n$ and $O : \mathbb{R}^n \to \mathbb{R}^n$ is orthogonal. Setting $T = \Phi(D)$ and $y_0 = \Phi(x_0)$, it is easy to see that T is a regular domain and the outer normal $\nu(y_0)$ of T at y_0 is $O(\nu(x_0))$, where $\nu(x_0)$ is the outer normal of D at x_0. Since we can

always choose $a \in \mathbb{R}^n$ and $O : \mathbb{R}^n \to \mathbb{R}^n$ so that

$$y_{0i} > 0\,, \quad \nu_i(y_0) > 0\,, \qquad \forall\, i = 1, 2, \ldots, n,$$

the general case reduces to the case in which (12.26) holds.

Lemma 3 *There is a finite number of open sets A_i, $i = 1, 2, \ldots, N$, such that*

$$D \subset \bigcup_{i=1}^{N} A_i$$

and (12.22) holds whenever $w : \mathbb{R}^n \to \mathbb{R}^n$ is a C^1 function that vanishes outside a compact set contained in one of the A_i. Moreover, for any $i = 1, 2, \ldots, N$, there is a C^1 function $\eta_i : \mathbb{R}^n \to [0, 1]$, null outside a compact set contained in A_i, such that

$$\sum_{i=1}^{N} \eta_i(x) = 1\,, \qquad \forall\, x \in D. \tag{12.29}$$

In this case too (see the lemma in the previous section) the functions η_i, $i = 1, 2, \ldots, N$, form a *partition of unity* subordinated to the open covering $\{A_i\}$ of D.

Proof Note that if $\zeta : \mathbb{R} \to [0, 1]$ is a C^1 function such that $\zeta(t) = 1$ for $t \le 1/2$ and $\zeta(t) = 0$ for $t \ge 2/3$, then

$$x \to \zeta\left(\frac{|x - x_0|}{r}\right)$$

is of class C^1 on $\mathbb{R}^n$, it equals 1 on the ball $B_{r/2}(x_0)$ of centre x_0 and radius $r/2$, and it vanishes outside the ball $B_{(2r)/3}(x_0)$.

Fix $x \in D$. If $x \in \partial D$ we consider a ball of centre x and radius $r_x > 0$ such that (12.22) holds if w is zero outside a compact set contained in $B_{r_x}(x)$. This ball exists due to Lemma 2. If, instead, $x \in \overset{\circ}{D}$, fix a ball $B_{r_x}(x) \subseteq \overset{\circ}{D}$. By Remark , we still have (12.22) if w is zero outside a compact set contained in $B_{r_x}(x)$.

As D is compact, there is a finite number of points $x_i \in D_i$, $i = 1, 2, \ldots, N$, such that,

$$D \subset \bigcup_{i=1}^{N} B_{r_i/2}(x_i)$$

where $r_i = r_{x_i}$. The open sets $A_i = B_{r_i}(x_i)$ satisfy the requirements. At last, for any $i = 1, 2, \ldots, N$ and any $x \in \mathbb{R}^n$, set

$$\sigma_i(x) = \zeta\left(\frac{|x - x_0|}{r_i}\right), \qquad \sigma(x) = \sum_{i=1}^{N} \sigma_i(x)\,.$$

Immediately, the functions

$$\eta_i(x) = \frac{\sigma_i(x)}{\sigma(x)}$$

satisfy the desired properties and also (12.29).

Proof (of the Divergence Theorem) The previous three lemmas easily imply the divergence theorem, at least when $w(x)$ is defined everywhere in $\mathbb{R}^n$. If so, in fact, for $i = 1, 2, \ldots, N$ let A_i and $\eta_i : \mathbb{R}^n \to [0, 1]$ be as in Lemma 3. By (12.29),

$$\int_D \operatorname{div} w \, dx = \sum_{i=1}^{N} \int_D \operatorname{div}(w\eta_i) \, dx \,. \tag{12.30}$$

As the functions $w\,\eta_i$ are zero outside a compact set contained in A_i, for any $i = 1, 2, \ldots, N$, we have

$$\int_D \operatorname{div}(w\,\eta_i) \, dx = \int_{\partial D} \eta_i(w, \nu) \, dH^{n-1} \,. \tag{12.31}$$

Since $\sum_{i=1}^{N} \eta_i(x) = 1$ for any $x \in \partial D$, from (12.30), (12.31) the claim follows.

The general case is a consequence of the following proposition, whose proof we omit for the sake of brevity.

Proposition *Suppose $D \subset \mathbb{R}^n$ is a regular domain and $w : D \to \mathbb{R}^n$ a C^1 map. There exists an extension $\widetilde{w}$ of w to $\mathbb{R}^n$, i.e. a map $\widetilde{w} : \mathbb{R}^n \to \mathbb{R}^n$ of class C^1 such that*

$$\widetilde{w}(x) = w(x) \,, \qquad \forall\, x \in D.$$

12.5 Alternating Forms

In this section we shall present the fundamental results regarding the theory of alternating multilinear forms. Fix positive integers n, k and consider the Cartesian product $(\mathbb{R}^n)^k$. A map

$$\omega : (h_1, h_2, \ldots, h_k) \in (\mathbb{R}^n)^k \;\to\; \omega(h_1, h_2, \ldots, h_k) \in \mathbb{R}$$

is called an *alternating form of degree k* (or a *k-covector*) if ω is linear in each variable $h_1, h_2, \ldots, h_k$, and *alternating*:

$$\omega(h_{\sigma(1)}, h_{\sigma(2)}, \ldots, h_{\sigma(k)}) = \text{sign}(\sigma)\ \omega(h_1, h_2, \ldots, h_k)\,,$$

for any permutation $\sigma : \{1, 2, \ldots, k\} \to \{1, 2, \ldots, k\}$, where $\text{sign}(\sigma)$ is 1 or -1 depending on whether σ is the product of an even or odd number of transpositions.

Example 1 For $n = k$ the map

$$\omega : (h_1, h_2, \ldots, h_n) \in \mathbb{R}^n \times \mathbb{R}^n \times \ldots \times \mathbb{R}^n \ \to \det(e_i\,, h_j)_{ij}\,,$$

where $\{e_1, e_2, \ldots, e_n\}$ is the canonical basis of $\mathbb{R}^n$, is an alternating form of degree n. This is immediate since ω maps the *n-tuple* $(h_1, h_2, \ldots, h_n)$ to the determinant of the matrix whose columns are the vectors $h_1, h_2, \ldots, h_n$.

In general, given arbitrary n and k, and letting $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ be a k-dimensional vector of integers between 1 and n, the map

$$e^\lambda : (h_1, h_2, \ldots, h_k) \in (\mathbb{R}^n)^k \ \to \det(e_{\lambda_i}\,, h_j)_{ij}\,, \tag{12.32}$$

associating with the k-tuple $(h_1, h_2, \ldots, h_k)$ the $k \times k$ determinant whose columns are the λ_i-th components of the vectors h_j, is an alternating form of degree k.

Proposition 1 *Let n, k be positive integers and λ, λ' two k-tuples of integers lying between* 1 *and n. Then:*

(a) *if λ' is obtained from λ via r transpositions, then*

$$e^\lambda(h_1, h_2, \ldots, h_k) = (-1)^r\, e^{\lambda'}(h_1, h_2, \ldots, h_k)$$

for any $(h_1, h_2, \ldots, h_k) \in (\mathbb{R}^n)^k$;
(b) *if λ has two equal components, e^λ is identically zero;*
(c) *if $k > n$, the map e^λ is identically zero.*

Proof To show (a) it suffices to apply, r times, the property whereby a determinant changes sign if we interchange two rows. For (b), if a square matrix has two equal rows its determinant vanishes. As for (c), just note that for $k > n$ the vector λ necessarily has two equal components.

In the sequel we will denote with λ, μ, ν a generic k-tuple of *integers* between 1 and n, and we will call its components *indices* of the k-tuple.

By the above proposition it is evident that when considering the map e^λ it will be convenient to assume that the indices of λ are distinct (otherwise e^λ is the zero map).

Actually, up to a sign change in e^λ we may suppose that the indices are ordered increasingly.

Define then for any $1 \le k \le n$ (λ_i integer for all i)

$$\Lambda_{n,k} = \{\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k) : \ 1 \le \lambda_1 < \lambda_2 < \dots < \lambda_k \le n\}.$$

When $n, k \ge 1$ we shall write $\Lambda_k(\mathbb{R}^n)$ for the set of alternating forms on $(\mathbb{R}^n)^k$.

$\Lambda_k(\mathbb{R}^n)$ can be endowed with the structure of a real vector space by defining the sum of k-covectors ω, η and the multiplication by a scalar $t \in \mathbb{R}$ as follows

$$(\omega + \eta)(h_1, h_2, \dots, h_k) = \omega(h_1, h_2, \dots, h_k) + \eta(h_1, h_2, \dots, h_k)\,,$$
$$(t\omega)(h_1, h_2, \dots, h_k) = t\omega(h_1, h_2, \dots, h_k)\,.$$

Note that for $k = 1$, $\Lambda_1(\mathbb{R}^n)$ coincides with the dual $(\mathbb{R}^n)^*$ of $\mathbb{R}^n$, the space of linear maps on $\mathbb{R}^n$ (see Sect. 2.5) (sometimes referred to as *covectors*).

Theorem 1 *For $1 \le k \le n$ the set*

$$\left\{e^\lambda : \ \lambda \in \Lambda_{n,k}\right\}$$

is a basis of $\Lambda_k(\mathbb{R}^n)$, called canonical basis. Hence given $\omega \in \Lambda_k(\mathbb{R}^n)$, for any $\lambda \in \Lambda_{n,k}$ there is a real number ω_λ such that

$$\omega = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda \, e^\lambda\,, \tag{12.33}$$

and the representation is unique.

We skip the proof of Theorem 1, since it is based on elementary Linear Algebra. We shall point out some consequences, instead.

When $1 \le k \le n$ the cardinality of the set $\Lambda_{n,k}$ is $\binom{n}{k}$, so this is the dimension of the space $\Lambda_k(\mathbb{R}^n)$.

Let us also point out that (12.33) says that every alternating form of degree k can be written uniquely as a linear combination of suitable $k \times k$ determinants, to be precise those defined by (12.32) with $\lambda \in \Lambda_{n,k}$.

Finally, for $n = k$, $\Lambda_n(\mathbb{R}^n)$ is a one-dimensional vector space, whose elements are multiples of the alternating form of degree n

$$e^{(1,2,\dots,n)}(h_1, h_2, \dots, h_n) = \det(e_i \, , h_j)_{ij}$$

(in the sequel the symbol $e^{12\dots n}$ will denote $e^{(1,2,\dots,n)}$). Any alternating form of degree n is therefore a multiple of the determinant.

By Proposition 1, (c) we have the following easy consequence:

Proposition 2 *For $1 \leq n < k$, $\Lambda_k(\mathbb{R}^n)$ consists of the zero functional only.*

A simple way to construct alternating forms is using the wedge product. Let e^λ, e^μ be elements of the canonical basis of $\Lambda_k(\mathbb{R}^n)$, $\Lambda_l(\mathbb{R}^n)$ respectively. The *wedge product* of e^λ and e^μ is the alternating form of degree $k+l$

$$e^\lambda \wedge e^\mu = e^\nu \in \Lambda_{k+l}(\mathbb{R}^n)\,, \tag{12.34}$$

where ν is the $(k+l)$-tuple obtained by appending the indices of μ to the vector λ, as follows: $\nu = (\lambda_1, \lambda_2, \ldots, \lambda_k, \mu_1, \mu_2, \ldots, \mu_l)$.

By property (a) in Proposition 1 it is clear that

$$e^\lambda \wedge e^\mu = (-1)^{k\,l}\, e^\mu \wedge e^\lambda\,. \tag{12.35}$$

In general, for

$$\omega = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda e^\lambda \in \Lambda_k(\mathbb{R}^n)\,, \qquad \eta = \sum_{\mu \in \Lambda_{n,l}} \eta_\mu e^\mu \in \Lambda_l(\mathbb{R}^n)$$

one defines

$$\omega \wedge \eta = \sum_{\substack{\lambda \in \Lambda_{n,k} \\ \mu \in \Lambda_{n,l}}} \omega_\lambda\, \eta_\mu\, e^\lambda \wedge e^\mu \in \Lambda_{k+l}(\mathbb{R}^n)\,. \tag{12.36}$$

Proposition 3 *For any $\omega, \overline{\omega} \in \Lambda_k(\mathbb{R}^n)$, $\eta \in \Lambda_l(\mathbb{R}^n)$, $\zeta \in \Lambda_m(\mathbb{R}^n)$ and any real number t,*

$$\begin{aligned} (\omega + \overline{\omega}) \wedge \eta &= \omega \wedge \eta + \overline{\omega} \wedge \eta\,, \\ (t\,\omega) \wedge \eta &= t\,(\omega \wedge \eta)\,, \\ \omega \wedge \eta &= (-1)^{kl} \eta \wedge \omega\,, \\ (\omega \wedge \eta) \wedge \zeta &= \omega \wedge (\eta \wedge \zeta)\,. \end{aligned} \tag{12.37}$$

The above properties are easy to check when $\omega, \overline{\omega}, \eta, \zeta$ are canonical basis elements, using Proposition 1 and (12.35). The general case follows from definition (12.36).

In particular, if e^λ, with $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$, is defined by (12.32), then

$$e^\lambda = e^{\lambda_1} \wedge e^{\lambda_2} \wedge \ldots \wedge e^{\lambda_k}\,, \tag{12.38}$$

where $e^{\lambda_i} \in (\mathbb{R}^n)^*$, $i = 1, 2, \ldots, k$, is the functional mapping any $h \in \mathbb{R}^n$ to its λ_i-th component.

It is customary to denote the map e^{λ_i} by dx_{λ_i}, and similarly the element e^λ defined by (12.38) will be written dx_λ. In these notations we can recast (12.38)

as

$$dx_\lambda = dx_{\lambda_1} \wedge dx_{\lambda_2} \wedge \ldots \wedge dx_{\lambda_k} .$$

Example 2 When $n = k = 2$, as observed earlier the space $\Lambda_2(\mathbb{R}^2)$ consists of multiples of the bilinear form

$$dx \wedge dy : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R} ,$$

which maps pairs $(h, k) \in \mathbb{R}^2 \times \mathbb{R}^2$, with $h = \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$, $k = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$, to

$$(dx \wedge dy)\,(h, k) = \det \begin{pmatrix} h_1 & k_1 \\ h_2 & k_2 \end{pmatrix} .$$

Example 3 When $n = 3, k = 2$, a basis of $\Lambda_2(\mathbb{R}^3)$ is given by the bilinear forms $dx \wedge dy$, $dy \wedge dz, dz \wedge dx$, which map pairs $h, k \in \mathbb{R}^3$ to

$$(dx \wedge dy)\,(h, k) = \det \begin{pmatrix} h_1 & k_1 \\ h_2 & k_2 \end{pmatrix} ,$$

$$(dy \wedge dz)\,(h, k) = \det \begin{pmatrix} h_2 & k_2 \\ h_3 & k_3 \end{pmatrix} ,$$

$$(dz \wedge dx)\,(h, k) = \det \begin{pmatrix} h_3 & k_3 \\ h_1 & k_1 \end{pmatrix}$$

respectively.

Example 4 For $n = k$, the space $\Lambda_n(\mathbb{R}^n)$ is spanned by the alternating form of degree n mapping the vector $(h_1, h_2, \ldots, h_n) \in \mathbb{R}^n$ to

$$(dx_1 \wedge dx_2 \wedge \ldots \wedge dx_k)\ (h_1, h_2, \ldots, h_n) = \det(e_i \, , h_j)_{ij} .$$

Example 5 Let us compute the wedge product of

$$\omega = e^1 + 3e^2 - e^4 \in \Lambda_1(\mathbb{R}^4)\,, \qquad \eta = 2e^{14} - e^{23} \in \Lambda_2(\mathbb{R}^4).$$

Using the wedge product's properties stated in Proposition 3,

$$\begin{aligned}\omega \wedge \eta &= (e^1 + 3e^2 - e^4) \wedge (2e^{14} - e^{23}) = \\ &= 2e^1 \wedge e^{14} - e^1 \wedge e^{23} + 6e^2 \wedge e^{14} - 3e^2 \wedge e^{23} - 2e^4 \wedge e^{14} + e^4 \wedge e^{23}.\end{aligned}$$

By Proposition 1 and (12.34) we have

$$\begin{aligned} &e^1 \wedge e^{14} = e^{114} = 0, \qquad e^1 \wedge e^{23} = e^{123}, \qquad e^2 \wedge e^{14} = e^{214} = -e^{124}, \\ &e^2 \wedge e^{23} = e^{223} = 0, \qquad e^4 \wedge e^{14} = e^{414} = 0, \qquad e^4 \wedge e^{23} = e^{423} = e^{234}, \end{aligned}$$

so we conclude that

$$\omega \wedge \eta = -e^{123} - 6\,e^{124} + e^{234}\,.$$

Example 6 Consider the bilinear forms $\omega = e^{12} + e^{35}$, $\eta = e^{25} + e^{34}$, $\zeta = e^{46} + e^{56}$ in $\Lambda_2(\mathbb{R}^6)$. Their wedge product is

$$\omega \wedge \eta \wedge \zeta = (e^{12} + e^{35}) \wedge (e^{25} + e^{34}) \wedge (e^{46} + e^{56}) = e^{1234} \wedge (e^{46} + e^{56}) = e^{123456}\,.$$

Example 7 Let us calculate the wedge product of

$$\omega = dx + 2\,dz \in \Lambda_1(\mathbb{R}^3)\,, \qquad \eta = dx \wedge dy + 2\,dy \wedge dz \in \Lambda_2(\mathbb{R}^3)\,.$$

We have

$$\begin{aligned}\omega \wedge \eta &= (dx + 2\,dz) \wedge (dx \wedge dy + 2\,dy \wedge dz) = \\ &= 2\,dx \wedge dy \wedge dz + 2\,dz \wedge dx \wedge dy = 4\,dx \wedge dy \wedge dz\,.\end{aligned}$$

We remark that the vector space $\Lambda_k(\mathbb{R}^n)$ has an *inner product*, defined by

$$(\omega, \eta) = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda\, \eta_\lambda\,,$$

for $\omega, \eta \in \Lambda_k(\mathbb{R}^n)$. As a result, it also inherits a norm that maps $\omega \in \Lambda_k(\mathbb{R}^n)$ to the real number

$$|\omega| = (\omega, \omega)^{1/2} = \sqrt{\sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda^2}\,. \tag{12.39}$$

The same procedure employed to define the space $\Lambda_k(\mathbb{R}^n)$ of alternating k-forms on $\mathbb{R}^n$ can be used to define *k-vectors* on $\mathbb{R}^n$, whose space we denote by $\Lambda_k^*(\mathbb{R}^n)$. We shall say that the map

$$\alpha : (a_1, a_2, \ldots, a_k) \in \left[(\mathbb{R}^n)^*\right]^k \to \mathbb{R}$$

is a *k-vector* if α is linear in each variable $a_1, a_2, \ldots, a_k$ and alternating:

$$\alpha(a_{\sigma(1)}, a_{\sigma(2)}, \ldots, a_{\sigma(k)}) = \operatorname{sign}(\sigma)\alpha(a_1, a_2, \ldots, a_k)$$

for any permutation $\sigma : \{1, 2, \ldots, k\} \to \{1, 2, \ldots, k\}$.

For any k-tuple of integers $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$, with $1 \le \lambda_i \le n$, we define the k-vector e_λ as follows: when evaluated on elements $a_1, a_2, \ldots, a_k \in (\mathbb{R}^n)^*$, it equals the determinant of the matrix whose ith row has entries given by the components $\lambda_1, \lambda_2, \ldots, \lambda_k$ of a_i, for all $i = 1, 2, \ldots, k$. That is,

$$e_\lambda : (a_1, a_2, \ldots, a_k) \in \left[(\mathbb{R}^n)^*\right]^k \to \det(a_i, e^{\lambda_j})_{ij}\,.$$

The definition is, in some sense, dual to (12.32). Given column vectors $h_1, h_2, \ldots, h_k$, the k-covector e^λ computes the minor with rows $\lambda_1, \lambda_2, \ldots, \lambda_k$ in the matrix whose columns are $h_1, h_2, \ldots, h_k$. The k-vector e_λ, on the other hand, given row vectors (or covectors) $a_1, a_2, \ldots, a_k$, computes the minor with columns $\lambda_1, \lambda_2, \ldots, \lambda_k$ in the matrix with rows $a_1, a_2, \ldots, a_k$.

As for k-covectors, it can be proved that the set

$$\left\{e_\lambda \in \Lambda_k^*(\mathbb{R}^n) : \ \lambda \in \Lambda_{n,k}\right\}$$

is a basis of the vector space $\Lambda_k^*(\mathbb{R}^n)$, $1 \le k \le n$.

Given $n, k \ge 1$, the map

$$(\omega, \alpha) \in \Lambda_k(\mathbb{R}^n) \times \Lambda_k^*(\mathbb{R}^n) \to \langle \omega, \alpha \rangle = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda\, \alpha_\lambda\,,$$

where ω_λ and α_λ are the components of ω and α in the canonical bases e^λ and e_λ, is called *duality*, or *pairing*, of the two spaces. When $k = 1$ the pairing

$$\langle a, h \rangle = \sum_{i=1}^{n} a_i h_i\,,$$

where $a \in (\mathbb{R}^n)^*$ and $h \in \mathbb{R}^n$, can be viewed as the matrix product of the row vector a and the column vector h.

The wedge product can be defined for k-vectors exactly as for k-covectors. We leave it to the reader to extend (12.35), (12.36) and Proposition 3 to k-vectors. In the same way k-vectors also come equipped with a norm (see (12.39)).

We finish with the following simple but useful result.

Theorem 2 *Fix $1 \le k \le n$. For any vectors $h_1, h_2, \ldots, h_k$ of $\mathbb{R}^n$,*

$$h_1 \wedge h_2 \wedge \ldots \wedge h_k = \sum_{\lambda \in \Lambda_{n,k}} \alpha_\lambda e_\lambda \,,$$

where

$$\alpha_\lambda = e^\lambda(h_1, h_2, \ldots, h_k) = \det(e_{\lambda_i}, h_j)_{ij} \,. \tag{12.40}$$

Consequently $|h_1 \wedge h_2 \wedge \ldots \wedge h_k|^2$ equals the sum of the squares of the $k \times k$ minors of the matrix with columns $h_1, h_2, \ldots, h_k$.

Proof Let us compute

$$h_1 \wedge h_2 \wedge \ldots \wedge h_k = \left(\sum_{i=1}^n (e_i, h_1)\, e_i \right) \wedge \ldots \wedge \left(\sum_{i=1}^n (e_i, h_k)\, e_i \right).$$

Given $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k) \in \Lambda_{n,k}$, the wedge product's properties immediately imply that the λth component α_λ of $h_1 \wedge h_2 \wedge \ldots \wedge h_k$ is

$$\alpha_\lambda = \sum_\sigma \operatorname{sign}(\sigma) \prod_{j=1}^n (e_{\sigma(\lambda_j)}, h_j) \,, \tag{12.41}$$

where σ is an arbitrary permutation of the indices $\lambda_1, \lambda_2, \ldots, \lambda_k$ in λ. Now (12.41) gives (12.40), by definition of determinant. The last claim follows from the definition of norm of a k-vector.

12.6 Differential k-Forms

Let $A \subseteq \mathbb{R}^n$ be an open subset and $k \ge 1$. A *differential form of degree k*, or *k-form*, is a map $\omega : A \to \Lambda_k(\mathbb{R}^n)$ sending $x \in A$ to the alternating form of degree k

$$\omega(x) = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda(x)\, dx_\lambda \,. \tag{12.42}$$

The functions $\omega_\lambda(x)$ are called coefficients of ω, and the latter is said of class C^r, $r = 0, 1, \ldots, \infty$, when its coefficients are of class $C^r(A)$.

For $k = 1$ definition (12.42) recovers differential 1-forms, introduced in Sect. 7.2.

In general, for $k \geq 1$, for any $x \in A$ and any k-tuple $(h_1, h_2, \ldots, h_k)$ of vectors in $\mathbb{R}^n$,

$$\omega(x)(h_1, h_2, \ldots, h_k) = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda(x) \det(e_{\lambda_i}, h_j)_{ij} .$$

The operations of sum, multiplication by a scalar and wedge product defined in the previous section for general alternating forms extend naturally to differential forms.

For the reader's benefit we summarise below the general expression of a k-form ω for special values of the degree. When $k = 1$, differential forms of degree 1 are precisely the 1-forms

$$\omega = a_1(x)\, dx_1 + a_2(x) dx_2 + \ldots + a_n(x) dx_n ;$$

2-forms are bilinear forms of type

$$\omega = \sum_{1 \leq i < j \leq n} a_{ij}(x)\, dx_i \wedge dx_j ;$$

$(n-1)$-forms are of type

$$\omega = \sum_{i=1}^{n} a_i(x)\, d\widehat{x_i} ,$$

where $d\widehat{x_i} = dx_1 \wedge \ldots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \ldots \wedge dx_n$, for any $i = 1, 2, \ldots, n$. Finally, n-forms are

$$\omega = a(x)\, dx_1 \wedge dx_2 \wedge dx_3 \wedge \ldots \wedge dx_n .$$

When $k > n$, differential k-forms are zero. At times we shall speak of a function $a : A \subseteq \mathbb{R}^n \to \mathbb{R}$ as a 0-form.

Example 1 Let A be an open subset of $\mathbb{R}^3$. The generic differential 2-form on A is

$$\omega = L(x, y, z)\, dy \wedge dz + M(x, y, z)\, dz \wedge dx + N(x, y, z)\, dx \wedge dy ,$$

where L, M, N are real functions defined on A.

Let $\omega : A \subset \mathbb{R}^n \to \Lambda_k(\mathbb{R}^n)$ be a k-form of class C^1

$$\omega(x) = \sum_{\lambda \in \Lambda_{n,k}} \omega_\lambda(x)\, dx_\lambda .$$

The *exterior derivative* $d\omega$ is the $(k+1)$-form

$$d\omega(x) = \sum_{\lambda \in \Lambda_{n,k}} d\omega_\lambda \wedge dx_\lambda \,. \tag{12.43}$$

To calculate $d\omega$, in other words, we have to compute the differential (a differential 1-form) of the coefficients ω_λ, take the wedge product with the canonical basis element dx_λ, thus obtaining a $(k+1)$-form, and then add over λ.

Clearly the exterior derivative of the constant form dx_λ is zero, while if $g : A \to \mathbb{R}$ is a C^1 function and $\lambda \in \Lambda_{n,k}$, by (12.43) we have

$$d(g\, dx_\lambda) = dg \wedge dx_\lambda \,. \tag{12.44}$$

If a is a function of class $C^1(A)$, its exterior derivative as a 0-form is the usual differential of a.

Example 2 Let $\omega : A \subset \mathbb{R}^2 \to \Lambda_1(\mathbb{R}^2)$ be the 1-form

$$\omega(x, y) = a(x, y)dx + b(x, y)dy$$

with $C^1(A)$ coefficients. Then

$$\begin{aligned} d\omega(x) &= da \wedge dx + db \wedge dy = \\ &= \left(\frac{\partial a}{\partial x}\, dx + \frac{\partial a}{\partial y}\, dy\right) \wedge dx + \left(\frac{\partial b}{\partial x}\, dx + \frac{\partial b}{\partial y}\, dy\right) \wedge dy = \\ &= \frac{\partial a}{\partial y}\, dy \wedge dx + \frac{\partial b}{\partial x}\, dx \wedge dy = \left(\frac{\partial b}{\partial x} - \frac{\partial a}{\partial y}\right) dx \wedge dy \,. \end{aligned}$$

Example 3 Let us calculate the exterior derivative of the 2-form in $\mathbb{R}^3$

$$\omega = xy\, dy \wedge dz + xy^2\, dz \wedge dx + z\, dx \wedge dy \,.$$

We have

$$\begin{aligned} d\omega &= d(xy) \wedge dy \wedge dz + d(xy^2) \wedge dz \wedge dx + dz \wedge dx \wedge dy = \\ &= (y\, dx + x\, dy) \wedge dy \wedge dz \\ &\quad + (y^2\, dx + 2xy\, dy) \wedge dz \wedge dx + dz \wedge dx \wedge dy = \\ &= (y + 2xy + 1)\, dx \wedge dy \wedge dz \,. \end{aligned}$$

Example 4 Let $\omega : A \subset \mathbb{R}^n \to \Lambda_n(\mathbb{R}^n)$ be the n-form

$$\omega(x) = f(x)\, dx_1 \wedge dx_2 \wedge \ldots \wedge dx_n\,,$$

with $f \in C^1(A)$. Since $d\omega$ is an $(n+1)$-form, its differential vanishes identically: $d\omega = 0$.

Proposition 1 *For any $\omega, \overline{\omega} : A \to \Lambda_k(\mathbb{R}^n)$ and $\eta : A \to \Lambda_l(\mathbb{R}^n)$ of class C^1 on the open set A in $\mathbb{R}^n$,*

$$d(\omega + \overline{\omega}) = d\omega + d\overline{\omega}\,, \tag{12.45}$$

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta\,, \tag{12.46}$$

$$\text{If } \omega \text{ is of class } C^2, \qquad d(d\omega) = 0\,. \tag{12.47}$$

Proof Property (12.45) is obvious. To prove (12.46), suppose $g : A \to \mathbb{R}$ is C^1 and take $\lambda \in \Lambda_{n,k}$, $\mu \in \Lambda_{n,l}$. Then

$$d(g\, dx_\lambda \wedge dx_\mu) = dg \wedge dx_\lambda \wedge dx_\mu\,, \tag{12.48}$$

since either $dx_\lambda \wedge dx_\mu$ is zero (and the previous relation is trivial) or it coincides, up to a sign, with one of the dx_ν in the canonical basis of $\Lambda_{k+l}(\mathbb{R}^n)$. In the latter case (12.48) follows from (12.44). So by (12.37),

$$\begin{aligned} d(\omega \wedge \eta) &= \sum_{\substack{\lambda \in \Lambda_{n,k} \\ \mu \in \Lambda_{n,l}}} d(\omega_\lambda\, \eta_\mu\, dx_\lambda \wedge dx_\mu) = \\ &= \sum_{\lambda,\mu} \eta_\mu\, d\omega_\lambda \wedge dx_\lambda \wedge dx_\mu + \sum_{\lambda,\mu} \omega_\lambda\, d\eta_\mu \wedge dx_\lambda \wedge dx_\mu = \\ &= d\omega \wedge \eta + (-1)^k \sum_{\lambda,\mu} \omega_\lambda\, dx_\lambda \wedge d\eta_\mu \wedge dx_\mu = d\omega \wedge \eta + (-1)^k\, \omega \wedge d\eta. \end{aligned}$$

At last, to prove (12.47) we consider a function $g : A \to \mathbb{R}$ of class C^2. By Schwarz's theorem (Sect. 3.4)

$$\begin{aligned} d(dg) &= d\left(\sum_{i=1}^n \frac{\partial g}{\partial x_i}\, dx_i\right) = \sum_{i,j=1}^n \frac{\partial^2 g}{\partial x_i \partial x_j}\, dx_j \wedge dx_i = \\ &= \sum_{1 \le i < j \le n} \left(\frac{\partial^2 g}{\partial x_j \partial x_i} - \frac{\partial^2 g}{\partial x_i \partial x_j}\right) dx_i \wedge dx_j = 0\,. \end{aligned}$$

In the general case, using (12.46) and $d(dx_\lambda) = 0$, we have

$$d(d\omega) = d\left(\sum_{\lambda\in\Lambda_{n,k}} d\omega_\lambda \wedge dx_\lambda\right) = \sum_{\lambda\in\Lambda_{n,k}} d(d\omega_\lambda)\wedge dx_\lambda = 0\,.$$

Let $f : U \to \mathbb{R}^n$ be a map defined on the open set $U \subseteq \mathbb{R}^m$ and of class $C^1(U)$. Consider the element $dx_\lambda = dx_{\lambda_1}\wedge dx_{\lambda_2}\wedge\ldots\wedge dx_{\lambda_k}$ of the canonical basis of $\Lambda_k(\mathbb{R}^n)$. The *pull-back* of the constant k-form dx_λ by f is the k-form $(dx_\lambda)^\# : U \to \Lambda_k(\mathbb{R}^m)$,

$$(dx_\lambda)^\# = \sum_{\mu\in\Lambda_{m,k}} \det\frac{\partial(f_{\lambda_1}, f_{\lambda_2},\ldots,f_{\lambda_k})}{\partial(y_{\mu_1}, y_{\mu_2},\ldots,y_{\mu_k})}\, dy_{\mu_1}\wedge dy_{\mu_2}\wedge\ldots\wedge dy_{\mu_k}. \tag{12.49}$$

In general, if $\omega : A \subset \mathbb{R}^n \to \Lambda_k(\mathbb{R}^n)$ is a k-form and $f(U) \subseteq A$, the form's *pull-back* by f is defined by

$$\begin{aligned}\omega^\#(y) &= \sum_{\lambda\in\Lambda_{n,k}} \omega_\lambda(f(y))\left(dx_{\lambda_1}\wedge dx_{\lambda_2}\wedge\ldots\wedge dx_{\lambda_k}\right)^\# = \\ &= \sum_{\substack{\lambda\in\Lambda_{n,k}\\ .\mu\in\Lambda_{m,k}}} \omega_\lambda(f(y))\det\frac{\partial(f_{\lambda_1}, f_{\lambda_2},\ldots,f_{\lambda_k})}{\partial(y_{\mu_1}, y_{\mu_2},\ldots,y_{\mu_k})}\, dy_{\mu_1}\wedge dy_{\mu_2}\wedge\ldots\wedge dy_{\mu_k}.\end{aligned} \tag{12.50}$$

Clearly $\omega^\# : U \to \Lambda_k(\mathbb{R}^m)$.

The *pull-back* of a 0-form $a : A \to \mathbb{R}$ is the 0-form defined by $a^\#(y) = a(f(y))$ on U.

Example 5 Let $\omega : A \subset \mathbb{R}^n \to \Lambda_1(\mathbb{R}^n)$ be the one-form

$$\omega(x) = \sum_{i=1}^n \omega_i(x)\, dx_i$$

defined on the open set A, and suppose $f : (a, b) \to \mathbb{R}^n$ a C^1 function with $f((a, b)) \subseteq A$. Then

$$\omega^\#(t) = \left(\sum_{i=1}^n \omega_i(f(t))\frac{df_i}{dt}\right) dt\,. \tag{12.51}$$

Example 6 Let $\omega : A \subset \mathbb{R}^3 \to \Lambda_2(\mathbb{R}^3)$ be the two-form

$$\omega = L\,dy \wedge dz + M\,dz \wedge dx + N\,dx \wedge dy\,,$$

and $f : U \subset \mathbb{R}^2 \to \mathbb{R}^3$ a C^1 function on the open set U such that $f(U) \subseteq A$. From (12.49) we have

$$\omega^{\#}(s,t) = \left(L(f(s,t)) \det \frac{\partial(f_2, f_3)}{\partial(s,t)} + \right.$$
$$\left. + M(f(s,t)) \det \frac{\partial(f_3, f_1)}{\partial(s,t)} + N(f(s,t)) \det \frac{\partial(f_1, f_2)}{\partial(s,t)} \right) ds \wedge dt. \tag{12.52}$$

Example 7 Let $\Phi : U \subset \mathbb{R}^n \to \mathbb{R}^n$ be a C^1 map on the open set U. The *pull-back* of the n-form $dx_{(1,2,\ldots,n)} = dx_1 \wedge dx_2 \wedge \ldots \wedge dx_n$ (we shall also write $dx_{12\ldots n}$) is

$$(dx_1 \wedge dx_2 \wedge \ldots \wedge dx_n)^{\#} = J_\Phi(y)\, dy_1 \wedge dy_2 \wedge \ldots \wedge dy_n\,.$$

We conclude the section observing that the *pull-back* operation satisfies the following properties, whose proof we skip for brevity.

Proposition 2 *Given differential forms $\omega, \overline{\omega} : A \to \Lambda_k(\mathbb{R}^n)$, $\eta : A \to \Lambda_l(\mathbb{R}^n)$ and a C^1 map $f : U \subseteq \mathbb{R}^m \to A$, the pull-back by f satisfies:*

$$(\omega + \overline{\omega})^{\#} = \omega^{\#} + \overline{\omega}^{\#}\,,$$
$$(\omega \wedge \eta)^{\#} = \omega^{\#} \wedge \eta^{\#}\,.$$

Moreover, if ω is of class C^1 and f is of class C^2 then

$$d\omega^{\#} = (d\omega)^{\#}\,. \tag{12.53}$$

The *pull-back*, in other words, commutes with the sum, with the wedge product and with the exterior derivative of forms.

12.7 Orientable Manifolds. Integration of k-Forms on Manifolds

In the present section we will discuss the notion of *orientation of a k-dimensional manifold M*. As we shall see, the notion is based on the possibility of choosing an

orientation on every tangent space $T_M(x)$. It is then necessary to analyse first how to endow vector subspaces in $\mathbb{R}^n$ with an orientation.

Theorem 1 *Fix $1 \leq k \leq n$. The vectors $h_1, h_2, \ldots, h_k \in \mathbb{R}^n$ are linearly independent if and only if*

$$h_1 \wedge h_2 \wedge \ldots \wedge h_k \neq 0\,. \tag{12.54}$$

Moreover, the two sets $\{h_1, h_2, \ldots, h_k\}$ and $\{h'_1, h'_2, \ldots, h'_k\}$ generate the same subspace V in $\mathbb{R}^n$ if and only if

$$h_1 \wedge h_2 \wedge \ldots \wedge h_k = c\,(h'_1 \wedge h'_2 \wedge \ldots \wedge h'_k)$$

for some constant $c \neq 0$.

We shall not prove this fact, which only involves elementary properties of determinants by the way. But we remark that it gives a simple criterion to decide whether a given set of vectors is linearly independent, and if two sets span the same subspace.

Example 1 Given $h_1 = (1, 2, -1)$, $h_2 = (1, -1, 1)$, $h_3 = (1, -4, 2)$ in $\mathbb{R}^3$,

$$\begin{aligned} h_1 \wedge h_2 \wedge h_3 &= (e_1 + 2e_2 - e_3) \wedge (e_1 - e_2 + e_3) \wedge (e_1 - 4e_2 + 2e_3) = \\ &= (-3e_{12} + 2\,e_{13} + e_{23}) \wedge (e_1 - 4e_2 + 2e_3) = 3e_{123}\,, \end{aligned} \tag{12.55}$$

so by (12.54) h_1, h_2, h_3 are linearly independent.

Example 2 The vectors $h_1 = (1, -1, 0)$, $h_2 = (-1, 0, 1)$, $k_1 = (1, 1, -2)$, $h_2 = (1, 2, -3)$ in $\mathbb{R}^3$ satisfy

$$h_1 \wedge h_2 = (e_1 - e_2) \wedge (-e_1 + e_3) = -e_{12} + e_{13} - e_{23}\,, \tag{12.56}$$

$$k_1 \wedge k_2 = (e_1 + e_2 - 2\,e_3) \wedge (e_1 + 2\,e_2 - 3\,e_3) = e_{12} - e_{13} + e_{23}\,, \tag{12.57}$$

and so $k_1 \wedge k_2 = -h_1 \wedge h_2$. Hence $\{h_1, h_2\}$ and $\{k_1, k_2\}$ generate the same subspace $V \subset \mathbb{R}^3$.

Let V be a vector subspace of dimension k contained in $\mathbb{R}^n$ and $\{h_1, h_2, \ldots, h_k\}$ a basis. We shall refer to the unit k-vector

$$\alpha = \frac{h_1 \wedge h_2 \wedge \ldots \wedge h_k}{|h_1 \wedge h_2 \wedge \ldots \wedge h_k|}$$

as the *orientation of V induced by the basis* $\{h_1, h_2, \ldots, h_k\}$. By Theorem 1, if $\{h'_1, h'_2, \ldots, h'_k\}$ is another basis of V inducing the orientation β, then $\beta = \alpha$ or $\beta = -\alpha$.

Because of the wedge product's properties, the orientation induced by $\{h_1, h_2, \ldots, h_k\}$ depends on the order of the basis elements.

Example 3 The orientation induced by the canonical basis $\{e_1, e_2, \ldots, e_n\}$ of $\mathbb{R}^n$ is $e_1 \wedge e_2 \wedge \ldots \wedge e_n = e_{12\ldots n}$. On the other hand, the basis $\{e_2, e_1, \ldots, e_n\}$ induces $-e_{12\ldots n}$. We shall say that the orientation on $\mathbb{R}^n$ induced by the canonical basis is *positive*, while the opposite orientation is *negative*.

Example 4 The vectors $\{h_1, h_2, h_3\}$ of Example 1 form a basis of $\mathbb{R}^3$. By (12.55) this basis induces the positive orientation on $\mathbb{R}^3$.

Example 5 Let V be the subspace of $\mathbb{R}^3$ generated by the set $\{h_1, h_2\}$ from Example 2. Due to (12.56), (12.57) the induced orientation on V is

$$\alpha = \frac{1}{\sqrt{3}} \left(-e_{12} + e_{13} - e_{23}\right),$$

while the orientation induced by $\{k_1, k_2\}$, also in Example 2, is $-\alpha$.

A k-manifold M in $\mathbb{R}^n$ is called *orientable* if there exists a continuous map

$$\alpha : M \to \Lambda_k^*(\mathbb{R}^n)$$

such that, for any $x \in M$, $\alpha(x)$ is an orientation on the tangent space $T_M(x)$.

Equivalently, the manifold M is orientable if we can equip every tangent space with an orientation that varies with continuity from point to point.

Let us examine a few special instances.

Let $M \subset \mathbb{R}^n$ be a k-manifold and $\varphi : U \subset \mathbb{R}^k \to \mathbb{R}^n$ a parametrisation of M such that $\varphi(U) = M$. If $\varphi(y)$, for $y \in U$, is the generic point of M, the tangent space to M at $\varphi(y)$ is spanned by $\{\varphi_{y_1}(y), \varphi_{y_2}(y), \ldots, \varphi_{y_k}(y)\}$. The induced orientation is

$$\alpha(y) = \frac{\varphi_{y_1}(y) \wedge \varphi_{y_2}(y) \wedge \ldots \wedge \varphi_{y_k}(y)}{\left|\varphi_{y_1}(y) \wedge \varphi_{y_2}(y) \wedge \ldots \wedge \varphi_{y_k}(y)\right|}. \tag{12.58}$$

and is called *orientation induced on M* by the parametrisation φ. In such a case, clearly, (12.58) is a continuous map and M is orientable.

Moreover, by Proposition 2, Sect. 12.3 and Theorem 2, Sect. 12.5, we have

$$\left|\varphi_{y_1}(y) \wedge \varphi_{y_2}(y) \wedge \ldots \wedge \varphi_{y_k}(y)\right| = \widehat{J}_\varphi(y)\,,$$

so the orientation induced by φ is

$$\alpha(y) = \frac{1}{\widehat{J}_\varphi(y)}\left[\varphi_{y_1}(y) \wedge \varphi_{y_2}(y) \wedge \ldots \wedge \varphi_{y_k}(y)\right]. \tag{12.59}$$

Proposition 1 *Let $V \subset \mathbb{R}^n$ be the subspace spanned by linearly independent vectors $h_1, h_2, \ldots, h_{n-1}$, and*

$$\alpha = \sum_{i=1}^{n} \alpha_i\, \widehat{e}_i$$

the orientation they induce on V, where $\widehat{e}_i = e_1 \wedge \ldots \wedge e_{i-1} \wedge e_{i+1} \wedge \ldots \wedge e_n$. Then

$$h = {}^*\alpha = \sum_{i=1}^{n} \alpha_i (-1)^{i-1} e_i \tag{12.60}$$

does not belong to V. Moreover, $\{h, h_1, h_2, \ldots, h_{n-1}\}$ is a basis of $\mathbb{R}^n$ inducing the positive orientation.

Proof Note that

$$\begin{aligned} h \wedge \frac{h_1 \wedge h_2 \wedge \ldots \wedge h_{n-1}}{|h_1 \wedge h_2 \wedge \ldots \wedge h_{n-1}|} &= h \wedge \alpha = \\ &= \left(\sum_{i=1}^{n} (-1)^{i-1} \alpha_i e_i\right) \wedge \left(\sum_{i=1}^{n} \alpha_i \widehat{e_i}\right) = \sum_{i=1}^{n} (-1)^{i-1} \alpha_i^2 e_i \wedge \widehat{e_i} = \\ &= \left(\sum_{i=1}^{n} \alpha_i^2\right) e_{1,2,\ldots,n} = e_{1,2,\ldots,n}\,. \end{aligned}$$

The claim follows from Theorem 1 and the definition of positive orientation of $\mathbb{R}^n$.

The vector h defined by (12.60) is called *Hodge dual* of the $(n-1)$-vector α. Observe that h has norm 1, just like α.

Let D be a regular domain in $\mathbb{R}^n$. We say

$$\alpha : \partial D \to \Lambda_{n-1}(\mathbb{R}^n)$$

is a *positive orientation* of the boundary ∂D if

$${}^*\alpha(x) = \nu(x)\,, \qquad \forall\, x \in \partial D, \tag{12.61}$$

where $\nu(x)$ is the outer normal of D at the point x.

This notion is well defined, because the boundary ∂D of a regular domain D is an orientable manifold. It is in fact possible to prove, more generally, that any compact submanifold M of $\mathbb{R}^n$ is orientable.

Example 6 Let $\gamma : [a, b] \to \mathbb{R}^n$ be a regular simple curve such that $M = \gamma((a, b))$ is a one-dimensional manifold in $\mathbb{R}^n$ (recall that a regular simple curve may not be a manifold, globally – see the remark at the end of Sect. 12.1). By (12.58), the orientation induced on M by the parametrisation γ at the point $\gamma(t) \in M$ is

$$\alpha(t) = \frac{\gamma'(t)}{|\gamma'(t)|} ,$$

so it coincides with the unit tangent vector $T(t)$ of the curve γ.

Example 7 Let $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a regular surface such that $M = \varphi(\overset{\circ}{D})$ is a two-dimensional manifold in $\mathbb{R}^3$. The orientation induced on M by φ at a point $\varphi(s, t) \in M$ is the 2-vector

$$\alpha(s, t) = \frac{\varphi_s(s, t) \wedge \varphi_t(s, t)}{|\varphi_s(s, t) \wedge \varphi_t(s, t)|} . \tag{12.62}$$

If φ has equations $\varphi(s, t) = (x(s, t), y(s, t), z(s, t))$, then

$$\begin{aligned} \varphi_s \wedge \varphi_t &= (x_s e_1 + y_s e_2 + z_s e_3) \wedge (x_t e_1 + y_t e_2 + z_t e_3) = \\ &= (x_s y_t - x_t y_s) e_{12} + (x_s z_t - x_t z_s) e_{13} + (y_s z_t - y_t z_s) e_{23} . \end{aligned}$$

Therefore by (12.60)

$${}^*\alpha(s, t) = \nu(s, t) = \frac{\varphi_s(s, t) \wedge \varphi_t(s, t)}{|\varphi_s(s, t) \wedge \varphi_t(s, t)|} , \tag{12.63}$$

where $\nu(s, t)$ is the normal to the surface M determined by the parametrisation φ.

The reader should note that in (12.62) the symbol $\wedge$ denotes the *wedge product* of vectors, while in (12.63) the same symbol is the *cross product*.

The previous two examples establish the equivalence, in the cases examined, between the notion of orientation given in this section and the analogous concepts for curves and surfaces in $\mathbb{R}^3$ (Chaps. 6 and 10). In particular, the reader can easily show that the notion of positive orientation of the boundary ∂D of a regular domain of $\mathbb{R}^n$ reduces, in the plane and in space, to the notions already seen in Chaps. 8 and 10. For an example of non-orientable manifold see the discussion on the *Möbius strip* in Sect. 10.5.

In the sequel we shall refer to an open subset $A \subseteq \mathbb{R}^n$ as an n-manifold. At every point $x \in A$ the tangent space to A is $\mathbb{R}^n$, and the *positive orientation* of A is the positive orientation of $\mathbb{R}^n$, i.e. $e_{12\ldots n}$.

Let $M \subset \mathbb{R}^n$ be an orientable k-manifold, $1 \leq k \leq n$, and $\alpha : M \to \Lambda_k^*(\mathbb{R}^n)$ its orientation. Suppose $B \subset M$ is a bounded measurable subset and $\omega : A \to \Lambda_k(\mathbb{R}^n)$ is a continuous and bounded k-form on $B \subseteq A$. *The integral of ω over the set B*, with orientation α, is

$$\int_B \omega = \int_B \langle \omega(x), \alpha(x) \rangle \, dH^k = \sum_{\lambda \in \Lambda_{n,k}} \int_B \omega_\lambda(x) \alpha_\lambda(x) \, dH^k \,. \tag{12.64}$$

Let us denote by $-B$ the set B oriented by $-\alpha$. Then obviously

$$\int_{-B} \omega = -\int_B \omega \,.$$

Moreover, since α is a unit k-vector,

$$\left| \int_B \omega \right| \leq \sup_B |\omega| \cdot H^k(B) \,.$$

Example 8 Suppose $A \subset \mathbb{R}^n$ is an open bounded set and $\omega = f \, dx_1 \wedge dx_2 \wedge \ldots \wedge dx_k$ is a continuous and bounded n-form on A. If we choose on A the positive orientation, by (12.64) we have

$$\int_A \omega = \int_A f(x) \, dx \,. \tag{12.65}$$

Proposition 2 provides a useful formula for computing the integral (12.64).

Proposition 2 *Let $M \subset \mathbb{R}^n$ be a k-manifold, $1 \leq k < n$, $\varphi : U \subset \mathbb{R}^k \to M$ a local parametrisation and $B \subseteq \varphi(U)$ a bounded measurable subset of M, with orientation induced by φ. Then*

$$\int_B \omega = \int_{\varphi^{-1}(B)} \omega^\# \,, \tag{12.66}$$

where $\omega^\#$ is the pull-back by φ, and $\varphi^{-1}(B) \subset \mathbb{R}^k$ has the positive orientation of $\mathbb{R}^k$.

Proof Suppose $\omega(x) = \sum_{\lambda\in\Lambda_{n,k}} \omega_\lambda(x)\, dx_\lambda$, so by (12.50) and (12.65) we obtain

$$\begin{aligned}\int_{\varphi^{-1}(B)} \omega^{\#} &= \int_{\varphi^{-1}(B)} \sum_{\lambda\in\Lambda_{n,k}} \omega_\lambda(\varphi(y)) \det \frac{\partial(\varphi_{\lambda_1}, \varphi_{\lambda_2}, \ldots, \varphi_{\lambda_k})}{\partial(y_1, y_2, \ldots, y_k)}\, dy_1 \wedge dy_2 \wedge \ldots \wedge dy_k = \\ &= \int_{\varphi^{-1}(B)} \sum_{\lambda\in\Lambda_{n,k}} \omega_\lambda(\varphi(y)) \det \frac{\partial(\varphi_{\lambda_1}, \varphi_{\lambda_2}, \ldots, \varphi_{\lambda_k})}{\partial(y_1, y_2, \ldots, y_k)} dy.\end{aligned} \tag{12.67}$$

Now, (12.59) and (12.64) imply

$$\int_B \omega = \int_B \langle \omega(x), \varphi_{y_1} \wedge \varphi_{y_2} \wedge \ldots \wedge \varphi_{y_k} \rangle \frac{1}{\widehat{J}_\varphi(y)}\, dH^k .$$

By Theorem 2, Sect. 12.5,

$$\varphi_{y_1} \wedge \varphi_{y_2} \wedge \ldots \wedge \varphi_{y_k} = \sum_{\lambda\in\Lambda_{n,k}} \det \frac{\partial(\varphi_{\lambda_1}, \varphi_{\lambda_2}, \ldots, \varphi_{\lambda_k})}{\partial(y_1, y_2, \ldots, y_k)}\, e_\lambda ,$$

so this and (12.16) give

$$\begin{aligned}\int_B \omega &= \int_B \sum_{\lambda\in\Lambda_{n,k}} \omega_\lambda \det \frac{\partial(\varphi_{\lambda_1}, \varphi_{\lambda_2}, \ldots, \varphi_{\lambda_k})}{\partial(y_1, y_2, \ldots, y_k)} \frac{1}{\widehat{J}_\varphi(y)}\, dH^k = \\ &= \int_{\varphi^{-1}(B)} \sum_{\lambda\in\Lambda_{n,k}} \omega_\lambda(y) \det \frac{\partial(\varphi_{\lambda_1}, \varphi_{\lambda_2}, \ldots, \varphi_{\lambda_k})}{\partial(y_1, y_2, \ldots, y_k)}\, dy .\end{aligned} \tag{12.68}$$

Equalities (12.67) and (12.68) allow to conclude.

Example 9 Let M be the one-manifold considered in Example 6, with orientation induced by the parametrisation γ. If $\omega : M \to \Lambda_1(\mathbb{R}^n)$ is a 1-form on M, from (12.66) and (12.51) we have

$$\int_M \omega = \int_a^b \sum_{i=1}^n \omega_i(\gamma(t)) \frac{d\gamma_i}{dt}(t)\, dt ,$$

recovering the definition of integral of a one-form seen in Chap. 7.

Example 10 Let M be the two-dimensional manifold considered in Example 7, with orientation induced by the parametrisation φ. If

$$\omega = L(x, y, z)\, dy \wedge dz + M(x, y, z)\, dz \wedge dx + N(x, y, z)\, dx \wedge dy$$

is a continuous and bounded 2-form on M, by (12.66) and (12.52) it follows that

$$\int_M \omega = \iint_D \left[L \det \frac{\partial(\varphi_2, \varphi_3)}{\partial(s, t)} + M \det \frac{\partial(\varphi_3, \varphi_1)}{\partial(s, t)} + N \det \frac{\partial(\varphi_1, \varphi_2)}{\partial(s, t)} \right] ds\, dt\,.$$

Therefore the integral of ω over M coincides with the flux of the vector field (L, M, N) across the regular surface M, in the direction of the normal determined by φ.

Example 11 Let $M = \{(x, y, z, w) :\ x^2 + y^2 = 1,\ z^2 + w^2 = 1,\ x, z > 0\}$ be the two-dimensional submanifold of $\mathbb{R}^4$ considered in Example 4, Sect. 12.3. We have seen that the map

$$\varphi(u, v) = (\cos u, \sin u, \cos v, \sin v)\,, \qquad (u, v) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

is a parametrisation of the entire manifold M and $\widehat{J}_\varphi = 1$. Let us calculate

$$\int_M xy\, dx \wedge dy + xz\, dy \wedge dw\,,$$

where M is oriented by φ. In view of (12.66) and (12.49) the given integral is

$$\int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \left\{ \cos u \sin u \det \frac{\partial(\cos u, \sin u)}{\partial(u, v)} + \right.$$
$$\left. + \cos u \cos v \det \frac{\partial(\sin u, \sin v)}{\partial(u, v)} \right\} du\, dv =$$
$$= \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \cos^2 u \cos^2 v\, du\, dv = \frac{\pi^2}{4}\,.$$

Example 12 Let $f : U \subset \mathbb{R}^{n-1} \to \mathbb{R}$ be a C^1 function on the open set U, and let M denote the graph of f oriented coherently with the orientation induced by the parametrisation

$$\varphi(x') = (x', f(x'))\,, \qquad x' \in U.$$

The $(n-1)$-form $\omega = \sum_{i=1}^{n} \omega_i(x)\, d\widehat{x}_i$, where

(continued)

Example 12 (continued)

$$d\widehat{x_i} = dx_1 \wedge \ldots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \ldots \wedge dx_n\,, \qquad i = 1, 2, \ldots, n, \tag{12.69}$$

is continuous and bounded on M. The orientation induced by φ on M at $(x', f(x'))$ is

$$\alpha(x') = \frac{(e_1 + f_{x_1}(x')e_n) \wedge \ldots \wedge (e_{n-1} + f_{x_{n-1}}(x')e_n)}{\left|(e_1 + f_{x_1}(x')e_n) \wedge \ldots \wedge (e_{n-1} + f_{x_{n-1}}(x')e_n)\right|} =$$

$$= \frac{1}{\sqrt{1 + |Df(x')|^2}} \left(e_{12\ldots n-1} + \sum_{1=1}^{n-1} (-1)^{n-i-1} \frac{\partial f}{\partial x_i}(x') \widehat{e_i} \right),$$

where

$$\widehat{e_i} = e_1 \wedge \ldots \wedge e_{i-1} \wedge e_{i+1} \wedge \ldots \wedge e_n\,, \qquad i = 1, 2, \ldots, n-1. \tag{12.70}$$

Then from (12.64) and (12.15) we have

$$\int_M \omega = \int_U \left\{ \sum_{1=1}^{n-1} (-1)^{n-i-1} \omega_i(x', f(x')) \frac{\partial f}{\partial x_i}(x') + \omega_n(x', f(x')) \right\} dx'\,.$$

To close the section we prove the following result, which rephrases the divergence theorem for $(n-1)$-forms.

Theorem 2 *Let $D \subset \mathbb{R}^n$ be a regular domain and $+\partial D$ its positively oriented boundary. If $\omega : D \to \Lambda_{n-1}(\mathbb{R}^n)$ is an $(n-1)$-form of class C^1,*

$$\int_{+\partial D} \omega = \int_D d\omega\,.$$

Proof Write $\omega(x) = \sum_{i=1}^{n} \omega_i(x)\, d\widehat{x_i}$, with $d\widehat{x_i}$ as in (12.69). Call $\alpha(x)$ the positive orientation of the boundary ∂D of D. Then $\alpha(x) = \sum_{i=1}^{n} \alpha_i(x)\, \widehat{e_i}$, where $\widehat{e_i}$ is defined by (12.70). Using (12.61), (12.60) and (12.64) we obtain

$$\int_{+\partial D} \omega = \int_{\partial D} \sum_{1=1}^{n} \omega_i(x) \alpha_i(x)\, dH^{n-1} = \int_{\partial D} \sum_{1=1}^{n} (-1)^{i-1} \omega_i(x) \nu_i(x)\, dH^{n-1},$$

where $\nu_i(x)$ is the ith component of the outer normal of D at x. The divergence theorem (Sect. 12.4) tells that

$$\int_{+\partial D} \omega = \sum_{1=1}^{n} (-1)^{i-1} \int_D \frac{\partial \omega_i}{\partial x_i}\, dx\,. \tag{12.71}$$

On the other hand

$$d\omega = \sum_{i=1}^{n} d\omega_i \wedge d\widehat{x_i} = \sum_{i=1}^{n} \frac{\partial \omega_i}{\partial x_i}\, dx_i \wedge d\widehat{x_i} =$$

$$= \sum_{1=1}^{n} (-1)^{i-1} \frac{\partial \omega_i}{\partial x_i}\, dx_1 \wedge dx_2 \wedge \ldots \wedge dx_n\,,$$

and so

$$\int_D d\omega = \sum_{1=1}^{n} (-1)^{i-1} \int_D \frac{\partial \omega_i}{\partial x_i}\, dx\,. \tag{12.72}$$

The assertion follows from (12.71) and (12.72).

12.8 Manifolds with Boundary. Stokes's Formula

In this section we will extend to k-manifolds the notion of boundary and Stokes's formula, discussed in Chaps. 10 and 8.

We shall say that $M \subset \mathbb{R}^n$ is a regular k-manifold with boundary, for $1 \leq k < n$, if there exist a k-manifold M_0 and a local parametrisation $\varphi : U \subset \mathbb{R}^k \to M_0$ such that

$$M \subset M_0\,, \qquad M = \varphi(\overset{\circ}{D})\,, \tag{12.73}$$

where $D \subset U$ is a regular domain of $\mathbb{R}^k$ (see Fig. 12.9).

The set $\varphi(\partial D)$ will be called *boundary of the manifold* and denoted by ∂M. It is possible to show that this definition of boundary is independent of the manifold M_0 and of the parametrisation φ; moreover, ∂M is a manifold of dimension $k-1$.

Observe that the regular manifold with boundary M, being contained in $\varphi(U)$, where φ is a local parametrisation of M_0, is clearly orientable. Given an orientation $\alpha(x)$ on M, we wish to show how one can define an orientation on the boundary ∂M that is coherent with the given orientation.

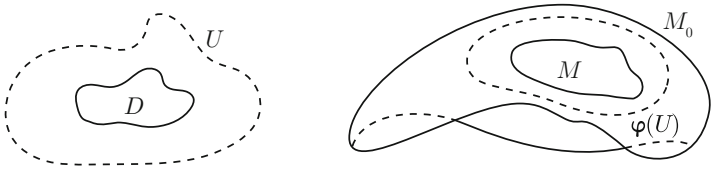


Fig. 12.9

Without loss of generality we may always assume that the orientation induced by the parametrisation φ on M coincides with $\alpha(x)$. In fact, if that were not the case, φ would induce on M the opposite orientation, and then the parametrisation

$$\widetilde{\varphi}(y) = (-y_1, y_2, \dots, y_k)\,,$$

defined on the open set $\widetilde{U} = \{y \in \mathbb{R}^k : (-y_1, y_2, \dots, y_k) \in U\}$, would induce on M the desired orientation $\alpha(x)$.

Let us then fix a point $x_0 \in \partial M$ and set $y_0 = \varphi^{-1}(x_0) \in \partial D$. As ∂D is a $(k-1)$-manifold, we can always find an open neighbourhood U_0 of y_0 with $U_0 \subseteq U$, an open set $W \subset \mathbb{R}^{k-1}$ and an injective C^1 map $\psi : W \to \partial D$ such that $D\psi$ has rank $k-1$ on W and $\psi(W) = \partial D \cap U_0$. Such ψ is then a local parametrisation of ∂D, and with the same argument used above, one proves that ψ can be chosen so to induce on $\partial D \cap U_0$ the positive orientation of the boundary of D.

It is easy to see that $\varphi \circ \psi : W \to \partial M$ is a local parametrisation of ∂M. Namely, $\varphi \circ \psi$ is 1-1, of class C^1 and the rank of $D(\varphi \circ \psi)$ is $k-1$. Moreover, it can be proved that there exists an open neighbourhood A_0 of x_0 such that $(\varphi \circ \psi)(W) = \partial M \cap A_0$.

The portion of boundary ∂M is said to be *coherently oriented* (with M) when its orientation coincides with the orientation induced by the local parametrisation $\varphi \circ \psi$. Repeating these considerations for every point of ∂M, one defines an orientation on the entire manifold ∂M, called *orientation induced* by the orientation of M or *orientation coherent* with M (see Fig. 12.10, and compare to Fig. 10.23 in Sect. 10.5). Note how this definition agrees with the positive orientation of the boundary of a regular surface.

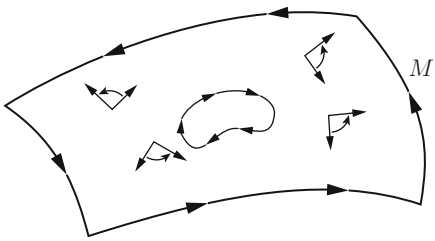


Fig. 12.10 The boundary ∂M of the manifold M is oriented coherently with M

We state, without proof, a useful result from Linear Algebra.

Proposition 1 *Let $L : \mathbb{R}^m \to \mathbb{R}^n$ be a linear map and $(\ell_{ij})_{ij}$, $i = 1, 2, \dots, n$, $j = 1, 2, \dots, m$, its matrix in the canonical bases. Given vectors $h_1, h_2, \dots, h_k$ in $\mathbb{R}^m$, with $1 \le k \le \min\{n, m\}$, define*

$$\alpha = h_1 \wedge h_2 \wedge \ldots \wedge h_k \in \Lambda_k^*(\mathbb{R}^m), \quad \beta = L(h_1) \wedge L(h_2) \wedge \ldots \wedge L(h_k) \in \Lambda_k^*(\mathbb{R}^n),$$

and $L_{\lambda,\mu} = \det(\ell_{\lambda_r,\mu_s})$, $r, s = 1, 2, \ldots, k$, *where* $\lambda \in \Lambda_{n,k}$, $\mu \in \Lambda_{m,k}$. *Then*

$$\beta_\lambda = \sum_{\mu \in \Lambda_{m,k}} L_{\lambda,\mu} \alpha_\mu \,, \qquad \forall\, \lambda \in \Lambda_{n,k}. \tag{12.74}$$

What this says is the following: in the matrix with columns $L(h_1), L(h_2), \ldots, L(h_k)$, the minor formed by the rows $\lambda_1, \lambda_2, \ldots, \lambda_k$ is a linear combination, with coefficients $L_{\lambda,\mu}$, of the minors α_μ formed by the rows $\mu_1, \mu_2, \ldots, \mu_k$ in the matrix with columns $h_1, h_2, \ldots, h_k$.

Proposition 2 *Let* M *be a manifold with boundary,* $D \subset U \subseteq \mathbb{R}^k$ *a regular domain and* $\varphi : U \to \mathbb{R}^n$ *a map satisfying* (12.73)*. If* M *is oriented by* φ*, and* ∂M *is coherently oriented, then any continuous* $(k-1)$*-form* $\omega : \partial M \to \Lambda_{k-1}(\mathbb{R}^n)$ *satisfies*

$$\int_{\partial M} \omega = \int_{+\partial D} \omega^{\#} \,, \tag{12.75}$$

where $\omega^{\#}$ *is the pull-back by* φ.

Proof As ∂M is compact, the reasoning used for the lemma on partitions of unity, Sect. 12.3, easily shows that there are finitely many maps $\psi_i : W_i \subset \mathbb{R}^{k-1} \to \partial D$ and continuous functions $\eta_i : \partial M \to [0, 1]$, $i = 1, 2, \ldots, N$, such that each ψ_i is a local parametrisation of ∂D, and

$$\bigcup_{i=1}^{N} \psi_i(W_i) = \partial D \qquad \text{and so} \quad \bigcup_{i=1}^{N} (\varphi \circ \psi_i)(W_i) = \partial M \,,$$

$$\eta_i(x) = 0 \qquad \text{if } x \in \partial M - (\varphi \circ \psi_i)(W_i) \,, \qquad i = 1, 2, \ldots, N, \tag{12.76}$$

$$\sum_{i=1}^{N} \eta_i(x) = 1 \,, \qquad \forall\, x \in \partial M. \tag{12.77}$$

The considerations made at the beginning of the section show that it is not restrictive to suppose that the ψ_i induce on $\partial D \cap \psi_i(W_i)$ the positive orientation. Hence the $\varphi \circ \psi_i$ induce on $\partial M \cap (\varphi \circ \psi_i)(W_i)$ the orientation coherent with the orientation induced by φ on M.

Call $\eta_i\, \omega$, $i = 1, 2, \ldots, N$, the $(k-1)$-form

$$(\eta_i\, \omega)(x) = \sum_{\lambda \in \Lambda_{n,k-1}} \eta_i(x) \omega_\lambda(x)\, dx_\lambda \,, \qquad x \in \partial M.$$

By (12.76), (12.77) and (12.66) we have

$$\int_{\partial M}\omega=\sum_{i=1}^{N}\int_{(\varphi\circ\psi_i)(W_i)}\eta_i\omega=$$

$$=\sum_{i=1}^{N}\int_{W_i}\sum_{\lambda\in\Lambda_{n,k-1}}(\eta_i\omega_\lambda)(\varphi\circ\psi_i)\det\frac{\partial(\varphi_{\lambda_1}\circ\psi_i,\varphi_{\lambda_2}\circ\psi_i,\ldots,\varphi_{\lambda_{k-1}}\circ\psi_i)}{\partial(z_1,z_2,\ldots,z_{k-1})}dz=$$

$$=\sum_{i=1}^{N}\int_{W_i}\sum_{\lambda\in\Lambda_{n,k-1}}\eta_i\omega_\lambda\left((D\varphi)\left(\frac{\partial\psi_i}{\partial z_1}\right)\wedge(D\varphi)\left(\frac{\partial\psi_i}{\partial z_2}\right)\wedge\ldots\wedge(D\varphi)\left(\frac{\partial\psi_i}{\partial z_{k-1}}\right)\right)_\lambda dz\,.$$

Applying (12.74) to $L=(D\varphi)(\psi_i(z))$, we then find

$$\int_{\partial M}\omega=$$

$$=\sum_{i=1}^{N}\sum_{\substack{\lambda\in\Lambda_{n,k-1}\\ \mu\in\Lambda_{k,k-1}}}\int_{W_i}\eta_i\omega_\lambda\det\frac{\partial(\varphi_{\lambda_1},\varphi_{\lambda_2},\ldots,\varphi_{\lambda_{k-1}})}{\partial(y_{\mu_1},y_{\mu_2},\ldots,y_{\mu_{k-1}})}\det\frac{\partial(\psi_{i,\mu_1},\psi_{i,\mu_2},\ldots,\psi_{i,\mu_{k-1}})}{\partial(z_1,z_2,\ldots,z_{k-1})}dz= \tag{12.78}$$

$$=\sum_{i=1}^{N}\sum_{\substack{\lambda\in\Lambda_{n,k-1}\\ \mu\in\Lambda_{k,k-1}}}\int_{\psi_i(W_i)}\eta_i\omega_\lambda\det\frac{\partial(\varphi_{\lambda_1},\varphi_{\lambda_2},\ldots,\varphi_{\lambda_{k-1}})}{\partial(y_{\mu_1},y_{\mu_2},\ldots,y_{\mu_{k-1}})}\,dy_{\mu_1}\wedge dy_{\mu_2}\wedge\ldots\wedge dy_{\mu_{k-1}}.$$

By (12.49) on the other hand, since $\eta_i(\varphi(y))=0$ if $y\in\partial D-\psi_i(W_i)$, $i=1,2,\ldots,N$ and $\sum\limits_{i=1}^{N}\eta_i(\varphi(y))=1$ for any $y\in\partial D$, we have

$$\int_{+\partial D}\omega^{\#}=\sum_{i=1}^{N}\int_{\psi_i(W_i)}(\eta_i\circ\varphi)\omega^{\#}=\sum_{i=1}^{N}\int_{\psi_i(W_i)}(\eta_i\omega)^{\#}= \tag{12.79}$$

$$=\sum_{i=1}^{N}\sum_{\substack{\lambda\in\Lambda_{n,k-1}\\ \mu\in\Lambda_{k,k-1}}}\int_{\psi_i(W_i)}\eta_i\omega_\lambda\det\frac{\partial(\varphi_{\lambda_1},\varphi_{\lambda_2},\ldots,\varphi_{\lambda_{k-1}})}{\partial(y_{\mu_1},y_{\mu_2},\ldots,y_{\mu_{k-1}})}\,dy_{\mu_1}\wedge$$

$$\wedge\,dy_{\mu_2}\wedge\ldots\wedge dy_{\mu_{k-1}}.$$

At this point (12.75) follows by comparing (12.78) and (12.79).

At last we prove the following

Stokes's Formula in $\mathbb{R}^n$ *Let M be a regular manifold with coherently oriented boundary ∂M. If $\omega : A \to \Lambda_{k-1}(\mathbb{R}^n)$ is a $(k-1)$-form of class C^1 on an open subset A of $\mathbb{R}^n$ containing M,*

$$\int_{\partial M} \omega = \int_M d\omega\,.$$

Proof Consider a parametrisation $\varphi : U \subset \mathbb{R}^k \to \mathbb{R}^n$ and a regular domain D satisfying (12.73). Assume the orientation induced by φ on M is the given one on M. By Proposition 2

$$\int_{\partial M} \omega = \int_{+\partial D} \omega^{\#}\,, \tag{12.80}$$

At the same time (12.53) implies

$$\int_M d\omega = \int_D (d\omega)^{\#} = \int_D d\omega^{\#}\,. \tag{12.81}$$

Theorem 2 in the previous section tells that

$$\int_{+\partial D} \omega^{\#} = \int_D d\omega^{\#}\,,$$

so keeping (12.80) and (12.81) into account, the claim follows.

Appendix to Chap. 12

12.9 Exact and Closed Differential Forms

Consider a differential form of class C^1 over the open set $A \subseteq \mathbb{R}^n$

$$\omega = \sum_{i=1}^{n} a_i(x)\, dx_i\,.$$

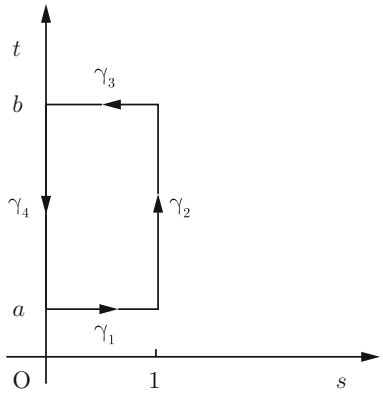
Recall that ω is called *closed* when

$$\frac{\partial a_i}{\partial x_j}(x) = \frac{\partial a_j}{\partial x_i}(x)\,, \qquad \forall\, i,\, j = 1, 2, \ldots, n,$$

for any $x \in A$, and *exact* if

$$df(x) = \omega(x)\,, \qquad \forall\, x \in A$$

Fig. 12.11



for some function $f : A \to \mathbb{R}$. In Sect. 7.6 we saw that every exact form is closed. Here we set out to provide conditions for the converse.

An open set A of $\mathbb{R}^n$ is called *simply connected* if every closed curve $\varphi : [a, b] \to A$ of class C^2 is *homotopic* to a point (see Sect. 7.6).

Theorem 1 *If $A \subseteq \mathbb{R}^n$ is a simply connected open set and ω is a closed 1-form of class C^1 on A, then ω is exact on A.*

Proof Due to the characterisation of exact forms of Sect. 7.3 it suffices to show that

$$\int_\gamma \omega = 0 \tag{12.82}$$

for any piecewise-regular, closed curve γ contained in A. We shall only prove (12.82) in case γ is of class C^2 (the general case can be then obtained by approximation). Fix a closed C^2 curve $\gamma : [a, b] \to A$. As A is simply connected, γ is homotopic to a point $x_0 \in A$, meaning that there exists a map $f : [0, 1] \times [a, b] \to A$, of class C^2, such that

$$f(0, t) = \gamma(t)\,, \qquad \forall\, t \in [a, b], \tag{12.83}$$

$$f(1, t) = x_0\,, \qquad \forall\, t \in [a, b], \tag{12.84}$$

$$f(s, a) = f(s, b)\,, \qquad \forall\, s \in [0, 1]. \tag{12.85}$$

Let R be the rectangle $[0, 1] \times [a, b]$ and $\gamma_1, \gamma_2, \gamma_3, \gamma_4$, the four edges of R oriented as in Fig. 12.11.

Let $\omega^\#$ be the one-form on R defined by

$$\omega^\#(s, t) = \sum_{i=1}^{n} a_i(f(s, t))\left(\frac{\partial f_i}{\partial s}\, ds + \frac{\partial f_i}{\partial t}\, dt\right), \qquad (s, t) \in R,$$

where $f_1, f_2, \dots, f_n$ are the components of f ($\omega^\#$ is the *pull-back* of ω by f, as defined in Sect. 12.6). By (12.85) we have

$$\begin{aligned}\int_{\gamma_1} \omega^\# &= \int_0^1 \sum_{i=1}^n a_i(f(s,a)) \frac{\partial f_i}{\partial s}(s,a)\, ds = \\ &= \int_0^1 \sum_{i=1}^n a_i(f(s,b)) \frac{\partial f_i}{\partial s}(s,b)\, ds = -\int_{\gamma_3} \omega^\#\end{aligned} \tag{12.86}$$

and from (12.84),

$$\int_{\gamma_2} \omega^\# = \int_a^b \sum_{i=1}^n a_i(f(1,t)) \frac{\partial f_i}{\partial t}(1,t)\, dt = 0\,. \tag{12.87}$$

At the same time (12.83) implies

$$\int_{\gamma_4} \omega^\# = -\int_\gamma \omega\,. \tag{12.88}$$

The Stokes theorem on the plane (see (8.43)) tells that

$$\int_{+\partial R} \omega^\# = \iint_R \left\{ \frac{\partial}{\partial s}\left(\sum_{i=1}^n a_i(f(s,t)) \frac{\partial f_i}{\partial t} \right) - \frac{\partial}{\partial t}\left(\sum_{j=1}^n a_j(f(s,t)) \frac{\partial f_j}{\partial s} \right) \right\} ds\, dt\,,$$

where $+\partial R$ is the boundary of R with the positive orientation.

Since ω is closed, the integrand in the right-hand side is zero, as the following relations show:

$$\begin{aligned}\frac{\partial}{\partial s}\left(\sum_{i=1}^n a_i(f(s,t)) \frac{\partial f_i}{\partial t} \right) &= \sum_{i,j=1}^n \frac{\partial a_i}{\partial x_j} \frac{\partial f_j}{\partial s} \frac{\partial f_i}{\partial t} - \sum_{i=1}^n a_i \frac{\partial^2 f_i}{\partial t \partial s}\,, \\ \frac{\partial}{\partial t}\left(\sum_{j=1}^n a_j(f(s,t)) \frac{\partial f_j}{\partial s} \right) &= \sum_{i,j=1}^n \frac{\partial a_j}{\partial x_i} \frac{\partial f_i}{\partial t} \frac{\partial f_j}{\partial s} - \sum_{j=1}^n a_j \frac{\partial^2 f_j}{\partial s \partial t}\,.\end{aligned}$$

Hence we obtain

$$\int_{+\partial R} \omega^\# = 0$$

and by (12.86), (12.87) and (12.88) we reach the conclusion (12.82).

The notions of closed and exact differential form introduced in Chap. 7 for 1-forms extend in a natural way to any degree k.

Let $\omega : A \subseteq \mathbb{R}^n \to \Lambda_k(\mathbb{R}^n)$, $n, k \geq 1$, be a differential form. We call it *exact on* A if there is a $(k-1)$-form η, differentiable on A, such that

$$d\eta = \omega$$

($d\eta$ is the exterior derivative of η introduced in Sect. 12.6. The differential form η is called a primitive of ω.)

Note how this definition generalises the one given for one-forms in Sect. 7.3, as we have agreed to call 0-*form* any function $f : A \to \mathbb{R}$ and to define its exterior derivative as the usual differential.

Example 1 The two-form

$$\omega = x\,dx \wedge dy + z\,dz \wedge dx$$

is exact on $\mathbb{R}^3$. In fact $\omega = d\eta$ (for the definition of exterior derivative $d\eta$ see Sect. 12.6), where

$$\eta = -xy\,dx - xz\,dz\,.$$

Example 2 If $\omega = f(x)\,dx_1 \wedge dx_2 \wedge \ldots \wedge dx_n : \mathbb{R}^n \to \Lambda_n(\mathbb{R}^n)$ is an n-form of class C^1, then it is exact on $\mathbb{R}^n$. In fact, defining

$$g(x) = \int_0^{x_1} f(t, x_2, \ldots, x_n)\,dt$$

we have $d\eta = \omega$, where η is the $(n-1)$-form

$$\eta(x) = g(x)\,dx_2 \wedge dx_3 \wedge \ldots \wedge dx_n\,, \qquad x \in \mathbb{R}^n.$$

Example 3 Consider the 1-form

$$\omega(x, y) = a(x, y)dx + b(x, y)dy\,,$$

with C^1 coefficients on an open set $A \subseteq \mathbb{R}^2$. We have seen in Example 2, Sect. 12.6, that the exterior derivative is

$$d\omega(x) = \left(\frac{\partial b}{\partial x} - \frac{\partial a}{\partial y}\right) dx \wedge dy\,.$$

(continued)

Example 3 (continued)
That ω is exact on A means that ω is the differential of a 0-form, i.e. of a function $f \in C^2(A)$. In that case

$$\omega = df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \,,$$

and so

$$d\omega = \left(\frac{\partial^2 f}{\partial y \partial x} - \frac{\partial^2 f}{\partial x \partial y} \right) dx \wedge dy = 0 \,.$$

In other terms $d(df) = 0$ on A. This property holds in general, and is expressed by saying that *an exact form is closed*, according to the following definition.

Let $\omega : A \to \Lambda_k(\mathbb{R}^n)$, $k \geq 1$, be a k-form of class C^1 on the open set A of $\mathbb{R}^n$. Then ω is said to be *closed on* A if

$$d\omega = 0$$

on A. If ω is exact on A and one of its primitives η is a k-form of class $C^2(A)$, then ω is closed on A. That is because (12.47) implies

$$d\omega = d(d\eta) = 0 \,.$$

An n-form ω of class $C^1(\mathbb{R}^n)$ is always closed on $\mathbb{R}^n$, since $d\omega$ is an $(n+1)$-form and hence identically zero. Example 2 above shows that ω is exact as well.

Example 4 The 3-form in $\mathbb{R}^4$

$$\omega = w \, dx \wedge dy \wedge dz + (x^2 w - y) \, dx \wedge dz \wedge dw \,,$$

is closed on $\mathbb{R}^4$, since

$$\begin{aligned} d\omega &= dw \wedge dx \wedge dy \wedge dz + (2xw \, dx + x^2 \, dw - dy) \wedge dx \wedge dz \wedge dw = \\ &= dw \wedge dx \wedge dy \wedge dz - dy \wedge dx \wedge dz \wedge dw = 0 \,. \end{aligned}$$

We conclude by mentioning that the theorem proved in Sect. 7.5 can be generalised to arbitrary k-forms. We will just state the following result.

Theorem 2 *Any closed k-form ω defined on a star-shaped open subset $A \subset \mathbb{R}^n$ is exact on A.*

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